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ASVAB

Armed Services Vocational
Aptitude Battery

Version 1.0

Sections

- 1. Arithmetic Reasoning**
- 2. Auto and Shop Information**
- 3. Electronic Information**
- 4. General Science**
- 5. Mathematical Knowledge**
- 6. Paragraph Comprehension**
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Section 1: Arithmetic Reasoning

QUESTION: 1

If a car is towed 12 miles to the repair shop, and the tow charge is \$3.50 per mile, how much did the tow cost?

- A. \$12.00
- B. \$3.50
- C. \$42.00
- D. \$100.00

Answer(s): C

Explanation:

Multiply 12 miles by \$3.50 per mile to get \$42.00.

QUESTION: 2

The sum of two numbers is 70. One number is 8 more than the other. What's the smaller number?

- A. 31
- B. 33
- C. 35
- D. 36

Answer(s): A

Explanation:

Let x = the smaller number and $x + 8$ equal the larger number. Because the sum of the two numbers equals 70, you can express this mathematically as $x + (x + 8) = 70$. Now all you have to do is solve for x . Combine the like terms: $2x + 8 = 70$. Then subtract 8 from both sides of the equation: $2x + 8 - 8 = 70 - 8$, or $2x = 62$. Divide both sides of the equation by two, and you find that x is equal to 31.

QUESTION: 3

A sales manager buys antacid in bottles by the gross. If he goes through 3 bottles of antacid every day, how long will the gross last?

- A. 144 days
- B. 3 days
- C. 20 days
- D. 48 days

Answer(s): D

Explanation:

144 bottles are in a gross. $144 \div 3$ (bottles per day) = 48 days.

QUESTION: 4

Jenny's test grades are 93, 89, 96, and 98.

If she wishes to raise her average to 95, what does she need to score on her next test?

- A. 100
- B. 99
- C. 97
- D. 95

Answer(s): B

Explanation:

To determine Jenny's average, add the test scores and divide the sum by the number of tests taken.

You want to know what she would need to make on the next test in order to achieve an average of 95.

The formula can be set up as $(93 + 89 + 96 + 98 + x) \div 5 = 95$.

Combining the like terms results in $(376 + x) \div 5 = 95$.

Multiplying both sides by 5 results in $376 + x = 475$. So, $x = 99$.

QUESTION: 5

A waitress earns an average tip of 12% of the cost of the food she serves.

If she serves \$375 worth of food in one evening, how much money in tips will she earn on average?

- A. \$37
- B. \$45
- C. \$42
- D. \$420

Answer(s): B

Explanation:

Multiply the total amount spent on drinks, \$375, by 12% to determine the amount of tips. $\$375 \times 0.12 = \45 .

QUESTION: 6

How many square feet of carpeting are needed to carpet a 12-foot $\times$ 12-foot room?

- A. 24
- B. 120
- C. 48
- D. 144

Answer(s): D

Explanation:

Square footage is determined by multiplying length by width, or $12 \times 12 = 144$.

QUESTION: 7

Carpet stain protector costs \$0.65 per square yard to apply.
How much will it cost to apply the protectant to a 16-foot × 18-foot carpet?

- A. \$187.20
- B. \$62.40
- C. \$20.80
- D. \$96.00

Answer(s): C

Explanation:

To determine the number of square yards to be protected, multiply 16 feet by 18 feet to determine the number of square feet, 288. Then divide 288 by 9 to convert square feet to square yards (1 square yard = 3 feet × 3 feet = 9 square feet). Multiply the quotient, 32 square yards, by the cost of protection per square yard, \$0.65, to get the correct answer, \$20.80. Remember to perform all the steps in a calculation.

QUESTION: 8

A printing plant that produces baseball cards has a monthly overhead of \$6,000. It costs 18 cents to print each card, and the cards sell for 30 cents each.
How many cards must the printing plant sell each month in order to make a profit?

- A. 30,000
- B. 40,000
- C. 50,000
- D. 60,000

Answer(s): C

Explanation:

Let x = the number of cards printed and sold each month. Therefore, cost is equal to $6000 + 18x$, and revenue is equal to $0.30x$. You're looking for the point where revenue is greater than the cost (revenue > cost). The inequity is $0.30x > 6000 + 18x$. Subtracting $18x$ from both sides of the inequity results in $0.12x > 6000$. Divide both sides by 0.12. The result is that $x > 50,000$. The printing plant would have to print and sell 50,000 cards per month to make a profit.

QUESTION: 9

Joe received an hourly wage of \$8.15. His boss gave him a 7% raise.
How much does Joe make per hour now?

- A. \$0.57
- B. \$8.90
- C. \$8.72
- D. \$13.85

Answer(s): C

Explanation:

To calculate the new wage, start off by multiplying $\$8.15 \times 0.07 = \0.57 . Then add that number (the amount of Joe's raise) to his original hourly wage. Joe's new hourly wage is $\$8.15 + \$0.57 = \$8.72$.

QUESTION: 10

Alice leaves her house, driving east at 45 miles per hour (mph). Thirty minutes later, her husband Dave notices she forgot her cell phone and sets off after her. How fast must Dave travel in order to catch up with Alice 3 hours after he leaves?

- A. 49 mph
- B. 50.5 mph
- C. 52.5 mph
- D. 54 mph

Answer(s): C

Explanation:

By the time Dave leaves, Alice has already been traveling for half an hour. Three hours later, she would've been traveling for $3\frac{1}{2}$ hours at 45 mph, or 157.5 miles. In order to travel 157.5 miles in 3 hours, Dave would have to travel at 52.5 mph (divide 157.5 by 3).

QUESTION: 11

A baker made 20 pies. A Boy Scout troop buys one-fourth of his pies, a preschool teacher buys one-third of his pies, and a caterer buys one-sixth of his pies. How many pies does the baker have left?

- A. $\frac{3}{4}$
- B. 15
- C. 12
- D. 5

Answer(s): D

Explanation:

Convert the different denominators to a common denominator that all the denominators can divide into evenly.

4, 3, and 6 all divide evenly into 12.

To convert $\frac{1}{4}$ to $x/12$, divide 12 (the new common denominator) by 4 (the old common denominator) to get 3.

Then multiply $\frac{1}{4}$ by $\frac{3}{3}$ (another way of saying 1).

Do the same calculation for the other fractions: $\frac{1}{3} = \frac{4}{12}$ and $\frac{1}{6} = \frac{2}{12}$.

Then add the new numerators together: $3 + 4 + 2 = 9$.

This gives you your new added numerator.

Place the added numerator over the new denominator, and you can see that $\frac{9}{12}$ of the pies have been sold.

$\frac{9}{12}$ can be reduced to $\frac{3}{4}$.

$\frac{3}{4}$ or 75% of the pies have been sold.

$20 \times 0.75 = 15$.
 15 of 20 pies have been sold.
 $20 - 15 = 5$ pies remaining

QUESTION: 12

Miriam bought five cases of motor oil on sale. A case of motor oil normally costs \$24.00, but she was able to purchase the oil for \$22.50 a case.
 How much money did Miriam save on her entire purchase?

- A. \$7.50
- B. \$1.50
- C. \$8.00
- D. \$22.50

Answer(s): A

Explanation:

Subtract the sale price from the regular price: $\$24.00 - \$22.50 = \$1.50$. Multiply the remainder by the number of cases to get your $\$1.50 \times 5 = \7.50 .

QUESTION: 13

A security guard walks the equivalent of six city blocks when he makes a circuit around the building.

If he walks at a pace of eight city blocks every 30 minutes, how long will it take him to complete a circuit around the building, assuming he doesn't run into any thieves?

- A. 20.00 minutes
- B. 3.75 minutes
- C. 22.50 minutes
- D. 24.00 minutes

Answer(s): C

Explanation:

Divide 30 by 8 to determine that the security guard takes 3.75 minutes to walk one city block. Multiply 3.75 by 6, the number of blocks it takes to complete the circuit, to arrive at 22.50, or $22\frac{1}{2}$ minutes.

QUESTION: 14

The population of Grand Island, Nebraska, grew by 600,000 people between 1995 and 2005, one-fifth more than the town council predicted.

The town council originally predicted the city's population would grow by _____.

- A. 400,000
- B. 500,000
- C. 300,000
- D. 200,000

Answer(s): B

Explanation:

Let x = the original estimate. An additional one-fifth would be $\frac{6}{5}x$, or 120% of x .

The equation can be expressed as $1.2x = 600000$.

To solve for x , divide both sides of the equation by 1.2.

$x = 500,000$.

QUESTION: 15

Joan is taking an admissions examination.

If she has to get at least 40 of the 60 questions right to pass, what percent of the questions does she need to answer correctly?

- A. 30%
- B. 40%
- C. $66\frac{1}{3}\%$
- D. $66\frac{2}{3}\%$

Answer(s): D

Explanation:

Divide the number of questions she must get right (40) by the total number of questions (60) to reach $66\frac{2}{3}\%$.

QUESTION: 16

A teacher deposited \$3,000 in a retirement fund.

If she didn't add any more money to the fund, which earns an annual interest rate of 6%, how much money would she have in 1 year?

- A. \$180
- B. \$3,006
- C. \$3,180
- D. \$6,000

Answer(s): C

Explanation:

To determine the amount of interest earned, multiply the principal (\$3,000) by the interest rate (6%) and the number of years interest accrues (1 year): $\$3,000 \times 0.06 \times 1 = \180 .

Add the interest earned to the principal to show how much total money the teacher would have: $\$180 + \$3,000 = \$3,180$.

QUESTION: 17

The high-school track measures one quarter of a mile around.

How many laps would you have to run in order to run three and a half miles?

- A. 12
- B. 14

- C. 16
- D. 18

Answer(s): B

Explanation:

Divide the total number of laps by the length of one lap. $31\frac{1}{2} \div \frac{1}{4}$. First, convert the mixed number to a fraction, then divide by $\frac{1}{4}$.
 $7\frac{1}{2} \div \frac{1}{4} = 28\frac{2}{2}$, which can be reduced to 14.

QUESTION: 18

Karl is driving in Austria, where the speed limit is posted in kilometers per hour. The car's speedometer shows that he's traveling at a rate of 75 kilometers per hour. Karl knows that a kilometer is about $\frac{5}{8}$ of a mile. Approximately how many miles per hour is Karl traveling?

- A. 47
- B. 120
- C. 50
- D. 53

Answer(s): A

Explanation:

A kilometer is $\frac{5}{8}$ of a mile, so multiply $75 \times \frac{5}{8}$, or $75/1 \times 5/8 = 375/8$. Divide 8 into 375 to reduce the fraction and determine that Karl was traveling at 47 miles per hour.

QUESTION: 19

A carpenter earns \$12.30 an hour for a 40-hour week. His overtime pay is $1\frac{1}{2}$ times his base pay. If he puts in a 46-hour week, how much is his weekly pay?

- A. \$602.70
- B. \$492.00
- C. \$565.80
- D. \$110.70

Answer(s): A

Explanation:

$\$12.30 \times 40$ hours = \$492, his base pay per week. $\$12.30 \times 1.5 = \18.45 , his overtime rate per hour. $\$18.45$ (overtime rate per hour) $\times 6$ (hours of overtime) = \$110.70 (overtime pay). $\$492.00$ (base pay) + \$110.70 (overtime pay) = \$602.70 (total pay for the week).

QUESTION: 20

An office building has 30 employees and allows 42 square feet of work space per employee. If five more employees are hired, how much less work space will each employee have?

- A. 6 square feet
- B. 7 square feet
- C. 7.5 square feet
- D. 8 square feet

Answer(s): A

Explanation:

The office has 1,260 square feet of space (multiply 42 square feet by 30 employees). With 35 employees, each employee will have 36 square feet of work space ($1,260 \div 35$), which is 6 square feet less than originally.

QUESTION: 21

Stan bought a monster truck for \$2,000 down and payments of \$450 a month for five years. What's the total cost of the monster truck?

- A. \$4,250
- B. \$29,000
- C. \$27,000
- D. \$34,400

Answer(s): B

Explanation:

Five years contain 60 months, so multiply \$450 (monthly payment) $\times 60 = \$27,000$ (total payments). Then add \$27,000 (total payments) + \$2,000 (down payment) = \$29,000 (total cost).

QUESTION: 22

Darla spent \$120.37 on groceries in January, \$108.45 in February, and \$114.86 in March. What was the average monthly cost of Darla's groceries?

- A. \$343.68
- B. \$110.45
- C. \$114.86
- D. \$114.56

Answer(s): D

Explanation:

Add the three monthly amounts to determine the total amount Darla spent on groceries: $\$120.37 + \$108.45 + \$114.86 = \343.68 .

Divide the total by 3 to determine the average monthly cost: \$114.56.

QUESTION: 23

Keith is driving from Reno to Kansas City to meet his girlfriend. The distance between the two cities is 1,650 miles.

If Keith can average 50 miles per hour, how many hours will it take him to complete his trip?

- A. 8 hours
- B. 30 hours
- C. 33 hours
- D. 82 hours

Answer(s): C

Explanation:

1,650 miles (total distance) ÷ 50 miles per hour (average speed) = 33 hours.

QUESTION: 24

Michael needs 55 gallons of paint to paint an apartment building. He would like to purchase the paint for the least amount of money possible.

Which of the following should he buy?

- A. two 25-gallon buckets at \$550 each
- B. eleven 5-gallon buckets at \$108 each
- C. six 10-gallon buckets at \$215 each
- D. fifty-five 1-gallon buckets at \$23 each

Answer(s): B

Explanation:

Determine the cost of each option.

Choice (two 25-gallon buckets at \$550 each): doesn't provide enough paint (2×25 gallons = 50 gallons).

Choice (eleven 5-gallon buckets at \$108 each): $11 \times \$108 = \$1,188$.

Choice (six 10-gallon buckets at \$215 each): $6 \times \$215 = \$1,290$.

Choice (fifty-five 1-gallon buckets at \$23 each): $55 \times \$23 = \$1,265$.

The lowest price is \$1,188.

QUESTION: 25

As a member of FEMA, you're required to set up a contingency plan to supply meals to residents of a town devastated by a tornado. A breakfast ration weighs 12 ounces and the lunch and dinner rations weigh 18 ounces each.

Assuming a food truck can carry 3 tons and that each resident will receive 3 meals per day, how many residents can you feed from one truck during a 10-day period?

- A. 150 residents
- B. 200 residents
- C. 250 residents
- D. 300 residents

Answer(s): B

Explanation:

One ton = 2,000 pounds, so one truck can carry 6,000 pounds.

There are 16 ounces in a pound, so one truck can carry 96,000 ounces.
 The total daily ration for each resident is 12 ounces + 18 ounces + 18 ounces, or 48 ounces.
 The number of daily rations supplied can be expressed as $96,000 \div 48 = 2,000$.
 Dividing 2,000 by 10 days results in 200 residents who can be fed by one truck during this 10-day period.

QUESTION: 26

A train headed south for Wichita left the station at the same time a train headed north for Des Moines left the same station. The train headed for Wichita traveled at 55 miles per hour. The train headed for Des Moines traveled at 70 miles per hour.
 How many miles apart were the trains at the end of 3 hours?

- A. 210 miles
- B. 165 miles
- C. 125 miles
- D. 375 miles

Answer(s): D

Explanation:

The train headed for Wichita traveled 55 miles per hour $\times$ 3 hours = 165 total miles. The train headed for Des Moines traveled 70 miles per hour $\times$ 3 hours = 210 total miles. Adding the distances together gives you the number of miles apart the two trains were after three hours: $210 + 165 = 375$. Another option: You can add the two rates of speed ($55 + 70$) and multiply the sum by 3 hours (125×3 hours = 375).

QUESTION: 27

A carpenter needs to cut four sections, each 3-feet 8-inches long, from a piece of molding. If the board is only sold by the foot, what's the shortest length of board she can buy?

- A. 15 feet
- B. 14 feet
- C. 16 feet
- D. 12 feet

Answer(s): A

Explanation:

Convert the mixed number to inches. 3 feet 8 inches equals 44 inches (12 inches per foot $\times$ 3 feet = 36 inches + 8 inches = 44 inches). 44 inches (length each section needs to be) $\times$ 4 (number of sections needed) = 176 inches (total molding needed). To determine the amount of molding needed in feet, convert 176 inches into feet by dividing 176 inches by 12 inches. You get $14\frac{2}{3}$ feet, so the shortest board length is 15 feet.

QUESTION: 28

Kiya had a coupon for 10% off one frozen turkey breast. The turkey breasts cost \$8.50 each, and Kiya bought two.
 How much did she pay?

- A. \$16.15
- B. \$17.00
- C. \$15.30
- D. \$7.65

Answer(s): A

Explanation:

One turkey breast costs \$8.50 minus 10% of \$8.50, or $\$8.50 - \$0.85 = \$7.65$. The other turkey breast is full price. $\$7.65 + \$8.50 = \$16.15$.

QUESTION: 29

A recruiter travels 1,100 miles during a 40-hour workweek.

If she spends $\frac{2}{5}$ of her time traveling, how many hours does she spend traveling?

- A. 22
- B. $5\frac{1}{2}$
- C. 16
- D. 8

Answer(s): C

Explanation:

Don't let the number of miles traveled confuse you - you don't use them to solve the problem. $\frac{2}{5}$ of a 40-hour workweek is $\frac{2}{5} \times \frac{40}{1} = \frac{80}{5}$. Reduce the fraction: $80 \div 5 = 16$ hours per week spent traveling.

QUESTION: 30

Your car uses gasoline at the rate of 21 miles per gallon.

If gasoline costs \$2.82 per gallon, and you drive for 7 hours at a speed of 48 miles per hour, how much will you pay for gasoline for the trip?

- A. \$38.18
- B. \$45.12
- C. \$47.73
- D. \$59.27

Answer(s): B

Explanation:

Your first step is to determine the number of miles traveled. Multiply the rate of travel by the time. $48 \times 7 = 336$ miles. The amount of gas used is the total miles driven, divided by the number of miles per gallon. $336 \div 21 = 16$ gallons of gasoline used. At the price of \$2.82 per gallon, you spent \$45.12 for gas ($\$2.82 \times 16 = \45.12).

QUESTION: 31

If a hexahedral die is rolled two times, what's the probability of NOT rolling a five both times?

- A. $\frac{1}{36}$
- B. $\frac{1}{6}$
- C. $\frac{4}{36}$
- D. $\frac{25}{36}$

Answer(s): D

Explanation:

The probability of rolling a 5 is 1 out of 6 ($\frac{1}{6}$), so the probability of NOT rolling a 5 is $1 - \frac{1}{6}$, or $\frac{5}{6}$. The probability of not rolling a 5 twice is $\frac{5}{6} \times \frac{5}{6}$, or $\frac{25}{36}$.

QUESTION: 32

Jack loaned Bob \$1,500 at an annual interest rate of 7%.
After one year, how much will Bob owe Jack?

- A. \$105
- B. \$1,500
- C. \$1,605
- D. \$1,507

Answer(s): C

Explanation:

Multiply loaned money, \$1,500, by 7% to determine how much is the annual interest rate.
 $\$1,500 \times 0.07 = \105 .
Then add the annual interest state to the total amount of debt $\$105 + \$1,500 = \$1,605$.

QUESTION: 33

A 2-ton truck is taxed at a rate of \$0.12 per pound.
How much is the total tax bill?

- A. \$480
- B. \$240
- C. \$120
- D. \$600

Answer(s): A

Explanation:

2 tons = 4,000 pounds. $4,000 \times \$0.12 = \480.00 .

QUESTION: 34

If $ab = 10$, and $a^2 + b^2 = 30$, solve for y in the equation, $y = (a + b)^2$.

- A. 40
- B. 45
- C. 50

D. 55

Answer(s): C

Explanation:

$y = (a + b)^2$. Expanding the equation results in $y = a^2 + b^2 + 2ab$. You know that $a^2 + b^2 = 30$, and $ab = 10$. When you substitute these known values into the equation, you get $y = 30 + 2(10)$. Solving for y , results in $y = 50$.

QUESTION: 35

A half-pint of cream is what part of a gallon?

- A. $\frac{1}{8}$
- B. $\frac{1}{4}$
- C. $\frac{1}{16}$
- D. $\frac{1}{6}$

Answer(s): C

Explanation:

8 pints make up a gallon, so a gallon contains 16 half-pints. One half-pint equals $\frac{1}{16}$ of a gallon.

QUESTION: 36

The cost of a protein bar increased from \$2.50 to \$2.80. How much was the percent increase to the \$2.80 rate?

- A. 16%
- B. 10%
- C. 15%
- D. 12%

Answer(s): D

Explanation:

First subtract the old cost from the new cost: $\$2.80 - \$2.50 = \$0.30$. Then divide the difference by the old cost to find the percent difference: $\$0.30 \div \$2.50 = 0.12 = 12\%$.

QUESTION: 37

An aircraft flies over Boondock Air Force Base at 10:20 a.m. At 10:32 a.m., the plane passes over Sea Side Naval Air Station, 120 miles away. How fast is the aircraft traveling?

- A. 400 mph
- B. 500 mph
- C. 600 mph
- D. 700 mph

Answer(s): C

Explanation:

The aircraft travels 120 miles in 12 minutes, which is $\frac{1}{5}$ of an hour. Therefore, in $\frac{5}{5}$ (one) hour, it would travel 5×120 , or 600 miles. The aircraft is traveling 600 miles per hour.

QUESTION: 38

Last year, Margot grew 50 bushels of corn in her backyard. This year, the yield has increased 8%.

How many bushels of corn did Margot grow this year?

- A. 56
- B. 52
- C. 60
- D. 54

Answer(s): D

Explanation:

Multiply 50 bushels by 8% to find the yield increase in bushels: $50 \times 0.08 = 4$. Add 4 bushels (the amount of the increase) to 50 bushels (the original yield) to determine that an 8% increase equals 54 bushels.

QUESTION: 39

Junior has saved money in his piggybank over the winter. He wants to buy a \$30 computer game.

If he has 14 one-dollar bills, 16 half dollars, 12 quarters, 8 dimes, 25 nickels, and 10 pennies, how much more does he need to borrow from Dad to buy the game?

- A. \$27.15
- B. \$2.85
- C. \$2.95
- D. \$1.85

Answer(s): B

Explanation:

Convert the change to dollars or fractions of dollars and add:

14 dollars	= \$14.00
16 half dollars	= \$8.00
12 quarters	= \$3.00
8 dimes	= \$0.80
25 nickels	= \$1.25
10 pennies	= \$0.10
	= \$27.15

Subtract the total from \$30.00 to determine how much money Junior has to borrow:
 $\$30.00 - \$27.15 = \$2.85$.

QUESTION: 40

Debbie receives a weekly salary of \$80, plus a 5% commission on any sales. During the week, she has \$800 in total sales.

What's the ratio of her commission to her salary?

- A. 2/1
- B. 1/2
- C. 3/1
- D. 1/3

Answer(s): B

Explanation:

Her commission for the week was \$40 ($0.05 \times 800 = 40$). The ratio of her commission to her salary is 40/80, which can be reduced to 1/2.

QUESTION: 41

How many quart cans can be filled from a 25-gallon bucket of paint?

- A. 50
- B. 75
- C. 100
- D. 80

Answer(s): C

Explanation:

A gallon consists of 4 quarts. $4 \times 25 = 100$.

QUESTION: 42

If a crew of four people can paint the barn in three days, how long will it take a crew of two people?

- A. 4 days
- B. $11\frac{1}{2}$ days
- C. 8 days
- D. 6 days

Answer(s): D

Explanation:

Dividing 4 crewmembers by 2 crewmembers shows that 4 members are twice as many as 2 members. Multiply the number of days it would take 4 people to paint by 2 (3×2) to determine how long it would take 2 people to do the same task.

QUESTION: 43

Brian works for five hours and is paid \$24. Christina works for three hours and is paid \$10.95. How much more per hour does Brian make than Christina?

- A. \$1.15
- B. \$1.25
- C. \$1.35
- D. \$1.37

Answer(s): A

Explanation:

Brian's hourly wage is $24/5$ (\$4.80). Christina's hourly wage is $10.95/3$ (\$3.65).
 $\$4.80 - \$3.65 = \$1.15$.

QUESTION: 44

Margaret is getting married and must be ready by 11:15 a.m. If it's now 8:30 a.m., how much time does she have to get ready?

- A. $11\frac{1}{2}$ hours
- B. $21\frac{1}{2}$ hours
- C. $23\frac{1}{4}$ hours
- D. $21\frac{1}{3}$ hours

Answer(s): C

Explanation:

The amount of time from 8:30 a.m. to 11:15 a.m. is 2 hours and 45 minutes, or $245/60$. To

reduce the fraction, divide 45 by 60 to get 0.75, or $\frac{3}{4}$. The total amount of time Margaret has to get ready is $2\frac{3}{4}$ hours.

QUESTION: 45

An accounting-firm employee is asked to shred 900 documents.

If he can shred documents at a rate of 7 per minute, the number of documents remaining after $11\frac{1}{2}$ hours of shredding is _____.

- A. 630
- B. 90
- C. 270
- D. 810

Answer(s): C

Explanation:

At a rate of 7 documents per minute, the employee can shred 630 documents in 90 minutes.

How do you come up with that number? Multiply 7 by 90 (the number of minutes in $11\frac{1}{2}$ hours).

Subtract 630 from 900 total documents to determine that after $11\frac{1}{2}$ hours of shredding, 270 documents remain.

QUESTION: 46

A home stereo depreciates by 20% each year.

What's the value of a stereo, purchased new for \$1,200, after two years?

- A. \$768
- B. \$693
- C. \$827
- D. \$654

Answer(s): A

Explanation:

The stereo depreciates \$240 the first year ($0.2 \times 1200 = 240$). Therefore, the value of the stereo after the first year is $\$1200 - \$240 = \$960$. The second year the stereo would depreciate by \$192 (0.2×960). The value of the stereo after the second year is $\$960 - \$192 = \$768$.

QUESTION: 47

Janet's old pickup truck can only reach a speed of 45 miles per hour.

If she drives at top speed, how long will it take her to reach a city 135 miles away?

- A. 3 hours
- B. 2 hours
- C. 4 hours
- D. $2\frac{1}{2}$ hours

Answer(s): A

Explanation:

Divide the distance (135 miles) by the speed (45 miles per hour) to determine that Janet will take 3 hours to reach the city.

QUESTION: 48

A blouse normally costs \$18.50.

How much money is saved if the blouse is purchased at a 20% discount?

- A. \$1.85
- B. \$14.80
- C. \$4.50
- D. \$3.70

Answer(s): D

Explanation:

Multiply the price of the blouse by the amount of the discount: $\$18.50 \times 0.20 = \3.70 .

QUESTION: 49

A clerk's weekly salary of \$320 is increased to \$360.

The percent increase is _____.

- A. 101/2%
- B. 11%
- C. 121/2%
- D. 12%

Answer(s): C

Explanation:

Subtract the original salary from the new salary to get the difference in salary: $\$360 - \$320 = \$40$. Then divide the difference in salary (\$40) by the original salary (\$320) to determine the percent increase: $40 \div 320 = 0.125 = 12.5\%$.

QUESTION: 50

Two go-carts are racing on a circular track with a perimeter of 360 feet. Camera One is following Go-Cart One, and Camera Two is following Go-Cart Two.

If the angle between the two cameras is 40 degrees, how far apart are the two go-carts?

- A. 30 feet
- B. 40 feet
- C. 300 feet
- D. 400 feet

Answer(s): B

Explanation:

A circle is 360 degrees, so 40 degrees is $\frac{1}{9}$ of a circle ($360 \div 40 = 9$). To get the answer of 40

feet, multiply the perimeter of the track $\frac{1}{9}$ ($360 \times \frac{1}{9} = 40$).

QUESTION: 51

Dinner at a nice restaurant cost \$35.98.

If Joan gave the cashier \$40.00, how much change should she get back?

- A. \$5.02
- B. \$4.02
- C. \$3.92
- D. \$1.02

Answer(s): B

Explanation:

Subtract \$35.98 from \$40.00 to get \$4.02.

QUESTION: 52

A balloonist circumnavigated the globe in 13 days, 12 hours, 16 minutes, and 13 seconds. A plane circumnavigates the globe in 4 days, 10 hours, 15 minutes, and 7 seconds.

How much longer did it take for the balloon to go around the world?

- A. 12 days, 7 hours, 11 minutes, and 35 seconds
- B. 9 days, 2 hours, 1 minute, and 6 seconds
- C. 8 days, 14 hours, 16 minutes, and 6 seconds
- D. 9 days, 7 hours, 3 minutes, and 20 seconds

Answer(s): B

Explanation:

Subtract the speed of the plane from the speed of the balloon to determine how much longer it took the balloonist:

13 days 12 hours 16 minutes 13 seconds

4 days 10 hours 15 minutes 7 seconds

9 days 2 hours 1 minute 6 seconds

QUESTION: 53

Darlene bought 12 boxes of cookies for \$48.00.

What was the cost of each box of cookies?

- A. \$4.00
- B. \$0.48
- C. \$0.40
- D. \$4.80

Answer(s): A

Explanation:

Divide the total cost by the number of boxes purchased to determine the cost per box: $\$48 \div 12$

= \$4.

QUESTION: 54

A tune-up increases a car's fuel efficiency by 5%.

If a car averaged 20 miles per gallon before the tune-up, how many miles per gallon will it average after the tune-up?

- A. 25
- B. 22
- C. $20\frac{1}{2}$
- D. 21

Answer(s): D

Explanation:

Multiply 20×0.05 to determine how many more miles per gallon the car will get. The answer is 1. Then add the number of additional miles per gallon the car will get to the original number of miles per gallon the car gets to reach the new average: $1 + 20 = 21$.

QUESTION: 55

If you pay \$22.90 for a DVD that includes a 7% sales tax, what is the price of the DVD before the sales tax?

- A. \$21.40
- B. \$21.30
- C. \$21.50
- D. \$20.95

Answer(s): A

Explanation:

If y represents the price of the DVD before the tax. Then price with tax would be $y + 0.07y = \$22.90$. Therefore, $1.07y = 22.90$, or $y = 22.90/1.07 = \$21.40$. Be careful not to just take 7% of \$22.90 and subtract. That is incorrect.

QUESTION: 56

A bin of hard candy holds $10\frac{1}{2}$ pounds.

How many $\frac{3}{4}$ -pound boxes of candy can be filled from the bin?

- A. 30 boxes
- B. $15\frac{1}{4}$ boxes
- C. $7\frac{7}{8}$ boxes
- D. 14 boxes

Answer(s): D

Explanation:

Divide $10\frac{1}{2}$ by $\frac{3}{4}$. You can perform this operation by multiplying $10\frac{1}{2}$ by the inverse of $\frac{3}{4}$:

$10\frac{1}{2} \times \frac{4}{3} = 21\frac{1}{2} \times \frac{4}{3} = \frac{84}{6}$. This fraction, reduced, becomes 14.

QUESTION: 57

A patio measures 12 feet by 14 feet.

How many 8-inch-square paving stones are needed to pave the patio?

- A. 21
- B. 252
- C. 32
- D. 168

Answer(s): B

Explanation:

First multiply length $\times$ width to determine the area of the patio: 12 feet $\times$ 14 feet = 168 square feet. Next determine the area in square inches that needs to be covered: 168 square feet $\times$ 12 inches = 2,016 square inches. Finally, divide that answer by the size of the stones to determine the number of stones needed: $2,016 \div 8 = 252$.

QUESTION: 58

A computer programmer is making \$25,000 per year. 28% of her salary is withheld for federal and state deductions.

How much is the computer programmer's net pay?

- A. \$20,000
- B. \$7,000
- C. \$18,750
- D. \$18,000

Answer(s): D

Explanation:

Calculate the amount of the deduction by multiplying her salary by the percent deducted: $\$25,000 \times 28\% = \$25,000 \times 0.28 = \$7,000$. Subtract that product from the salary to determine the net pay: $\$25,000 - \$7,000 = \$18,000$.

QUESTION: 59

Pam cuts a pie in half in a straight line. She then cuts a line from the center to the edge, creating a 55-degree angle.

What's the supplement of that angle?

- A. 55 degrees
- B. 125 degrees
- C. 70 degrees
- D. 130 degrees

Answer(s): B

Explanation:

When the sum of two angles is 180 degrees, the angles are said to be supplemental to each other. To find the supplement, subtract 55 from 180 ($180 - 55 = 125$).

QUESTION: 60

A stack of lumber is 6-feet high.

If each piece of lumber is 4-inches thick, how many pieces of lumber are in the stack?

- A. 72
- B. 12
- C. 18
- D. 10

Answer(s): C

Explanation:

Multiply the height of the stack in feet by 12 to determine the height of the stack in inches: $6 \times 12 = 72$ inches.

Divide that number by 4 inches, the thickness of each board to determine the number of pieces of lumber in the stack: $72 \div 4 = 18$.

QUESTION: 61

A baker sells a dozen donuts for \$3.99.

The cost to make three donuts is \$0.45. How much is the total profit on 5-dozen donuts?

- A. \$17.70
- B. \$13.20
- C. \$2.19
- D. \$10.95

Answer(s): D

Explanation:

Multiply \$0.45 (the cost of making 3 donuts) by 4 to find the cost of making a dozen donuts: $\$0.45 \times 4 = \1.80 .

Then multiply the cost of making a dozen donuts by 5 to determine the cost per 5 dozen: $\$1.80 \times 5 = \9.00 .

Next, multiply the selling price per dozen times 5, the number of dozens sold: $\$3.95 \times 5 = \19.95 . Finally subtract the cost of making 5-dozen donuts from the price the baker sells them for to determine the profit:

$\$19.95 - \$9.00 = \$10.95$.

QUESTION: 62

Your piggy bank contains \$19.75 in dimes and quarters. There are 100 coins in all.

How many dimes are there?

- A. 25
- B. 30

- C. 35
- D. 40

Answer(s): C

Explanation:

Let x equal the number of dimes. Then $100 - x$ represents the number of quarters.

There is $\$0.10x$ in dimes and $\$0.25(100 - x)$ in quarters.

$$0.10x + 0.25(100 - x) = 19.75$$

$$0.10x + 25 - 0.25x = 19.75$$

$$-0.15x = -5.25$$

$$x = -5.25 / -0.15$$

$$x = 35$$

QUESTION: 63

A bricklayer charges \$8 per square foot to lay a patio.

How much would it cost for the bricklayer to lay a 12-foot $\times$ 16-foot patio?

- A. \$960
- B. \$192
- C. \$224
- D. \$1,536

Answer(s): D

Explanation:

First determine the square footage of the patio: 12 feet $\times$ 16 feet = 192 square feet. Then multiply this number by the cost per square foot to determine what the brick layer charges:

$$192 \times \$8 = \$1,536.$$

QUESTION: 64

Terry earns three times more per hour than Tim. Tim earns \$2 more per hour than Angie. As a group they earn \$43 per hour.

What's Angie's hourly wage?

- A. \$7.00
- B. \$8.00
- C. \$9.00
- D. \$10.00

Answer(s): A

Explanation:

Let x equal Angie's hourly wage. $x + 2$ would then represent Tim's hourly wage, and $3(x + 2)$ would represent Terry's hourly wage.

$$x + (x + 2) + 3(x + 2) = 43$$

$$x + x + 2 + 3x + 6 = 43$$

$$5x + 8 = 43$$

$$5x = 35$$

$$x = 35/5$$

$$x = 7$$

QUESTION: 65

If 4 people can run 8 machines, how many machines can 2 people run?

- A. 2
- B. 4
- C. 1
- D. 3

Answer(s): B

Explanation:

Two people is $1/2$ as many as 4 people: $2 \div 4 = 1/2$. Multiply the number of machines 4 people can run by $1/2$ to determine how many machines 2 people can run: $8 \times 1/2 = 4$.

QUESTION: 66

The price of daily admission at an amusement park is \$36. The park sells an unlimited season pass for \$240.

How many trips would you need to make with the season ticket in order for it to cost less than paying the daily admission rate?

- A. 6
- B. 7
- C. 8
- D. 9

Answer(s): B

Explanation:

Let x equal the number of daily tickets you would purchase. $36x =$ the daily ticket cost.

$$240 < 36x$$

$$240/36 < x$$

$$62/3 < x$$

You would need to use the ticket more than $62/3$ times (or 7 times) in order for it to be cheaper to use the season ticket.

QUESTION: 67

A plumber needs four lengths of pipe, each 3-feet, 6-inches long. Pipes are sold by the foot. How many feet does he need to buy?

- A. 15
- B. 16
- C. 14
- D. 12

Answer(s): C

Explanation:

Convert the pipe length to inches: 3 feet, 6 inches = 42 inches. Multiply 42 inches by the number of pipes needed to find the number of inches of pipe needed: $42 \times 4 = 168$. Divide the total amount of pipe needed in inches by 12 to determine how many feet of pipes are needed: $168 \div 12 = 14$.

QUESTION: 68

The product of two consecutive odd numbers is 399.
What are the numbers?

- A. 17 and 19
- B. 19 and 21
- C. 21 and 23
- D. 25 and 27

Answer(s): B

Explanation:

The fastest way to solve this would be to simply multiply the possible choices together ($19 \times 21 = 399$). You can also solve this with algebra.

QUESTION: 69

A personal trainer earns a 65% commission on her training sales.
If she sells \$530.00 worth of training, how much commission does she make?

- A. \$874.50
- B. \$34.45
- C. \$344.50
- D. \$185.50

Answer(s): C

Explanation:

Multiply her total sales by her percent commission to find her commission: $\$530.00 \times 0.65 = \344.50 .

QUESTION: 70

A rectangle is one inch longer than it is wide. Its diameter is five inches.
What's the width of the rectangle?

- A. 2 inches
- B. 3 inches
- C. 4 inches
- D. 5 inches

Answer(s): B

Explanation:

The diagonal formula for a rectangle is $D^2 = L^2 + W^2$. In this case, $D = 5$, and $L = W + 1$. Substituting the known values into the formula results in $5^2 = (W + 1)^2 + W^2$.

$$25 = (W + 1)(W + 1) + W^2$$

$$25 = W^2 + 2W + 1 + W^2$$

$$25 = 2W^2 + 2W + 1$$

(Note: This equation is a quadratic equation and can be solved by setting it equal to zero and factoring.) $0 = 2W^2 + 2W - 24$

$$1/2(0) = 1/2(2W^2 + 2W - 24)$$

$$0 = W^2 + W - 12$$

$$0 = (W - 3)(W + 4)$$

$$W - 3 = 0 \quad W + 4 = 0$$

$$W = 3$$

$$W = -4 \text{ (not a possible solution)}$$

QUESTION: 71

A treasure map is drawn to a scale of 2 inches = 3 miles. On the map, the distance between Point A and X- marks-the-spot is $9\frac{1}{2}$ inches.

How many actual miles does this represent?

- A. $28\frac{1}{2}$ miles
- B. $14\frac{1}{4}$ miles
- C. $6\frac{1}{3}$ miles
- D. 19 miles

Answer(s): B

Explanation:

If 2 inches = 3 miles, then 1 inch equals $1\frac{1}{2}$ miles: $3 \div 2 = 1.5$. Multiply $1\frac{1}{2}$ miles $\times 9\frac{1}{2}$ inches to determine the actual distance: $1.5 \times 9.5 = 14.25$ or $14\frac{1}{4}$ miles.

QUESTION: 72

A painter has painted a picture on a piece of canvas that measures 10×14 inches. In order to accommodate a frame, he has left an unpainted margin of 1 inch all the way around.

What part of the canvas has been painted?

- A. 80%
- B. 75%
- C. 25%
- D. 66%

Answer(s): A

Explanation:

The area of the entire piece of canvas = $10 \text{ inches} \times 14 \text{ inches} = 140$ square inches. The portion painted on equals $8 \text{ inches} \times 12 \text{ inches} = 112$ square inches. (This is determined by subtracting 1 inch from the length of each side to account for the margin.) The portion used for painting can be expressed as a fraction: $112/140$.

Reduce this fraction (divide 112 by 140) to determine that 80% of the canvas is covered with

paint.

QUESTION: 73

A dog trainer is building a dog run that measures 9×16 feet.

If she wants to fence the perimeter of the run, how many feet of chain link fence will she need?

- A. 144 feet
- B. 25 feet
- C. 32 feet
- D. 50 feet

Answer(s): D

Explanation:

Calculate perimeter by adding the lengths of all four sides of a quadrilateral: $9 + 9 + 16 + 16 = 50$ feet.

QUESTION: 74

A rectangle is $1\frac{1}{2}$ times as long as it is wide. The perimeter of the rectangle is 100 inches. What's the length of the rectangle?

- A. 20 inches
- B. 30 inches
- C. 40 inches
- D. 45 inches

Answer(s): B

Explanation:

The formula for the perimeter of a rectangle is $P = 2L + 2W$. In this case, $P = 100$ and $L = 1.5W$.

$$100 = 2(1.5W) + 2W$$

$$100 = 3W + 2W$$

$$100 = 5W$$

$$W = 100/5$$

$$W = 20.$$

The width of the rectangle is 20 inches. Because the length is $1\frac{1}{2}$ the width, $1.5 \times 20 = 30$.

QUESTION: 75

Miguel passed seven of his history class quizzes and failed three.

The fraction of quizzes he passed is correctly expressed as _____.

- A. $\frac{7}{3}$
- B. $\frac{3}{7}$
- C. $\frac{7}{10}$
- D. $\frac{3}{5}$

Answer(s): C

Explanation:

The total number of quizzes is 10. If he passed seven of them, the fraction would be expressed as $\frac{7}{10}$.

QUESTION: 76

A 3-yard-long ribbon was used to trim four dresses. Each dress used the same amount of ribbon.

How much ribbon was used for each dress?

- A. 1 yard
- B. $\frac{2}{3}$ yard
- C. $\frac{1}{2}$ yard
- D. $\frac{3}{4}$ yard

Answer(s): D

Explanation:

Convert the measurement to inches: 1 yard = 36 inches; $36 \text{ inches} \times 3 = 108 \text{ inches}$.

Divide the total number of inches by the number of dresses being trimmed to determine the length of each piece of ribbon: $108 \text{ inches} \div 4 = 27 \text{ inches}$. Convert the quotient (27 inches) to a fraction of a yard: $\frac{27}{36} = 27 \div 36 = 75\% = \frac{3}{4}$ of a yard.

QUESTION: 77

Sarah found a wallet containing \$500 in the street. She returned the wallet to its owner, who gave her a \$30 reward.

What percentage of the \$500 was the reward?

- A. 5%
- B. 6%
- C. 7%
- D. 4%

Answer(s): B

Explanation:

Divide \$30 by \$500 to determine the percentage of \$500 that the reward comprised.

QUESTION: 78

A bin of bolts at the hardware store contains 7-dozen bolts when full. The stock clerk is supposed to reorder bolts when the bin is $\frac{1}{6}$ full.

How many bolts are in the bin when it's time to reorder?

- A. 14 bolts
- B. 1 bolt
- C. 84 bolts
- D. 12 bolts

Answer(s): A

Explanation:

First find how many bolts a full bin contains: $7 \times 12 = 84$ bolts. Then multiply the total number of bolts in a full bin by $1/6$ to find how many bolts are in the bin when it's $1/6$ full:

$$84/1 \times 1/6 = 84/6 = 14 \text{ bolts.}$$

QUESTION: 79

Two bicyclists head toward each other from the opposite ends of Main Street, which is six miles long. The first biker started at 2:05 going 12 mph. The second biker began peddling 4 minutes later at a rate of 14 mph.

What time will they meet?

- A. 2:13
- B. 2:24
- C. 2:29
- D. 2:34

Answer(s): C

Explanation:

The first bike got a $4/5$ mile head start ($12 \times 4/60$). Therefore, by the time the second bike leaves, there are $51/5$ miles between them ($6 - 4/5$). Their combined rate of travel is $12 + 14 = 26$ mph. Let t = the number of hours the second bike travels.

$$26t = 51/5$$

$$26t = 26/5$$

$$t = 26/5 \div 26/1$$

$$t = 26/5 \times 1/26$$

$$t = 1/5$$

$1/5$ of an hour = 20 minutes. The second bike left at 2:09, so both bikes will meet at 2:29.

QUESTION: 80

A recipe calls for 8 ounces of black beans or red beans.

The cheapest option to buy and use would be _____.

- A. two 4-ounce cans of black beans at \$0.79 each
- B. one 8-ounce can of red beans at \$1.49
- C. two 3-ounce cans of black beans at \$0.59 each
- D. three 3-ounce cans of red beans at \$0.65 each

Answer(s): B

Explanation:

Choice (one 8-ounce can of red beans at \$1.49) is the cheapest option. Calculate each answer option and compare:

Choice (two 4-ounce cans of black beans at \$0.79 each): $2 \times \$0.79 = \1.58 .

Choice (one 8-ounce can of red beans at \$1.49): \$1.49.

Choice (two 3-ounce cans of black beans at \$0.59 each): Two 3-ounce cans don't equal 8 ounces, so this answer can't be correct.

Choice (three 3-ounce cans of red beans at \$0.65 each): $3 \times \$0.65 = \1.95 .

QUESTION: 81

A street vendor sells \$25.70 worth of pretzels on Friday, \$32.30 on Saturday, and \$31.80 on Sunday. He spends a fourth of the money over the weekend. How much money does he have left?

- A. \$89.80
- B. \$22.45
- C. \$44.90
- D. \$67.35

Answer(s): D

Explanation:

Add the sales amounts together: $\$25.70 + \$32.30 + \$31.80 = \89.80 . Then multiply the total sales by $\frac{3}{4}$ to determine how much money he has left: $\$89.80 \times 0.75 = \67.35 .

QUESTION: 82

A recruit has \$30.00. He saw some camouflage socks for \$3.95 a pair. How many pairs of socks can he buy?

- A. 9
- B. 7
- C. 6
- D. 4

Answer(s): B

Explanation:

Divide \$30.00 by \$3.95. The whole number is the number of pairs of socks he could buy: $\$30.00 \div \$3.95 = 7.59$ or 7 pairs of socks.

QUESTION: 83

A crate containing a puppy weighs 60 pounds, 5 ounces. The puppy weighs 43 pounds, 7 ounces. How much does the crate alone weigh?

- A. 16 pounds, 8 ounces
- B. 16 pounds, 2 ounces
- C. 17 pounds
- D. 16 pounds, 14 ounces

Answer(s): D

Explanation:

16 ounces make a pound. Subtract 43 pounds, 7 ounces (the weight of the puppy) from 59 pounds, 21 ounces (the weight of the crate). 59 pounds, 21 ounces is the same as 60 pounds, 5 ounces, but converting an additional pound to ounces makes the subtraction possible.

59 pounds, 21 ounces - 43 pounds, 7 ounces = 16 pounds, 14 ounces

QUESTION: 84

In a manufacturing plant that produces new computers, a 0.15 probability exists that a computer will be defective.

If five computers are manufactured, what's the probability that all of them will be defective?

- A. 7.6
- B. 0.60
- C. 0.00042
- D. 0.000076

Answer(s): D

Explanation:

The probability that all five computers will be defective is $0.15 \times 0.15 \times 0.15 \times 0.15 \times 0.15 = 0.0000759$ (round up to 0.000076)

QUESTION: 85

A house contains one 12-foot $\times$ 14-foot bedroom, one 12-foot $\times$ 10-foot bedroom, and one 8-foot $\times$ 12-foot bedroom.

What's the total amount of carpeting needed to carpet all three bedrooms?

- A. 384 square yards
- B. 128 square yards
- C. 216 square yards
- D. 88 square yards

Answer(s): B

Explanation:

Find the area of each bedroom and add them together: $12 \times 14 = 168$; $12 \times 10 = 120$; $8 \times 12 = 96$. $168 + 120 + 96 = 384$ square feet. Then, because 3 feet make up a yard, divide the total area in square feet by 3 to determine the number of square yards needed: $384 \div 3 = 128$ square yards.

QUESTION: 86

Rafael can type 9 pages an hour.

How long will it take him to type 126 pages?

- A. 14 hours
- B. 9 hours
- C. 7 hours
- D. 16 hours

Answer(s): A

Explanation:

Divide the total number of pages to be typed by the number of pages Rafael can type per hour to find the number of hours it will take him to type the pages: $126 \text{ pages} \div 9 \text{ pages per hour} = 14 \text{ hours}$.

QUESTION: 87

In a 60-minute gym class, 40 girls want to play volleyball, but only 10 can play at a time. For each player to get the same amount of playing time, how many minutes should each person play?

- A. $11\frac{1}{2}$ minutes
- B. 6 minutes
- C. 30 minutes
- D. 15 minutes

Answer(s): D

Explanation:

Divide the group of 40 girls by the number of girls who can play at the same time: $40 \div 10 = 4$. This means 4 groups of girls have to share the 60 minutes or $60 \text{ minutes} \div 4 = 15 \text{ minutes}$. Thus, each girl plays for 15 minutes.

QUESTION: 88

The video-rental store charges \$2.00 for the first day a rented video is overdue and \$1.25 for each day after that.

If a person paid \$8.25 in late fees, how many days was the video overdue?

- A. 7 days
- B. 6 days
- C. 4 days
- D. 5 days

Answer(s): B

Explanation:

Subtract the first day's late charge from the total: $\$8.25 - \$2.00 = \$6.25$.

Then divide the remainder by \$1.25 to determine the number of additional days the video was overdue: $\$6.25 \div \$1.25 = 5$.

Add those 5 days to the first day the video was late, to find that the video was 6 days overdue.

QUESTION: 89

Janet is trying to watch her weight. A $\frac{1}{2}$ cup of pudding has 150 calories. The same amount of broccoli has 60 calories.

How much broccoli can Janet eat to equal the same number of calories in the $\frac{1}{2}$ cup of pudding?

- A. 2 cups
- B. $2\frac{1}{2}$ cups
- C. $1\frac{1}{2}$ cups

D. $1\frac{1}{4}$ cups

Answer(s): D

Explanation:

Divide the number of calories in the pudding by the number of calories in the broccoli: $150 \div 60 = 2\frac{1}{2}$. Janet can eat $2\frac{1}{2}$ times the amount of broccoli as she can eat pudding for the same number of calories. Multiply $2\frac{1}{2}$ by $\frac{1}{2}$ cup (the amount of pudding that contains 150 calories) to find how many cups of broccoli she can eat for 150 calories: $2.5 \times 0.5 = 1.25$ or $1\frac{1}{4}$ cups.

QUESTION: 90

The neighbor's dog barks at a squirrel every 15 minutes at night.

If he first barks at 10 p.m., when you're trying to fall asleep, how many times will he bark by 2 a.m., when you give up trying to sleep and decide to read a book instead?

- A. 16 times
- B. 132 times
- C. 17 times
- D. 15 times

Answer(s): C

Explanation:

The time between 10 p.m. and 2 a.m. is 4 hours or 240 minutes. Divide the total number of minutes in the time period by 15 minutes - the interval that the dog barks. Then add 1 because the dog started barking at the beginning of the period: $(240 \div 15) + 1 = 17$.

QUESTION: 91

If a barber is capable of cutting the hair of 35 people per day, and he works 7 days per week, how many haircuts could he give during the months of April, May, and June?

- A. 3185
- B. 3150
- C. 2545
- D. 2555

Answer(s): A

Explanation:

There are 30 days in April, 31 days in May, and 30 days in June for a total of 91 days. $91 \times 35 = 3185$.

QUESTION: 92

If you typed 45 words per minute, how many words would you be able to type in 12 minutes?

- A. 490
- B. 540
- C. 605

D. 615

Answer(s): B

Explanation:

Multiply the number of words you can type per minute (45) by the number of minutes you will be typing (12). $45 \times 12 = 540$.

QUESTION: 93

Tom is flying a kite at the end of a 500-foot string. His friend Kathy is standing directly under the kite 300 feet away from Tom.

How high is the kite flying?

- A. 300 feet
- B. 350 feet
- C. 400 feet
- D. 450 feet

Answer(s): C

Explanation:

Visualize a triangle, where the string represents the hypotenuse and the line between Tom and Kathy represents one of the legs. The Pythagorean theorem states that if one knows the length of two sides of a triangle, the length of the third side can be determined, using the formula $a^2 + b^2 = c^2$. In this case, $300^2 + b^2 = 500^2$.

$$90,000 + b^2 = 250,000$$

$$b^2 = 250,000 - 90,000$$

$$b^2 = 160,000$$

$$b = 400$$

$$b = 400$$

QUESTION: 94

Amy wants to fence in a yard using 400 feet of fencing.

If she wants the yard to be 30-feet wide, how long will it be?

- A. 170 feet
- B. 175 feet
- C. 180 feet
- D. 185 feet

Answer(s): A

Explanation:

The formula used to determine the perimeter of a rectangle is $P = 2(L + W)$. The width is 30, and the parameter is 400.

$$400 = 2(L + 30)$$

$$400 = 2L + 60$$

$$340 = 2L$$

$$L = 170$$

QUESTION: 95

A three-digit code must be used to access a computer file. The first digit must be an A or a B. The second digit must be a number between 0 and 9. The final digit is a single letter from the alphabet from A to Z.

How many possible access codes can there be?

- A. 38
- B. 468
- C. 520
- D. 640

Answer(s): C

Explanation:

There are 2 possibilities for the first digit (A or B), 10 possibilities for the second digit (0 to 9) and 26 possibilities for the third digit. Using the multiplication principle, $2 \times 10 \times 26 = 520$

QUESTION: 96

The sun is 93 million miles from Earth and light travels at a rate of 186,000 miles per second. How long does it take for light from the sun to reach the Earth?

- A. 5 minutes
- B. $6\frac{1}{2}$ minutes
- C. 7 minutes
- D. $8\frac{1}{2}$ minutes

Answer(s): D

Explanation:

Use the time formula $T = S \div V$, where T = time, S = displacement, V = velocity. Time (T) = 93 million miles (S) $\div$ 186,000 miles per second (V) = 500 second. 500 second = $8\frac{1}{2}$ minutes.

QUESTION: 97

A tanning-bed pass for unlimited tanning costs \$53 per month this year, but it was only \$50 per month last year.

What was the percentage of increase?

- A. 5%
- B. 5.5%
- C. 6%
- D. 6.5%

Answer(s): C

Explanation:

The difference in the price is \$3. $\$3 \div \$50 = 0.06$ or 6%

QUESTION: 98

Eleven plus forty-one is divided by a number.
If the result is 13, what's the number?

- A. 2
- B. 4
- C. 6
- D. 8

Answer(s): B

Explanation:

$$11 + 41 = 52$$

Calculate each answer option to figure out the correct answer:

$$52 \div 2 = 26$$

$$52 \div 4 = 13$$

$$52 \div 6 = 8,67$$

$$52 \div 8 = 6,5$$

QUESTION: 99

Mark received an hourly wage of \$9.25. His boss gave him a 4% raise.
How much does Mark make per hour now?

- A. \$9.29
- B. \$9.62
- C. \$9.89
- D. \$9.99

Answer(s): B

Explanation:

To calculate the new wage, start off by taking $\$9.25 \times 0.04 = \0.37 . Then add that number (the amount of Mark's raise) to his original hourly wage. Mark's new hourly wage is $\$9.25 + \$0.37 = \$9.62$.

QUESTION: 100

How many pounds of nails costing \$7 per pound must be mixed with 6 pounds of nails costing \$3 per pound to yield a mixture costing \$4 per pound?

- A. 2 pounds
- B. 2.5 pounds
- C. 3 pounds
- D. 3.5 pounds

Answer(s): A

Explanation:

Let x = number of nails costing \$7 per pound. The total cost of the mixture equals the sum of the

cost for each type of nail or $M = A + B$, where $A = 7x$, $B = 3(6)$, and $M = 4(6 + x)$. Substitute the known values into the equation. $4(6 + x) = 7x + 18$.

$$24 + 4x = 7x + 18$$

$$24 - 18 = 7x - 4x$$

$$3x = 6$$

$$x = 2$$

QUESTION: 101

A product originally costs \$125.

If the product goes on sale for 20% off, and then goes on sale for an additional 30% off, what is the price of the product?

- A. \$62.50
- B. \$70
- C. \$65
- D. \$90

Answer(s): B

Explanation:

You need to apply the discounts successively not just add the discounts. So $\$125 \times 80\% = \100 (price after original 20% discount). $\$100 \times 70\% = \70 the price after the 2nd 30% discount.

QUESTION: 102

A digital video recorder is programmed to record a TV show that lasts for a half hour.

If the digital video recorder has enough memory left to hold 180 minutes of recording, what percentage of the remaining capacity is used for this recording?

- A. $16\frac{2}{3}\%$
- B. 30%
- C. 60%
- D. $33\frac{1}{3}\%$

Answer(s): A

Explanation:

A half-hour is 30 minutes.

The portion of 180 minutes represented by 30 minutes is $30/180$ or $1/6$. As a percentage, $1/6$ is $16\frac{2}{3}\%$.

QUESTION: 103

On a trip to the beach you travel 200 miles in 300 minutes.

How fast did you travel?

- A. 30 mph
- B. 40 mph
- C. 50 mph
- D. 60 mph

Answer(s): B

Explanation:

First convert the 300 minutes to hours by dividing by 60 ($300 \div 60 = 5$ hours). Use the distance formula ($d = rt$) and substitute the known values. $300 = 5r$. $r = 40$.

QUESTION: 104

21 students, or 60% of the class, passed the final exam.
How many students are in the class?

- A. 45
- B. 40
- C. 35
- D. 30

Answer(s): C

Explanation:

Let x = the number of people in the class.
60% of $x = 21$, so $0.60x = 21$.
 $x = 35$.

QUESTION: 105

Joan invests \$4,000 in an account that earns 3% simple interest.
How much will Joan have in the account in 10 years?

- A. \$4,500
- B. \$4,800
- C. \$5,200
- D. \$5,400

Answer(s): C

Explanation:

Use the interest formula ($I = Prt$) to determine the amount of interest earned, where the principle (P) is 4,000, the rate (r) is 0.03 (3%) and the time (t) is 10. $I = 4,000(0.03)(10)$, or $I = \$1,200$.
Add the interest earned to the original amount invested. $\$4,000 + \$1,200 = \$5,200$.

QUESTION: 106

A rectangle has a perimeter of 36 inches. Its length is 3 inches greater than twice the width.
What's the rectangle's length?

- A. 5 inches
- B. 13 inches
- C. 18 inches
- D. 20 inches

Answer(s): B

Explanation:

A rectangle's perimeter is determined by the formula $P = 2(l + w)$. The length of this rectangle is $3 + 2w$.

Substituting the known values into the formula results in $36 = 2(w + 3 + 2w)$.

$$36 = 2(3w + 3)$$

$$18 = 3w + 3$$

$$15 = 3w$$

$$w = 5$$

As the length is $3 + 2w$, then $l = 3 + 2(5)$, or $l = 13$.

QUESTION: 107

A back yard is 50 feet by 100 feet.

What's its area?

- A. 150 square feet
- B. 500 square feet
- C. 2,500 square feet
- D. 5,000 square feet

Answer(s): D

Explanation:

The area of a rectangle is the length $\times$ the width of the rectangle. $50 \times 100 = 5,000$.

QUESTION: 108

Eric is driving a car in which the speedometer is calibrated in kilometers per hour (kph). He notes that his car is traveling at a rate of 75 kph, when he passes a speed limit sign stating the limit is 40 miles per hour (mph). He knows that a kilometer is about $\frac{5}{8}$ of a mile.

If a police officer stops him at this point, how many mph over the limit will the ticket read?

- A. 5
- B. 7
- C. 9
- D. 11

Answer(s): B

Explanation:

A kilometer is $\frac{5}{8}$ of a mile, so multiply $75 \times \frac{5}{8}$, or $75/1 \times 5/8 = 375/8$. Divide 8 into 375 to reduce the fraction and determine that Eric was traveling at 47 miles per hour, 7 mph over the 40 mph posted limit.

QUESTION: 109

Three apples and twice as many pears add up to one-half the number of grapes in a fruit basket.

How many grapes are in the basket?

- A. 8
- B. 18
- C. 28
- D. 38

Answer(s): B

Explanation:

Let x = the number of grapes. 3 apples and 6 pears equals $\frac{1}{2}$ of x or $\frac{1}{2}(x) = 9$. $x = 2(9)$ or $x = 18$.

QUESTION: 110

Apples are on sale for "Buy 2 get 1 free."

How many pounds must Janet purchase to get 2 pounds free?

- A. 2 pounds
- B. 4 pounds
- C. 6 apples
- D. 3 pounds

Answer(s): B

Explanation:

If Janet must purchase 2 pounds of apples to get 1 free pound, to get 2 free pounds, she would need to purchase twice as many apples or 4 pounds of apples.

QUESTION: 111

If 4 pipes of equal length measure 44 feet when they're connected together, how long is each pipe?

- A. 11 feet
- B. 4 feet
- C. 22 feet
- D. 9 feet

Answer(s): A

Explanation:

Divide the total length, 44 feet, by the total number of pipes, 4, because all the pipes are equal in length. The quotient, 11, is the length of each individual pipe. You can check this answer by multiplying: $4 \times 11 = 44$.

QUESTION: 112

A German shepherd and an Alaskan Malamute are both headed toward the same fire hydrant. The German Shepherd is 120 feet away from the hydrant and the Alaskan Malamute is 75 feet away from the hydrant.

How much closer to the hydrant is the Alaskan Malamute?

- A. 45 feet
- B. 25 feet
- C. 75 feet
- D. 195 feet

Answer(s): A

Explanation:

Subtract the Malamute's distance from the German Shepherd's distance (120 - 75) to determine how much closer the Malamute is to the hydrant.

QUESTION: 113

A recruit reporting to boot camp took a bus from her home to the military processing center in another city. The trip took 14 hours.

If she left at 6 a.m., what time did she arrive at the processing center?

- A. 7 p.m.
- B. 12 a.m.
- C. 8 p.m.
- D. 9 p.m.

Answer(s): C

Explanation:

Simply add 14 hours to 6 a.m. to reach 8 p.m.

QUESTION: 114

A farmer sold 3 pints of strawberries for \$1.98 each, 5 pints of raspberries for \$2.49 each, and a bushel of peaches for \$5.50 at his roadside stand.

How much money did the farmer make?

- A. \$9.97
- B. \$23.89
- C. \$18.39
- D. \$18.91

Answer(s): B

Explanation:

Multiply three pints of strawberries at \$1.98 ($3 \times \$1.98 = \5.94); 5 pints of raspberries at \$2.49 ($5 \times \$2.49 = \12.45) and 1 bushel of peaches at \$5.50 ($1 \times \$5.50 = \5.50). Add the products together to determine the amount of cash the farmer earned: $\$5.94 + \$12.45 + \$5.50 = \23.89 .

QUESTION: 115

A librarian wants to shelve 532 books.

If 4 books fit on a 1-foot length of shelving, how many feet of shelving does she need to shelve all the books?

- A. 13
- B. 45
- C. 33
- D. 133

Answer(s): D

Explanation:

Divide 532 by 4 to determine how many feet of shelving will be needed.

QUESTION: 116

A student buys a science textbook for \$18.00, a math textbook for \$14.50, and a dictionary for \$9.95.

What's the total cost of the books?

- A. \$27.95
- B. \$42.45
- C. \$41.95
- D. \$38.50

Answer(s): B

Explanation:

Simply add the cost of all the books: $\$18.00 + \$14.50 + \$9.95 = \42.45 .

QUESTION: 117

Debra works an 8-hour shift on Friday.

How many minutes does she work on Friday?

- A. 480 minutes
- B. 800 minutes
- C. 240 minutes
- D. 400 minutes

Answer(s): A

Explanation:

8 hours $\times$ 60 minutes per hour = 480 minutes.

QUESTION: 118

Six people can run 3 machines in the factory.

How many machines can 18 people run?

- A. 7
- B. 9
- C. 6
- D. 8

Answer(s): B

Explanation:

If 6 people can run 3 machines, then 18 people can run 3 times the number of machines 6 people can run because $18 = 3 \times 6$ (divide 18 by 6). 3×3 machines = 9 machines. 18 people can run 9 machines.

QUESTION: 119

On a map drawn to scale, $\frac{1}{2}$ inch equals 1 mile.
What length on the map equals 5 miles?

- A. 2.5 inches
- B. 5.0 inches
- C. 10.0 inches
- D. 1.5 inches

Answer(s): A

Explanation:

Multiply the scale measurement for 1 mile ($\frac{1}{2}$ inch per mile) by 5 miles: $\frac{1}{2} \times 5$ or $\frac{1}{2} \times \frac{5}{1} = \frac{5}{2}$. Reduce this fraction, and you get 2.5 inches.

QUESTION: 120

A man bought a pair of jeans for \$23.00, a shirt for \$14.95, and two ties for \$7.98 each.
What was the total cost of his clothing?

- A. \$53.91
- B. \$45.93
- C. \$51.99
- D. \$54.50

Answer(s): A

Explanation:

Simply add the cost of all the items: $\$23.00 + \$14.95 + \$7.98 + \$7.98 = \$53.91$.

QUESTION: 121

A rectangular block of brown metal folding chairs is set up on the floor of Reed Arena for the annual freshman convocation. The block of seats has x rows, with y folding chairs in each row.
How many folding chairs are there on the floor of the arena?

- A. $x + y$
- B. $x - y$
- C. xy
- D. $y - x$

Answer(s): C

Explanation:

To calculate the number of total chairs on the floor of the arena, multiply the number of rows (x) by the number of chairs in each row (y). This equation is written as xy .

QUESTION: 122

Jolie wants to get a new car before her second year of college. The sticker price of the kind of car she wants went up from \$22,399 last year to \$23,999 for this year's model. What percentage did the price increase?

- A. 1.071%
- B. 7.1%
- C. 9.3%
- D. 71.4%

Answer(s): B

Explanation:

First, calculate the amount of the price increase.

$$\$23,999 - \$22,399 = \$1,600$$

Then, calculate it as a percentage of the original price by multiplying it by 100 and then dividing by the original price:

$$(\$1,600 \times 100) / \$22,399 = 7.1\% \text{ increase}$$

QUESTION: 123

During his senior year a high school football wide receiver was thrown 82 passes and caught 57 of them.

What is his reception percentage (what percentage of the passes thrown to him did he catch)?

- A. 30.4%
- B. 69.5%
- C. 43.8%
- D. 81.7%

Answer(s): B

Explanation:

Multiply the number of caught passes by 100, then divide it by the number of passes thrown to the receiver. The result will be the percentage.

$$57 \times 100 = 5700$$

$$5700 / 82 = 69.5 = 69.5\%$$

QUESTION: 124

Amanda has a 20-year term life insurance policy for \$100,000. The annual premium is \$ 12.00 per thousand.

How much is her total premium payment every six months?

- A. \$600

- B. \$1,200
- C. \$100
- D. \$2,400

Answer(s): A

Explanation:

There are 100 units of \$1,000 in \$100,000. Thus, Amanda pays $100 \times \$12$ (or \$1,200) every year in premiums, or \$100 every month. Therefore, every six months, she pays $1/2$ of \$1,200 (or six times \$100), which equals \$600.

QUESTION: 125

If two pounds of salmon cost \$13.98, what is the cost of five ounces of the same brand of salmon?

- A. \$0.44
- B. \$0.87
- C. \$140
- D. \$2.20

Answer(s): D

Explanation:

There are 16 ounces in one pound. Therefore, if two pounds of salmon cost \$13.98, then one pound costs \$6.99. 1 ounce costs $\$6.99 \div 16 = \0.44 . 5 ounces cost $\$0.44 \times 5 = \2.20

QUESTION: 126

If five SUV tires and four rims cost \$1,730 and each rim costs \$120, what is the cost of one tire?

- A. \$150.00
- B. \$220.60
- C. \$320.20
- D. \$250.00

Answer(s): D

Explanation:

First, find the total cost of four rims:

$4 \times \$120 = \480 Then, calculate the total cost of the five SUV tires:

$\$1,730$ (cost of tires and rims) - $\$480$ (cost of rims) = $\$1,250$ Finally, find the cost of one SUV tire:

$\$1,250 \div 5 = \250

QUESTION: 127

Calculate the volume of a semi-truck trailer that is 23 feet long, 15 feet wide, and 11 feet high.

- A. 2,530 sq ft
- B. 3,450 cu ft

- C. 3,795 sq ft
- D. 3,795 cu ft

Answer(s): D

Explanation:

The formula for the volume of an object with parallel sides and right-angle comers is Length $\times$ Width $\times$ Height 23 feet of length $\times$ 15 feet of width $\times$ 11 feet of height = 3,795 cu ft (cubic feet)

QUESTION: 128

A college student spent \$119 on tickets for family and friends to a sporting event. If tickets were \$7 and \$10, and she bought an equal number of both kinds of tickets, how many \$7 tickets did she buy?

- A. 4
- B. 5
- C. 7
- D. 11

Answer(s): C

Explanation:

Let v be the number of tickets bought at each price (remember, the student bought the same number of both kinds of rickets).

So, $7v + 10v = 119$

Now combine the terms and continue solving the equation for v .

$17v = 119$

$v = 7$

QUESTION: 129

Amanda earns an average of \$22 an hour in tips as a waitress at the best steak restaurant in town.

If her hourly wage is \$2.50 and she has to pay a 10% tip share to the hostesses and busboys, how much does she take home at the end of a day when she worked from 10:00 a.m. to 5:00 p.m.?

- A. \$32.90
- B. \$121.11
- C. \$138.60
- D. \$156.10

Answer(s): D

Explanation:

First, calculate the amount of Amanda's hourly wages for a shift of 10:00 a.m. to 5:00 p.m., which is seven hours.

$7 \times \$2.50 = \17.50

Next, calculate the amount of Amanda's tips for her seven-hour shift.

$7 \times \$22 = \154

Now, calculate the amount of her tips she has to share with the busboys and hostesses.

$$\$154 \times 0.10 = \$15.40$$

Now add everything up.

Wages	\$17.50
Tips	\$154.00
Tip share	\$15.40
Net pay	\$156.10

QUESTION: 130

During Friday night's high school performance of the play *Seven Brides for Seven Brothers*, 76% of the seats in the performing arts center (with a seating capacity of 500 people) were occupied, and three-quarters of those attending were parents.

How many high school students attended the play?

- A. 95
- B. 100
- C. 76
- D. 285

Answer(s): A

Explanation:

The first step is to calculate the number of total playgoers by multiplying the total seats available by the percentage.

$$500 \times 0.76 = 380$$

Since three quarters (75%) of those attending were parents, that means that one quarter, or 25%, were high school students; this is the answer we are going to calculate. Therefore, multiply the number of total playgoers by the percentage of students to find out how many students attended the performance.

$$380 \times 0.25 = 95 \text{ students}$$

QUESTION: 131

Jonathan spent four hours doing calculus problems, one hour playing Web-based video games, half an hour talking to his girlfriend, and two hours lifting weights and exercising.

What percentage of his time was spent on math problems?

- A. 46.6%
- B. 50.0%
- C. 53.3%
- D. 57.1%

Answer(s): C

Explanation:

Add all the blocks of time together.

$$4 + 1 + 0.5 + 2 = 7.5 \text{ hours total}$$

Now multiply the time spent doing calculus (four hours) by 100 and divide it by the total time to get the percentage of time spent:

$$4 \times 100 = 400$$

$$400 \div 7.5 = 53.3\%$$

QUESTION: 132

Mrs. H.B. found a display model of a notebook computer to use for her business for \$1,400, including software. However, since the model had been discontinued and the display had no packaging material or instructions, the store manager discounted the price to \$1,150. What was the percentage of the reduction?

- A. 1.78%
- B. 13.0%
- C. 15.0%
- D. 17.9%

Answer(s): D

Explanation:

Subtract the discounted price from the original price.

$$\$1,400 - \$1,150 = \$250$$

Now multiply the amount of the discount by 100 and divide it by the original price.

$$\$250 \times 100 = 25,000$$

$$25,000 \div 1,400 = 17.857142 = 17.9\% \text{ discount}$$

QUESTION: 133

Heather bought a rectangular Persian rug with an outer perimeter measurement of 45 feet. If the long sides measure 15 feet each, how long is each short side of the rug?

- A. 7.5 ft
- B. 10 ft
- C. 12.5 ft
- D. 15 ft

Answer(s): A

Explanation:

The formula for the perimeter of a rectangle is $P = 2L + 2W$

If the long sides of the rug measure 15 feet each, then both long sides together equal 30 feet.

Subtract the length of both long sides from the total perimeter measurement.

$$45 - (2 \times 15) = 45 - 30 = 15\text{ft}$$

Now we have the part of the perimeter made up by the short sides, but we need to divide it by two to get the length of each short side.

$$15 \text{ ft} \div 2 = 7.5 \text{ ft}$$

QUESTION: 134

In the circulation department of a city newspaper, four customer service representatives each receive \$320 a week, while two assistant managers earn \$12 an hour each plus an average \$100 per day commission.

What is the average total weekly compensation paid to these six employees for a five-day, 40-hour work week?

- A. \$2,420
- B. \$3,240
- C. \$2,260
- D. \$4,520

Answer(s): B

Explanation:

First, add up the wages made by the four customer service reps.

$$\$320 \times 4 = \$1,280$$

Now calculate how much the two assistant managers make each week.

$$2 \text{ assistant managers} \times [(\$12 \text{ per hour} \times 40 \text{ hour}) + (5 \text{ days per week} \times \$100)] = 2 \text{ assistant managers} \times (\$480 + \$500)$$

$$= 2 \text{ assistant managers} \times \$980 = \$1,960 \text{ per week}$$

Now add the pay for the four customer service representatives to the pay for the two assistant managers.

$$\$1,280 + \$1,960 = \$3,240 \text{ total}$$

QUESTION: 135

Carpet costs \$3 per square yard. The room that will be carpeted is 12 feet $\times$ 18 feet.

How much will the carpet cost?

- A. \$648
- B. \$216
- C. \$72
- D. \$324

Answer(s): C

Explanation:

12 feet $\times$ 18 feet is equivalent to 4 yards $\times$ 6 yards = 24 square yards. 24 square yards $\times$ \$3 = \$72. Remember 3 feet = 1 yard, but 9 sq feet = 1 sq yard.

QUESTION: 136

If a train can travel 500 miles in five hours, how far can it travel in 15 minutes?

- A. 25 miles
- B. 30 miles
- C. 7 miles
- D. 125 miles

Answer(s): A

Explanation:

First find the number of miles per hour the train travels.

Now calculate how many miles the train travels in one minute.

$100 \text{ mph} \div 60 \text{ minutes} = 1.67 \text{ miles per minute}$

Now multiply the train's speed in miles per minute by the number of minutes in question.

$1.67 \text{ miles per minute} \times 15 \text{ minutes} = 25 \text{ miles}$

QUESTION: 137

An employee of a chocolate factory earns \$7.20 an hour. She is paid time and a half for overtime beyond a 40- hour week.

How much will she earn in a week in which she works 43 hours?

A. \$295.20

B. \$320.40

C. \$432.00

D. \$464.40

Answer(s): B

Explanation:

For the regular 40 hours, the worker earns $\$7.20 \times 40 = \288.00 . If the regular wage is \$7.20 per hour, then overtime paid at time and a half is $\$7.20 \times 1.5 = \10.80 . The three hours of

overtime earn the worker a total of \$32.40. Add the regular time wages to the overtime wages:

$\$288.00 + \$32.40 = \$320.40$ total pay

QUESTION: 138

A man died suddenly from a heart attack brought on by too much work and not enough exercise. Upon his death, his life insurance policies paid \$750,000 to his wife and three children.

The policies were set up to pay the wife and children in the ratio of 5:1:1:1.

How much did the three children receive altogether?

A. \$150,000

B. \$200,000

C. \$468,750

D. \$281,250

Answer(s): D

Explanation:

Start by letting* equal one share of the insurance money. According to the ratio, the wife received five shares (5x) and the children received one share (.x) a piece for a total of eight shares. Divide the total amount by the total number of shares.

$\$750,000 \div 8 = \$93,750$ per share

Now multiply the amount of each share by the total number of shares received by the children.

$\$93,750 \times 3 = \$281,250$

QUESTION: 139

B Company ("Bulldogs"), 3rd Battalion, 144th Infantry, is authorized a total of 131 soldiers. However, Bulldog Company only has 125 total soldiers and officers assigned. Of those assigned, four percent are officers. How many enlisted soldiers are assigned to the company?

- A. 114
- B. 123
- C. 120
- D. 121

Answer(s): C

Explanation:

If four percent of the assigned strength of the unit are officers, then the percent of enlisted men is $100\% - 4\% = 96\%$

To find the number of enlisted men, multiply the total number assigned by 96% or 0.96.

$125 \text{ total soldiers} \times 0.96 = 120 \text{ enlisted soldiers}$

QUESTION: 140

A technology reseller purchasing agent procured 20 mid-brand DVD players for the listed wholesale price of \$80 each, but receives a 25% discount because he is an established customer and friend of the wholesale dealer. His store then sells these DVD players at a 20% markup above the original wholesale price.

What is the store's profit margin on each DVD player?

- A. \$16
- B. \$72
- C. \$20
- D. \$36

Answer(s): D

Explanation:

The first step in solving this problem is to find the discount received from the wholesale dealer for each DVD player by multiplying the listed price by the discount percentage:

$\$80 \times 0.25 = \20

Next, find the actual price per DVD player paid to the wholesale dealer:

$\$80 - \$20 = \$60$

Now find the retail margin by multiplying the markup percentage by the original wholesale price:

$\$80 \times 0.20 = \16

Next, find the retail price for which the technology store sells the DVD players to its customers by adding the markup amount to the listed wholesale amount:

$\$80 + \$16 = \$96$

To find the profit margin, subtract the retail price from the actual price paid to the wholesale dealer.

$\$96 - \$60 = \$36 \text{ profit per DVD player}$

QUESTION: 141

A special-purpose cartographer (mapmaker) receives an order requiring a customized map with a scale of one inch = 50 miles.

If the actual straight-line ground distance between two points is 120 miles, how far apart should the cartographer show them on the map?

- A. 0.4 inches
- B. 2.4 inches
- C. 1.2 inches
- D. 4.8 inches

Answer(s): B

Explanation:

Since one inch on the map represents 50 miles, divide the ground distance (120 miles) by the scale (50 miles to the inch) to find the number of inches needed to represent 120 miles.

$$120 \div 50 = 2.4$$

QUESTION: 142

In Hendricks on Township, houses are assessed at 80% of the purchase price.

If Mr. Thomas buys a house in Hendricks on for \$120,000 and real estate taxes are \$4.75 per \$100 of assessed valuation, how much property tax must he pay per year?

- A. \$3,648
- B. \$5,472
- C. \$4,560
- D. \$4,845

Answer(s): C

Explanation:

Multiply the purchase price of the home by the assessment rate to find the assessed value.

$$120,000 \times 0.8 = \$96,000 \text{ (assessed value)}$$

Find the number of hundreds in the assessed value.

$$\$96,000 \div 100 = 960 \text{ (hundreds)}$$

Multiply the number of hundreds by the tax rate.

$$960 \times \$4.75 = \$4,560 \text{ (property tax)}$$

QUESTION: 143

The fuel tank of a gasoline generator holds enough fuel to operate the generator for one hour and 15 minutes.

About how many times must the fuel tank be filled to run the generator from 6:15 p.m. to 7:00 a.m.?

- A. 9.4
- B. 10.2
- C. 10.8
- D. 11.5

Answer(s): B

Explanation:

First find the total time needed to run the generator = 12 hours and 45 minutes or 12.75 minutes. Now divide the total time needed to run the generator by the amount of time each tank of fuel will last.

$$12.75 / 1.25 = 10.2 \text{ tanks of fuel}$$

QUESTION: 144

When Highway 2011 was converted from toll-free to a toll road, the traffic density per day declined from 11,200 vehicles to 10,044.

What was the percentage decline in traffic?

- A. 10.3%
- B. 0.5%
- C. 10.1%
- D. 8.9%

Answer(s): A

Explanation:

First find the amount of the decline in traffic. 11,200 cars - 10,044 cars = 1,156 fewer cars per day. Now multiply the amount of traffic decrease by 100 and then divide it by the amount of original traffic. This will yield the percentage decrease without further calculation.

$$1,156 \times 100 = 115,600$$

$$115,600 \div 11,200 = 10.3\% \text{ decrease}$$

QUESTION: 145

June drove for two hours at an average speed of 70 miles per hour, then drove another 90 minutes at an average speed of 65 miles per hour.

If her vehicle gets 25 miles per gallon on the highway in this speed range, about how many gallons of gas did she use for the trip?

- A. 5.4 gallons
- B. 8.2 gallons
- C. 8.4 gallons
- D. 9.5 gallons

Answer(s): D

Explanation:

First calculate the distances driven during the two legs of the trip.

$$\text{Leg 1: } 2 \text{ hours} \times 70 \text{ mph} = 140 \text{ miles}$$

$$\text{Leg 2: } 1.5 \text{ hours} \times 65 \text{ mph} = 97.5 \text{ miles}$$

$$140 \text{ miles} + 97.5 \text{ miles} = 237.5 \text{ miles}$$

Now divide the total distance traveled by the car's gas mileage rate.

$$237.5 \text{ miles} \div 25 \text{ mpg} = 9.5 \text{ gallons}$$

QUESTION: 146

Jonathan earns \$350 before taxes every two weeks at his part-time job. His withholdings are \$27.75 for federal income taxes, \$5.65 for Social Security (FICA), and \$12.87 for Medicare tax. How much will his net paycheck be?

- A. \$314.73
- B. \$304.73
- C. \$303.73
- D. \$313.73

Answer(s): C

Explanation:

First, total up all the withholding amounts.

$$\$27.75 + \$5.65 + \$12.87 = \$46.27$$

Jonathan's net pay is his salary for the pay period minus the total of all the withholdings.

$$\text{Net pay} = \$350.00 - \$46.27 = \$303.73$$

QUESTION: 147

A training certificate composed on a personal computer that was designed to be nine inches long and six inches wide has to be proportionately enlarged so that its length will be 12 inches. How many inches wide will the enlarged certificate be?

- A. 8
- B. 6
- C. 9
- D. 10

Answer(s): C

Explanation:

Since the certificate and its enlargement are similar (i.e., proportional) figures, the lengths will have the same ratio as the widths; if the length is increased by a given proportion, then the width will be increased by the same proportion or percentage of its original size.

QUESTION: 148

A basketball team won 70% of the 40 games it played. How many games did it lose?

- A. 28
- B. 30
- C. 22
- D. 12

Answer(s): D

Explanation:

To find the number of games won, multiply the number of games played, 40, by 70% or, put differently, 0.7.

$$40 \times 0.7 = 28$$

The number of games lost is the total number played minus the number won.

$$\text{Number of games lost} = 40 - 28 = 12$$

QUESTION: 149

The two members of Hoosiephaffer LLC, a limited liability company, operate their business such that it shows a profit for the year of \$63,000. Their LLC operating agreement calls for them to share the profits in the ratio 5:4.

How much of the profit should go to the member who gets the larger share?

- A. \$35,000
- B. \$28,000
- C. \$32,000
- D. \$36,000

Answer(s): A

Explanation:

If the profits are shared in the ratio 5:4, one partner gets 5/9 of the profits and the other gets 4/9.

Note that $5/9 +$

$4/9 = 1$, the whole profit. The larger share is 5/9 of the profit, \$35,000.

QUESTION: 150

A corporate purchaser paid \$17.16 for an item that had recently been increased in price by the wholesaler by four percent.

What was the price of the item before the increase?

- A. \$17.00
- B. \$17.12
- C. \$16.50
- D. \$16.47

Answer(s): C

Explanation:

Consider the original price as 100%. Then the price after an increase of 4% is 104%. To find the original price, divide the price after the increase, \$17.16, by 104% (or 1.04).

$$\$17.16 \div 1.04 = \$16.50$$

$$\$16.50 = \text{original price}$$

QUESTION: 151

A fire team is made up of four junior enlisted men and one noncommissioned officer as the team leader. A group of 64 privates fresh from basic training is to be divided into fire teams.

How many team leaders will be needed?

- A. 8
- B. 10
- C. 12

D. 16

Answer(s): D

Explanation:

The number of privates in the group who have just graduated from basic training divided by the number in each fire team gives you the number of fire teams and therefore the number of team leaders needed.

$$64 \div 4 = 16$$

16 team leaders are needed, one for each position.

QUESTION: 152

A tree 36 feet high casts a shadow 8 feet long. At the same time, another tree casts a shadow 6 feet long.

How tall is the second tree?

- A. 30 feet
- B. 27 feet
- C. 24 feet
- D. 32 feet

Answer(s): B

Explanation:

The ratio of the heights of the two trees will be the same as the ratio of the lengths of their shadows. The ratio of the length of the shadow of the second tree to the length of the shadow of the first tree is $\frac{6}{8}$ or $\frac{3}{4}$.

Thus, the height of the second tree is $\frac{3}{4}$ of the height of the 36 feet tree.

$$\frac{3}{4} \times 36 = 27 \text{ feet}$$

QUESTION: 153

Temperature readings on a certain day ranged from a low of -4°F to a high of 16°F .

What was the average temperature for the day?

- A. 10°F
- B. 6°F
- C. 12°F
- D. 8°F

Answer(s): B

Explanation:

The average is the sum of the high and low temperatures divided by 2.

$$\text{Average} = \frac{-4 + 16}{2} = \frac{12}{2} = 6^{\circ}$$

QUESTION: 154

A plane left San Francisco at 6:15 pm PST and arrived in Philadelphia at 2:45 am EST.

How long was the flight?

- A. $8\frac{1}{2}$ hours
- B. $5\frac{1}{2}$ hours
- C. $6\frac{1}{2}$ hour
- D. $9\frac{1}{2}$ hours

Answer(s): B

Explanation:

First, convert 6:15 PST to 9:15 EST. The flight then took from 9:15 pm EST to 2:45 am EST. That means that the flight took $5\frac{1}{2}$ hours.

QUESTION: 155

An automobile manufacturer offers a 15% rebate on the list price of a new car. What would be the rebate on a car that lists for \$13,620?

- A. \$900.00
- B. \$11,577.00
- C. \$204.30
- D. \$2,043.00

Answer(s): D

Explanation:

The rebate will be 15% (or 0.15) of the list price, \$13,600.
 $\$13,620 \times 0.15 = \$2,043.00 = \text{rebate}$

QUESTION: 156

A toll is \$2 for a car and the driver and 75 cent for each additional passenger. If the toll comes to \$4.25, how many total people are in the car?

- A. 6
- B. 5
- C. 3
- D. 4

Answer(s): D

Explanation:

The toll of \$4.25 includes \$2 for the driver/car. So \$2.25 (\$4.25 - \$2) is left for the passengers.
 $\$2.25 / 0.75 = 3$ passengers, so the total amount of people in the car is 1 driver and 3 passengers.

QUESTION: 157

A night watchman is required to check in every 40 minutes while making his rounds, to list tour of duty extends from 9:00 P.M. to 5:00 A.M.
 If he checks in at the start of his rounds and also when lie finishes, how many times does he check in during the night?

- A. 12
- B. 13
- C. 8
- D. 20

Answer(s): B

Explanation:

From 9:00 P.M. to 5:00 A.M. is a tour of duty of 8 hours. Since 60 minutes = 1 hour, when the watchman checks in every 40 minutes he checks in every $40/60$, or every $2/3$, of an hour.

$$8/1 \div 2/3 = 8/1 \times 3/2 = 12$$

The watchman must check in once at the beginning of each of the twelve 40-minute periods, or 12 times. But he must also check in at the end of the last period. Thus, he checks in $12 + 1$, or 13, times.

QUESTION: 158

A flight is scheduled for departure at 3:50 P.M.

If the flight takes 2 hours and 55 minutes, at what time is it scheduled to arrive at its destination?

- A. 5:05 P.M.
- B. 6:05 P.M.
- C. 6:15 P.M.
- D. 6:45 P.M.

Answer(s): D

Explanation:

The time of arrival is 2 hours and 55 minutes after the departure time of 3:50 P.M. By 4:00 P.M., the flight has taken 10 minutes of the total flight time of 2 hours and 55 minutes. 2 hours and 45 minutes remain, and 2 hours and 45 minutes after 4:00 P.M. is 6:45 P.M.

QUESTION: 159

How many 4-ounce candy bars are there in a 3-pound package of candy?

- A. 12
- B. 16
- C. 48
- D. 9

Answer(s): A

Explanation:

There are 16 ounces in 1 pound. Therefore, 4 of the 4-ounce candy bars will make 1 pound. A 3-pound package will hold 3 times 4 or 12 bars.

QUESTION: 160

In a restaurant, a diner orders an entree with vegetables for \$10.50, dessert for \$1.50, and

coffee for \$0.50.

If the tax on meals is 6%, what tax should be added to his check?

- A. \$0.06
- B. \$0.63
- C. \$0.72
- D. \$0.75

Answer(s): D

Explanation:

First add the prices of the three items ordered to get the cost of the meal before the tax. $\$10.50 + \$1.50 + \$0.50 = \12.50 . The tax is 6% (or 0.06) of the cost of the meal. $0.06 \times \$12.50 = \0.75

QUESTION: 161

A 55-gallon drum of oil is going to be used to fill cans that hold 2 quarts each.

How many cans can be filled from the drum?

- A. 55
- B. 27
- C. 110
- D. 220

Answer(s): C

Explanation:

There are 4 quarts in 1 gallon. A 55-gallon drum holds 4×55 quarts.

4×55 quarts = 220 quarts. If each can holds 2 quarts, the number of cans filled is 220 divided by 2.

$220 / 2 = 110$ cans

QUESTION: 162

In a factory that makes wooden spindles, a lathe operator takes 45 minutes to do the finishing work on nine spindles.

How many hours will it take him to finish 96 spindles at the same rate?

- A. 8
- B. 72
- C. 100
- D. 10

Answer(s): A

Explanation:

The number of minutes required to finish 1 spindle is the number of minutes, 45, taken for all 9 spindles, divided by their number, 9.

$45 / 9 = 5$ minutes per spindle. The number of minutes to finish 96 spindles will be 96 times as much. $5 \times 96 = 480$ minutes There are 60 minutes in 1 hour. Divide the number of minutes by 60 to convert their number to hours.

$$480 / 60 = 8 \text{ hours}$$

QUESTION: 163

A triangle has two equal sides. The third side has a length of 13 feet, 2 inches. If the perimeter of the triangle is 40 feet, what is the length of one of the equal sides?

- A. 13 feet, 4 inches
- B. 26 feet, 10 inches
- C. 13 feet, 11 inches
- D. 13 feet, 5 inches

Answer(s): D

Explanation:

The perimeter, 40 feet, is the sum of the lengths of all three sides. The sum of the lengths of the two equal sides is the difference between the perimeter and the length of the third side. Sum of lengths of 2 equal sides = 40 feet - 13 feet, 2 inches = 39 feet, 12 inches - 13 feet, 2 inches = 26 feet, 10 inches. The length of one side is obtained by dividing the sum by 2. Length of one equal side: 26 feet, 10 inches / 2 = 13 feet, 5 inches

QUESTION: 164

A lawn is 21 feet wide and 39 feet long. How much will it cost to weed and feed it if a gardening service charges \$0.40 per square yard for this treatment?

- A. \$109.20
- B. \$36.40
- C. \$327.60
- D. \$24.00

Answer(s): B

Explanation:

Since 3 feet = 1 yard, convert the length and width to yards by dividing their dimensions in feet by 3.

$$\text{Width} = 39/3 = 7 \text{ yards;}$$

$$39 \text{ length} = 39/3 = 13 \text{ yards}$$

The area of a rectangle is the product of its length and width.

Area = $7 \times 13 = 91$ square yards. The cost for the entire lawn is obtained by multiplying the area in square yards by the cost per square yard.

$$91 \times \$0.40 = \$36.40$$

QUESTION: 165

A bonded courier drove for 7 hours on back roads at an average speed of 48 miles per hour. Her car gets 21 miles per gallon of gas. How many gallons of gas did she use?

- A. 24

- B. 11
- C. 18
- D. 16

Answer(s): D

Explanation:

Find the number of miles traveled by multiplying the rate, 48 miles per hour, by the time, 7 hours.

$$48 \times 7 = 336 \text{ miles}$$

The number of gallons of gas used is the number of miles driven divided by the number of miles per gallon.

$$336 / 21 = 16 \text{ gallons}$$

QUESTION: 166

A wall is 7 feet 8 inches in height; four vertical pieces of wallpaper are needed to cover the wall. Assuming there is no waste, what is the minimum length that a roll of wallpaper must be to cover the wall?

- A. 29 feet, 2 inches
- B. 31 feet, 2 inches
- C. 31 feet
- D. 30 feet, 8 inches

Answer(s): D

Explanation:

If there are four pieces of wallpaper, each 7 feet, 8 inches long, the total length of the wallpaper is 4×7 feet, 8 inches, or 28 feet, 32 inches. Since 12 inches = 1 foot, 32 inches = 2 feet, 8 inches. Thus, 28 feet, 32 inches = 30 feet, 8 inches.

QUESTION: 167

We bought 5 pounds of cashew nuts worth \$1.80 per pound, mixed with 4 pounds of peanuts worth \$1.62 per pound.

What is the value per pound of the mixture?

- A. \$1.72
- B. \$1.71
- C. \$3.42
- D. \$1.82

Answer(s): A

Explanation:

5 pounds of cashew nuts at \$1.80 per pound is worth $5 \times \$1.80 = \9.00 4 pounds of peanuts at \$1.62 per pound is worth $4 \times \$1.62 = \6.48 .

The total value of the 9-pound mixture is $\$9.00 + \$6.48 = \$15.48$. The value of the mixture per pound is found by dividing the total value by the number of pounds.

$$\$15.48 / 9 = \$1.72$$

QUESTION: 168

In a clothing factory, 5 workers finish production of 6 garments each per day, 3 others turn out 4 garments each per day, and 1 worker turns out 12 per day.

What is the average number of garments produced per worker per day?

- A. $22/9$
- B. 6
- C. 4
- D. $71/3$

Answer(s): B

Explanation:

5 workers making 6 garments each = 30 garments per day

3 workers making 4 garments each = 12 garments per day

$1/9$ workers making 12 garments = $12/54$ garments per day

Add the number of workers and the number of garments per day.

9 workers make 54 garments per day.

To find the average number of garments per worker per day, divide the number of garments per day by the number of workers.

$54/9 = 6$ garments per worker per day.

QUESTION: 169

A man makes a 255-mile trip by car. He drives the first 2 hours at 45 miles per hour.

At what speed must he then travel for the remainder of the trip in order to arrive at his destination 5 hours after he started?

- A. 31 miles per hour
- B. 50 miles per hour
- C. 51 miles per hour
- D. 55 miles per hour

Answer(s): D

Explanation:

The distance traveled by driving at 45 mph for 2 hours is 2×45 , or 90 miles.

The remainder of the 255 mile trip is $255 - 90$, or 165 miles.

To finish the trip in 5 hours, the man has $5 - 2$, or 3, hours still to drive.

To find the rate of travel for a distance of 165 miles driven in 3 hours, divide the distance by the time.

$165/3 = 55$ miles per hour

QUESTION: 170

A contractor bids \$300,000 as her price for erecting a building. She estimates that $1/10$ of this amount will be spent for masonry materials and labor, $1/3$ for lumber and carpentry, $1/5$ for plumbing and heating, and $1/6$ for electrical and lighting work. The balance will be her profit. How much profit does she expect to make?

- A. \$24,000
- B. \$80,000
- C. \$60,000
- D. \$50,000

Answer(s): C

QUESTION: 171

The list price of a TV set is \$325, but the retailer offers successive discounts of 20% and 30%. What price does a customer actually pay?

- A. \$182.00
- B. \$270.00
- C. \$162.50
- D. \$176.67

Answer(s): A

Explanation:

The first discount of 20% means that a customer actually pays 80% (or 4/5) of the list price. The second successive discount of 30% means that a customer actually pays 70% (or 3/10) of the price determined after the first discount.

The price the customer actually pays is the list price multiplied by the portions determined from each discount.

QUESTION: 172

A certain brand of motor oil is regularly sold at a price of two cans holding a quart each for \$1.99. On a special sale, a carton containing six quart cans is sold for \$5.43. What is the saving per quart if the oil is bought at the special sale price?

- A. \$0.27
- B. \$0.09
- C. \$0.54
- D. \$0.541/4

Answer(s): B

Explanation:

At 2 quarts for \$1.99, each quart costs \$0.995. At 6 quarts for \$5.43, each quart costs \$0.905. $\$0.995 - \$0.905 = \$0.095$

QUESTION: 173

A company spends 3/5 of its advertising budget on newspaper ads and 1/3 on radio commercials.

What portion of the advertising budget is left for TV commercials?

- A. 1/2

- B. $7/30$
- C. $1/15$
- D. $1/8$

Answer(s): C

Explanation:

Add the portion spent on newspaper ads and radio commercials.

$$\frac{3}{5} + \frac{1}{3} = \frac{9}{15} + \frac{5}{15} = \frac{14}{15}.$$

The portion left for TV commercials is the entire budget, 1, minus portion spent on newspaper and radio $14/15$.

$$1 - \frac{14}{15} = \frac{15}{15} - \frac{14}{15} = \frac{1}{15}$$

QUESTION: 174

At Joe's Restaurant, one-fourth of the patrons are male and one-fifth of the patrons are from out of town.

What proportion would you expect to be male and out of town?

- A. $1/5$
- B. $1/10$
- C. $1/20$
- D. $1/25$
- E. $1/50$

Answer(s): C

Explanation:

If one-fourth of the patrons are male, then you would expect one-fourth of the out-of-town patrons to be male.

One fourth of one-fifth is $1/20$.

QUESTION: 175

There are 112 males and 78 females working at a hospital.

What percentage of the hospitals workforce is female?

- A. 22%
- B. 32%
- C. 38%
- D. 41%
- E. 78%

Answer(s): D

Explanation:

Add the number of men and women to get the total number: 190. The number of women, 78, so divide 78 by 190 ($78/190$) to get 0.4105. Then multiply by 100 to convert it to a percentage and you have 41%.

QUESTION: 176

What is the estimated product when 620 and 374 are rounded to the nearest hundred and multiplied?

- A. 12,000
- B. 24,000
- C. 60,000
- D. 120,000
- E. 240,000

Answer(s): E

Explanation:

620 is rounded to 600; 374 is rounded to 400. $600 \times 400 = 240,000$.

QUESTION: 177

If a patient doesn't receive treatment for a certain injury within 2 hours after injury, the patient's chance of survival starts decreasing by 15% each hour.

If no treatment has been given within 4 hours after injury, roughly, what chance of survival will the patient have?

- A. 70%
- B. 72%
- C. 75%
- D. 77%
- E. 80%

Answer(s): B

Explanation:

At the end of two hours, the patient still has a 100% chance of survival. After three hours, he has an 85% of survival, or 0.85. After four hours, he has an 85% of the 85% he had at the end of three hours: $0.85 \times 0.85 = 72.25\%$ chance of survival.

QUESTION: 178

William is from England and is visiting the United States. On his trip he visits Texas and notices that it is very hot. He looks at a thermometer and it reads 95 degrees Fahrenheit.

William wants to know what this converts to in Celsius? Assume $C = (F - 32) / 1.8$

- A. 30 degrees C
- B. 35 degrees C
- C. 40 degrees C
- D. 45 degrees C
- E. 50 degrees C

Answer(s): B

Explanation:

Substituting 95 for F in the equation yields $C = (95 - 32) / 1.8 = 35$ degrees C.

QUESTION: 179

A student scores 74, 87, 68, 63, 76 on his last five exams. His teacher will drop the lowest grade to compute the student's average.

What is the students average in the class?

- A. 76.3%
- B. 79.4%
- C. 84.9%
- D. 89.4%
- E. 68.2%

Answer(s): A

Explanation:

The teacher will drop the 63, and only count the four best scores. These are 74, 87, 68, and 76. To find the average, add the 4 numbers and divide the sum by 4.

QUESTION: 180

$21/4 + 63/8 = \underline{\hspace{2cm}}$.

- A. 8
- B. $83/8$
- C. $85/8$
- D. 9

Answer(s): C

Explanation:

Convert the $1/4$ to $2/8$ and then add $2/8$ to $3/8$ to get $5/8$. Finally add $6 + 2$ and you have $85/8$

QUESTION: 181

A world map has a scale of $1/4$ of an inch equals 250 miles. You measure the distance between New York, NY and Los Angeles, California and it equals 3 inches.

What is the real distance between New York and LA?

- A. 2,500 miles
- B. 2,750 miles
- C. 2,800 miles
- D. 3,000 miles
- E. 3,250 miles

Answer(s): D

QUESTION: 182

Of 1,000 dogs, 72% are male. Among the males, 3 out of 5 were over the age 5.

How many of the dogs were males over the age of 5?

- A. 312
- B. 432
- C. 468
- D. 512
- E. 720

Answer(s): B

Explanation:

First multiply 1000×0.72 to find out how many dogs are male. This results in 720 male dogs of which $\frac{3}{5}$ are over the age of 5. Therefore, multiply $720 \times \frac{3}{5}$ (or 0.6) which results in 432.

QUESTION: 183

There are 78 female and 54 male patrons at a local diner. What percentage of the diner's patrons are female?

- A. 37%
- B. 41%
- C. 52%
- D. 59%
- E. 67%

Answer(s): D

Explanation:

Add 78 to 54 to get 132 total patrons, then divide the number of females into the total number of patrons.

Therefore, $78/132 = 0.5909 = 59\%$.

QUESTION: 184

A new cancer treatment provides a 50% success rate for patients with lung tumors. Lung tumors make up 10% of all cancer patients.

What percentage of all cancer patients will this new drug work on?

- A. 5%
- B. 20%
- C. 50%
- D. 80%
- E. 95%

Answer(s): A

Explanation:

A simple way to solve this is to multiply the two percentages. Therefore, multiply 0.50×0.10 and you get 0.05 or 5%.

QUESTION: 185

A person who has a heart attack can be saved 90% of the time if a defibrillator is used within 5 minutes of the heart attack. Of the people that have heart attacks, defibrillators are within the vicinity 40% of the time; however, they are only used 60% of the time even when they are in the surrounding area.

How many people who suffer a heart attack are saved because of the use of a defibrillator?

- A. 18%
- B. 22%
- C. 40%
- D. 54%
- E. 60%

Answer(s): B

Explanation:

If a defibrillator is used, we know that the patient will have a 90% success rate; however, only 40% of the people have this option available.

Therefore, we take 0.9×0.4 and know that only 36% of the people have this opportunity.

However, we also know that of the 36% of suffers, only 60% of the time is a defibrillator used.

Therefore, we must take 0.6×0.36 to find out how many patients are saved by the use of a defibrillator.

This results in 0.216 or 22% of patients are saved by the use of a defibrillator.

QUESTION: 186

A paint company mixes its paint with water to create more paint per gallon. They currently have ten gallon jugs which contain 10% water.

How much water must they add to create an 80/20 mix of paint to water?

- A. 1 gallon
- B. 1.25 gallons
- C. 1.5 gallons
- D. 1.75 gallons
- E. 2 gallons

Answer(s): B

Explanation:

Currently the mixture is 90% paint, 10% water or 9 gallons of paint for every 1 gallon of water.

For the company to create an 80% / 20% mix, the company must add 1.25 gallons to their current 10 gallon mixtures. This will create 2.25 gallons of water for every 11.25 total gallons.

This results in a 2.25/11.25 or 20% water mix.

QUESTION: 187

A mail carrier can deliver mail to 1,400 different homes in one day. In the city of Utopia, there are 52,800 residents.

How many mail carriers must be hired to deliver mail to all of the residents of Utopia each day?

- A. 38
- B. 43

- C. 48
- D. 54
- E. 60

Answer(s): A

Explanation:

To find the number of required mail carriers, simply divide 52,800 by 1,400 to get 37.714. Therefore, you must round up and you have 38 mail carriers.

QUESTION: 188

How much change would you get back from a \$100.00 bill if you purchased eight DVDs costing \$6.49 each?

- A. \$24.24
- B. \$48.08
- C. \$51.92
- D. \$64.90

Answer(s): B

QUESTION: 189

Fencing costs \$1.50 per picket. 2 pickets are used per foot. Posts cost \$13.75 per post. How much will it cost to fence a yard if 10 posts and 46 feet of pickets are needed?

- A. \$125.50
- B. \$138.00
- C. \$275.50
- D. \$380.75

Answer(s): C

QUESTION: 190

Louis ran a race in 2.4 minutes. Laura ran the same race in 124 seconds. What is the difference between the two times?

- A. 2 seconds
- B. 20 seconds
- C. 40 seconds
- D. 1 minute

Answer(s): B

Explanation:

2.4 minutes is 144 seconds

QUESTION: 191

A grocery store has a special on sodas. Three 12 packs for \$9.00.
How much does five 12 packs of soda cost?

- A. \$9.00
- B. \$12.00
- C. \$15.00
- D. \$18.00

Answer(s): C

QUESTION: 192

A family of four have a dinner bill for \$44.00. A 15% tip is added after an 8% tax is added to the amount.
How much is the tip?

- A. \$4.05
- B. \$6.60
- C. \$6.71
- D. \$7.13

Answer(s): D

Explanation:

According to the question, this consumer added the 15% tip on top of the bill and the tax. Therefore, you must first multiply $\$44.00 \times 1.08$ which equals \$47.52. Now you will multiply $\$47.52 \times 15\%$ or 0.15. This results in \$7.128 which rounds up to \$7.13.

QUESTION: 193

Jane takes out a loan for \$100,000.00 over 10 years and agrees to pay interest only at an annual rate of 8%.
What will her monthly payment be?

- A. \$333.33
- B. \$666.67
- C. \$1,000
- D. \$8,000

Answer(s): B

Explanation:

This question is asking you to only determine the monthly payment of an interest only loan. Thus, you do not need to concern yourself with the 10-year term or the principal being repaid as that is irrelevant. The only numbers you need to work with are the principal amount, the interest rate and the number of months in a year.
The formula for calculating an interest only loan would be $(\text{Principal} \times \text{Annual Rate}) / \text{Number of months in a year}$. Therefore: $(\$100,000 \times 0.08) / 12 = \666.66666 or \$666.67. A payment of \$666.67 per month would result in payments of \$8000.04 per year which is 8% of \$100,000.00.

QUESTION: 194

Karen is 6 years older than her sister Michelle, and Michelle is 2 years younger than their brother David.

If the sum of their ages is 62, how old is Michelle?

- A. 20
- B. 16
- C. 18
- D. 24

Answer(s): C

QUESTION: 195

A barista averages making 16 coffees per hour.

At this rate, how many hours will it take until she's made 1,200 coffees?

- A. 70 hours
- B. 75 hours
- C. 80 hours
- D. 65 hours

Answer(s): B

QUESTION: 196

Camille uses a 20% off coupon when buying a sweater that costs \$47.99.

If she also pays 6% sales tax on the purchase, how much does she pay?

- A. \$50.87
- B. \$40.69
- C. \$38.39
- D. \$43.64

Answer(s): B

QUESTION: 197

At Michael's smoothie shop smoothies cost \$3.50 during the week and \$4.00 on weekends.

If Jonathan bought a smoothie every day except Saturday how much did he spend on smoothies that week?

- A. \$21.50
- B. \$25.50
- C. \$17.50
- D. \$21.00

Answer(s): A

QUESTION: 198

In a classroom of 32 students, 14 are male.
What percentage of the class is female?

- A. 56%
- B. 52%
- C. 44%
- D. 46%

Answer(s): A

QUESTION: 199

Cory's average score after 8 tests is 91%.
If he gets a 78% on his 9th test, what will his new average be?

- A. 90.5%
- B. 87.5%
- C. 89.6%
- D. 89.5%

Answer(s): C

QUESTION: 200

Julie gives 7 pieces of candy to each of her friends.
If Julie gives all of her candy away, which amount of candy could have been the amount she distributed?

- A. 128
- B. 161
- C. 145
- D. 162

Answer(s): B

QUESTION: 201

Four out of 28 students had to go to summer school.
What is the ratio of students who did not have to go to summer school expressed, in its lowest terms?

- A. $\frac{4}{5}$
- B. $\frac{7}{8}$
- C. $\frac{9}{10}$
- D. $\frac{6}{7}$

Answer(s): D

QUESTION: 202

If doughnuts are usually 75 cents each, but there is a sale on Friday advertising them as 1

dozen + 1/2 dozen for \$10.80, what is the new cost for just one?

- A. 55 cents
- B. 50 cents
- C. 60 cents
- D. 65 cents

Answer(s): C

QUESTION: 203

A cross country coach just announced that, by the end of the quarter, all students should run the 10 miles in less than 70 minutes.

There are 15 students in the group.

6 of them ran 10 miles in 1 hour + 5 minutes

4 of them ran 10 miles in 1 hour + 3 minutes

2 of them ran 10 miles in 1 hour + 9 minutes

3 of them ran 10 miles in 1 hour and 10 minutes.

How many students made the goal?

- A. 3
- B. 10
- C. 4
- D. 12

Answer(s): D

QUESTION: 204

You just went on a blind date and wanted to impress the girl by adding an 18% tip to the total cost of the meal. Two entrees came to a total of \$35.90. Desserts cost \$7.50 and beverages cost \$6.60.

What is the 18% tip going to cost?

- A. \$5.00
- B. \$7.75
- C. \$8.50
- D. \$9.00

Answer(s): D

QUESTION: 205

You are asked to buy bags of cement to pour a cement pad for a hot tub. You need to fill the pad frame which has an 8-foot square base and one-foot high sides. Each bag of cement mixed will fill 5 cubic feet in the frame.

How many bags will you need to buy to ensure enough to fill the entire frame?

- A. 8 bags
- B. 10 bags
- C. 13 bags

D. 12 bags

Answer(s): C

QUESTION: 206

You are asked to chart the temperature during an 8-hour period to give the average.

These are your results:

7am: -10 degrees

8am: 0 degrees

9 am: 30 degrees

10 am: 36 degrees

11 am: 40 degrees

12 pm: 42 degrees

1pm: 42 degrees

2 pm: 41 degrees

What is the average temperature?

A. about 32 degrees

B. about 40 degrees

C. about 28 degrees

D. about 30 degrees

Answer(s): C

QUESTION: 207

Your uncle passed away and left you and your siblings his private property. He owned a 180-acre ranch and there are 3 of you. Each acre is valued at approximately \$3,000.

This means the monetary value for each sibling will be _____.

A. \$60,000

B. \$18,000

C. \$180,000

D. \$54,000

Answer(s): C

QUESTION: 208

You are planning your vacation and trying to figure out the total number of miles you will be traveling on a round trip from Portland, Oregon to Denver, Colorado, and back to Portland, Oregon. The scale for the map you are consulting shows that 1 inch = 4 miles. Measuring, you find that you will be traveling $314\frac{1}{2}$ inches for the one- way trip.

What is the total number of miles you will be traveling during the entire round trip?

A. 314.5 miles

B. 2,516 miles

C. 1258 miles

D. 629 miles

Answer(s): B

QUESTION: 209

Pete and his partner Mike made a huge profit in their software business together this year and have decided to use \$20,000 of the profit for a Christmas donation to their favorite charities. Using the investment ratio each of them contributed to starting the business, Pete invested the most with 4:1 ratio to Mike.
How much money will Mike get to donate to his charity?

- A. \$5,000
- B. \$4,000
- C. \$3,000
- D. \$16,000

Answer(s): B

QUESTION: 210

What is the sixth number in the series: 3, 6, 12, 24?

- A. 48
- B. 30
- C. 90
- D. 96

Answer(s): D

QUESTION: 211

At boarding school, students spend $\frac{1}{4}$ of their time eating and sleeping, $\frac{3}{5}$ of their time in class, and $\frac{1}{12}$ studying. The rest of the time is considered free time, which the student can spend doing activities of their own choosing.
What fraction is considered free time?

- A. $\frac{1}{15}$
- B. $\frac{1}{20}$
- C. $\frac{1}{10}$
- D. $\frac{1}{12}$

Answer(s): A

QUESTION: 212

You just priced what it would cost to buy vinyl floor tiles at the local tile store. They quoted you \$3.29 per tile. When you went home, you found the same tile for sale at an online outlet for \$2.89 per tile.
As a percentage, how much more expensive is the tile store's quote?

- A. The tile store's price costs about 14% more than the online outlet store.
- B. The tile store's price costs about 11% more than the online outlet store.

- C. The tile store's price costs about 16% more than the online outlet store.
- D. The tile store's price costs about 10% more than the online outlet store.

Answer(s): A

QUESTION: 213

Susan went to the Spa and got a manicure for \$25.00, a pedicure for \$28.75 and a massage for \$60.00. She tipped a total of \$8.50.

What is the total Susan spent at the Spa?

- A. \$113.75
- B. \$120.25
- C. \$122.25
- D. \$97.25

Answer(s): C

QUESTION: 214

Steve is mailing a small package. Postage rates are: \$0.46 for the first ounce and \$0.20 for each additional ounce. The UPS store is offering a special for \$3.45 on a small box if your items fit, the weight not being a factor.

Which of the following weights would make it cheaper for Steve to ship his package by the post office rather than by UPS?

- A. 18 ounces or higher
- B. 16 ounces
- C. 15 ounces
- D. 17 ounces

Answer(s): C

QUESTION: 215

Mike's motorcycle stalled at the beach and he called the towing company. They charged him \$4.75 per mile for the first 25 miles and then \$5.25 per mile for each mile over 25. Mike was 29 miles from the motorcycle repair shop.

How much was Mike's towing bill?

- A. \$139.75
- B. \$137.75
- C. \$118.75
- D. \$131.25

Answer(s): A

QUESTION: 216

The sum of two numbers is 84. One number is 6 more than the other.

What is the smaller of the two numbers?

- A. 32
- B. 39
- C. 33
- D. 35

Answer(s): B

QUESTION: 217

The grocery store orders ibuprofen in bottles by the gross. The first day after stocking the shelves with a new shipment they sold 5 bottles. The next day they sold 8 and the third day 7. How many bottles remain from the original order?

- A. 124
- B. 131
- C. 104
- D. 100

Answer(s): A

QUESTION: 218

Cindy's test scores on her first five finals were: 90, 92, 98, 96, and 94. In order for Cindy to keep her scholarship she needs to maintain a finals' average of 94 or higher. Which of these scores on her sixth test (the final) will guarantee her average of 94 or higher?

- A. 89
- B. 90
- C. 96
- D. 88

Answer(s): C

QUESTION: 219

A waiter earns \$11.00 an hour and approximately 10% of what he serves in a shift. If he works a 6 hour shift and takes \$425.00 in orders, his total earnings for the six hours would be _____.

- A. \$66.00
- B. \$106.50
- C. \$42.50
- D. \$108.50

Answer(s): D

QUESTION: 220

How many square feet of tile is needed for a 24 foot × 24 foot room?

- A. 576 square feet
- B. 288 square feet
- C. 96 square feet
- D. 144 square feet

Answer(s): A

QUESTION: 221

John is going to waterproof his carpeting which is 18 feet $\times$ 24 feet. The rate is 57 cents per square foot to apply the waterproofing solution.
John's total cost will be _____.

- A. \$244.24
- B. \$200.04
- C. \$246.24
- D. \$239.94

Answer(s): C

QUESTION: 222

A pen factory has a monthly overhead of \$5,000. It costs them 21 cents to make a pen and they sell each pen for 99 cents.
How many pens will they need to sell each month in order to start making a profit?

- A. 5678
- B. 6411
- C. 7422
- D. 5416

Answer(s): B

QUESTION: 223

You are working in a hospital supply room and make \$13.65 an hour. The union negotiates a new contract giving each employee a 4.3% cost of living raise.
What is your new hourly rate?

- A. \$14.59 an hour
- B. \$14.33 an hour
- C. \$14.24 an hour
- D. \$14.43 an hour

Answer(s): C

QUESTION: 224

You just left for vacation 2 hours ago. You have driven about 50 miles per hour on the 2-lane road and take an exit to get dinner. You call your Dad to check in. Your Dad has decided to join you after initially turning down your invitation.

How long will you have to wait for him if he drives 40 miles per hour?

- A. 2 hours + 30 minutes
- B. 2 hours + 40 minutes
- C. 3 hours
- D. 2 hours + 20 minutes

Answer(s): A

QUESTION: 225

A pizzeria caters parties and just finished baking 60 pizzas. The church youth group just picked up their order of $\frac{1}{4}$ of the pizzas baked. A school graduation party is picking up $\frac{1}{3}$ of the pizzas baked. A martial arts club stopped in and bought $\frac{1}{20}$ of the pizzas baked. How many pizzas are left of the 60 that were baked?

- A. 20
- B. 18
- C. 15
- D. 22

Answer(s): D

QUESTION: 226

Jill bought 2 cases of computer paper (10 reams per case) on sale for \$85.00. The regular rate is: 1 case = \$52.50. What was her savings amount on the cost of each ream?

- A. \$1.00 per ream savings
- B. \$1.25 per ream savings
- C. \$0.75 per ream savings
- D. \$0.50 per ream savings

Answer(s): A

QUESTION: 227

You take your dog for a walk in the evenings and it takes about 25 minutes to walk 8 city blocks. If you decide to do a "marathon walk" of 32 city blocks on a Saturday, how many minutes will it take you to complete this?

- A. 1 hour + 40 minutes
- B. 1 hour + 25 minutes
- C. 1 hour
- D. 1 hour + 32 minutes

Answer(s): A

QUESTION: 228

School enrollments have fluctuated quite a bit in Oregon this year. The state department of education predicted there would be 300,000 students enrolled state-wide. The enrollment ended up being 375,000.

How much of a percentage increase over the projected amount is this?

- A. A 10% increase in the projected amount.
- B. A 20% increase in the projected amount.
- C. A 25% increase in the projected amount.
- D. A 15% increase in the projected amount.

Answer(s): C

QUESTION: 229

You're applying for a job that has a skills test requirement. In order to pass you need to get 48 of the 55 questions correct.

What percent of the questions do you need to answer correctly?

- A. 87%
- B. 75%
- C. 55%
- D. 67%

Answer(s): A

QUESTION: 230

You just started a retirement fund by depositing \$7,000 in an account. The annual interest rate for an account with an "early withdrawal" penalty is 5.75%.

How much money will you have in your account if you leave it for 2 years?

- A. \$7402.50
- B. \$7080.50
- C. \$7500.28
- D. \$7828.14

Answer(s): D

QUESTION: 231

You're getting in shape for a marathon and are going to use the local school track to build up to more mileage.

The track measures one quarter mile around.

If you ran 18 laps, how many miles would this be?

- A. 4 miles
- B. 3.50 miles
- C. 4.50 miles
- D. 3.75 miles

Answer(s): C

QUESTION: 232

You just drove 280 miles and it took you approximately 5 hours. How many miles per hour was your average speed?

- A. about 56 miles per hour
- B. about 60 miles per hour
- C. about 55 miles per hour
- D. about 58 miles per hour

Answer(s): A

QUESTION: 233

You work in a hospital and make \$15.28 per hour for a 40 hour a week. If you work the night shift you get a slight differential added at \$2.32 per hour worked during this shift. You just completed your first week and worked 34 hours on a regular shift and 6 hours on the night shift. What is the total amount you earned this week?

- A. \$519.52
- B. \$615.20
- C. \$611.20
- D. \$625.12

Answer(s): D

QUESTION: 234

Your local library branch has 28 employees and currently provides approximately 48 square feet per employee for a work station. The library just hired 2 new employees. To make room to add these 2 new people, each employee's work station will be reduced to about:

- A. 28.8 square feet
- B. 44.8 square feet
- C. 42.8 square feet
- D. 30.8 square feet

Answer(s): B

QUESTION: 235

Tom has been saving for a new car and finally has the money to make the deal. He put down \$3,800 and will be paying \$325 a month for 5.5 years. What is the total cost of his purchase?

- A. \$19,500
- B. \$21,450
- C. \$25,250
- D. \$23,300

Answer(s): C

QUESTION: 236

Eight girls in a club are comparing their weekly allowance. Jan and Sue each receive \$10 a week. Lori and Tina each receive \$5 a week. Julie, Christie and Faith each receive \$12 a week. Cheryl does not get an allowance but her parents often buy her things as she needs them. What is the average amount of allowance given among these 8 girls?

- A. \$7.00
- B. \$8.25
- C. \$9.42
- D. \$6.75

Answer(s): B

QUESTION: 237

The distance between cities A and B is approximately 1,800 miles. If you drive an average of 55 miles per hour, how many hours will it take you to drive from city A to city B?

- A. approximately 22 hours
- B. approximately 24 hours
- C. approximately 36 hours
- D. approximately 33 hours

Answer(s): D

QUESTION: 238

You are hosting a graduation party and you need 32 gallons of lemonade. Which is the best buy?

- A. three 10-gallon jugs at \$208 each + two 1-gallon jugs at \$15 each
- B. one 25-gallon jug at \$542 + one 5-gallon jug at \$100 + two 1-gallon jugs at \$15 each
- C. one 25-gallon jug at \$542 + one 10-gallon jug at \$208
- D. six 5-gallon jugs at \$100 each + two 1-gallon jugs at \$15 each

Answer(s): D

QUESTION: 239

Chris is driving a truck that can hold 4 tons maximum. He has a shipment of food weighing 26,000 pounds. How many trips will he need to make to deliver all of the food?

- A. 1 trip
- B. 3 trips
- C. 4 trips

D. 2 trips

Answer(s): C

QUESTION: 240

Two monorails depart their respective stations at the same time. The monorail headed south is traveling 65 miles an hour, while the monorail headed north is traveling 55 miles per hour. Given this information, how many miles will both monorails travel in 4 hours?

- A. 480 miles
- B. 465 miles
- C. 570 miles
- D. 508 miles

Answer(s): A

QUESTION: 241

Your new remodel is almost done. The contractor needs to put the final framing on and needs 8 sections, measuring 4 feet, 2 inches each. The framing is available from the supplier by 10-foot lengths only.

What is the total length of the lumber that needs to be purchased?

- A. 34 feet
- B. 40 feet
- C. 48 feet
- D. 30 feet

Answer(s): B

QUESTION: 242

You just bought 4 take-out Mexican dinners. They were priced \$14.96, \$14.96, \$15.75, and \$17.85. You had a coupon to "buy one, get a second one for half-price on all items that are the same price." (Only 1 coupon per purchase allowed).

What is the total amount you paid?

- A. \$55.25
- B. \$58.04
- C. \$56.04
- D. \$63.52

Answer(s): C

QUESTION: 243

Bob just got hired for on-the-road sales and will travel about 1,500 miles a week during a 40-hour work week.

If the time spent traveling is $\frac{2}{3}$ of his week, how many hours a week will he be on the road?

- A. Bob spends about 20 hours of his 40-hour work week on the road.
- B. Bob spends about 27 hours of his 40-hour work week on the road.
- C. Bob spends about 22 hours of his 40-hour workweek on the road.
- D. Bob spends about 13 hours of his 40-hour work week on the road.

Answer(s): B

QUESTION: 244

If you invest \$10,000 at an annual rate of 7%, how much interest will you earn after one year?

- A. \$7
- B. \$700
- C. \$10,700
- D. \$70

Answer(s): B

Explanation:

Interest = Principal $\times$ Rate = \$10,000 $\times$ 0.07 = \$700

QUESTION: 245

How many pounds does four tons of gravel weigh?

- A. 4000
- B. 16000
- C. 8000
- D. 12000

Answer(s): C

Explanation:

One ton = 2,000 pounds. Therefore, 4 tons = 8,000 pounds

QUESTION: 246

If the price of a gallon of gas increases from \$3.20 to \$3.68, what percent did it increase?

- A. 15%
- B. 10%
- C. 115%
- D. 12%

Answer(s): A

Explanation:

To find the percent of increase, subtract the second number from the original/first number ($3.68 - 3.2 = 0.48$) and divide that result by the original number ($0.48 / 3.20 = 0.15 = 15\%$)

QUESTION: 247

If 5 apples cost \$2.00, how much will 15 apples cost?

- A. \$8
- B. \$30
- C. \$10
- D. \$6

Answer(s): D

Explanation:

15 apples are three times as many as 5 apples, so 15 apples will cost $3 \times \$2 = \6

QUESTION: 248

If 4 people can stain a deck in 2 days, how many people would be required to stain the deck in only 1 day?

- A. 8
- B. 2
- C. 4
- D. 12

Answer(s): A

Explanation:

If it requires 4 people for 2 days, then it would require 8 people for 1 day.

QUESTION: 249

How much time is there between 7:45 am today and 3:15 pm tomorrow?

- A. $7\frac{1}{2}$ hours
- B. $30\frac{1}{2}$ hours
- C. $31\frac{1}{2}$ hours
- D. $32\frac{1}{2}$ hours

Answer(s): C

Explanation:

From 7:45 am today to 7:45 am tomorrow is 24 hours. From 7:45 am tomorrow to 3:15 pm tomorrow is $7\frac{1}{2}$ hours. Total is $31\frac{1}{2}$ hours.

QUESTION: 250

If a gas tank can hold 30 gallons, how many gallons does it contain when it is three-fifths full?

- A. 15
- B. 12
- C. 24
- D. 18

Answer(s): D

Explanation:

$$30 \text{ gallons} \times 3/5 = 18$$

QUESTION: 251

How long will it take to reach a town 250 miles away if you drive 50 miles per hour?

- A. 4 hours
- B. 5 hours
- C. 10 hours
- D. 6 hours

Answer(s): B

Explanation:

$$250 \text{ miles} / (50 \text{ miles per hour}) = 5 \text{ hours}$$

QUESTION: 252

Tina bought a scarf for \$16.65. She gave the clerk \$20.
How much change should she get back?

- A. \$4.35
- B. \$3.45
- C. \$3.35
- D. \$3.25

Answer(s): C

Explanation:

$$\$20 - \$16.65 = \$3.35$$

QUESTION: 253

Sam bought a 12 pack of soda for \$5.40.
How much did he pay per can of soda?

- A. \$0.45
- B. \$0.54
- C. \$0.40
- D. \$2.22

Answer(s): A

Explanation:

$$\$5.40 / 12 = \$0.45$$

QUESTION: 254

A stack of bricks is 3 1/2 feet high.

If each brick is 13/4 inches high, how many bricks are in the stack?

- A. 2
- B. 12
- C. 24
- D. 28

Answer(s): C

Explanation:

$3\frac{1}{2} \text{ feet} \times 12 \text{ in/ft} = 42 \text{ inches.}$

$42 \text{ inches} / 1.75 \text{ inches} = 24 \text{ bricks}$

QUESTION: 255

A book regularly sells for \$28.

If the book is on sale for "20% off", how much is the sales price of the book?

- A. \$5.60
- B. \$25.20
- C. \$21.20
- D. \$22.40

Answer(s): D

Explanation:

$20\% \text{ of } \$28 = 28 \times 0.2 = \$5.60.$

The sales price is then $\$28 - \$5.6 = \$22.40$

QUESTION: 256

A scale on a map says that 1 inch represents 60 miles.

If the distance between two points on the map is 2 1/2 inches, how many miles are between the two points?

- A. 150
- B. 120
- C. 24
- D. 180

Answer(s): A

Explanation:

$2\frac{1}{2} \times 60 = 150$

QUESTION: 257

A car has a 15-gallon gas tank.

If a car gets 32 miles per gallon and travels 384 miles, how much gas will be left in the tank?

- A. 12
- B. 6
- C. 3
- D. 4

Answer(s): C

Explanation:

The car will have used 12 gallons (384 miles / 32 gallons). Therefore, there will be 3 gallons left in the tank (15 gallons - 12 gallons).

QUESTION: 258

A man's hourly wage increased from \$12.50 to \$13.50.
What percent raise did he get?

- A. 10%
- B. 13.50%
- C. 9%
- D. 8%

Answer(s): D

Explanation:

To find the percent of increase, subtract the second number from the original/first number ($13.5 - 12.5 = 1.0$) and divide that result by the original number ($1 / 12.5 = 0.08 = 8\%$).

Section 2: Auto and Shop Information

QUESTION: 259

Overheating the engine can cause all of the following problems EXCEPT _____.

- A. burned engine bearings
- B. enlarged pistons
- C. melted engine parts
- D. improved fuel efficiency

Answer(s): D

QUESTION: 260

If an alternator overcharges the battery, a likely is _____.

- A. The governor has malfunctioned.
- B. The voltage regulator isn't working properly.
- C. The ignition coil has overheated.
- D. The battery-acid solution is low.

Answer(s): B

QUESTION: 261

A primary advantage of the electronic ignition system over conventional ignition systems is _____.

- A. the electronic ignition system is less expensive to repair.
- B. the electronic ignition system provides a higher voltage.
- C. the electronic ignition system allows for use of a lower octane fuel.
- D. All of the above.

Answer(s): B

QUESTION: 262

The primary purpose of piston rings is to _____.

- A. seal the combustion chamber and allow the pistons to move freely.
- B. connect the piston to the crankshaft.
- C. allow fuel to enter the piston cylinder.
- D. provide lubrication to the piston cylinder.

Answer(s): A

QUESTION: 263

Connecting rods connect the piston to the _____.

- A. flywheel
- B. fuel pump
- C. crankshaft
- D. battery

Answer(s): C

QUESTION: 264

In an overhead valve system (OHV), what mechanism opens and closes the valves?

- A. rocker arms
- B. camshaft
- C. valve rotator
- D. electrical energy from the alternator

Answer(s): A

QUESTION: 265

If a car's ignition system, lights, and radio don't work, the part that's probably malfunctioned is the _____.

- A. cylinder block
- B. water pump
- C. carburetor
- D. battery

Answer(s): D

QUESTION: 266

A gauge shows the complete loss of oil pressure while driving. The best action is to _____.

- A. Stop by the gas station when convenient to top off the oil.
- B. Pull over immediately and investigate the problem.
- C. Drive directly to a repair garage.
- D. Assume everything is fine and continue driving as usual.

Answer(s): B

QUESTION: 267

A two-cycle engine will normally be found on _____.

- A. small cars
- B. large diesel trucks
- C. trucks, vans, and some cars
- D. snowmobiles, chainsaws, and some motorcycles

Answer(s): D

QUESTION: 268

The difference between a single-acting and an opposed piston engine is _____.

- A. Single-acting piston engines wear longer.
- B. Opposed piston engines have cylinders set in a V-shape.
- C. Single-acting piston engines have one piston per cylinder and opposed piston engines have two.
- D. Single-acting piston engines are used with carburetors and opposed piston engines are used with fuel injectors.

Answer(s): C

QUESTION: 269

A car equipped with limited-slip differential _____.

- A. can be readily put into all-wheel (four-wheel) drive.
- B. won't lock up when the brakes are applied steadily.
- C. transfers the most driving force to the wheel with the greatest amount of traction.
- D. is rated for off-road driving.

Answer(s): C

QUESTION: 270

Pouring cold water on an overheated engine _____.

- A. reduces damage caused by overheating.
- B. makes no difference.
- C. should only be done by a qualified mechanic.
- D. could cause the engine block to crack.

Answer(s): D

QUESTION: 271

Soft brake-pedal movement can be caused by _____.

- A. air in the hydraulic brake system
- B. malfunctioning brake shoes
- C. loss of brake fluid
- D. worn rotors

Answer(s): A

QUESTION: 272

The safest way to make a hole in sheet metal is to _____.

- A. burn it with a soldering iron
- B. cut it using snips
- C. punch it using a metal punch
- D. get someone else to do it

Answer(s): C

QUESTION: 273

Rebar is used to _____.

- A. measure the depth of concrete
- B. reinforce concrete
- C. stir concrete
- D. smooth concrete

Answer(s): B

QUESTION: 274

Nail sets are used to _____.

- A. protect your fingers from the hammer.
- B. set nails below the surface of wood.
- C. complete projects requiring sets of nails.
- D. mark the position where the nail should go.

Answer(s): B

QUESTION: 275

A rip saw cuts _____.

- A. against the grain of the wood
- B. with the grain of the wood
- C. most materials, including metal
- D. only plastic

Answer(s): B

QUESTION: 276

Sledges can be used to drive _____.

- A. nails
- B. screws
- C. staples
- D. bolts and chisels

Answer(s): D

QUESTION: 277

To check for horizontal trueness, the best tool to use is a _____.

- A. steel tape rule
- B. plumb bob
- C. level
- D. sliding T-bevel

Answer(s): C

QUESTION: 278

A bucking bar is used to _____.

- A. pull nails
- B. pry wood apart
- C. drive rivets
- D. drive screws

Answer(s): C

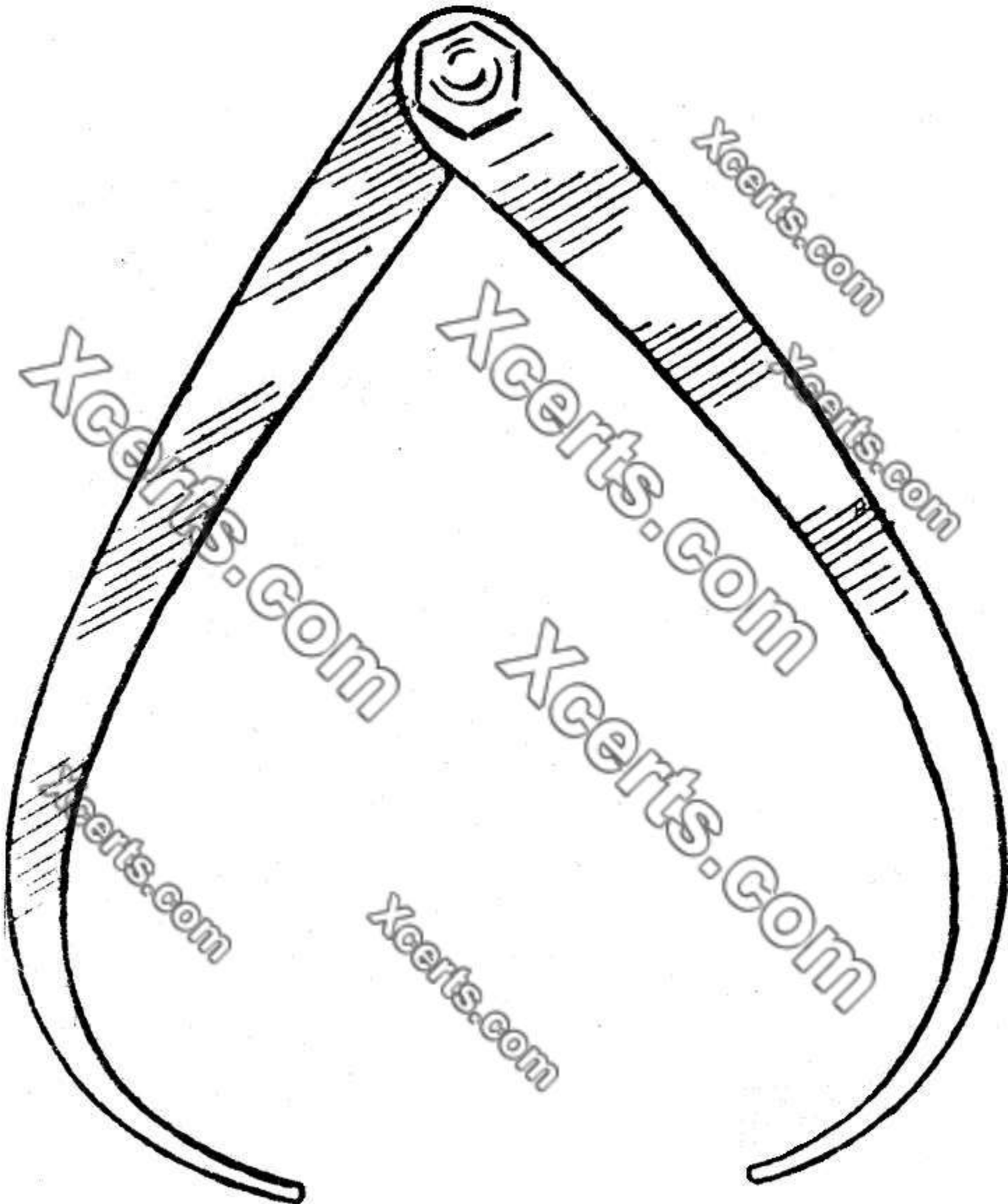
QUESTION: 279

Washers that have teeth all around the circumference to prevent them from slipping are called _____.

- A. shake proof washers
- B. jaw washers
- C. flat washers
- D. split-lock washers

Answer(s): A

QUESTION: 280

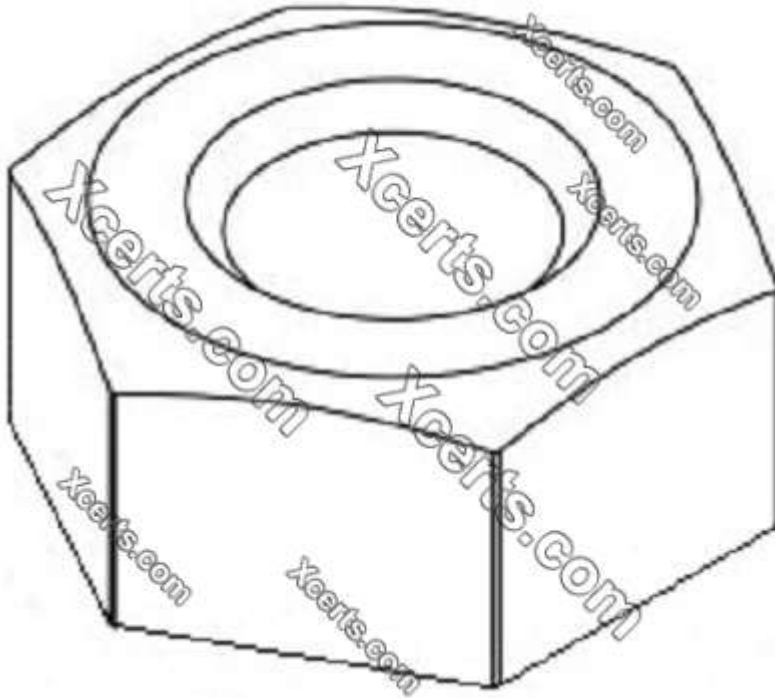


The tool above measures:

- A. an inside curve
- B. an outside curve
- C. the depth of a hole
- D. the thickness of wire

Answer(s): B

QUESTION: 281

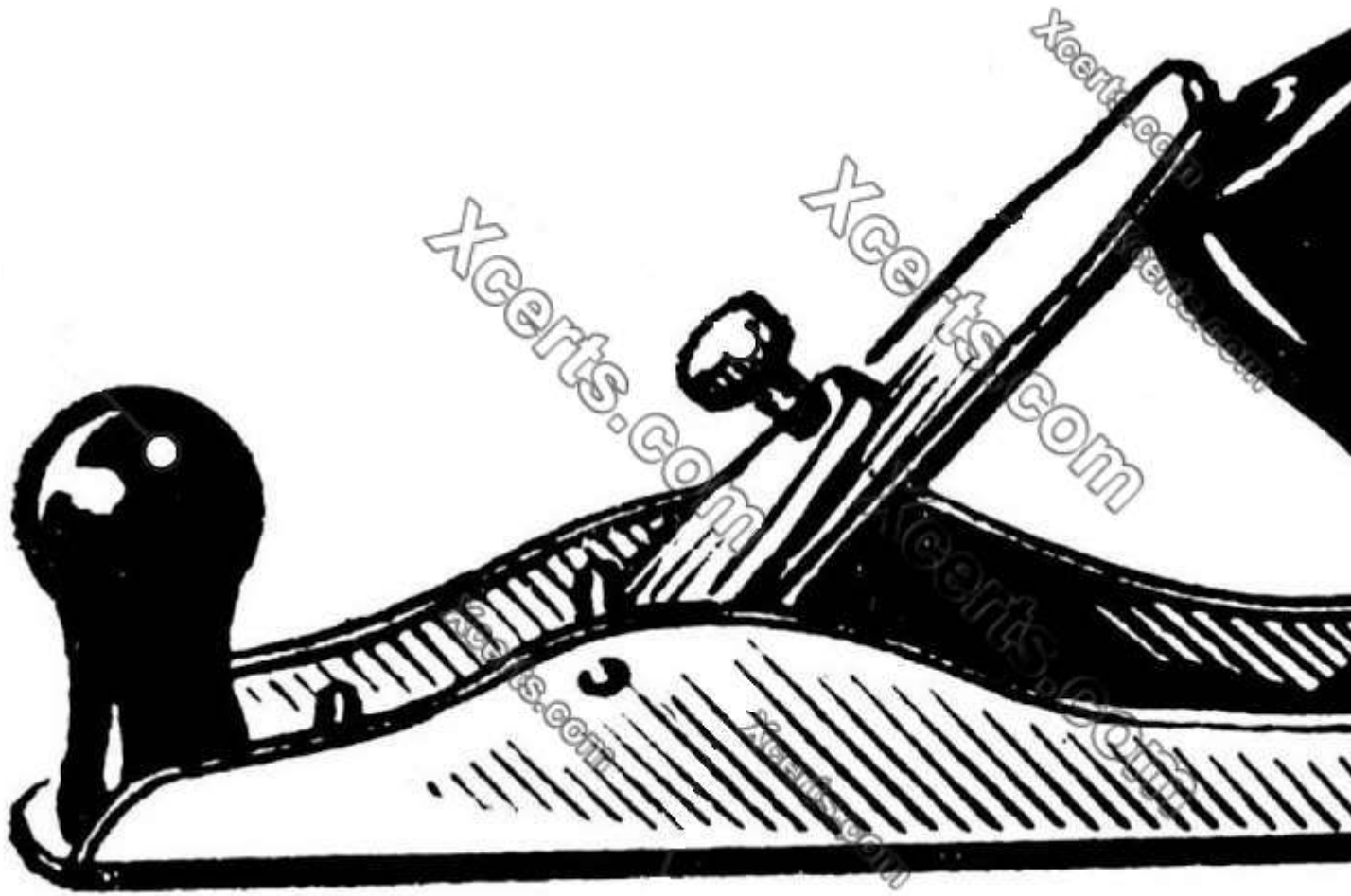


The object above is a type of:

- A. nut
- B. washer
- C. screw
- D. bolt

Answer(s): A

QUESTION: 282



The tool above is used to:

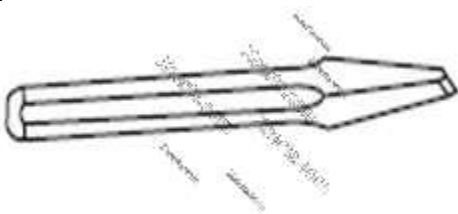
- A. finish concrete
- B. spread joint compound
- C. smooth wallpaper
- D. dress wood

Answer(s): D

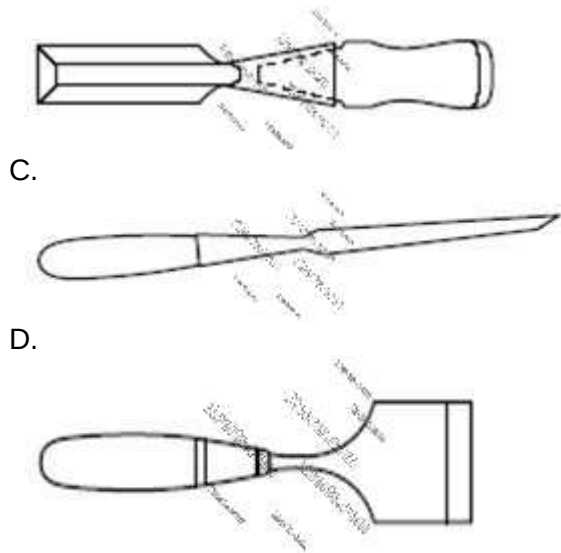
QUESTION: 283

The chisel used to cut metal is:

A.



B.



C.

D.

Answer(s): A

QUESTION: 284

What tool is used by electricians to route new wiring through walls?

- A. Spike maul
- B. Wall snake
- C. Fish tape
- D. Bradawl

Answer(s): C

Explanation:

Fish tape (or draw wire/draw tape) is used to route wiring through walls and conduit.

QUESTION: 285

On which automotive system would a "stall test" be performed?

- A. transmission
- B. exhaust
- C. engine
- D. suspension

Answer(s): C

QUESTION: 286

The alternator _____.

- A. starts the engine
- B. supplies power to the battery

- C. connects the ignition system to the engine
- D. can be used as an alternative to motor oil

Answer(s): B

QUESTION: 287

In which automotive system would you find a "wishbone"?

- A. transmission
- B. engine
- C. exhaust
- D. suspension

Answer(s): A

QUESTION: 288

If the electrolyte solution in a battery is too low, you should add _____.

- A. sulfuric acid
- B. antifreeze
- C. distilled water
- D. gasoline

Answer(s): C

QUESTION: 289

What area of your car should be flushed periodically to maintain optimum performance?

- A. exhaust system
- B. brake system
- C. cooling system
- D. ignition system

Answer(s): D

QUESTION: 290

In a hydraulic brake system, if brake action is soft, a likely is _____.

- A. the rotors are worn
- B. air has gotten into the system
- C. the master cylinder needs to be replaced
- D. the brake calipers need adjustment

Answer(s): B

QUESTION: 291

Car restorers often seek NOS parts.
What does NOS stand for?

- A. Near Original Specifications.
- B. NASCAR Operating Standards.
- C. New Old Stock.
- D. None of the above.

Answer(s): D

QUESTION: 292

To make spark plugs work effectively, the coil and breaker _____.

- A. provide a gap between the electrodes.
- B. ignite the spark.
- C. transfer the electricity to the correct spark plug.
- D. create a very high voltage of electricity.

Answer(s): D

QUESTION: 293

Schrader valves can be found in your car's _____.

- A. tires
- B. engine
- C. transmission
- D. electronic ignition

Answer(s): B

QUESTION: 294

A bent frame causes _____.

- A. improper tracking
- B. auto accidents
- C. poor ride
- D. excessive rust

Answer(s): A

QUESTION: 295

In the tire designation 205/55 R 15 92 H, what does the "H" signify?

- A. tread type
- B. tire height
- C. maximum speed
- D. turning radius

Answer(s): D

QUESTION: 296

A sheet of aluminum should be cut with _____.

- A. a pair of kitchen scissors
- B. a handsaw
- C. tin snips
- D. a coping saw

Answer(s): C

QUESTION: 297

Hammer faces are commonly made of each of the following materials EXCEPT _____.

- A. steel
- B. brass
- C. plastic
- D. lead

Answer(s): C

QUESTION: 298

Hammers, mallets, and sledges are all striking tools, but mallets and sledges don't have _____.

- A. a claw
- B. metal parts
- C. as much durability
- D. wedges

Answer(s): A

QUESTION: 299

Round objects can be measured most exactly using a _____.

- A. rigid steel rule
- B. folding rule
- C. caliper
- D. depth gage

Answer(s): C

QUESTION: 300

The best chisel to use when making a circular cut in metal is a _____.

- A. cold chisel
- B. socket chisel
- C. butt chisel
- D. round chisel

Answer(s): D

QUESTION: 301

The term penny is used to designate the _____.

- A. number of threads per inch on a screw
- B. length of a nail
- C. cost of a nail
- D. material a nail is made from

Answer(s): B

QUESTION: 302

Painting on a surface with too much moisture _____.

- A. causes no problems
- B. causes bubbling
- C. requires an extra coat of paint
- D. takes longer

Answer(s): B

QUESTION: 303

To tighten a loose handle on a hammer _____.

- A. soak the handle in water
- B. use wood glue to secure the handle more firmly to the head
- C. drive another wedge into the head
- D. replace the hammer because the problem can't be fixed

Answer(s): C

QUESTION: 304

An 8-point saw _____.

- A. has 8 teeth per inch
- B. weighs 8 ounces
- C. can saw 8 kinds of material
- D. is 8-inches long

Answer(s): A

QUESTION: 305

Concrete is made by mixing _____.

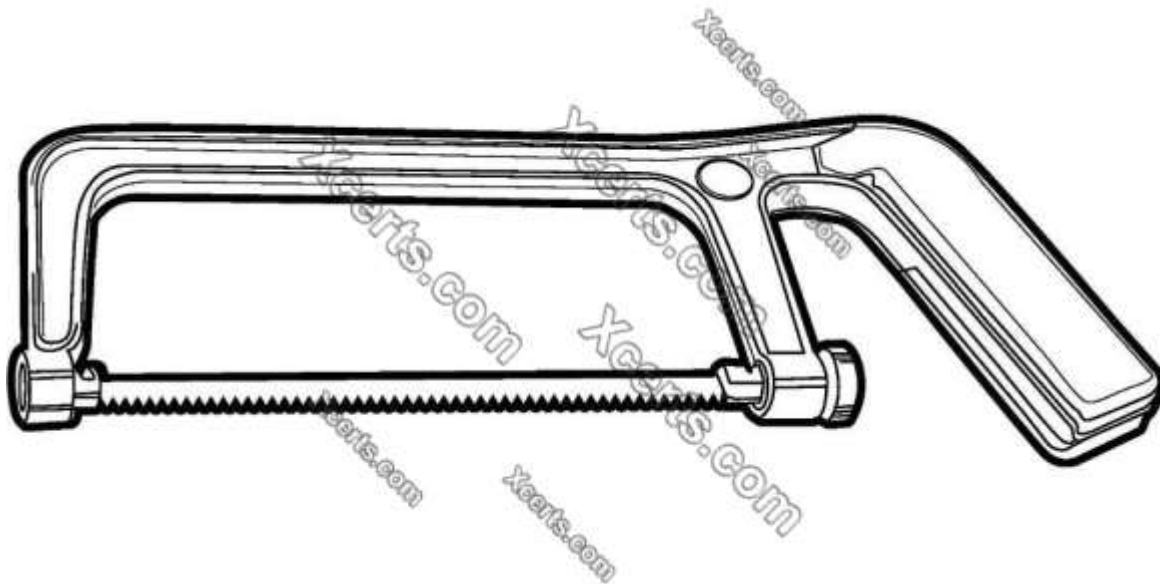
- A. cement and sand
- B. cement, sand, and water
- C. cement and water
- D. cement, sand, gravel, and water

Answer(s): D

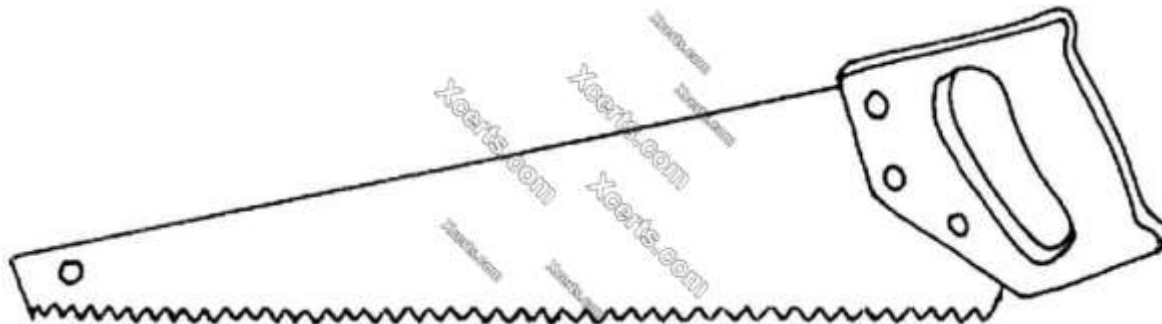
QUESTION: 306

Which of the following tools isn't used to cut metal?

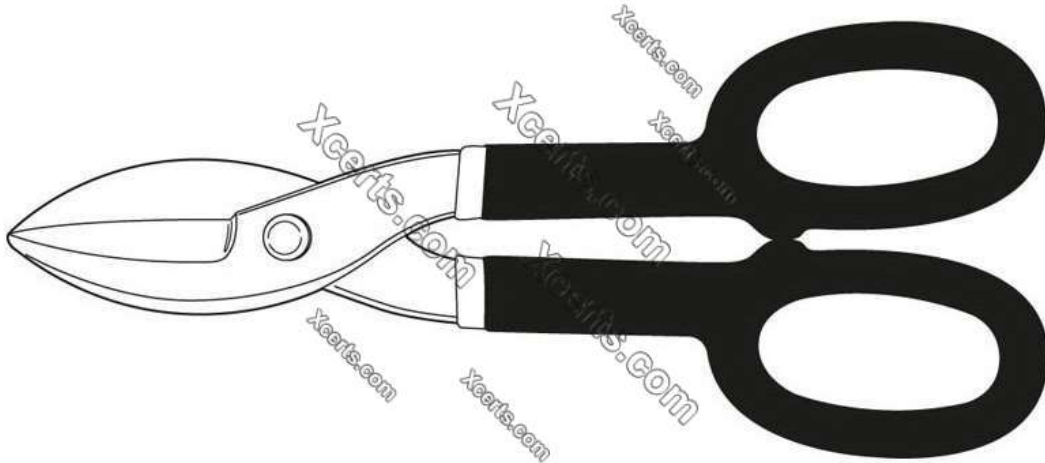
A.



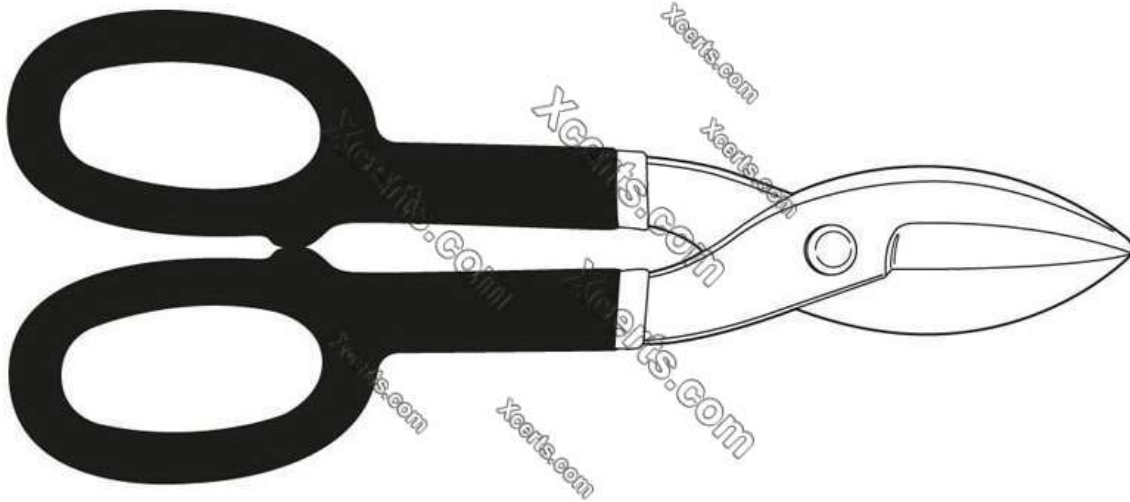
B.



C.

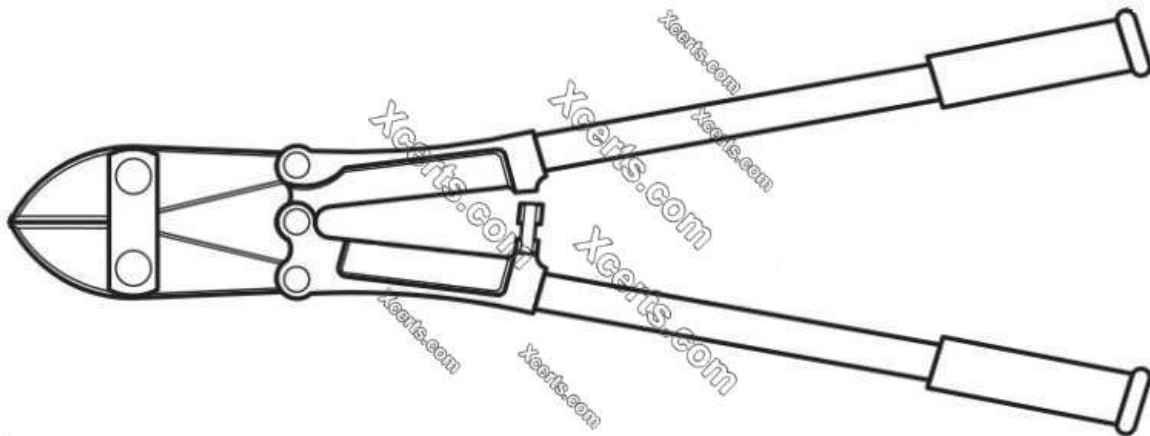


D.



Answer(s): B

QUESTION: 307



The tool above is used to _____.

- A. cut tile
- B. cut wire
- C. turn screws
- D. cut bolts

Answer(s): D

QUESTION: 308



The tool above is a(n) _____.

- A. Phillips screwdriver
- B. Allen wrench
- C. socket wrench

D. offset screwdriver

Answer(s): B

QUESTION: 309

A symptom of worn piston rings is _____.

- A. a knocking and pinging sound when driving
- B. soft and spongy acceleration
- C. the smell of exhaust in the car
- D. an engine using excessive amounts of oil

Answer(s): D

QUESTION: 310

What term refers to the rebuilding of an engine to precise factory specifications?

- A. blueprinting
- B. specing
- C. gold rebuild
- D. silver rebuild

Answer(s): A

QUESTION: 311

The number of cranks a crankshaft has on a V-8 engine is _____.

- A. 6
- B. 4
- C. 3
- D. 8

Answer(s): D

QUESTION: 312

When an engine runs on after the ignition key is turned off, it's called _____.

- A. auto-ignition
- B. sputtering
- C. ignition recharge
- D. ignition malfunction

Answer(s): A

QUESTION: 313

If a radiator fails, the engine _____.

- A. will idle roughly
- B. may burn fuel less efficiently
- C. works hard to maintain speed
- D. can quickly overheat

Answer(s): D

QUESTION: 314

On modern automobile engines, what's the purpose of the intake manifold?

- A. regulate air flow to the cooling system
- B. provide air flow to the air conditioner and heater
- C. connect the air/fuel management device to the head
- D. regulate fuel pump pressure

Answer(s): C

QUESTION: 315

Brake systems work by _____.

- A. applying friction to the wheels to stop their rotation
- B. reversing power to the wheels
- C. applying pressure to the axle
- D. interrupting power to the transmission

Answer(s): A

QUESTION: 316

Which of the following isn't a component of the cooling system?

- A. freeze plugs
- B. radiator
- C. thermostat
- D. hydrator

Answer(s): D

QUESTION: 317

A catalytic converter _____.

- A. combines the fuel-air mixture
- B. reduces dangerous exhaust emissions
- C. converts the up-and-down motion of the pistons to rotary motion
- D. charges the battery when the engine is in operation

Answer(s): B

QUESTION: 318

If the steering wheel vibrates at high speeds, the most likely problem is _____.

- A. front end alignment
- B. front tire balance
- C. cracked steering column
- D. over-inflated tires

Answer(s): B

QUESTION: 319

During the compression stroke on a four cycle engine _____.

- A. The intake valve opens to fill the cylinder with fuel.
- B. The burning fuel mixture forces the piston to the bottom of the cylinder.
- C. The intake valve closes, and the piston moves to the top of the cylinder.
- D. The exhaust valve releases the burned gas.

Answer(s): C

QUESTION: 320

On older cars, the air filter can be found _____.

- A. on top of the engine
- B. under the engine
- C. behind the engine
- D. on the left or right side of the engine

Answer(s): A

QUESTION: 321

Glazing is the process of _____.

- A. cutting glass to size
- B. using putty to hold glass to a window frame
- C. polishing glass before using
- D. removing glass from a window

Answer(s): B

QUESTION: 322

A wrench with fixed, open jaws is called a(n) _____.

- A. adjustable wrench
- B. Allen wrench

- C. socket wrench
- D. open-end wrench

Answer(s): D

QUESTION: 323

All hammers have a _____.

- A. head, face, and handle
- B. head, toe, and handle
- C. head and foot
- D. head and claw

Answer(s): A

QUESTION: 324

To determine the number of threads per inch on a fastener, use a _____.

- A. depth gauge
- B. thread gauge
- C. thickness gauge
- D. wire gauge

Answer(s): B

QUESTION: 325

To chip or cut wood in close, the best tool is a _____.

- A. screwdriver
- B. butt chisel
- C. framing chisel
- D. mortising chisel

Answer(s): B

QUESTION: 326

Machine screws _____.

- A. are made by machines
- B. can be used interchangeably with wood screws
- C. fasten metal parts
- D. are machined to fine tolerances

Answer(s): C

QUESTION: 327

Double-headed nails are used _____.

- A. to reinforce a joint
- B. on temporary construction
- C. to make frames for furniture
- D. when a larger striking surface is needed

Answer(s): B

QUESTION: 328

To thin paint, use _____.

- A. turpentine
- B. mineral spirits
- C. benzene
- D. varnish

Answer(s): A

QUESTION: 329

When finishing a piece of wood, it's best to sand _____.

- A. diagonal to the grain
- B. against the grain
- C. with the grain
- D. in small circles

Answer(s): C

QUESTION: 330

To measure an angle other than a 90-degree angle, the best tool to use is a _____.

- A. square
- B. caliper
- C. level
- D. sliding T-bevel

Answer(s): D

QUESTION: 331



The above tool is a(n) _____.

- A. pipe wrench
- B. socket wrench
- C. adjustable crescent wrench
- D. box-end wrench

Answer(s): C

QUESTION: 332

Which of the following screw heads requires a Phillips screwdriver?

A.



B.



C.



D.



Answer(s): B

QUESTION: 333



The above tool is used to _____.

- A. punch holes
- B. drive nails
- C. measure thickness
- D. set nails

Answer(s): A

QUESTION: 334

Which is considered the left side of an engine?

- A. The left side when you stand facing the car in front.
- B. The side that the starter motor is on.
- C. Depends on the manufacturer.
- D. The left side when you sit in the driver's seat and face forward.

Answer(s): D

Explanation:

When viewed from the driver's seat, the side to the driver's left is the left side of an engine, and of the entire vehicle, as well.

QUESTION: 335

The letter and numbers G78-15 printed on a tire refer to _____.

- A. load, width, and diameter.
- B. diameter and air pressure.
- C. cubic inch displacement.
- D. price code.

Answer(s): A

Explanation:

"G" indicates the load-carrying capacity of a tire. 78 is the aspect ratio, which is the relationship between the tire height and tire width. When a tire is marked 78, the height is 78 percent of the width. 15 indicates that the tire will fit a 15-inch rim.

QUESTION: 336

"Tracking," in terms of front-end alignment, means that _____.

- A. each wheel travels independently of the other.
- B. the wheels follow well on ice or in snow.
- C. the front tires leave stronger tracks than the rear tires.
- D. the rear wheels follow the front wheels correctly.

Answer(s): D

Explanation:

Tracking is correct when both rear wheels are parallel to, and the same distance from, the vehicle centerline, which is an imaginary line drawn through the center of the vehicle. A bent frame or twisted body structure will cause improper tracking.

QUESTION: 337

A car with worn-out shock absorbers _____.

- A. cannot carry a heavy load.
- B. will bounce a lot on a rough road.
- C. sags very low to the ground.
- D. will have no traction on a wet road.

Answer(s): B

Explanation:

Shock absorbers control spring oscillation. Original equipment shocks that do not use air or spring assists are not designed to control vehicle height.

QUESTION: 338

In a three-speed transmission, the cluster gear is supported by the _____.

- A. pinion shaft
- B. countershaft
- C. main shaft
- D. output shaft

Answer(s): B

Explanation:

The countershaft is pressed into the transmission case and supports the cluster gear. The main shaft or output shaft is splined with the driveshaft.

QUESTION: 339

A clutch release bearing _____.

- A. rotates whenever the engine is turning.
- B. rotates when only the clutch is engaged.
- C. rotates whenever the clutch pedal is depressed.
- D. holds the clutch shaft in alignment.

Answer(s): C

Explanation:

When a clutch pedal is depressed, the release bearing comes in contact with the fingers of the clutch pressure plate and rotates.

QUESTION: 340

The general procedure to follow when adjusting a band on an automatic transmission is to _____.

- A. loosen, tighten to a specified torque, loosen a specified number of turns, and lock.
- B. tighten to a specified torque with a torque wrench and lock.
- C. tighten to a specified torque, loosen five turns, and lock.
- D. loosen, tighten to a specified torque, and lock.

Answer(s): A

Explanation:

To properly adjust an automatic transmission band, loosen the adjusting screw locknut. Then tighten the adjusting screw to the specified torque with a torque wrench. Next, loosen the adjusting screw the specified number of turns. Complete the job by holding the adjusting screw and tightening the locknut.

QUESTION: 341

The purpose of the drive shaft is to connect the _____.

- A. piston to the transmission
- B. differential to the transmission
- C. flywheel to the crankshaft
- D. camshaft to the crankshaft

Answer(s): B

Explanation:

On a rear-wheel-drive vehicle, the driveshaft transmits power from the transmission to the differential. The flywheel is bolted directly to the crankshaft.

QUESTION: 342

The order in which an engine's spark plugs fire is established by the order in which the _____.

- A. plugs are mounted in the engine.
- B. plug leads are connected to the distributor cap.
- C. condenser is connected.
- D. contact points are opened.

Answer(s): B

Explanation:

Spark plug leads must be installed in the correct tower of the cap to ensure the proper firing order. The firing order is the order that the spark plugs ignite the air/fuel mixture in the combustion chamber. A typical firing order for a six-cylinder engine is 1-5-3-6-2-4.

QUESTION: 343

The terminals exposed on the ignition induction coil are _____.

- A. two secondary and one primary
- B. one secondary and one primary
- C. one secondary and two primary
- D. two secondary and two primary

Answer(s): C

Explanation:

On a conventional ignition coil, the center tower is the secondary or high-voltage terminal. The two small studs at the top of the coil are primary or low-voltage terminals.

QUESTION: 344

Battery electrolyte is a mixture of distilled water and _____.

- A. baking soda
- B. sulfuric acid
- C. lead peroxide
- D. carbon particles

Answer(s): B

Explanation:

Battery electrolyte is a mixture of approximately 60% distilled water and 40% sulfuric acid.

QUESTION: 345

The brushes in an alternator ride on _____.

- A. the commutator
- B. the stator
- C. slip rings
- D. the heat sink

Answer(s): C

Explanation:

Two slip rings are mounted on the rotor shaft. Two brushes ride on the slip rings and deliver electrical energy to the rotor windings.

QUESTION: 346

When all other lights show normal illumination but one headlight is dim, it is a good sign that the _____.

- A. battery is weak.
- B. headlight switch is defective.
- C. headlight unit needs adjustment.
- D. headlight has a poor ground connection.

Answer(s): D

Explanation:

A loose or corroded headlight ground connection will cause a dim light. If the headlight switch is faulty, all lights will be affected, not just the headlight.

QUESTION: 347

If air gets into the hydraulic brake system, _____.

- A. brake application will be hard.
- B. it will have no effect.
- C. brake pedal action will be spongy.
- D. the brake pedal will stick.

Answer(s): C

Explanation:

Air is compressible and will cause a spongy or soft brake pedal action.

QUESTION: 348

The proportioning valve in a hydraulic brake system _____.

- A. reduces pressure to the front brakes.
- B. controls the brake warning switch.
- C. is used only on drum-type brake systems.
- D. reduces pressure to the rear brakes.

Answer(s): D

Explanation:

A proportioning valve is commonly used on a vehicle equipped with front disc and rear drum-type brakes. The valve reduces hydraulic pressure to the rear drum brakes to prevent rear wheel lockup and skidding during heavy brake pedal application.

QUESTION: 349

To fuse metals together, gas welding uses _____.

- A. oxygen and acetylene to produce a flame.
- B. oxygen and fluorine to produce a flame.
- C. nitrogen and acetylene to produce a chemical reaction.
- D. flux and acetylene to create an arc.

Answer(s): A

Explanation:

Gas welding uses an oxygen and acetylene mixture in a welding torch that produces a flame hot enough to melt and fuse metals.

QUESTION: 350

Tin snips are a type of _____.

- A. gas welding aid
- B. shear
- C. metal roof measuring tool
- D. nipper

Answer(s): B

Explanation:

The cutting blades of tin snips cut metal with a shearing action as they slide past each other. The cutting blades of nippers come together and don't slide past each other. Tin snips have nothing to do with either welding or measuring.

QUESTION: 351

A combination wrench has _____.

- A. an adjustable wrench on one end and an open-end wrench on the other.
- B. a box-end wrench on one end and an open-end wrench on the other.
- C. two box-end wrenches of successively larger sizes.
- D. two open-end wrenches of successively smaller sizes.

Answer(s): B

Explanation:

A combination wrench has a box-end wrench on one end and an open-end wrench on the other.

QUESTION: 352

Auger bits are used to _____.

- A. make a dimple in metal to locate the bit as starts to drill.
- B. make holes in any surface.
- C. make holes in metal.
- D. make holes in wood.

Answer(s): D

Explanation:

Auger bits are used to make holes in wood.

QUESTION: 353

A carpenter's square is used to _____.

- A. tell if something is level.
- B. tell if something is vertical.
- C. draw lines at right angles (90°) to each other.
- D. draw lines at right angles (180°) to each other.

Answer(s): C

Explanation:

A carpenter's square is used to draw lines at right angles (90°) to each other. A level or spirit level is used to tell if something is level or vertical; a plumb can also be used to tell if something is vertical.

QUESTION: 354

In arc welding, TIG stands for _____.

- A. Turner Insertion Gas
- B. Tungsten Inert Gas
- C. Torch Inert Gasification

D. Tin Impressible Galvanisation

Answer(s): B

Explanation:

In arc welding, TIG stands for Tungsten Inert Gas.

QUESTION: 355

An Allen wrench _____.

- A. is made exclusively by the Allen Manufacturing Company, LLC.
- B. fits all kinds of screws.
- C. fits screws with an octagonal recess in the head.
- D. fits screws with a hexagonal recess in the head.

Answer(s): D

Explanation:

An Allen wrench fits screws with a hexagonal recess in the head of the screw.

QUESTION: 356

A machine screw _____.

- A. cannot cut threads in wooden surfaces.
- B. is only used for steel building construction.
- C. will work with any nut that has the correct diameter.
- D. can be used to cut threads in sheet metal.

Answer(s): A

Explanation:

A machine screw cannot cut threads in wooden surfaces. Machine screws are used in a wide variety of applications, not just building or construction; the diameter and pitch of the screw and the nut must be the same for them to work together; and a machine screw cannot be used to cut threads in sheet metal.

QUESTION: 357

Uncontrolled, rapid drying of wood (for instance, lumber) causes _____.

- A. warping
- B. harmless discoloration, unless it's green, which indicates rot
- C. checking and cracking
- D. an increase in wood strength

Answer(s): C

Explanation:

Uncontrolled, rapid drying of wood (for instance, lumber) causes checking and cracking. This is

why new lumber sometimes is kept submerged in water until it's time for it to be cut.

QUESTION: 358

A finishing nail has a than a common nail _____.

- A. larger head and larger diameter
- B. larger head and smaller diameter
- C. smaller head and smaller diameter
- D. smaller head and larger diameter

Answer(s): C

Explanation:

A finishing nail has a smaller head and smaller diameter than a common nail.

QUESTION: 359

The piston of an internal combustion engine fits inside the _____.

- A. crankshaft
- B. cylinder
- C. radiator
- D. brake drum

Answer(s): B

Explanation:

The piston is the object inside the engine that moves up and down inside the cylinder to produce the power needed.

QUESTION: 360

The flywheel is attached to the _____.

- A. crankshaft
- B. carburetor
- C. radiator
- D. differential

Answer(s): A

Explanation:

The flywheel is used to keep the engine running smoothly inasmuch as it has a tendency to smooth out the individual power surges generated by the various cylinders firing the fuel charge. The flywheel is attached to the crankshaft.

QUESTION: 361

There are four types of fuel injection used in cars today. They are throttle body injection, multi-port injection, direct injection, and _____.

- A. a manual continuous electronic flow injection
- B. sequential fuel injection
- C. automatic circuitous injection
- D. manual intermittent injection

Answer(s): B

Explanation:

The four types of fuel injection used in cars today are throttle body injection, multi-port injection, direct injection, and sequential fuel injection.

QUESTION: 362

Most automobile engines are _____.

- A. two-stroke cycle engines
- B. three-stroke cycle engines
- C. four-stroke cycle engines
- D. six-stroke cycle engines

Answer(s): C

Explanation:

Lawn mowers and outboard engines are two-cycle. They are used wherever horsepower is needed in a lightweight package. They run very fast in order to produce power. Four-cycle engines are used in automobiles, where various conditions exist and load variations are greater. One exception is the rotary or Wankel engine, which uses a rotary design to convert pressure into rotating motion instead of using reciprocating pistons.

QUESTION: 363

When a spark plug fires, it is part of the _____.

- A. power stroke
- B. exhaust stroke
- C. intake stroke
- D. compression stroke

Answer(s): A

Explanation:

The spark plug fires, driving the piston downward to produce power. The intake stroke allows the fuel and air mixture to enter the cylinder. The compression stroke compresses the air/fuel mixture. The exhaust stroke allows the burnt gases to be removed from the cylinder.

QUESTION: 364

If the universal joints are broken and the car is standing still, the car will _____.

- A. not move when the engine is revved up

- B. increase in speed
- C. work as usual
- D. become impossible to push

Answer(s): A

Explanation:

Universal joints are needed to allow the drive shaft to flex when the car goes over a bump in the road. If the universal joints are broken, the car will be unable to move since the drive shaft will not be connected to the power source.

QUESTION: 365

When the clutch is engaged on a manual transmission, the clutch pedal will be _____.

- A. down
- B. up
- C. halfway down
- D. halfway up

Answer(s): B

Explanation:

The clutch pedal is pushed to the floorboard in order to release the engine from the drive train. Once the pedal is released and spring action returns it to the upward position, the clutch is engaged and the engine is connected to the drive train again.

QUESTION: 366

What gear is the transmission in when the engine is driving the car directly?

- A. first gear
- B. second gear
- C. third gear
- D. fourth gear

Answer(s): C

Explanation:

In a three-speed car first gear is used to get the car moving. Second gear is used to get it moving a little faster.

Then it is placed in third gear when it has reached a higher speed and needs less power to make it move even faster. When the third gear is engaged, no gear reduction is taking place and the engine speed is reflected in the speed at which the wheels turn.

QUESTION: 367

When the clutch is disengaged, the car will _____.

- A. be in reverse
- B. be in third gear

- C. be in neutral
- D. move quickly

Answer(s): C

Explanation:

Disengaging the clutch means placing the car in neutral since the engine is disconnected from the drive train.

QUESTION: 368

A four-wheel vehicle driving on all four wheels is known as a _____.

- A. 4×2
- B. 2×4
- C. 4×4
- D. 4×6

Answer(s): C

Explanation:

A four-wheel vehicle is called a 4×4 if all wheels are driven. One with six wheels with all six being driven is called a 6×6 , and one with six wheels (like a truck) but only four being driven is designated as a 4×6 .

QUESTION: 369

A squaring shear is used to _____.

- A. cut metal off to a straight line.
- B. square a piece of metal.
- C. fold a piece of sheet metal.
- D. punch a piece of sheet metal.

Answer(s): A

Explanation:

A squaring shear is used to cut sheet metal - usually keeping it square to one edge if properly aligned with the side of the unit. To shear means to cut.

QUESTION: 370

If the engine will not start, the problem could be _____.

- A. the clutch slipping
- B. a clogged oil filter
- C. too much carbon in the drain holes or slots
- D. a weak battery

Answer(s): D

Explanation:

A weak battery is the first thing to check when the engine does not start or turn over.

QUESTION: 371

When the mixture of fuel and air burns in the cylinder, it creates a temperature of _____.

- A. 4000°F to 4500°F
- B. 400°F to 450°F
- C. 1000°F to 2500°F
- D. 1250°F to 1500°F

Answer(s): A

Explanation:

When the mixture of fuel and air burns in the cylinder, it creates a temperature of 4000° to 4500°F.

QUESTION: 372

The first emission control applied to automobile engines was called the _____.

- A. preventive combustion venturi system
- B. positive crankcase ventilation system
- C. pollution control ventilation system
- D. pollution combustion vertical system

Answer(s): B

Explanation:

The first emission control device for automobiles was the positive crankcase ventilation (PCV) system.

QUESTION: 373

A tire that bears the code "P 185/70 R 14" is _____.

- A. a passenger-car tire that is 70 millimeters high.
- B. a passenger-car tire that is 185 millimeters wide.
- C. a radial tire that is 70 millimeters high.
- D. a P-metric tire that is 70 millimeters wide.

Answer(s): B

QUESTION: 374

In concrete work, to butter a concrete block means to _____.

- A. put a finishing trowel to it
- B. put a hawk on it
- C. use a jointer on it

D. put mortar on it

Answer(s): D

Explanation:

Buttering a concrete block means to put the mortar on it. This is usually done by taking the mortar from the hawk and applying it to the block's edges.

QUESTION: 375

On a woodworking plane, the "frog" refers to the _____.

- A. handle on the rear of the plane
- B. the sliding iron wedge that holds the plane iron at the proper angle
- C. the opening at the bottom of the plane
- D. the lever used to adjust the iron

Answer(s): B

Explanation:

The frog is the sliding iron wedge that holds the plane iron at the proper angle.

QUESTION: 376

Which of the following is a softwood?

- A. poplar
- B. maple
- C. pine
- D. oak

Answer(s): C

Explanation:

Softwood comes from conifers, or trees with needles, usually called evergreens. Such trees are the pines, firs, and spruces.

QUESTION: 377

To prevent rapid drying, freshly cut logs are often _____.

- A. submerged in oil
- B. placed in kilns
- C. submerged in water
- D. placed in sheds to dry slowly

Answer(s): C

Explanation:

Sometimes logs are not cut into lumber immediately and must be kept from checking and splitting. This can be done by keeping them wet. At the sawmill the logs are kept floating in a

pond until they are about ready to be used. This way they do not dry out too quickly and check or split.

QUESTION: 378

A bumping hammer is used to smooth out dents by a(n) _____.

- A. sheet metal duct worker
- B. auto body repairman
- C. bumper repairman
- D. none of the above

Answer(s): B

Explanation:

A collision auto repair worker uses a bumper hammer to take out some of the dents in a car's sheet metal. This hammer is ideally suited for the job.

QUESTION: 379

A groover is used to _____.

- A. lock seams together.
- B. make sheet metal seams.
- C. open a seam.
- D. fold a seam.

Answer(s): A

Explanation:

A groover is designed to cause the seams to become flattened and make a watertight seal.

QUESTION: 380

Sheet metal is formed into round cylinders by using a _____.

- A. squaring shear
- B. slip roll
- C. bar folder
- D. punch press

Answer(s): B

Explanation:

A squaring shear is used to cut sheet metal along a straight line. A bar folder is used to bend sheet metal, and a punch press makes holes in sheet metal. The slip roll is used to form the metal into round, cylindrical shapes.

QUESTION: 381

The tip of a soldering iron is coated with solder before it is used. This process of coating the tip

is called _____.

- A. welding
- B. brazing
- C. soldering
- D. tinning

Answer(s): D

Explanation:

The tip of a soldering iron is tinned or coated with solder to make it more efficient in transferring the heat at the tip to the object being soldered. It also prevents copper oxide buildup on the tip.

QUESTION: 382

Punches are used in sheet metal work to _____.

- A. flatten the metal
- B. make holes in the metal
- C. groove the metal
- D. snip the metal

Answer(s): B

Explanation:

Punches are used in sheet metal for the same reason punches are used elsewhere - to punch holes.

QUESTION: 383

A chisel is used for _____.

- A. twisting
- B. prying
- C. grinding
- D. cutting

Answer(s): D

QUESTION: 384

Which of the following items is used to measure angles?

- A. a protractor
- B. a lever
- C. a tachometer
- D. a gear

Answer(s): A

Explanation:

A protractor is typically a half-circle made of metal or plastic that has tick marks spaced at one-degree intervals around the edge.

QUESTION: 385

Which system on an automobile uses shock absorbers?

- A. the braking system
- B. the exhaust system
- C. the electrical system
- D. the suspension system

Answer(s): D

Explanation:

The shock absorbers are the components in the suspension that dampen the vibrations in the system.

QUESTION: 386

Which automotive system uses universal joints, a drive shaft, and a clutch?

- A. the transmission system
- B. the braking system
- C. the steering system
- D. the cooling system

Answer(s): A

Explanation:

The clutch is used to engage and disengage the transmission. The drive shaft is used to transfer power from the engine to the wheels. Universal joints are special linkages in the transmission system.

QUESTION: 387

Of which automotive system does the driver have the most direct control while driving?

- A. the exhaust system
- B. the braking system
- C. the cooling system
- D. the suspension system

Answer(s): B

Explanation:

The braking system can be controlled by the driver by pressing the brake pedal. The other systems listed cannot typically be directly controlled while operating the automobile.

QUESTION: 388

Which of the following systems connects directly to the internal combustion engine in an automobile?

- A. the braking system
- B. the suspension system
- C. the cooling system
- D. the steering system

Answer(s): C

Explanation:

The cooling system circulates water around the engine to prevent it from overheating. None of the other systems is connected directly to the engine.

QUESTION: 389

In the typical workshop, a plane is used for what purpose?

- A. removing thin strips of wood
- B. welding metal
- C. curing concrete
- D. loosening bolts

Answer(s): A

Explanation:

A plane is a hand tool that has a sharp blade extending slightly below the bottom surface of the tool. The plane is pushed across the surface of a piece of wood and the blade removes a thin layer of wood from the surface.

QUESTION: 390

Which of the following woodworking objects would most likely be created with a lathe?

- A. a building block
- B. a door
- C. a sign
- D. a table leg

Answer(s): D

Explanation:

A lathe is a tool which is typically used to carve long, slender pieces of wood. The wood is attached to the lathe and an electric motor spins the piece of wood. A cutting blade is then used to remove parts of the wood, giving a decorative shape. A lathe would not be useful in making the other objects listed.

QUESTION: 391

The function of a catalytic converter is primarily to _____.

- A. change pollutants into substances that are less harmful
- B. improve the starting power of the automobile
- C. force air into the engine for better acceleration
- D. control the amount of gasoline that flows into the engine

Answer(s): A

Explanation:

The catalytic converter converts pollutants into less harmful substances.

QUESTION: 392

Which automotive system uses fuses and an alternator?

- A. the engine
- B. the cooling system
- C. the electrical system
- D. the steering system

Answer(s): C

Explanation:

The fuses are used as links in the electrical system to prevent damage to other key components. The alternator is used to recharge the battery when the car is running.

QUESTION: 393

Which of the following items is part of the braking system in an automobile?

- A. master cylinder
- B. calipers
- C. drums
- D. all of the above

Answer(s): D

Explanation:

When the brake pedal is pressed, it operates the master cylinder, which forces the brake fluid through the lines out to each wheel. The pressure in the brake lines causes the calipers to move and press the brake shoes against the brake drum. This causes the automobile to slow down.

QUESTION: 394

To make a wide groove in a board with a table saw, what type of blade would you use?

- A. mitre
- B. cross-cut
- C. rip
- D. dado

Answer(s): D

Explanation:

A dado head blade is used for making wide grooves.

QUESTION: 395

The function of a brake drum or rotor is to act as _____.

- A. a heat extinguisher
- B. a heat sink
- C. a friction element
- D. all of the above

Answer(s): D

Explanation:

A brake system works on the principle of friction, which develops heat in the process of braking. The drum absorbs and dissipates the heat developed.

QUESTION: 396

The strongest bond for a wood joint would be _____.

- A. tack nails
- B. carpenter's glue
- C. corrugated fasteners
- D. grooved dowels and glue

Answer(s): D

QUESTION: 397

A typical car engine has how many strokes per cycle?

- A. 1
- B. 6
- C. 8
- D. 4

Answer(s): D

QUESTION: 398

What does "EFI" stand for in EFI computer?

- A. Electronic Fuel Insertion
- B. Electronic Fuel Injection
- C. Express Fuel Injection
- D. Engineered Focal Injection

Answer(s): B

QUESTION: 399

When a driver presses the accelerator, the throttle _____.

- A. breaks down to signal that maximum speed has been reached
- B. tightens to maintain fuel efficiency
- C. remains neutral, as it has no connection to the accelerator
- D. opens, which allows more air into the engine

Answer(s): D

QUESTION: 400

The function of the tachometer is to _____.

- A. denote car mileage
- B. measure revolutions per minute
- C. check engine heat levels
- D. measure speed of wheels

Answer(s): B

QUESTION: 401

Which of these is not a stroke?

- A. intake
- B. compression
- C. combustion
- D. power

Answer(s): C

QUESTION: 402

Ron had his dad's permission to use his car, even though the predicted weather was frigid and a snowstorm was expected. Ron remembered to check the oil and refill the cooling system with water the night before he left. He also positioned the car near the end of the driveway so he could easily get out if the snow started overnight. The next morning, the car started fine, but steam began to come from under the hood after he had driven for a while.

What is the most likely cause of this?

- A. He forgot to put the oil cap back on.
- B. The engine is old and needs to be replaced.
- C. The battery was dead.
- D. He forgot to add antifreeze, as well as water, to the cooling system.

Answer(s): D

QUESTION: 403

The purpose of the oil pump is to _____.

- A. allow excess or dirty gas to be pumped out of the gas tank
- B. lubricate the wheels to ensure they keep on rotating
- C. reserve oil for the car to run in case the gas tank becomes empty
- D. provide lubrication and prevent friction between metal parts of the engine

Answer(s): D

QUESTION: 404

If a driver inserts the key to start his car and finds that it does not work, then there is a problem with the ignition system.

Seeing that the car has a full tank of gas, the next logical thing to do is _____.

- A. replace the gas in the gas tank immediately
- B. call a tow truck since there is no possible way of moving the car at that point
- C. check the batteries by turning on the lights or radio
- D. check to see if the wheels are flat

Answer(s): C

QUESTION: 405

The _____ is the part of the electrical system in charge of absorbing excess current.

- A. distributor
- B. voltage regulator
- C. shock absorber
- D. coil

Answer(s): B

QUESTION: 406

With an automatic transmission, which is the best as to why is there no need for a clutch?

- A. There is a torque converter that automatically changes the torque needed.
- B. The clutch is responsible for the turning of the wheels but in automatic transmissions, it is replaced by the universal joint.
- C. The clutch is already built into the accelerator.
- D. The clutch is solely to change gears from 1-5. Automatic cars only use the D or drive gear.

Answer(s): A

QUESTION: 407

Joe loves to go on wild nature adventures. He always brings his four-wheel drive truck. In difficult terrain such as deep snow or loose sand, why would Joe's truck do better than a 2-wheel drive SUV?

- A. It is capable of running faster on straight, dry pavements.
- B. It can carry much heavier loads.
- C. More driven wheels increase the likelihood of gaining traction.
- D. The drive shaft extends from the transmission to the front axle.

Answer(s): C

QUESTION: 408

The two types of brakes are _____.

- A. major and minor
- B. front and rear
- C. drum and disc
- D. first and second

Answer(s): C

QUESTION: 409

The _____ is the emission-control system that forces unburned or partially-burned fuel back into the cylinder in order to be burned.

- A. electronic fuel injector
- B. alternator
- C. positive-crankcase ventilation
- D. fuel pump

Answer(s): C

QUESTION: 410

The primary purpose of emission-control systems is to _____.

- A. reduce emissions
- B. reduce fuel consumption
- C. reuse exhaust
- D. inject fuel

Answer(s): A

QUESTION: 411

Generally speaking, a framing hammer is different from a mallet or a sledge in that _____.

- A. framing hammers cannot be used to destroy or break surfaces
- B. a framing hammer is the most difficult to control
- C. framing hammers have softer heads than mallet and sledge hammers
- D. a framing hammer has a claw to pull out nails

Answer(s): D

QUESTION: 412

The fastening tool used to turn nuts and bolts is called a _____.

- A. mallet
- B. hammer
- C. stapler
- D. wrench

Answer(s): D

QUESTION: 413

Which is not a real type of wrench?

- A. pipe
- B. torque
- C. box-end
- D. closed-end

Answer(s): D

QUESTION: 414

The usual shape of the tip of a Phillips screwdriver is _____.

- A. round-shaped
- B. T-shaped
- C. cross-shaped
- D. heart-shaped

Answer(s): C

QUESTION: 415

Another term for needle-nose pliers is _____.

- A. curved-nose pliers
- B. long-nose pliers
- C. wrench
- D. offset screwdriver

Answer(s): B

QUESTION: 416

The term for a pair of pliers made specifically for cutting wires is _____.

- A. long-nose pliers

- B. vice grip pliers
- C. slip-joint pliers
- D. cutting pliers

Answer(s): D

QUESTION: 417

The crosscut saw has a unique shape and tooth angle enabling it to _____.

- A. cut manufactured wood
- B. cut wood with the grain
- C. cut wood against the grain
- D. cut out intricate shapes

Answer(s): C

QUESTION: 418

Which of these is not a cutting tool?

- A. allen wrench
- B. circle snips
- C. hacksaw
- D. shears

Answer(s): A

QUESTION: 419

The significance of a saw having more teeth or points is that _____.

- A. it can cut more efficiently on thick and rough surfaces
- B. it can cut on smaller objects
- C. it prevents the material being cut from breaking apart
- D. it can cut more finely

Answer(s): D

QUESTION: 420

The cold chisel called "cold" because _____.

- A. it can cut metal without the need for a heat source
- B. it is used in areas with colder temperatures
- C. it is used to cut through giant blocks of ice
- D. it is used to cut cold metal only

Answer(s): A

QUESTION: 421

To bore holes into surfaces, a _____ is attached to a drill.

- A. drill bit
- B. fastener
- C. screw
- D. nail

Answer(s): A

QUESTION: 422

Phineas decided to build a tree house in the tree in his backyard. He had finished making the walls and the roof and the last step was to put in the door, but he was having a difficult time fitting it perfectly onto the door frame.

The door sides kept chafing on the door frame.

To fix this, he should _____.

- A. replace the door
- B. use a finishing tool (plane) to fine the sides of the door
- C. make sure the wood material is intended for door purposes
- D. varnish the sides of the door very well

Answer(s): B

QUESTION: 423

Which of these is not a type of file?

- A. single-cut
- B. half-round
- C. double-cut
- D. triple-cut

Answer(s): D

QUESTION: 424

The clamp used for holding things stationary, that has a screw which loosens and tightens the clamp, is called a _____.

- A. pliers
- B. handscrew vise
- C. bench vise
- D. C clamp

Answer(s): D

QUESTION: 425

The function of a wire gauge is to _____.

- A. measure if an old wire is still useable
- B. determine the type of wire if it is undetermined
- C. measure the electrical capacity of the wire
- D. measure thickness of wires

Answer(s): D

QUESTION: 426

The two arms of a square are called _____.

- A. hour and minute
- B. blade and tongue
- C. left and right
- D. south and east

Answer(s): B

QUESTION: 427

The material contained inside small tubes in levels to make sure that a surface is true or even usually consists of _____.

- A. mercury
- B. water and oil
- C. a type of liquid and an air bubble
- D. sand

Answer(s): C

QUESTION: 428

A square tests the accuracy of an angle. A level tests the evenness of a surface. A plumb bob tests _____.

- A. vertical trueness
- B. horizontal trueness
- C. height
- D. weight limit

Answer(s): A

QUESTION: 429

The best type of nail to use when building a temporary wall that is certain to be torn down after a while is the _____.

- A. common nail
- B. double-headed nail
- C. finishing nail

D. screw

Answer(s): B

QUESTION: 430

ABS is a modern four-wheel system that prevents the wheels of the car from locking up. ABS stands for _____.

- A. anti-break system
- B. anti-lock brake system
- C. automatic breakaway system
- D. armed blocking system

Answer(s): B

QUESTION: 431

Screws and bolts are types of fasteners which can be easily taken out of the wood surface with little damage.

A screw with a square or hexagon-shaped head is called a _____.

- A. wood screw
- B. lag screw
- C. machine screw
- D. bolt

Answer(s): B

QUESTION: 432

The small, flat, metal item responsible for protecting the surface from the bolt digging into it and additionally for keeping the bolt tight in place is called a _____.

- A. washer
- B. nut
- C. screw
- D. flat bolt

Answer(s): A

QUESTION: 433

The tool used to back up a rivet is called a _____.

- A. screwdriver
- B. bucking bar
- C. clamp
- D. hammer

Answer(s): B

QUESTION: 434

When talking about nails, a 10d nail for example, the d stands for _____.

- A. penny
- B. pounds
- C. inches
- D. nail head size

Answer(s): A

QUESTION: 435

A single row of cylinders in a car engine is called _____.

- A. a cylinder group
- B. cycle
- C. inline
- D. a piston

Answer(s): C

QUESTION: 436

The twisting force that causes rotation is properly referred to as _____.

- A. energy
- B. horsepower
- C. MPH
- D. torque

Answer(s): D

QUESTION: 437

Talking about the radiator, it is more common for a car to overheat during hot days than on cold days because _____.

- A. with direct sunlight, the cooling system does not work
- B. the air that is supposed to cool the fluid or the coolant that flows through the radiator is not cool enough to do the job, hence overheating can happen
- C. the radiator automatically shuts off when the outside temperature is too high
- D. hot days have the tendency to melt the radiator

Answer(s): B

QUESTION: 438

The acoustical chamber that is part of the car exhaust system that lessens engine noise is called _____.

- A. a silencer
- B. the acoustic pad
- C. a muffler
- D. the noise-reduction system

Answer(s): C

QUESTION: 439

The gauge that measures the electrical charge in a car is called the _____.

- A. speedometer
- B. pressure gauge
- C. ammeter
- D. tachometer

Answer(s): C

QUESTION: 440

The flow meter, a type of gauge, measures _____.

- A. amount of exhaust flowing out
- B. volume of flow in a piping system
- C. air pressure
- D. fuel levels

Answer(s): B

QUESTION: 441

The tool that can be used to remove or tighten the spark plugs of an engine is called _____.

- A. a hacksaw
- B. clamps
- C. long nose pliers
- D. a spark plug wrench

Answer(s): D

QUESTION: 442

In order to change a tire, a _____ must be used to lift the bottom of the car.

- A. jack
- B. socket wrench
- C. hoist
- D. boom truck

Answer(s): A

QUESTION: 443

If a car has difficulty starting up, despite having a full tank of gas and no problems with the electrical system, the next possible reason is because _____.

- A. the battery is dead or missing
- B. the hand brake is engaged
- C. the air filter is very dirty
- D. the pressure in the wheels is low

Answer(s): C

QUESTION: 444

If a car starts but it does not move, there may be a problem with the _____.

- A. transmission system
- B. emission-control system
- C. brake system
- D. cooling system

Answer(s): A

QUESTION: 445

A special type of ruler that has a half-moon shape to measure angles is called a _____.

- A. plumb bob
- B. simple ruler
- C. protractor
- D. level

Answer(s): C

QUESTION: 446

Which of these is true regarding tire plies?

- A. A "ply rating" tells you approximately how long the tire should last.
- B. They connect the bead to the tread.
- C. Plies can be radial, bias, or belted bias.
- D. Their job is to protect the tire from curb impact.

Answer(s): C

QUESTION: 447

If your car has a fuel injection system, which of the following will it not need?

- A. Distributor
- B. Carburetor
- C. Manifold
- D. Oil filter

Answer(s): B

Explanation:

A carburetor is not need with a fuel injection system since the fuel injection system forces the gas/air mixture into each of the cylinders.

QUESTION: 448

Which component distributes high voltage to the spark plugs?

- A. Camshaft
- B. Ignition Coil
- C. Valves
- D. Rotor

Answer(s): D

Explanation:

The key term in the question is "distributes". A rotor is part of a car's distributor which distributes high voltage from the coil to the spark plugs.

QUESTION: 449

Which type of bolt is has a square section that grips into the part being fixed (typically wood) to prevent the bolt from turning when the nut is tightened?

- A. Hex bolt
- B. Eye bolt
- C. Carriage bolt
- D. Set bolt

Answer(s): C

Explanation:

A carriage bolt has a domed or countersunk head and a shank that is topped by a short square section under the head.

QUESTION: 450

What is the correct sequence of a four-stroke cycle gas engine?

- A. intake stroke, compression stroke, power stroke, exhaust stroke
- B. compression stroke, intake stroke, exhaust stroke, power stroke
- C. intake stroke, power stroke, compression stroke, exhaust stroke
- D. power stroke, compression stroke, intake stroke, exhaust stroke

Answer(s): A

Explanation:

The correct sequence is intake stroke, compression stroke, power stroke and exhaust stroke.

QUESTION: 451

Which would be the last woodworking plane that you would use to prepare a rough sawn board for finishing?

- A. Jointer plane
- B. Scrub plane
- C. Jack plane
- D. Smoothing plane

Answer(s): D

Explanation:

A typical order to use these planes to prepare a rough sawn board would be: scrub plane, jack plane, jointer plane and smoothing plane.

QUESTION: 452

Which type of wrench is used to turn screw or bolt heads designed with a hexagonal socket (recess) to receive the wrench?

- A. Bristol wrench
- B. Allen wrench
- C. Box-end wrench
- D. Combination wrench

Answer(s): B

Explanation:

The Allen wrench is used to turn screws or bolts with a recessed hexagonal socket.

QUESTION: 453

The best way to sand the end grain of wood is _____.

- A. with the grain
- B. with a circular motion
- C. crosswise
- D. up and down

Answer(s): C

Explanation:

The best way to sand end grain is crosswise.

QUESTION: 454

Which type of saw is best used for cutting small, often awkward features in building materials such as drywall?

- A. Keyhole saw
- B. Crosscut saw
- C. Hacksaw
- D. Rip saw

Answer(s): A

Explanation:

The keyhole saw has a long narrow blade and is often used to cut small openings in drywall and other building materials.

QUESTION: 455

Which component controls the amount of current that the alternator sends to the battery?

- A. Distributor
- B. Regulator
- C. Inverter
- D. Ignition Coil

Answer(s): B

Explanation:

The regulator controls the output of the alternator to insure that the battery gets the proper amount of current to keep it charged.

QUESTION: 456

Which of the following will NOT occur from setting a spark plug gap wider?

- A. a smaller weaker spark
- B. bigger spark
- C. better combustion
- D. potential misfire at high speeds

Answer(s): A

Explanation:

Setting a spark plug gap narrower can result in a smaller weaker spark. Setting it wider may result in a bigger spark, better combustion but a potential misfire.

QUESTION: 457

What should you do if your temperature gauge on your car indicates that the car is overheating (in the red zone)?

- A. Open radiator cap and pour in cold water immediately

- B. Open radiator cap and pour in antifreeze immediately
- C. Turn on the car's heater
- D. Pull over and let the car cool down

Answer(s): D

Explanation:

You should immediately pull over and let the car cool down.

QUESTION: 458

Which of the following can cause water sludge in engine crankcase oil?

- A. water condensation in the crankcase
- B. changing your oil too frequently
- C. using synthetic oil
- D. using a low viscosity oil

Answer(s): A

Explanation:

Water condensation in the crankcase is a leading cause of oil sludge. Changing your oil frequently and using synthetic oil can help eliminate sludge.

QUESTION: 459

Which of the following components prevents a car's battery from being overcharged?

- A. Rotor
- B. Voltage Regulator
- C. Inverter
- D. Governor

Answer(s): B

Explanation:

The voltage regulator controls the current that is sent to the battery to prevent overcharging.

QUESTION: 460

Which of the following is probably NOT a cause of an engine overheating?

- A. Faulty thermostat
- B. Inoperative fan motor
- C. Cracked distributor cap
- D. External coolant leaks

Answer(s): C

Explanation:

A faulty thermostat, inoperative fan motor and external coolant leaks can all cause an engine to

overheat.

QUESTION: 461

When you try and start an engine, you hear a clicking sound but nothing happens. What is the potential problem?

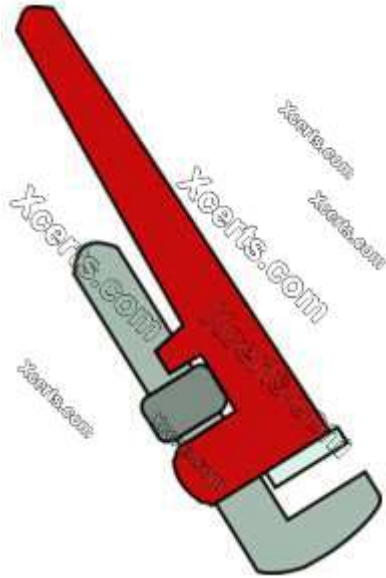
- A. Dead battery
- B. Excessive resistance in the solenoid
- C. Faulty starter
- D. All of the above

Answer(s): D

Explanation:

If you hear clicking but nothing happens, the problem could be the battery, solenoid or starter.

QUESTION: 462



What type of wrench is pictured?

- A. Crescent wrench
- B. Pipe wrench
- C. Ratchet wrench
- D. Combination wrench

Answer(s): B

Explanation:

The figure is a pipe wrench.

QUESTION: 463



What type of clamp is this?

- A. Bar clamp
- B. Pipe clamp
- C. Miter clamp
- D. Parallel clamp

Answer(s): D

Explanation:

The picture is a parallel clamp.

QUESTION: 464

Which type of saw is designed to make accurate crosscuts at specific angles?

- A. Miter saw
- B. Swing saw
- C. Rip saw
- D. Pad saw

Answer(s): A

Explanation:

The miter saw is used from making accurate cross cuts and miter cuts.

Section 3: Electronic Information

QUESTION: 465

Ohm's law states:

- A. Voltage = Current $\times$ Resistance
- B. Amperes = Current $\times$ Resistance
- C. Voltage = Resistance $\div$ Amperes
- D. Ohms = Voltage $\div$ Current

Answer(s): A

Explanation:

Ohm's law states that Voltage (V) = Current (I) $\times$ Resistance (R). All other answers are incorrect expressions of this law.

QUESTION: 466

Which of the following materials usually has an electromagnetic-induction device in its core?

- A. brass
- B. silver
- C. aluminum
- D. iron

Answer(s): D

Explanation:

Iron is easily magnetized and demagnetized, so it works well for this device.

QUESTION: 467

How many diodes should you expect to find in a bridge rectifier?

- A. 0
- B. 4
- C. 8
- D. 10

Answer(s): C

Explanation:

A bridge rectifier is also known as a full wave rectifier, usually containing 4 diodes.

QUESTION: 468

Another name for cycles per second is _____.

- A. watts
- B. voltage

- C. hertz
- D. amperes

Answer(s): C

Explanation:

The number of times alternating current changes direction in one second is known as its frequency, which is measured in hertz.

QUESTION: 469

Newer cell phones contain a removable memory card, which is often called a _____.

- A. SIM card
- B. DIM chip
- C. PIN card
- D. Pin chip

Answer(s): A

Explanation:

SIM stands for Subscriber Identity Module. The card contains information such as your phone number, your billing information, and your address book. It makes it easier to switch from one cell phone to another.

QUESTION: 470

When current flows through a wire, the following influences are present _____.

- A. amperes and ohms only
- B. voltage, watts, and ohms only
- C. voltage and amperes only
- D. voltage, ohms, and amperes

Answer(s): D

Explanation:

Voltage, ohms, and amperes are always present when current flows through a wire.

QUESTION: 471

Millihenries are related to _____.

- A. capacitors
- B. inductors
- C. relays
- D. transformers

Answer(s): B

Explanation:

Inductors are rated in millihenries.

QUESTION: 472

Radar can operate at frequencies as high as _____.

- A. 100,000 Hz
- B. 100,000 kHz
- C. 100,000 MHz
- D. 500,000 MHz

Answer(s): C

Explanation:

Radar can operate as high as 100,000 MHz (megahertz).

QUESTION: 473

Radio waves travel _____.

- A. at the speed of light
- B. at the speed of sound
- C. faster than the speed of light
- D. faster than the speed of sound but slower than the speed of light

Answer(s): A

Explanation:

Radio waves travel at the speed of light. The speed of sound is much slower.

QUESTION: 474

Changing alternating current to direct current is called _____.

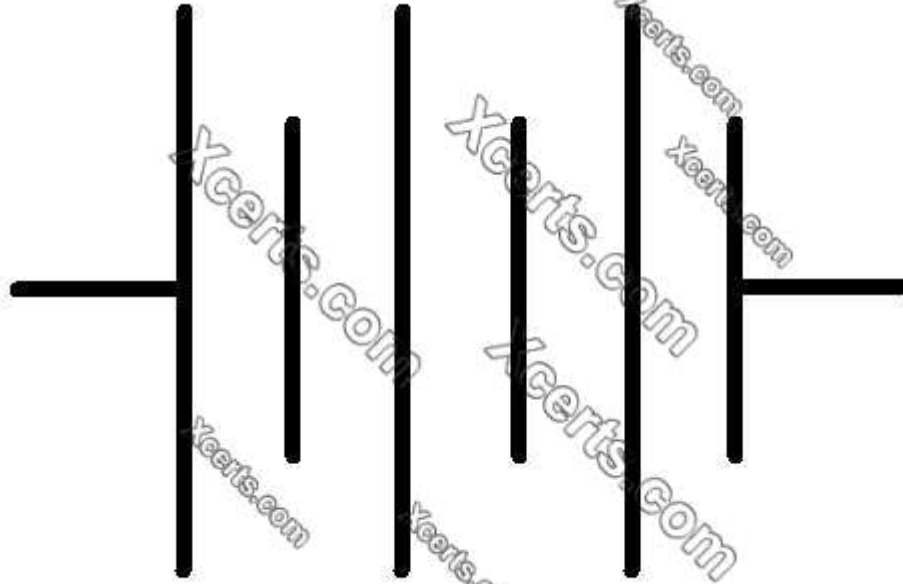
- A. capacitance
- B. impedance
- C. rectification
- D. induction

Answer(s): C

Explanation:

Changing AC to DC is a process called rectification.

QUESTION: 475



Insulated fittings can be used to splice wires, thus eliminating the need for _____.

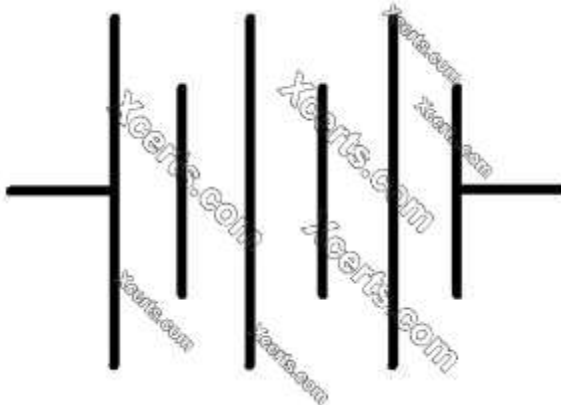
- A. cleaning the wires
- B. removing the plastic coating from the wires
- C. twisting the wires together
- D. soldering the wires together

Answer(s): D

Explanation:

Insulated fittings replace soldering.

QUESTION: 476



The symbol shown above stands for _____.

- A. battery
- B. transformer

- C. capacitor
- D. resistor

Answer(s): A

Explanation:

This symbol stands for battery.

QUESTION: 477

How many wires do serial cables used on computers have?

- A. 3
- B. 9
- C. 15
- D. 25

Answer(s): B

Explanation:

This is true, even if the cable has a 25-pin connector. Serial cables are often used to connect computers to perpetual devices.

QUESTION: 478

To produce greater storage of electrons and more capacitance, capacitors should _____.

- A. be connected in parallel
- B. be connected in series
- C. have more voltage applied to them
- D. be eliminated

Answer(s): A

Explanation:

Connecting capacitors in parallel produces more capacitance.

QUESTION: 479

A light bulb is 60 watts.

Operated at 120 volts, how much current does it draw?

- A. 0.5 ampere
- B. 5.0 amperes
- C. 50.0 amperes
- D. 7.200 amperes

Answer(s): A

Explanation:

Power = Current × Voltage or, written another way, Current = Power/Voltage.

$60 / 120 = 0.5$ ampere.

QUESTION: 480

A number-12 wire, compared to a number-6 wire _____.

- A. is longer
- B. is shorter
- C. is smaller in diameter
- D. is larger in diameter

Answer(s): C

Explanation:

The larger the number, the smaller (in diameter) the wire.

QUESTION: 481

A fuse with a higher-than-required rating used in an electrical circuit _____.

- A. improves safety
- B. increases maintenance
- C. may not work properly
- D. is less expensive

Answer(s): C

Explanation:

Because fuses are designed to prevent current overload at a specific level, a fuse with a high rating may not work properly for a circuit rated at a lower level.

QUESTION: 482

Neutral wire is always _____.

- A. whitish or natural
- B. black
- C. green
- D. blue

Answer(s): A

Explanation:

Neutral wire is always whitish or natural colored.

QUESTION: 483

To measure electrical power, you would use a(n) _____.

- A. ammeter
- B. ohmmeter

- C. voltmeter
- D. wattmeter

Answer(s): D

Explanation:

Electrical power is measured in watts, so you use a wattmeter.

QUESTION: 484

If you operate an incandescent light bulb at less than its rated voltage _____.

- A. the bulb will burn brighter and last longer.
- B. the bulb will burn dimmer and last longer.
- C. the bulb will burn brighter but won't last as long.
- D. the bulb will burn dimmer but won't last as long.

Answer(s): B

Explanation:

The bulb will burn dimmer because its full potential isn't used; it will last longer for the same reason.

QUESTION: 485

Two electrical wires run to a machine. One wire is 6mm thick, and the other is 3mm thick. For the two wires to carry the same current, the larger wire _____.

- A. requires less voltage
- B. requires more voltage
- C. requires the same voltage
- D. It can't be determined from the information given.

Answer(s): A

Explanation:

Larger wires require less electrical pressure to run current.

QUESTION: 486

Which of the following isn't a component of a DC motor?

- A. rotor bars
- B. armature
- C. field poles
- D. yoke

Answer(s): A

Explanation:

Rotor bars are found only on AC induction motors.

QUESTION: 487

A current passed through a muscle in the human body causes the muscle to contract. This is known as _____.

- A. the chemical effect of current
- B. the biological effect of current
- C. the magnetic effect of current
- D. the physiological effect of current

Answer(s): D

Explanation:

The ability of current to contracts muscles is called the physiological effect.

QUESTION: 488

In a closed electrical circuit _____.

- A. one terminal is always positive, and one terminal is always negative.
- B. both terminals can be positive.
- C. both terminals can be negative.
- D. terminals are neither positive nor negative.

Answer(s): A

Explanation:

In a closed circuit, one terminal is always positive, and the other is always negative.

QUESTION: 489

What does RMS stand for?

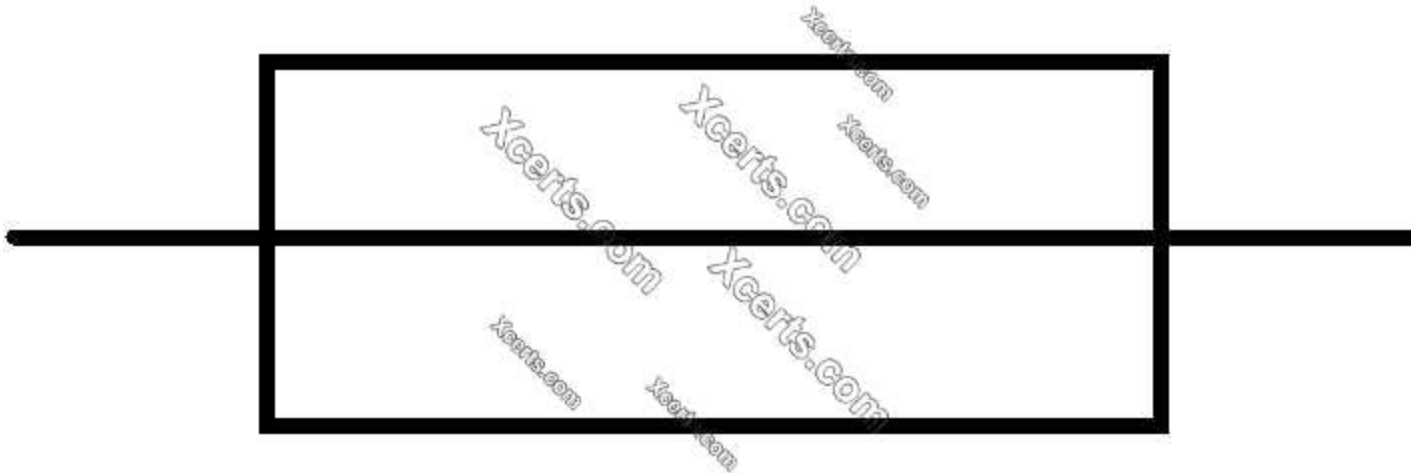
- A. Root Mean Square
- B. Resistance Measurement System
- C. Real Metric Standards
- D. Realistic Matrix Stems

Answer(s): A

Explanation:

Root Mean Square is the peak value of voltage multiplied by 0.707 (sine 45).

QUESTION: 490



The symbol above is a/an _____.

- A. resistor
- B. fuse
- C. capacitor
- D. inductor

Answer(s): B

Explanation:

The symbol is a fuse. Fuses are designed to "blow" (melt) if the current flowing through it exceeds a specified value.

QUESTION: 491

The television broadcast standard used in the United States is _____.

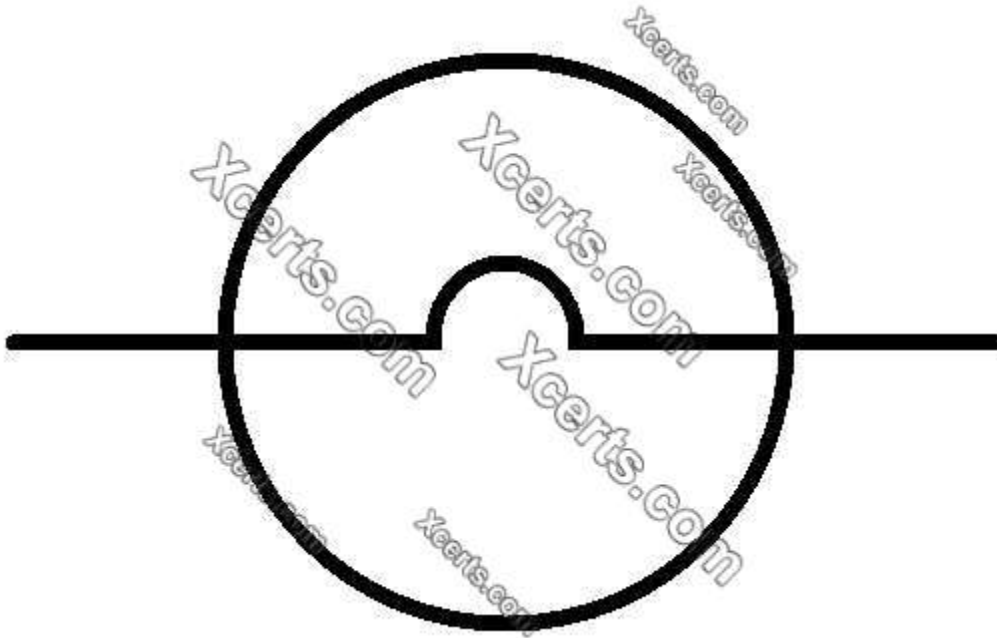
- A. PAL
- B. NTSC
- C. SECAM
- D. SSTC

Answer(s): B

Explanation:

NTSC stands for National Television Standards Committee. The system is also used in Japan. PAL and SECAM are used in other countries.

QUESTION: 492



The symbol above is a/an _____.

- A. lamp
- B. fuse
- C. inductor
- D. bell

Answer(s): A

Explanation:

The symbol is a lamp. A lamp is a transducer that converts electrical energy to light.

QUESTION: 493

When a circuit breaker trips, in what position will you find the operating handle?

- A. on position
- B. off position
- C. half way between on and off
- D. three-fourths of the way between the on position and the off position

Answer(s): C

Explanation:

Conventional circuit breaker handles have four positions: on, off, trip, and reset. When tripped, the handle moves to the middle position.

QUESTION: 494

Which wire is smallest?

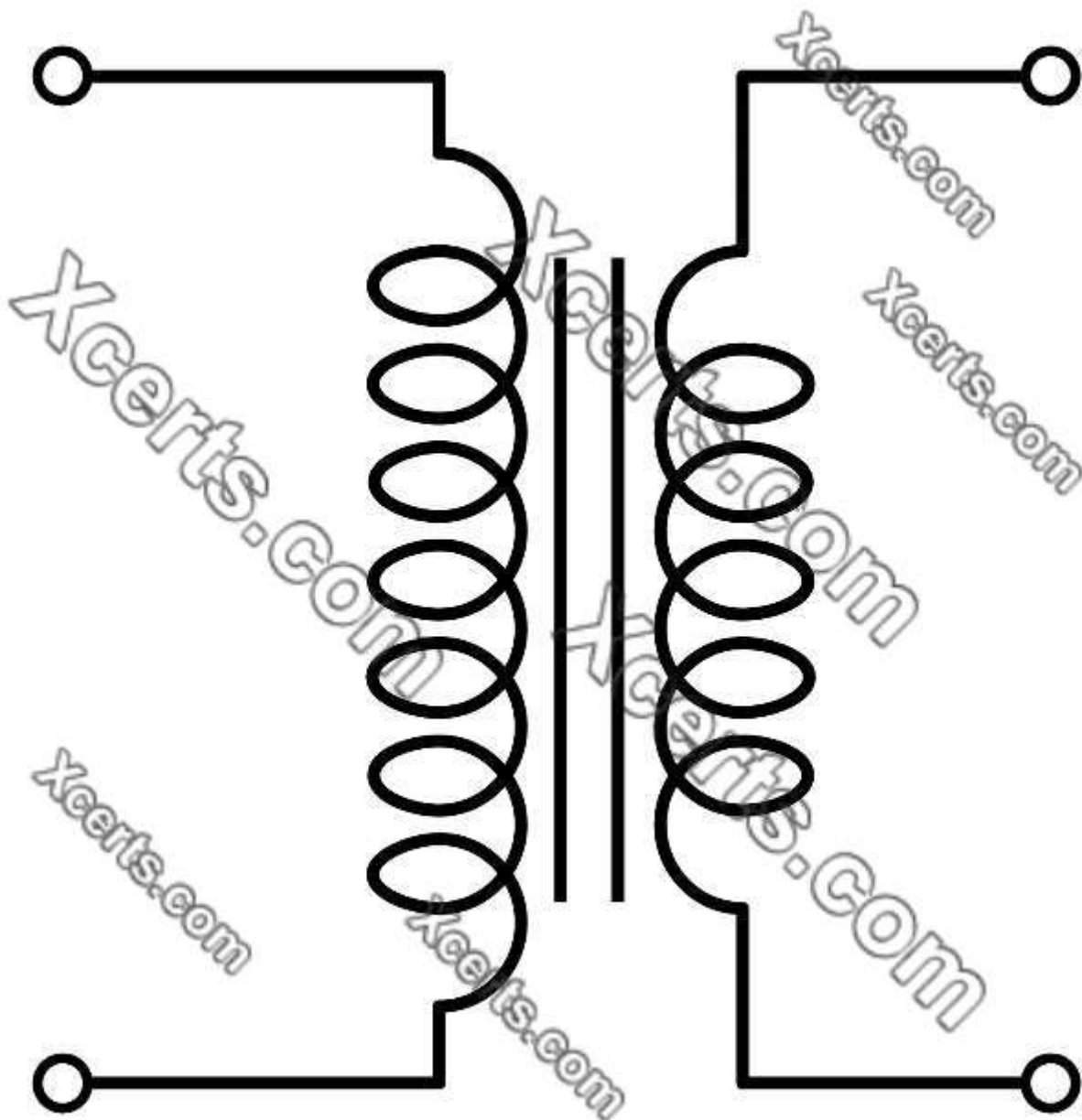
- A. 00 AWG
- B. 4 AWG
- C. 10 AWG
- D. 12 AWG

Answer(s): D

Explanation:

The smaller the wire, the larger the number

QUESTION: 495



The symbol above stands for _____.

- A. battery
- B. inductor
- C. transformer
- D. diode

Answer(s): C

Explanation:

The symbol stands for transformer. Transformers are used to step up (increase) and step down (decrease) AC voltages.

QUESTION: 496

How many paths of electrical flow can be found in a series circuit?

- A. one
- B. two
- C. two or more
- D. It can't be determined from the information given.

Answer(s): A

Explanation:

An example of a series circuit would be a string of Christmas lights. If you break the circuit's path at any point, electricity will stop flowing and the lights will not work.

QUESTION: 497

A microwave is rated at 1,200 watts.

At 120 volts, how much current does it draw?

- A. 1 amp
- B. 10 amps
- C. 100 amps
- D. 1,440 amps

Answer(s): B

Explanation:

$I \text{ (current)} = \text{Power (watts)} / \text{Effort (volts)}$. In this case, $I = 1200 / 120 = 10$ amperes.

QUESTION: 498

Electricians use the term low potential to refer to _____.

- A. electrical circuits with a low potential for overload
- B. building codes that reduce the risk of fire
- C. the likelihood of getting a raise this year
- D. 600 watts or less

Answer(s): D

Explanation:

Potential equals voltage; low potential is anything less than 600 watts.

QUESTION: 499

Which of the following isn't a conductor of electricity?

- A. water
- B. graphite
- C. gold
- D. glass

Answer(s): D

Explanation:

Glass is an insulator. Other insulators are plastics, paper, and rubber.

QUESTION: 500

The ground wire is always _____.

- A. green
- B. black
- C. whitish
- D. blue

Answer(s): A

Explanation:

Ground wires are always green.

QUESTION: 501

To cause a current of 10 amperes to flow through 20-ohm resistance, the voltage needed is _____.

- A. 20 volts
- B. 1 volt
- C. 10 volts
- D. 200 volts

Answer(s): D

Explanation:

Ohm's law states that $V = IR$. $V = 10 \times 20 = 200$.

QUESTION: 502

Silver is a better conductor than copper. But copper is more often used because of

_____.

- A. the cost of silver
- B. the strength of copper
- C. the low melting point of silver
- D. the tendency of silver to tarnish

Answer(s): A

Explanation:

Silver is a better conductor, but it's prohibitively expensive.

QUESTION: 503

Electronic circuits that produce high frequency AC are called _____.

- A. amplifiers
- B. regulators
- C. transformers
- D. oscillators

Answer(s): D

Explanation:

Oscillators produce high-frequency AC.

QUESTION: 504

If you plug an appliance designed for AC into a DC power source, the appliance _____.

- A. will operate normally
- B. will produce excessive heat
- C. won't operate
- D. will explode into tiny pieces

Answer(s): B

Explanation:

When DC is applied to an AC appliance, the amount of resistance is less, so more current flows through the wire and heat builds up.

QUESTION: 505

What effect does a speaker wire's gauge have on speaker sound quality?

- A. higher gauge wires are thicker with better sound quality
- B. lower gauge wires are thicker with better sound quality
- C. lower gauge wires are thicker with lesser sound quality
- D. hire gauge wires are thicker with lesser sound quality

Answer(s): B

Explanation:

Unless a specific gauge is specified by the speaker manufacturer, you should always choose lower gauges for better sound quality.

QUESTION: 506

What's the primary advantage of a quad band cell phone over a dual-band cell phone?

- A. transmission strength
- B. coverage area
- C. reception strength
- D. smaller phone size

Answer(s): B

Explanation:

There are four frequency bands used throughout the world. A quad-cell phone would be able to access any of these frequency bands.

QUESTION: 507

When working with electricity, you should assume that all electrical equipment is alive unless you know for certain otherwise.

This prevents _____.

- A. damage to circuits
- B. personal injury
- C. unnecessary labor
- D. overheating the equipment

Answer(s): B

Explanation:

The greatest concern when dealing with electricity is personal injury.

QUESTION: 508

The heat effect of current is created _____.

- A. when the pressure of the current in the wire breaks up impurities in the wire, creating heat.
- B. when the current in the wire decays electrons, causing them to move more quickly, creating heat.
- C. when the current overcomes resistance in the wire, creating heat.
- D. The heat effect of current is only theoretical; it has never been proven to exist.

Answer(s): C

Explanation:

Heat effect occurs when electrical current must overcome the resistance of the wire. Heat effect can be quite obvious or very subtle.

QUESTION: 509

What special type of diode is commonly used to regulate voltage?

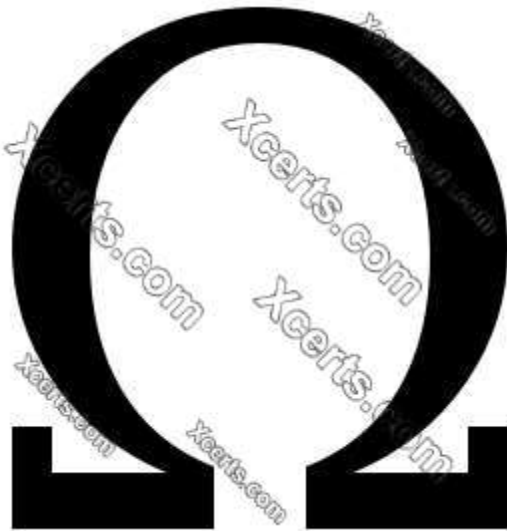
- A. capacitor
- B. transistor
- C. Zener diode
- D. LED

Answer(s): C

Explanation:

Zener diodes are available in a variety of voltages.

QUESTION: 510



This symbol means _____.

- A. ohm
- B. ampere
- C. high voltage
- D. wattage

Answer(s): A

Explanation:

The symbol stands for ohm.

QUESTION: 511

Electromotive force is another way of saying _____.

- A. frequency

- B. watts
- C. cycles per second
- D. voltage

Answer(s): D

Explanation:

Electromotive force is the pressure of the current, so the term is another way of saying voltage.

QUESTION: 512

A primary advantage of using a Li-Ion battery instead of a NiMH battery in your cell phone is _____.

- A. Li-Ion batteries are lighter
- B. Li-Ion batteries last longer
- C. Li-Ion batteries don't interfere with signal quality
- D. none of the above

Answer(s): A

Explanation:

Lithium-Ion batteries are much lighter than Nickel Metal Hydride batteries.

QUESTION: 513

Transistors contain at least three terminals called the _____.

- A. base, emitter, and collector
- B. base, positive terminal, and negative terminal
- C. emitter, amplifier, and collector
- D. base and two gates

Answer(s): A

Explanation:

The three terminals a transistor must have are the base, emitter, and collector.

QUESTION: 514

To control a light fixture from two different wall switches, you should use _____.

- A. a single-pole switch and a four-way switch
- B. two three-way switches
- C. two four-way switches
- D. two single-pole switches

Answer(s): B

Explanation:

To control a light fixture from two different positions, use two three-way switches.

QUESTION: 515

A transistor is also called a _____.

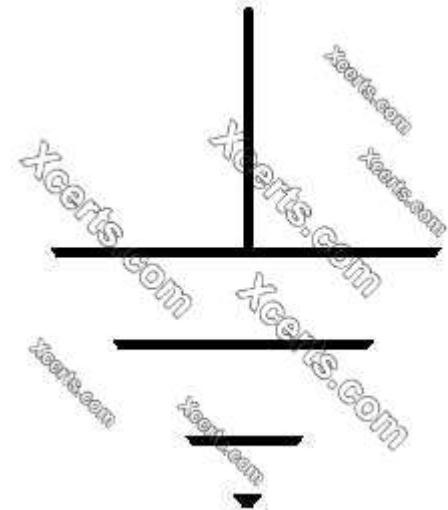
- A. rectifier
- B. cathode
- C. crystal amplifier
- D. semiconductor

Answer(s): C

Explanation:

Crystal amplifier is another name for transistor.

QUESTION: 516



This symbol means _____.

- A. ground
- B. resistor
- C. diode
- D. battery

Answer(s): A

Explanation:

The symbol means ground.

QUESTION: 517

To decrease capacitance, capacitors _____.

- A. should have less voltage applied to them
- B. should be connected in parallel

- C. should be connected in series
- D. should be eliminated

Answer(s): C

Explanation:

Capacitors connected in series reduce the amount of capacitance.

QUESTION: 518

A resistor marked 2.5K ohms has the value of _____.

- A. 2.5 ohms
- B. 250 watts
- C. 2,500 ohms
- D. 25,000 ohms

Answer(s): C

Explanation:

2.5K ohms is 2,500 ohms. K = one kilo or 1,000.

QUESTION: 519

A 9-volt transistor contains _____.

- A. 1 cell
- B. 6 cells
- C. 9 cells
- D. 3 cells

Answer(s): B

Explanation:

A cell is equal to about 1.5 volts, so $9 \div 1.5 = 6$.

QUESTION: 520

The hot wire is always _____.

- A. blue
- B. green
- C. whitish
- D. black

Answer(s): D

Explanation:

Live wires are black.

QUESTION: 521

How wide is the full AT motherboard?

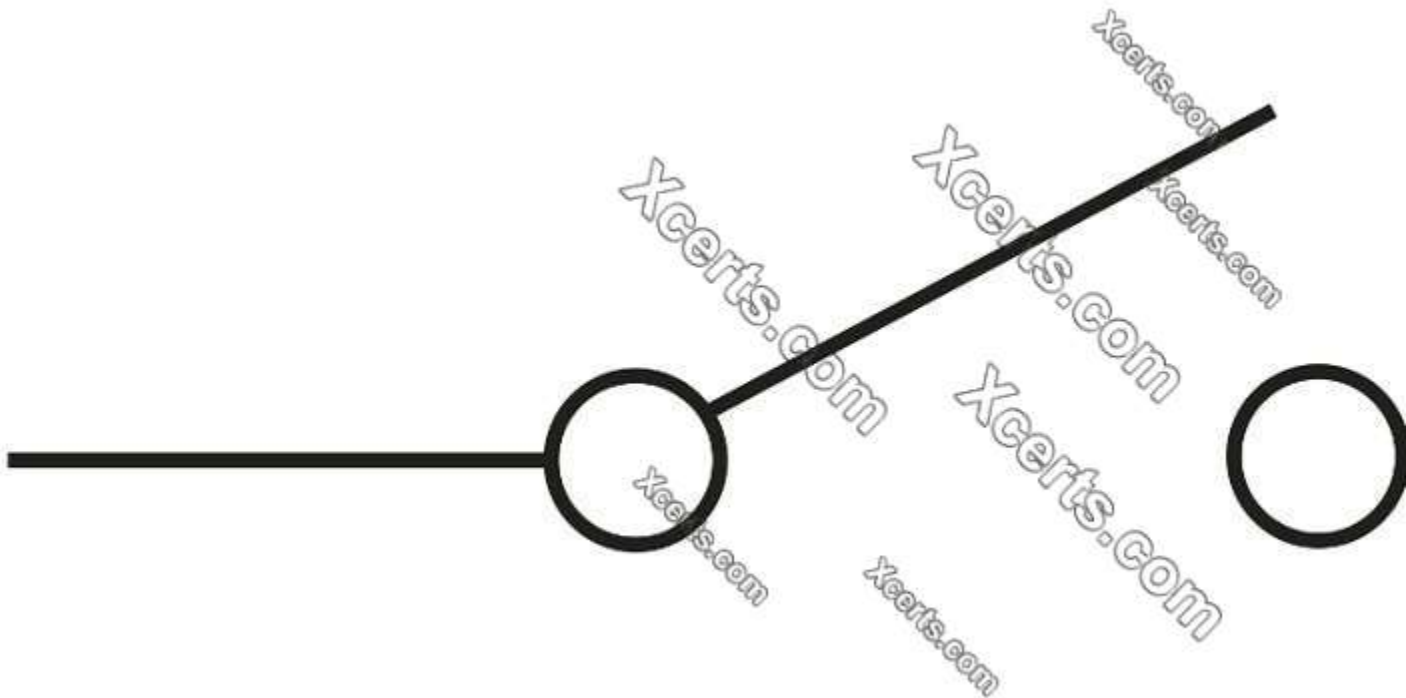
- A. 11 inches
- B. 11.5 inches
- C. 12 inches
- D. 12.5 inches

Answer(s): C

Explanation:

A motherboard is the physical arrangement in a computer that contains the computer's basic circuitry and components.

QUESTION: 522



The above symbol represents a/an _____.

- A. relay
- B. on-off switch
- C. push switch
- D. connected wire

Answer(s): B

Explanation:

An on-off switch allows current to flow only when it's in the closed (on) position.

QUESTION: 523

If a 120-volt current is protected by a 25-amp circuit breaker, what's the largest number of watts an appliance can safely use?

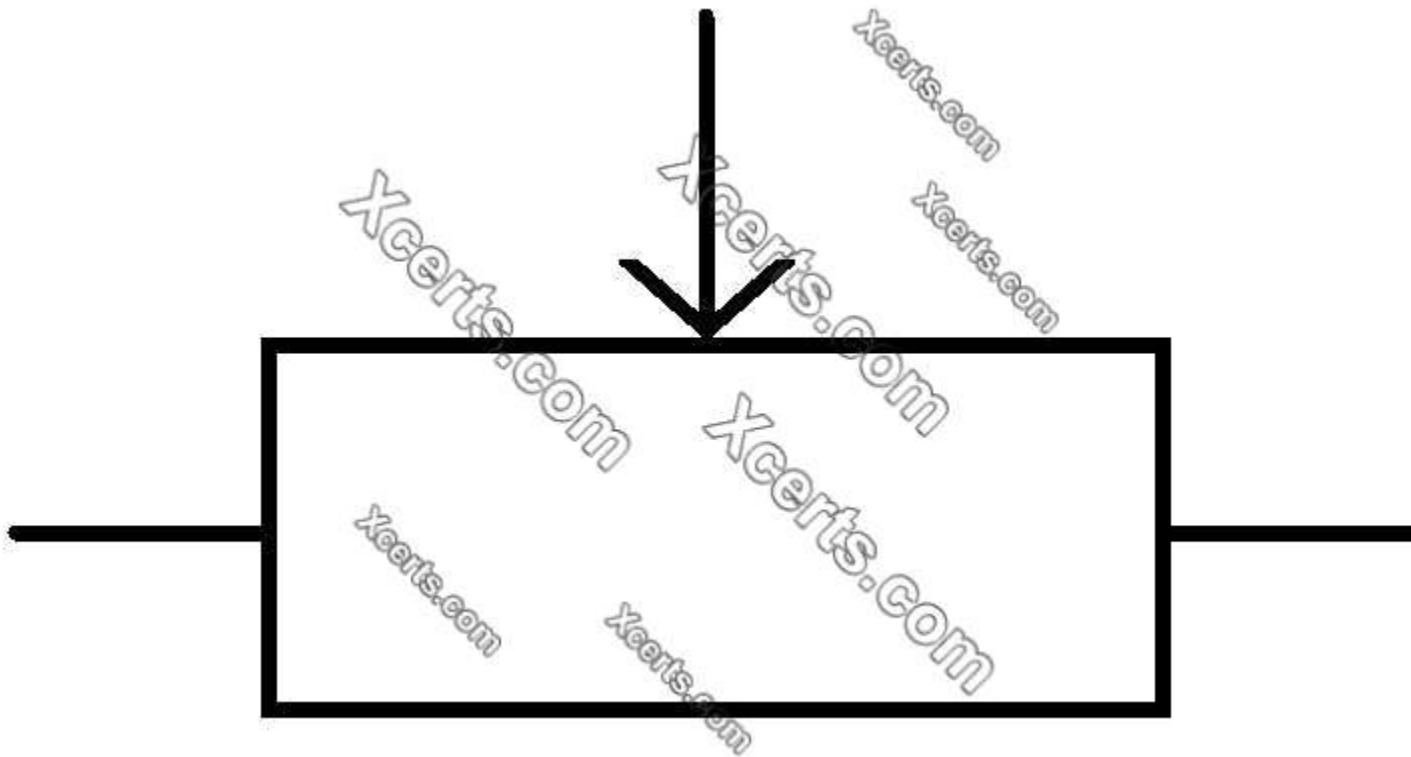
- A. 1,200 watts
- B. 1,800 watts
- C. 3,000 watts
- D. 3,600 watts

Answer(s): C

Explanation:

Determine the wattage that could cause the circuit breaker to trip with this formula:
Watts = Amperes × Volts or $25 \times 120 = 3,000$ watts.

QUESTION: 524



The above symbol represents a _____.

- A. rheostat
- B. capacitor
- C. relay
- D. potentiometer

Answer(s): A

Explanation:

This type of variable resistor is usually used to control voltage.

QUESTION: 525

16-gauge wire is _____ than 12-gauge wire.

- A. less flexible
- B. more flexible
- C. thinner
- D. thicker

Answer(s): C

Explanation:

16-gauge wire is thinner than 12-gauge wire - the larger the gauge number, the thinner the wire.

QUESTION: 526

A 120-volt circuit has a resistance of 12 ohms.
How much current is present in the circuit?

- A. 0.1 amperes
- B. 10 amperes
- C. 20 amperes
- D. 1440 amperes

Answer(s): B

Explanation:

A 120-volt circuit has a resistance of 12 ohms. A current of 10 amperes is flowing through the circuit (120 volts divided by 12 ohms = 10 amps).

QUESTION: 527

When the secondary side of a transformer has more turns than the primary side, the output voltage will be _____ the input voltage.

- A. lower than
- B. higher than
- C. the same as
- D. the square root of

Answer(s): B

Explanation:

When the secondary side of a transformer has more turns than the primary side, the output voltage will be higher than the input voltage.

QUESTION: 528

How much power is being carried in a 220-volt circuit when the current is 10 amps?

- A. 22 watts
- B. 220 watts
- C. 2,200 watts
- D. 4,400 watts

Answer(s): C

Explanation:

How much power is being carried in a 220-volt circuit when the current is 10 amps? 2,200 watts (220 volts multiplied by 10 amps = 2,200 watts).

QUESTION: 529

Capacitors connected in parallel _____.

- A. produce less capacitance.
- B. produce more capacitance.
- C. are capable of handling more voltage.
- D. produce a higher WVDC rating.

Answer(s): B

Explanation:

Capacitors connected in parallel effectively increase the plate area, which allows for more storage of electrons and more capacity or capacitance.

QUESTION: 530

Another name for a transistor is _____.

- A. integrated circuit
- B. semiconductor
- C. transfer resistor
- D. rathbone connector

Answer(s): C

Explanation:

The word "transistor" is a shortening of the term "transfer resistor". The transistor's original name when it was invented in 1933 was the "crystal amplifier".

QUESTION: 531

Current that changes polarity many times per second is called _____.

- A. direct current
- B. oscillating current
- C. alternating current
- D. AC/DC current

Answer(s): C

Explanation:

Current that changes polarity many times per second is called alternating current.

QUESTION: 532

The measurement "mil" refers to _____.

- A. a quarter of an inch
- B. one thousandth of an inch
- C. a millionth of an inch
- D. a hundredth of an inch

Answer(s): B

Explanation:

A "mil" (from milli) refers to one thousandth.

QUESTION: 533

A filter is used in a power supply to _____.

- A. convert alternating current to direct current.
- B. change direct current to alternating current.
- C. even out voltage variations.
- D. smooth out power surges.

Answer(s): C

Explanation:

A filter is placed in the power supply to even out any voltage variations. In some cases, a choke may be used in the filter to smooth out the current variations.

QUESTION: 534

Which of the following contains an electromagnet?

- A. Light switch
- B. Doorbell
- C. Fluorescent lightbulb
- D. Incandescent lightbulb

Answer(s): B

Explanation:

Of the items listed, only a doorbell contains an electromagnet.

QUESTION: 535

Radio frequency amplifiers are used in _____.

- A. audio amplifiers
- B. differential amplifiers
- C. operational amplifiers
- D. receivers and transmitters

Answer(s): D

Explanation:

Radio frequency amplifiers are used in receivers and transmitters to amplify frequencies above the range of sounds audible to humans.

QUESTION: 536

A crystal microphone uses _____.

- A. the piezoelectric effect
- B. the Doppler effect
- C. magnetic wavicles to operate
- D. a permanent magnet for resonance

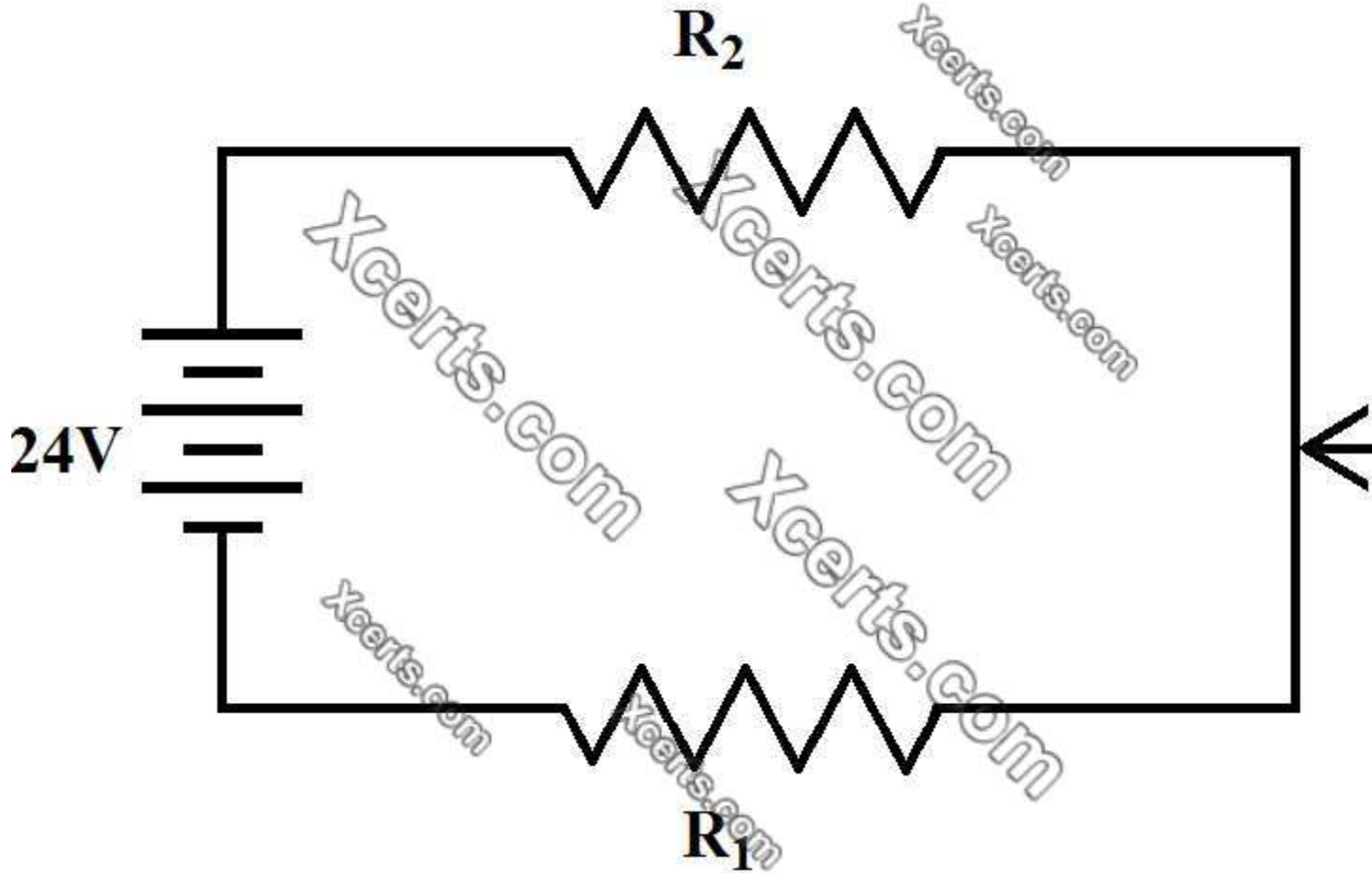
Answer(s): A

Explanation:

A crystal microphone relies on the piezoelectric effect, where a pressure on a crystal produces an electric current.

QUESTION: 537

If both the resistors shown (R1 and R2) in the diagram below are of equal size, what is the voltage at the arrow?



- A. 9V
- B. 12V
- C. 18V
- D. 24V

Answer(s): B

Explanation:

The total voltage drop of the circuit must equal the voltage across the battery, which is 24V. Because the resistors are both the same size, each resistor creates half of the total voltage drop: $24/2 = 12$. Therefore, the voltage at the arrow is 12V.

QUESTION: 538

A crossover network is used to _____.

- A. direct the proper frequency range to the right speaker
- B. eliminate noise
- C. filter out high frequencies
- D. filter out low frequencies

Answer(s): A

Explanation:

A crossover network is designed to make sure that the right frequencies get to a speaker so it can reproduce them better.

QUESTION: 539

The klystron is used in _____.

- A. low-frequency transmitters
- B. radar units
- C. audio frequency amplifiers
- D. frequency detectors

Answer(s): B

Explanation:

The klystron is used as a frequency source in microwave units - from a microwave oven for the kitchen to a radar unit for an aircraft

QUESTION: 540

The two basic types of oscillators used for microwave generation are the magnetron and the _____.

- A. klystron
- B. op amp
- C. clapp oscillator
- D. multivibrator

Answer(s): A

Explanation:

The two basic types of oscillators used in microwave installations are the klystron and the magnetron.

QUESTION: 541

The two types of diodes used in electronics work are the semiconductor diode and the _____.

- A. vacuum tube diode
- B. full-wave diode
- C. half-wave diode
- D. zener diode

Answer(s): A

Explanation:

The two general types of diodes are the semiconductor and the vacuum tube diode. Both types

have their own particular applications and advantages.

QUESTION: 542

Analog color television relies on three colors to produce the full color range needed for a good picture: red, blue, and _____.

- A. gray
- B. orange
- C. yellow
- D. green

Answer(s): D

Explanation:

Color television uses red, blue, and green "guns" to direct electron streams toward the phosphors in the front of the picture tube, producing the full color spectrum when combined properly.

QUESTION: 543

The folded dipole antenna has an impedance of _____.

- A. 72 ohms
- B. 300 ohms
- C. 600 ohms
- D. 75 ohms

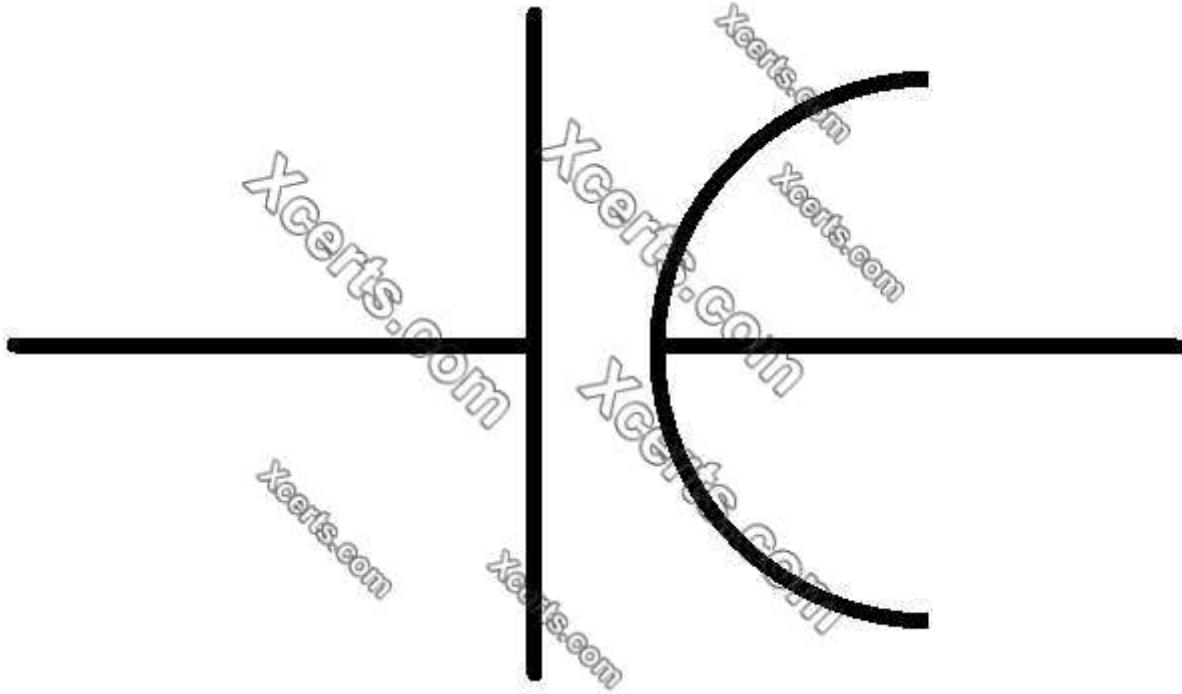
Answer(s): B

Explanation:

The folded dipole antenna - the one most commonly used for home TV reception - has an impedance of 300 ohms. The dipole has an impedance of 72 ohms.

QUESTION: 544

The symbol below is used in electrical diagrams to represent a _____.

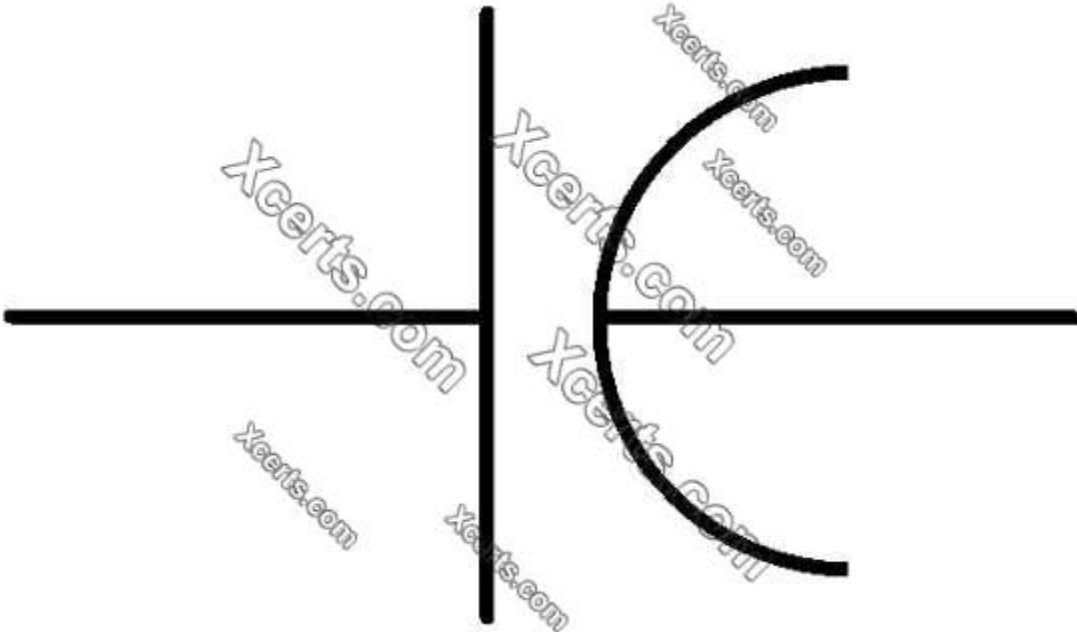


- A. resistor
- B. capacitor
- C. transformer
- D. transistor

Answer(s): B

Explanation:

The symbol below is used in electrical diagrams to represent a capacitor.



QUESTION: 545

Electricity is defined as _____.

- A. the flow of electrons along a conductor
- B. the flow of ions along a conductor
- C. the movement of charges
- D. static electricity

Answer(s): A

QUESTION: 546

A conductor is any material that _____.

- A. has no free electrons
- B. has many free electrons
- C. has free ions
- D. has free protons

Answer(s): B

Explanation:

Conductors have free electrons that are easily moved along with the right kind of push or force. There are a number of ways to cause these electrons to progress along a conductor. Heat, light, magnetism, chemicals, pressure, and friction can cause the generation of an EMF useful in causing electrons to move along a conductor.

QUESTION: 547

Convention current flow is from _____.

- A. + to +
- B. - to +
- C. + to -
- D. - to -

Answer(s): C

Explanation:

Conventional current flow is said to be from + to -. This idea was originally used by Ben Franklin to explain the conduct of lightning. This conventional flow is still used today by electrical engineers and people working in physics.

QUESTION: 548

Electron flow is from _____.

- A. + to +
- B. - to +

- C. + to -
- D. - to -

Answer(s): B

Explanation:

Electron flow can be demonstrated to flow from - to +. The action that takes place in a vacuum tube indicates that the + attracts the - electron to cause it to move from - to +.

QUESTION: 549

An insulator is a material with _____.

- A. many free electrons
- B. few free electrons
- C. many free protons
- D. many free ions

Answer(s): B

Explanation:

An insulator is a material that has tightly held electrons and very few, if any, free electrons. This means that it can be used to protect the conductor since the insulator will not allow free electrons from the conductor to knock lose any tightly held electrons of the insulator material.

QUESTION: 550

Opposition to current flow is measured in _____.

- A. ohms
- B. amps
- C. volts
- D. watts

Answer(s): A

Explanation:

The unit of measure for resistance or opposition to current flow is the ohm. It was agreed on many years ago and has aided in making the electrical field progress so rapidly.

QUESTION: 551

A watt is the power that gives rise to the production of energy at the rate of _____.

- A. 1 joule per minute
- B. 1 volt per second
- C. 1 amp per second
- D. 1 joule per second

Answer(s): D

Explanation:

The watt is the unit of power in electrical work. It is defined as the production of energy at the rate of one joule per second (per second is the basis for all electrical work).

QUESTION: 552

The Greek letter phi (ϕ) is used to _____.

- A. indicate ohms
- B. indicate angles or phases
- C. indicate volts
- D. indicate dielectric flux

Answer(s): B

Explanation:

The Greek alphabet is used almost in its entirety in the electrical field to represent various quantities and effects. The Greek letter phi (ϕ) is familiar if you have been looking at three-phase power. It is used to represent the phases of electrical power.

QUESTION: 553

A parallel circuit with resistors of 10 ohms, 10 ohms, and 5 ohms has a total resistance of _____.

- A. 10 ohms
- B. 5 ohms
- C. 25 ohms
- D. 2.5 ohms

Answer(s): D

Explanation:

Total resistance of a parallel circuit can be found two ways: by using the reciprocal formula or by using the product of two resistors divided by the sum of the two resistors divided by the sum of the two and then using it again to obtain the final answer in the case of three in parallel. In this case the two 10-ohm resistors will reduce to an equivalent of 5 ohms. This leaves, then, two 5-ohm resistors in parallel, which reduce to 2.5 or half of the value of one.

QUESTION: 554

Which of these is a symbol for a fuse?

A.



B.



C.



D.



Answer(s): B

Explanation:

The symbols shown are for the resistor, the fuse, and the capacitor, as well as for a lamp or light bulb. The one for the fuse is shown as in B.

QUESTION: 555

How much voltage has a typical home outlet?

- A. 120 volts
- B. 200 volts
- C. 60 volts
- D. 100 volts

Answer(s): A

Explanation:

A typical residential electrical outlet has 120 volts.

QUESTION: 556

A circuit contains three parallel resistors. Each resistor has 3 amps flowing through them. What is the total current flowing through the circuit?

- A. 1 A
- B. 3 A

- C. 9 A
- D. 1.5 A

Answer(s): C

Explanation:

For resistors in parallel, the total current is equal to the sum of the currents through each resistor.

QUESTION: 557

A capacitor of 100 micro farads has a capacity of _____.

- A. 100 picofarads
- B. 10 picofarads
- C. 100,000,000 picofarads
- D. 10,000 picofarads

Answer(s): C

Explanation:

A microfarad is 1 million picofarads. Therefore, if you wish to convert micro farads to picofarads, you have to multiply by 1 million. That means the answer is "100,000,000 picofarads" because 100 times 1 million is 100 million.

QUESTION: 558

Impedance is represented by the symbol _____.

- A. XL
- B. R
- C. XC
- D. Z

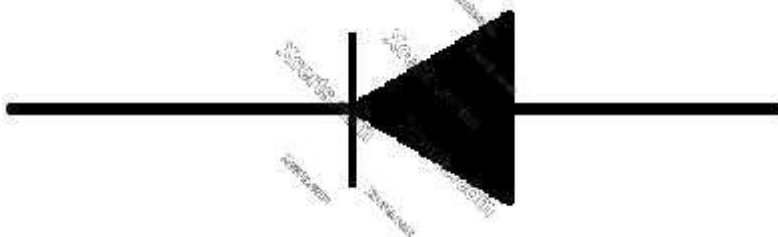
Answer(s): D

Explanation:

Impedance is the total opposition to current flow in a circuit. It has its own symbol, Z.

QUESTION: 559

In electronics this symbol represents a _____.



- A. cathode ray tube
- B. diode
- C. resistor
- D. insulator

Answer(s): B

Explanation:

The symbol shown represents a diode.

QUESTION: 560

A junction diode is used for _____.

- A. rectifying power-line frequencies and lower currents.
- B. rectifying power-line frequencies and higher currents.
- C. rectifying higher frequencies at lower currents.
- D. rectifying higher frequencies at higher currents.

Answer(s): B

Explanation:

A junction diode has a larger surface area than a point-contact diode, therefore it can be used to handle larger currents at power-line frequencies.

QUESTION: 561

The SCR is a _____.

- A. specialized rectifier or diode.
- B. special colored rectifier.
- C. special controlled rectifier.
- D. silicon-controlled relay.

Answer(s): A

Explanation:

The SCR is a special type of rectifier. It is used in control circuits. The name SCR was a descriptive one given it by General Electric Company engineers and it caught on in the literature. Its correct name is not silicon- controlled rectifier but thyristor.

QUESTION: 562

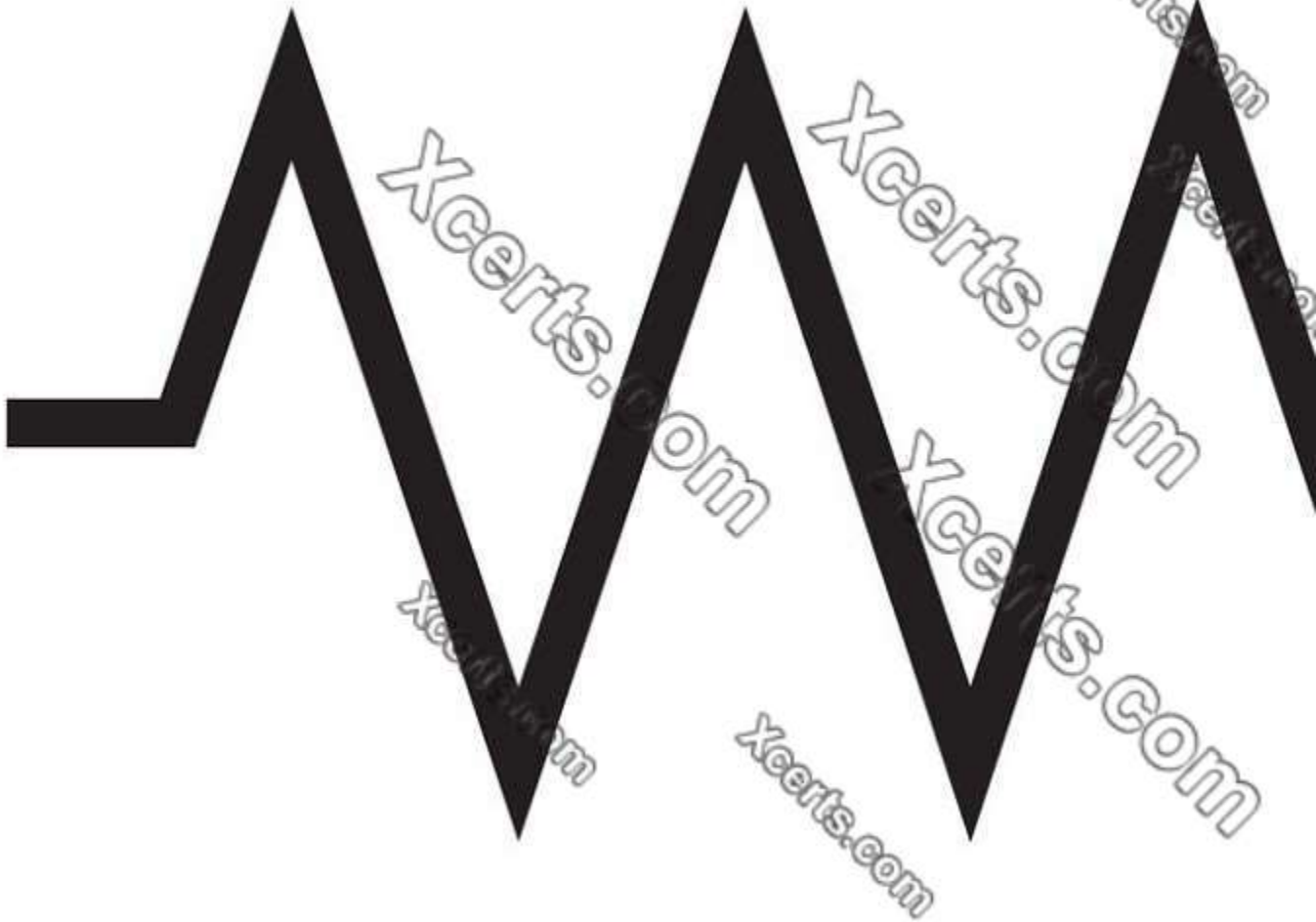
In an FM receiver the AFC is _____.

- A. automatic frequency control for stabilizing the IF.
- B. automatic frequency control for stabilizing the local oscillator.
- C. automatic frequency control for tuning the incoming frequency.
- D. automatic frequency control for the radio frequency amplifier.

Answer(s): B

Explanation:

On an FM receiver the AFC is automatic frequency control. It keeps the receiver from drifting from the station it is tuned to. This is done by having part of the signal fed back to the local oscillator to keep it operating on the correct frequency and not drifting.

QUESTION: 563

The symbol above represents a _____.

- A. thyristor
- B. resistor
- C. transistor
- D. fuse

Answer(s): B

Explanation:

The symbol shown represents a resistor.

QUESTION: 564

The FM band covers _____.

- A. 550 to 1,600 kHz
- B. 88 to 108 MHz
- C. 100 to 200 MHz
- D. 550 to 1600 MHz

Answer(s): B

Explanation:

The FM band covers 88 to 108 MHz. The AM band covers 550 to 1600 kilohertz.

QUESTION: 565

The two most commonly used metals in solder are _____.

- A. tin and gold
- B. tin and lead
- C. gold and nickel
- D. gold and lead

Answer(s): B

Explanation:

Typical solders are almost equal parts tin and lead.

QUESTION: 566

A properly soldered connection will appear _____.

- A. shiny
- B. dull
- C. dark
- D. greasy

Answer(s): A

Explanation:

Shiny joints indicate a properly soldered connection.

QUESTION: 567

Which electronic component can store charge?

- A. a transistor
- B. a transformer
- C. a resistor
- D. a capacitor

Answer(s): D

Explanation:

A capacitor can store a charge. A capacitor stores energy in an electric field. Positive charges collect on one plate and negative charges on the other plate.

QUESTION: 568

Most wires are made from _____.

- A. copper
- B. gold
- C. aluminum
- D. silver

Answer(s): A

Explanation:

Copper has a high conductivity and is financially cheaper than most metals.

QUESTION: 569

Most 12V car battery are made up of six individual cells connected in series. What is the voltage of each cell in a car battery?

- A. 2V
- B. 3.5V
- C. 6V
- D. 12V

Answer(s): A

Explanation:

Voltages in series are added to determine the total voltage, so divide 12 volts by 6 cells to get 2 volts. To be exact, each cell in a car battery provides 2.1 volts for a total of 12.6 volts when connected in a series and at full charge.

QUESTION: 570

What is the frequency of the alternating voltage and current typically used in the United States?

- A. 40hz
- B. 50hz
- C. 60hz
- D. 110hz

Answer(s): C

Explanation:

Electricity in the U.S. has a frequency of 60hz.

QUESTION: 571

What should the temperature of a typical soldering iron be when joining two wires together?

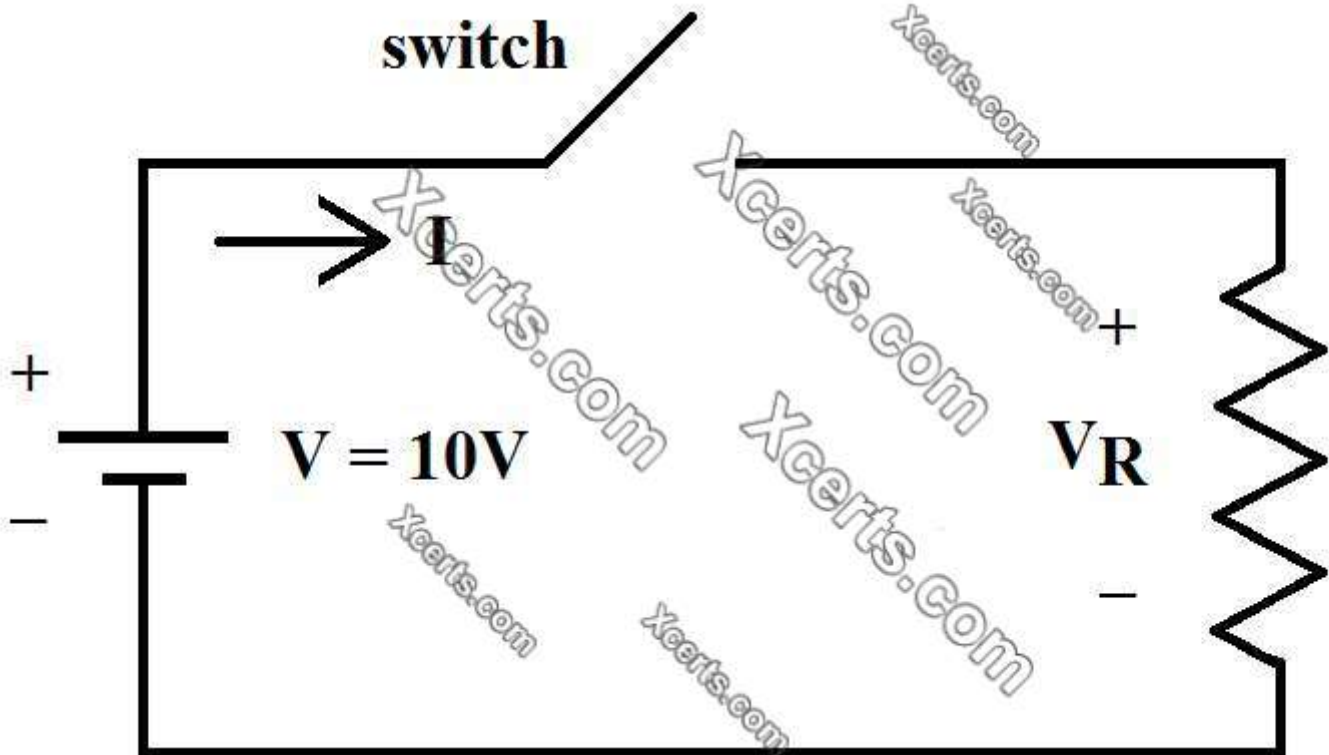
- A. 200 to 300 degrees Fahrenheit
- B. 400 to 500 degrees Fahrenheit
- C. 500 to 600 degrees Fahrenheit
- D. 700 to 800 degrees Fahrenheit

Answer(s): C

Explanation:

To properly melt solder, you should use a soldering iron heated between 500 and 600 degrees Fahrenheit.

QUESTION: 572



What is the voltage across the load R and the current through R when the switch is open in the following circuit?

- A. The current is 0A; the voltage is 10V.
- B. The current is 2A; the voltage is 10V.
- C. The current is 2A; the voltage is 2V.
- D. The current is 0A; the voltage is 0V.

Answer(s): D

Explanation:

When the switch is open, no current flows. The voltage V_R is I times R , which equals $0(5) = 0$.

QUESTION: 573

Which of the following is NOT needed to produce electrical power from fossil fuels?

- A. a dam
- B. steam
- C. heat
- D. a turbine

Answer(s): A

Explanation:

Dams are used to produce power from water.

QUESTION: 574

An AA battery produces a potential difference of 1.5V. Four AA batteries are placed in series to power a portable FM radio.

What is the total voltage of the radio?

- A. 0.75V
- B. 1.5V
- C. 3.0V
- D. 6.0V

Answer(s): D

Explanation:

Voltages in series are added to determine the total voltage.

$1.5V + 1.5V + 1.5V + 1.5V = 6.0V$.

QUESTION: 575

Two 8-ohm speakers are connected in parallel to an amplifier with a 24-volt output.

What is the voltage across each speaker?

- A. 6 volts
- B. 12 volts
- C. 18 volts
- D. 24 volts

Answer(s): D

Explanation:

Resistances in parallel have the same voltage across them.

QUESTION: 576

Which function does the receiving antenna provide in radio communication?

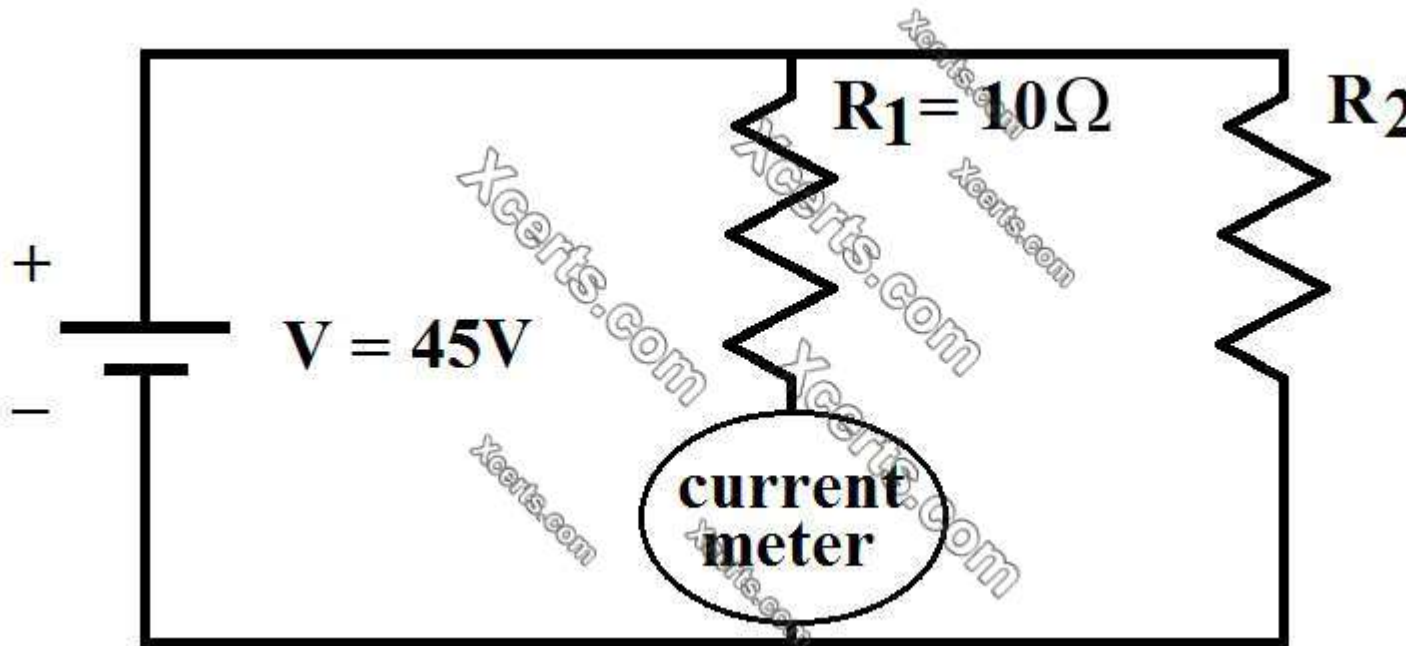
- A. It catches the signal out of the air.
- B. It separates the carrier wave from the signal.
- C. It mixes the signal and the carrier wave.
- D. It broadcasts the signal over the air.

Answer(s): A

Explanation:

The broadcast antenna is used to broadcast the signal over the air. The mixer combines the signal and carrier wave. The de-mixer separates the carrier and the signal.

QUESTION: 577



A current meter is connected to a circuit as shown. What does the current meter read?

- A. 3.0 amperes
- B. 4.5 amperes
- C. 13.5 amperes
- D. 15 amperes

Answer(s): B

Explanation:

The current through the meter is equal to the current through R1. The current through R1 is $45V / 10 = 4.5A$

QUESTION: 578

An oscilloscope is typically used to analyze _____.

- A. resistance
- B. power in a circuit
- C. complex waveforms
- D. simple circuits

Answer(s): C

Explanation:

An oscilloscope is used primarily to look at complex waveforms.

QUESTION: 579

Printed circuit boards are typically made out of _____.

- A. laminated paper
- B. fiberglass
- C. copper
- D. glass

Answer(s): B

Explanation:

Fiberglass is used to make most printed circuit boards.

QUESTION: 580

The rate of electrons through a conductor is measured in units of _____.

- A. current
- B. electricity
- C. volts
- D. amperes

Answer(s): D

QUESTION: 581

Which of these is an acceptable symbol used for amperes?

- A. Ampo
- B. M
- C. Am
- D. A

Answer(s): D

QUESTION: 582

Current is measured using _____.

- A. wires
- B. an ammeter
- C. a currentometer
- D. a spectrometer

Answer(s): B

QUESTION: 583

The two types of electrical current are _____.

- A. direct and changing
- B. backwards and forwards
- C. simple and complex
- D. direct and alternating

Answer(s): D

QUESTION: 584

32 cycles per second is equivalent to _____.

- A. 32 ohms (Ω)
- B. 32 positons (pe)
- C. 64 hertz (Hz)
- D. 32 hertz (Hz)

Answer(s): D

QUESTION: 585

Electrical resistance is best defined as _____.

- A. opposition to the flow of current
- B. the time it takes for electricity to travel 10,000 miles
- C. ease of an electrical current through a conductor
- D. oscillation of the flow of an electrical current

Answer(s): A

QUESTION: 586

To measure voltage, one would use _____.

- A. a volt gauge
- B. a voltmeter
- C. a volta
- D. an electrical difference machine

Answer(s): B

QUESTION: 587

Voltage is also known as _____.

- A. electrical potential
- B. current charge
- C. shock factor
- D. electronic strength

Answer(s): A

QUESTION: 588

Electrical "pressure" is also known as _____.

- A. charge
- B. conductivity
- C. voltage
- D. ohms

Answer(s): C

QUESTION: 589

An ampere is defined as one _____ flowing past a point in one second.

- A. coulomb
- B. hertz
- C. ohm
- D. volt

Answer(s): A

QUESTION: 590

A galvanometer is usually used to measure _____.

- A. volts
- B. current
- C. conductance
- D. resistance

Answer(s): B

QUESTION: 591

Electromotive force is another word for _____.

- A. ohms
- B. volts
- C. amps
- D. current

Answer(s): B

QUESTION: 592

Ohm's Law is defined as _____.

- A. $E = I \times R$ (Electrons = Integrated circuits $\times$ Resistance)
- B. $E = I \times R$ (Ohms = Intensity $\times$ Regulated flow)
- C. $E = I \times R$ (Volts = Current $\times$ Resistance)
- D. $E = I \times R$ (Electromotive force = Ions $\times$ Resistance)

Answer(s): C

QUESTION: 593

Electrical power is expressed in _____.

- A. ohms
- B. volts
- C. watts
- D. current

Answer(s): C

QUESTION: 594

A series circuit would best be described as _____.

- A. an electrical circuit with only a single path for the electrical current
- B. an electrical circuit with multiple paths for the electrical current
- C. an electrical circuit with no path for the electrical current
- D. an electrical circuit with two paths for the electrical current

Answer(s): A

QUESTION: 595

Given a series circuit, a break at any point in the circuit would cause _____.

- A. the current to double in power
- B. the current to jump the break and continue flowing
- C. the current to stop flowing
- D. the current to go in reverse

Answer(s): C

QUESTION: 596

In residential wiring, the internal grounds will be connected to an external ground, also known as _____.

- A. extra ground
- B. Earth ground
- C. safe ground
- D. dirt ground

Answer(s): B

QUESTION: 597

In an electrical circuit, if one wanted to reduce the current and/or divide the voltage, one would use a _____.

- A. rectifier
- B. reducer
- C. reflector
- D. resistor

Answer(s): D

QUESTION: 598

A capacitor is also known as _____.

- A. a resistor
- B. a dielectric
- C. a condenser
- D. an alternator

Answer(s): C

QUESTION: 599

A capacitor is used to perform which of these in an electrical circuit?

- A. reduce current
- B. store an electrical charge
- C. increase voltage
- D. prevent shock

Answer(s): B

QUESTION: 600

The farad is the unit of measurement of _____.

- A. capacitance

- B. voltage
- C. resonance
- D. ohms

Answer(s): A

QUESTION: 601

Induction is measured in _____.

- A. Henries
- B. ohms
- C. Franks
- D. joules

Answer(s): A

QUESTION: 602

A semiconductor describes elements that have _____ electrons in their valance shell.

- A. 3
- B. 4
- C. 2
- D. 1

Answer(s): B

QUESTION: 603

Which of these elements is considered a semiconductor?

- A. sulfur
- B. magnesium
- C. argon
- D. silicon

Answer(s): D

QUESTION: 604

Transistors are known as being solid-state.

This means _____.

- A. transistors have no moving parts and all of their electricity exists in their material
- B. transistors are very reliable and are generally "solid"
- C. transistors never change state once set
- D. transistors carry no state information between charges

Answer(s): A

QUESTION: 605

The two types of transistors are _____.

- A. PNP and NPN
- B. PPP and NNN
- C. ABC and BCA
- D. CBC and PRN

Answer(s): A

QUESTION: 606

The primary components in a bipolar junction transistor are the collector, the base and the _____.

- A. emitter
- B. charger
- C. dispenser
- D. emulcifier

Answer(s): A

QUESTION: 607

When an electric current passes through a wire, a _____ field is generated around the wire.

- A. charged
- B. magnetic
- C. positive
- D. negative

Answer(s): B

QUESTION: 608

Winding wire into a coil will have what effect on the magnetic field in the wire?

- A. The magnetic field will be removed.
- B. There will be no change.
- C. It will make it stronger.
- D. It will make it weaker.

Answer(s): C

QUESTION: 609

Components of an electrical circuit that resist a change in the flow of current are known as _____.

- A. self-inducing agents
- B. selfers
- C. inductors
- D. inducers

Answer(s): C

QUESTION: 610

1.5 volts sent through a 20K resistor would result in _____ of current.

- A. 7.5 milliamperes
- B. 0.075 milliamperes
- C. 75 milliamperes
- D. 0.000075 milliamperes

Answer(s): B

QUESTION: 611

How many volts does it take to send 99 amps of current through 99,000 ohms of resistance?

- A. 0.009801 volts
- B. 980.1 megavolts
- C. 9.801 megavolts
- D. 98.01 volts

Answer(s): C

QUESTION: 612

The largest subatomic particle with a positive charge is called the _____.

- A. neutron
- B. positron
- C. electron
- D. proton

Answer(s): D

QUESTION: 613

The nucleus of an atom contains what two subatomic particles?

- A. protons and gamma particles
- B. neutrons and electrons
- C. protons and electrons
- D. protons and neutrons

Answer(s): D

QUESTION: 614

Carbon has 6 protons and 6 neutrons in its nucleus.
How many electrons does it have?

- A. 0
- B. 24
- C. 6
- D. 12

Answer(s): C

QUESTION: 615

_____ is the name given to the orbit an electron resides in around an atom.

- A. circulation
- B. shell
- C. field
- D. level

Answer(s): B

QUESTION: 616

The electron shell closest to the nucleus of an atom can hold _____ electrons.

- A. 2
- B. 8
- C. 4
- D. 6

Answer(s): A

QUESTION: 617

The outer shell of an atom is known as the _____.

- A. suborbital shell
- B. valence shell
- C. final shell
- D. core shell

Answer(s): B

QUESTION: 618

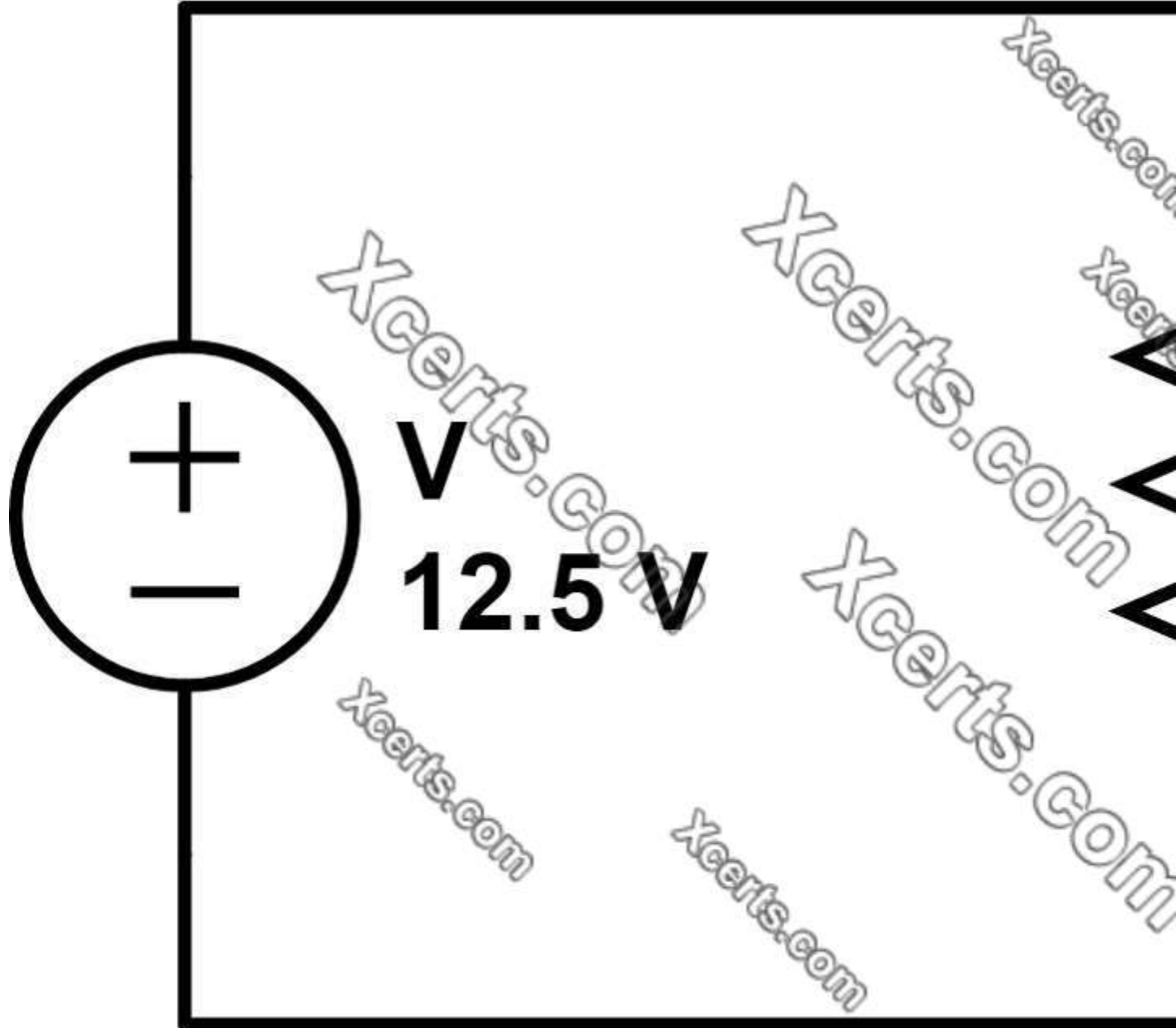
Given Ohm's law, if voltage increases and current remains the same, then _____.

- A. induction increases
- B. resistance decreases

- C. power decreases
- D. resistance increases

Answer(s): D

QUESTION: 619

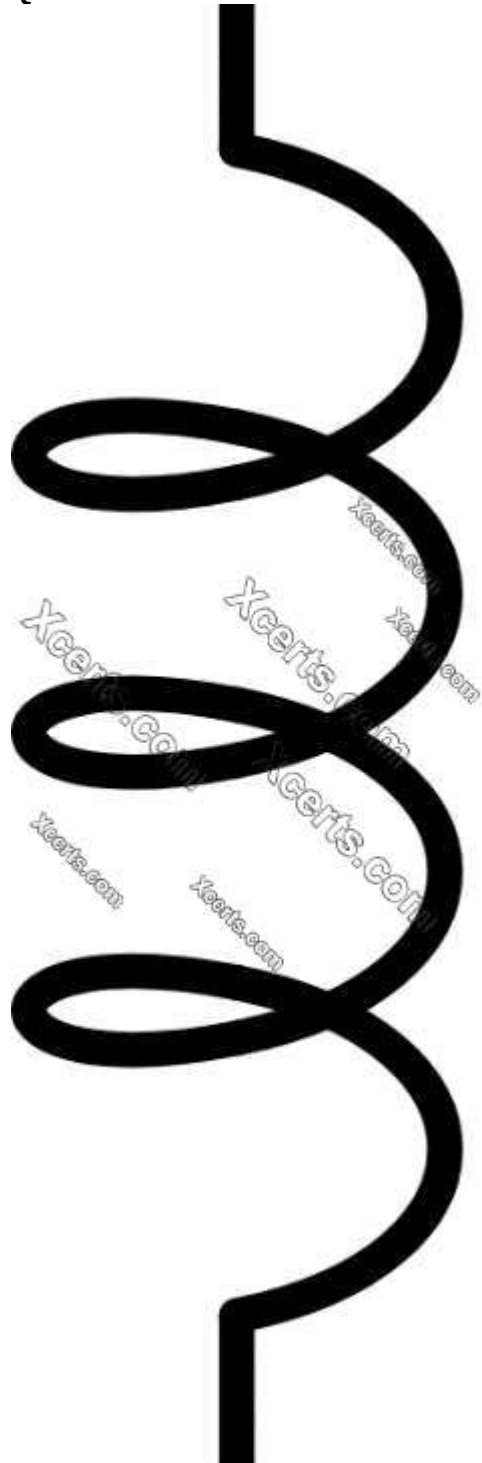


In this diagram, how much current is flowing through the circuit?

- A. 0.625 milliamperes
- B. 0.625 amperes
- C. 62.5 milliamperes
- D. 625 amperes

Answer(s): A

QUESTION: 620

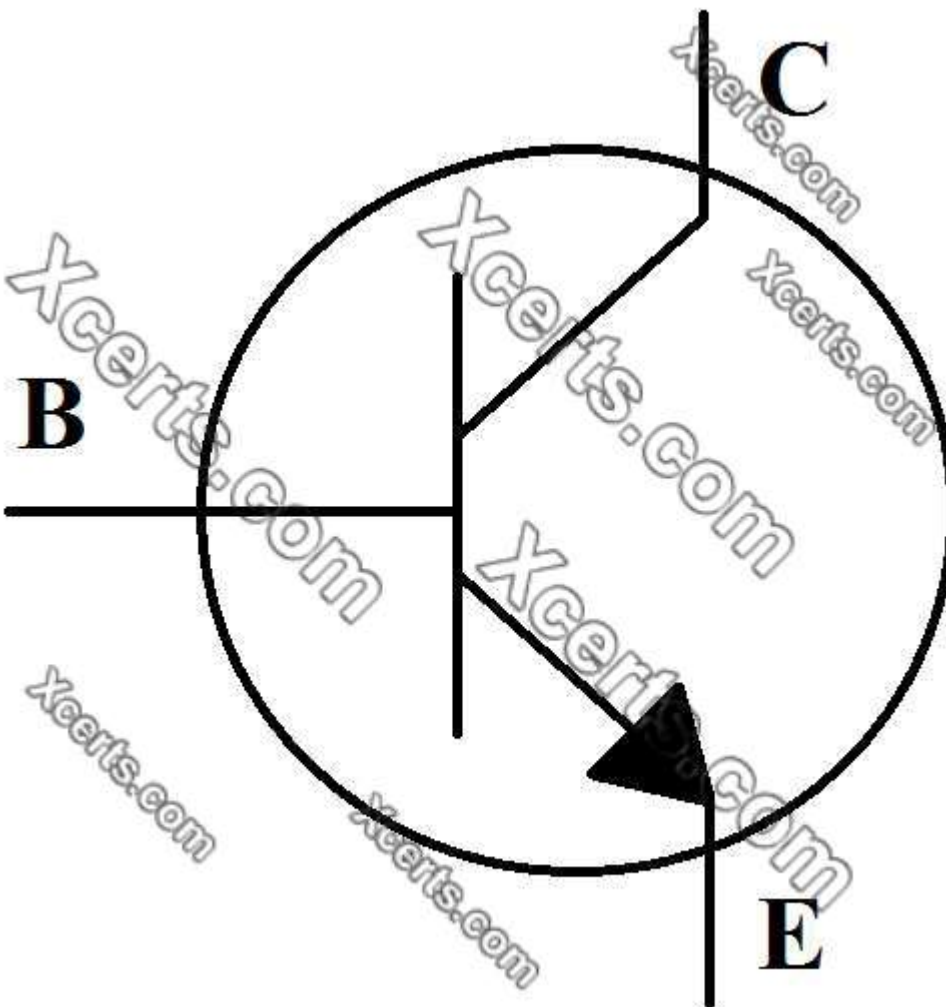


This symbol is the symbol for which of these electrical circuit components?

- A. inductor
- B. elevator
- C. conductor
- D. resistor

Answer(s): A

QUESTION: 621

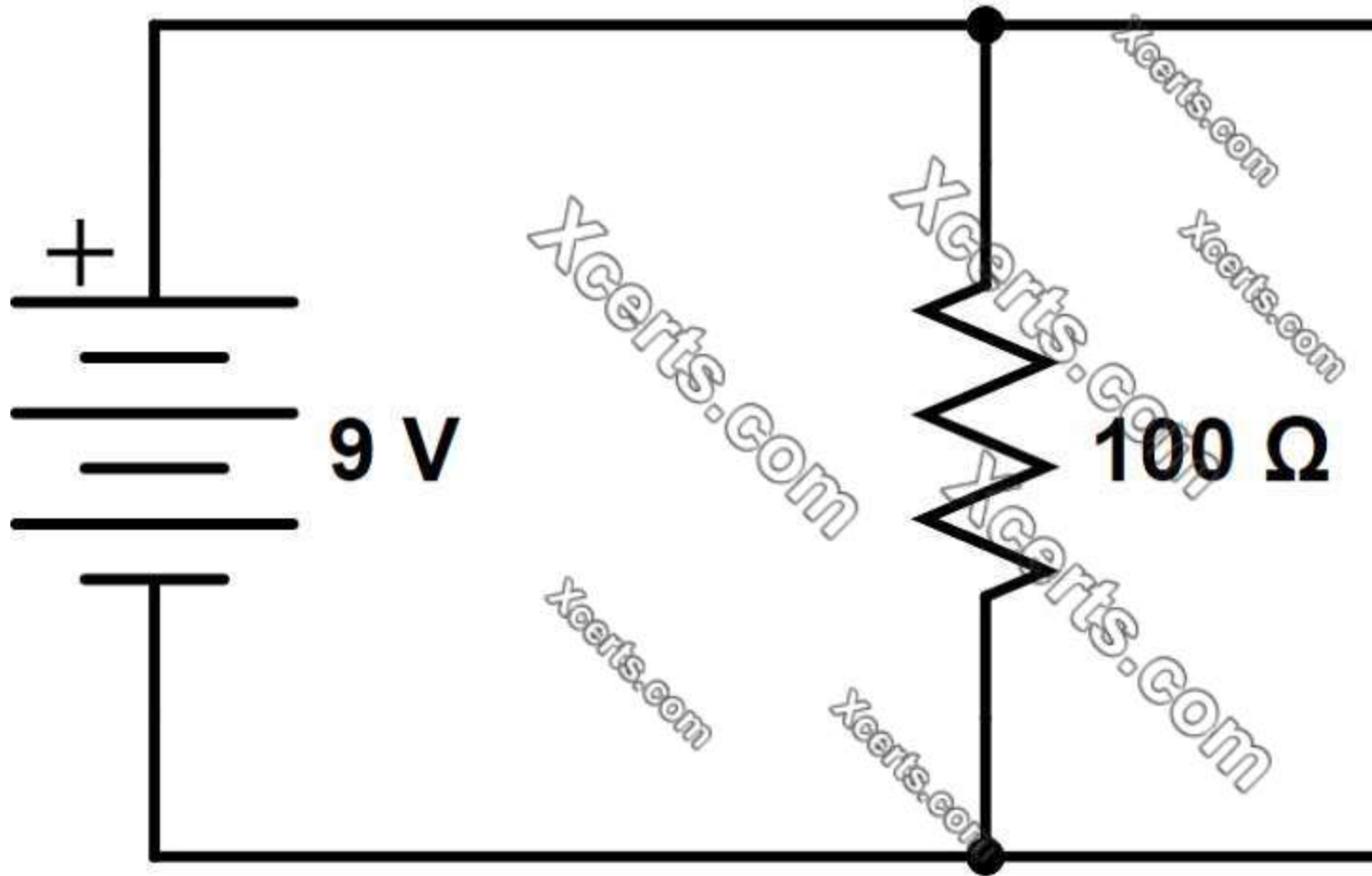


This symbol represents what electronic circuit component?

- A. power source
- B. load
- C. transistor
- D. conductor

Answer(s): C

QUESTION: 622

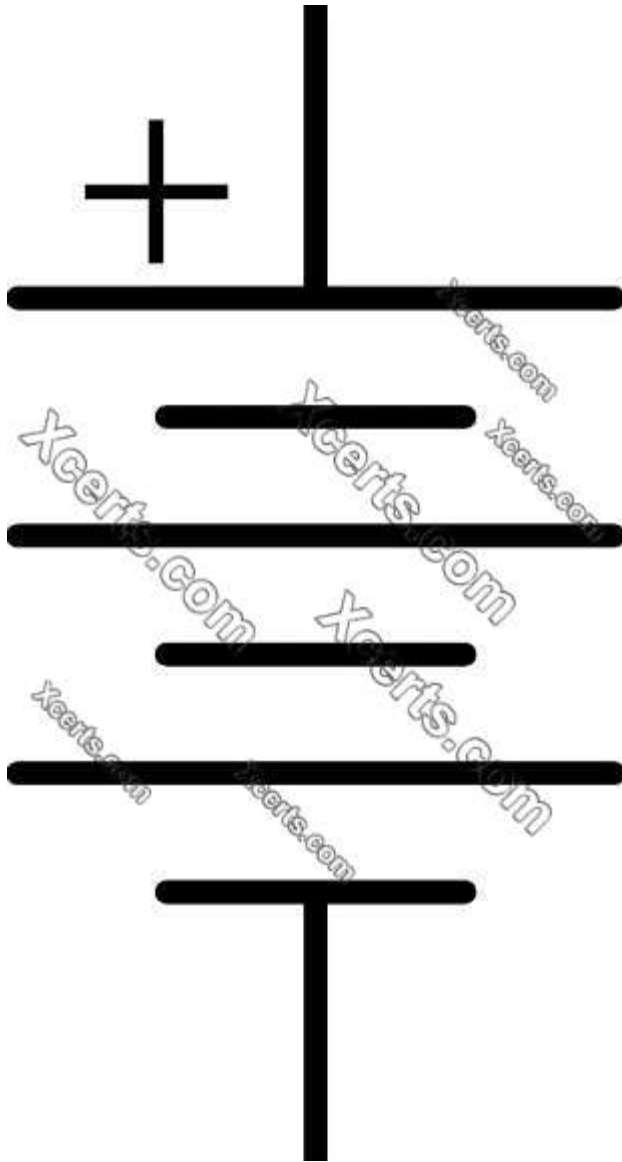


This diagram illustrates an electrical circuit wired in _____.

- A. series-parallel
- B. parallel-parallel
- C. series
- D. parallel

Answer(s): D

QUESTION: 623

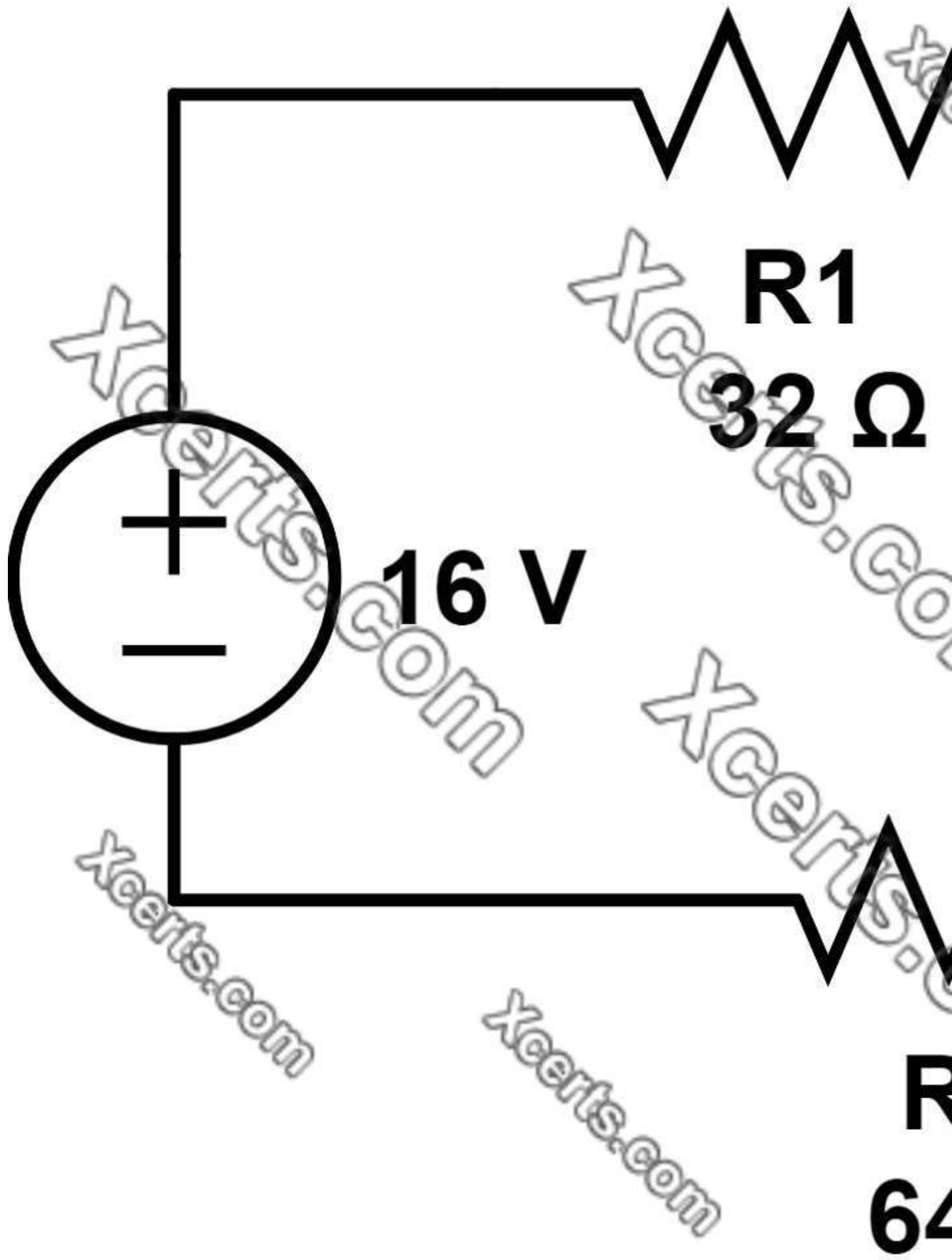


The symbol in this image represents a _____ in an electrical circuit.

- A. relay
- B. switch
- C. fuse
- D. battery

Answer(s): D

QUESTION: 624

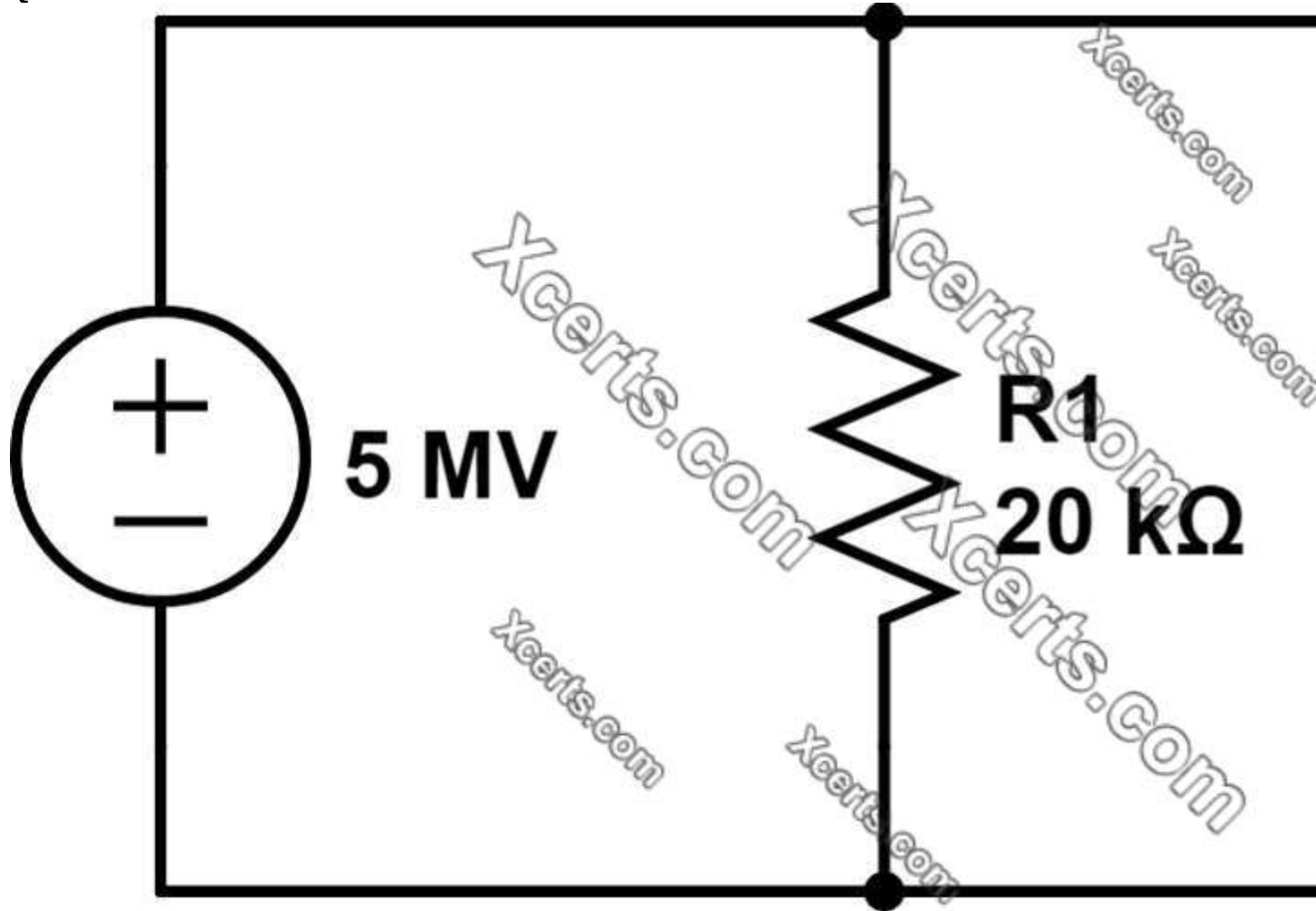


Given the electrical circuit in the image provided, what is the current in resistor R1?

- A. 0.5 amperes
- B. 7.66666666661 amperes
- C. 0.25 amperes
- D. 0.16666666667 amperes

Answer(s): D

QUESTION: 625

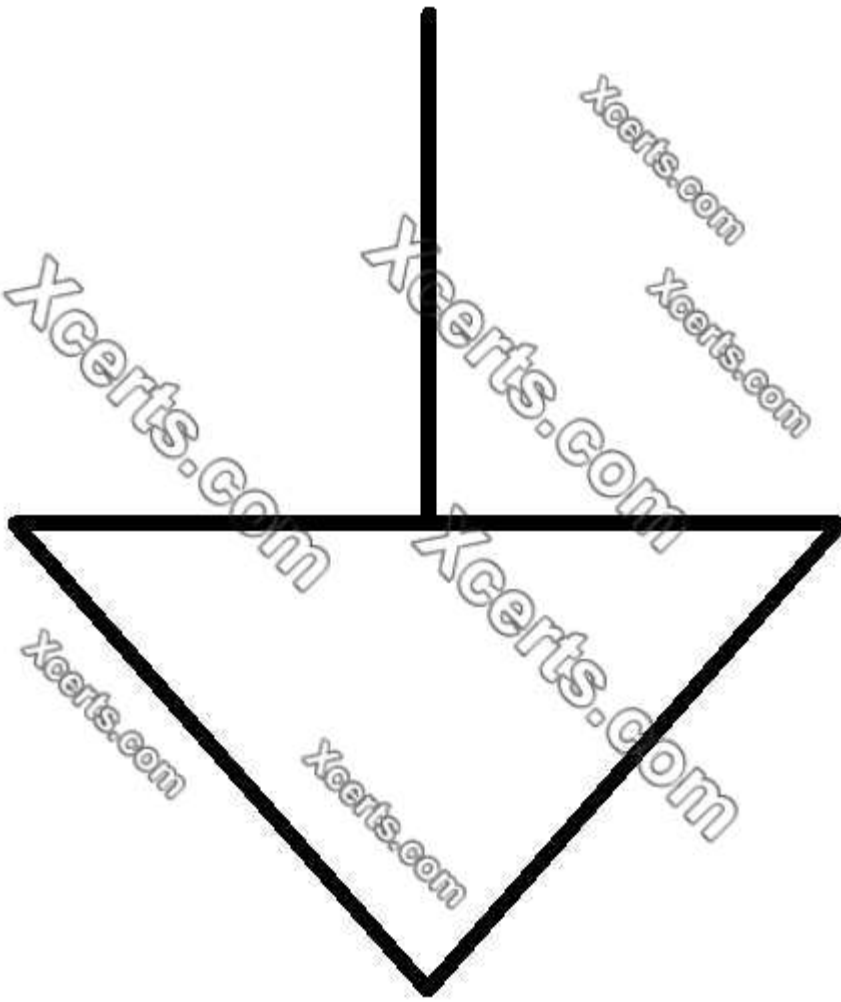


Given the electronic circuit diagram, what is the current flowing through resistor at position R2?
(MV = megavolts, k = kilo)

- A. 250 amperes
- B. 0 amperes
- C. 100 milliamperes
- D. 125 amperes

Answer(s): A

QUESTION: 626

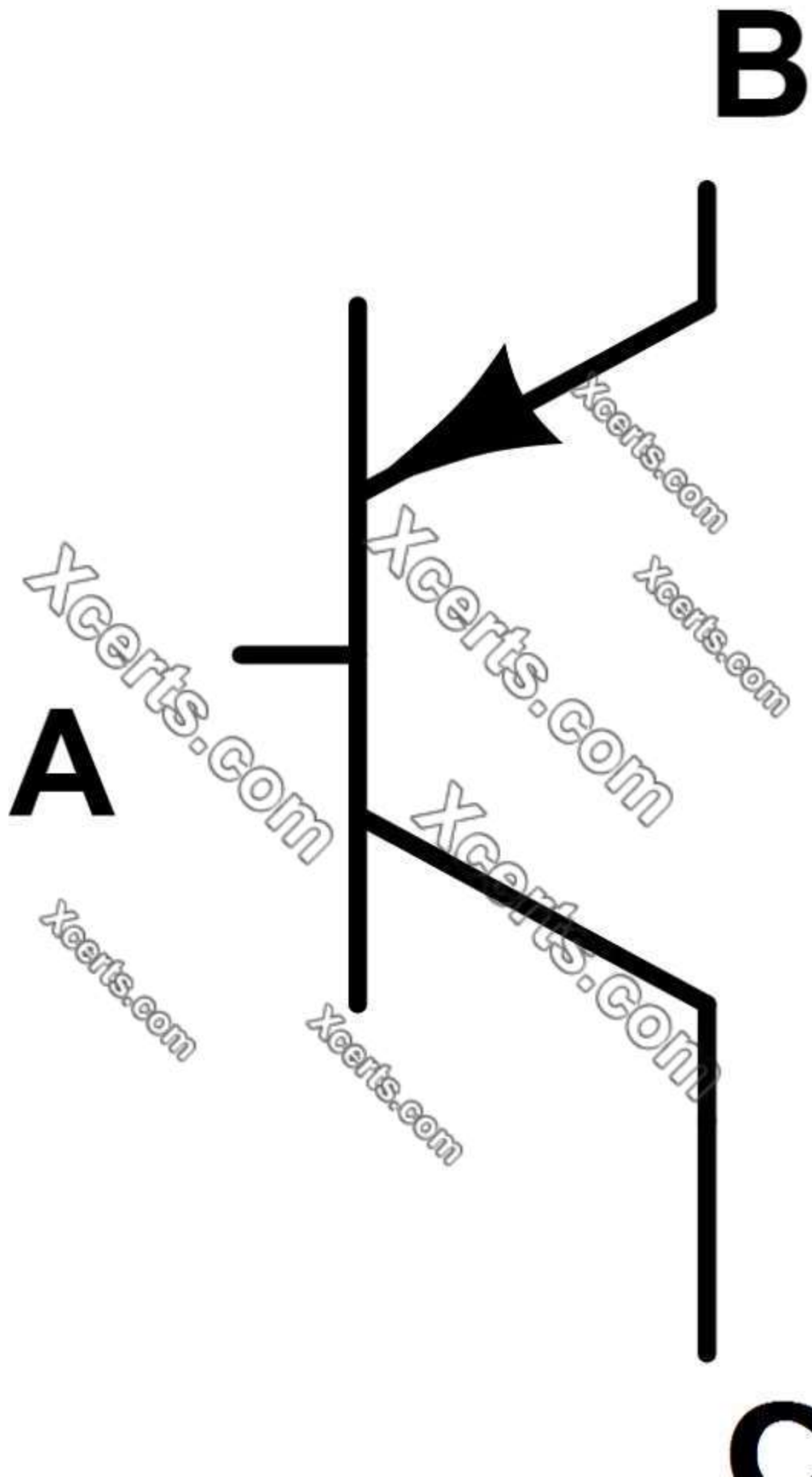


The symbol in this diagram is the symbol for _____.

- A. lightning
- B. Earth ground
- C. power source
- D. signal ground

Answer(s): D

QUESTION: 627



In the diagram of a transistor provided, the letter B corresponds with the _____.

- A. controller
- B. emitter
- C. base
- D. collector

Answer(s): B

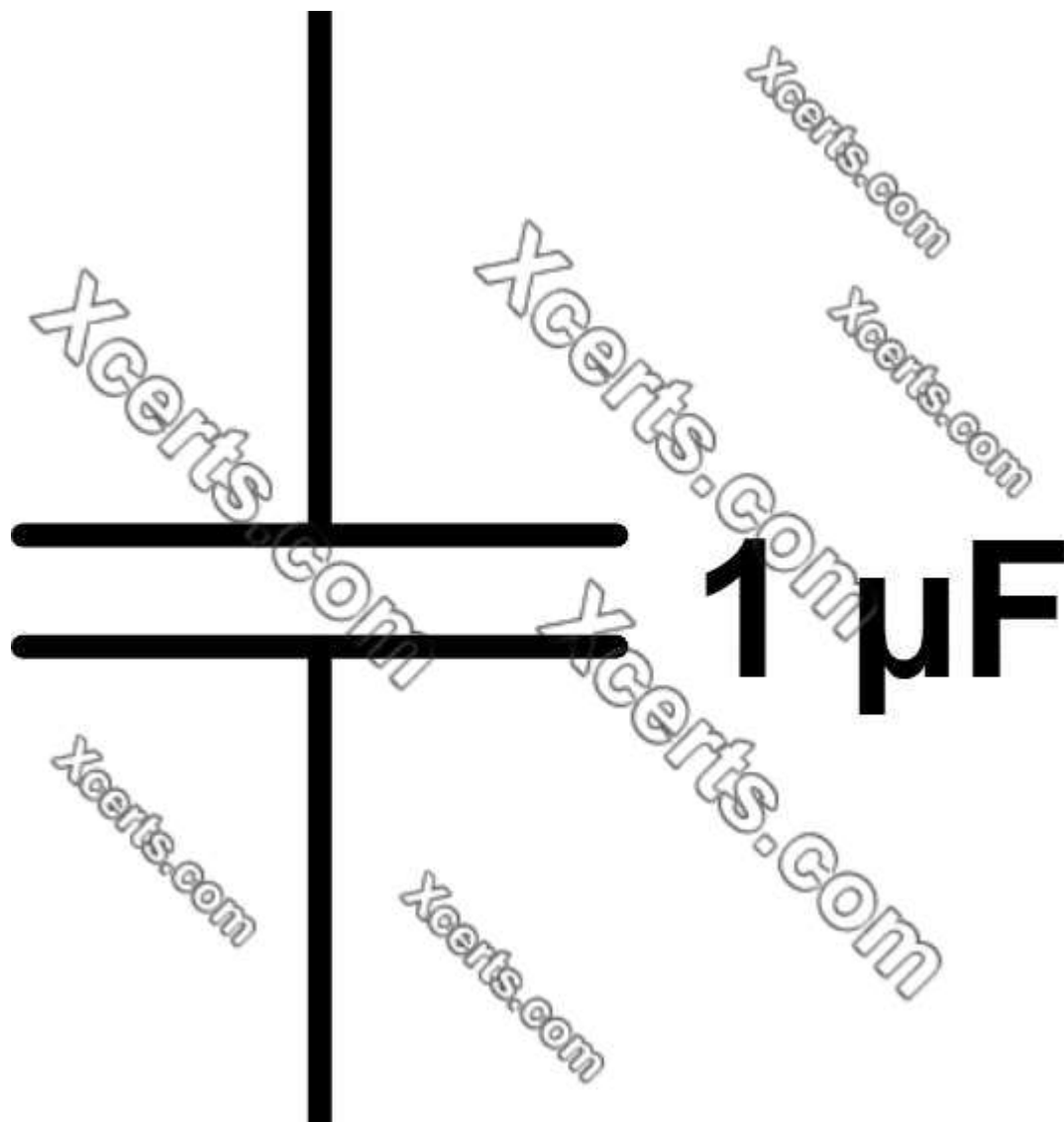
QUESTION: 628

Regarding the material found in a transistor, _____ is the name given to the object created by the joining of P-type and N-type material.

- A. semiconductor
- B. integrated circuit
- C. diode
- D. cathode

Answer(s): C

QUESTION: 629



The symbol in the given diagram represents a _____.

- A. transducer
- B. resistor
- C. power source
- D. capacitor

Answer(s): D

QUESTION: 630

An ammeter measures _____.

- A. Resistance
- B. Current
- C. Voltage
- D. Capacitance

Answer(s): B

Explanation:

An ammeter measures current.

QUESTION: 631

A 120V light bulb is rated for 40 watts.
What is the light bulb's current rating?

- A. 3A
- B. 1/3A
- C. 2A
- D. 1A

Answer(s): B

Explanation:

$P = IV$, so $I = P/V$ or $I = 40 \text{ Watts}/120V = 1/3A$

QUESTION: 632

What color is Neutral wire?

- A. Blue
- B. Green
- C. Black
- D. White

Answer(s): D

Explanation:

Neutral wire is whitish (or natural) colored.

QUESTION: 633

What voltage will cause an 8 Amp current to flow through a 24 ohm resistor?

- A. 3 volts
- B. 192 volts
- C. 1/3 volt
- D. 32 volts

Answer(s): B

Explanation:

Ohm's law is $V = IR$, so $V = 8 \times 24 = 192 \text{ volts}$

QUESTION: 634

The electronic symbol shown below represents a(n) _____.



- A. Capacitor
- B. Resistor
- C. Inductor
- D. Diode

Answer(s): C

Explanation:

The symbol is a standard representation of an inductor.

QUESTION: 635

AC is an abbreviation for _____.

- A. Alternating Charge
- B. Ampere Coil
- C. Alternating Current
- D. Amp Current

Answer(s): C

Explanation:

AC is an abbreviation for Alternating Current.

QUESTION: 636

Two 16 ohm speakers are connected in parallel to an amplifier that provides a 32 volt output. The voltage across each speaker is _____.

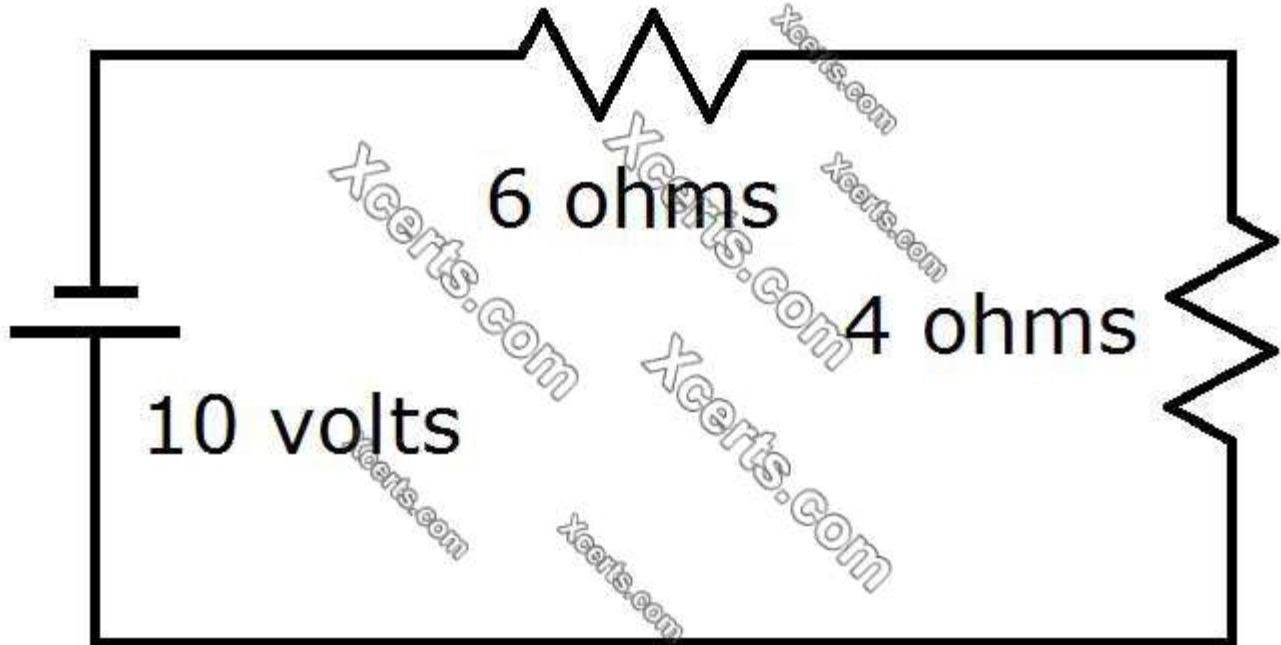
- A. 2 volts
- B. 16 volts
- C. 64 volts
- D. 32 volts

Answer(s): D

Explanation:

The voltage across parallel resistors is the same.

QUESTION: 637



The current in the 4 ohm resistor is _____.

- A. 2 amps
- B. 2.5 amps
- C. 0.4 amps
- D. 1 amp

Answer(s): D

Explanation:

Use Ohm's law - $V=IR$. The total resistance in the circuit is 10 ohms (add the resistance for serial resistors).

Therefore, the total current across either of the resistors is $I = V/R = 10/10 = 1$ amp.

QUESTION: 638

What is the reading on the kilowatt-hour meter shown below?



- A. 9282

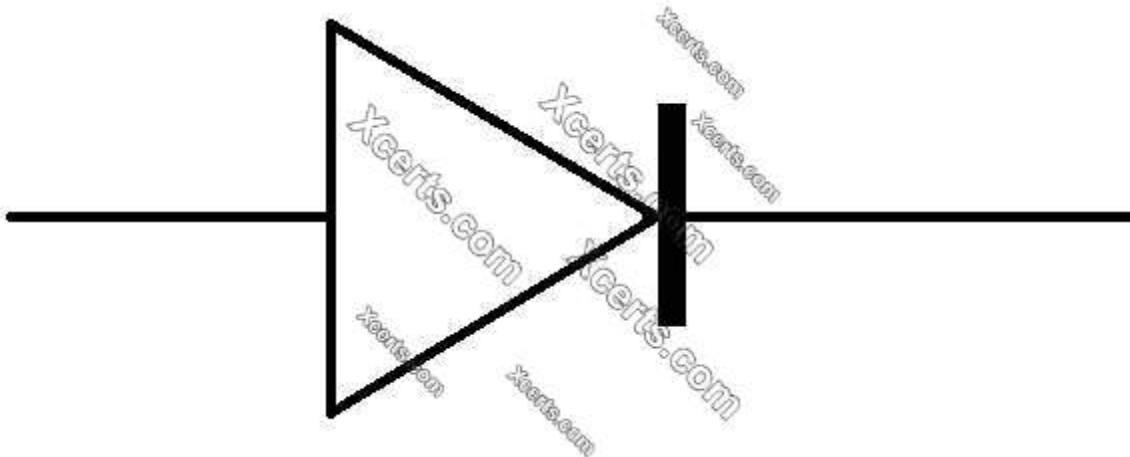
- B. 2828
- C. 8282
- D. 9292

Answer(s): C

Explanation:

The meter has four dials with the numbers 0 to 9. The pointers rotate in different directions. Read the dials from the left, using the number the dial pointer just passed.

QUESTION: 639



The electronic symbol shown below represents a(n) _____.

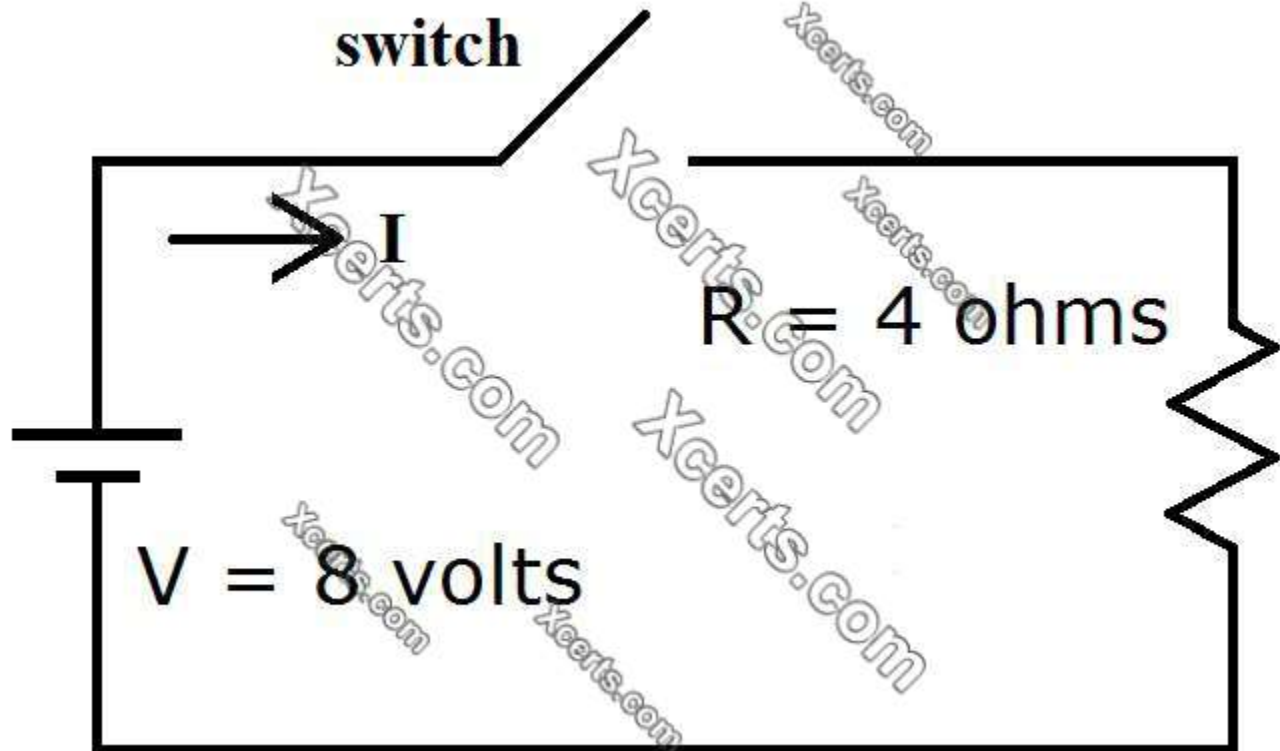
- A. Capacitor
- B. Transistor
- C. Inductor
- D. Diode

Answer(s): D

Explanation:

The symbol is a standard representation of a diode.

QUESTION: 640



What is the voltage across and current through the resistor R when the switch is open?

- A. 0A Current; 8V voltage
- B. 2A Current; 2V voltage
- C. 0A Current; 0V voltage
- D. 2A Current; 8V voltage

Answer(s): C

Explanation:

When the switch is open, there is no voltage or current.

QUESTION: 641

What is the reading on the kilowatt-hour meter shown below?



- A. 5703
- B. 5692
- C. 5693
- D. 5793

Answer(s): B

Explanation:

The meter has four dials with the numbers 0 to 9. The pointers rotate in different directions. Read the dials from the left, using the number the dial pointer just passed.

QUESTION: 642

Resistance is measured in _____.

- A. Watts
- B. Volts
- C. Ohms
- D. Farads

Answer(s): C

Explanation:

Resistance is measure in ohms.

QUESTION: 643

What is the total resistance when two 10 ohm resistors are connected in series?

- A. 5 ohms
- B. 10 ohms
- C. 20 ohms
- D. 100 ohms

Answer(s): C

Explanation:

When resistors are in series, you simply add their resistance together to find a total resistance.
 $10 + 10 = 20$ ohms.

QUESTION: 644

Using a fuse rated higher than the required current rating of a circuit can _____.

- A. protect your circuit better
- B. cause the fuse to blow more often
- C. lower your power usage for that circuit
- D. result in serious damage to your circuit

Answer(s): D

Explanation:

Using a higher rated fuse is dangerous and can result in serious damage to your circuit.

Section 4: General Science

QUESTION: 645

A series of cell divisions that results in the formation of an embryo is called _____.

- A. mitosis
- B. meiosis
- C. osmosis
- D. cleavage

Answer(s): D

QUESTION: 646

An animal that eats only plants is called a(n) _____.

- A. omnivore
- B. herbivore
- C. carnivore
- D. voracious

Answer(s): B

QUESTION: 647

The process by which energy is provided at the cellular level is called _____.

- A. respiration
- B. recreation
- C. oxidation
- D. metabolism

Answer(s): D

QUESTION: 648

All of the following are domains except _____.

- A. Regelia
- B. Eukarya
- C. Bacteria
- D. Archaea

Answer(s): A

QUESTION: 649

Light waves travel at a rate of about _____.

- A. 186,000 miles per hour

- B. 186,000 miles per minute
- C. 18,600 miles per hour
- D. 186,000 miles per second

Answer(s): D

QUESTION: 650

The largest moon in the solar system is _____.

- A. Ganymede
- B. Titan
- C. Io
- D. Charon

Answer(s): A

QUESTION: 651

The spinal cord is part of the _____.

- A. circulatory system
- B. nervous system
- C. respiratory system
- D. digestive system

Answer(s): B

QUESTION: 652

Joints that hold bones firmly together are called _____.

- A. hinge joints
- B. ball and socket joints
- C. fixed joints
- D. pivot joints

Answer(s): C

QUESTION: 653

The top or broadest level of the classification system for living organisms is called _____.

- A. class
- B. phylum
- C. kingdom
- D. genus

Answer(s): C

QUESTION: 654

If there are two full moons in a single month, the second full moon is called _____.

- A. new moon
- B. full moon
- C. blue moon
- D. secondary moon

Answer(s): C

QUESTION: 655

The brainstem connects the brain to the _____.

- A. heart
- B. lungs
- C. neck
- D. spinal cord

Answer(s): D

QUESTION: 656

Red blood cells _____.

- A. produce antibodies
- B. fight infections
- C. carry oxygen and carbon dioxide
- D. are few in number

Answer(s): C

QUESTION: 657

Protein can be found in all of the following foods EXCEPT _____.

- A. eggs
- B. meat
- C. peas
- D. apples

Answer(s): D

QUESTION: 658

Which inorganic substance is present in the greatest quantity inside animal cells?

- A. protein
- B. oxygen
- C. sodium chloride
- D. water

Answer(s): D

QUESTION: 659

Cell protoplasm is made up mostly of _____.

- A. water
- B. oxygen
- C. sugar
- D. protein

Answer(s): A

QUESTION: 660

A meter consists of _____.

- A. 10 centimeters
- B. 100 millimeters
- C. 100 centimeters
- D. 10 millimeters

Answer(s): C

QUESTION: 661

It's impossible for the sun to turn into a black hole because _____.

- A. it's too large
- B. it's too small
- C. it's a yellow star
- D. it has planets

Answer(s): B

QUESTION: 662

Electrons are particles that are _____.

- A. positively charged
- B. neutral
- C. able to move freely
- D. negatively charged

Answer(s): D

QUESTION: 663

The chamber of the heart that pumps blood to the lungs is called the _____.

- A. right ventricle
- B. left ventricle
- C. right atrium
- D. left atrium

Answer(s): A

QUESTION: 664

The atomic number of an atom is determined by _____.

- A. the size of its nucleus
- B. the number of protons
- C. the number of electrons
- D. its location in the periodic table

Answer(s): B

QUESTION: 665

The smallest part of an element that still acts like an element is _____.

- A. the nucleus
- B. a compound
- C. the element itself
- D. the atom

Answer(s): D

QUESTION: 666

How many planets in the solar system have rings?

- A. one
- B. two
- C. three
- D. four

Answer(s): D

QUESTION: 667

The temperature at which a solid becomes a liquid is its _____.

- A. melting point
- B. boiling point
- C. freezing point
- D. concentration point

Answer(s): A

QUESTION: 668

The surface of the earth is called the _____.

- A. mantle
- B. core
- C. shawl
- D. crust

Answer(s): D

QUESTION: 669

Not counting the sun, the closest star to the Earth is _____.

- A. Rigel
- B. Proxima Centauri
- C. Antares
- D. Betel

Answer(s): B

QUESTION: 670

The ovaries produce _____.

- A. androgen
- B. estrogen
- C. adrenaline
- D. growth hormone

Answer(s): B

QUESTION: 671

How many times would an earthquake that is measured 4 on the Richter Scale be stronger than an earthquake that is measured 2?

- A. 2 times stronger
- B. 4 times stronger
- C. 10 times stronger
- D. 100 times stronger

Answer(s): D

QUESTION: 672

Muscles attach to bone with _____.

- A. connective tissue
- B. ligaments

- C. tendons
- D. rubber bands

Answer(s): C

QUESTION: 673

Atoms are most tightly packed in _____.

- A. solids
- B. gas
- C. liquids
- D. all of the above

Answer(s): A

QUESTION: 674

Blood leaving the lungs is _____.

- A. hydrogenated
- B. coagulated
- C. watery
- D. oxygenated

Answer(s): D

QUESTION: 675

The longest river in the world is _____.

- A. Amazon
- B. Nile
- C. Colorado
- D. Congo

Answer(s): B

QUESTION: 676

Mitosis is a process of _____.

- A. creating energy
- B. sexual reproduction
- C. destroying mass
- D. asexual reproduction

Answer(s): D

QUESTION: 677

When added to a cloud's name, which of the following means rain?

- A. Nimbus
- B. Cirrus
- C. Strato
- D. Alto

Answer(s): A

QUESTION: 678

The movement of particles from an area of high concentration to an area of lower concentration is called _____.

- A. metabolic action
- B. diffusion
- C. photosynthesis
- D. catalysis

Answer(s): B

QUESTION: 679

The instrument used to measure wind speed is _____.

- A. barometer
- B. anemometer
- C. altimeter
- D. fanometer

Answer(s): B

QUESTION: 680

Electric charges can be _____.

- A. positive or negative
- B. positive or neutral
- C. negative or neutral
- D. neutral only

Answer(s): A

QUESTION: 681

Which planet in the solar system has the most moons?

- A. Neptune
- B. Venus
- C. Saturn
- D. Uranus

Answer(s): C

QUESTION: 682

Hydrogen has the atomic number 1.

This means that hydrogen has _____.

- A. one electron
- B. one nucleus
- C. one proton
- D. one neutron

Answer(s): C

QUESTION: 683

Which U.S. space program is responsible for putting 12 men on the moon?

- A. Gemini
- B. Titan
- C. Voyager
- D. Apollo

Answer(s): D

QUESTION: 684

Organisms that eat other organisms are called _____.

- A. prey
- B. plants
- C. predators
- D. insects

Answer(s): C

QUESTION: 685

Unlike most other fish, sharks have no _____.

- A. gills
- B. bones
- C. liver
- D. heart

Answer(s): B

QUESTION: 686

What human organ is responsible for cleaning the blood?

- A. liver
- B. kidneys
- C. intestines
- D. stomach

Answer(s): B

QUESTION: 687

Kinetic energy is the energy that _____.

- A. is produced by sound waves
- B. an object potentially has
- C. is possessed by a moving object
- D. results from the attraction of two magnets

Answer(s): C

QUESTION: 688

In what year did the first woman travel in space?

- A. 1963
- B. 1969
- C. 1974
- D. 1985

Answer(s): A

QUESTION: 689

A step-up transformer _____.

- A. increases the voltage in a power line
- B. decreases the voltage in a power line
- C. doesn't affect the voltage in a power line
- D. measures the voltage in a power line

Answer(s): A

QUESTION: 690

What animal has the heaviest brain?

- A. human
- B. elephant
- C. rhinoceros
- D. sperm whale

Answer(s): D

QUESTION: 691

Passing sunlight through a prism _____.

- A. separates the colors into a spectrum
- B. reflects the light rays
- C. creates incident rays
- D. determines the focal point of a ray

Answer(s): A

QUESTION: 692

Molecules are created when _____.

- A. matter is created
- B. matter is destroyed
- C. atoms combine together
- D. atoms are separated

Answer(s): C

QUESTION: 693

The first genetically engineered organism was _____.

- A. sheep
- B. tobacco
- C. rats
- D. wheat

Answer(s): B

QUESTION: 694

The vernal equinox is _____.

- A. the first day of winter
- B. near the equator
- C. the first day of spring
- D. a lunar eclipse

Answer(s): C

QUESTION: 695

The moon completes a revolution around the earth approximately every _____.

- A. 28 days
- B. 365 days

- C. 24 hours
- D. 7 days

Answer(s): A

QUESTION: 696

Carcinogens are chemicals that cause _____.

- A. high blood pressure
- B. gene mutations
- C. blood clots
- D. diabetes

Answer(s): B

QUESTION: 697

A paramecium is _____.

- A. a one-celled organism
- B. algae
- C. bacteria
- D. a many-celled organism

Answer(s): A

QUESTION: 698

What large halogen is essential for the function of the thyroid gland?

- A. potassium chloride (salt)
- B. hemoglobin
- C. calcium
- D. iodine

Answer(s): D

QUESTION: 699

The brainstem controls _____.

- A. vision
- B. most muscle movements
- C. your sense of balance
- D. some involuntary activities

Answer(s): B

QUESTION: 700

What element is the most abundant one in the atmosphere?

- A. oxygen
- B. nitrogen
- C. helium
- D. hydrogen

Answer(s): B

QUESTION: 701

Minerals are necessary for _____.

- A. respiration
- B. eliminating waste
- C. preventing night blindness
- D. metabolic function

Answer(s): D

QUESTION: 702

What's the only metallic element commonly found as a liquid?

- A. bromine
- B. tellurium
- C. mercury
- D. silver

Answer(s): C

QUESTION: 703

Which of the following isn't a type of telescope?

- A. Newtonian
- B. Copernican
- C. Gregorian
- D. Hershelian

Answer(s): B

QUESTION: 704

A dekagram _____.

- A. is larger than a kilogram
- B. is smaller than a kilogram
- C. is the same as a kilogram
- D. doesn't exist

Answer(s): B

QUESTION: 705

The Aurora Borealis can be seen only in the _____.

- A. winter
- B. summer
- C. Southern Hemisphere
- D. Northern Hemisphere

Answer(s): D

QUESTION: 706

The three important properties of sound waves are _____.

- A. wavelength, speed, and crest
- B. speed, frequency, and reflection
- C. wavelength, frequency, and vibration
- D. wavelength, frequency, and speed

Answer(s): D

QUESTION: 707

Most of the asteroids in the solar system can be found between two planets: _____.

- A. Mars and Jupiter
- B. Saturn and Jupiter
- C. Earth and Mars
- D. Mercury and Venus

Answer(s): A

QUESTION: 708

At room temperature, an element is a _____.

- A. gas
- B. liquid or gas
- C. gas or solid
- D. liquid, gas, or solid

Answer(s): D

QUESTION: 709

The elements hydrogen and helium comprise what percentage of almost all matter in the Universe?

- A. 75 percent
- B. 82 percent
- C. 90 percent
- D. 98 percent

Answer(s): D

QUESTION: 710

Compounds are created when _____.

- A. atoms of two or more like elements are combined
- B. atoms of two or more different elements are combined
- C. two or more compounds are combined
- D. a compound decomposes

Answer(s): B

QUESTION: 711

What theory suggests the Universe will come to an end when its ever-increasing rate of expansion causes all matter to fly apart?

- A. The Big Rip
- B. The Big Bang
- C. The Big Crunch
- D. The Big Easy

Answer(s): A

QUESTION: 712

A watt measures _____.

- A. how much electricity is consumed
- B. the number of electrons moving past a specific point
- C. resistance
- D. voltage

Answer(s): A

QUESTION: 713

Which of the following planets, known as gas giants have no rings?

- A. Neptune
- B. Jupiter
- C. Uranus
- D. they all have rings

Answer(s): D

QUESTION: 714

Gas particles move _____.

- A. more slowly than liquid particles
- B. more slowly than solid particles
- C. more quickly than liquid particles
- D. at the same rate as all other particles

Answer(s): C

QUESTION: 715

Absolute zero is _____.

- A. 0 degrees Fahrenheit
- B. 0 degrees Celsius
- C. -263 degrees Celsius
- D. -32 degrees Fahrenheit

Answer(s): C

QUESTION: 716

What does the term forensic mean?

- A. pertaining to law
- B. pertaining to death
- C. pertaining to medicine
- D. pertaining to life

Answer(s): A

QUESTION: 717

Which of the following statements is not true?

- A. The human female chin is usually more rounded or pointed than the human male chin.
- B. The human female pelvis is usually deeper than the human male pelvis.
- C. The human male skull is usually larger than the human female skull.
- D. The human male skull has a larger brow ridge than the human female skull.

Answer(s): B

QUESTION: 718

A lunar eclipse occurs when _____.

- A. the earth moves into the moon's shadow
- B. the sun blocks the moon from view

- C. the earth moves into the sun's shadow
- D. the moon moves into the earth's shadow

Answer(s): D

QUESTION: 719

What chemical can be used to detect blood, even if it's been wiped from a surface?

- A. luminol
- B. cyanide
- C. ninhydrin
- D. alcohol

Answer(s): A

QUESTION: 720

The function of an enzyme is to _____.

- A. provide energy for a chemical reaction
- B. speed up a chemical reaction
- C. become hydrolyzed during a chemical reaction
- D. serve as an inorganic catalyst

Answer(s): B

Explanation:

Enzymes are organic (protein) catalysts that affect the speed of chemical reactions without being used up themselves.

QUESTION: 721

The DNA of a cell is found mainly in its _____.

- A. membrane
- B. cytoplasm
- C. vacuoles
- D. chromosomes

Answer(s): D

Explanation:

Chromosomes are composed of DNA and protein and are usually found in the nucleus.

QUESTION: 722

Most cells without a cell wall also lack _____.

- A. mitochondria
- B. chloroplasts

- C. cell membranes
- D. vacuoles

Answer(s): B

Explanation:

The cell wall, mainly composed of cellulose, is an important characteristic of plant cells. Chloroplasts contain energy-accumulating pigments and are found only in plant cells.

QUESTION: 723

The cells that can easily be scraped from the lining of the mouth make up _____.

- A. connective tissue
- B. epithelial tissue
- C. supporting tissue
- D. voluntary tissue

Answer(s): B

Explanation:

Epithelial cells form the skin and inner lining for organs.

QUESTION: 724

The _____ of an isotope is the time in which it takes half of an initial amount of an isotope to decay into something else.

- A. decay period
- B. half-life
- C. covalence coefficient
- D. midlife decay point

Answer(s): B

Explanation:

The half-life of an isotope is the time in which it takes half of an initial amount of an isotope to decay into something else.

QUESTION: 725

The end products of protein digestion consists of _____.

- A. glucose
- B. fatty acids
- C. glycerol
- D. amino acids

Answer(s): D

Explanation:

Proteins are synthesized from a variety of amino acid "building blocks." Digestion breaks down the complex protein structures.

QUESTION: 726

Blood circulation from the lungs goes directly to the _____.

- A. legs
- B. head
- C. heart
- D. digestive system

Answer(s): C

Explanation:

The pulmonary circulation returns oxygenated blood from the lungs through the pulmonary veins to the left atrium of the heart.

QUESTION: 727

The glucose content of the blood is regulated by _____.

- A. insulin and adrenaline
- B. nerve stimulation
- C. ACTH
- D. progesterone

Answer(s): A

Explanation:

The relative concentration of glucose is maintained by the combined action of insulin from the pancreas and adrenaline from the adrenal medulla.

QUESTION: 728

Erosion and depletion are problems associated with _____.

- A. blood circulation
- B. baldness
- C. soil conservation
- D. cardiovascular exercise

Answer(s): C

Explanation:

Soil is eroded by the action of wind, water, and ice and/or is depleted by the removal of organic matter or minerals.

QUESTION: 729

The beneficial relationship between termites and the protozoa that inhabit their digestive

systems is called _____.

- A. symbiosis
- B. parthenogenesis
- C. mitosis
- D. phagocytosis

Answer(s): A

Explanation:

Symbiosis is a nutritive relationship in which organisms of different species live together and benefit each other.

QUESTION: 730

A viral infection in humans that has a very long time between the initial exposure and the exhibiting of symptoms is _____.

- A. AIDS
- B. measles
- C. pneumonia
- D. chicken pox

Answer(s): A

Explanation:

A person exposed to the AIDS virus will take up to 6 months to show antibodies in the blood serum and 2 to 5 years to show externally observable symptoms.

QUESTION: 731

Which substance can be used to treat drinking water of questionable purity?

- A. ammonia
- B. chlorine
- C. hydrogen chloride
- D. iodine

Answer(s): D

Explanation:

Iodine crystals in a small bottle of water make a saturated aqueous solution. A few milliliters of this solution added to a quart of water will destroy most organisms in an hour.

QUESTION: 732

An Anemometer measures _____.

- A. Temperature
- B. Humidity
- C. Wind Speed

D. Air Pressure

Answer(s): C

Explanation:

An anemometer measures wind speed.

QUESTION: 733

The reaction between sodium, metal, and water can be classified as _____.

- A. decomposition
- B. synthesis
- C. single replacement
- D. double replacement

Answer(s): C

QUESTION: 734

Nitrates are very useful as _____.

- A. detergents
- B. fertilizers
- C. fuels
- D. solvents

Answer(s): B

Explanation:

The nitrogen in the soil must be replaced by nitrogen-fixing bacteria in the roots of legumes or by artificial fertilizer containing nitrates.

QUESTION: 735

A gaseous product of the fermentation of sugar is _____.

- A. carbon monoxide
- B. carbon dioxide
- C. sulfur dioxide
- D. oxygen

Answer(s): B

Explanation:

The fermentation of sugar by yeast produces ethyl alcohol and carbon dioxide.

QUESTION: 736

The half-life of iodine-131 is 8 days.

Approximately how long would it take for 1 milligram of 1-131 to decay to 1/2 milligram of 1-

131?

- A. 4 days
- B. 8 days
- C. 2 weeks
- D. 1 month

Answer(s): D

Explanation:

If the amount of 1-131 is reduced by one-half every 8 days: 8 mg (8 days), 4 mg (8 days), 2 mg (8 days), 1/2 mg. It would take 32 days (approximately 1 month).

QUESTION: 737

A magnetic compass points in the direction of the _____.

- A. true north
- B. geographic North Pole
- C. magnetic North Pole
- D. local magnetic field

Answer(s): D

Explanation:

A magnetic compass is influenced by all magnetic sources in its vicinity and aligns itself according to the net magnetic field at that location.

QUESTION: 738

The mass of 1 cubic meter of water is approximately _____.

- A. 1 kilogram
- B. 10 kilograms
- C. 100 kilograms
- D. 1,000 kilograms

Answer(s): D

Explanation:

One meter equals 100 centimeters. Therefore, 1 cubic meter equals 1,000,000 cubic centimeters. Since each cubic centimeter has a mass of 1 gram, the approximate total mass is 1,000 kilograms.

QUESTION: 739

A 1-kilogram mass is dropped from a height of 1 meter. Its kinetic energy just before it hits the ground is _____.

- A. 1 joule

- B. 10 joules
- C. 100 joules
- D. 1,000 joules

Answer(s): B

Explanation:

Gravitational potential energy is converted to kinetic energy. The kinetic energy is equal to the mass of the object times the force of gravity times the height. Thus, $mgh = 1 \text{ kilogram} \times 9.8 \text{ meters/second} \times 1 \text{ meter} = (\text{approximately}) 10 \text{ joules}$.

QUESTION: 740

A magnifying glass is what kind of lens used on objects how far away?

- A. convex lens used closer than one focal length
- B. concave lens used closer than one focal length
- C. convex lens used beyond one focal length
- D. concave lens used beyond one focal length

Answer(s): A

Explanation:

An object held closer than one focal length from a convex lens will produce a virtual magnified image.

QUESTION: 741

Frictional forces usually do NOT _____.

- A. oppose motion
- B. decrease kinetic energy
- C. increase potential energy
- D. produce wear

Answer(s): C

Explanation:

The friction force will tend to oppose motion, slow the object down, and produce wear between contacting surfaces.

QUESTION: 742

A 120-volt electrical power supply produces 1/2 ampere to the load. The power delivered is _____.

- A. 60 watts
- B. 60 ohms
- C. 240 watts
- D. 240 ohms

Answer(s): A

Explanation:

Power = voltage × current (in watts) = 120 volts × 1/2 amp = 60 watts.

QUESTION: 743

The category of weather system with the highest velocity winds is the _____.

- A. thunderstorm
- B. tornado
- C. hurricane
- D. sea breeze

Answer(s): B

Explanation:

Wind velocities of tornados have been measured in excess of 250 miles per hour.

QUESTION: 744

The member of the solar system with the shortest period of solar revolution is a _____.

- A. planetary satellite
- B. asteroid
- C. planet
- D. comet

Answer(s): C

Explanation:

The planet Mercury orbits the sun in 88 days.

QUESTION: 745

The ocean current that extends from Florida across the Atlantic Ocean to Europe is _____.

- A. El Nino
- B. the Japan Current
- C. the Gulf Stream
- D. the Southwest countercurrent

Answer(s): C

QUESTION: 746

The temperature of the atmosphere is less during the nighttime hours because _____.

- A. the side of the earth not feeling the sun loses heat by convection
- B. the side of the earth feeling the sun loses heat by evaporation

- C. the side of the earth not feeling the sun loses heat by radiation
- D. people set their thermostats lower when they go to bed

Answer(s): C

Explanation:

The temperature of the atmosphere falls during the nighttime hours because the side of the earth not facing the sun loses heat by radiation.

QUESTION: 747

The basic unit of heat is the calorie, which is the amount of heat needed to _____.

- A. raise the temperature of one ounce of water one degree Fahrenheit
- B. raise the temperature of one gram of water one degree Celsius
- C. change the temperature of one kilogram of water one degree Kelvin
- D. lower the temperature of one quart of water one degree Celsius

Answer(s): B

Explanation:

The basic unit of heat is the calorie, which is the amount of heat needed to raise the temperature of one gram of water one degree Celsius.

QUESTION: 748

When atoms and molecules pass thermal energy by direct contact, it is known as _____.

- A. conduction
- B. convection
- C. insulation
- D. swapping paint

Answer(s): A

Explanation:

When atoms and molecules pass thermal energy by direct contact, it is known as conduction. Convection is a current in a fluid (such as the atmosphere or ocean) caused by uneven distribution of heat. Insulation is a substance or property that tends to keep heat (or lack of it) in a certain place, rather than passing it on. Swapping paint is what race cars do when they bump or scrape against each other during a race.

QUESTION: 749

The speed of light in a vacuum is about _____.

- A. 186,000 miles per hour
- B. 186,000 kilometers per hour
- C. 186,000 miles per second
- D. 186,000 feet per second squared

Answer(s): C

Explanation:

The speed of light in a vacuum is about 186,000 miles per second.

QUESTION: 750

When a plant performs photo synthesis, it produces _____ as a byproduct.

- A. carbon dioxide
- B. oxygen
- C. water
- D. methane

Answer(s): B

Explanation:

When a plant performs photo synthesis, it produces oxygen as a byproduct.

QUESTION: 751

A person's appearance resulting from his or her genetic makeup is called his or her _____.

- A. genotype
- B. individual apparent mutation
- C. phenotype
- D. source of heredity

Answer(s): C

Explanation:

A person's appearance resulting from his or her genetic makeup is called their phenotype. A person's (or organism's) genotype is their genetic makeup itself (i.e., the combination of alleles that determines a what that person or organism looks like).

QUESTION: 752

Which is the correct and complete sequence of objects in the solar system moving outward from the sun?

- A. Mercury, Venus, Earth, asteroid belt. Mars, Jupiter
- B. Mars, asteroid belt, Jupiter, Saturn, Neptune
- C. Mars, asteroid belt, Jupiter, Saturn, Uranus
- D. Mars, asteroid belt, Saturn, Jupiter, Neptune

Answer(s): C

Explanation:

Mars, asteroid belt, Jupiter, Saturn, Uranus. The order of the major inhabitants of the Solar

System is (from the Sun outward): Mercury, Venus, Earth, Mars, the asteroid belt, Jupiter, Saturn, Uranus, Neptune, and Pluto. The International Astronomical Union reclassified Pluto as a dwarf planet in 2006. However, in 2008 Pluto was again reclassified as a plutoid, a new class of objects that currently has four members (the others being Eris, Haumea, and Makemake).

QUESTION: 753

What is the type of blood vessels called that carries blood from the other parts of the body to the heart?

- A. Capillaries
- B. Veins
- C. Arteries
- D. Aortas

Answer(s): B

Explanation:

Veins. Capillaries are the tiny blood vessels that form an intricate network throughout the body for the interchange of substances such as oxygen and carbon dioxide between blood and tissue cells. Arteries are the blood vessels that carry blood from the heart to the various parts of the body, and the aorta is a part of the heart itself.

QUESTION: 754

Spiders are _____.

- A. marsupials
- B. insects
- C. arachnids
- D. annelids

Answer(s): C

Explanation:

Spiders are arachnids, part of a large class of arthropods having four pairs of walking legs and no wings, unlike insects (which have three pairs of walking legs and a pair of wings) and annelids (which are a type of segmented worm with no jointed legs). Marsupials are a type of non-placental mammal whose young are born very underdeveloped and then continue developing outside their mother's body attached to one of her nipples, usually in a pouch; examples are kangaroos, koalas, and possums.

QUESTION: 755

A substance that speeds up chemical reactions is a(n) _____.

- A. accelerator
- B. adipose
- C. compound
- D. catalyst

Answer(s): D

Explanation:

A substance that speeds up chemical reactions is a catalyst.

QUESTION: 756

Which of the following is not usually a gas at room temperature?

- A. Oxygen
- B. Fluorine
- C. Sulfur
- D. Argon

Answer(s): C

Explanation:

Which of the following is not usually a gas at room temperature? Sulfur is normally a soft, bright-yellow solid at room temperature. The others are gaseous at room temperature.

QUESTION: 757

What are the major layers of Earth's atmosphere in order from the Earth's surface upward?

- A. Troposphere, stratosphere, mesosphere, thermosphere, exo sphere
- B. Troposphere, stratosphere, ionosphere, exo sphere
- C. Troposphere, tropopause, ionosphere, stratosphere, exo sphere
- D. Meso sphere, stratosphere, troposphere, tropopause

Answer(s): A

Explanation:

The five major layers of the atmosphere from the Earth's surface upward are the troposphere, stratosphere, mesosphere, thermosphere, and exosphere.

The troposphere starts at the Earth's surface and extends to between 23,000 ft (7 km) at the poles and 60,000 ft (17 km) at the equator, with some variations due to weather factors.

The stratosphere extends from the top of the troposphere (23,000-60,000 ft or 7-17 km) to about 160,000 ft (60 km).

The stratosphere contains the ozone layer, which is mainly located in the lower part of the stratosphere from about 50,000-115,000 ft (15-35 km) above the surface, although the thickness varies somewhat across both the seasons and the Earth's geographical regions.

The mesosphere extends from about 160,000 ft (50 km) to about 265,000-285,000 ft (50-54 miles or 80-85 km).

This is also where most meteors burn up when entering the atmosphere

QUESTION: 758

What are chloroplasts?

- A. Structures found in plant cells that assist in photo synthesis.
- B. Organelles found in plant cells that conduct photo synthesis.

- C. Substances found in primitive cyanobacteria.
- D. Substances found in green plants only after they blossom.

Answer(s): B

Explanation:

Chloroplasts are organelles found, in plant cells that conduct photosynthesis.

QUESTION: 759

What is the scientific study of human culture called?

- A. Sociology
- B. Archaeology
- C. Anthropology
- D. Scientology

Answer(s): C

Explanation:

The scientific study of human culture is called anthropology, and includes study of the origin and behavior of humankind as well as its physical, social, and cultural development. Sociology is the study of human society, including patterns of social relations, social interaction, and culture. Archaeology is the study of past human life and culture by the recovery or discovery and examination of material evidence such as graves, buildings, tools, weapons, and other artifacts.

QUESTION: 760

Humans have ____ pairs of chromosomes.

- A. 13
- B. 23
- C. 46
- D. 47

Answer(s): B

Explanation:

Humans have 23 pairs of chromosomes, or 46 chromosomes in all.

QUESTION: 761

The average distance of the Sun from the Earth is about _____.

- A. 240,000 miles
- B. 10 million miles
- C. 93 million miles
- D. 240 million miles

Answer(s): C

Explanation:

The average distance of the Sun from the Earth is about 93 million miles. The Earth's moon is about 240,000 miles from the Earth.

QUESTION: 762

What is the smallest part of matter that retains the chemical properties characteristic of a particular element?

- A. Isotope
- B. Molecule
- C. Electron
- D. Atom

Answer(s): D

Explanation:

The smallest part of matter that retains the chemical properties characteristic of a particular element is an atom. Molecules are made up of two or more atoms, and electrons are particles within the atom. An isotope is one of two or more atoms of the same element that have the same atomic number (i.e., the same number of protons) but a different number of neutrons.

QUESTION: 763

A gamete is a cell that _____.

- A. fuses with a zygote during fertilization
- B. provides nutritional support to a developing fetus
- C. fuses with another gamete during fertilization
- D. fuses with a spermatid before fertilization

Answer(s): C

Explanation:

A gamete is a cell that fuses with another gamete during fertilization.

QUESTION: 764

The nucleus of an atom consists of _____.

- A. protons, which have a positive electrical charge, and neutrons, which have no charge.
- B. electrons, which have a positive electrical charge, and neutrons, which have no charge.
- C. protons, which have no electrical charge, and neutrons, which have a positive charge.
- D. protons (positive charge), electrons (negative charge), and neutrons (no charge).

Answer(s): A

Explanation:

The nucleus of an atom consists of protons, which have a positive electrical charge, and neutrons, which have no charge - Electrons, which have a negative charge, circle the nucleus in

the orbital shell.

QUESTION: 765

Antibodies _____.

- A. identify and neutralize foreign objects inside the body.
- B. bond with antigens to keep the body's hormones balanced.
- C. prevent infection by blocking access to the circulatory system.
- D. are produced by the endocrine system.

Answer(s): A

Explanation:

Antibodies identify and neutralize foreign objects inside the body. Humans and other vertebrates have an immune system that uses antibodies to fight bacteria, viruses, toxins, and other foreign objects. Antibodies are complex, Y-shaped protein molecules that develop from plasma cells, which in turn develop from "B lymphocytes" produced by the bone marrow. The plasma cells can generate a very large array of different antibodies, each of which can combine with and destroy an antigen (foreign molecule).

Antibodies react to very specific characteristics of different antigens, binding them to the top ends of their Y- shaped form. Once the antibody and antigen combine, the antibodies deactivate the antigen or lead it to macrophages (a type of white blood cell) that ingest and destroy it.

High counts of a particular antibody may persist for months after an invasion before eventually diminishing.

However, the B cells can quickly manufacture more of the same antibody if exposure to that particular antigen reoccurs.

Vaccines work by "training" B cells to recognize and react quickly to potential disease molecules.

QUESTION: 766

A lens surface that is _____ bulges outward from the lens, while a lens surface that is _____ is depressed into the lens.

- A. planar, convex
- B. concave, convex
- C. convex, concave
- D. orthogonal, planar

Answer(s): C

Explanation:

A lens surface that is convex bulges outward from the lens, while a lens surface that is concave is depressed into the lens.

QUESTION: 767

Electrons have a _____ charge.

- A. positive

- B. negative
- C. neutral
- D. variable

Answer(s): B

Explanation:

Electrons have a negative charge.

QUESTION: 768

The four forces that act on an aircraft in flight are _____.

- A. lift, thrust, friction, and inertia
- B. lift, gravity, thrust, and drag
- C. thrust, vector, speed, and fluid density
- D. lift, downdraft, thrust, and vector

Answer(s): B

Explanation:

The four forces that act on an aircraft in flight are lift, gravity, thrust, and drag.

QUESTION: 769

The particle with a _____ charge in the nucleus of an atom is a(n) _____.

- A. positive, electron
- B. positive, proton
- C. negative, neutron
- D. neutral, positron

Answer(s): B

Explanation:

The particle with a positive charge in the nucleus of an atom is a proton. The nucleus of an atom contains protons, (which have a positive electrical charge) and neutrons, (which have no electrical charge).

Electrons, which have a negative charge, circle the nucleus in what is known as the orbital shell.

QUESTION: 770

What is the scientific notation for 832,000?

- A. 8.32×10^{-2}
- B. 8.32×10^2
- C. 8.32×10^{-5}
- D. 8.32×10^5

Answer(s): D

Explanation:

To express a number in scientific notation, you move the decimal as many places as necessary until there is only one digit to the left of the decimal. For 832,000, you move the decimal to the left by 5 decimal places.

QUESTION: 771

_____ is the reason it is harder to stop a baseball flying at 100 MPH than it is to stop a baseball flying at 50 MPH.

- A. Velocity
- B. Momentum
- C. Mass
- D. Volume

Answer(s): B

Explanation:

Momentum equals mass (amount of matter in an object) times velocity (speed in a given direction).

QUESTION: 772

Which of the following ecosystems could be described as housing thousands of species and has tall trees that form a ceiling from the the Sun above?

- A. a deciduous forest
- B. a tropical rain forest
- C. a tundra
- D. a taiga

Answer(s): B

Explanation:

Tropical rain forests are one of the most important areas on Earth. These special ecosystems are home to thousands of species animals and plants.

Rain forests are not only densely packed with plants, but are also full of tall trees that form a ceiling from the Sun above. This ceiling keeps smaller plants from growing. Areas where sunlight can reach the surface are full of interesting plants.

QUESTION: 773

Via which of the following unoxygenated blood travels to the heart?

- A. arteries
- B. veins
- C. capillaries
- D. aorta

Answer(s): B

Explanation:

Veins carry blood devoid of oxygen to the right atrium of the heart. Arteries, capillaries, and the aorta carry oxygenated blood from the heart to blood cells throughout the body.

QUESTION: 774

In vertebrates, which substance gives red blood cells their color?

- A. carotene
- B. dopamine
- C. melanin
- D. hemoglobin

Answer(s): D

Explanation:

Hemoglobin, used by all vertebrates and some invertebrates to transport oxygen and carbon dioxide in the body, gives red blood cells their color.

QUESTION: 775

How many different elements are present in glucose, $C_6H_{12}O_6$?

- A. 1
- B. 3
- C. 5
- D. 7

Answer(s): B

Explanation:

$C_6H_{12}O_6$ contains 3 elements: carbon, hydrogen, and oxygen.

QUESTION: 776

On the pH scale, any substance below what number is considered acid?

- A. 0
- B. 5
- C. 7
- D. 15

Answer(s): C

Explanation:

The pH scale ranges from 0 to 14, with 0 being acidic and 14 being alkaline (or basic). A pH of 7 is considered neutral. Anything below 7, then, is acidic.

QUESTION: 777

Which of the following structures has DNA?

- A. a double helix
- B. a helix
- C. a parabola
- D. a hexagon

Answer(s): A

Explanation:

The Watson-Crick model of DNA is a double-stranded twisted ladder, or double helix.

QUESTION: 778

Which of the following plants lacks a vascular system?

- A. fern
- B. peanut plant
- C. liverworts
- D. oak tree

Answer(s): C

Explanation:

Liverworts are bryophytes, which are characterized by their lack of a vascular system.

QUESTION: 779

Which of the following symbols represents one of the elements present in NaCl?

- A. C
- B. Na
- C. NH
- D. N

Answer(s): B

Explanation:

The elements that make up NaCl are sodium (Na) and chlorine (Cl).

C is the symbol for carbon, which is not present at all.

N is the symbol for nitrogen.

NH is a compound including more than one element.

QUESTION: 780

Heparin is an anticoagulant.

What is the primary purpose of drugs like Heparin?

- A. to cause nausea
- B. to prevent the heart from stopping
- C. to cause drowsiness
- D. to prevent the blood from clotting

Answer(s): D

Explanation:

An anticoagulant is a substance that stops blood from clotting.

QUESTION: 781

Which of the following is an organic compound?

- A. salt (NaCl)
- B. glucose (C₆H₁₂O₆)
- C. potassium (K)
- D. water (H₂O)

Answer(s): B

Explanation:

All organic compounds contain carbon (C).

Potassium (K) is only a single element, and it does not contain carbon.

QUESTION: 782

What is the latitude of any point on earth's equator?

- A. 0
- B. 90
- C. 180
- D. 270

Answer(s): A

Explanation:

All points on the equator have a latitude of 0 degrees. The North Pole has a latitude of 90 degrees North. The South Pole has a latitude of 90 degrees South.

QUESTION: 783

What does an anemometer measure?

- A. temperature
- B. wind speed
- C. humidity
- D. air pressure

Answer(s): B

Explanation:

An anemometer is a device for measuring wind speed. Humidity is measured by a hygrometer. Atmospheric or air pressure is measured by a barometer. Temperature is measured by a thermometer.

QUESTION: 784

Ten thousand centimeters equals _____ kilometers.

- A. 0.001
- B. 0.01
- C. 0.1
- D. 1

Answer(s): C

Explanation:

One hundred centimeters equals one meter. 1,000 meters equal one kilometer, so ten thousand centimeters equals one-tenth of a kilometer or 0.1 kilometers.

QUESTION: 785

At which temperature would water freeze if recorded on a Fahrenheit thermometer?

- A. 32 degrees
- B. 15 degrees
- C. 0 degrees
- D. 21 degrees

Answer(s): A

QUESTION: 786

What blood type is considered the universal recipient?

- A. B
- B. O
- C. AB
- D. A

Answer(s): C

QUESTION: 787

During its circulation throughout the body, blood from the left ventricle then goes to which of these locations?

- A. left atrium
- B. right atrium
- C. aorta
- D. pulmonary artery

Answer(s): C

QUESTION: 788

Which of these substances has the lowest pH?

- A. baking soda
- B. lemon juice
- C. water
- D. bleach

Answer(s): B

QUESTION: 789

Which color of visible light has the lowest frequency?

- A. yellow
- B. red
- C. indigo
- D. violet

Answer(s): B

QUESTION: 790

The energy possessed by a moving object is called _____ energy.

- A. acceleration
- B. potential
- C. kinetic
- D. true

Answer(s): C

QUESTION: 791

Which part of the atom helps determine its atomic number?

- A. neutron
- B. electron
- C. anion
- D. proton

Answer(s): D

QUESTION: 792

Which part of the brain is responsible for involuntary actions such as breathing and heart beat?

- A. cerebrum
- B. medulla
- C. pons
- D. cerebellum

Answer(s): B

QUESTION: 793

Of these options, which is the most specific category of biological classification?

- A. genus
- B. phylum
- C. domain
- D. species

Answer(s): D

QUESTION: 794

Which of these is not a unit of length?

- A. cubic centimeter
- B. millimeter
- C. kilometer
- D. meter

Answer(s): A

QUESTION: 795

Which gas primarily makes up the Earth's atmosphere?

- A. carbon dioxide
- B. oxygen
- C. argon
- D. nitrogen

Answer(s): D

QUESTION: 796

Which element has no neutrons?

- A. nitrogen
- B. hydrogen
- C. carbon
- D. helium

Answer(s): B

QUESTION: 797

How many chromosomes are found in a human sperm cell?

- A. 21
- B. 23
- C. 2
- D. 46

Answer(s): B

QUESTION: 798

Shale, gypsum, coal, and sandstone are examples of which type of rock?

- A. gneiss
- B. metamorphic
- C. sedimentary
- D. igneous

Answer(s): C

QUESTION: 799

What structure connects muscle to bone?

- A. tendons
- B. joints
- C. cartilage
- D. ligaments

Answer(s): A

QUESTION: 800

Which of these has the most influence on oceanic tides?

- A. the gravitational pull of the Sun on the Earth
- B. the gravitational pull of the Moon on the Earth
- C. the rotational turn of the Earth on its axis
- D. the shifting of tectonic plates

Answer(s): B

QUESTION: 801

Earth is made of several layers.
Which makes up most of Earth's volume?

- A. the outer core
- B. the inner core
- C. the mantle
- D. the crust

Answer(s): C

QUESTION: 802

Which of these is the best conductor of heat?

- A. plastic
- B. styrofoam
- C. wood
- D. metal

Answer(s): D

QUESTION: 803

Which of these statements is false about the periodic table?

- A. The rows of elements are called periods.
- B. The columns of elements are called a group.
- C. Elements are arranged in order of increasing atomic number.
- D. The far left column contains a group called the noble gases.

Answer(s): D

QUESTION: 804

What is the pH of water?

- A. 1
- B. 0
- C. 6
- D. 7

Answer(s): D

QUESTION: 805

Which of these would be a normal resting heart rate of an adult?

- A. 84 beats per minute
- B. 50 beats per minute
- C. 118 beats per minute
- D. 42 beats per minute

Answer(s): A

QUESTION: 806

Which blood cells are responsible for delivering oxygen to the body?

- A. white blood cells
- B. red blood cells

- C. platelets
- D. thrombocytes

Answer(s): B

QUESTION: 807

Simple eukaryotes, such as protozoa, would be categorized as _____.

- A. Protista
- B. Fungi
- C. Plante
- D. Monera

Answer(s): A

QUESTION: 808

Gene expression in physical characteristics such as eye or hair color is referred to as _____.

- A. allele
- B. genotype
- C. phenotype
- D. homozygous allele

Answer(s): C

QUESTION: 809

Which of these best defines the term "biosphere"?

- A. the part of Earth where life occurs on land
- B. the part of Earth where life occurs on land and in the water
- C. the part of Earth where life occurs, which includes land, water, and air
- D. a collection of populations in a habitat

Answer(s): C

QUESTION: 810

An animal that eats both plants and meat is called _____.

- A. a piscivore
- B. an omnivore
- C. an herbivore
- D. a carnivore

Answer(s): B

QUESTION: 811

Skin is considered part of the _____ system.

- A. endocrine
- B. integumentary
- C. muscular
- D. skeletal

Answer(s): B

QUESTION: 812

What is the closest definition of "osmosis"?

- A. the combination of a substance with oxygen
- B. the diffusion of a fluid across a semi-permeable membrane
- C. a type of cell division that results in four daughter cells
- D. the process by which a cell separates the chromosomes in its nucleus

Answer(s): B

QUESTION: 813

Which planet is closest to the sun?

- A. Venus
- B. Mercury
- C. Mars
- D. Neptune

Answer(s): B

QUESTION: 814

What cell structure is not shared by both animal and plant cells?

- A. ribosome
- B. cell wall
- C. mitochondria
- D. nucleus

Answer(s): B

QUESTION: 815

Cold air is _____ than warm air.

- A. less dense
- B. generally faster
- C. more dense
- D. lower-pressured

Answer(s): C

QUESTION: 816

Given that iron has 26 protons, 26 electrons, 30 neutrons, and an atomic weight of 55.845 amu, what is its atomic number?

- A. 52
- B. 30
- C. 55.845
- D. 26

Answer(s): D

QUESTION: 817

What number is referred to as "absolute zero"?

- A. 0 degrees Farenheit
- B. 0 Kelvin
- C. -100 degrees Celsius
- D. 0 degrees Celsius

Answer(s): B

QUESTION: 818

The study of the environment and organisms' interaction within it is called _____?

- A. botany
- B. entomology
- C. ecology
- D. geology

Answer(s): C

QUESTION: 819

A tendon connects which two structures?

- A. ligament and bone
- B. muscle and bone
- C. muscle and muscle
- D. bone and bone

Answer(s): B

QUESTION: 820

If a trait can only be passed from father to son, it can be assumed the trait is linked to what

chromosome?

- A. dominant chromosome
- B. recessive chromosome
- C. Y chromosome
- D. X chromosome

Answer(s): C

QUESTION: 821

Which is considered the earliest stage of human development?

- A. blastocyst
- B. fetus
- C. zygote
- D. embryo

Answer(s): C

QUESTION: 822

When exactly two atoms share electrons, they are said to have which type of bond?

- A. metallic
- B. polar
- C. covalent
- D. ionic

Answer(s): C

QUESTION: 823

"Sublimation" involves _____.

- A. the change from a liquid to a gas
- B. the change from a gas to a liquid
- C. the change from a liquid to a solid
- D. the change from a solid to a gas

Answer(s): D

QUESTION: 824

Which of these is not an exocrine gland?

- A. pancreas
- B. salivary glands
- C. adrenal glands
- D. mammary glands

Answer(s): C

QUESTION: 825

Which of these would be considered a primary consumer?

- A. rabbit
- B. human
- C. aloe plant
- D. snake

Answer(s): A

QUESTION: 826

Which scale is used to measure the intensity of an earthquake?

- A. the Likert scale
- B. the Kinsey scale
- C. the Richter scale
- D. the magnitude scale

Answer(s): C

QUESTION: 827

Given that a kilogram is 2.2 pounds, what is the approximate weight of a person weighing 137 pounds expressed in kilograms?

- A. 93 kg
- B. 74 kg
- C. 301 kg
- D. 62 kg

Answer(s): D

QUESTION: 828

According to Newton's first law of motion, an object in motion tends to _____.

- A. stop eventually
- B. decelerate quickly
- C. stay in motion
- D. move forever

Answer(s): C

QUESTION: 829

According to the electromagnetic spectrum, which is the highest frequency (shortest wavelength) wave known to exist?

- A. gamma rays
- B. infrared waves
- C. ultraviolet light waves
- D. radio waves

Answer(s): A

QUESTION: 830

The heat we get from the sun is what type of heat transfer?

- A. radiation
- B. convection
- C. conduction
- D. direct

Answer(s): A

QUESTION: 831

Which type of clouds are the lowest and responsible when fog occurs?

- A. trophic
- B. stratus
- C. cumulus
- D. cirrus

Answer(s): B

QUESTION: 832

Most scientists agree the Earth is approximately how old?

- A. 2.4 million years old
- B. 4.6 billion years old
- C. 5.8 million years old
- D. 10.4 billion years old

Answer(s): B

QUESTION: 833

The compound NaCl is more commonly known as _____.

- A. gold
- B. baking soda
- C. table salt
- D. aspirin

Answer(s): C

QUESTION: 834

Which organ is responsible for regulating blood sugar levels by secreting insulin and glucagon?

- A. gall bladder
- B. spleen
- C. liver
- D. pancreas

Answer(s): D

QUESTION: 835

The flat, colored, ring-shaped membrane of the eye is called _____.

- A. lens
- B. cornea
- C. iris
- D. pupil

Answer(s): C

Explanation:

The iris is a flat, colored, ring-shaped membrane that sits behind the cornea.
A. It has an adjustable circular opening in the center called the pupil.

QUESTION: 836

Which of the following is not one of the 6 basic animal groups?

- A. Birds
- B. Mammals
- C. Humans
- D. Invertebrates

Answer(s): C

Explanation:

The six basic animal groups are birds, mammals, invertebrates, fishes, reptiles, and amphibians.

QUESTION: 837

Which of the following cloud types occur at the lowest altitude?

- A. Cirrus
- B. Altocumulus
- C. Cirrocumulus
- D. Stratus

Answer(s): D

Explanation:

Stratus clouds occur below 6,000 feet.

QUESTION: 838

How long does it take the earth to rotate 90 degrees?

- A. 6 hours
- B. 4 hours
- C. 12 hours
- D. 8 hours

Answer(s): A

Explanation:

The earth rotates 360 degrees in 24 hours, therefore it rotates 90 degrees in 6 hours.

QUESTION: 839

Which of the planets has the longest revolutionary period around the sun?

- A. Mercury
- B. Neptune
- C. Earth
- D. Saturn

Answer(s): B

Explanation:

Neptune is farther away than the other planets and has the longest period of revolution (164.81 earth years).

QUESTION: 840

Which of the following is a positively charged subatomic particle?

- A. Electron
- B. Proton
- C. Neutron
- D. Nucleus

Answer(s): B

Explanation:

A Proton is a stable subatomic particle that occurs in all atomic nuclei. It has a positive electric charge.

QUESTION: 841

A Celsius temperature of 100 degrees is equivalent to a Fahrenheit temperature of _____.

- A. 100 degrees
- B. 32 degrees
- C. 0 degrees
- D. 212 degrees

Answer(s): D

Explanation:

Celsius temperature of 100 degrees = 212 degree Fahrenheit

QUESTION: 842

The process of water molecules escaping the Earth's surface and entering the atmosphere is known as _____.

- A. Condensation
- B. Precipitation
- C. Evaporation
- D. Collection

Answer(s): C

Explanation:

Evaporation is when the sun heats up water and turns it into vapor/stream which then goes into the air.

QUESTION: 843

The majority of Earth's air is composed of _____.

- A. Nitrogen
- B. Oxygen
- C. Carbon
- D. Dioxide

Answer(s): A

Explanation:

Earth's air is approximately 80% nitrogen.

QUESTION: 844

About how long does it take the Earth to complete an orbit of the sun?

- A. 1 day
- B. 1 year
- C. 1 month
- D. 1 season

Answer(s): B

Explanation:

It takes the earth 1 year to complete one trip around the sun.

QUESTION: 845

If two forces act on an object in opposite directions, the force that results is _____.

- A. the difference between the two forces
- B. the sum of the two forces
- C. the product of the two forces
- D. the ratio of the two forces

Answer(s): A

Explanation:

When the two forces are in opposite directions, the resulting force is the difference between the two forces.

QUESTION: 846

Of which two parts does the formal binomial nomenclature (or scientific name) of a living organism consist?

- A. Family and Species
- B. Class and Family
- C. Order and Class
- D. Genus and Species

Answer(s): D

Explanation:

An organism's scientific name is comprised of its Genus and Species.

QUESTION: 847

Which of the following is NOT a fruit?

- A. Corn
- B. Tomatoes
- C. Peppers
- D. Carrots

Answer(s): D

Explanation:

Fruits have seeds.

Carrots do not have seeds and are considered a vegetable.

Corn is scientifically a fruit since it is the seed of the corn plant.

QUESTION: 848

Watts are a unit of measurement for _____.

- A. Power
- B. Pressure
- C. Energy
- D. Speed

Answer(s): A

Explanation:

Power is measured by Watts.

QUESTION: 849

On a Celsius thermometer, how many degrees are there between the boiling point of water and the freezing point of water?

- A. 212 degrees
- B. 100 degrees
- C. 180 degrees
- D. 32 degrees

Answer(s): B

Explanation:

In Celsius, the boiling point of water is 100 degrees and the freezing point of water is 0 degrees.

QUESTION: 850

Force can be represented by a vector.
A vector represents _____.

- A. Speed
- B. Direction
- C. Magnitude and Direction
- D. Magnitude

Answer(s): C

Explanation:

A force represented by a vector consists of magnitude and direction.

QUESTION: 851

Ascorbic acid is also known as _____.

- A. Vitamin B12
- B. Vitamin E

- C. Vitamin C
- D. Vitamin A

Answer(s): C

Explanation:

Vitamin C is also known as ascorbic acid.

QUESTION: 852

A pot sitting on a hot burner is an example of _____.

- A. Convection
- B. Conduction
- C. Insulation
- D. Radiation

Answer(s): B

Explanation:

Conduction is where heat is directly transmitted from a hot object to a cooler object.

QUESTION: 853

The product of force and distance is _____.

- A. Momentum
- B. Power
- C. Velocity
- D. Work

Answer(s): D

Explanation:

Force $\times$ Distance = Work

Section 4: Mathematical Knowledge

QUESTION: 854

If $x = 8$, what's the value of y in the equation:

$$y = (x^2 \div 4) - 2?$$

- A. 14
- B. 16
- C. 18
- D. 20

Answer(s): A

Explanation:

$$y = (x^2 \div 4) - 2$$

$$y = (8^2 \div 4) - 2$$

$$y = (64 \div 4) - 2$$

$$y = 16 - 2 = 14$$

QUESTION: 855

The cube of 5 is _____.

- A. 125
- B. 25
- C. 15
- D. 50

Answer(s): A

Explanation:

$$\text{The cube of } 5 = 5 \times 5 \times 5 = 125.$$

QUESTION: 856

$$2.5 \times 3^3 = \underline{\hspace{2cm}}.$$

- A. 22.5
- B. 75.0
- C. 67.5
- D. 675.0

Answer(s): C

Explanation:

$$2.5 \times 3^3 = 2.5 (3 \times 3 \times 3) = 2.5 \times 27 = 67.50.$$

QUESTION: 857

The fourth root of 16 is _____.

- A. 4
- B. 1
- C. 3
- D. 2

Answer(s): D

Explanation:

$2^4 = 16$; the fourth root of 16 is 2.

QUESTION: 858

What's the equation of a line that passes through points (0, -1) and (2, 3)?

- A. $y = 2x - 1$
- B. $y = 2x + 1$
- C. $x = 2y - 1$
- D. $x = 2y + 1$

Answer(s): A

Explanation:

The slope of the line is equal to the change in y values divided by the change in x values.

The change in y values is $4(3 - -1)$.

The change in x values is $2(2 - 0)$. $4/2 = 2$.

To find the intercept, substitute 0 for x in the equation $y = 2x + b - 1 = 2(0) + b$.

Therefore, $b = -1$, so the equation is $y = 2x - 1$.

QUESTION: 859

$(12 \text{ yards} + 14 \text{ feet}) \div 5 = \underline{\hspace{2cm}}$.

- A. 12 feet
- B. $5\frac{1}{5}$ feet
- C. 10 feet
- D. $2\frac{1}{2}$ yards

Answer(s): C

Explanation:

Convert 12 yards and 14 feet to feet: $(12 \text{ yards} \times 3 \text{ feet per yard}) + 14 \text{ feet} = 36 \text{ feet} + 14 \text{ feet} = 50 \text{ feet}$. Divide by 5 as instructed: $50 \text{ feet} \div 5 = 10 \text{ feet}$.

QUESTION: 860

$x^3 \times x^4 = \underline{\hspace{2cm}}$.

- A. x^{12}
- B. $2x^7$
- C. $2x^{12}$

D. x^7

Answer(s): D

Explanation:

If two powers have the same base, they can be multiplied by keeping the base and adding the powers together.

QUESTION: 861

$(x + 4)(x + 2) = \underline{\hspace{2cm}}$.

- A. $x^2 + 6x + 6$
- B. $x^2 + 8x + 8$
- C. $x^2 + 8x + 6$
- D. $x^2 + 6x + 8$

Answer(s): D

Explanation:

Multiply the first variable in the first set of parentheses with the first variable in the second set of parentheses ($x \times x = x^2$).

Next, multiply the first variable in the first set of parentheses with the second number in the second set of parentheses ($x \times 2 = 2x$). So far, the results are $x^2 + 2x$.

Now, multiply the second number in the first set of parentheses to the first variable in the second set of parentheses ($4 \times x = 4x$).

Next, multiply the second variable in the first set of parentheses to the second number in the second set of parentheses ($4 \times 2 = 8$).

The solution is $x^2 + 2x + 4x + 8$.

Combining the like terms results in $x^2 + 6x + 8$.

QUESTION: 862

$1.5 \times 10^3 = \underline{\hspace{2cm}}$.

- A. 45
- B. 150
- C. 1,500
- D. 15

Answer(s): C

Explanation:

$1.5 \times 10^3 = 1.5 \times (10 \times 10 \times 10) = 1.5 \times 1,000 = 1,500$.

QUESTION: 863

Which of the following is a prime number?

- A. 27
- B. 11

- C. 8
- D. 4

Answer(s): B

Explanation:

A prime number is a number that can be divided evenly by itself or by one, but not by any other number.

Choices 27, 8 and 4 can all be divided evenly by other numbers.

QUESTION: 864

What's the mode of the following series of numbers?

4, 4, 8, 8, 8, 10, 10, 12, 12

- A. 9
- B. 8
- C. 11
- D. 10

Answer(s): B

Explanation:

The mode of a series of numbers is the number that appears in the series the most frequently.

In this case, it's 8.

QUESTION: 865

If $a = 4$, then $a^3 \div a =$ _____.

- A. 4
- B. 12
- C. 64
- D. 16

Answer(s): D

Explanation:

$$(4 \times 4 \times 4) \div 4 = 64 \div 4 = 16$$

QUESTION: 866

Solve for the factorial of 5 (5!).

- A. 25
- B. 125
- C. 120
- D. 15

Answer(s): C

Explanation:

The factorial (!) of a number is the number multiplied by the next smallest whole number, then by the next smallest whole number, and so on (down to 1). $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

QUESTION: 867

$(900 \times 2) \div 6 = \underline{\hspace{2cm}}$.

- A. 30
- B. 300
- C. 150
- D. 3,000

Answer(s): B

Explanation:

$(900 \times 2) \div 6 = 1,800 \div 6 = 300$.

QUESTION: 868

If $x = 2$, then $xx \times x = \underline{\hspace{2cm}}$.

- A. 8
- B. $2xx$
- C. 4
- D. 6

Answer(s): A

Explanation:

Substitute 2 for all x's in the problem. $22 \times 2 = 4 \times 2 = 8$.

QUESTION: 869

If $(5 + 1)(6 \div 3)(8 - 5) = (3 + 3)x$, then $x = \underline{\hspace{2cm}}$.

- A. 12
- B. 3
- C. 4
- D. 6

Answer(s): D

Explanation:

Solve the first half of the equation. $(6)(2)(3) = 36$. Therefore, $36 = (3 + 3)x$, which turns into $36 = 6x$. Isolate x.

$$36 \div 6 = 6x \div 6$$

$$6 = x$$

To check your answer, substitute 6 for x.

QUESTION: 870

$49 \times 64 = \underline{\hspace{2cm}}$.

- A. 56
- B. 15
- C. 42
- D. 3,136

Answer(s): A

Explanation:

The square root of 49 is 7; the square root of 64 is 8.
 $7 \times 8 = 56$.

QUESTION: 871

Which of the following fractions is the largest?

- A. $\frac{2}{5}$
- B. $\frac{3}{8}$
- C. $\frac{7}{10}$
- D. $\frac{13}{16}$

Answer(s): D

Explanation:

Find a common denominator for the fractions. In this case, 80 work for all the fractions. Convert all the fractions: $\frac{2}{5} = \frac{32}{80}$; $\frac{3}{8} = \frac{30}{80}$; $\frac{7}{10} = \frac{56}{80}$; and $\frac{13}{16} = \frac{65}{80}$. Comparing the fractions, you can see that $\frac{13}{16}$ ($\frac{65}{80}$) is the largest fraction.

QUESTION: 872

If $2 + x = 4$, what is the value of x ?

- A. 6
- B. 2
- C. 4
- D. $\frac{1}{2}$

Answer(s): B

Explanation:

Solve as you would solve for any unknown. $2 + x = 4 \Rightarrow 2 + x - 2 = 4 - 2$. Or, $x = 2$. To check your answer, substitute 2 for x .
 $2 + 2 = 4$. True, so the answer is correct.

QUESTION: 873

If a circle has a radius of 12 feet, what's its circumference most nearly?

- A. 24 feet

- B. 72 feet
- C. 75 feet
- D. 36 feet

Answer(s): C

Explanation:

Circumference equals π $\times$ diameter, and diameter is equal to two times the radius. Or $C = \pi \times d$, and $d = 2r$. $C = \pi \times 24$.

If you round π to 3.14, the answer is about 75.36 or about 75 feet.

QUESTION: 874

An aquarium measures 16-inches long $\times$ 8-inches deep $\times$ 18-inches high.

What's its volume?

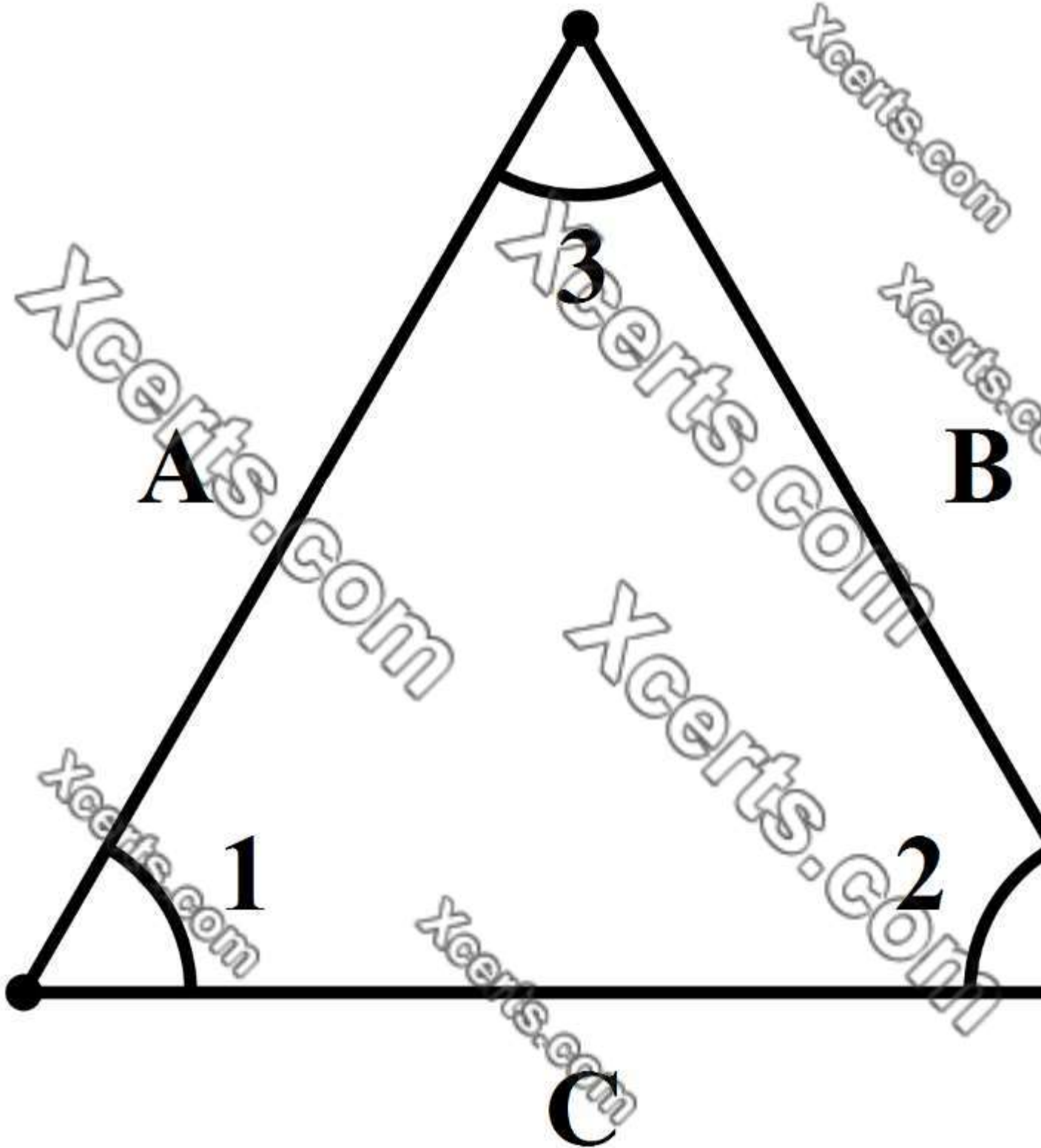
- A. 2,304 cubic inches
- B. 128 cubic inches
- C. 42 cubic inches
- D. 288 cubic inches

Answer(s): A

Explanation:

Volume equals length $\times$ width $\times$ height ($V = lwh$). $16 \times 8 \times 18 = 2,304$ cubic inches.

QUESTION: 875



Triangle ABC (shown above) is a(n) _____.

- A. right triangle
- B. obtuse triangle

- C. equilateral triangle
- D. isosceles triangle

Answer(s): C

Explanation:

In an equilateral triangle, all sides are equal, and all angles are equal.

QUESTION: 876

The sum of the measures of the angles of a trapezoid is _____.

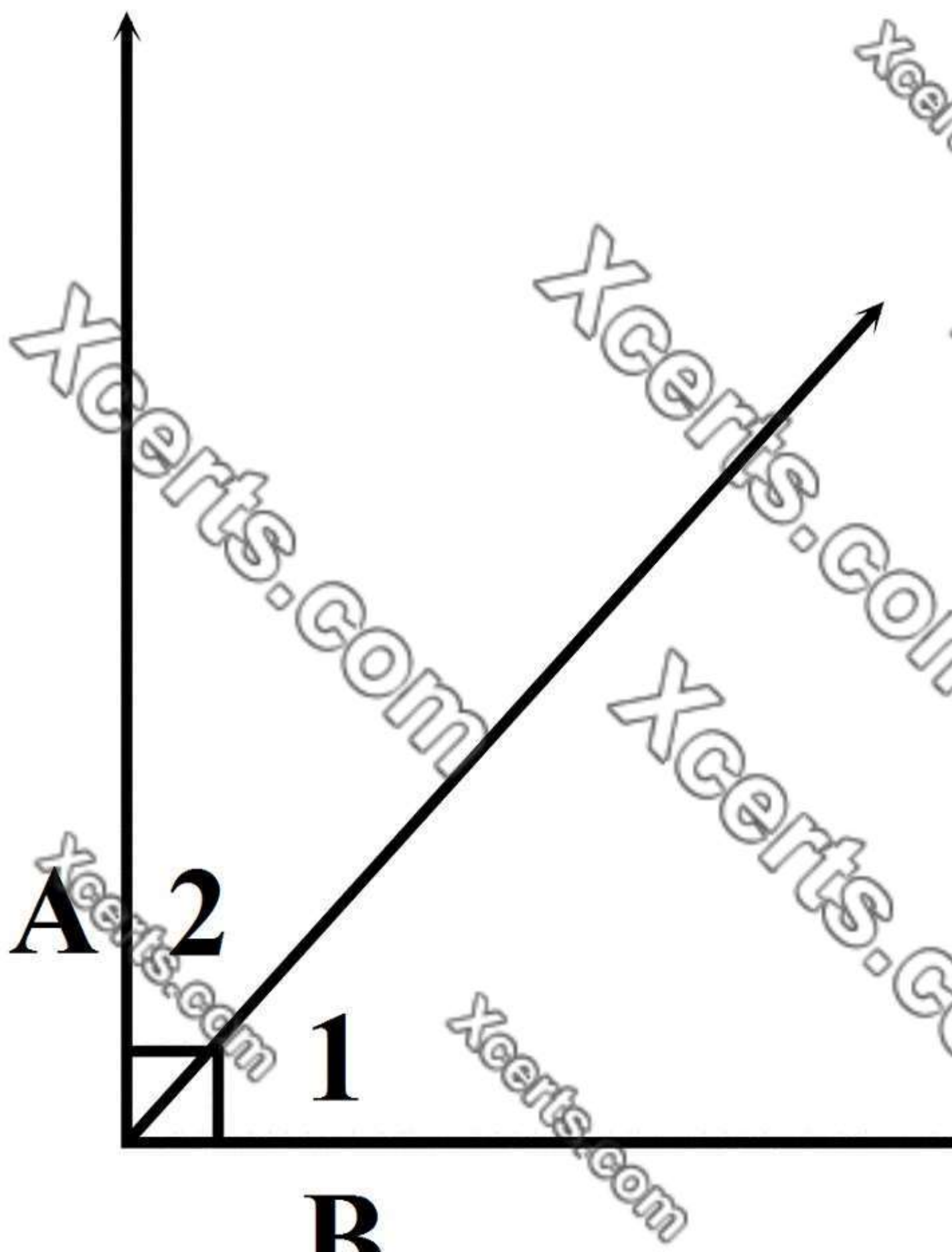
- A. 360 degrees
- B. 540 degrees
- C. 180 degrees
- D. 720 degrees

Answer(s): A

Explanation:

All quadrilaterals have angles that total 360 degrees.

QUESTION: 877



In the Angle AB (shown above), Angles 1 and 2 are _____.

- A. supplementary
- B. complimentary
- C. both obtuse
- D. both right angles

Answer(s): B

Explanation:

If the sum of two angles equals 90 degrees, they're called complementary angles.

QUESTION: 878

Convert 24% to a fraction.

- A. 6/25
- B. 1/25
- C. 6/24
- D. 1/24

Answer(s): A

Explanation:

24% = 24/100. This fraction can be further reduced to 6/25.

QUESTION: 879

$x^2 \times x^4 =$ _____.

- A. x^6
- B. x^8
- C. $2x^6$
- D. $2x^8$

Answer(s): A

Explanation:

If two exponents have the same base, you can multiply them by keeping the base and adding the exponents together.

QUESTION: 880

If a rectangle has a perimeter of 36 feet, and it's 4-feet wide, what's its area?

- A. 56 square feet
- B. 128 square feet
- C. 112 square feet
- D. 16 square feet

Answer(s): A

Explanation:

To find area, multiply length times width ($A = lw$). To determine the length, subtract two times the width from the perimeter: $36 - 2(4) = 36 - 8 = 28$.

Then divide the remainder by 2 to determine the length of one side: $28 \div 2 = 14$.

Then multiply length times width to determine the area: $14 \times 4 = 56$.

QUESTION: 881

The cube root of 64 is _____.

- A. 3
- B. 9
- C. 2
- D. 4

Answer(s): D

Explanation:

The cube of 4 = $4 \times 4 \times 4 = 64$, so 4 is the cube root of 64.

QUESTION: 882

Convert 314,000 to scientific notation.

- A. 3.14×10^5
- B. 3.14×10^{-5}
- C. 314×10
- D. 31.4×100

Answer(s): A

Explanation:

To convert this number to scientific notation, move the decimal point to the left until it's to the immediate right of the first number, while counting the number of moves. In this case, you move it 5 places.

The result is then multiplied by 10, raised to the power of the number of places the decimal point was moved.

QUESTION: 883

The reciprocal of $1/6$ is _____.

- A. 1
- B. 3
- C. 6
- D. $1/3$

Answer(s): C

Explanation:

A reciprocal is the number by which a number can be multiplied to produce 1. So the reciprocal of $\frac{1}{6}$ is 6 because $\frac{1}{6} \times 6 = 1$.

QUESTION: 884

If $0.05 \div x = 1$, then $x =$ _____.

- A. 0.05
- B. 0.5
- C. 50.0
- D. 5.0

Answer(s): A

Explanation:

Multiply both sides of the equation by 0.05 to isolate x : $0.05 \div x \times 0.05 = 1 \times 0.05$ or $x = 0.05$. Check by substituting 0.05 for x in the original equation.

QUESTION: 885

Factor $x^2 - 6x + 9$.

- A. $(x + 6)(x + 6)$
- B. $(x - 6)(x + 6)$
- C. $(x - 3)^2$
- D. $(x + 3)^2$

Answer(s): C

Explanation:

$$x^2 - 6x + 9 = (x - 3)(x - 3) = (x - 3)^2$$

QUESTION: 886

$(3 \times 2)(7 - 2)(6 + 2) = (6 \times 4)x$.
What's the value of x ?

- A. -5
- B. 5
- C. 10
- D. 1

Answer(s): C

Explanation:

Solve the left side of the equation first. $(6)(5)(8) = 240$. Therefore, $240 = (6 \times 4)x$, which equals $240 = 24x$.

Now isolate x by dividing both sides of the equation by 24:

$$240 \div 24 = 24x \div 24 \text{ or } 10 = x.$$

Check your answer by substituting 10 for x in the original equation.

QUESTION: 887

Solve for x: $2x - 6 = x + 5$

- A. 3
- B. 11
- C. 7
- D. 5

Answer(s): B

Explanation:

Isolate x on one side of the equation:

$$2x - 6 = x + 5$$

$$2x - 6 - x = x + 5 - x$$

$$x - 6 = 5$$

$$x - 6 + 6 = 5 + 6$$

$$x = 11$$

Check by substituting 11 for x in the original equation.

QUESTION: 888

If $I = prt$, and $p = \$1,000$, $r = 7\%$, and $t = 1$, what does I equal?

- A. \$35
- B. \$1,000
- C. \$700
- D. \$70

Answer(s): D

Explanation:

Solve for I.

$$I = (1,000) (7\%) (1) \text{ or } I = (1,000) (0.07) (1) = 70.$$

QUESTION: 889

A circle has a radius of 5 inches.

What's its approximate area?

- A. 78.5 inches
- B. 70.0 inches
- C. 314.0 inches
- D. 25.0 inches

Answer(s): A

Explanation:

The area of a circle is $A = r^2$. $A = 52$. is approximately 3.14, so 3.14×25 means A is approximately 78.5 inches.

QUESTION: 890

If $F = M \times A$, then what does $A =$ _____.

- A. $F \times M$
- B. $F - M$
- C. F/M
- D. M/F

Answer(s): C

Explanation:

If you divide both sides of the formula by M , then $A = F/M$

QUESTION: 891

A tube has a radius of 3 inches and a height of 5 inches.
What's its approximate volume?

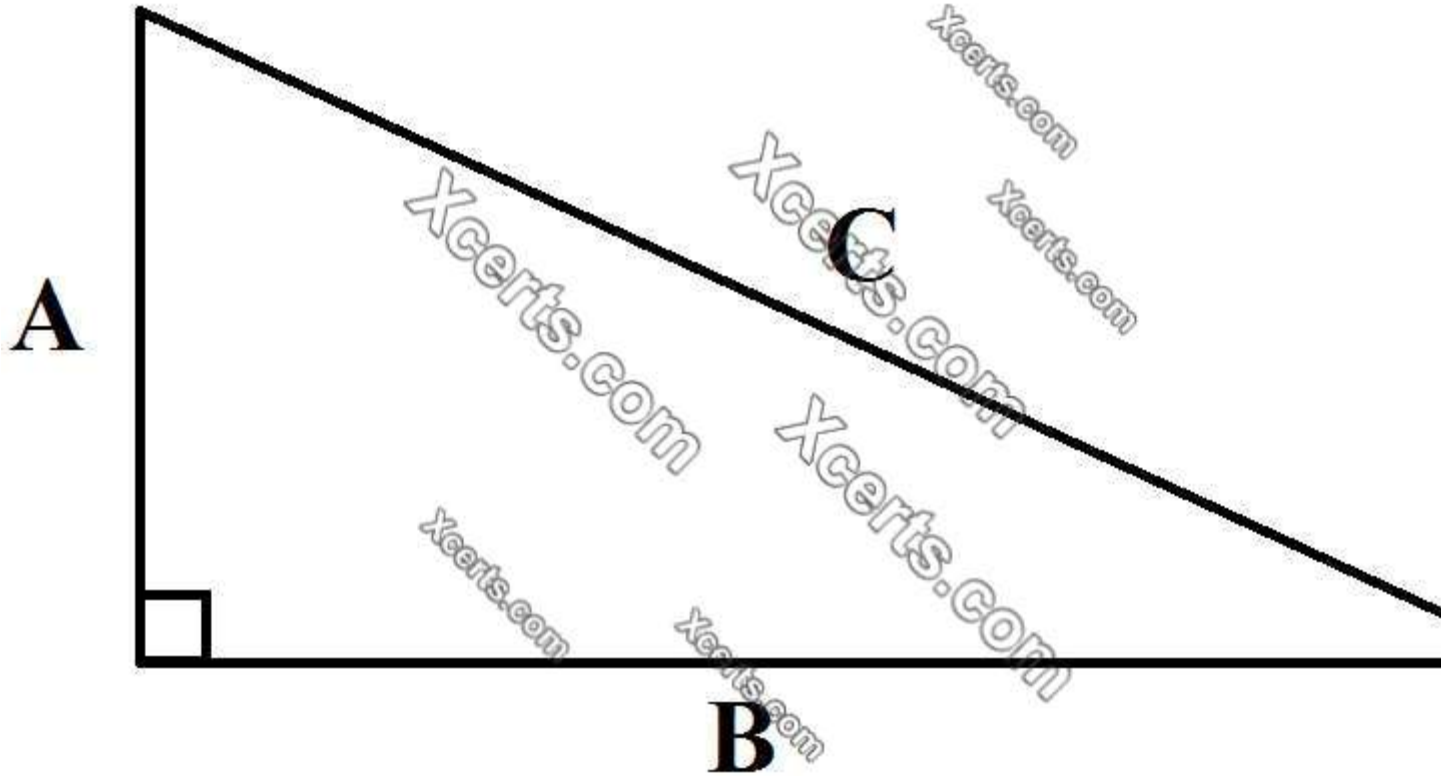
- A. 34 cubic inches
- B. 141 cubic inches
- C. 565 cubic inches
- D. 45 cubic inches

Answer(s): B

Explanation:

For cylinders, $\text{Volume} = r^2(h)$. In this problem, $V = (3^2)(5)$. Assume π is approximately 3.14.
 V is approximately equal to $(3.14)(9)(5)$ or 141 cubic inches.

QUESTION: 892



Triangle ABC (shown above) is a(n) _____.

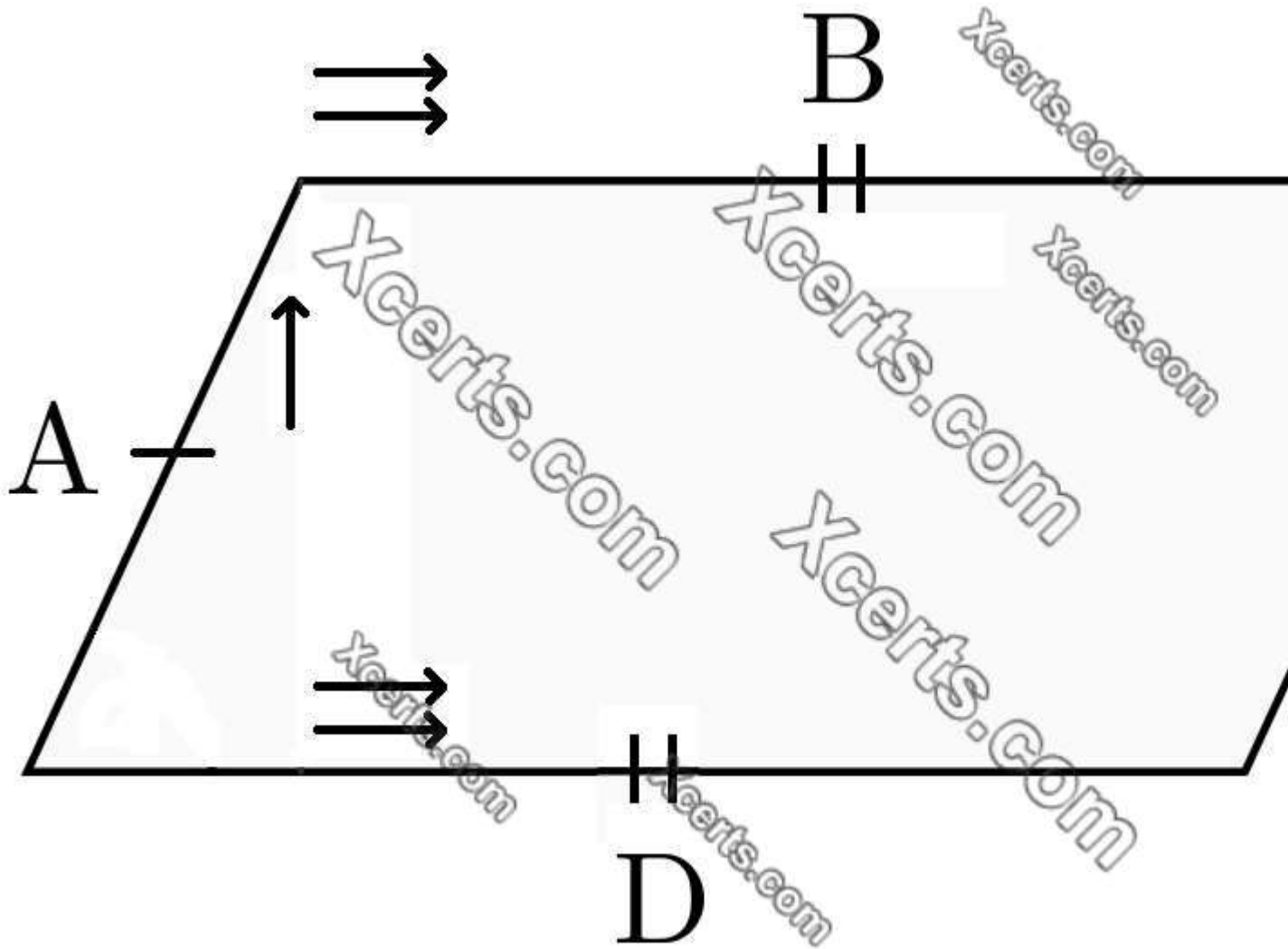
- A. right triangle
- B. equilateral triangle
- C. scalene triangle
- D. isosceles triangle

Answer(s): A

Explanation:

A right triangle has one right angle (one 90-degree angle).

QUESTION: 893



What type of quadrilateral is the figure above?

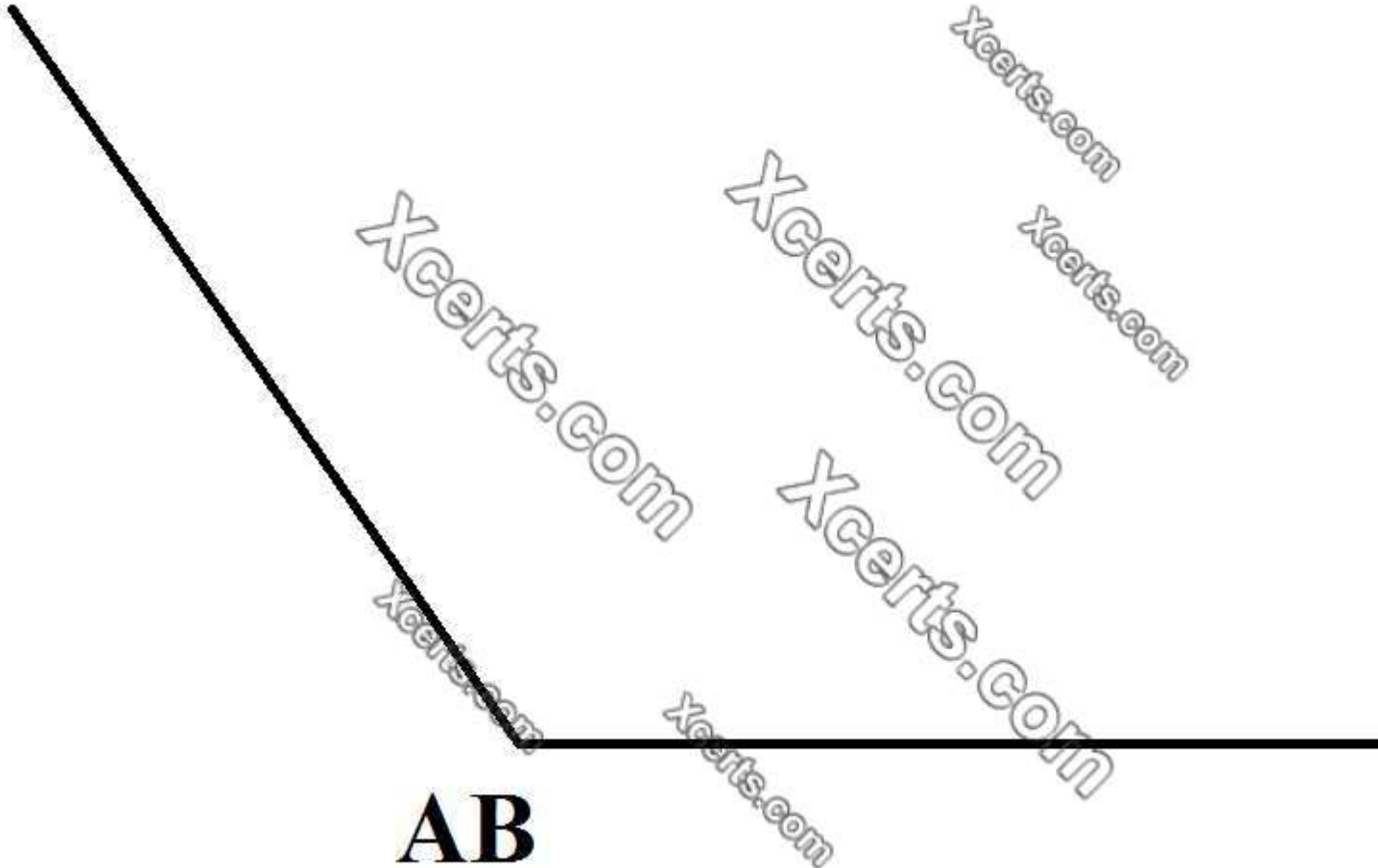
- A. square
- B. rhombus
- C. trapezoid
- D. parallelogram

Answer(s): D

Explanation:

Parallelograms have opposite sides of equal length.

QUESTION: 894



Angle AB (shown above) is a(n) _____.

- A. complementary angle
- B. supplementary angle
- C. acute angle
- D. obtuse angle

Answer(s): D

Explanation:

Angles measuring more than 90 degrees are obtuse angles.

QUESTION: 895

Solve for x: $-x^2 - x + 30 = 0$

- A. 4, -8
- B. -6, 5
- C. -4, 5
- D. 6, -3

Answer(s): B

Explanation:

This is a quadratic equation, which is solved by factoring.

$$-x^2 - x + 30 = 0$$

$$-(-x^2 - x + 30) = 0$$

$$x^2 + x - 30 = 0$$

$$(x + 6)(x - 5) = 0$$

$$x + 6 = 0 \text{ or } x - 5 = 0$$

$$x = -6 \text{ or } x = 5$$

QUESTION: 896

A square box has a volume of 64 cubic inches.

What's its perimeter?

- A. 8 inches
- B. 16 inches
- C. 64 inches
- D. 32 inches

Answer(s): B

Explanation:

Volume equals length times width times height ($V = lwh$). In this case, $V = 64$, so one side of the box is 4-inches long.

The cube root of 64 produces this number: $64 = 4 \times 4 \times 4$.

Find the perimeter by adding the four sides together: $4 + 4 + 4 + 4 = 16$.

QUESTION: 897

A cube has a volume of 64 cubic inches.

What's its surface area?

- A. 16 square inches
- B. 64 square inches
- C. 96 square inches
- D. 32 square inches

Answer(s): C

Explanation:

Volume is calculated by multiplying length times width times height ($V = lwh$).

Because the edges are equal on a cube, each edge is 4 inches.

The cube root of 64 produces this number: $64 = 4 \times 4 \times 4$.

The area of one side of the cube is $4 \times 4 = 16$, and because a cube has 6 sides, you multiply 16×6 to find the surface area of the cube, 96 inches.

QUESTION: 898

$$(x^2)^3 = \underline{\hspace{2cm}}.$$

- A. $3x^3$

- B. x^6
- C. x^9
- D. $2x^6$

Answer(s): B

Explanation:

$(x^2)^3$ is the same as $(x^2)(x^2)(x^2)$. Multiply exponents with the same base by keeping the base and adding the exponents: $(x^2)(x^2)(x^2) = x^6$.

QUESTION: 899

$4! =$ _____.

- A. 16
- B. 40
- C. 0
- D. 24

Answer(s): D

Explanation:

$4!$ (4 factorial) $= 4 \times 3 \times 2 \times 1 = 24$.

QUESTION: 900

If $a^3 + b^3 = a^3 + x^3$, then $b =$ _____.

- A. $b^3 - a^3$
- B. x
- C. $a^3 - b^3$
- D. a

Answer(s): B

Explanation:

To solve, subtract a^3 from both sides of the equation:

$$a^3 + b^3 - a^3 = a^3 + x^3 - a^3$$

$$b^3 = x^3$$

$$b = x$$

QUESTION: 901

What's the sum of the integers from 1 to 300?

- A. 38,243
- B. 45,150
- C. 49,923
- D. 52,024

Answer(s): B

Explanation:

The formula to find the sum of a finite arithmetic sequence is $S = n/2(a+b)$, where n is the number of terms, a is the first term in the sequence, and b is the last term in the sequence. In this case there are 300 terms (n), and the first term is 1 and the final term is 300.

$$S = n/2(a + b)$$

$$S = 300/2(1 + 300)$$

$$S = 150(301)$$

$$S = 45,150$$

QUESTION: 902

$$(y^2)^3 = \underline{\hspace{2cm}}.$$

- A. y^5
- B. y^6
- C. y^3
- D. $3y^2$

Answer(s): B

Explanation:

$(y^2)^3$ is the same as $(y^2)(y^2)(y^2)$. Multiply exponents with the same base by keeping the base and adding the exponents: $(y^2)(y^2)(y^2) = y^6$.

QUESTION: 903

$$\text{If } y = 6, \text{ then } 2y \times y = \underline{\hspace{2cm}}.$$

- A. 12
- B. 72
- C. 18
- D. 242

Answer(s): B

Explanation:

Substitute 6 for y the equation: $2(6) \times 6 = 12 \times 6 = 72$.

QUESTION: 904

$$\text{If } 0.05x = 1, \text{ then } x \text{ equals } \underline{\hspace{2cm}}.$$

- A. $1/20$
- B. 20
- C. 10
- D. 5

Answer(s): B

Explanation:

To check your answer, substitute 20 for x in the original equation.

QUESTION: 905

$$25x^2 = \underline{\hspace{2cm}}.$$

- A. x
- B. x^2
- C. $5x^2$
- D. $-5x^2$

Answer(s): C

Explanation:

$$25x^2 = (5x)^2 = 5x^2$$

QUESTION: 906

Factor $9x^3 + 18x^2 - x - 2$.

- A. $(9x^2 - 1)(x + 2)$
- B. $(9x^2 + 1)(x - 2)$
- C. $(9x^2 + 2)(x - 1)$
- D. $(9x^2 - 2)(x + 1)$

Answer(s): C

Explanation:

$$9x^3 + 18x^2 - x - 2 = 9x^2(x + 2) - 1(x + 2) = (9x^2 - 1)(x + 2)$$

QUESTION: 907

Solve for x: $5x + 7 = 6(x - 2) - 4(2x - 3)$

- A. 1
- B. -1
- C. 2
- D. -2

Answer(s): B

Explanation:

$$5x + 7 = 6(x - 2) - 4(2x - 3)$$

$$5x + 7 = 6x - 12 - 8x + 12$$

$$5x + 7 = -2x$$

$$7x + 7 = 0$$

$$7x = -7$$

$$7/7x = -7/7$$

$$x = -1$$

QUESTION: 908

$x \times x^2 = \underline{\hspace{2cm}}$.

- A. x^2
- B. $2x$
- C. $2x^2$
- D. x^3

Answer(s): D

Explanation:

If two powers have the same base, they can be multiplied by keeping the base and adding the exponents together. In this case, x is the same as x^1 .

QUESTION: 909

$(5 + x)^2 = \underline{\hspace{2cm}}$.

- A. $5 - x$
- B. $5 + x$
- C. $5 - x$
- D. $5 + x$

Answer(s): B

Explanation:

This is so easy that it may tempt you to think that the correct answer is too obvious. The square root of $(5 + x)^2$ is simply $5 + x$.

QUESTION: 910

$(3 \times 3)(5 - 3)(6 + 2) = x^2$.

What's the value of x ?

- A. 6
- B. 12
- C. 144
- D. 64

Answer(s): B

Explanation:

First solve the left side of the equation: $(9)(2)(8) = 144$. So $x^2 = 144$. Find the square root of each side: $x = 12$.

QUESTION: 911

If $-5x = 25$, x equals $\underline{\hspace{2cm}}$.

- A. -5
- B. 5

- C. 10
- D. 0

Answer(s): A

Explanation:

Isolate x by dividing each side of the equation by -5 :

$$-5x \div -5 = 25 \div -5$$

$$x = -5$$

QUESTION: 912

A circle measures 12 feet in diameter.
What's its area to the nearest foot?

- A. 452
- B. 24
- C. 113
- D. 48

Answer(s): C

Explanation:

The area of a circle equals π times the radius squared. The radius is $1/2$ the diameter.

$A = \pi r^2$. $A = 62 = 36$. If π is approximately 3.14, the area of the circle is approximately 3.14×36 or 113 feet.

QUESTION: 913

A square box has 6-inch sides.
What's its volume?

- A. 18 cubic inches
- B. 216 cubic inches
- C. 12 cubic inches
- D. 36 cubic inches

Answer(s): B

Explanation:

Volume equals length $\times$ width $\times$ height ($V = lwh$): $6 \times 6 \times 6 = 216$ cubic inches.

QUESTION: 914

A cylinder has a diameter of 10 inches.
What's its approximate area?

- A. (102)
- B. (25)
- C. (5)
- D. (102)(10)

Answer(s): B

Explanation:

The area of a cylinder is $A = r^2$. In this problem, $A = 52 = (25)$. Radius is half the diameter.

QUESTION: 915

A cylinder has a diameter of 12 inches and a height of 10 inches.
What's its approximate volume?

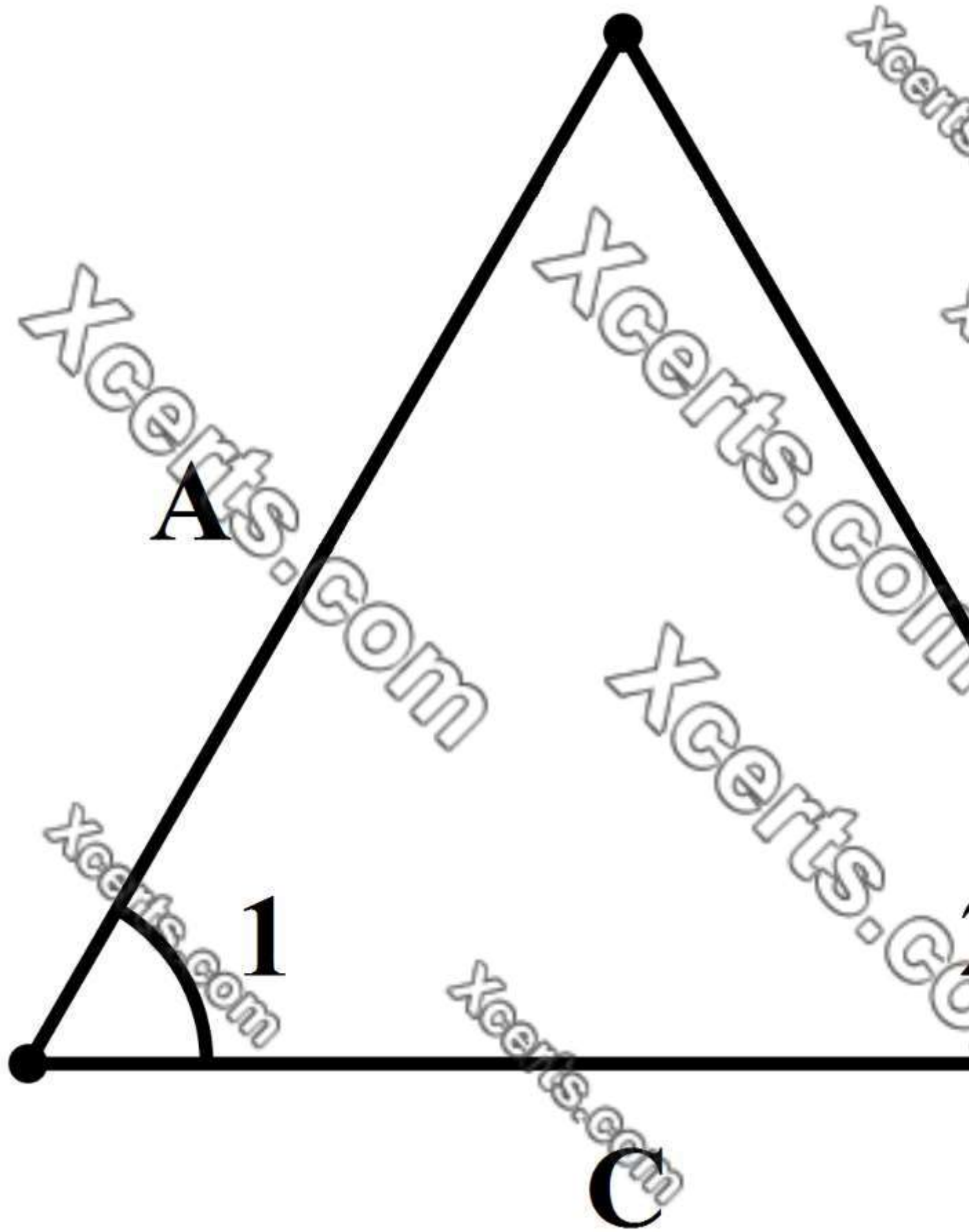
- A. 4,521 cubic inches
- B. 120 cubic inches
- C. 1,130 cubic inches
- D. 1,440 cubic inches

Answer(s): C

Explanation:

For cylinders, $\text{Volume} = r^2(h)$. Since the radius is half the diameter, the problem can be calculated this way: $V = (6^2)10$. $V = (36)10$. If π is approximately 3.14, then $3.14 \times 36 \times 10 = 1,130$ cubic inches.

QUESTION: 916



Triangle ABC as shown above is a(n) _____.

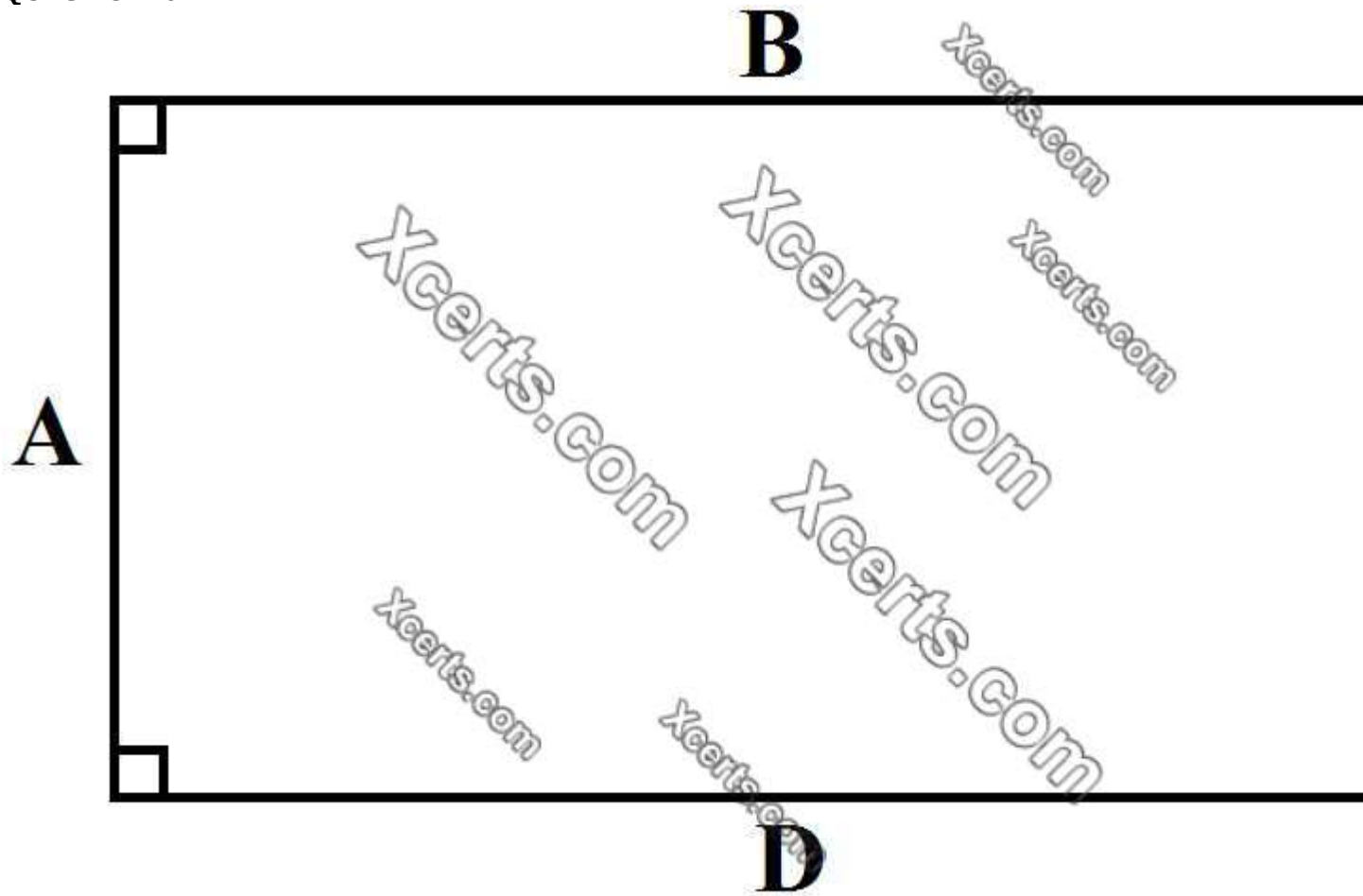
- A. equilateral triangle
- B. right triangle
- C. scalene triangle
- D. isosceles triangle

Answer(s): D

Explanation:

In an isosceles triangle, Sides A and C are equal, and Angles 1 and 2 are equal.

QUESTION: 917



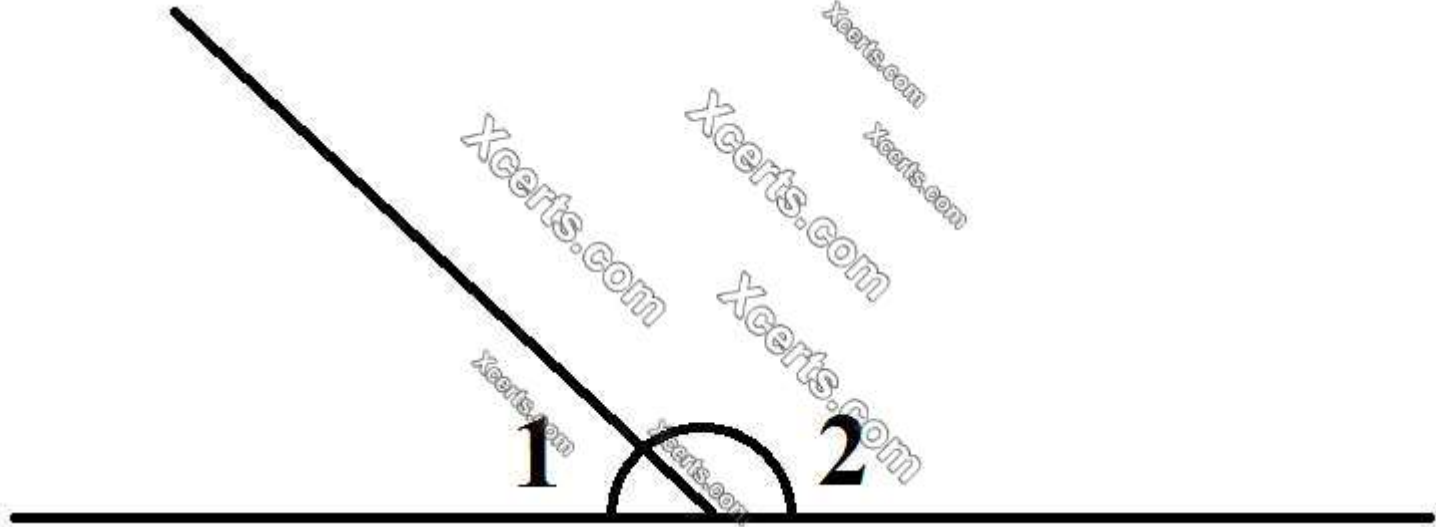
The angles of the quadrilateral above _____.

- A. are all right angles
- B. each equal 45 degrees
- C. are all unequal
- D. total 180 degrees

Answer(s): A

Explanation:

Rectangles have four equal angles, and all angles are right angles.

QUESTION: 918

In the above figure, the sum of Angles 1 and 2 equals _____.

- A. 180 degrees
- B. 90 degrees
- C. 45 degrees
- D. 360 degrees

Answer(s): A

Explanation:

Supplementary angles always equal 180 degrees.

QUESTION: 919

$$3(2x - 5) - 2(4x + 1) = -5(x + 3) - 2$$

- A. 0
- B. 1
- C. 2
- D. 3

Answer(s): A

Explanation:

$$3(2x - 5) - 2(4x + 1) = -5(x + 3) - 2$$

$$6x - 15 - 8x - 2 = -5x - 15 - 2$$

$$-2x - 17 = -5x - 17$$

$$3x - 17 = -17$$

$$3x = 0$$

$$\frac{3}{3}x = \frac{0}{3}$$

$$x = 0$$

QUESTION: 920

A cube has a volume of 64-cubic inches.
What's the length of one side of the cube?

- A. 4 inches
- B. 16 inches
- C. 8 inches
- D. 32 inches

Answer(s): A

Explanation:

Volume equals length $\times$ width $\times$ height ($V = lwh$). Finding the cube root of 64 shows that each edge measures 4 inches.

QUESTION: 921

$$(x^3)^2 = \underline{\hspace{2cm}}.$$

- A. x^5
- B. x^6
- C. x^9
- D. $2x^3$

Answer(s): B

Explanation:

$(x^3)^2$ is the same as $(x^3)(x^3)$. To multiply exponents with the same base, keep the base and add the exponents:
 $(x^3)(x^3) = x^6$.

QUESTION: 922

If I inches of rain fall in one minute, how many inches fall in H hours?

- A. $IH \div 60$
- B. $60I - H$
- C. IH
- D. $60IH$

Answer(s): D

Explanation:

To find out how much rain falls in an hour, multiply the amount that falls in one minute by 60 because 60 minutes make up an hour. In H hours, the amount of rain is $60IH$.

QUESTION: 923

If $x = y$, $6 + 4(x - y) =$ _____.

- A. $6xy + 4$
- B. $6 + 4xy$
- C. $10x - 10y$
- D. 6

Answer(s): D

Explanation:

$6 + 4(x - y) = 6 + 4x - 4y$. Because $x = y$, $4x = 4y$.
Therefore, $4x - 4y = 4x - 4x = 0$, and
 $6 + 0 = 6$.

QUESTION: 924

820 is a number between _____.

- A. 20 and 30
- B. 10 and 20
- C. 80 and 90
- D. 40 and 50

Answer(s): A

Explanation:

$202 = 400$, and $302 = 900$, so the range of 20 to 30 is correct.

QUESTION: 925

$(x + 2)(x + 2) =$ _____.

- A. $x^2 + 2x + 4$
- B. $x^2 + 4x + 4$
- C. $x^2 + 4x + 2$
- D. $x^2 + 2x + 0$

Answer(s): B

Explanation:

$(x + 2)(x + 2)$
 $x^2 + 2x + 2x + 4$
 $x^2 + 4x + 4$

QUESTION: 926

Evaluate the expression $6a - 3x - 2y$, if $a = 3$, $x = 7$, and $y = 4$.

- A. -5
- B. -40

- C. 31
- D. 40

Answer(s): A

Explanation:

Replace the unknowns with the numbers given. $(6 \times -3) - (3 \times -7) - (2 \times 4) = -18 + 21 - 8 = -5$.

QUESTION: 927

$(x + 4)(3x + 5) = \underline{\hspace{2cm}}$.

- A. $3x^2 + 9x + 20$
- B. $3x^2 + 17x + 15$
- C. $3x^2 + 17x + 20$
- D. $3x^2 + 9x + 20$

Answer(s): C

Explanation:

$(x + 4)(3x + 5)$
 $3x^2 + 12x + 5x + 20$
 $3x^2 + 17x + 20$

QUESTION: 928

Solve for x: $5x - 2x = 7x + 2x - 24$

- A. 2
- B. -2
- C. 4
- D. -4

Answer(s): C

Explanation:

$5x - 2x = 7x + 2x - 24$
 $3x = 9x - 24$
 $-6x = -24$
 $6x = 24$
 $x = 4$

QUESTION: 929

The cube of 6 is $\underline{\hspace{2cm}}$.

- A. 125
- B. 225
- C. 216
- D. 238

Answer(s): C

Explanation:

The cube of 6 = $6 \times 6 \times 6 = 216$.

QUESTION: 930

In the equation $3x + 7y = 21$, at what point is the x-axis intersected?

- A. (7, 0)
- B. (0, 7)
- C. (0, 4)
- D. (4, 0)

Answer(s): A

Explanation:

The x access is intersected at the point where the y-coordinate is 0 ($y = 0$). Substitute 0 for the y variable in the equation. $3x + 7y = 21 = 3x + 7(0) = 21 = 3x = 21$.

Therefore, $x = 7$. The point's coordinates are (7, 0).

QUESTION: 931

$x + y = 6$ and $x - y = 4$.

Solve for x.

- A. 3
- B. 5
- C. 7
- D. 8

Answer(s): B

Explanation:

$x + y = 6$, so $y = 6 - x$. Substitute this known value for y in the second equation.

$$x - y = 4$$

$$x - (6 - x) = 4$$

$$x - 6 + x = 4$$

$$2x - 6 = 4$$

$$2x = 10$$

$$x = 5$$

QUESTION: 932

Solve for y:

$$4(y + 3) + 7 = 3$$

- A. 2
- B. -2
- C. 4
- D. -4

Answer(s): D

Explanation:

$$4(y + 3) + 7 = 3$$

$$4y + 12 + 7 = 3$$

$$4y + 19 = 3$$

$$4y = -16$$

$$y = -4$$

QUESTION: 933

12 yards + 14 feet ÷ by 2 = _____.

- A. 25 feet
- B. 12 feet
- C. 32 feet
- D. 8 feet

Answer(s): A

Explanation:

Convert 12 yards and 14 feet to feet: (12 yards × 3 feet per yard) + 14 feet = 36 feet + 14 feet = 50 feet.

Divide by 2 as instructed: 50 feet ÷ 2 = 25 feet.

QUESTION: 934

$x^3 \times x^3 =$ _____.

- A. x^9
- B. $2x^9$
- C. $2x^6$
- D. x^6

Answer(s): D

Explanation:

If two powers have the same base, the numbers can be multiplied by keeping the base and adding the powers together.

QUESTION: 935

$\frac{41}{5} + \frac{12}{5} + \frac{33}{10} =$ _____.

- A. $\frac{61}{5}$
- B. $\frac{89}{10}$
- C. $\frac{51}{2}$
- D. $\frac{71}{5}$

Answer(s): B

Explanation:

Convert to the lowest common denominator (which is 10), then add.
 $41/2 + 12/5 + 33/10 = 42/10 + 14/10 + 33/10 = 89/10$.

QUESTION: 936

$1.5 \times 102 = \underline{\hspace{2cm}}$.

- A. 45
- B. 150
- C. 1,500
- D. 15

Answer(s): B

Explanation:

$1.5 \times 102 = 1.5 \times (10 \times 10) = 1.5 \times 100 = 150$.

QUESTION: 937

The average of 0, 54, 61, 70, and 75 is .

- A. 50
- B. 52
- C. 55
- D. 58

Answer(s): B

Explanation:

Add the number and then divide by the number of terms.
 $54 + 61 + 70 + 75 = 260$.
 $260 \div 5 = 52$.

QUESTION: 938

2 feet 4 inches + 4 feet 8 inches = .

- A. 6 feet 8 inches
- B. 7 feet
- C. 7 feet 2 inches
- D. 8 feet

Answer(s): B

Explanation:

2 feet + 4 feet = 6 feet, and 4 inches plus 8 inches = 12 inches (the equivalent to 1 foot).

QUESTION: 939

If $x = 4$, then $x^4 \div x =$ _____.

- A. 12
- B. 36
- C. 64
- D. 72

Answer(s): C

Explanation:

Substitute 4 for all x's in the problem. $4^4 \div 4 = (4 \times 4 \times 4 \times 4) \div 4 = 256 \div 4 = 64$.

QUESTION: 940

Solve for x: $5 - 3x = 14 + 6x$

- A. $x = -1$
- B. $x = -1$
- C. $x > -1$
- D. $x < -1$

Answer(s): B

Explanation:

$$5 - 3x = 14 + 6x$$

$$5 - 3x - 6x = 14$$

$$-9x = 14 - 5$$

$$-9x = 9$$

$$x = -1$$

QUESTION: 941

$(900 \times 3) \div 6 =$ _____.

- A. 45
- B. 450
- C. 55
- D. 550

Answer(s): B

Explanation:

$$(900 \times 3) \div 6 = 2,700 \div 6 = 450$$

QUESTION: 942

If $x = 2$, then $x^x \times x^x =$ _____.

- A. 16
- B. 2^{xx}
- C. 8

D. 24

Answer(s): A

Explanation:

Substitute 2 for all the x's. $xx \times xx = (2 \times 2) \times (2 \times 2) = 4 \times 4 = 16$

QUESTION: 943

Solve for x: $x^2 - 2x - 15 = 0$

A. 4, -2

B. 3, -3

C. 5, -3

D. -1, 1

Answer(s): C

Explanation:

This is a quadratic equation that can be solved by factoring and setting each factor equal to zero.

$$x^2 - 2x - 15 = 0$$

$$(x - 5)(x + 3) = 0$$

$$x - 5 = 0 \quad x + 3 = 0$$

$$x = 5 \quad x = -3$$

QUESTION: 944

$$49 \div 64 = \underline{\hspace{2cm}}.$$

A. $1/4$

B. $1/2$

C. $1/3$

D. $7/8$

Answer(s): D

Explanation:

The square root of 49 is 7; the square root of 64 is 8. $7 \div 8 = 7/8$.

QUESTION: 945

If $5y^2 = 80$, y is .

A. a positive number

B. a negative number

C. either a positive or negative number

D. an imaginary number

Answer(s): C

Explanation:

The square root of a squared positive number can either be positive or negative.

QUESTION: 946

If $2 + x \leq 15$, what's the value of x ?

- A. $x < 13$
- B. $x > 13$
- C. $x \leq 13$
- D. $x \geq 13$

Answer(s): C

Explanation:

Solving this equation doesn't require multiplying or dividing by a negative number, so the inequality sign remains the same.

QUESTION: 947

If a circle has a radius of 15 feet, what is its circumference most nearly?

- A. 24 feet
- B. 72 feet
- C. 94 feet
- D. 36 feet

Answer(s): C

Explanation:

Circumference equals π times diameter, and diameter is equal to two times the radius. Or $C = \pi d$, and $d = 2r$. $C = 30\pi$. If you round π to 3.14, the answer is about 94.2 or about 94 feet.

QUESTION: 948

What's the volume of a box measuring 12- inches long by 8-inches deep by 10-inches high?

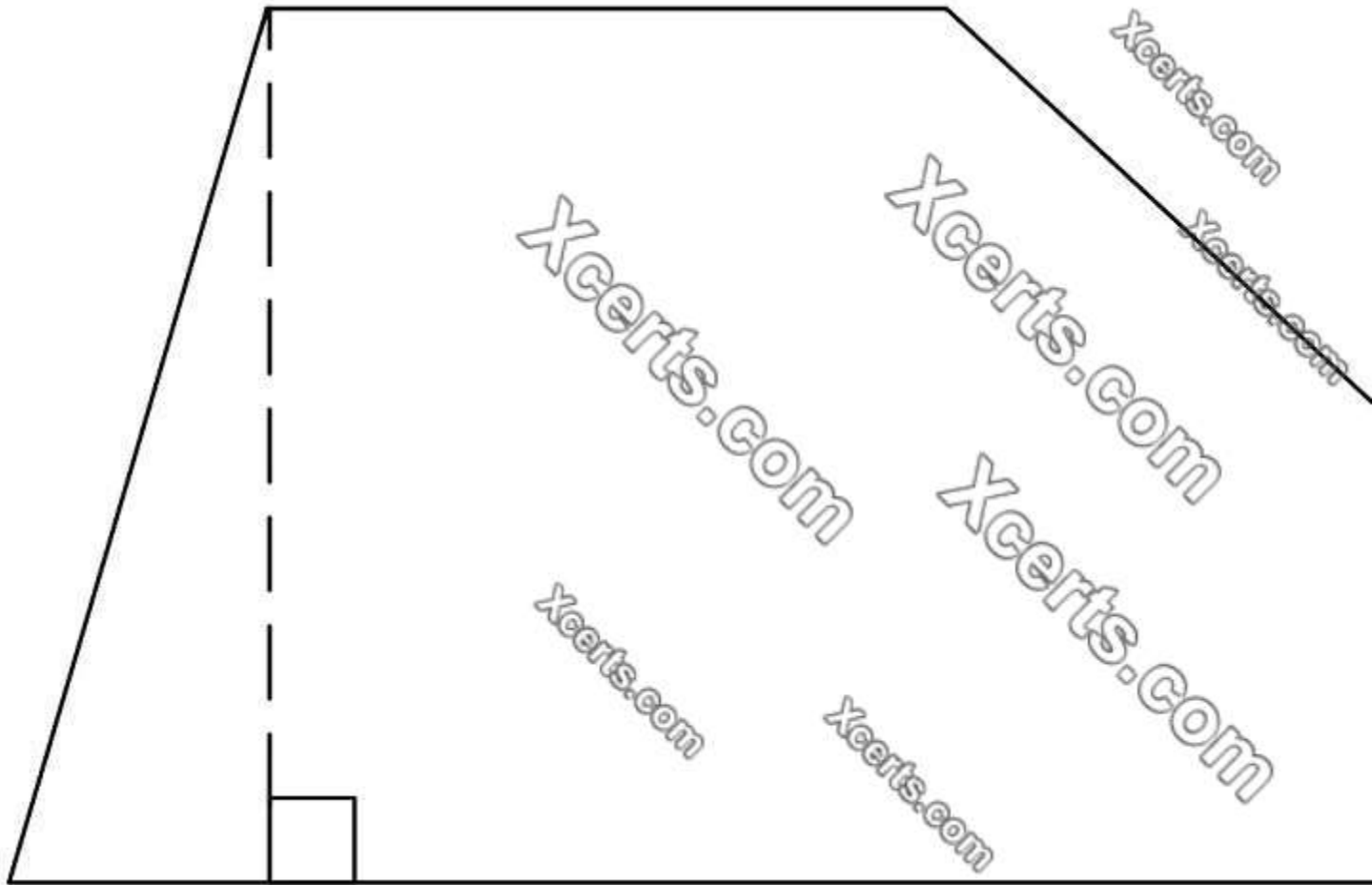
- A. 960 cubic inches
- B. 128 cubic inches
- C. 42 cubic inches
- D. 288 cubic inches

Answer(s): A

Explanation:

Volume equals length $\times$ width $\times$ height ($V = lwh$). $12 \times 8 \times 10 = 960$ cubic inches.

QUESTION: 949



The figure above is a(an) _____.

- A. parallelogram
- B. obtuse triangle
- C. trapezoid
- D. rectangle

Answer(s): C

Explanation:

Trapezoid is a convex quadrilateral with at least one pair of parallel sides.

QUESTION: 950

The sum of the measures of the angles of a parallelogram is _____.

- A. 360 degrees
- B. 540 degrees
- C. 180 degrees
- D. 720 degrees

Answer(s): A

Explanation:

All quadrilaterals have angles that total 360 degrees.

QUESTION: 951

What is the prime factorization of 100?

- A. 2×50
- B. 22×52
- C. 4×25
- D. 252

Answer(s): B

Explanation:

$100 = 4 \times 25 = (2 \times 2) \times (5 \times 5) = 22 \times 52$.

QUESTION: 952

-9 is an example of a(an) _____.

- A. real number
- B. imaginary number
- C. irrational number
- D. sloping number

Answer(s): B

Explanation:

The square root of a negative number doesn't exist as far as real numbers are concerned. In mathematics, this is called an imaginary number.

QUESTION: 953

If $b - 3 = 7$, then b is equal to _____.

- A. 10
- B. 4
- C. 21
- D. 8

Answer(s): A

Explanation:

This equation means "some number, decreased by 3, is equal to 7." $b - 3 = 7$

To arrive at a true statement for b, we want to eliminate -3 on the left side of the equation. We do this by adding 3 to both sides of the equation, undoing the subtraction on the left side. When we add 3 to both sides, we are keeping the statement an equation.

$$(b - 3) + 3 = 7 + 3$$

By simplifying both sides, we isolate b and thus find the solution.

$$b = 10$$

QUESTION: 954

What is the product of $(z + 2)(2z - 3)$?

- A. $3z - 6$
- B. $z + 4z - 3$
- C. $z^2 + 4z - 6$
- D. $2z^2 + z - 6$

Answer(s): D

Explanation:

An easy way to perform the multiplication is to do four separate multiplications. Then the procedure looks like ordinary multiplication in arithmetic.

Multiply $(2z - 3)$ by 2

Multiply $(2z - 3)$ by z

Add the partial products as you do in arithmetic.

$$\begin{array}{r}
 2z - 3 \\
 z + 2 \\
 \hline
 4z - 6 \\
 2z^2 - 3z \\
 \hline
 2z^2 + z - 6
 \end{array}$$

QUESTION: 955

The suburb of Hewitt has a private subdivision that has a pool in the shape of a quadrilateral. If the subdivision wants to put a lifeguard on each side of the pool, how many lifeguards will be needed?

- A. eight
- B. six
- C. four
- D. three

Answer(s): C

Explanation:

Since a quadrilateral is a four-sided figure, the subdivision will need four lifeguards. A quadrilateral, by definition, has four sides.

Whether the sides are parallel - a parallelogram - or if all four sides are equal and all four angles are right angles - a square - there are still four sides, and, therefore, four lifeguards.

QUESTION: 956

If you cut the largest possible circular tabletop from a square whose sides measure two feet

each, how much wood is wasted?
Use 3.14 for π .

- A. 1 square foot
- B. 1.86 square feet
- C. 5.86 square feet
- D. 0.86 square feet

Answer(s): D

Explanation:

Step 1: Find the area of the square.

$$2 \text{ feet} \times 2 \text{ feet} = 4 \text{ square feet}$$

Step 2: Find the area of the circle, using the formula $A = \pi r^2$. (The radius equals half the diameter; the diameter of this circle is 2 feet - the same length as one side of the square.) $3.14 \times r^2 = 3.14 \times 1^2 = 3.14$ square feet
Step 3: Subtract 3.14 square feet from 4 square feet to find the wood that is wasted, 0.86 square feet.

QUESTION: 957

Solve for x:

$$3x + 2 = -13$$

- A. $x = 13$
- B. $x = -41/3$
- C. $x = 8$
- D. $x = -5$

Answer(s): D

Explanation:

Step 1: Subtract 2 from each side of the equation in order to eliminate -2 from the left side. (You are undoing the addition.)

$$3x + 2 - 2 = -13 - 2$$

$$3x = -15$$

Step 2: Now divide each side by 3 to find x. (You are undoing the multiplication.) $3x/3 = -15/3$
 $x = -5$

QUESTION: 958

An artist sold 4 of his paintings. These represented 0.05 of all the artwork he had done. How many paintings had he made at that point?

- A. 100
- B. 80
- C. 50
- D. 20

Answer(s): B

Explanation:

Let p stand for the total number of paintings the artist has made. The 4 paintings he sold are equal to 0.05 of all his paintings. This can be expressed as an equation:

$$0.05p = 4$$

To solve for p , divide both sides by 0.05. This undoes the multiplication of 0.05 and p and gives you the value of p so that you know the total number of paintings the artist has painted.

QUESTION: 959

One of the equal angles of an isosceles triangle is 40 degrees.
What is the angle opposite the unequal side?

- A. 40 degrees
- B. 90 degrees
- C. 100 degrees
- D. 140 degrees

Answer(s): C

Explanation:

In an isosceles triangle, two of the sides are equal. This means that the angles opposite them are equal, too. If one is 40 degrees, then so is the other. To find the angle opposite the unequal side, begin by adding the equal angles.

$$40 + 40 = 80 \text{ degrees}$$

To find the third angle, subtract this amount from 180 (the number of degrees in any triangle)

$$180 - 80 = 100 \text{ degrees (third angle).}$$

QUESTION: 960

If you divide $24x^3 + 16x^2 - 8x$ by $8x$, how many x 's will be there in the quotient?

- A. 0
- B. 5
- C. 2
- D. -1

Answer(s): C

Explanation:

An easy way to do this example is to break it into three examples, dividing each by $8x$. Divide the numbers first and then the letters. However, to divide the exponents of x , just find the difference between them. (Thus, $x^3 \div x = x^{(3-1)} = x^2$)

$$24x^3/8x + 16x^2/8x - 8x/8x = 3x^2 + 2x - 1$$

Since the question asks only how many x 's there are in the quotient (not how many x^2 's), the answer is 2.

QUESTION: 961

A room is 19 feet long, 10 feet wide, and 8 feet high.

If you want to paint the walls and ceiling, how many square feet of surface will you have to cover with paint?

- A. 232 square feet
- B. 422 square feet
- C. 464 square feet
- D. 654 square feet

Answer(s): D

Explanation:

First, find the area (surface) of the ceiling. Since it is opposite the floor, it has the same length and width $A = lw$. $19 \text{ feet} \times 10 \text{ feet} = 190 \text{ square feet}$ (ceiling).

Next find the combined area of two matching (opposite) walls. Start with the walls formed by the length and height of the room. (Clear the decimal in the divisor.) $19 \text{ feet} \times 8 \text{ feet} = 152 \text{ square feet}$ (first wall). $152 \text{ feet} \times 2 = 304 \text{ square feet}$ (matching walls).

Then find the area of the walls formed by the width and height of the room. $10 \text{ feet} \times 8 \text{ feet} = 80 \text{ square feet}$ (second wall). $80 \text{ feet} \times 2 = 160 \text{ square feet}$ (matching walls).

Finally, combine all surfaces to be painted.

$$190 + 304 + 160 = 654 \text{ square feet}$$

QUESTION: 962

If a car traveled 200 miles at an average rate of speed of r miles per hour, the time it took for the trip could be written as _____.

- A. $200/r$
- B. $r/200$
- C. $200r$
- D. $r/60$

Answer(s): A

Explanation:

The basic formula for travel is "distance equals rate multiplied by time," or $D = rt$.

The car traveled 200 miles (D); therefore $200 = rt$.

To solve for time, divide both sides of the equation by r . (You are undoing the multiplication.)

$$200/r = rt/r$$

$$200/r = t \text{ (time it took for trip)}$$

QUESTION: 963

An equilateral triangle has the same perimeter as a square whose side is 12 inches. What is the length of a side of the triangle?

- A. 9 inches
- B. 12 inches
- C. 18 inches
- D. 16 inches

Answer(s): D

Explanation:

The perimeter of a square is 4 times a side.

Therefore, the perimeter of this square is 4×12 inches or 48 inches.
 The equilateral triangle has the same perimeter as the square.
 Since the 3 sides of an equilateral triangle are equal, divide by 3 to find the length of one side.
 $(48 \text{ inches}) \div 3 =$
 16 inches (length of one side).

QUESTION: 964

What is the value of $(0.1)^3$?

- A. 0.3
- B. 0.003
- C. 0.1
- D. 0.001

Answer(s): D

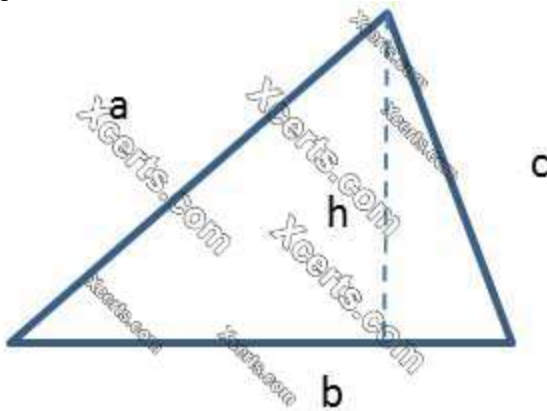
Explanation:

The exponent in $(0.1)^3$ means you use 0.1 as a multiplier 3 times.

$$(0.1)^3 = (0.1) (0.1) (0.1) = 0.001$$

When multiplying decimals, count off one decimal place in the answer for each decimal place in the numbers you multiply.

QUESTION: 965



What is the area of this shape?

- A. $\frac{1}{2}bh$
- B. $a^2 + b^2$
- C. abc
- D. bh

Answer(s): A

Explanation:

The area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$. In this case, the base is b and the height is h .

QUESTION: 966

How many different combinations of shirts and ties are possible if you have 4 shirts and 5 ties?

- A. 120
- B. 9
- C. 30
- D. 20

Answer(s): D

Explanation:

With each of the 4 shirts you can wear one of the 5 ties, so the total number of combinations is $4 \times 5 = 20$

QUESTION: 967

What is the value of $(+2)(-5x + 3)(-3)$?

- A. +90
- B. +60
- C. -13
- D. -3

Answer(s): A

Explanation:

To find the product of more than two numbers, work on only two numbers at a time. If both of these numbers have plus signs (+), their product has a plus sign, to both have minus signs (-), their product has a plus (not a minus) sign. But if their signs are different, the product has a minus sign.

$$\begin{aligned} & (+2)(-5)(+3)(-3) \\ & = (-10)(+3)(-3) \\ & = (-30)(-3) \\ & = +90 \end{aligned}$$

QUESTION: 968

Solve for x:

$$x^2 = 3x + 10.$$

- A. $x = 3$; $x = 10$
- B. $x = -3$; $x = -10$
- C. $x = -2$; $x = 5$
- D. $x = 2$; $x = -5$

Answer(s): C

Explanation:

To solve the equation $x^2 = 3x + 10$, turn it into an equation equal to 0, find the two factors of the new equation, and then set each factor equal to 0, to solve for x.

Step 1: Move all expressions to one side of the equal sign. Change the signs of terms that are

moved.

$$x^2 - 3x - 10 = 0$$

Step 2: Find the two factors that you would multiply to produce this polynomial. Do this one expression at a time. What gives you x^2 ? The answer is x times x . Therefore, place an x at the beginning of each factor.

$$(x)(x)$$

Next find the two numbers you would multiply to get 10. The numbers could be 10 and 1, or 5 and 2, but

remember that the two numbers also have to produce 3, the middle term in the polynomial. The difference between 5 and 2 is 3. Therefore 5 and 2 are the numbers that complete the factors $(x - 5)(x + 2)$

Now decide the signs that belong in each factor. The appearance of -10 in the polynomial means that 5 and 2 have different signs. The $-3x$ in the polynomial indicates that 5 (the larger number) has the minus sign, and that 2 has a plus sign. Thus

$$(x - 5)(x + 2)$$

Step 3: Set each factor equal to zero and solve the equations.

$$x - 5 = 0 \quad x + 2 = 0$$

$$x = 5$$

$$x = -2$$

QUESTION: 969

Which of these is a cylinder?

- A. a stick of butter
- B. an orange
- C. a hula hoop
- D. a can of vegetables

Answer(s): D

Explanation:

A cylinder is a solid figure whose upper and lower bases are circles; a can of vegetables is one example of a cylinder. An orange is a sphere; a stick of butter is a rectangular solid; and a hula hoop is (for all practical purposes) a two-dimensional figure, a circle.

QUESTION: 970

What is the reciprocal of $5/3$?

- A. 0.6
- B. $12/3$
- C. 2
- D. 1

Answer(s): A

Explanation:

When the product of two numbers is 1, each number is the reciprocal of the other. In the following equation, r is the reciprocal you want to find.

$$r \times 5/3 = 1$$

To isolate r , divide each side by $5/3$. This is undoing the multiplication.

$$r \times 5/3 \div 5/3 = 1 \div 5/3$$

$$r \times 5/3 \div 3/5 = 1 \times 3/5$$

$$r = 3/5 \text{ (reciprocal of } 5/3)$$

QUESTION: 971

What is the value of $(13)^2$?

- A. 26
- B. 13
- C. 87
- D. 169

Answer(s): B

Explanation:

This problem does not have to be computed, because the two symbols cancel each other. The radical sign $()$ in (13) tells you to find the square root of 13. But the exponent (2) tells you to square the answer - that is, to multiply the square root of 13 by itself. This would get you back to 13.

QUESTION: 972

Mr. Langendorfer drove his car steadily at 40 miles per hour for 120 miles. He then increased his speed and drove the next 120 miles at 60 miles per hour. What was his average speed?

- A. 48 miles per hour
- B. 52 miles per hour
- C. 50 miles per hour
- D. 46 miles per hour

Answer(s): A

Explanation:

To find the average rate of speed (mph), divide the distance he covered by the time he spent traveling ($R = D / T$). In this example, begin by finding the distance traveled.

$$120 + 120 = 240 \text{ miles (distance)}$$

Next find the length of time he traveled.

At the beginning of his trip, he drove 120 miles at 40 mph.

$$120 / 40 = 3 \text{ hours (first part of trip)}$$

Later, he increased his speed.

$$120 / 60 = 2 \text{ hours (second part of trip)}$$

Altogether, he traveled for 5 hours. Now apply the formula for finding his average rate of speed

$$D/T = 240/5 = 48 \text{ mph (rate of speed)}$$

QUESTION: 973

An architect designs two walls of a museum to meet at an angle of 120 degrees. What is an angle of this size called?

- A. acute
- B. obtuse
- C. right
- D. straight

Answer(s): B

Explanation:

An angle of 180 degrees is a straight angle. An angle of 90 degrees is a right angle. An angle greater than 90 degrees but less than 180 degrees is an obtuse angle. An angle less than 90 degrees is an acute angle.

QUESTION: 974

Solve the following equations for x.

$$5x + 4y = 27$$

$$x - 2y = 11$$

- A. $x = 3$
- B. $x = 9$
- C. $x = 4.5$
- D. $x = 7$

Answer(s): D

Explanation:

To solve these equations for x, begin by finding a way to eliminate y.

Multiply both sides of the second equation by 2.

$$2(x - 2y) = 2 \times 11$$

$$2x - 4y = 22$$

Add the new form of the second equation to the first equation, and solve for x.

$$5x + 4y = 27$$

$$2x - 4y = 22$$

(+4 cancels -4)

$$7x = 49$$

$$x = 7$$

QUESTION: 975

If a is a negative number, and ab is a positive number, then which of the following must be true?

- A. b is positive
- B. a is greater than b
- C. b is negative
- D. b is greater than a

Answer(s): C

Explanation:

The product of a negative number and a positive number is always negative. The product of two

negative numbers is always a positive number. Since ab is positive and a is negative, b must be negative, too.

QUESTION: 976

Solve the following inequality.

$$x + 5 > 7$$

- A. $x = 2$
- B. $x > 2$
- C. $x - 7 > 5$
- D. $x - 5 > 7$

Answer(s): B

Explanation:

The expression $(x + 5 > 7)$ is a statement of inequality, it means that x plus 5 is greater than 7 - not equal to it.

To solve this inequality, subtract 5 from both sides of the statement.

$$x + 5 > 7$$

$$x + 5 - 5 > 7 - 5$$

$$x > 2$$

Thus the statement of inequality is true for any value of x that is greater than 2. Try it with 3, for example.

$$3 + 5 > 7$$

$8 > 7$ a true statement

QUESTION: 977

What is 4% of 0.0375?

- A. 0.0015
- B. 0.9375
- C. 0.0775
- D. 0.15

Answer(s): A

Explanation:

Multiply 0.0375 by 4% (or 0.04).

$$0.0375 \times 0.04 = 0.001500 = 0.0015$$

Note that the product has as many decimal places as there are in the multiplicand and multiplier combined, but the two final zeros may be dropped since they are to the right of the decimal point and at the end.

QUESTION: 978

What number multiplied by $\frac{2}{3}$ will give a product of 1?

- A. $-\frac{2}{3}$
- B. $-\frac{3}{2}$

- C. $\frac{3}{2}$
- D. $\frac{4}{6}$

Answer(s): C

Explanation:

If a number is multiplied by its reciprocal (also called its multiplicative inverse), the product is 1. The reciprocal of a fraction is found by inverting the fraction. The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$, that is, $\frac{2}{3} \times \frac{3}{2} = 1$

QUESTION: 979

What is the value of the expression $x^2 - 5xy + 2y$ if $x = 3$ and $y = -2$?

- A. -25
- B. -27
- C. 32
- D. 35

Answer(s): D

Explanation:

Substitute 3 for x and -2 for y in the expression $x^2 - 5xy + 2y$ $(3)^2 - 5(3)(-2) + 2(-2)$
 $9 - 15(-2) - 4$
 $9 + 30 - 4$

QUESTION: 980

Solve the following inequality:
 $x - 3 < 14$.

- A. $x < 11$
- B. $x < 17$
- C. $x = 11$
- D. $x > 17$

Answer(s): B

Explanation:

$x - 3 < 14$ is an inequality that states that x minus 3 is less than 14. To solve the inequality, add 3 to both sides of it.
 $x - 3 + 3 < 14 + 3$
 $x < 17$

QUESTION: 981

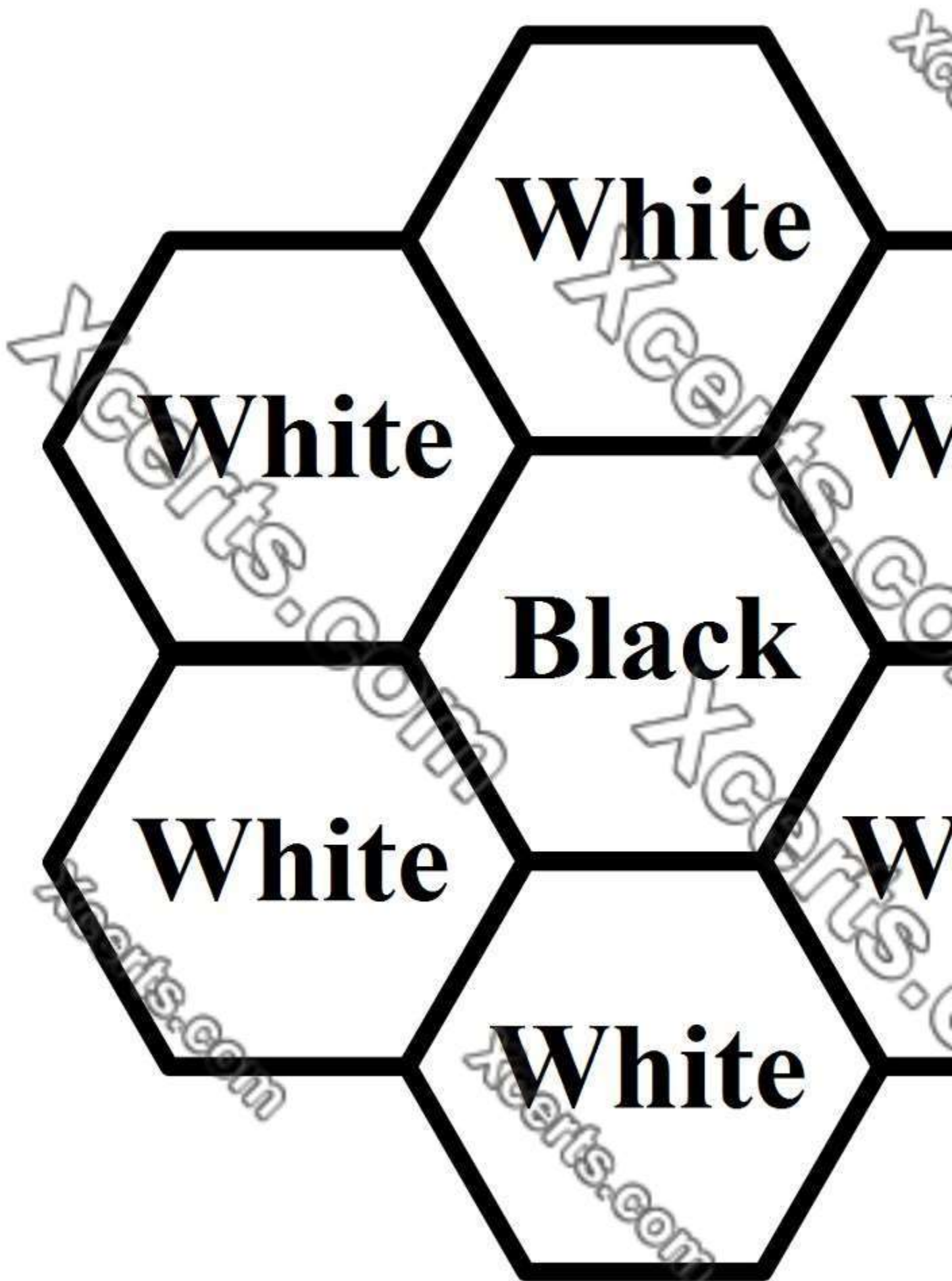
A floor is made up of hexagonal tiles, some of which are black and some of which are white. Every black tile is completely surrounded by white tiles. How many white tiles are there around each black tile?

- A. four

- B. five
- C. six
- D. eight

Answer(s): C

Explanation:



A hexagon has six sides. Each of the six sides of the black tile must touch a side of a white tile, so there are six white tiles surrounding black tiles.

QUESTION: 982

Solve the following problem:

What does $96/92 =$ _____.

- A. 94
- B. 93
- C. 184
- D. 95

Answer(s): A

Explanation:

When dividing two powers with the same base, you keep the base and subtract the exponents.

QUESTION: 983

If $2x - 3 = 37$, what is the value of x ?

- A. 17
- B. 38
- C. 20
- D. 80

Answer(s): C

Explanation:

The equation $2x - 3 = 37$ means that "twice a number minus 3 is equal to 37." To arrive at a value for x , we first eliminate -3 on the left side. This can be done by adding 3 to both sides of the equation, thus undoing the subtraction.

$$2x - 3 + 3 = 37 + 3$$

$$2x = 40$$

To eliminate the 2 that multiplies x , we undo the multiplication by dividing both sides of the equation by 2.

$$2x/2 = 40/2 = 20$$

QUESTION: 984

An audience consists of M people, and $2/3$ of the audience are adults. C the adults, $1/2$ are males.

How many adult males are in the audience?

- A. $1/6 M$
- B. $M - 2/3 - 1/2$
- C. $1/3 M$
- D. $M - 1/3$

Answer(s): C

Explanation:

If $\frac{2}{3}$ of M people are adults, then $20M$ represents the number of adults. If $\frac{1}{2}$ of $(\frac{2}{3}) M$ are males, then $\frac{1}{2} \times (\frac{2}{3}) M$ represents the number of adult males.

$$\frac{1}{2} \times \frac{2}{3} M = \frac{1}{3} M$$

QUESTION: 985

What is the value of $(0.2)^3$?

- A. 0.008
- B. 0.8
- C. 0.006
- D. 0.6

Answer(s): A

Explanation:

$(0.2)^3$ means $0.2 \times 0.2 \times 0.2$

Multiply the first two numbers:

$$0.2 \times 0.2 \times 0.2 = 0.04 \times 0.2$$

Now multiply the remaining two numbers:

$$0.04 \times 0.2 = 0.008$$

QUESTION: 986

If $x^2 + x = 6$, what is the value of x ?

- A. 6 or -1
- B. 1 or -6
- C. 2 or -3
- D. 3 or -2

Answer(s): C

QUESTION: 987

What is the number of square inches in the area of a circle whose diameter is 28 inches?

Use $\frac{22}{7}$ for the value of π .

- A. 616
- B. 88
- C. 44
- D. 1,232

Answer(s): A

Explanation:

The area of a circle is πr^2 , where $r = \frac{22}{7}$ and represents the length of the radius. A radius is one-half the length of a diameter. Therefore, if the diameter is 28 inches, the radius of the circle is 14 inches.

$$\begin{aligned}
 \text{Area of circle} &= \frac{22}{7} \times \frac{14}{1} \times \frac{14}{1} \\
 &= \frac{22}{1} \times \frac{14}{1} \times \frac{14}{1} \\
 &= \frac{44}{1} \times \frac{14}{1} \\
 &= \frac{616}{1} \\
 &= 616 \text{ square inches}
 \end{aligned}$$

QUESTION: 988

The expression $x^2 + 2x - 3 / x + 3$ cannot be evaluated if x has a value of _____.

- A. 0
- B. -1
- C. 3
- D. -3

Answer(s): D

Explanation:

If $x = -3$, the denominator of $x^2 + 2x - 3 / x + 3$ will equal 0. Division of 0 is undefined, so x cannot equal -3.

QUESTION: 989

If $x = y$, what does $4(x - y) + 7 =$ _____.

- A. $4x - 4y + 7$
- B. 7
- C. $4xy + 7$
- D. $11x - 11y$

Answer(s): B

Explanation:

If $x = y$, then you can substitute x for y in the 2nd equation - so $4(x - x) + 7 = 4(0) + 7 = 7$

QUESTION: 990

For a special operations air assault mission, one extra soldier for the security element is to be chosen at random from among three soldiers from the 327th Infantry Regiment, two soldiers from the 502nd Infantry Regiment, and five soldiers from the 187th Infantry Regiment. What is the probability that a soldier from the 327th Infantry Regiment will be chosen?

- A. $\frac{3}{10}$
- B. $\frac{1}{10}$
- C. $\frac{1}{3}$
- D. $\frac{3}{7}$

Answer(s): A

Explanation:

The probability of an event occurring is the number of "favorable" outcomes divided by the total possible number of outcomes.

Since there are three soldiers from the 327th from which to choose, there are three possible favorable outcomes for choosing a soldier from that unit.

Since the choice will be made from among a total of 10 soldiers, there are 10 different possible outcomes.

The probability of choosing a soldier from the 327th Infantry Regiment is $\frac{3}{10}$.

QUESTION: 991

A cylindrical post has a cross section that is a circle with a radius of 3 inches. A piece of cord can be wound around it exactly seven times.

How long is the piece of cord? Use $\frac{22}{7}$ as the value of π .

- A. 66 inches
- B. 42 inches
- C. 198 inches
- D. 132 inches

Answer(s): D

Explanation:

A length of cord that will wind around once is equal to the circumference of the circle whose radius is 3 inches.

The circumference of a circle equals $2\pi r$ where $\pi = \frac{22}{7}$ and r is the radius.

Circumference = $2\pi r = 2 \times \frac{22}{7} \times 3 = \frac{132}{7}$ inches

If the cord can be wound around the post seven times, its length is seven times the length of one circumference.

Length of cord $\frac{132}{7} \times 7 = 132$ inches

QUESTION: 992

A naval task force is to be made up of a destroyer, a supply ship, and a submarine.

If four destroyers, two supply ships, and three submarines are available from which to choose, how many different combinations are possible for the task force?

- A. 9

- B. 24
- C. 8
- D. 12

Answer(s): B

Explanation:

There are four possible choices for the destroyer.
Each of these choices may be coupled with any of the two choices for the supply ship.
Each such destroyer-supply ship combination may in turn be coupled with any of the three possible choices for the submarine.
Thus, there are $4 \times 2 \times 3$, or 24, different combinations possible.

QUESTION: 993

The base of a cylindrical can is a circle whose diameter is 2 inches. Its height is 7 inches. How many cubic inches are there in the volume of the can? Use $\frac{22}{7}$ for the value of π .

- A. $\frac{124}{7}$
- B. 22
- C. 44
- D. 88

Answer(s): B

Explanation:

The volume of a cylinder is equal to the product of its height and the area of its base. The base is a circle. The area of a circle is πr^2 , where $\pi = \frac{22}{7}$ and r is the radius. Since the diameter is 2 inches, the radius (which is one-half the diameter) is 1 inch.

Area of circular base =

$$\frac{22}{7} \times \frac{1}{1} \times \frac{1}{1} = \frac{22}{7}$$

The height is 7 inches.

$$\frac{22}{7} \times \frac{7}{1} = 22 \text{ cubic inches}$$

QUESTION: 994

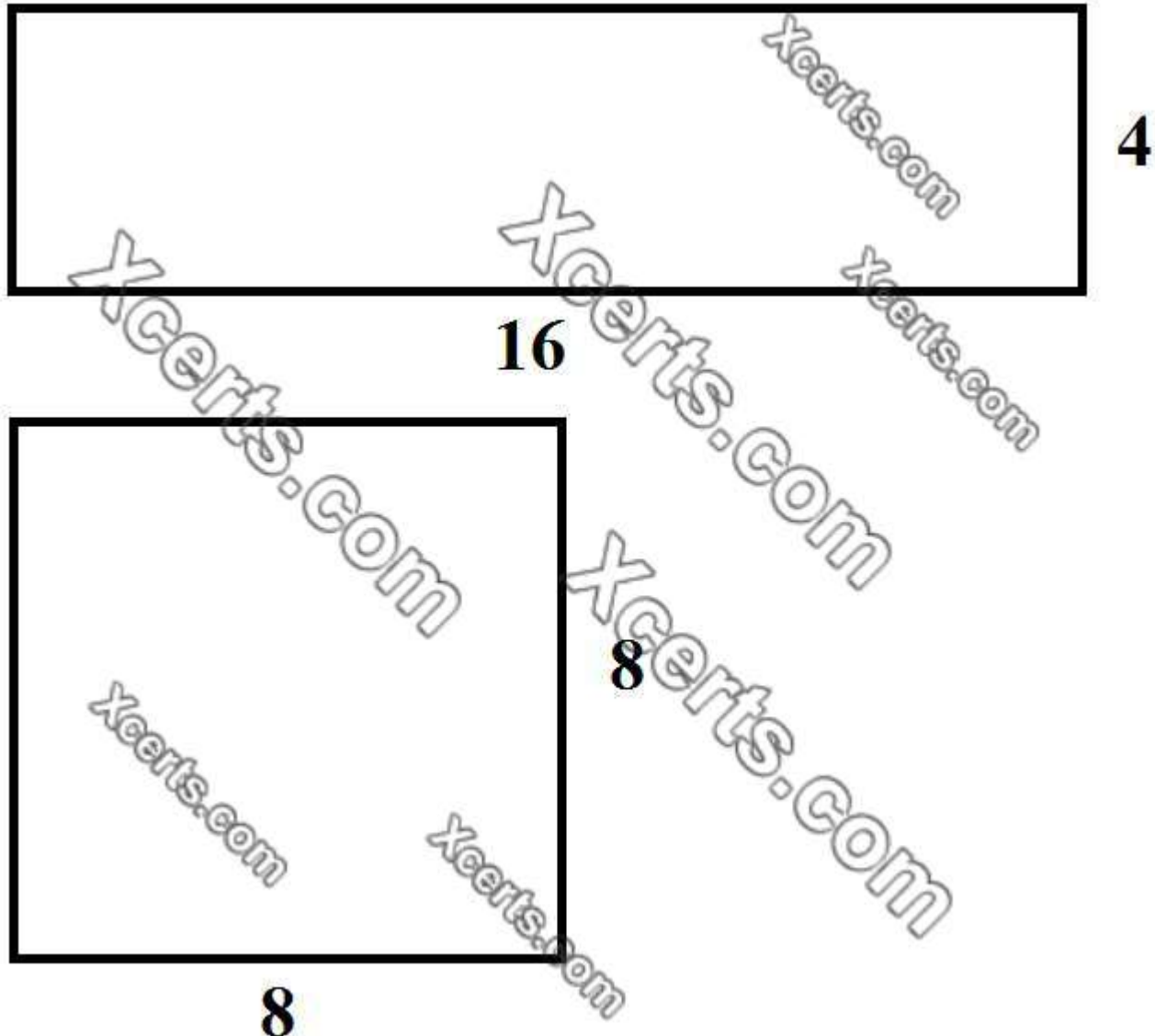
A rectangular vegetable garden 16 yards long and 4 yards wide is completely enclosed by a fence. To reduce the amount of fencing used, the owner replaced the garden with a square one having the same area.

How many yards of fencing did he save?

- A. 4
- B. 6
- C. 8
- D. 16

Answer(s): C

Explanation:



The area of the rectangular garden is equal to the product of its length and width. Area of rectangle = $16 \times 4 = 64$ square yards.

In order for the square to have the same area, 64 square yards, its sides must each be 8 yards long, since $8 \times 8 = 64$ square yards. The fence around the rectangular garden has a length of $16 + 4 + 16 + 4$, or 40, yards.

The fence around the square garden has a length of 4×8 , or 32, yards.

The saving in fencing is $40 - 32$, or 8, yards.

QUESTION: 995

What is the maximum number of boxes, each measuring 3 inches by 4 inches by 5 inches, that can be packed into a storage space measuring 1 foot by 2 feet by 2 feet, 1 inch?

- A. 120
- B. 60
- C. 15
- D. 48

Answer(s): A

Explanation:

The storage space measurements of 1 foot by 2 feet by 2 feet, 1 inch can be converted to inches as 12 inches by 24 inches by 25 inches. Boxes measuring 3 inches by 4 inches by 5 inches can be stacked so that four of the 3-inch sides make up the 12-inch storage dimension, six of the 4-inch sides fill the 24-inch storage dimension, and five of the 5-inch sides fill the 25-inch storage dimension.

There will be $4 \times 6 \times 5$, or 120, boxes packed into the storage space.

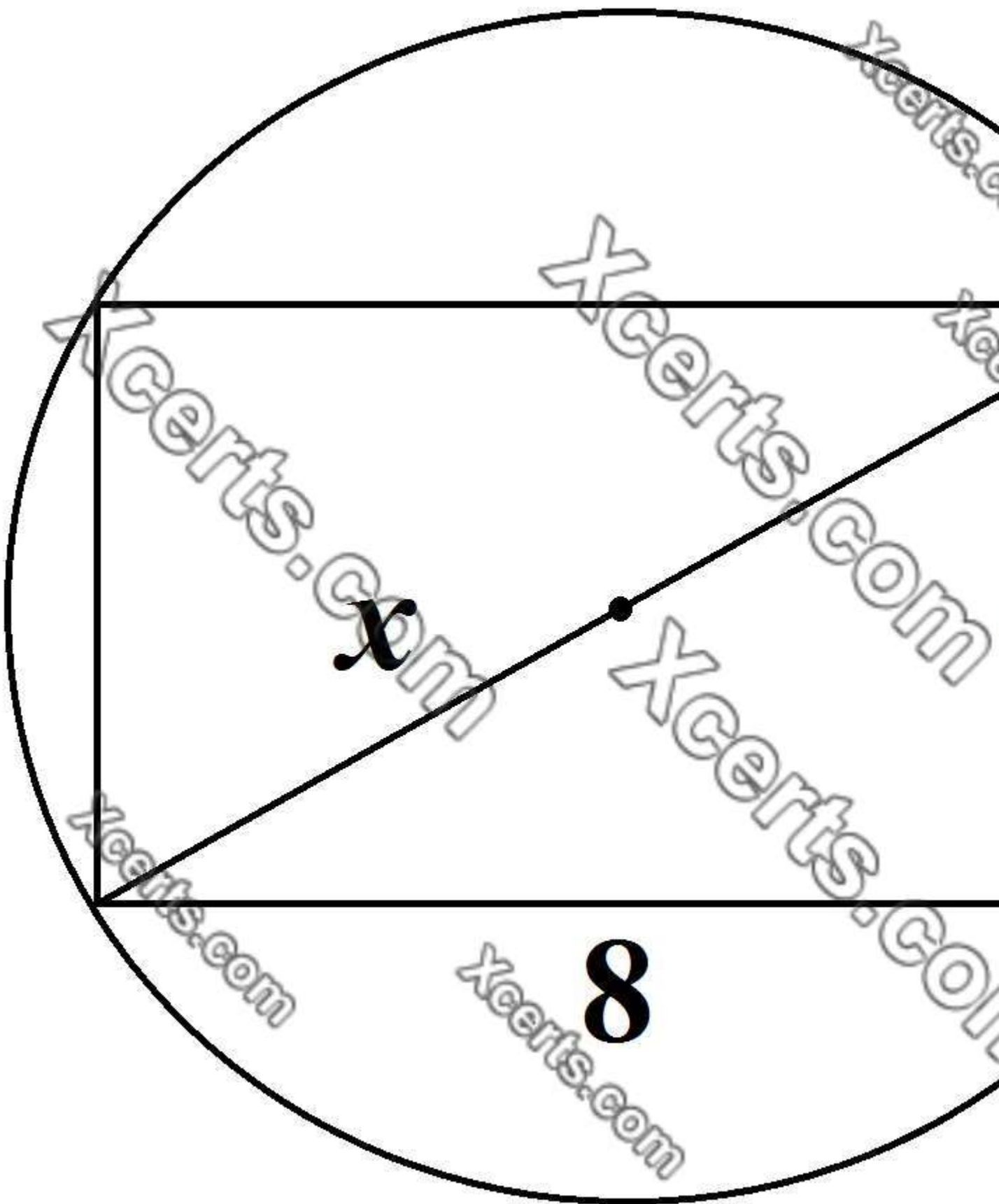
QUESTION: 996

A circle passes through the four vertices of a rectangle that is 8 feet long and 6 feet wide. How many feet are there in the radius of the circle?

- A. 14
- B. $2\frac{1}{4}$
- C. 10
- D. 5

Answer(s): D

Explanation:



If the circle passes through all four vertices of the rectangle, its diameter will be a diagonal of the rectangle.

Since a rectangle's angles are right angles, the diagonal for n is a right triangle with two sides of the rectangle.

Let x = the length of the diagonal. By the Pythagorean Theorem, in a right triangle the square of the length of the hypotenuse is equal to the sum of the squares of the legs.

$$x^2 = 8^2 + 6^2$$

$$x^2 = 64 + 36$$

$$x^2 = 100$$

$$x^2 = 100$$

means that x times $x=100$.

Therefore, $x = 10$ feet.

But 10 is a diameter of the circle. A radius of a circle is one-half the diameter. Hence, the radius is 5 feet.

QUESTION: 997

$$(x + 3)(x + 5) = \underline{\hspace{2cm}}.$$

- A. $x^2 + 16x + 8$
- B. $x^2 + 16x + 15$
- C. $x^2 + 8x + 15$
- D. $x^2 + 8x + 8$

Answer(s): C

Explanation:

Use the FOIL method (First, Outside, Inside, Last) to solve.

$$x^2 + 3x + 5x + 3 \times 5$$

QUESTION: 998

25 percent of what number is equal to 5?

- A. 5
- B. 20
- C. 40
- D. 50

Answer(s): B

Explanation:

Divide 5 by 25 and then multiply the answer by 100 to find the percent.

QUESTION: 999

Which of the following is the equivalent of $12/25$?

- A. 0.36
- B. 0.42
- C. 0.48

D. 0.52

Answer(s): C

Explanation:

The fraction $12/25$ is equivalent to $48/100$ or 0.48.

QUESTION: 1000

Which of the following means $3n + 7 = 16$?

- A. 7 more than 3 times a number is 16
- B. 3 more than 7 times a number is 16
- C. 7 less than 3 times a number is 16
- D. 10 times a number is 16

Answer(s): A

Explanation:

The expression $3n$ means 3 times n . The addition sign before the 7 indicates the phrase more than.

QUESTION: 1001

Which number sentence is true?

- A. $62 < 6.2$
- B. $0.62 < 0.062$
- C. $62.62 < 6.262$
- D. $0.006262 < 0.062$

Answer(s): D

Explanation:

The more zeroes to the right of the decimal point, the smaller the number.

QUESTION: 1002

What is 0.3172 rounded to the nearest tenth?

- A. 0.3
- B. 0.32
- C. 0.317
- D. 0.3172

Answer(s): A

Explanation:

Choice 0.32 is rounded to the hundredths place. Choice 0.317 is rounded to the thousandths place. Choice 0.3172 is rounded to the nearest ten-thousandths place.

QUESTION: 1003

What is another way to write 43?

- A. 36
- B. 48
- C. 56
- D. 64

Answer(s): D

Explanation:

$4 \times 4 \times 4 = 64$.

QUESTION: 1004

$61/3 + 53/10 =$ _____.

- A. $114/13$
- B. $113/10$
- C. $1119/30$
- D. $111/3$

Answer(s): C

Explanation:

You must convert both fractions to thirtieths before adding. Thus, with the fractions converted the numbers are $610/30 + 59/30 = 1119/30$.

QUESTION: 1005

$12(82 - 3) - (2 \times 81) =$ _____.

- A. 382
- B. 786
- C. 851
- D. 1,137

Answer(s): B

Explanation:

Perform the operations in the parentheses first: $(12 \times 79) - 162 = 948 - 162 = 786$.

QUESTION: 1006

Which is equivalent to $43/8$?

- A. $12/8$
- B. $3/32$
- C. $3/20$
- D. $35/8$

Answer(s): D

Explanation:

Multiply 8×4 to get $32/8 + 3/8 = 35/8$

QUESTION: 1007

What is 12% of 102?

- A. 1.2
- B. 12
- C. 24
- D. 120

Answer(s): B

Explanation:

First find $10 \times 10 = 100$ and then multiply by 12%.

QUESTION: 1008

Which is of the following numbers is NOT a prime number?

- A. 53
- B. 93
- C. 67
- D. 127

Answer(s): B

Explanation:

93 is not a prime number since it is equal to 31×3 . If you add up the digits of a number and they are divisible by 3, then the number is also divisible by 3.

QUESTION: 1009

What is the value of y when $x = 1$ and $y = 3 + 2x$?

- A. 2
- B. 3
- C. 5
- D. 17

Answer(s): C

Explanation:

First multiply 2×1 and then add 3.

QUESTION: 1010

What is 50 percent of 78?

- A. 24
- B. 29
- C. 39
- D. 44

Answer(s): C

Explanation:

Multiply 78×0.5

QUESTION: 1011

Which of the following is equivalent to $4(z^2)$?

- A. $4(z + z)$
- B. $4(z - z)$
- C. $4(z \times z)$
- D. $4(z \div z)$

Answer(s): C

Explanation:

z^2 is equivalent to $z \times z$.

QUESTION: 1012

How much water must be added to 1 liter of a 5% saline solution to get a 2% saline solution?

- A. 1L
- B. 1.5L
- C. 2L
- D. 2.5L

Answer(s): B

Explanation:

Use the equation $(0.05)(1) = (0.02)x$:

the left side represents 5% of 1 liter; the right side represents 2% of some amount of water.

From the equation, $x = 2.5$. Subtracting the 1 liter of water already present in the 5% solution, you will find that 1.5 liters need to be added.

QUESTION: 1013

What is another way to write 312?

- A. 63
- B. 34
- C. 123
- D. 12

Answer(s): A

Explanation:

The square root of 12 is the same as the square root of (4×3) , which is the same as the square root of 4 times the square root of 3. The square root of 4 is 2. So 3 times the square root of 12 is the same as 3 times 2 times the square root of 3.

QUESTION: 1014

What is another way to write 2.25×1002 ?

- A. 225
- B. 2,250
- C. 22,500
- D. 225,000

Answer(s): C

Explanation:

$1002 = (100)(100)$, or 10,000
 $(10,000)(2.25) = 22,500$

QUESTION: 1015

Choose the answer to the following problem: $107 \div 104$?

- A. 103
- B. 104
- C. 107
- D. 1011

Answer(s): A

Explanation:

In a division problem like this, leave the whole number the same and subtract the exponents.

QUESTION: 1016

What is $51/5\%$ of 325?

- A. 1.69
- B. 169
- C. 16.9
- D. 32.5

Answer(s): C

Explanation:

First, change the percent to a decimal: $51/5\% = (0.052)$
 Thus, $(0.052)(325) = 16.9$

QUESTION: 1017

$\frac{6}{9} + \frac{12}{9} = \underline{\hspace{2cm}}$.

- A. $\frac{12}{9}$
- B. $\frac{24}{9}$
- C. 1
- D. 2

Answer(s): D

Explanation:

$$6 + 12 = 18$$

$$\frac{18}{9} = 2$$

QUESTION: 1018

Which of these fractions is the largest?

- A. $\frac{1}{4}$
- B. $\frac{7}{8}$
- C. $\frac{3}{8}$
- D. $\frac{2}{5}$

Answer(s): B

QUESTION: 1019

What is the product of the square root of 64 and the square root of 25?

- A. 40
- B. 1600
- C. 20
- D. 14

Answer(s): A

QUESTION: 1020

A pizza maker has y pounds of flour to make pizzas. After he has used 60 pounds of flour, how much flour is left?

The expression that correctly represents the quantity of flour left is $\underline{\hspace{2cm}}$.

- A. $60 - y$
- B. $y/60$
- C. $60 + y$
- D. $y - 60$

Answer(s): D

QUESTION: 1021

If $x = 12$, what is the value of p in this equation?

$$p = x^2/6 - 4$$

- A. 24
- B. 22
- C. 28
- D. 20

Answer(s): D

QUESTION: 1022

The cube of 8 is _____.

- A. 24
- B. 64
- C. 512
- D. 80

Answer(s): C

QUESTION: 1023

What is 4.7×53 ?

- A. 587.5
- B. 235
- C. 5875
- D. 70.5

Answer(s): A

QUESTION: 1024

The sixth root of 64 is:

- A. 6
- B. 2
- C. 8
- D. 4

Answer(s): B

QUESTION: 1025

$(12 \text{ feet} + 14 \text{ yards})/6 = \underline{\hspace{2cm}}$.

- A. 4 feet
- B. 9 feet

- C. 28 feet
- D. 7 feet

Answer(s): B

QUESTION: 1026

$$(y^5) \times (y^3) = \underline{\hspace{2cm}}.$$

- A. y^{15}
- B. y^8
- C. $2y^{15}$
- D. $2y^8$

Answer(s): B

QUESTION: 1027

$$(y + 6)(y + 4) = \underline{\hspace{2cm}}.$$

- A. $y^2 + 10y + 24$
- B. $2y + 10y + 10$
- C. $y^2 + 24y + 10$
- D. $y^2 + 10y + 10$

Answer(s): A

QUESTION: 1028

$$2.8 \times 10^2 = \underline{\hspace{2cm}}.$$

- A. 280
- B. 56
- C. 2,800
- D. 28

Answer(s): A

QUESTION: 1029

Which of the following is a prime number: 13, 15, 4, or 12?

- A. 4
- B. 15
- C. 13
- D. 12

Answer(s): C

QUESTION: 1030

Which of the following is the mode in this series of numbers?
2, 2, 6, 6, 6, 6, 8, 8, 8, 10, 10, 12

- A. 8
- B. 10
- C. 6
- D. 12

Answer(s): C

QUESTION: 1031

If $a = 8$, then $84/a =$ _____.

- A. 32
- B. 12
- C. 512
- D. 4096

Answer(s): C

QUESTION: 1032

Solve this:

$6! =$ _____.

- A. 72
- B. 12
- C. 720
- D. 24

Answer(s): C

QUESTION: 1033

$(675 \times 3)/5 =$ _____.

- A. 400
- B. 40
- C. 200
- D. 405

Answer(s): D

QUESTION: 1034

If $x = 3$, then $xx \times x =$ _____.

- A. 27
- B. 30
- C. $3x$

D. 81

Answer(s): D

QUESTION: 1035

If a circle has a radius of 18 feet, what is its circumference (to the nearest whole foot)?

- A. 36 feet
- B. 113 feet
- C. 54 feet
- D. 108 feet

Answer(s): B

QUESTION: 1036

If a rectangular swimming pool has a perimeter of 132 feet and is 21 feet wide, what is its area?

- A. 2772 square feet
- B. 1890 square feet
- C. 111 square feet
- D. 945 square feet

Answer(s): D

QUESTION: 1037

Convert 520,000 to scientific notation.

- A. 5.20×1000
- B. 5.20×100
- C. 5.20×105
- D. 5.20×10^{-5}

Answer(s): C

QUESTION: 1038

If $0.08/x = 1$, then $x =$ _____.

- A. 8
- B. 80
- C. 0.8
- D. 0.08

Answer(s): D

QUESTION: 1039

A circle has a radius of 8 inches.

What is its approximate area?

- A. 200.9 square inches
- B. 25.1 square inches
- C. 50.2 square inches
- D. 64 square inches

Answer(s): A

QUESTION: 1040

If $y^3 + x^3 = y^3 + p^3$, then $x =$ _____.

- A. p
- B. y
- C. $x^3 - y^3$
- D. $y^3 - x^3$

Answer(s): A

QUESTION: 1041

If $-7x = 49$, then $x =$ _____.

- A. 7
- B. 0
- C. -7
- D. 14

Answer(s): C

QUESTION: 1042

A square box has 10-inch sides.
What is its volume?

- A. 1,000 cubic inches
- B. 10 cubic inches
- C. 100 cubic inches
- D. 10,000 cubic inches

Answer(s): A

QUESTION: 1043

A cube has a volume of 512 cubic inches.
What is the length of one side of the cube?

- A. 8 inches
- B. 64 inches
- C. 128 inches

D. 16 inches

Answer(s): A

QUESTION: 1044

$(x5)3 = \underline{\hspace{2cm}}$.

- A. $x25$
- B. $x15$
- C. $x8$
- D. $2x5$

Answer(s): B

QUESTION: 1045

If y inches of rain falls in one minute, how many inches will fall in p hours?

- A. $60y/p$
- B. $60y$
- C. $60py$
- D. $60p$

Answer(s): C

QUESTION: 1046

$(x + 6)(4x + 8) = \underline{\hspace{2cm}}$.

- A. $4x^2 + 14x + 48$
- B. $4x^2 + 32x + 14$
- C. $4x^2 + 32x + 32$
- D. $4x^2 + 32x + 48$

Answer(s): D

QUESTION: 1047

Solve for x :

$$6(x + 5) + 8 = 2$$

- A. 6
- B. 36
- C. -6
- D. -2

Answer(s): C

QUESTION: 1048

The average of 80, 46, 32, and 22 is _____.

- A. 48
- B. 42
- C. 36
- D. 45

Answer(s): D

QUESTION: 1049

4 feet, 8 inches + 6 feet, 10 inches = how many inches?

- A. 142 inches
- B. 138 inches
- C. 148 inches
- D. 160 inches

Answer(s): B

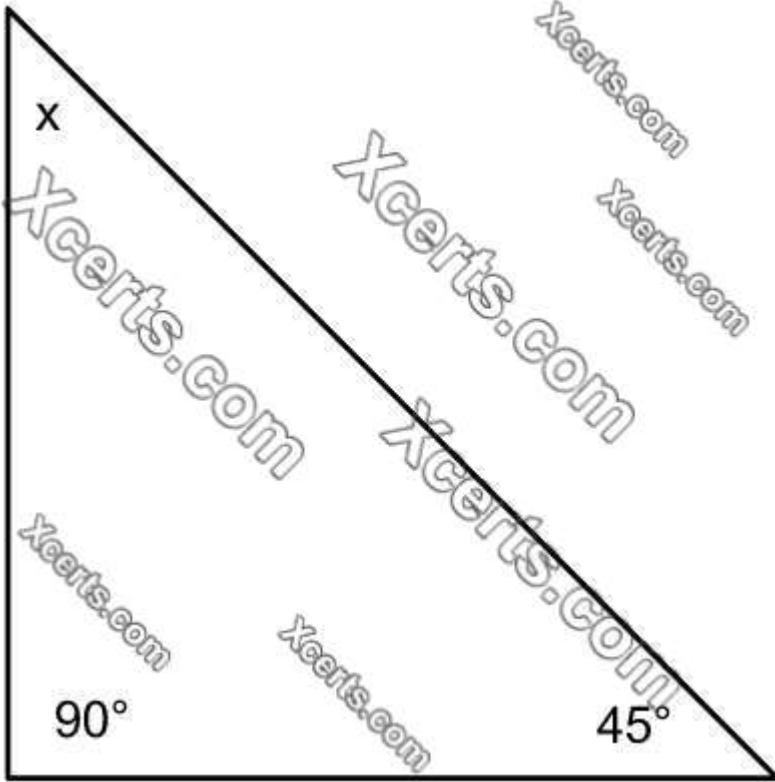
QUESTION: 1050

If $(8) \times 2 = 128$, then x is _____.

- A. either a positive or negative number
- B. a negative number
- C. a prime number
- D. a positive number

Answer(s): A

QUESTION: 1051

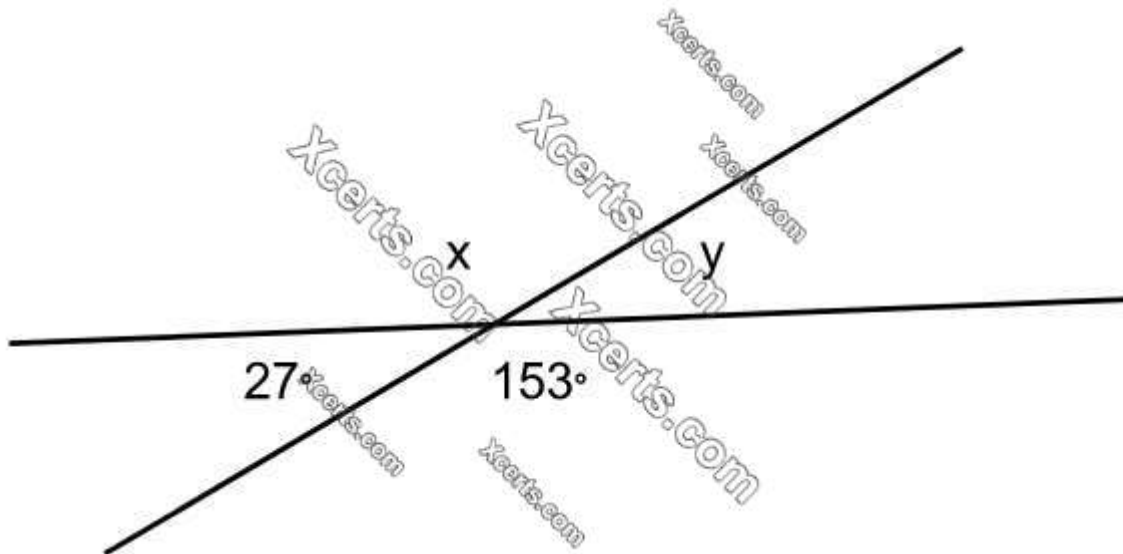


In the attached diagram what is the value of x ?

- A. 45°
- B. 90°
- C. 60°
- D. 15°

Answer(s): A

QUESTION: 1052

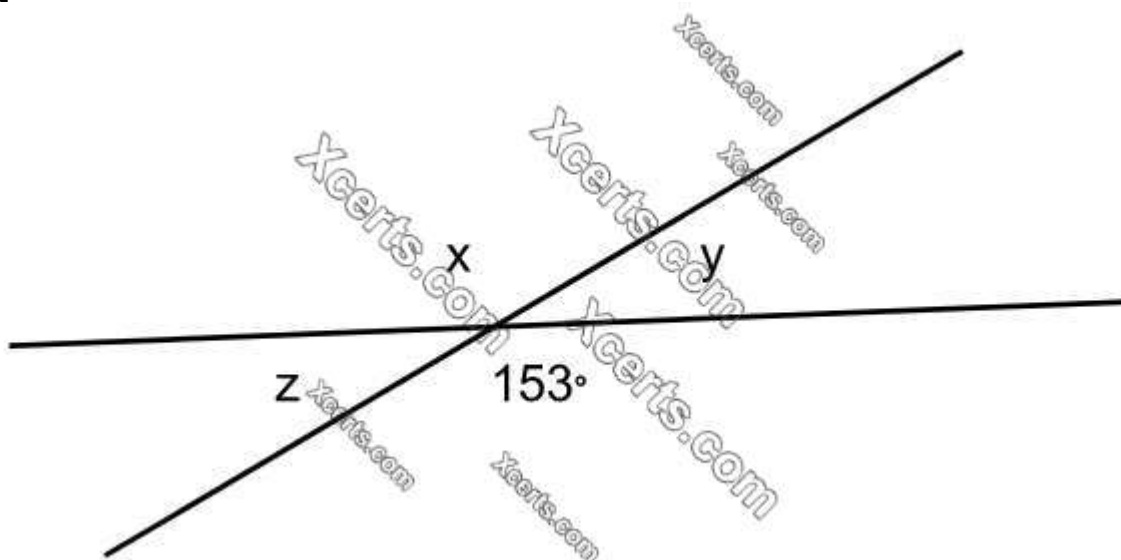


In the diagram provided, what is the value of x ?

- A. 153°
- B. 333°
- C. 27°
- D. 207°

Answer(s): A

QUESTION: 1053

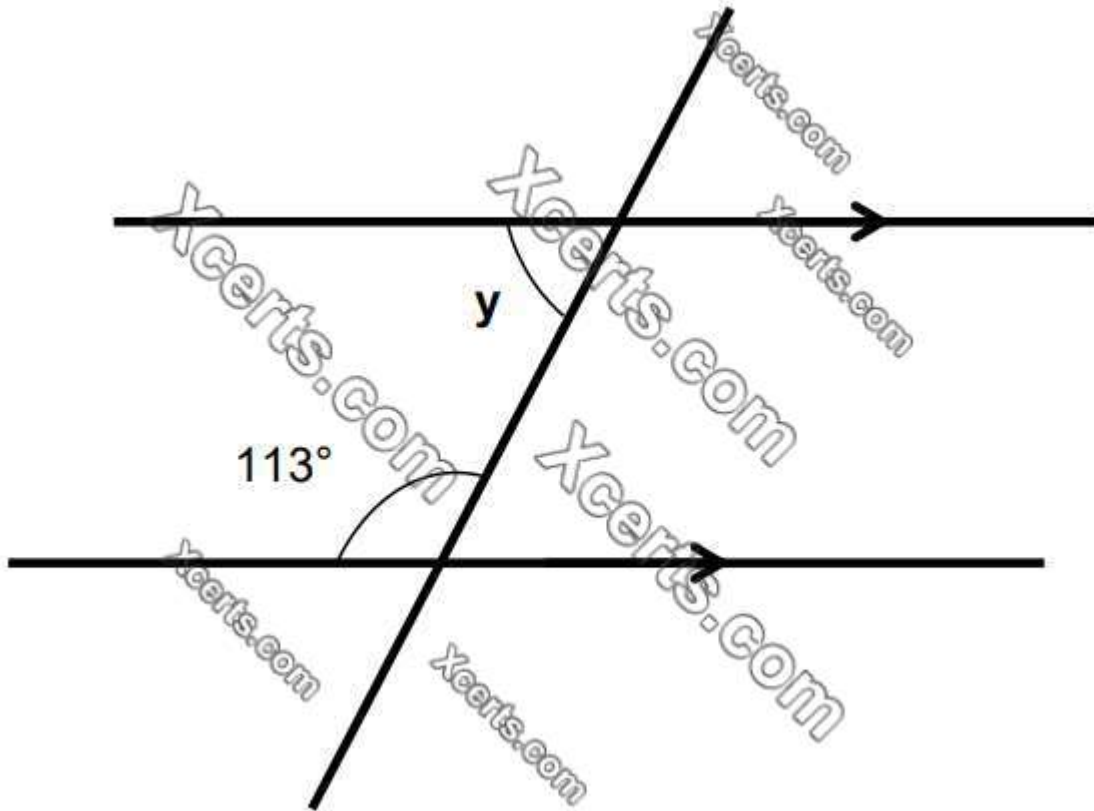


In the diagram provided, what is the value of y ?

- A. 153°
- B. 126°
- C. 27°
- D. 180°

Answer(s): C

QUESTION: 1054

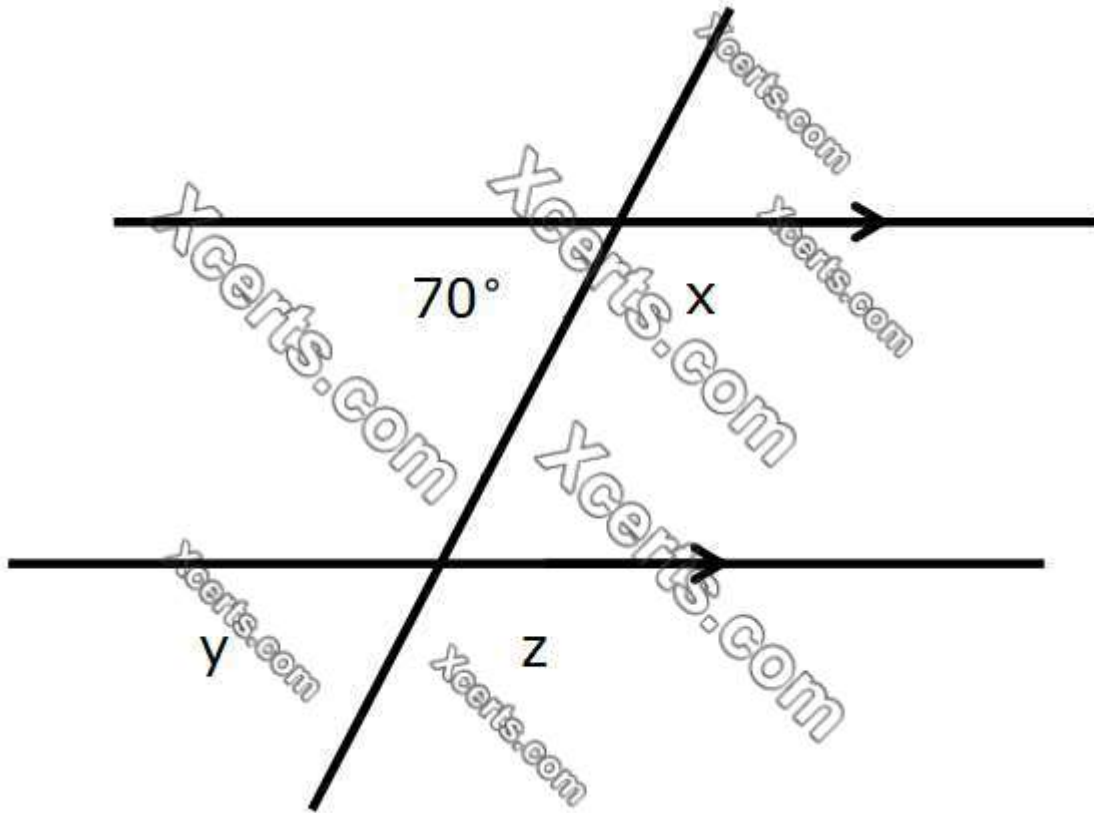


Given the diagram of parallel lines, what is the value of y ?

- A. 247°
- B. 67°
- C. 46°
- D. 293°

Answer(s): B

QUESTION: 1055



In the diagram of parallel lines provided, what is the measure of angle z ?

- A. 180°
- B. 90°
- C. 110°
- D. 70°

Answer(s): C

QUESTION: 1056



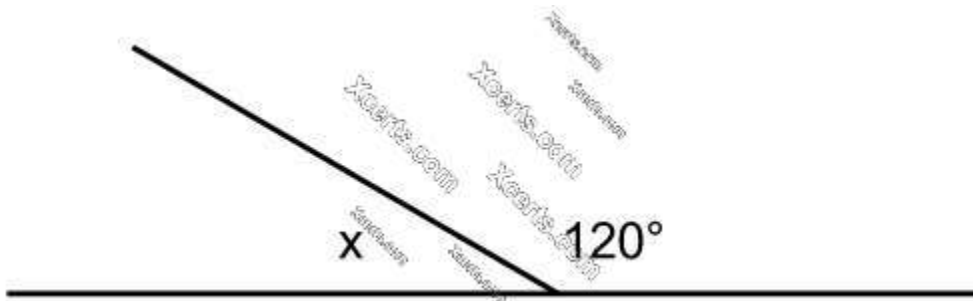
Use the diagram provided as a reference.

If the length between point A and C is 43, and the length between point A and B is 15, what is the length between point B and C?

- A. 18
- B. 33
- C. 58
- D. 28

Answer(s): D

QUESTION: 1057

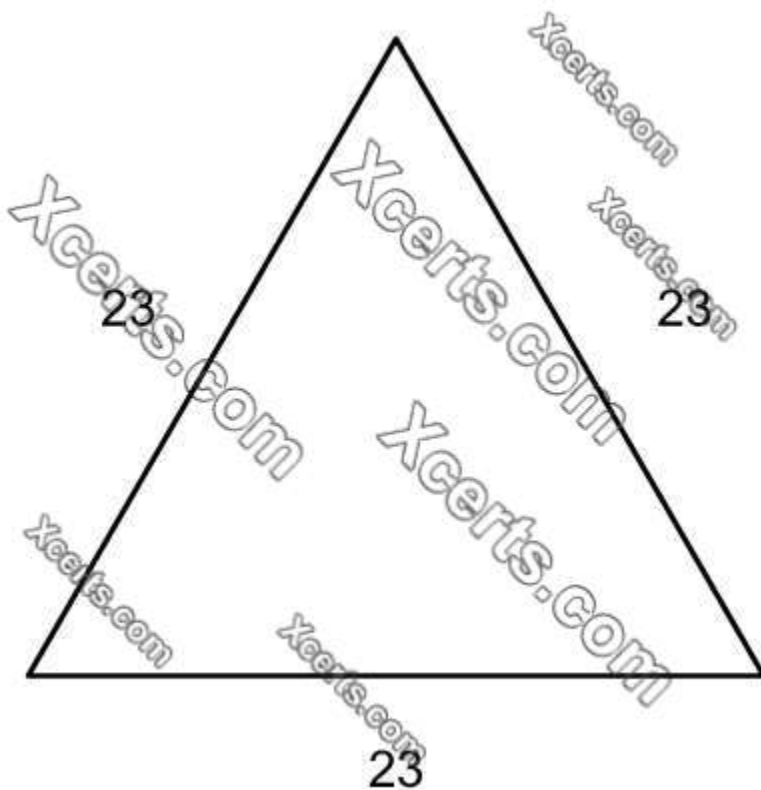


In the diagram, the straight line is divided by one angled line at 120° . Solve for the value of x .

- A. 60°
- B. 90°
- C. 120°
- D. 180°

Answer(s): A

QUESTION: 1058



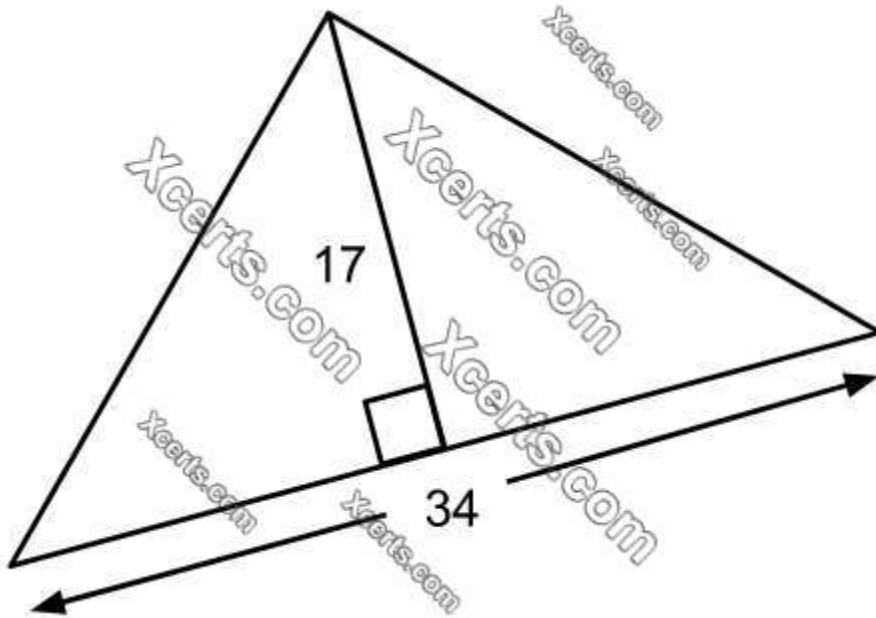
What is the perimeter of the triangle in the provided diagram?

- A. 529

- B. 23
- C. 69
- D. 12167

Answer(s): C

QUESTION: 1059

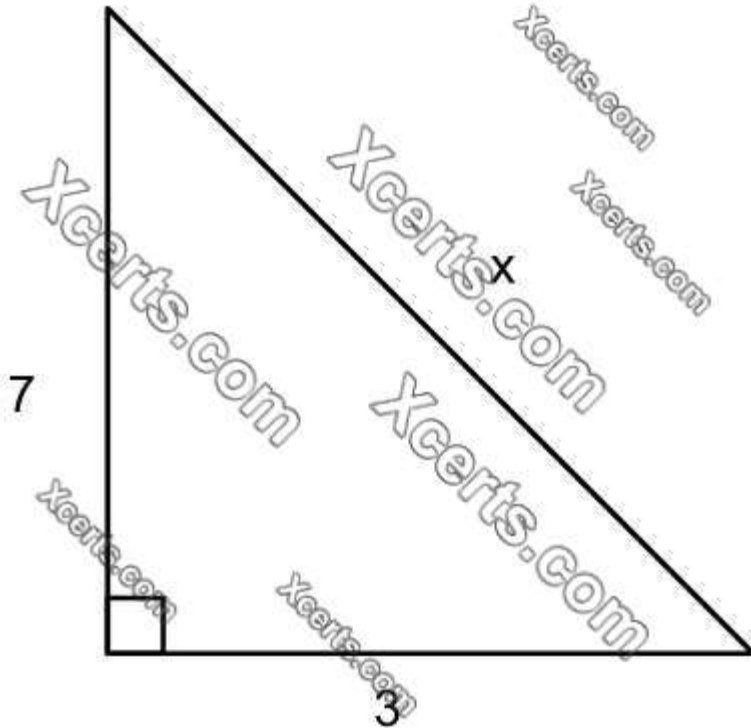


In the given diagram, what is the area of the triangle?

- A. 26
- B. 51
- C. 289
- D. 102

Answer(s): C

QUESTION: 1060

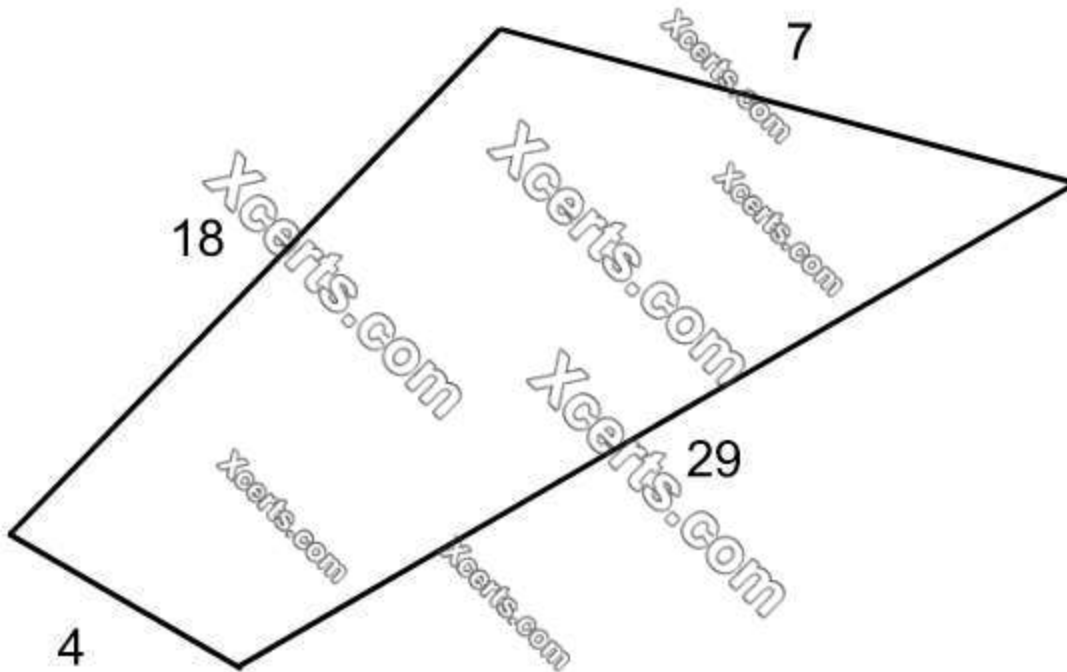


In the diagram provided, what is x ?

- A. 10.2893644
- B. 2.333333333
- C. 7.61577310586
- D. 21.23947643

Answer(s): C

QUESTION: 1061

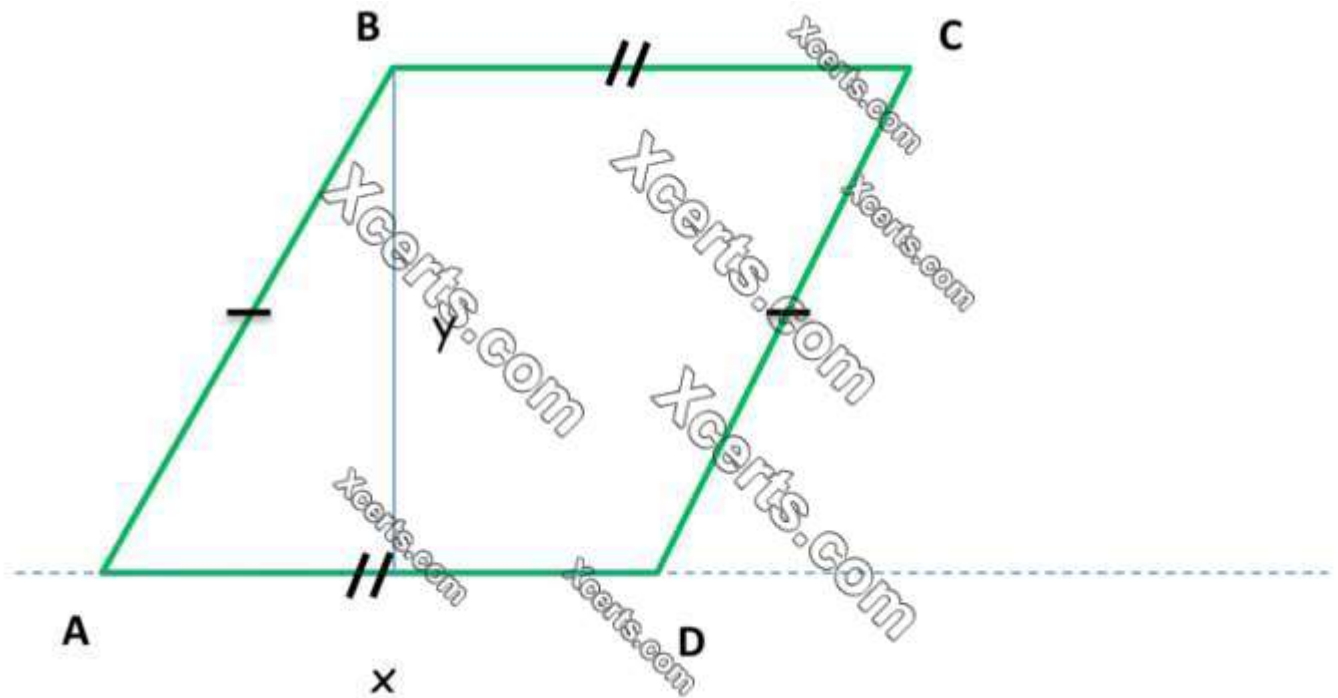


Given the diagram, what is the perimeter of the quadrilateral?

- A. 14616
- B. 28
- C. 517
- D. 58

Answer(s): D

QUESTION: 1062

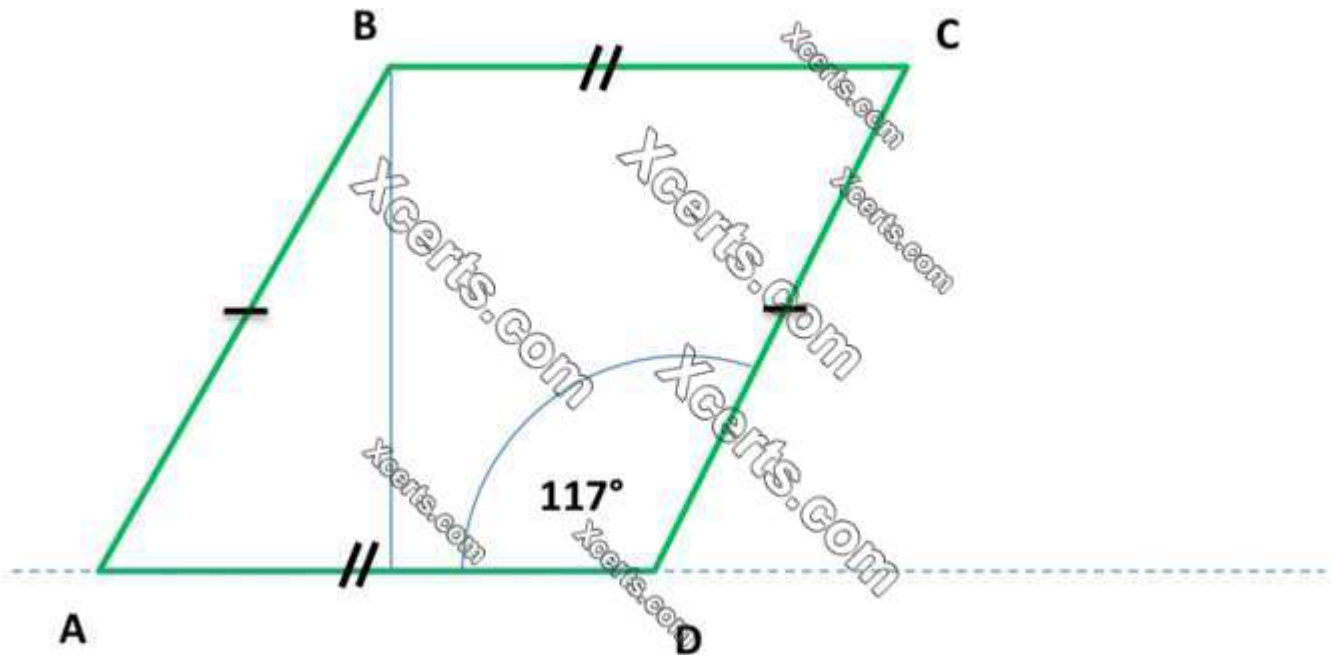


Given the diagram provided, what is the area of parallelogram ABCD if the height y is 7 and the base x is 5?

- A. 12
- B. 16807
- C. 35
- D. 74

Answer(s): C

QUESTION: 1063

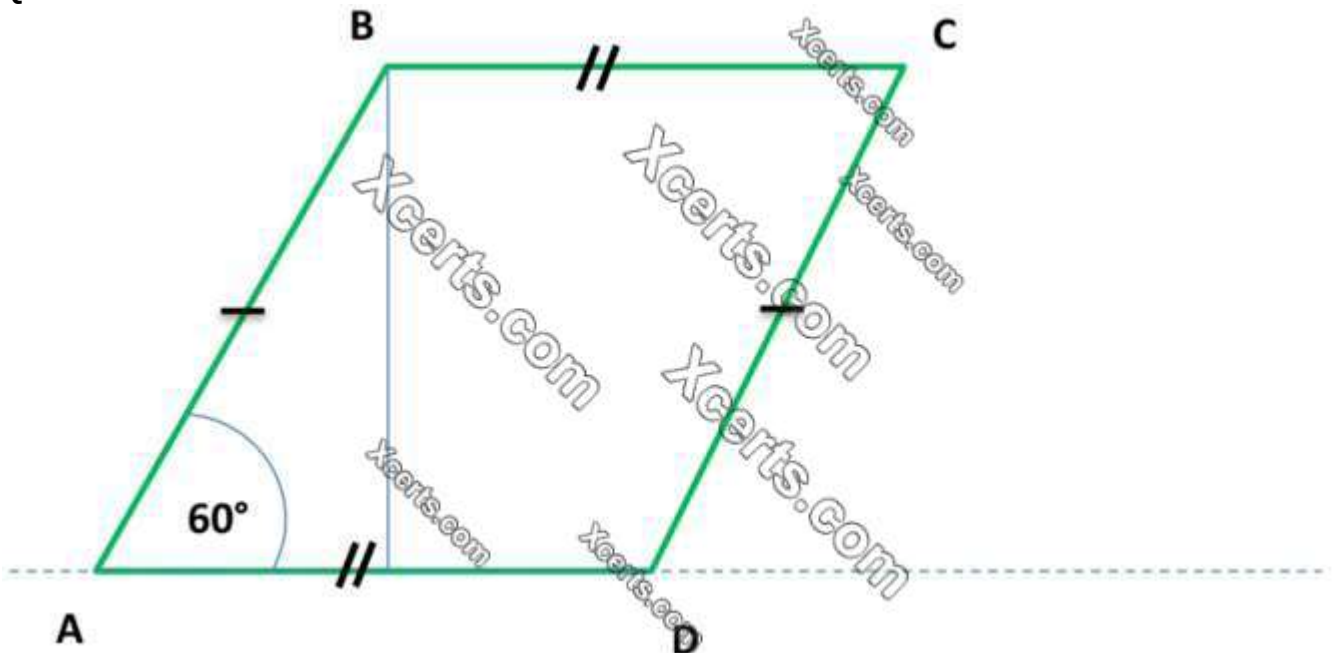


The diagram shows parallelogram ABCD.
What is the measure of angle A if angle D measures 117° ?

- A. 63°
- B. 117°
- C. 180°
- D. 297°

Answer(s): A

QUESTION: 1064

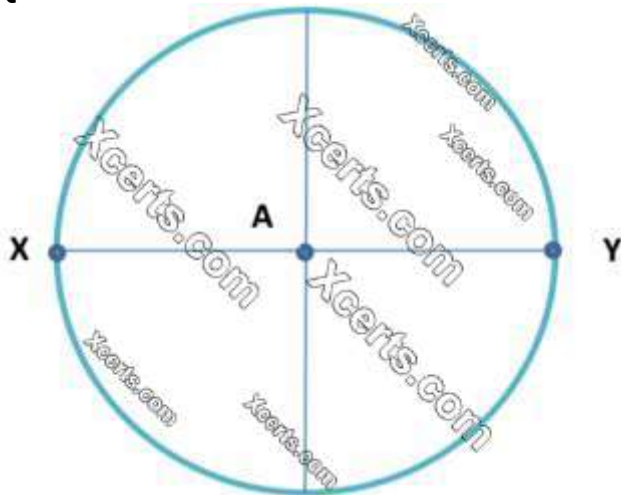


Given the diagram of parallelogram ABCD and the measure of A, what is the measurement of angle of C?

- A. 120°
- B. 90°
- C. 60°
- D. 240°

Answer(s): C

QUESTION: 1065

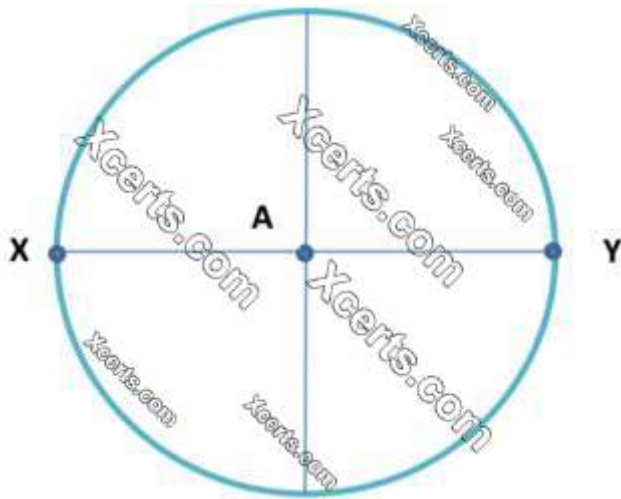


What is the circumference of a circle with center at point A if the distance from point X to Y is 42?

- A. 131.88
- B. 126.63
- C. 9.86
- D. 136.93

Answer(s): A

QUESTION: 1066

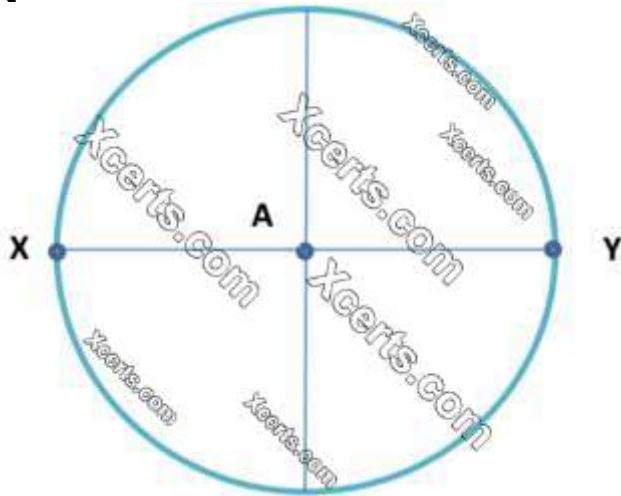


What is the diameter of the circle with center at point A if the distance from point A to point Y is 7?

- A. 14
- B. 7
- C. 21
- D. 49

Answer(s): A

QUESTION: 1067



What is the area of the circle with center at point A if the distance between X and Y is 1.62 and points X, A, and Y lie on a straight line? Round your answer to 2 decimal places.

- A. 2.060154
- B. 2.06
- C. 5.05
- D. 2.07

Answer(s): B

QUESTION: 1068

If 40 is divided by 0.08, the result is _____.

- A. 5000
- B. 500
- C. 50
- D. 5

Answer(s): B

Explanation:

40 divided by 0.08 = 500

QUESTION: 1069

What is 4 cubed?

- A. 12
- B. 81
- C. 128
- D. 64

Answer(s): D

Explanation:

4 cubed is $4 \times 4 \times 4 = 64$

QUESTION: 1070

$(x + 4)(x + 4) =$ _____.

- A. $x^2 + 16x + 8$
- B. $x^2 + 16x + 16$
- C. $x^2 + 8x + 16$
- D. $x^2 + 8x + 8$

Answer(s): C

Explanation:

Use the FOIL method (First, Outside, Inside, Last) to solve.

$x^2 + 4x + 4x + 4 \times 4$

QUESTION: 1071

Solve the following equation.

$y^3 \times y^4 =$ _____.

- A. y^7

- B. y^{12}
- C. $2y^7$
- D. $2y^{12}$

Answer(s): A

Explanation:

When multiplying two powers with the same base, you keep the base and add the exponents.

QUESTION: 1072

In a bag of candy, there are 4 Tootsie Rolls, 3 Lollipops, 5 Hershey Kisses and 6 Starbursts. If you select one piece of candy at random, what is the probability that it will be a lollipop?

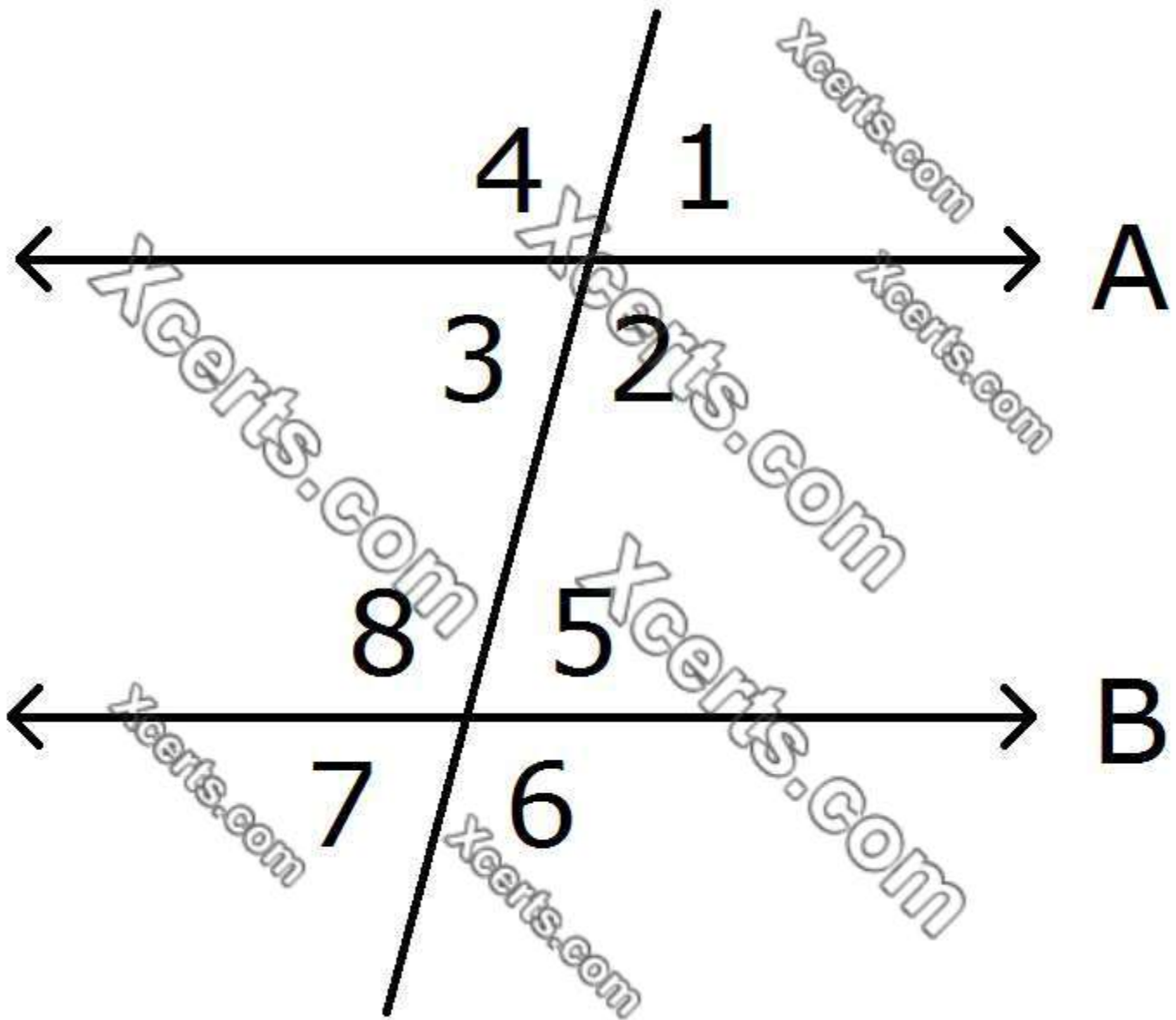
- A. $1/18$
- B. $3/10$
- C. $1/6$
- D. $1/4$

Answer(s): C

Explanation:

There are 18 total pieces of candy and 3 lollipops. Therefore, the probability of picking one lollipop is $3/18$ which simplifies to $1/6$.

QUESTION: 1073



Lines A and B are parallel. If Angle 4 is 100 degrees, then angle 1 is _____.

- A. 100 degrees
- B. 90 degrees
- C. 80 degrees
- D. Unknown

Answer(s): C

Explanation:

Two angles that form a straight line add up to 180 degrees. If Angle 4 is 100 degrees then Angle 1 must be 80 degrees.

QUESTION: 1074

A woman has Z dollars to spend. After she spends \$500, how much money does she have left?

- A. $500 - Z$
- B. $Z + 500$
- C. $500Z$
- D. $Z - 500$

Answer(s): D

Explanation:

You need to subtract the money that she spent (\$500) from the original money that she had (Z).

QUESTION: 1075

If $\frac{1}{3}$ of an office staff is on vacation and $\frac{1}{4}$ of those not on vacation are out sick, what fraction of the staff is in the office?

- A. $\frac{1}{2}$
- B. $\frac{1}{12}$
- C. $\frac{1}{6}$
- D. $\frac{1}{4}$

Answer(s): A

Explanation:

If $\frac{1}{3}$ are on vacation, then $\frac{2}{3}$ are not. Of those $\frac{2}{3}$ not on vacation, $\frac{3}{4}$ are NOT sick.

Therefore $\frac{2}{3} \times \frac{3}{4}$ are in the office (not on vacation and not sick) = $(2 \times 3)/(3 \times 4) = 6/12 = \frac{1}{2}$

QUESTION: 1076

If $0.05z = 1$, then $z =$ _____.

- A. 2
- B. 20
- C. 200
- D. 0.05

Answer(s): B

Explanation:

Divide both sides by 0.05. Therefore, $z = 1/0.05 = 20$

QUESTION: 1077

A cube has a volume of 64 cubic feet.

The surface area of the cube is _____.

- A. 64 square feet
- B. 384 square feet
- C. 96 square feet
- D. 128 square feet

Answer(s): C

Explanation:

A cube is made up of equal length sides (s). The volume of a cube is $s \times s \times s$. Therefore, $s = 4$ in this case ($4 \times 4 \times 4 = 64$). A cube is made up of 6 squares, each with an area of $s \times s$. Therefore, the surface area of a cube is $6 \times s \times s = 6 \times 4 \times 4 = 96$

QUESTION: 1078

A stack of sports cards contains 3 baseball cards, 3 football cards and 6 hockey cards. If one card is picked at random, what is the probability that it is a hockey card?

- A. $\frac{1}{3}$
- B. $\frac{1}{6}$
- C. $\frac{1}{4}$
- D. $\frac{1}{2}$

Answer(s): D

Explanation:

There are 12 cards in total. Therefore, the probability of picking a hockey card is $\frac{6}{12}$ or $\frac{1}{2}$.

QUESTION: 1079

If $x + y + 15 = 30$ and $x + y = z$, then $40 - z =$ _____.

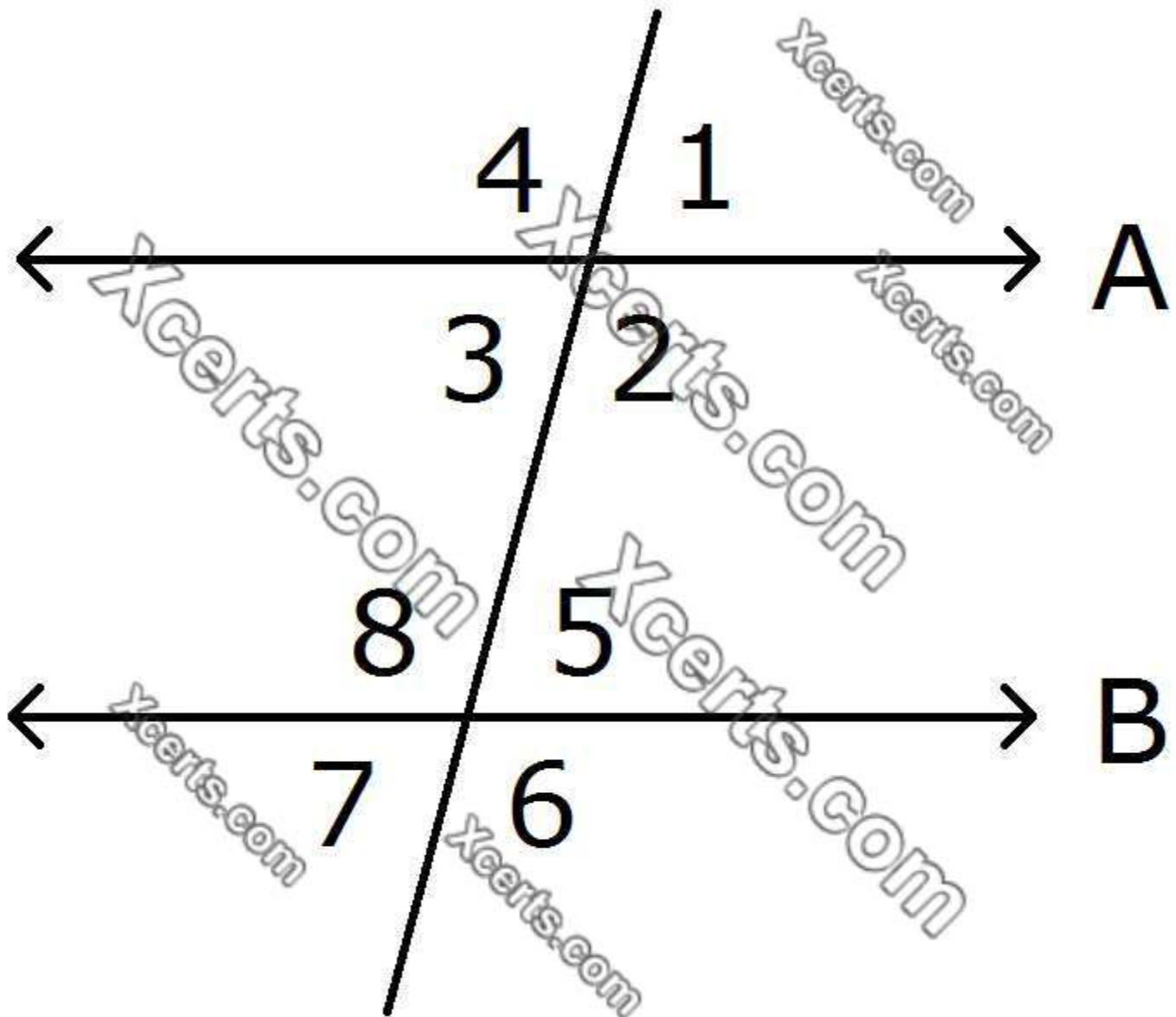
- A. 25
- B. 15
- C. 30
- D. 20

Answer(s): A

Explanation:

If $x + y + 15 = 30$, then $z + 15 = 30$ (substitute $x + y = z$) which means that $z = 15$. If $z = 15$, then $40 - z = 40 - 15 = 25$

QUESTION: 1080



Lines A and B are parallel.
If Angle 1 is 70 degrees, then angle 8 is _____.

- A. 70 degrees
- B. Unknown
- C. 110 degrees
- D. 140 degrees

Answer(s): C

Explanation:

If angle 1 is 70 degrees, then Angle 4 is 110 degrees since they combine to make a straight line. Since Lines A and B are parallel, Angles 4 and 8 are the same. Therefore, Angle 8 is 110 degrees.

Section 6: Paragraph Comprehension

QUESTION: 1081

An important stage of personal time management is to take control of appointments. Determined by external obligation, appointments constitute interaction with other people and an agreed-on interface between your activities and those of others. Start with a simple appointment diary. List all appointments including regular and recurring ones. Now, be ruthless and eliminate the unnecessary. There may be committees where you can't productively contribute or where a subordinate may be able to participate. Eliminate the waste of your time. Effectively managing your appointments allows you to _____.

- A. spend more time with your subordinates
- B. delegate responsibility to subordinates
- C. make more efficient use of your time
- D. attend only the most important meetings

Answer(s): C

Explanation:

Effective appointment management eliminates the waste of your time, as explained in the last sentence of the passage.

QUESTION: 1082

The U.S. Congress consists of 100 senators and 435 representatives. Two senators are elected from each state. The number of representatives from each state is based on population, although each state has at least one representative. Senators serve six-year terms and representatives serve two-year terms. According to this passage, _____.

- A. there are an equal number of senators and representatives
- B. the number of representatives from each state is decided by a lottery
- C. it's possible for a state to have no representatives
- D. senators and representatives have different term lengths

Answer(s): D

Explanation:

The passage gives the numbers of senators and representatives, so Choice "there are an equal number of senators and representatives" is incorrect.

The passage states that each state's population determines the number of representatives a state has, so Choice "the number of representatives from each state is decided by a lottery" is incorrect.

As stated in the passage, each state has at least one representative, so Choice "it's possible for a state to have no representatives" is incorrect.

QUESTION: 1083

Indo-European languages consist of those languages spoken by most of Europe and in those parts of the world that Europeans have colonized since the 16th century (such as the United

States). Indo-European languages are also spoken in India, Iran, parts of western Afghanistan, and in some areas of Asia.

The author of this passage would agree that _____.

- A. Indo-European languages are spoken in areas all over the world
- B. Indo-European languages include all the languages spoken in the world
- C. only Europeans speak Indo-European languages
- D. Indo-European language speakers can easily understand one another

Answer(s): A

Explanation:

Many languages are excluded from the Indo-European language group, so Choice "Indo-European languages include all the languages spoken in the world" is incorrect.

Indians, Iranians, Asians, and Afghans aren't Europeans, so Choice "only Europeans speak Indo-European languages" is incorrect.

The passage gives no evidence to support Choice "Indo-European language speakers can easily understand one another", which isn't true.

QUESTION: 1084

In privatization, the government relies on the private sector to provide a service. However, the government divests itself of the entire process, including all assets. With privatized functions, the government may specify quality, quantity, and timeliness requirements, but it has no control over the operations of the activity. Also, the government may not be the only customer. Whoever the government chooses to provide the services would likely provide the same services to others.

This paragraph best supports the statement that _____.

- A. the government must closely supervise privatized functions
- B. privatized functions consist of a mixture of government employees, military personnel, and private contractors
- C. privatized functions are those institutions that provide services only to a government agency
- D. privatized functions provide essential services to the government

Answer(s): D

Explanation:

Privatized functions operate independently of the government, making Choices "the government must closely supervise privatized functions" and "privatized functions consist of a mixture of government employees, military personnel, and private contractors" incorrect.

The passage states that privatized functions may sell goods and services to other customers as well as the government, so Choice "privatized functions are those institutions that provide services only to a government agency" is also incorrect.

Choice "privatized functions provide essential services to the government" is the correct answer, as privatized functions do perform essential services to government agencies.

QUESTION: 1085

The success or failure of a conference lies largely with its leader. A leader's zest and enthusiasm must be real, apparent, and contagious. The leader is responsible for getting the

ball rolling and making the attendees feel as if the meeting is theirs and its success depends on their participation. A good, thorough introduction helps establish the right climate.

A good title to this paragraph would be _____.

- A. Lead by Example
- B. The Importance of Proper Introductions
- C. Leading a Successful Conference
- D. Conference Participation Basics

Answer(s): C

Explanation:

One can assume that causing rain or snow would end a drought.

Nothing in the passage has to do with expense, so Choice "cloud seeding is prohibitively expensive" is incorrect.

The passage says nothing about how frequently the process is used, so Choice "cloud seeding is rarely used" is incorrect.

The passage specifies that dry ice (solid carbon dioxide) is used; solid water (regular ice) is a different chemical, so Choice "cloud seeding can be accomplished by using regular ice" is wrong.

QUESTION: 1086

Cloud seeding is accomplished by dropping particles of dry ice (solid carbon dioxide) from a plane onto super- cooled clouds. This process encourages condensation of water droplets in the clouds, which usually, but not always, results in rain or snow.

From this passage, it's reasonable to assume that _____.

- A. cloud seeding could be used to end a drought
- B. cloud seeding is prohibitively expensive
- C. cloud seeding is rarely used
- D. cloud seeding can be accomplished by using regular ice

Answer(s): A

Explanation:

Choices "Written records are important as they provide detailed documentation." and "More business people should invest time and energy improving their writing skills." may be true in certain situations, but they're not the point of this particular paragraph.

The passage doesn't say anything about working to improve writing skills being a waste of time, so Choice " It's pointless for business people to spend time improving their writing skills " is incorrect.

The main point of the paragraph is that writing may not be the most efficient way of communicating, depending on the situation.

QUESTION: 1087

To write or not to write - that is the question. If assigned a writing task, there's no option.

However, if someone is looking for a specific answer, find out if they need a short answer or a detailed one. Can the requirement be met with a telephone call, e-mail, or short note, or is something more necessary? A former CEO of a major corporation once commented that he had

looked at 13,000 pieces of paper in a 5-day period. Think how much easier and more economical it would be if people would use the telephone, send an e-mail, or write a short note. The main point of this passage is _____.

- A. Written records are important as they provide detailed documentation.
- B. More business people should invest time and energy improving their writing skills.
- C. Writing may not be the best way to communicate information.
- D. It's pointless for business people to spend time improving their writing skills.

Answer(s): C

Explanation:

Choices "Written records are important as they provide detailed documentation." and "More business people should invest time and energy improving their writing skills." may be true in certain situations, but they're not the point of this particular paragraph.

The passage doesn't say anything about working to improve writing skills being a waste of time, so Choice "It's pointless for business people to spend time improving their writing skills" is incorrect.

The main point of the paragraph is that writing may not be the most efficient way of communicating, depending on the situation.

QUESTION: 1088

The transistor, a small, solid-state device that can amplify sound, was invented in 1947. At first, it was too expensive and too difficult to produce to be used in cheap, mass-market products. By 1954, though, these cost and production problems had been overcome, and the first transistor radio was put on the market.

According to this passage, _____.

- A. there was no market for transistors before 1954
- B. when transistors could be produced cheaply and easily, the transistor radio was put on the market
- C. transistors were invented in 1947 by order of the Department of Defense
- D. transistors are still expensive to produce

Answer(s): B

Explanation:

Products with transistors weren't widely sold before 1954 because of the expense and difficulty of production, not because markets didn't exist; so Choice "there was no market for transistors before 1954" is incorrect.

Choices "transistors were invented in 1947 by order of the Department of Defense" and "transistors are still expensive to produce" aren't supported in the passage.

The passage states that the problem of transistors being expensive and difficult to produce was solved by 1954.

QUESTION: 1089

I returned from the City about three o'clock on that May afternoon pretty well disgusted with life. I had been three months in the Old Country and was fed up with it. If people had told me a year ago that I would've been feeling like that I should've laughed at them; but there was the fact.

The weather made me liverish, the talk of the ordinary Englishman made me sick, I couldn't get enough exercise, and the amusements of London seemed as flat as soda water that had been standing in the sun.

The author is speaking of his travels in _____.

- A. Spain
- B. Great Britain
- C. Germany
- D. Scotland

Answer(s): B

Explanation:

The words London and Englishmen make it clear that the author is speaking of his travels in England (Great Britain).

QUESTION: 1090

Surveys show that the average child under the age of 18 watches four hours of television per day. Although some of the programming may be educational, most isn't. Spending this much time watching television interferes with a child's ability to pursue other interests, such as reading, participating in sports, and playing with friends.

The author of this passage would agree that _____.

- A. television viewing should be restricted
- B. parents who let their children watch this much television is neglectful
- C. reading, participating in sports, playing with friends, and watching television should all be given equal time
- D. adults over 18 can watch as much television as they want

Answer(s): A

Explanation:

The author makes no reference to parents in the passage, so Choice "parents who let their children watch this much television is neglectful" is incorrect.

The author doesn't imply anything about all these interests requiring equal time, so Choice "reading, participating in sports, playing with friends, and watching television should all be given equal time" is incorrect.

The passage is about children under 18; no conclusion can be drawn about what the author thinks people over 18 should do, so Choice "adults over 18 can watch as much television as they want" is incorrect.

QUESTION: 1091

High-school and college graduates attempting to find jobs should participate in mock job interviews. These mock interviews help students prepare for the types of questions they'll be asked, make them more comfortable with common interview formats, and help them critique their performance before facing a real interviewer. Because they're such a valuable aid, schools should organize mock job interviews for all of their graduating students.

The above passage states that mock job interviews _____.

- A. frighten students
- B. should be offered to the best students
- C. help prepare students for real job interviews
- D. should be organized by students

Answer(s): C

Explanation:

The passage doesn't say anything about mock job interviews being frightening, so Choice "frighten students" is wrong.

The passage says that mock job interviews should be available to all students, so Choice "should be offered to the best students" is wrong.

The passage says that schools, not students, should organize mock interviews, so Choice "should be organized by students" is incorrect.

QUESTION: 1092

High-school and college graduates attempting to find jobs should participate in mock job interviews. These mock interviews help students prepare for the types of questions they'll be asked, make them more comfortable with common interview formats, and help them critique their performance before facing a real interviewer. Because they're such a valuable aid, schools should organize mock job interviews for all of their graduating students.

From the above passage, it is reasonable to assume that _____.

- A. mock interviews can increase a student's confidence when he or she goes into a real job interview
- B. mock interviews are expensive to organize
- C. few students are interested in mock interviews
- D. students don't need job interview preparation

Answer(s): A

Explanation:

Choices "mock interviews are expensive to organize", "few students are interested in mock interviews", "students don't need job interview preparation" are the opposite of what the paragraph states and implies.

QUESTION: 1093

Due process, the guarantee of fairness in the administration of justice, is part of the 5th Amendment to the U.S. Constitution. The 14th Amendment further requires states to abide by due process. After this amendment was enacted, the U.S. Supreme Court struck down many state laws that infringed on the civil rights guaranteed to citizens in the Bill of Rights.

According to the above passage, due process _____.

- A. is an outdated concept
- B. guarantees fairness in the justice system
- C. never became part of the U.S. Constitution
- D. is the process by which winning lottery tickets are selected

Answer(s): B

Explanation:

Nothing in the paragraph supports Choice "is an outdated concept", which is incorrect. When an amendment is passed, it becomes part of the Constitution, so Choice "never became part of the U.S.

Constitution" is incorrect.

The passage doesn't support Choice "is the process by which winning lottery tickets are selected".

QUESTION: 1094

Due process, the guarantee of fairness in the administration of justice, is part of the 5th Amendment to the U.S. Constitution. The 14th Amendment further requires states to abide by due process. After this amendment was enacted, the U.S. Supreme Court struck down many state laws that infringed on the civil rights guaranteed to citizens in the Bill of Rights. According to the above passage, it's reasonable to assume that the 5th Amendment _____.

- A. is about taxes
- B. guarantees due process in all criminal and civil cases
- C. guarantees due process in federal law
- D. should never have become part of the Bill of Rights

Answer(s): C

Explanation:

Because the 14th Amendment guarantees due process in states' laws, the 5th Amendment must guarantee due process only in federal law.

Nothing in the passage implies that the 5th Amendment is about taxes, so Choice "is about taxes" is wrong.

Because the passage states that the 14th Amendment had to be enacted to require states to abide by due process, Choice "guarantees due process in all criminal and civil cases" is incorrect.

Choice "should never have become part of the Bill of Rights" is neither stated nor implied in the passage.

QUESTION: 1095

Due process, the guarantee of fairness in the administration of justice, is part of the 5th Amendment to the U.S. Constitution. The 14th Amendment further requires states to abide by due process. After this amendment was enacted, the U.S. Supreme Court struck down many state laws that infringed on the civil rights guaranteed to citizens in the Bill of Rights.

The author of the above passage would agree that _____.

- A. without the passage of the 14th Amendment, many laws restricting civil rights would still exist in various states
- B. the Supreme Court overstepped its jurisdiction when it struck down laws infringing on citizens' civil rights
- C. the Supreme Court had every right to strike down state laws before the passage of the 14th Amendment
- D. the 14th Amendment was opposed by all states

Answer(s): A

Explanation:

Because the Supreme Court struck down many state laws after the 14th Amendment was enacted, it's probably true that these laws would still exist if there had been no 14th Amendment.

The passage doesn't support Choices "the Supreme Court overstepped its jurisdiction when it struck down laws infringing on citizens' civil rights", "the Supreme Court had every right to strike down state laws before the passage of the 14th Amendment" or "the Supreme Court had every right to strike down state laws before the passage of the 14th Amendment."

QUESTION: 1096

Scientists believe that a black hole is created when a supernova from a large star collapses on itself. This collapse causes a gravitational field that grows more and more intense until nothing can escape from its pull, not even light. It's thought that the universe may end as a black hole. According to this passage, _____.

- A. a black hole emits light
- B. a supernova is a black hole
- C. the gravitational field of a black hole allows nothing to escape
- D. the universe was created by a black hole

Answer(s): C

Explanation:

Nothing escapes from a black hole, including light, so Choice "a black hole emits light" is incorrect.

A black hole occurs when a supernova collapses, but they're not the same thing, so Choice "a supernova is a black hole" is incorrect.

The passage states that the universe might end as a black hole, not that a black hole created the universe, so Choice "the universe was created by a black hole" is incorrect.

QUESTION: 1097

Military customs and courtesies are proven traditions that explain what should and shouldn't be done in many situations. They are acts of respect and courtesy when dealing with other people and have evolved as a result of the need for order, as well as the mutual respect and sense of fraternity that exists among military personnel. Military customs and courtesies go beyond basic politeness; they play an extremely important role in building morale, esprit de corps, discipline, and mission effectiveness. Customs and courtesies ensure proper respect for the chain of command and build the foundation for self-discipline.

According to this passage, _____.

- A. military customs and courtesies are enforced by regulation
- B. military customs and courtesies are nothing more than basic politeness
- C. military customs and courtesies are the building blocks to self-discipline
- D. military customs and courtesies aren't applicable to the Coast Guard

Answer(s): C

Explanation:

The passage makes no reference about the enforceability of customs and courtesies, so Choice "military customs and courtesies are enforced by regulation" would be incorrect. According to the passage, military customs and courtesies go beyond basic politeness, making Choice "military customs and courtesies are nothing more than basic politeness" incorrect. Choice "military customs and courtesies aren't applicable to the Coast Guard" is incorrect because the passage doesn't reference any individual military service. The last sentence of the passage contains the correct answer.

QUESTION: 1098

The Panama Canal is a ship canal that cuts through the Isthmus of Panama, connecting the Atlantic and Pacific oceans. Although several foreign companies tried to build the canal throughout the 19th century, none were successful. After the U.S. helped Panama revolt against Columbia, the U.S. was given rights to the land the canal occupied. The U.S. government finished the canal in 1914.

According to this passage, _____.

- A. Panama and Columbia fought a war over the Panama Canal
- B. the U.S. was given rights to the canal land
- C. foreign companies built the canal before the U.S. stepped in
- D. Panama built the canal in 1914

Answer(s): B

Explanation:

The passage states that Panama revolted against Columbia, not that they fought over the canal, so Choice "Panama and Columbia fought a war over the Panama Canal" is incorrect. The passage states that the foreign companies were unsuccessful in building the canal, so Choice "foreign companies built the canal before the U.S. stepped in" is incorrect. The United States, not Panama, built the canal, so Choice "Panama built the canal in 1914" is wrong.

QUESTION: 1099

Extreme care must be exercised to ensure proper handling and cleaning of soiled U.S. flags. A torn flag may be professionally mended, but a badly torn or tattered flag should be destroyed. When the flag is in such a condition that it's no longer a fitting emblem for display, destroy it in a dignified manner, preferably by burning.

According to this passage, torn flags should be _____.

- A. mended
- B. burned
- C. destroyed
- D. all of the above

Answer(s): D

Explanation:

According to the passage a torn U.S. Flag can be professionally mended, but a severely torn

flag should be destroyed. The preferred method of destruction is by burning.

QUESTION: 1100

Medieval guilds were similar to modern-day labor unions. These groups of merchants or craftspeople set rules regarding economic activity in order to protect themselves. Some guilds held considerable economic power, but even small guilds protected members. Guilds also served a social purpose.

According to this passage, guilds _____.

- A. had only one purpose
- B. had little in common with modern labor unions
- C. exploited workers
- D. held considerable economic power

Answer(s): D

Explanation:

The passage states that guilds had economic and social purposes, so Choice "had only one purpose" is incorrect.

The passage states that guilds were similar to labor unions, so Choice "had little in common with modern labor unions" is incorrect.

The passage states that guilds protected merchants and craftspeople; it says nothing about exploiting workers, so Choice "exploited workers" is incorrect.

QUESTION: 1101

After a series of well-publicized failures by various inventors, Orville and Wilbur Wright succeeded in flying and controlling a heavier than-air craft on December 17, 1903. The War Department, stung by its investment in a failed effort by Samuel Langley and compounded by the Wright's own secretiveness, initially rejected the brothers' overtures toward the government to buy the aircraft. Prevailing sentiments held that the immediate future still belonged to the balloon. In August 1908, the two brothers delivered the first Army aircraft to the U.S. Government. That the U.S. government managed to purchase an airplane was a minor miracle. For more than four years after the Wright brothers' successful flight at Kitty Hawk, North Carolina, the government refused to accept the fact that man had flown in a heavier than air machine.

Which of the following statements is not supported by the above passage?

- A. The U.S. Government felt that balloons were more practical than airplanes.
- B. The Wright Brother's own secretiveness contributed to their problems in getting the government interested in their aircraft.
- C. The historic flight took place on the East Coast.
- D. It took more than six years for the Wright Brothers to interest the U.S. Government in their airplane.

Answer(s): D

Explanation:

According to the passage it took over four years for the Government to believe that anyone had flown a heavier- than-air craft.

The historic flight was in December 1903, and the Wright Brothers delivered the first aircraft to the government in August 1908, 4 1/2 years later.

All of the other statements are supported by the passage.

QUESTION: 1102

An ex-post-facto law is a law applied retroactively. Usually it makes an act illegal that was legal at the time it was performed. The U.S. Constitution prohibits ex-post-facto criminal laws, although the British Commonwealth allows them.

According to this passage, it can be assumed that _____.

- A. ex-post-facto laws are a problem in the United States
- B. laws applied retroactively are fair
- C. a person could be prosecuted for an act that was not a crime when he or she did it
- D. retroactive laws are illegal in the British Commonwealth

Answer(s): C

Explanation:

Because the U.S. Constitution prohibits these kinds of laws, Choice "ex-post-facto laws are a problem in the United States" is incorrect.

The passage doesn't support Choice "laws applied retroactively are fair" The passage states that the British Commonwealth allows ex-post-facto laws, so Choice "retroactive laws are illegal in the British Commonwealth" is incorrect.

QUESTION: 1103

Troy weight is based on a pound of 12 ounces and an ounce of 480 grains. Common, or avoirdupois, weight is based on a pound having 16 ounces and an ounce having 437.5 grains. A common pound has 7,000 grains while a troy pound has 5,760.

According to this passage, _____.

- A. a troy pound is smaller than a common pound
- B. a troy pound and a common pound are the same weight
- C. common weight and avoirdupois weight are different measures
- D. a troy ounce is smaller than a common ounce

Answer(s): A

Explanation:

The passage describes how troy and common weights are different, so Choice "a troy pound and a common pound are the same weight" is incorrect.

Common and avoirdupois are the same system, so Choice "common weight and avoirdupois weight are different measures" is incorrect.

A troy ounce is larger than a common ounce, so Choice "a troy ounce is smaller than a common ounce" is incorrect.

QUESTION: 1104

Good leaders get involved in their subordinates' careers. People merely obey arbitrary commands and orders, but they respond quickly and usually give extra effort for leaders who

genuinely care for them. An often neglected leadership principle in today's environment of technology and specialization is knowing the workers and showing sincere interest in their problems, career development, and welfare. Leadership is reflected in the degree of efficiency, productivity, morale, and motivation demonstrated by subordinates. Leadership involvement is the key ingredient to maximizing worker performance.

A key leadership principle that's often ignored is _____.

- A. leading by example
- B. showing sincere interest in the problems of the workers
- C. ensuring workers have access to the most modern technology
- D. maximizing worker performance

Answer(s): B

Explanation:

The passage doesn't address leading by example or use of technology by workers, so Choices "leading by example" and "ensuring workers have access to the most modern technology" would be incorrect.

Maximizing worker performance is a result of leadership involvement, not a principle of leadership, making Choice "maximizing worker performance" incorrect.

The correct answer is found in the third sentence of the passage.

QUESTION: 1105

Leukemia is a blood disease in which white blood cells in the blood or bone marrow reproduce rapidly, interfering with the body's ability to produce red blood cells. Red blood cells are needed to perform vital bodily functions.

According to this passage, _____.

- A. white blood cells perform no vital function in the body
- B. no treatment for leukemia exists
- C. leukemia makes it hard for the body to produce red blood cells
- D. white blood cells are found only in the blood

Answer(s): C

Explanation:

The passage doesn't support Choices "white blood cells perform no vital function in the body" or "no treatment for leukemia exists."

The passage states that white blood cells are found in blood and bone marrow, so Choice "white blood cells are found only in the blood" is wrong.

QUESTION: 1106

Any discussion of distinctive military capabilities would be incomplete without looking at their relationship to the Joint Service vision of the future. JV 2020 guides all the Services into the next century with its vision of future war fighting. JV 2020 sets forth four overarching operational concepts: dominant maneuver, precision engagement, focused logistics, and full-dimensional protection. Each of these operational concepts reinforces the others. The aggregate of these four concepts, along with their interaction with information superiority and innovation, allows joint forces to dominate the full range of military operations from humanitarian assistance

through peace operations to the highest intensity conflict.

According to the passage above, which of the following is not an operational concept?

- A. dominant maneuver
- B. focused logistics
- C. high intensity conflict
- D. precision engagement

Answer(s): C

Explanation:

High intensity conflict is listed as a type of military operation.

QUESTION: 1107

Any discussion of distinctive military capabilities would be incomplete without looking at their relationship to the Joint Service vision of the future. JV 2020 guides all the Services into the next century with its vision of future war fighting. JV 2020 sets forth four overarching operational concepts: dominant maneuver, precision engagement, focused logistics, and full-dimensional protection. Each of these operational concepts reinforces the others. The aggregate of these four concepts, along with their interaction with information superiority and innovation, allows joint forces to dominate the full range of military operations from humanitarian assistance through peace operations to the highest intensity conflict.

The document discussed in the above passage is primarily about _____.

- A. military operations of the past
- B. present military operations
- C. military operations in the future
- D. training for future military operations

Answer(s): C

Explanation:

The JV 2020 guides all of the military services with its vision of future war fighting.

While Choice "training for future military operations" would be close, the passage doesn't specifically reference military training.

QUESTION: 1108

Genetics is a branch of science dealing with heredity. The field is concerned with how genes operate and the way genes are transmitted to offspring. Subdivisions in the field include cytogenetics, which is the study of the cellular basis of inheritance; microbial genetics, the study of inheritance in microbes; molecular genetics, the study of the biochemical foundation of inheritance; and human genetics, the study of how people inherit traits that are medically and socially important. Genetic counselors are primarily concerned with human genetics. They advise couples and families on the chances of their offspring having specific genetic defects. In the passage above, cytogenetics is defined as _____.

- A. the study of the psychological impact of genetics
- B. the study of the cellular foundation of genetics
- C. the study of molecular genetics

D. the study of human genetics

Answer(s): B

Explanation:

Cytogenetics is the study of the cellular basis of inheritance; the text doesn't support Choices "the study of the psychological impact of genetics", "the study of molecular genetics" or "the study of human genetics."

QUESTION: 1109

Genetics is a branch of science dealing with heredity. The field is concerned with how genes operate and the way genes are transmitted to offspring. Subdivisions in the field include cytogenetics, which is the study of the cellular basis of inheritance; microbial genetics, the study of inheritance in microbes; molecular genetics, the study of the biochemical foundation of inheritance; and human genetics, the study of how people inherit traits that are medically and socially important. Genetic counselors are primarily concerned with human genetics. They advise couples and families on the chances of their offspring having specific genetic defects. According to the passage above, genetics _____.

- A. concerns how genes operate and how they're passed along
- B. is a field of study populated by quacks, fakes, and frauds
- C. is a field of study only concerned with human genetics
- D. is a new field of study

Answer(s): A

Explanation:

Nothing in the passage supports Choices "is a field of study populated by quacks, fakes, and frauds" or "is a new field of study."

Although human genetics is an important subfield of genetics, nothing in the passage suggests that it's the only concern of geneticists.

Microbial genetics, as mentioned in the passage, is a subfield in genetics that has nothing to do with humans, so Choice "is a field of study only concerned with human genetics" is incorrect.

QUESTION: 1110

Genetics is a branch of science dealing with heredity. The field is concerned with how genes operate and the way genes are transmitted to offspring. Subdivisions in the field include cytogenetics, which is the study of the cellular basis of inheritance; microbial genetics, the study of inheritance in microbes; molecular genetics, the study of the biochemical foundation of inheritance; and human genetics, the study of how people inherit traits that are medically and socially important. Genetic counselors are primarily concerned with human genetics. They advise couples and families on the chances of their offspring having specific genetic defects. According to the passage above, it's reasonable to assume that genetic counseling _____.

- A. is restricted to the very rich
- B. is used to diagnose diseases
- C. can be used by parents to learn if their offspring are likely to inherit a disease one of the parents has

D. can be used by parents to prevent their offspring from inheriting a specific genetic defect

Answer(s): C

Explanation:

Nothing in the passage supports Choices "is restricted to the very rich", "is used to diagnose diseases", or "can be used by parents to prevent their offspring from inheriting a specific genetic defect."

QUESTION: 1111

Because leadership is charged with bringing new ideas, methods, or solutions into use, innovation is inextricably connected with the process of being an effective leader. Innovation means change, and change requires leadership. Leaders must be the chief transformation officers in their organizations and learn everything there is to know about the change before it even takes place. Furthermore, they must learn how to deal with the emotions that result from the chaos and fear associated with change.

According to the passage, _____.

- A. leaders should resist making changes that subordinates are likely to resist
- B. innovation and change are distinctly different processes
- C. it's not necessary for the leader to know everything about a change before it's implemented
- D. change is often associated with panic and disorder

Answer(s): D

Explanation:

The passage states that leaders must learn to deal with negative emotions connected with change, making Choice "leaders should resist making changes that subordinates are likely to resist" incorrect.

The second sentence makes it clear that innovation means change, so Choice "innovation and change are distinctly different processes" is incorrect.

The third sentence clearly states that leaders must learn everything there is to know about the change, making Choice "it's not necessary for the leader to know everything about a change before it's implemented" the wrong choice.

QUESTION: 1112

Cougars are the most wide-ranging big cats in North America, inhabiting a wide variety of environments. A cougar, also called a puma or a mountain lion, lives about 18 years in the wild, can jump 20 feet (in distance) at a time, and can range 50 miles when on the prowl for food.

According to this passage, _____.

- A. a cougar isn't the same thing as a mountain lion
- B. cougars are an endangered species
- C. cougars live in many areas of North America
- D. cougars live only a few years in the wild

Answer(s): C

Explanation:

The passage states that pumas, mountain lions, and cougars are the same thing, so Choice "a cougar isn't the same thing as a mountain lion" is incorrect. Nothing in the passage supports Choice "cougars are an endangered species." The passage states that cougars live about 18 years in the wild, so Choice "cougars live only a few years in the wild" is incorrect.

QUESTION: 1113

Conan Doyle's Sherlock Holmes saga has enjoyed - or in some cases suffered - countless reinventions since its original publication from 1887 to 1927. The BBC's current television version starring Benedict Cumberbatch is perhaps one of the most successful, not least as its scriptwriters combine a deep knowledge of the original with a flair for departing wittily from it: the show's strategy of allusion, transformation, and up-to-dateness gives it both freshness and familiarity. The new series begins with "The Empty Hearse", its title alluding playfully to Conan Doyle's 1903 story "The Empty House" in which Holmes returns from apparent death at the hands of Professor Moriarty in Switzerland.

The word *flair*, as used in the above passage, most nearly means _____.

- A. dash
- B. aptitude
- C. style
- D. poise

Answer(s): B

QUESTION: 1114

Many small cities and towns rely on volunteer fire departments to put out fires. A professional fire department, however, has more training, more expertise, and more experience in fighting fires and investigating their causes. In many cases, it's worthwhile for even very small towns to hire professional firefighters.

According to this passage, it's reasonable to assume that _____.

- A. volunteer firefighters have less training, expertise, and experience than professional firefighters
- B. volunteer firefighters have the skills and resources to investigate the causes of fires
- C. professional firefighters don't know what causes fires
- D. a professional fire department is cost prohibitive for small towns

Answer(s): A

Explanation:

The passage says that professionals, not volunteers, have the skills needed to investigate fires, so Choice "volunteer firefighters have the skills and resources to investigate the causes of fires" is incorrect.

The passage states that professional firefighters have more experience investigating the causes of fires, so Choice "professional firefighters don't know what causes fires" is incorrect.

The passage states that hiring professional firefighters is worthwhile, so Choice "a professional fire department is cost prohibitive for small towns" is incorrect.

QUESTION: 1115

The idea being an alarming one, he scrambled out of bed and groped his way to the window. He was obliged to rub the frost off with the sleeve of his dressing-gown before he could see anything and could see very little then. All he could make out was that it was still very foggy and extremely cold and that there was no noise of people running to and fro and making a great stir, as there unquestionably would've been if night had beaten off bright day, and taken possession of the world.

This story takes place _____.

- A. in England
- B. on a calm summer evening
- C. on a winter night
- D. in England on a winter night

Answer(s): C

Explanation:

The passage doesn't state the locale of the story, so Choices "in England" and "in England on a winter night" are incorrect.

The references to extreme cold and lack of light makes Choice "on a calm summer evening" an incorrect answer.

QUESTION: 1116

Epidemiology is the study of what causes diseases, injuries, and other physiological damage to humans and why such problems occur. Epidemiologists examine where and when disease outbreaks occur. By using statistics and other scientific methods, epidemiologists determine what factors affect the frequency and severity of disease patterns. The primary goal of epidemiology

is to control or prevent outbreaks of disease - other goals are subordinate.

What would be the best title for this passage?

- A. Epidemiology: The Study of Disease Patterns.
- B. Goals for the Future of Epidemiology.
- C. Using Statistical Methods in Epidemiology.
- D. Employment Outlook for Epidemiologists.

Answer(s): A

Explanation:

The main point of the passage is to define epidemiology. Choices "Goals for the Future of Epidemiology," "Using Statistical Methods in Epidemiology," and "Employment Outlook for Epidemiologists" aren't the main points of the passage.

QUESTION: 1117

Buddhism is a religion that must be viewed from many angles. Its original form, as preached by Gautama in India and developed in the early years succeeding and as embodied in the sacred literature of early Buddhism, isn't representative of the actual Buddhism of any land today.

According to this passage, _____.

- A. most Buddhists live in India
- B. Buddhist teachings have changed over the years
- C. Buddhism draws its teachings from early Christianity
- D. Buddhist temples can be found in any land of the world

Answer(s): B

Explanation:

The only statement that's supported by the passage is Choice "Buddhist teachings have changed over the years."

In fact, this sentence is the primary theme of the passage.

The other choices aren't supported by information contained in the paragraph.

QUESTION: 1118

Many criminal-law statutes permit more severe punishment of a person convicted of a crime if he or she intended to harm another person. For example, voluntary manslaughter carries a heavier penalty than involuntary manslaughter in most states. Planned crimes are also punished more severely than spur-of-the-moment crimes. The problem is that juries find it difficult to know what the intent of a person was at the time he or she committed a crime. Many defendants will deny that they intended to harm the other person and claim that any harm that occurred was "accidental." The law asks too much of juries when it expects them to determine what a person was thinking. Juries should only be asked to weigh objective evidence.

The author of this passage would agree that _____.

- A. laws should not punish people based on intention
- B. juries aren't intelligent enough to weigh evidence
- C. more laws should distinguish between crimes committed with intent and crimes committed on the spur of the moment
- D. lawyers will lie about anything

Answer(s): A

Explanation:

Choice "juries aren't intelligent enough to weigh evidence" isn't supported by the passage.

Choice "more laws should distinguish between crimes committed with intent and crimes committed on the spur of the moment" is the opposite of what the author argues.

The text doesn't support Choice "lawyers will lie about anything".

QUESTION: 1119

Many criminal-law statutes permit more severe punishment of a person convicted of a crime if he or she intended to harm another person. For example, voluntary manslaughter carries a heavier penalty than involuntary manslaughter in most states. Planned crimes are also punished more severely than spur-of-the-moment crimes. The problem is that juries find it difficult to know what the intent of a person was at the time he or she committed a crime. Many defendants will deny that they intended to harm the other person and claim that any harm that occurred was "accidental." The law asks too much of juries when it expects them to determine what a person was thinking. Juries should only be asked to weigh objective evidence.

According to this passage, _____.

- A. most states don't distinguish between voluntary and involuntary manslaughter
- B. punishing people more severely for voluntary manslaughter is unconstitutional
- C. it's difficult for juries to determine a defendant's intentions at the time a crime was committed
- D. prosecutors can, through careful questioning, show a defendant's intention at the time a crime was committed

Answer(s): C

Explanation:

The passage says that most states punish voluntary manslaughter more severely than involuntary manslaughter, so Choice "most states don't distinguish between voluntary and involuntary manslaughter" is incorrect.

The argument that punishing people more severely for voluntary manslaughter is unconstitutional isn't made in the passage, so Choice "punishing people more severely for voluntary manslaughter is unconstitutional" is incorrect.

The passage doesn't support Choice "prosecutors can, through careful questioning, show a defendant's intention at the time a crime was committed".

QUESTION: 1120

Ergonomics is the science of designing and arranging workspaces so that people and objects interact efficiently and safely. Lack of attention to ergonomics causes thousands of workers to suffer repetitive stress injury, eye fatigue, muscle soreness, and many other medical problems each year. Adequate lighting, well-designed chairs, and clutter-free work areas contribute to effective ergonomic design. The opportunity to take short breaks every hour or two, especially for deskbound workers, is also helpful. It's also important for workers to avoid performing the same movements over and over for hours at a time. Variety in the type of work being done can decrease the chance of injury.

According to this passage, _____.

- A. ergonomics can cause injuries
- B. ergonomics is about designing and arranging workspaces efficiently and safely
- C. ergonomics is expensive and time consuming
- D. few people experience problems due to poor ergonomics

Answer(s): B

Explanation:

Lack of attention to ergonomics, not ergonomics itself, can cause injury, so Choice "ergonomics can cause injuries" is incorrect.

The passage doesn't support Choice "ergonomics is expensive and time consuming." The passage states that many people suffer injuries when sufficient attention isn't paid to ergonomics, so Choice "few people experience problems due to poor ergonomics" is incorrect.

QUESTION: 1121

Ergonomics is the science of designing and arranging workspaces so that people and objects interact efficiently and safely. Lack of attention to ergonomics causes thousands of workers to suffer repetitive stress injury, eye fatigue, muscle soreness, and many other medical problems each year. Adequate lighting, well-designed chairs, and clutter-free work areas contribute to

effective ergonomic design. The opportunity to take short breaks every hour or two, especially for deskbound workers, is also helpful. It's also important for workers to avoid performing the same movements over and over for hours at a time. Variety in the type of work being done can decrease the chance of injury.

According to this passage, _____.

- A. adequate lighting and well-designed chairs, although important, have nothing to do with ergonomics
- B. repetition in the type of work people do helps them accomplish their tasks safely and efficiently
- C. short breaks aren't important for deskbound employees because they do little heavy labor
- D. ergonomic design also includes keeping work areas well lit and clutter free

Answer(s): D

Explanation:

The passage states that adequate lighting and well-designed chairs are part of ergonomic design, so Choice "adequate lighting and well-designed chairs, although important, have nothing to do with ergonomics" is incorrect.

The passage states that repetitious work can cause injury, so Choice "repetition in the type of work people do helps them accomplish their tasks safely and efficiently" is incorrect.

The passage states that desk-bound workers should take breaks, so Choice "short breaks aren't important for deskbound employees because they do little heavy labor" is incorrect.

QUESTION: 1122

Ergonomics is the science of designing and arranging workspaces so that people and objects interact efficiently and safely. Lack of attention to ergonomics causes thousands of workers to suffer repetitive stress injury, eye fatigue, muscle soreness, and many other medical problems each year. Adequate lighting, well-designed chairs, and clutter-free work areas contribute to effective ergonomic design. The opportunity to take short breaks every hour or two, especially for deskbound workers, is also helpful. It's also important for workers to avoid performing the same movements over and over for hours at a time. Variety in the type of work being done can decrease the chance of injury.

According to this passage, it's reasonable to assume that _____.

- A. employers should invest in ergonomic design to protect workers
- B. lack of ergonomic design isn't dangerous
- C. labor unions have opposed ergonomic design
- D. poor design is responsible for most employee accidents

Answer(s): A

Explanation:

The passage makes it clear that lack of ergonomic design is dangerous, so Choice "lack of ergonomic design isn't dangerous" is incorrect.

Nothing in the passage supports Choice "labor unions have opposed ergonomic design."

Although the passage claims that lack of ergonomic design causes injury, nothing in the passage supports Choice "poor design is responsible for most employee accidents."

QUESTION: 1123

Electricity is the most inefficient and costly way to heat a home. One kilowatt-hour of electricity creates about 3,400 British thermal units (BTUs). (BTUs are a standard heat measurement.) The price of electricity per kilowatt-hour is between \$0.10 and \$0.25 or between \$29.35 and \$73.13 per million BTUs. In contrast, fuel oil, which produces 140,000 BTUs per gallon, costs about \$8.33 to \$13.89 per million BTUs. Natural gas, which produces 100,000 BTUs per therm, can be purchased for \$5.00 to \$22.50 per million BTUs. Oak firewood, which produces 26,000,000 BTUs per cord, costs \$5.77 to \$13.46 per million BTUs. Choosing the right heating method for your home, based on the cost of fuel, may be more expensive at installation but will be cheaper in the long run.

According to the above passage, a BTU _____.

- A. is an unusual method of measuring heat
- B. stands for "British thermal unit"
- C. is the abbreviation for a "big thermal unit"
- D. can heat a 9 × 12 room

Answer(s): B

Explanation:

The passage says that BTUs are the standard measure of heat, so Choice "is an unusual method of measuring heat" is incorrect.

BTU stands for British thermal unit, so Choice "is the abbreviation for a "big thermal unit" is incorrect. Nothing in the passage supports Choice "can heat a 9 × 12 room."

QUESTION: 1124

Electricity is the most inefficient and costly way to heat a home. One kilowatt-hour of electricity creates about 3,400 British thermal units (BTUs). (BTUs are a standard heat measurement.) The price of electricity per kilowatt-hour is between \$0.10 and \$0.25 or between \$29.35 and \$73.13 per million BTUs. In contrast, fuel oil, which produces 140,000 BTUs per gallon, costs about \$8.33 to \$13.89 per million BTUs. Natural gas, which produces 100,000 BTUs per therm, can be purchased for \$5.00 to \$22.50 per million BTUs. Oak firewood, which produces 26,000,000 BTUs per cord, costs \$5.77 to \$13.46 per million BTUs. Choosing the right heating method for your home, based on the cost of fuel, may be more expensive at installation but will be cheaper in the long run.

According to the above passage, _____.

- A. heating with fuel oil is always cheaper than other methods
- B. oak firewood produces fewer BTUs per dollar than the other types of fuel
- C. natural gas costs more than all other fuels except oak firewood
- D. electricity is always the most expensive way to heat a house

Answer(s): D

Explanation:

The passage shows that fuel oil can be more expensive than other heating methods, so Choice "heating with fuel oil is always cheaper than other methods" is incorrect.

Oak firewood is sometimes less expensive than other types of fuel, so Choice "oak firewood produces fewer BTUs per dollar than the other types of fuel" is incorrect.

Natural gas can sometimes cost less than firewood, so Choice "natural gas costs more than all

other fuels except oak firewood" is incorrect.

QUESTION: 1125

Electricity is the most inefficient and costly way to heat a home. One kilowatt-hour of electricity creates about 3,400 British thermal units (BTUs). (BTUs are a standard heat measurement.) The price of electricity per kilowatt-hour is between \$0.10 and \$0.25 or between \$29.35 and \$73.13 per million BTUs. In contrast, fuel oil, which produces 140,000 BTUs per gallon, costs about \$8.33 to \$13.89 per million BTUs. Natural gas, which produces 100,000 BTUs per therm, can be purchased for \$5.00 to \$22.50 per million BTUs. Oak firewood, which produces 26,000,000 BTUs per cord, costs \$5.77 to \$13.46 per million BTUs. Choosing the right heating method for your home, based on the cost of fuel, may be more expensive at installation but will be cheaper in the long run.

The title of this passage should be _____.

- A. Choosing the Right Heating Method
- B. Heating Methods for Houses
- C. Know Your BTUs
- D. Price List for Fuel

Answer(s): A

Explanation:

The main point of this passage deals with choosing the right fuel based on price; only Choice "Choosing the Right Heating Method" summarizes this point. The remaining are less important points.

QUESTION: 1126

On June 22, 1944, President Franklin Delano Roosevelt signed into law one of the most significant pieces of legislation ever produced by the United States government: The Service members' Readjustment Act of 1944, commonly known as the GI Bill of Rights. By the time the original GI Bill ended in July 1956, 7.8 million World War II veterans had participated in an education or training program, and 2.4 million veterans had home loans backed by the Veterans Administration (VA).

The G.I. Bill provided _____.

- A. free housing, training, and education
- B. medical coverage, education, and assistance to veterans
- C. home loan guarantees, training, and education for many former military members
- D. a means to exempt veterans from social security taxes

Answer(s): C

Explanation:

According to the passage millions of veterans received home loan guarantees, education, and training, making Choice "home loan guarantees, training, and education for many former military members" the correct answer.

Be careful here, as Choice "free housing, training, and education" is tempting, but nothing in the passage indicates that the housing, education, and training were totally free.

QUESTION: 1127

You can put up to \$3,000 a year into an individual retirement account (IRA) on a tax deductible basis if your spouse isn't covered by a retirement plan at work or as long as your combined incomes aren't too high. You also can put the same amount tax-deferred into an IRA for a nonworking spouse if you file your income tax return jointly.

The maximum amount that a married couple could possibly save in a tax-deferred IRA during a year is _____.

- A. \$3,000
- B. \$6,000
- C. \$9,000
- D. The question can't be answered based on the information contained in the passage.

Answer(s): B

Explanation:

The paragraph states that the maximum amount one can place into a tax-deferred IRA is \$3,000, plus an additional \$3,000 if the spouse isn't employed.

QUESTION: 1128

Common sense says that the shade of a tree will help retain snow, and snow exposed to sunlight in open areas will melt. This typically is the case in regions where winter temperatures are below freezing, such as the Northeast, Midwest and most of central and eastern Canada.

A. But in Mediterranean climates - where the average winter temperatures usually are above 30 degrees Fahrenheit - a different phenomenon occurs. Snow tends to melt under the tree canopy and stay more intact in open meadows or gaps in a forest. This happens in part because trees in warmer, maritime forests radiate heat in the form of long-wave radiation to a greater degree than the sky does. Heat radiating from the trees contributes to snow melting under the canopy first.

Which of these statements is true, given the information in the above passage?

- A. There is always snow under trees in Canada.
- B. Average winter temperatures in the Northeast, Midwest, and most of central and eastern Canada are generally below 30 degrees Fahrenheit.
- C. Average winter temperatures in the Northeast, Midwest, and most of central and eastern Canada are generally above 30 degrees Fahrenheit.
- D. It does not snow in Mediterranean climates.

Answer(s): B

QUESTION: 1129

A link between advertising and alcohol consumption is intuitively compelling but hasn't been consistently supported by research. Because alcohol advertising is pervasive, econometric studies may not be sensitive to change or assess in a range where change actually makes a difference. In dealing with advertising, partial bans aren't likely to be effective, and total bans aren't practical. Advertising bans in one medium also are weakened by substitution of increased advertising in alternative media and/or other promotions.

The author of this passage believes _____.

- A. advertisement of alcoholic beverages should be illegal
- B. partial bans on alcohol advertising could be effective in some cases
- C. bans on alcohol advertising aren't likely to work
- D. clear links have been established between alcohol consumption and advertising

Answer(s): C

Explanation:

The author specifically states that partial bans on alcohol advertising isn't likely to be effective and total bans wouldn't be practical.

QUESTION: 1130

The etymology of the word or name, Alabama, has evoked much discussion among philological researchers. It was the name of a noted southern Indian tribe whose habitat when first known to Europeans was in what is now central Alabama

A. One of the major waterways in the state was named for this group and from this river, in turn, the name of the state was derived. According to some investigations, the tribal name Alabama must be sought in the Choctaw tongue, because it isn't uncommon for tribes to accept a name given them by a neighboring tribe.

The state of Alabama was named after _____.

- A. a Choctaw Indian tribe
- B. European settlers
- C. a river
- D. an Indian Chief

Answer(s): C

Explanation:

According to the passage, a river was named after the Alabama Indian Tribe, and the state derived its name from this river.

QUESTION: 1131

An employer may offer, as part of its health plan, a limited incentive (in the form of a reward or penalty) to an employee whose spouse (1) is covered under the employee's health plan; (2) receives health or genetic services offered by the employer, including as part of a wellness program; and (3) provides information about his or her current or past health status. Information about current or past health status usually is provided as part of a health risk assessment (HRA), which may include a questionnaire or medical examination, such as a blood pressure test or blood test to detect high cholesterol or high glucose levels. The total incentive for an employee and spouse to participate in a wellness program that is part of a group health plan and that collects information about current or past health status may not exceed 30 percent of the total cost of the plan in which the employee and any dependents are enrolled. The incentive may be financial or in-kind (e.g., time-off awards, prizes, and other items of value).

What is the best title for this passage?

- A. "Health Risks and Incentives"
- B. "Spousal Insurance Cost"

- C. "Incentives for Employee Health Plans"
- D. "Employee Wellness Plans"

Answer(s): C

QUESTION: 1132

The U.S. Department of Justice has prepared a report about hate crimes in the United States between 1997 and 1999. In 60% of hate crime incidents, the most serious offense was a violent crime, most commonly intimidation or simple assault. The majority of incidents motivated by race, ethnicity, sexual orientation, or disability involved a violent offense, while two-thirds of incidents motivated by religion involved a property offense, most commonly vandalism. Younger offenders were responsible for most hate crimes. Thirty-one percent of violent offenders, and 46% of property offenders were under age 18.

Most property offense hate crimes were motivated by _____.

- A. religion
- B. race
- C. sexual orientation
- D. abortion

Answer(s): A

Explanation:

The second sentence states that most violent crimes were motivated by such factors as race and sexual orientation, while most property crimes were motivated by religion.

QUESTION: 1133

The U.S. Department of Justice has prepared a report about hate crimes in the United States between 1997 and 1999. In 60% of hate crime incidents, the most serious offense was a violent crime, most commonly intimidation or simple assault. The majority of incidents motivated by race, ethnicity, sexual orientation, or disability involved a violent offense, while two-thirds of incidents motivated by religion involved a property offense, most commonly vandalism. Younger offenders were responsible for most hate crimes. Thirty-one percent of violent offenders, and 46% of property offenders were under age 18.

The majority of hate crimes during this period can be classified as _____.

- A. property offenses
- B. violent crimes
- C. assault
- D. intimidation

Answer(s): B

Explanation:

60% of all hate crimes during the period were violent crimes. Assault and intimidation are examples of this category.

QUESTION: 1134

Linewatch operations are conducted near international boundaries and coastlines in areas of Border Patrol jurisdiction to prevent the illegal entry and smuggling of aliens into the United States and to intercept those who do enter illegally before they can escape from border areas. Signcutting is the detection and the interpretation of any disturbances in natural terrain conditions that indicate the presence or passage of people, animals, or vehicles. The operation that's designed to detect changes in the natural environment, which may indicate passage of illegal aliens is called _____.

- A. Linewatching
- B. Signcutting
- C. Border Patrol Operations
- D. Terrain Observation

Answer(s): B

Explanation:

The last sentence in the passage describes the signcutting operation.

QUESTION: 1135

Lasers and incandescent lights are different in an important way. Incandescent light from light bulbs emits light of different wavelengths and moves in many directions. On the other hand, laser beams are monochromatic, consisting of a single wavelength. They can travel great distances without spreading.

Any comparison of laser and incandescent lights must include a discussion of _____.

- A. small and large light bulbs
- B. wavelengths
- C. soundwaves
- D. echoes

Answer(s): B

Explanation:

The purpose of the paragraph is to make a comparison between laser and incandescent light. The two are compared in one way, that is, in terms of their different wavelengths.

QUESTION: 1136

Wales was in ancient times divided into three parts nearly equal, consideration having been paid, in this division, more to the value than to the just quantity or proportion of territory. They were Venedotia, now called North Wales; Demetia, or South Wales, which in British is called Deheubarth, that is, the southern part; and Powys, the middle or eastern district. Roderic the Great, or Rhodri Mawr, who was king over all Wales, was the cause of this division. He had three sons, Mervin, Anarawt, and Cadell, amongst whom he partitioned the whole principality. Wales was divided into divisions because _____.

- A. Natural boundaries, such as rivers and mountains made the division necessary
- B. Wales was too large for the King to oversee personally
- C. The King of Wales wanted his sons to rule
- D. all of the above

Answer(s): C

Explanation:

The rationale for the division is explained in the final sentence.

The passage makes no reference to the size of Wales or the natural boundaries.

QUESTION: 1137

The fierce and warlike tribe, called the Huns, who'd driven the Goths to seek new homes, came from Asia into Southeastern Europe and took possession of a large territory lying north of the River Danube. During the first half of the fifth century, the Huns had a famous king named Attila. He was only 21 years old when he became their king. But although he was young, he was very brave and ambitious, and he wanted to be a great and powerful king. As soon as his army was ready, he marched with it into countries, which belonged to Rome. He defeated the Romans in several great battles and captured many of their cities. The Roman Emperor Theodosius had to ask for terms of peace. Attila agreed that there should be peace, but soon afterwards he found out that Theodosius had formed a plot to murder him. He was so enraged at this that he again began war. He plundered and burned cities wherever he went, and at last the emperor had to give him a large sum of money and a portion of the country south of the Danube.

A good title for the above paragraph would be _____.

- A. The Burning of Rome
- B. Emperor Theodosius
- C. Attila the Hun
- D. Rome for Dummies

Answer(s): C

Explanation:

The primary theme of this paragraph is about Attila, who was King of the Huns.

QUESTION: 1138

The fierce and warlike tribe, called the Huns, who'd driven the Goths to seek new homes, came from Asia into Southeastern Europe and took possession of a large territory lying north of the River Danube. During the first half of the fifth century, the Huns had a famous king named Attila. He was only 21 years old when he became their king. But although he was young, he was very brave and ambitious, and he wanted to be a great and powerful king. As soon as his army was ready, he marched with it into countries, which belonged to Rome. He defeated the Romans in several great battles and captured many of their cities. The Roman Emperor Theodosius had to ask for terms of peace. Attila agreed that there should be peace, but soon afterwards he found out that Theodosius had formed a plot to murder him. He was so enraged at this that he again began war. He plundered and burned cities wherever he went, and at last the emperor had to give him a large sum of money and a portion of the country south of the Danube.

After terms of peace were offered, Attila resumed the war against Rome because _____.

- A. he discovered the Emperor wanted to assassinate him
- B. he wanted to further expand his kingdom

- C. the Emperor of Rome offered too little money in the peace terms
- D. Danube, his second-in-charge, advised him not to accept the peace terms

Answer(s): A

Explanation:

Attila agreed to peace but soon after discovered that the Roman Emperor had launched a plot to kill him.

QUESTION: 1139

In the military, as in all professions, the issue of competence is directly relevant to professional integrity. Because human life, national security, and expenditures from the national treasury are so frequently at issue when the military acts, the obligation to be competent isn't merely prudential. That obligation is a moral one, and culpable incompetence here is clearly a violation of professional integrity. Part of the social aspect of professional integrity involves the joint responsibility for conduct and competence shared by all members of the profession. Only fellow professionals are capable of evaluating competence in some instances; hence, fellow professionals must accept the responsibility of upholding the standards of the profession. Fellow military members can spot derelictions of duty, failures of leadership, failures of competence, and the venalities of conduct that interfere with the goals of the military mission. Often, the obligations of professional integrity may be pitted against personal loyalties or friendships; and, where the stakes for society are so high, professional integrity should win out. One word that best describes the primary theme of the above passage would be _____.

- A. proficiency
- B. equality
- C. evaluations
- D. relationships

Answer(s): A

Explanation:

The primary theme of the passage is stated in the first sentence. Proficiency is closest in meaning to the word competence, which is the primary theme of the passage.

QUESTION: 1140

In the military, as in all professions, the issue of competence is directly relevant to professional integrity. Because human life, national security, and expenditures from the national treasury are so frequently at issue when the military acts, the obligation to be competent isn't merely prudential. That obligation is a moral one, and culpable incompetence here is clearly a violation of professional integrity. Part of the social aspect of professional integrity involves the joint responsibility for conduct and competence shared by all members of the profession. Only fellow professionals are capable of evaluating competence in some instances; hence, fellow professionals must accept the responsibility of upholding the standards of the profession. Fellow military members can spot derelictions of duty, failures of leadership, failures of competence, and the venalities of conduct that interfere with the goals of the military mission. Often, the obligations of professional integrity may be pitted against personal loyalties or friendships; and, where the stakes for society are so high, professional integrity should win out.

Professional competence is _____.

- A. a moral obligation
- B. directly relevant to professional integrity
- C. essential because military operations impact human life, national security, and use of taxpayer funds
- D. all of the above

Answer(s): D

Explanation:

All of the statements are directly supported by the passage.

QUESTION: 1141

In the military, as in all professions, the issue of competence is directly relevant to professional integrity. Because human life, national security, and expenditures from the national treasury are so frequently at issue when the military acts, the obligation to be competent isn't merely prudential. That obligation is a moral one, and culpable incompetence here is clearly a violation of professional integrity. Part of the social aspect of professional integrity involves the joint responsibility for conduct and competence shared by all members of the profession. Only fellow professionals are capable of evaluating competence in some instances; hence, fellow professionals must accept the responsibility of upholding the standards of the profession. Fellow military members can spot derelictions of duty, failures of leadership, failures of competence, and the venalities of conduct that interfere with the goals of the military mission. Often, the obligations of professional integrity may be pitted against personal loyalties or friendships; and, where the stakes for society are so high, professional integrity should win out. The author of the above passage would agree that _____.

- A. friendship must often take a back seat to professional integrity
- B. only fellow professionals should evaluate competence
- C. professional competence is a direct result of effective training programs
- D. all of the above

Answer(s): A

Explanation:

The author specifically states that when pitted against friendship, professional integrity must often win out.

The author explains that only fellow professionals can evaluate other professionals in some (not all) cases, making Choice "only fellow professionals should evaluate competence" incorrect. Choice "professional competence is a direct result of effective training programs" isn't supported by information in the passage.

QUESTION: 1142

Black Americans served in every major U.S. military conflict from the Revolutionary War onward except the 1848 Mexican War. However, they were almost always restricted to segregated, black-only units - usually with white officers - until after the Korean War. President Harry S. Truman issued an executive order in 1948 that mandated "equality of treatment... in the armed services without regard to race, color, religion or national origin." By October 1953, the Army

announced that 95 percent of black soldiers were serving in integrated units. By when was segregation of black soldiers effectively ended?

- A. In 1848
- B. In 1948
- C. In 1953
- D. During the Korean War

Answer(s): C

Explanation:

Even though President Truman ordered an end to segregation in 1948, it was not effectively ended until 1953, by which time the Army announced that 95 percent of black soldiers were serving in integrated units.

QUESTION: 1143

A stranger meets you and shows you a wad of cash he has just "found." He says he wants to split it with you, but says that, in order to show your "good faith," you have to put up some of your own money. When you agree to give him your money, the stranger finds a reason to leave for a while. Do you see him again? Probably not - you have just been conned. The main theme of this passage is _____.

- A. don't ever speak to strangers
- B. be careful of "get-rich-quick" schemes
- C. how to make lots of money
- D. don't believe anything a stranger says

Answer(s): B

Explanation:

This passage warns the reader of one particular get-rich-quick scheme, and by extension cautions them to be wary of all such ideas.

It does not say to never speak to a stranger, nor does it say not to ever believe anything a stranger says - if we did that, we couldn't go in a convenience store on a long trip to buy gas or go to the restroom.

The passage also does not describe how to make lots of money, since the subject of the con loses money rather than getting rich.

QUESTION: 1144

The new Army Service Uniform or ASU is a hybrid that combines elements of the green Class A and B uniforms and the dress blue uniform. Many of the policies governing the green uniform have been integrated with or adapted to the blue uniform, now known as the ASU. The ASU replaces the green Army Class A and B uniforms. What uniform change is happening?

- A. The Army is changing its Class A uniform from green to blue.
- B. The Army is changing its Class A uniform from blue to green.
- C. The Army is changing its combat uniform.
- D. The Army is governing the elements of its uniform policies.

Answer(s): A

Explanation:

The passage states that the blue uniform is now known as the Army Service Uniform or ASU, which replaces the green Class A and B uniforms.

QUESTION: 1145

LLC stands for "Limited Liability Corporation," meaning the business owners' liability is limited. Without the idea of "limited liability," the owners of a business can be held personally responsible for the debts of their business. The idea of "limited liability" originated in Germany in the late 19th century. By the 1940s, the idea had taken hold in 17 countries, but not the United States. In 1977, Wyoming became the first U.S. state to pass an LLC law modeled after the German statute. Many other states eventually passed their own version, though most of them have variations from other states.

Why is the idea of limited liability helpful to business owners?

- A. Because it can limit their personal financial risk.
- B. Because it is a tradition dating back to the 19th century.
- C. Because it is now legal in all 50 states.
- D. Because it holds them personally responsible for their business' debts.

Answer(s): A

Explanation:

The idea of a limited liability company or LLC is helpful to business owners because it can limit the financial risk to them personally, rather than making business owners personally responsible for the debts of their business.

This idea is expressed in the first two sentences of the passage; it is left for the reader to understand that decreased personal financial risk is better, but that's a conclusion most people would easily reach on their own.

QUESTION: 1146

We are focused so much on requiring diversity that we have lost sight of why diversity is a good thing. The good part about having different kinds of people working together is that they bring their various abilities and insights to help the organization accomplish their shared goals. The different parts of the whole are important because, when joined together, they make a stronger, more capable whole. It is the strengthened unity resulting from diversity that is important, not just diversity for its own sake.

What does the writer say about different kinds of people working together?

- A. Working together is important for its own sake.
- B. Working together is too difficult to be important.
- C. Having different types of people working together is important because it strengthens the entire organization or group.
- D. Working together makes people lose their focus on their individual goals.

Answer(s): C

Explanation:

Having different types of people working together is important because it strengthens the entire organization or group. The writer of this passage is trying to get across the idea that the advantage of diversity is the increased strength and capability that the diverse group has over the non-diverse group, not that diversity itself is the only goal.

QUESTION: 1147

Specialized warships are relatively recent in origin. They have always been expensive to build and need to be manned by crews with specialized skills. Their construction and operation therefore requires a great deal of the ruler's wealth. The earliest form of fighting at sea was probably piratical as opposed to political; even a pirate needs resources to get his start. The first navies may or may not have been pirate hunters but navies, by definition, cost more than individual ships. Whatever perspective one takes, fighting on water has cost more than fighting on land from the beginnings.

According to this passage, navies _____.

- A. all started out as pirates
- B. are more expensive to have and maintain than individual ships
- C. normally cooperate with pirates to avoid attack
- D. are suitable to be built by specialists

Answer(s): B

Explanation:

According to this passage, navies are more expensive to have and maintain than individual ships. The passage states that, "navies, by definition, cost more than individual ships."

QUESTION: 1148

A noted communication researcher has found that 7 percent of our feelings and attitudes are communicated with words, 38 percent through tone of voice, and 55 percent with nonverbal expressions. These facts tell us that the communications channels over which we have the most control - and understand the best - have the least amount of impact, and that the channels over which we have the least control - and understand the least - have the most impact.

How much of our communication with other people is with words?

- A. 7 percent
- B. 38 percent
- C. 55 percent
- D. Depends upon which communication channel it goes through.

Answer(s): A

Explanation:

According to the passage, only 7 percent of our feelings and attitudes are communicated with words.

QUESTION: 1149

The process by which the Roman legions came to serve so far from the Roman army's

birthplace began in earnest during the Punic wars with Carthage. The Phoenician colony first clashed with the Romans when Rome's success in subduing its Italian neighbors drew it south to Sicily, which Carthage regarded as within its sphere of influence. In 265 B.C., after Rome's expensive victory over Pyrrhus, Rome and Carthage found themselves at war over Sicily. The war rapidly extended by both land and sea until the Carthaginians had to admit defeat and allow the establishment of Roman control over the island. Rome soon after added Corsica and Sardinia to these beginnings of its overseas empire.

What area were the Romans and Carthaginians fighting over in 265 B.C.?

- A. Phoenicia
- B. Carthage
- C. Sicily
- D. Corsica

Answer(s): C

Explanation:

The passage states, "In 265 B.C. ...Rome and Carthage found themselves at war over Sicily."

QUESTION: 1150

It is vitally important in strategic planning to select the "right" goals and objectives. Some experts maintain that this is the primary role and purpose of strategic planning. Planning also includes determining the best means of achieving objectives, facilitating effective communication, and reviewing results as the plan is executed. In order to select the truly "right" objectives, however, planners must keep in mind the overall mission of the organization. How must strategic planners choose appropriate goals?

- A. By selecting the "right" objectives.
- B. By facilitating effective communication.
- C. While keeping in mind the overall mission of the organization.
- D. By reviewing results as the plan is executed.

Answer(s): C

Explanation:

The passage states that, "In order to select the truly 'right' objectives, however, planners must keep in mind the overall mission of the organization."

QUESTION: 1151

The traditions of Freemasonry have evolved over many centuries and from many sources. These traditions are a powerful force that cements a Lodge and Freemasons into one sacred band. They can be modified - but only slowly - to fit the needs of the organization. As individual Freemasons become aware of the traditions, they soon learn that they are part of something larger than themselves.

The traditions of Freemasonry teach new members that _____.

- A. They are part of something larger than themselves.
- B. New members can quickly change traditions to meet the organization's needs.
- C. Traditions are never written in cement.

D. There is only one source for the traditions of Freemasonry.

Answer(s): A

Explanation:

The passage states, "As individual Freemasons become aware of the traditions, they soon learn that they are part of something larger than themselves."

QUESTION: 1152

Captain Agnew had located the Italian naval convoy by means of radar, and then brought his Force K into the most favorable attacking position. To the Italians, who possessed no such technology, the darkness remained impenetrable, and their surprise was consequently complete.

What happened to the Italian convoy?

- A. It used radar to locate Captain Agnew's Force K.
- B. It was surprised by a nighttime attack.
- C. It surprised the Force K nighttime convoy.
- D. It got lost in the impenetrable darkness.

Answer(s): B

Explanation:

The Italian convoy did not have radar or anything like it, so they could not see Captain Agnew's Force K in the darkness and Force K surprised them with an attack.

QUESTION: 1153

A major part of safe driving is driving at the right speed. But what is the "right" speed? Is it 20 miles per hour, or 45, or 60? That question may be hard to answer. On some city streets and in heavy traffic, 20 miles per hour could be too fast, even if it's within the posted speed limit. On an interstate highway, 45 miles per hour could be too slow. Of course, a good driver must follow the speed limit, but he must also use good judgment. The "right speed will vary depending on the number of cars, the road surface and its condition, and the driver's visibility.

The general theme of this passage is that a good driver _____.

- A. drives at 45 miles an hour
- B. adjusts to different driving conditions
- C. always drives at the same speed
- D. always follows the speed limit

Answer(s): B

Explanation:

The general theme of this passage is that a good driver adjusts to different driving conditions.

QUESTION: 1154

About three-fourths of the Earth's surface is covered with water. Of these 336 million cubic miles of water, most (97.2 percent) is found in the oceans and is salty. Glaciers at both poles contain

another 2 percent of the total amount. Less than 1 percent (0.8 percent) is available as fresh water for people to use - and much of that is not near the people who need it the most. The amount of fresh water available for people to use is _____.

- A. 97.2 percent
- B. 0.8 percent
- C. two-tenths
- D. 2.0 percent

Answer(s): B

Explanation:

The amount of fresh water available for people to use is, as stated in the passage, 0.8 percent.

QUESTION: 1155

Nucleic acids are found in all living organisms, from viruses to man. They received their name because of their discovery in the nuclei of white blood cells and fish sperm by Swiss physiologist Johann Mescher in 1869. However, it is now well established that nucleic acids occur outside the cell nucleus as well. Nucleic acids are found _____.

- A. only in human cells
- B. only in viruses
- C. in all living cells
- D. only in the cell nucleus

Answer(s): C

Explanation:

Nucleic acids are found in all living organisms, as stated in the first sentence of the passage.

QUESTION: 1156

The Tao-te-ching (The Way and Its Power) is the basic text of Taoism, a Chinese religion and philosophy. The Tao-te-ching is composed of 81 short chapters or poems that describe a way of life marked by quiet effortlessness and freedom from desire. Taoism teaches that this can be achieved by following the creative, spontaneous life force of the universe, called the Tao. The book is attributed to Lao-tzu, but it was probably written in the third century B.C. The Tao-te-ching _____.

- A. is the basic text of Chinese poetry
- B. describes a life way marked by quiet freedom
- C. is made up of 81 chapters and a short poem
- D. was probably not written by Lao-tzu, even though it is attributed to him

Answer(s): D

Explanation:

The passage says that the Tao-te-ching "is attributed to Lao-tzu, but it was probably written in the third century

E. c." The use of the word "but" clearly implies that Lao-tzu (who actually lived three centuries earlier, during the sixth century b.c.) did not write the book.

QUESTION: 1157

History tells us that Christopher Columbus once saved his own life because he knew from his almanac that there was going to be an eclipse of the moon. In 1504 Columbus was marooned in Jamaica with no food and unfriendly natives confronting him. The explorer threatened to turn off the light in the sky if the natives refused to give him food. Columbus kept his word; the natives gave him food.

The main idea of this paragraph is that _____.

- A. natives rarely own almanacs or understand scientific facts
- B. Christopher Columbus was in Jamaica in 1504 on his third voyage from Spain
- C. on his arrival in Jamaica, Christopher Columbus had no food
- D. Columbus saved his own life because he knew when there was going to be an eclipse of the moon

Answer(s): D

Explanation:

The main idea is stated in the first sentence of the paragraph.

All of the other sentences contribute to the main idea by describing the circumstances under which Columbus was able to save his own life.

QUESTION: 1158

One implied warranty required by state law is the "warranty of merchantability." This means that the seller promises that the product will do what it is supposed to do.

For example, _____.

- A. a car will never need repairs
- B. a lawnmower will also cut bushes
- C. a toaster will toast
- D. your roof will last 15 to 100 years

Answer(s): C

Explanation:

When you apply the "warranty of merchantability," that is, that the product will do what it is supposed to do, you see that only one answer mentions a product (a toaster) and what it does (toasts).

QUESTION: 1159

To get full value for your energy dollars, first arrange for an energy audit of your home. An audit will reveal such things as energy-sapping drafts, insufficient insulation, improper weather stripping, and a need for storm windows and doors.

Installing storm windows may be energy-saving but should be preceded by _____.

- A. an energy audit

- B. storm doors
- C. insulation
- D. proper weather stripping

Answer(s): A

Explanation:

The main idea of the paragraph is that an energy audit should be done as a first step toward saving energy dollars.

The details that follow are some of the items an audit concentrates on.

The question, however, asks you to identify what precedes, or comes before, the installation of an energy- saving device.

QUESTION: 1160

Railroads originating on both coasts of the United States were joined on May 10, 1869, when a golden spike was driven to join the last rail at Promontory Point, Utah. Great numbers of immigrants moved west to take up free homestead land or purchase tracts at low prices. According to the passage, the settling of the American West _____.

- A. made immigrants very wealthy from land ownership
- B. followed the completion of the railroad
- C. made it impossible for anyone to purchase land
- D. was finished before the summer of 1869

Answer(s): B

Explanation:

According to the passage, the settling of the American West took place after both coasts were joined by completion of the railroad.

QUESTION: 1161

The most disadvantaged type in both countries was the single-parent, female-headed household, of which 38% in Canada and 61% in the United States were in need. Moreover, 44% of single females living alone in Canada and 51% in the United States were in housing need. By contrast, male single parents experienced about half the housing need of their female equivalents.

You can conclude from the paragraph that needy male single parents in housing need in the United States numbered approximately _____.

- A. 61%
- B. 30%
- C. 44%
- D. 22%

Answer(s): B

Explanation:

The paragraph states that 61% of single female parents were in housing need and that single male parents experienced half that housing need. Half of 61% is approximately 30%.

QUESTION: 1162

In 1973, when the OPEC nations embargoed oil exports to the United States, their action signaled an unprecedented rise in oil prices. In fact, crude oil prices quadrupled, sending oil-dependent countries into a recession.

One of the results of the oil embargo was that _____.

- A. the United States embargoed oil exports of OPEC nations
- B. the price of crude oil declined
- C. the price of crude oil doubled
- D. oil-dependent countries experienced a recession

Answer(s): D

Explanation:

The embargo caused the price of crude oil to quadruple.

This increase, in turn, caused oil-dependent countries to experience a recession.

Answers "the United States embargoed oil exports of OPEC nations", "the price of crude oil declined" and "the price of crude oil doubled" are all incorrect details.

QUESTION: 1163

Although they last no longer than a fraction of a second, voltage variations can cause problems with sensitive electronic equipment. Variation in voltage causes digital clocks to blink and timers to be thrown off schedule.

Computers are especially _____.

- A. useful when you transfer data to memory
- B. easy to protect from variations in voltage
- C. useful in publishing
- D. sensitive to voltage variations

Answer(s): D

Explanation:

The paragraph talks about the effect of voltage variations on sensitive equipment.

It is logical to conclude that computers, which are also sensitive equipment, would be affected by voltage variation.

QUESTION: 1164

In 1948, the United Nations General Assembly adopted the resolution entitled the Universal Declaration of Human Rights. This resolution stresses personal, civil, political, social, cultural, and economic rights of people.

In so doing, the article promotes _____.

- A. financial well-being only
- B. fundamental freedoms
- C. social unrest
- D. only one political party

Answer(s): B

Explanation:

Answer "fundamental freedoms". Fundamental freedoms, summarizes all the rights listed in the paragraph.

QUESTION: 1165

DNA, a nucleic acid contained in all living material, orders every cell to reproduce in a certain manner. In fact, DNA molecules determine both the form and function of all living cells. DNA is _____.

- A. found only in animals
- B. one kind of cell
- C. a nucleic acid
- D. a nuclear device

Answer(s): C

Explanation:

The fact that DNA is a nucleic acid is stated in the first sentence.

Answers "found only in animals", "one kind of cell", and "a nuclear device" are all incorrect details.

QUESTION: 1166

Garbage has become an alarming problem - and it won't go away. State landfills, as they are presently designed, are quickly filling up. Consequently, since 30% of trash being dumped can be recycled, it is time for communities to set up recycling programs. Not all trash needs to be dumped.

We know that _____.

- A. a third of what we dump could be recycled
- B. people could start their own landfills
- C. recycling will never take place
- D. 70% of trash can be recycled

Answer(s): A

Explanation:

The paragraph states that 30% of trash being dumped can be recycled.

Another way of stating this detail is to say that a third of what we dump could be recycled.

QUESTION: 1167

Sometimes a state legislature asks the voters to approve or reject a proposal called a referendum. The proposal may be a constitutional amendment, a plan for long-term borrowing, or a special law affecting a city. In effect, the question is "referred" to the voters. The Constitution of the United States was submitted to state conventions for approvals. From the definition above, you can say that it was _____.

- A. approved unanimously
- B. approved by a form of referendum
- C. rejected by a majority of the states
- D. referred to the president of the United States

Answer(s): B

Explanation:

According to the paragraph's definition of a referendum, submitting the Constitution of the United States to the state conventions for approvals was a form of referendum.

QUESTION: 1168

Some pollutants that render air unsafe to breathe cannot be seen or smelled. Nitric oxide, carbon monoxide, and radon gas are examples. Other pollutants, such as formaldehyde, can be smelled at high levels only but are still unsafe to breathe even at lower levels.

You can conclude from the paragraph that formaldehyde _____.

- A. is safer to breathe than nitric oxide
- B. can never be seen or smelled
- C. cannot be smelled at lower levels
- D. is not one of the pollutants that renders air unsafe to breathe

Answer(s): C

Explanation:

The paragraph states that unlike the other pollutants mentioned, formaldehyde can be smelled only at high levels. You can conclude that it cannot be smelled at lower levels.

QUESTION: 1169

Pneumonia is caused by bacteria, viruses, fungi, or mycoplasmas. Bacterial pneumonia is the main type of pneumonia that follows an attack of the flu and a major precipitator (beginning cause) of pneumonia deaths.

Which kind of pneumonia is the most deadly?

- A. bacterial
- B. viral
- C. fungi
- D. flu

Answer(s): A

Explanation:

Sentence 2 states that bacterial pneumonia is a major precipitator (cause) of pneumonia deaths. Answers "viral", "fungi", and "flu" are all incorrect details.

QUESTION: 1170

Because the temperature of ocean water changes more slowly than the temperature of air,

oceans affect the world's climate. As ocean water moves past land masses, it affects the climate by warming the air in winter and cooling it in summer.

The key idea in this paragraph is that _____.

- A. ocean water moves past land masses
- B. ocean water cools the air in winter
- C. ocean water warms land masses in summer
- D. the world's climate is affected by the temperature of the oceans

Answer(s): D

Explanation:

The key idea is found in the first sentence: The temperature of ocean water changes more slowly than that of the air; therefore, the oceans affect the world's climate.

QUESTION: 1171

There are several differences between reptiles and birds. One key difference involves the skeletal structure between the two and the fact that birds have wings. Reptiles have solid bones that are dense and they travel on land or in the water. Birds have hollow or pneumatic bones filled with air that are lighter and allow them to fly. In addition, birds have a lightweight beak instead of teeth, or a heavy jaw.

Birds _____.

- A. can travel faster than reptiles
- B. may have a heavy jaw
- C. do not have teeth
- D. have a solid bone structure

Answer(s): C

QUESTION: 1172

The maggot larvae may be more than just a repugnant animal. Maggot larvae, called *Phaenicia sericata*, have recently been used in a treatment known as Maggot Therapy. Twice a week, larvae are placed on a wound and left there for 48 to 72 hours. The maggots only eat the dead tissue, leaving healthy tissue intact, a process called debridement. In a recent study, wounds that got presurgical maggot therapy developed no infections after surgery. While about one-third of wounds that didn't get maggot therapy developed an infection.

In the context, repugnant means _____.

- A. distasteful
- B. cute
- C. friendly
- D. jovial

Answer(s): A

QUESTION: 1173

Doctors and experts recommend that an aspirin a day may cut the risk of developing colon

cancer. However, a smaller dose of baby aspirin may be more effective at this than a full size aspirin.

A full size aspirin _____.

- A. does not cut the risk of developing colon cancer
- B. prevents colon cancer
- C. may cut the risk of colon cancer
- D. is not as effective as a baby aspirin

Answer(s): C

Explanation:

Of the following choices only, the only one that is true is that a full size aspirin may cut the risk of colon cancer.

The last choice is not correct, because it does not limit the statement to cutting the risk of colon cancer.

QUESTION: 1174

A new study suggests that 1 in 4 Americans do not get enough sleep. Sleep experts recommend 8 hours of sleep per night and most Americans report an average of less than 7 hours of sleep on weeknights and 7.5 hours of sleep on weekends.

The average American sleeps _____.

- A. 6 hours on weeknights
- B. 8 hours on weekend nights
- C. the recommended hours of sleep
- D. 7.5 hours on weekend nights

Answer(s): D

QUESTION: 1175

Regular weight lifting has been shown to improve muscle mass in people of all ages. Even 80-year-old subjects in weight lifting studies show signs of benefits by supplementing their muscle mass.

It can be inferred from the passage that _____.

- A. young people do not benefit from weight lifting
- B. weight lifting even once a week will produce benefits for all ages
- C. older people show the most improvements in muscle mass in weight lifting studies
- D. any able body person can begin weight lifting at any age and see benefits

Answer(s): D

QUESTION: 1176

Aquaculture, otherwise known as "fish farming, " is a rapidly growing industry. Instead of catching fish in the wild, organizations around the world are developing aquaculture resources so that they can raise and harvest fish in controlled environments. This technique allows people to provide large supplies of fresh fish for marketing.

The best title for this passage is _____.

- A. The Science of Aquamarines
- B. Aquaculture - Developing Fish Farms
- C. Fishing Is Part of Our Culture
- D. Fish Farming Is the Next Best Thing to Deep-sea Excursions

Answer(s): B

QUESTION: 1177

If you are thirsty, most likely you are already dehydrated. Most people underrate the importance of drinking water all day long. They wait until their throats are parched or they eat meals before they drink any fluids - and they usually don't choose water. Since the human body is about two-thirds water, drinking it is essential to maintaining good health. Just a few of the benefits of staying hydrated are better memory throughout the day, increased energy, and less tiredness. In this context, parched means _____.

- A. hydrated
- B. sore
- C. dry
- D. wet

Answer(s): C

QUESTION: 1178

If you are thirsty, most likely you are already dehydrated. Most people underrate the importance of drinking water all day long. They wait until their throats are parched or they eat meals before they drink any fluids - and they usually don't choose water. Since the human body is about two-thirds water, drinking is essential to maintaining good health. Just a few of the benefits of staying hydrated are better memory throughout the day, increased energy, and less tiredness. The best title for this selection is _____.

- A. Dehydration and the Benefits of Drinking Water
- B. The Benefits of Drinking Water
- C. Better Memory with Water
- D. How to stay Hydrated

Answer(s): A

Explanation:

The selection discusses both hydration and the benefits of drinking water.

QUESTION: 1179

If you are thirsty, most likely you are already dehydrated. Most people underrate the importance of drinking water all day long. They wait until their throats are parched or they eat meals before they drink any fluids - and they usually don't choose water. Since the human body is about two-thirds water, drinking is essential to maintaining good health. Just a few of the benefits of staying hydrated are better memory throughout the day, increased energy, and less tiredness.

It can be inferred from this selection that most people are _____.

- A. dehydrated quite often without even knowing it
- B. dehydrated despite drinking large quantities of water
- C. often hospitalized for severe dehydration
- D. rarely ever hydrated

Answer(s): A

QUESTION: 1180

General Dwight Eisenhower was one of the most significant military leaders in the history of the United States. Although he was a brilliant strategist and had the heart of a lion, his concern was for the safety and well-being of his "boys." All who served under General Eisenhower were devoted to him, and looked to him as a father, devoted more to their leader than to the cause. For that reason, every casualty struck General Eisenhower as if it were his own kin. According to the passage, what kind of man can we assume that General Dwight Eisenhower was?

- A. Aloof and cold.
- B. Timid and uncertain.
- C. Militarily aggressive at all costs.
- D. Intelligent and compassionate.

Answer(s): D

QUESTION: 1181

Obesity has become a national epidemic. Recent studies have shown that obesity is more serious than previously thought. In fact, obesity is harder on the health than cigarette smoking. Since 27% of Americans are currently obese, and 61% are overweight, this weight problem is exacting a huge cost from the medical community. The percentage of Americans who are currently overweight is _____.

- A. 27%
- B. 38%
- C. 61%
- D. 78%

Answer(s): C

QUESTION: 1182

Although carjacking has become more common in the past 10 years, there are several preventative measures that drivers can take. The first way to prevent carjacking is to never walk alone to your car at night. Another means of prevention is always driving with the windows rolled up and the doors locked. Also, driving on well-lit and often traveled roads is another preventative measure that drivers can take to ensure their safety from carjacking. Which of the following is not mentioned as a preventative measure against carjacking?

- A. Driving with windows rolled up.
- B. Never walking to your car alone at night.
- C. Carrying pepper spray.
- D. Driving on well-lit roads.

Answer(s): C

QUESTION: 1183

Although carjacking has become more common in the past 10 years, there are several preventative measures that drivers can take. The first way to prevent carjacking is to never walk alone to your car at night. Another means of prevention is always driving with the windows rolled up and the doors locked. Also, driving on well-lit and often traveled roads is another preventative measure that drivers can take to ensure their safety from carjacking. The author probably believes that _____.

- A. carjacking never happens during the day
- B. carjacking can often be avoided by employing simple preventive measures
- C. it is almost impossible to avoid carjacking
- D. carjacking only happen at night.

Answer(s): B

QUESTION: 1184

Although carjacking has become more common in the past 10 years, there are several preventative measures that drivers can take. The first way to prevent carjacking is to never walk alone to your car at night. Another means of prevention is always driving with the windows rolled up and the doors locked. Also, driving on well-lit and often traveled roads is another preventative measure that drivers can take to ensure their safety from carjacking. The best title for this selection is _____.

- A. Carjacking are more common than you think
- B. Driving Safety Tips
- C. Night Driving Safety
- D. Preventing Carjacking

Answer(s): D

QUESTION: 1185

"There was an air of calm and reserved opulence about the Weightman mansion that spoke not of money squandered, but of wealth prudently applied. " - Henry Van Dyke, The Mansion. In this passage, what is the inferred meaning of "opulence"?

- A. Peacefulness
- B. Affluence
- C. Extravagance
- D. Age

Answer(s): B

QUESTION: 1186

From Alaska to the southern tip of Chile lies the Ring of Fire. In between, it travels along the western coast of North and South America and along the eastern coast of Asia.

A. At its northernmost points, the Ring goes through Russia, Japan, the Philippines, and New Guinea.

A. Also going through Australia, the Ring of Fire marks a unique area.

A. It is here that some of the most destructive recorded volcanoes have erupted.

According to the above passage, where would a destructive volcanic eruption be most likely to happen?

- A. anywhere along the Ring of Fire
- B. along the east coast of South America
- C. anywhere in Asia
- D. along the east coast of North America

Answer(s): A

QUESTION: 1187

Every four years, athletes from all over the world gather in a different place for the Summer Olympic Games. In 2012, the site was London, England, where 10,500 participants gathered. They represented 204 countries and territories and earned 302 medals in 26 different sports. According to the above passage, all of these are years of the Summer Olympics, except _____.

- A. 2016
- B. 2009
- C. 2000
- D. 2012

Answer(s): B

QUESTION: 1188

The American Flag began with a flag created in 1777 and used until 1795. Congress then ordered, with passage of an act, that the new flag be composed of 15 alternating red and white stripes and contain a blue field with 15 stars. Another change was made by Congress in 1818. This edition of the flag was to alternate only 13 stripes, but a new star was to be added for each state that joined the Union.

The word alternate, as used in the above passage, most nearly means _____.

- A. rotate
- B. swap
- C. substitute
- D. exchange

Answer(s): A

QUESTION: 1189

What is the best title for this passage?

Geologists in Rome have a new venture. Underneath the city's streets and architecture lies a system of quarries and tunnels. It is believed that these passages were formed in the early days of the ancient city. The main concern for geologists is that the city may sink into the below-ground areas. These professionals hope their studies of the tunnels will help prevent that occurrence.

- A. "Geologists Use Tunnels"
- B. "Studying Tunnels to Avoid Disasters"
- C. "Rome's Ancient Tunnels"
- D. "Mapping Rome's Tunnels"

Answer(s): B

QUESTION: 1190

Comparable to the decline of the great Mayan culture, the fall of the ancient Indus Valley people yields interesting information about their daily lives. It seems that this civilization is the oldest recorded civilization in India, yet many of their practices seem advanced. For example, there is evidence of careful and creative design of bathroom facilities, including sewage disposal. What is evident in other ancient societies seems to be missing from the Indus Valley one, namely social conflict, slavery, armed forces, and various vices. The cause of their sudden demise is a mystery.

According to the above passage, what is the oldest known civilization of India?

- A. Mayan people
- B. Afghanistan people
- C. Indus Valley people
- D. Aztec people

Answer(s): C

QUESTION: 1191

One would not think that evidence of the existence of European people 2,000 years ago could be found in China, but Victor Mair might beg to differ. A group of archaeologists in western China came upon some mummified bodies. A college professor, Mair, helped examine them and was stupefied. What he found on the Tarim mummies were signs of blond hair and long noses. In 1993, after the mummies had been placed in a museum, Mair returned to take DNA samples from the specimens. The test results on these samples confirmed what Mair had suspected: the genes were most certainly European.

The word stupefied, as used in the above passage, most nearly means _____.

- A. doubtful
- B. foolish
- C. impressed
- D. puzzled

Answer(s): D

QUESTION: 1192

When is a book unreadable? Well, when nobody knows the language in which it is written, this might be the case. In 1912, the Voynich Manuscript was found at a library in Rome. Among the 240 pages were pictures, accompanied by text written in an unknown language. Efforts by the most noted language experts failed and some people wrote off the 500-year-old manuscript as a hoax. As our methods of language analysis have improved, however, it has been determined that the mystery language does indeed have a working language pattern and structure. Various experts are now claiming to be able to decode a few words within this mysterious text. From the above passage, the reader can infer that the Voynich Manuscript is believed to have originated around _____.

- A. 1812
- B. 1912
- C. 1412
- D. 500

Answer(s): C

QUESTION: 1193

Jeans from gold miners? It seems that one of today's most popular pants styles actually came out of the California gold rush in the 19th century. Levi Strauss, a shopkeeper in California, started making something called "waist overalls," his name for these early jeans. Gold miners loved them. Strauss added the familiar brown leather patch to the back pocket in 1886. The label featured two horses pulling a pair of jeans between themselves. Then, with the Hollywood Western craze of the 1930s, the popularity of jeans grew, as everyone wanted to look like a movie star. Since the 1990s, jeans have been worn by nearly everyone, from child to professional.

What is the main idea of this paragraph?

- A. Levi Strauss created the original design for jeans and it has changed.
- B. Jeans were most popular during the Gold Rush era.
- C. Jeans have evolved over the years but remain popular.
- D. Jeans have evolved over the years and are no longer as popular as they once were.

Answer(s): C

QUESTION: 1194

If you want to visit the deepest lake in the United States, you'll need to go to the southwestern part of Oregon. Since its establishment in 1902, Crater Lake has occupied 183,000 acres there and is 1,958 feet deep. This makes it one of the world's deepest lakes. It all began when the Mount Mazama volcano erupted over 7,000 years ago. There were glaciers around the eruption site and clear blue water from them filled the volcano basin, or calder

A. When you visit the Crater Lake National Park to see this wonder of nature, you will also find 680 species of plants, 158 types of birds, and 74 different animals.

Based on the above passage, the author would agree that _____.

- A. Crater Lake is considered one of the deepest lakes in both the United States and the world
- B. Crater Lake is a man-made lake

- C. Crater Lake National Park has very few plant species
- D. Crater Lake is the only tourist attraction at Crater Lake National Park

Answer(s): A

QUESTION: 1195

In Africa, there lives an interesting tribe of people: the Pygmies. They make their home in what used to be called Zaire, but is now the Democratic Republic of Congo. Pygmies are nomadic, wandering around, hunting, and procuring their own food. They have light-colored skin of a yellowish-brown shade and can also be found in rain forests of the Philippines and India. The word nomadic, as used in the above passage, most nearly means _____.

- A. aggressive
- B. uneducated
- C. wandering
- D. barbarian

Answer(s): C

QUESTION: 1196

My cousin has an interesting job. He contacts museums, organizers of cultural events, and performing arts companies in San Francisco about interesting activities they are planning. Then he does all sorts of promotions for these activities for visitors to the city. He has a background in hospitality and tourism and is very good at multi-tasking. These things really help! Another part of his job is to create ads, including those for billboards, magazines, and Web sites in other cities.

From the above passage, the reader can infer that the cousin's job title is most likely _____.

- A. Director of Travel Marketing
- B. Director of a San Francisco outdoor advertising company
- C. Director of San Francisco's Art Museum
- D. Director of San Francisco's Art Gallery

Answer(s): A

QUESTION: 1197

In Wiltshire, England, there is a fascinating site. Near Salisbury stands a creation called Stonehenge. No one is sure of the meaning of the monument that has stood there since ancient times. Some people theorize that it was a temple for people who worshiped the sun. Others have dubbed it a center for healing, a burial site, or even a very large calendar. Many questions remain, such as how in the world the tremendous stones were moved and arranged by primitive people who had only the most basic of tools. In spite of all these questions, Stonehenge remains an impressive site.

Which of these titles would best summarize the point of the above passage?

- A. "An Unforgettable Mystery"
- B. "Stonehenge in England"

- C. "Visiting Wiltshire, England"
- D. "Visiting Stonehenge"

Answer(s): A

QUESTION: 1198

Simple machines are devices that make our lives easier. They increase the magnitude or change the direction of a force. Using simple machines makes it easier to do many kinds of work. For example, when trying to get a refrigerator onto the back of a truck, a worker will use a ramp or inclined plane. Instead of lifting something heavy a short distance, we can more easily push it over a longer distance but to the same height. The term inclined plane can be used to describe an escalator, staircase, or slide. Many shoppers use an inclined plane to get to the second level in a mall. According to the above passage, this would mean that they are most likely traveling by _____.

- A. access doors on upper parking lots
- B. escalator
- C. tunnel
- D. elevator

Answer(s): B

QUESTION: 1199

Too much exposure to the sun can damage not only your skin, but also other parts of your body. Everyone is exposed to some sunlight during daily activities. Being in the sun too much can burn the skin and cause skin cancer and premature aging. Many people do not know, however, that overexposure to the sun can also contribute to cataract formation, allergies, and immune system damage. Young people are the most at risk since 50% of one's sun exposure is accomplished by age 18. Older folks also need to be careful and monitor their sun exposure. The word immune, as used in the above passage, most nearly means _____.

- A. circulatory
- B. resistance
- C. nerve
- D. danger

Answer(s): B

QUESTION: 1200

Many Americans do not exercise at all and this lack of movement may be caused by lack of accurate information. There are those who think that they must be active for a long period of time each day. While more exercise is better, the minimum recommendation is only 30 minutes per day and you can split that into several shorter periods. Others think that they need a gym membership to get "proper" exercise. Actually, various activities can provide the needed movement, including gardening, walking, and dancing, if you move enough while doing them. There is a great deal of misinformation about exercise in magazines and on TV. Even friends can misrepresent what is actually needed to stay in shape. The author of the above passage would agree that _____.

- A. the cost of gym membership does not matter to most people
- B. Americans may be inactive because they are bored
- C. Americans are not influenced by infomercials, magazines, or videos
- D. Americans may be more likely to exercise 30 minutes a day if they broke it up into several intervals

Answer(s): D

QUESTION: 1201

Many people assume that, once the eye of a hurricane has passed, the danger is over. Actually, the wind and heavy rains that follow a hurricane can last for a long time after. The dangers of large storms, like hurricanes and tropical storms, include flash flooding, which can occur far inside the point at which a storm made land. In the U.S., records show that over 70% of the lives lost in hurricanes are due to the hazards of flooding. In 1979, a tropical storm named Claudette, caused 45 inches of rain to fall in Alvin, Texas.

What is the main idea of this passage?

- A. Alvin, Texas, is not a safe place to live in a hurricane.
- B. Wind and water damage can occur even after a hurricane has passed.
- C. Tropical Storm Claudette flooded Alvin, Texas.
- D. Hurricanes bring record amounts of rain.

Answer(s): B

QUESTION: 1202

Travelers should check passport and visa requirements with consular officials of the countries to be visited well in advance of their departure date. Application for a passport may be made at a passport agency, many federal and state courts, probate courts, some county and municipal offices, and some post offices. The 14 major cities with U.S. passport agencies are Boston, Chicago, Denver, Honolulu, Houston, Los Angeles, Miami, New Orleans, New York, Norwalk, Connecticut, Philadelphia, San Francisco, Seattle, and Washington DC. Text quoted from:

Passport: US Passport. (n.d.).

Text quoted from: Passport: US Passport. (n.d.).

According to the above passage, which of these offices is the least likely to issue a passport?

- A. your post office
- B. your public library
- C. your municipal offices
- D. your federal or state court

Answer(s): B

QUESTION: 1203

A hungry fox saw some fine bunches of grapes hanging from a vine that was trained along a high trellis, and did his best to reach them by jumping as high as he could into the air. But it was all in vain, for they were just out of reach, so he gave up trying and walked away with an air of dignity and unconcern, remarking, "I thought those grapes were ripe, but I see now they are

quite sour."

Aesop's Fable "The Fox and the Grapes". William Caxton, 1484.

The word vain, as used in the above passage, most nearly means _____.

- A. a vessel in your body that carries blood
- B. a hopeless situation
- C. a weather device placed on top of a house that shows wind directions
- D. being arrogant and excessively proud

Answer(s): B

QUESTION: 1204

There was once a dog who used to snap at people and bite them without any provocation, and who was a great nuisance to everyone who came to his master's house. So his master fastened a bell round his neck to warn people of his presence. The dog was very proud of the bell, and strutted about tinkling it with immense satisfaction. But an old dog came up to him and said, "The fewer airs you give yourself the better, my friend. You don't think, do you, that your bell was given to you as a reward of merit? On the contrary, it is a badge of disgrace."

Aesop's Fable "The Mischievous Dog". William Caxton, 1484.

From the above passage, the reader can infer that the Aesop fable is teaching that _____.

- A. being noticed does not always mean you are famous
- B. the old dog did not understand why the master had put a bell on the dog
- C. the dog was cleverer than he was given credit for
- D. the master put the bell around the dog's neck so if he got lost he would be returned

Answer(s): A

QUESTION: 1205

In recent years, there has been much talk about something called ozone. What is this stuff and why is it important? Ozone is an invisible gas. There is a thin layer of it about 15 miles above the surface of the Earth. Some of the sun's rays can be harmful to us and ozone helps protect us from damage. Unfortunately, the amount of ozone in our atmosphere has been reduced in the past few years and scientists think they know how this happened. Humans have produced chlorofluorocarbons (CFCs), which have caused a decrease in ozone. One other effect of reduced ozone involves climate change. If more of the sun's rays reach the Earth, they can cause changing weather patterns and a warmer climate. Possible disastrous effects of this change include the increased incidence of tornadoes and droughts.

What is the best title for this passage?

- A. "The Earth's Atmosphere and Human Life"
- B. "The Harmful Nature of CFCs"
- C. "The Ozone Layer Prevents Tornadoes"
- D. "The Varied Effects of Reduced Ozone"

Answer(s): D

QUESTION: 1206

In 1965, a television show aired that would become a Christmas classic for many families. It was called "A Charlie Brown Christmas." This television movie has the distinction of being the cartoon special to run for the longest time in television history and has been shown every year since its debut. This tradition began when an ad agent for Coca-Cola contacted producer, Lee Mendelson. Peanuts had just been featured on a 1965 cover of Time Magazine and the agent told Mendelson about a documentary about Peanuts creator, Charles Schulz. Mendelson was asked if he had ever thought about producing a Peanuts Christmas special. The producer, wanting to appear on top of his game, assured the agent that he had. Hurriedly contacting Schulz, Mendelson met with him the next day and they created the story.

From the above passage, the reader would know that 1965 was an important year because _____.

- A. "A Charlie Brown Christmas" first debuted and the Peanuts characters made the cover of Time Magazine.
- B. The passage does not give a stated reason.
- C. "A Charlie Brown Christmas" first debuted.
- D. The Peanuts characters made the front of Time Magazine.

Answer(s): A

QUESTION: 1207

It is not surprising that John Fitzgerald Kennedy entered public service. He came from a family that valued such commitment, including his father, Joseph P. Kennedy. The elder Kennedy served as ambassador to Great Britain from 1937 to 1940. John served his country in the U.S. Navy after he graduated from Harvard University. During that time, Kennedy escaped death while serving as skipper of a PT boat in the Pacific. He even rescued another Navy man by swimming, with him in tow, to the shore of a safe island.

The word ambassador, as used in the above passage, most nearly means _____.

- A. messenger
- B. diplomat
- C. dispatcher
- D. courier

Answer(s): B

QUESTION: 1208

In light of the increased incidence of security issues in recent years, attendees at the 2014 Super Bowl saw heightened regulations regarding transportation, parking, and other game-related activities. The 80,000 ticket holders were not allowed to tailgate or utilize taxis or car services for transportation to the game site. They could arrive via Fan Express, a charter bus, for a fee of \$51.00. The NJ Transit was another approved method of transportation. Finally, a lucky 13,000 people could use one of the parking passes that were available.

According to the above passage, what is the approximate number of ticket holders who will not have a parking pass to use for the NFL Super Bowl?

- A. 67,000
- B. 13,000

- C. 51
- D. 80,000

Answer(s): A

QUESTION: 1209

The United States has carried out intelligence activities since the days of George Washington, but only since World War II have they been coordinated on a government-wide basis. President Franklin D. Roosevelt appointed New York lawyer and war hero, William J. Donovan, to become first the Coordinator of Information, and then, after the US entered World War II, head of the Office of Strategic Services (OSS) in 1942. The OSS - the forerunner to the CIA - had a mandate to collect and analyze strategic information. What is the best title for this passage?

- A. "War Changes the Role of Intelligence Organizations"
- B. "Roosevelt Creates COI Job"
- C. "Joint Chiefs of Staff Conduct Special Operations"
- D. "America's Entry into World War II"

Answer(s): A

QUESTION: 1210

The CIA is separated into five basic components: the Directorate of Operations, the Directorate of Analysis, the Directorate of Science & Technology, the Directorate of Support, and the Directorate of Digital Innovation. They carry out "the intelligence cycle," the process of collecting, analyzing, and disseminating intelligence information to top US government officials. According to the above paragraph, which of these plays a major role in disseminating CIA intelligence information to top government officials?

- A. the Secret Service
- B. the White House
- C. the Directorate of Analysis
- D. State Intelligence Officers

Answer(s): C

QUESTION: 1211

The saying, "It's not over until it's over" certainly applies to football. In spite of a significant lead, a team cannot count on a win until the last whistle sounds. Any team, no matter how skilled, must maintain a sense of purpose throughout all four quarters and should not relax if they want to guarantee a mark in the win column. Double-digit leads have been magically erased during small time spans late in a game. It can happen when an opposing player intercepts the ball and makes a break for the goal line, scoring the decisive points. Which of these words most nearly means decisive as it is used in the above passage?

- A. earnest
- B. certain
- C. critical

D. determined

Answer(s): C

QUESTION: 1212

Apparently, two sheriffs in the state of New York are finding fault with Governor Cuomo's recent gun control efforts. The men have gone so far as to state that they will not enforce gun law packages Cuomo has recently signed into law. Public statements to this effect have been made by the two law enforcement officials and Cuomo warned that the precedent they set could be dangerous. The officers' intent does not appear to signify a new practice, however, as records show a general lack of enforcement of felony gun laws among sheriffs since 2010. State and local police have been the ones to file the majority of common gun-related charges (98%) during that time.

Based on the above passage, the reader can conclude that _____.

- A. Gov. Cuomo is unaffected by the sheriff reaction to the new gun laws
- B. sheriffs are not going to be the primary group to file felony gun charges
- C. state and local police do not agree with the new gun laws in New York
- D. Gov. Cuomo is going to take legal action against officers who do not comply with new laws

Answer(s): B

QUESTION: 1213

Using functional magnetic resonance imaging (fMRI), researchers have identified abnormalities in the brains of children with attention deficit/hyperactivity disorder (ADHD) that may serve as a biomarker for the disorder, according to a study presented November 28 at the annual meeting of the Radiological Society of North America (RSNA). ADHD is one of the most common childhood disorders, affecting an estimated five to eight percent of school-aged children. Symptoms, which may continue into adulthood, include inattention, hyperactivity and impulsivity behaviors that are out of the normal range for a child's age and development. What is the best title for this passage?

- A. "ADHD Occurrence in Children is Between 5% and 8%"
- B. "MRIs Aid in Earlier ADHD Diagnosis"
- C. "Brain Differences Found in Children with ADHD"
- D. "ADHD Kids Exhibit Abnormal Behavior"

Answer(s): C

QUESTION: 1214

Like so many other high-tech developments, GPS was designed by the U.S. military. The concept started in the late 60s, but the first satellite was not launched until February 1978. In 1989, the Magellan Corp. introduced the first hand-held GPS receiver. In 1992, GPS was used in Operation Desert Storm. On March 1996, the President decided to make GPS free for civilian users.

According to the above passage, when did the first idea for developing GPS surface?

- A. March 1996

- B. in the late 60s
- C. early in 1978
- D. in 1989

Answer(s): B

QUESTION: 1215

The Library of Congress' Law Library is a closed stacks library, which means that nearly 99% of its print collection is housed in compact shelving in cavernous underground storage rooms covering an area nearly the size of two U.S. football fields. Patrons desiring to review books held in the closed stacks simply submit an online request for them; requested materials are usually delivered to the Law Library Reading Room for use within approximately an hour. What is the best choice for the meaning of cavernous, as used in the above passage?

- A. hollow
- B. spacious
- C. containing cavities
- D. filled with caverns

Answer(s): B

QUESTION: 1216

Gerry Marpus is 78 years old and needs to use a cane for balance when walking. Many of her friends have given up driving and rely on their families to take them around town. Gerry, however, maintains that she is perfectly capable of driving herself. Knowing that her children were concerned about this, she decided to take advantage of a senior citizen driving course at the local community center. When she presented her completion certificate to her daughter and son, she could see the relief in their eyes and she knew she had done the right thing.

From the above passage, the reader can infer that Gerry _____.

- A. wanted the people at the community center to decide if she could drive because she was afraid
- B. wants to disregard whatever her children think and do what she wants to do
- C. cares about her children's concerns and is an independent thinker
- D. wants to be labeled disabled so she can get further treatments for balance issues

Answer(s): C

QUESTION: 1217

The Missile Gap was in essence a growing perception in the West, especially in the U.S.A., that the Soviet Union was quickly developing an intercontinental range ballistic missile (ICBM) capability earlier, in greater numbers, and with far more capability than that of the United States. Even as that perception was disproved, it became evident that the Soviets were placing their major effort toward developing strategic missiles against which, once launched, there was no defense. The perceived missile gap that ensued was based on a comparison between U.S. ICBM strength as then programmed, and reasonable, although erroneous estimates of prospective Soviet ICBM strength that were generally accepted.

What is the main idea of this paragraph?

- A. Despite erroneous thinking, the Soviet Union missile threat was still taken seriously.
- B. The Soviet Union was never a real threat for ballistic missiles.
- C. The Missile Gap had nothing to do with the perceptions of the U.S.
- A. regarding Soviet advances.
- D. The United States and Soviet Union worked together on the Missile Gap.

Answer(s): A

QUESTION: 1218

Mainstream news reports, that focus on averting global warming, have asked questions such as "How much will this cost?" and "How much will we have to cut back on economic growth?" Such irresponsible and naive questions highlight the scale of self-interests and ignorance that dominate national debate. If we are not part of the solution (attempting to seriously reduce carbon emissions), then we are part of the problem (ignoring the serious consequences of global warming). Whatever the `costs' are to save the planet, this will be our greatest challenge and our greatest investment and, if we fail in this task, then we will perish. The author of the above passage would probably support _____.

- A. people who ignore the consequences of global warming
- B. the efforts of industry to limit carbon emissions
- C. the idea that slowing global warming is simple
- D. the consequences of global warming

Answer(s): B

QUESTION: 1219

This is a portion of a position statement by the National Association for the Deaf (NAD) concerning American Sign Language (ASL): Developments in the field of linguistics and an emerging body of research in education have validated the historical views of the NAD on ASL and its value in the education of deaf and hard of hearing children. The inherent capability of children to acquire ASL should be recognized and used to enhance their cognitive, academic, social, and emotional development. Accordingly, the NAD supports the bilingual approach for deaf and hard of hearing children. Deaf and hard of hearing children must have the right to receive early and full exposure to ASL as a primary language, along with English. Studies have shown that when deaf and hard of hearing children are exposed to ASL at an early age, they are given the opportunity to reach their full potential. Within the context of the above passage, which of these most nearly means emerging as it is used in the first sentence?

- A. moving out
- B. transpiring
- C. coming into view
- D. becoming known

Answer(s): D

QUESTION: 1220

We think of chitchat and small talk as the things people say to pass the time or kill an awkward silence. But new research suggests these idle conversations could be a social-bonding tool passed down from our ape-like ancestors. Scientists with Princeton University studied ring tailed lemurs, animals that are considered among the more ancestral forms of primates - the evolutionary lineage that includes humans, apes and monkeys. The study concludes that lemurs call to each other essentially as a replacement for mutual grooming, or cleaning each other's fur, a widespread social-bonding behavior among non-human primates. From the above passage, the reader can conclude that _____.

- A. chatting cannot be used to pass time
- B. lemurs would rather groom each other than call out to each other
- C. lemurs are more like monkeys than apes
- D. some human behaviors may have evolved from those of our primate ancestors

Answer(s): D

QUESTION: 1221

Make a special appointment by calling the passport office if you're leaving in less than 14 days. If you're leaving within six business days, skip the appointment and head to the nearest passport office immediately and take along your airline tickets or airline-generated itinerary to prove your need for speed. Bring the photos and your old passport - or, if you don't have one, your birth certificate and valid driver's license. For each application, in addition to the standard fee (see travel.state.gov/newfees.html), you will need to pay a \$60 fee for expedited service plus fees for two-way overnight mail. What is the best title for this passage?

- A. "Obtaining a Passport in a Hurry"
- B. "Guidelines for Getting a Passport Photo"
- C. "Costs Involved in Getting a Passport"
- D. "Passport Office Locations"

Answer(s): A

QUESTION: 1222

Scientists have shown that people who have a variant of a longevity gene, called KLOTHO, have improved brain skills such as thinking, learning and memory, regardless of their age, sex, or whether they have a genetic risk factor for Alzheimer's disease. Increasing KLOTHO gene levels in mice made them smarter, possibly by increasing the strength of connections between nerve cells in the brain. The study was partly funded by the National Institutes of Health. According to the above passage, _____ could be a factor in the development of Alzheimer's disease.

- A. a longevity gene
- B. sex
- C. a genetic risk factor
- D. the KLOTHO gene

Answer(s): C

QUESTION: 1223

Switzerland is a small yet diverse country. It consists of 26 member states, the cantons, each of them with its own constitution, legislature, government - and a unique history. One of these is the canton of Solothurn. Its roots are in the city of the same name which can trace back its history to Roman times, when it was known as Salodurum. Becoming a Free Imperial City (Freie Reichsstadt) in the Holy Roman Empire in 1218 and joining the Old Swiss Confederacy in 1481, the city state of Solothurn grew in size through buying up available scraps of territory as well as some acquisitions by force, finally reaching its current shape in the early 16th century. Original author: Patrick Borer. This article, Ambassadors, milkmaids, and hot air balloons, was originally published in "The Public Domain Review. "

The word constitution, as used in the above passage, most nearly means _____.

- A. arrangement
- B. body
- C. laws
- D. construction

Answer(s): C

QUESTION: 1224

University of British Columbia (UBC) researchers have discovered a unique nerve structure in the mouth and tongue of rorqual whales that can double in length and then recoil like a bungee cord. The stretchy nerves explain how the massive whales are able to balloon an immense pocket between their body wall and overlying blubber to capture prey during feeding dives. In humans, stretching nerves usually damages them. In these whales, the nerve cells are packaged inside a central core in such a way that the individual nerve fibers are never really stretched, they simply unfold.

From the above passage, the reader can infer that _____.

- A. all whales have stretchy tongues
- B. human tongues can double in length
- C. stretching would not damage a whale's nerve cell fibers
- D. tongue construction is important to these whales' ability to survive

Answer(s): D

QUESTION: 1225

Doctors in many countries have gone back to using old antibiotics that were abandoned... because their toxic side effects were so frequent and so bad. But superbugs; like Acinetobacter, have challenged doctors all over the world by now be-coming resistant to these older and considered more dangerous medicines. Even colistin, an antibiotic discovered 60 years ago, has recently been used as a salvage remedy to treat patients with Acinetobacter infections. And it was successful for a while, but now it occasionally fails due to recent extensive use that has caused the bacteria to become resistant; leading to problem superbugs that are resistant to all available antibiotics.

Quote from Matthew Falagas of the Alfa Institute of Biomedical Sciences in Athens, Greece and Tufts University School of Medicine in Boston, Mass.

Based on the above passage, the author would agree that _____.

- A. newer drugs are more dangerous than old ones
- B. acinetobacter is the only "superbug"
- C. old antibiotics are not as strong as newer ones
- D. the widespread use of antibiotics for minor illnesses should be reduced

Answer(s): D

QUESTION: 1226

The U.S. Department of Homeland Security's Federal Emergency Management Agency announced that federal disaster aid has been made available to the State of Alaska to supplement state, tribal, and local recovery efforts in the area affected by a severe storm on August 27, 2015. The President's action makes federal funding available to state, tribal and eligible local governments and certain private nonprofit organizations on a cost-sharing basis for emergency work and the repair or replacement of facilities damaged by a severe storm in the North Slope Borough. Based on the above passage, which of these statements is true?

- A. States may not receive this aid.
- B. Aid recipients do not have to pay anything to receive funds.
- C. The funding for this aid will come from the states.
- D. A Native American tribal council may receive this aid.

Answer(s): D

QUESTION: 1227

All of the laws enforced by Equal Employment Opportunity Commission, except for the Equal Pay Act, require you to file a Charge of Discrimination before you can file a job discrimination lawsuit against your employer. In addition, an individual, organization, or agency may file a charge on behalf of another person in order to protect the aggrieved person's identity. There are time limits for filing a charge.

The word aggrieved, as used in the above passage, most nearly means _____.

- A. distressed
- B. wounded
- C. pained
- D. upset

Answer(s): A

QUESTION: 1228

Tucked away in Southeast Portland near the Sellwood Bridge is Oaks Amusement Park, one of the oldest continuously operated amusement parks in the United States. A modest operation, the park currently includes about two dozen rides that operate seasonally, a skating rink that is open year-round, and picnic grounds. The Oaks has been a part of the lives of many Portlanders and other Oregonians for nearly a hundred years, and many people hold fond memories of it.

From the above passage, the reader can infer that _____.

- A. Oaks Amusement Park is very easy to find
- B. Oaks Amusement Park is getting too old to maintain
- C. many people would be saddened if Oaks Amusement Park closed down
- D. there are limited activities from which to choose at Oaks Amusement Park

Answer(s): C

QUESTION: 1229

About 2 to 3 out of every 1,000 children in the United States are born deaf or hard of hearing. Over 90% of these children are born to parents who can hear. Early identification of children who are born deaf or hard of hearing is critical to ensure that their families have the resources they need to help their children acquire language, spoken and/or visual, and achieve age-appropriate communicative, cognitive, academic, social, and emotional development. The National Association for the Deaf (NAD) supported passage of the Early Hearing Detection and Intervention Act (EHDI) in 2000 and subsequent federal funding to establish state newborn hearing screening programs in hospitals. At the time, the average age of identification of deaf children was about 21/2 years old; children who were hard of hearing were often identified much later. The goals of the EHDI program include hearing screening of all newborns by age one month, confirmation of hearing status by three months, and enrollment in an early intervention program for deaf and hard of hearing babies and their families by six months.

From the information given in the above passage, the reader can assume the author would agree that _____.

- A. early identification of hearing problems can help match parents with resources
- B. a child's first hearing test should be done at 6 months of age
- C. intervention for hard-of-hearing children is not necessary until they reach school age
- D. most deaf children are born to deaf parents

Answer(s): A

QUESTION: 1230

For the seventh consecutive year, Intelligence Community employee job satisfaction ranks the IC as one of the "Best Places to Work in the Federal Government," according to an independent analysis of federal workers' job satisfaction and commitment. The IC finished in first place in national security and second overall - up from fourth place in 2014 - among large agencies that employ more than 15,000 full-time permanent employees, a category which includes 19 organizations. Rankings were compiled by the Partnership for Public Service and announced today. Additionally, the IC finished in first place in teamwork, second in pay, effective leadership, employee skills-mission match and third in strategic management.

The data for this ranking study was gathered by _____.

- A. an agency outside the Intelligence Community
- B. a large agency within the IC
- C. the Federal Government
- D. the Intelligence Community

Answer(s): A

QUESTION: 1231

Over the past few years, body mass index (BMI), a ratio of a person's height and weight, has effectively become a proxy for whether a person is considered healthy. Many U.S. companies use their employees' BMIs

as a factor in determining workers' health care costs. And people with higher BMIs could soon have to pay higher health insurance premiums, if a rule proposed in April by the Equal Employment Opportunity Commission is adopted. But a new study led by UCLA psychologists has found that using BMI to gauge health incorrectly labels more than 54 million Americans as "unhealthy," even though they are not.

The word proxy, as used in the above passage, most nearly means _____.

- A. authority
- B. go-between
- C. representation
- D. server

Answer(s): C

QUESTION: 1232

In 2013, a gigantic landslide at a Utah copper mine probably was the biggest non volcanic slide in North America's modern history, and included two rock avalanches that happened 90 minutes apart and surprisingly triggered 16 small earthquakes, University of Utah scientists discovered. The landslide - which moved at an average of almost 70 mph and reached estimated speeds of at least 100 mph - left a deposit so large it "would cover New York's Central Park with about 20 meters (66 feet) of debris," the researchers report in the January 2014 cover study in the Geological Society of America magazine GSA Today. While earthquakes regularly trigger landslides, the gigantic landslide the night of April 10, 2013, is the first known to have triggered quakes.

From the above passage, the reader can conclude that _____.

- A. the Utah copper mine landslide was the largest ever recorded in the world
- B. the 2013 earthquakes were triggered by a volcano
- C. the landslide was a slow-moving one
- D. scientists would be surprised if another landslide created earthquakes

Answer(s): D

Section 7: Word Knowledge

QUESTION: 1233

The gold was kept in a secure vault.

- A. locked
- B. safe
- C. unknown
- D. thick

Answer(s): B

QUESTION: 1234

Assimilate most nearly means _____.

- A. absorb
- B. react
- C. pretend
- D. lie

Answer(s): A

QUESTION: 1235

Theorize most nearly means _____.

- A. know
- B. speculate
- C. study
- D. travel

Answer(s): B

QUESTION: 1236

Symmetrical most nearly means _____.

- A. uplifted
- B. congruent
- C. handsome
- D. positive

Answer(s): B

QUESTION: 1237

The exchange student was proficient in French, German, and English.

- A. poor

- B. knowledgeable
- C. adept
- D. exacting

Answer(s): C

QUESTION: 1238

The students were scheduled to observe a plenary session of Congress.

- A. scheduled
- B. example
- C. special
- D. full

Answer(s): D

QUESTION: 1239

He tried to goad his audience.

- A. insult
- B. incite
- C. please
- D. bore

Answer(s): B

QUESTION: 1240

He ran headlong into the fight.

- A. headfirst
- B. reluctantly
- C. happily
- D. recklessly

Answer(s): D

QUESTION: 1241

Flagrant most nearly means _____.

- A. quiet
- B. amazing
- C. delayed
- D. glaring

Answer(s): D

QUESTION: 1242

The word most opposite in meaning to stimulate is _____.

- A. support
- B. arrest
- C. travel
- D. dislike

Answer(s): B

QUESTION: 1243

Legacy most nearly means _____.

- A. history
- B. bequest
- C. story
- D. will

Answer(s): B

QUESTION: 1244

The actions of the CEO were unconscionable.

- A. clever
- B. illegal
- C. excessive
- D. automatic

Answer(s): C

QUESTION: 1245

The sergeant gave his reasoned opinion.

- A. irate
- B. logical
- C. impressive
- D. uninformed

Answer(s): B

QUESTION: 1246

Laudable most nearly means _____.

- A. loud
- B. fluid
- C. commendable
- D. transparent

Answer(s): C

QUESTION: 1247

The brass was burnished.

- A. yellow
- B. old
- C. expensive
- D. polished

Answer(s): D

QUESTION: 1248

The commodity was sold.

- A. product
- B. stock
- C. idea
- D. table

Answer(s): A

QUESTION: 1249

Her motives were oblique.

- A. pure
- B. emotional
- C. obscure
- D. amusing

Answer(s): C

QUESTION: 1250

Superfluous most nearly means _____.

- A. superior
- B. unnecessary
- C. helpful
- D. expensive

Answer(s): B

QUESTION: 1251

The word most opposite in meaning to hypocrisy is _____.

- A. honesty
- B. happy
- C. angry
- D. threatening

Answer(s): A

QUESTION: 1252

The report indicated a significant hemorrhage of corporate earnings.

- A. gain
- B. payout
- C. trade
- D. loss

Answer(s): D

QUESTION: 1253

The army soldiers were ordered to immediate garrison duty.

- A. field
- B. combat
- C. latrine
- D. fort

Answer(s): D

QUESTION: 1254

Fiscal most nearly means _____.

- A. year
- B. financial
- C. calendar
- D. three months

Answer(s): B

QUESTION: 1255

Domicile most nearly means _____.

- A. office
- B. domestic
- C. home
- D. vacation

Answer(s): C

QUESTION: 1256

Abate most nearly means _____.

- A. recover
- B. aid
- C. foreclose
- D. end

Answer(s): D

QUESTION: 1257

To commission most nearly means _____.

- A. to give
- B. to rescind
- C. to earn
- D. to authorize

Answer(s): D

QUESTION: 1258

He gave a succinct account of the events.

- A. passionate
- B. lengthy
- C. uncensored
- D. concise

Answer(s): D

QUESTION: 1259

The vote resulted in the demise of the proposed new law.

- A. passage
- B. death
- C. postponement
- D. abatement

Answer(s): B

QUESTION: 1260

The politician exuded charisma.

- A. odors
- B. falseness
- C. charm

D. generosity

Answer(s): C

QUESTION: 1261

Burrow most nearly means _____.

- A. deepen
- B. hide
- C. nestle
- D. jump

Answer(s): C

QUESTION: 1262

That custom still prevails.

- A. angers
- B. persists
- C. surprises
- D. excites

Answer(s): B

QUESTION: 1263

Contravene most nearly means _____.

- A. invade
- B. obstruct
- C. argue
- D. reverse

Answer(s): B

QUESTION: 1264

Chasm most nearly means _____.

- A. hole
- B. sky
- C. mountain
- D. valley

Answer(s): A

QUESTION: 1265

Fundamental most nearly means _____.

- A. radical
- B. religious
- C. basic
- D. excessive

Answer(s): C

QUESTION: 1266

Susceptible most nearly means _____.

- A. travel
- B. resistant
- C. limited
- D. gullible

Answer(s): D

QUESTION: 1267

To emit most nearly means _____.

- A. to give off
- B. to smell
- C. to contain
- D. to admit

Answer(s): A

QUESTION: 1268

Ardent most nearly means _____.

- A. trustworthy
- B. passion
- C. diligent
- D. busy

Answer(s): B

QUESTION: 1269

It was a sturdy table.

- A. well-built
- B. ugly
- C. thick
- D. small

Answer(s): A

QUESTION: 1270

Listless most nearly means _____.

- A. gullible
- B. inattentive
- C. lazy
- D. practical

Answer(s): B

QUESTION: 1271

Brevity is the soul of wit.

- A. beauty
- B. intelligence
- C. terseness
- D. humor

Answer(s): C

QUESTION: 1272

Pare most nearly means _____.

- A. cut
- B. fruit
- C. two
- D. umbrella

Answer(s): A

QUESTION: 1273

He facilitated her promotion.

- A. hindered
- B. helped
- C. disliked
- D. ignored

Answer(s): B

QUESTION: 1274

The word most opposite in meaning to perpetuate is _____.

- A. kill
- B. preserve

- C. small
- D. structure

Answer(s): A

QUESTION: 1275

The spectator enjoyed the game.

- A. competitor
- B. observer
- C. referee
- D. organizer

Answer(s): B

QUESTION: 1276

Sally was such a precocious child.

- A. tall
- B. talkative
- C. shy
- D. smart

Answer(s): D

QUESTION: 1277

The teacher cited some examples.

- A. memorized
- B. finished
- C. specified
- D. examined

Answer(s): C

QUESTION: 1278

The word most opposite in meaning to sentinel is _____.

- A. guard
- B. prisoner
- C. pilot
- D. teacher

Answer(s): B

QUESTION: 1279

Illustrious most nearly means _____.

- A. illustrated
- B. famous
- C. foolish
- D. intelligent

Answer(s): B

QUESTION: 1280

Fallacious most nearly means _____.

- A. invalid
- B. delightful
- C. noisy
- D. overlooked

Answer(s): A

QUESTION: 1281

Tim had a penchant for engaging in subterfuge.

- A. religion
- B. evasion
- C. gambling
- D. danger

Answer(s): B

QUESTION: 1282

The goods were transported yesterday.

- A. donated
- B. conveyed
- C. destroyed
- D. trucked

Answer(s): B

QUESTION: 1283

Rigid most nearly means _____.

- A. strong
- B. weak
- C. pliable
- D. inflexible

Answer(s): D

QUESTION: 1284

The vassals engaged in hard work daily.

- A. employees
- B. slaves
- C. soldiers
- D. construction workers

Answer(s): B

QUESTION: 1285

Deplore most nearly means _____.

- A. accept
- B. insult
- C. disapprove
- D. salute

Answer(s): C

QUESTION: 1286

Sparse most nearly means _____.

- A. space
- B. meager
- C. brief
- D. thirsty

Answer(s): B

QUESTION: 1287

Weal most nearly means _____.

- A. happiness
- B. injury
- C. scream
- D. tire

Answer(s): A

QUESTION: 1288

The dog yearned for a bone.

- A. barked

- B. dug
- C. begged
- D. longed

Answer(s): D

QUESTION: 1289

The customs agent confiscated the goods.

- A. bought
- B. noticed
- C. seized
- D. stole

Answer(s): C

QUESTION: 1290

Eloquent most nearly means _____.

- A. long
- B. beautiful
- C. articulate
- D. dangerous

Answer(s): C

QUESTION: 1291

Illusion most nearly means _____.

- A. mirage
- B. distant
- C. sight
- D. perspective

Answer(s): A

QUESTION: 1292

The sight was fantastic.

- A. simple
- B. illusory
- C. typical
- D. strange

Answer(s): D

QUESTION: 1293

Enmity most nearly means _____.

- A. enemy
- B. hatred
- C. anger
- D. childish

Answer(s): B

QUESTION: 1294

Abridge most nearly means _____.

- A. native
- B. shorten
- C. travel
- D. delirious

Answer(s): B

QUESTION: 1295

They terminated his contract.

- A. bought
- B. extended
- C. sold
- D. ended

Answer(s): D

QUESTION: 1296

The textbook presented a finite number of solutions to the problem.

- A. unlimited
- B. limited
- C. variety
- D. unusual

Answer(s): B

QUESTION: 1297

Null most nearly means _____.

- A. zero
- B. dull
- C. unskilled
- D. rapid

Answer(s): A

QUESTION: 1298

She was a tennis fanatic.

- A. observer
- B. zealot
- C. player
- D. coach

Answer(s): B

QUESTION: 1299

Impertinence most nearly means _____.

- A. fun
- B. boring
- C. rude
- D. impatient

Answer(s): C

QUESTION: 1300

Accord most nearly means _____.

- A. belief
- B. accurate
- C. settle
- D. harmony

Answer(s): D

Explanation:

Accord, like harmony, means agreement.

QUESTION: 1301

Ridicule most nearly means _____.

- A. mock
- B. support
- C. ride
- D. silly

Answer(s): A

QUESTION: 1302

Pardon most nearly means _____.

- A. courtesy
- B. excuse
- C. believe
- D. respect

Answer(s): B

QUESTION: 1303

Isochronous most nearly means _____.

- A. fast
- B. slow
- C. equal
- D. different

Answer(s): C

QUESTION: 1304

Lackadaisical most nearly means _____.

- A. lazy
- B. listless
- C. promiscuous
- D. suitable

Answer(s): B

QUESTION: 1305

The fruit was edible.

- A. wax
- B. expensive
- C. foreign
- D. digestible

Answer(s): D

QUESTION: 1306

Universities and colleges should be designed to cater to the philomath.

- A. students
- B. scholars
- C. teachers
- D. faculty

Answer(s): B

QUESTION: 1307

Pretense most nearly means _____.

- A. polite
- B. dishonesty
- C. stress
- D. appearance

Answer(s): B

QUESTION: 1308

At an early age Jane showed a proclivity for music and dancing.

- A. predisposition
- B. interest
- C. dislike
- D. fever

Answer(s): A

QUESTION: 1309

Her conversation was incoherent.

- A. eloquent
- B. succinct
- C. unintelligible
- D. amusing

Answer(s): C

QUESTION: 1310

The week following Joe DiMaggio's death was filled with often mawkish eulogies.

- A. long
- B. sentimental
- C. boring
- D. detailed

Answer(s): B

QUESTION: 1311

She established proof.

- A. offered
- B. invented
- C. demanded
- D. demonstrated

Answer(s): D

QUESTION: 1312

Ephemeral most nearly means _____.

- A. short-lived
- B. mythical
- C. dead
- D. exceptional

Answer(s): A

QUESTION: 1313

Avocation most nearly means _____.

- A. hobby
- B. occupation
- C. vacation
- D. education

Answer(s): A

QUESTION: 1314

Kvetch most nearly means _____.

- A. assert
- B. yell
- C. complain
- D. argue

Answer(s): C

QUESTION: 1315

Her eyesight was acute.

- A. sharp
- B. poor
- C. unusual
- D. tested

Answer(s): A

QUESTION: 1316

Inamorata most nearly means _____.

- A. boyfriend
- B. mistress
- C. best friend
- D. acquaintance

Answer(s): B

QUESTION: 1317

Her thoughts on the matter were inconsequential.

- A. profound
- B. disturbing
- C. irrelevant
- D. confused

Answer(s): C

QUESTION: 1318

Debouch most nearly means _____.

- A. emerge
- B. fight
- C. relax
- D. capture

Answer(s): A

QUESTION: 1319

He was an amateur astronomer.

- A. veteran
- B. novice
- C. interested
- D. pleased

Answer(s): B

QUESTION: 1320

She had no idea how to react to her ludic boyfriend.

- A. playful
- B. cheating
- C. crazy

D. lazy

Answer(s): A

QUESTION: 1321

The rose was crimson.

- A. blooming
- B. colorful
- C. fragrant
- D. red

Answer(s): D

QUESTION: 1322

The word most opposite in meaning to benison is _____.

- A. theft
- B. replaceable
- C. curse
- D. heavy

Answer(s): C

QUESTION: 1323

She was exempt from gym class.

- A. banned
- B. excused
- C. tired
- D. refreshed

Answer(s): B

QUESTION: 1324

The eldritch light of the desert can play tricks on your eyes.

- A. bright
- B. wavering
- C. strange
- D. yellow

Answer(s): C

QUESTION: 1325

Defective most nearly means _____.

- A. flawed
- B. noticeable
- C. rare
- D. durable

Answer(s): A

QUESTION: 1326

Allot most nearly means _____.

- A. plow
- B. assign
- C. property
- D. test

Answer(s): B

QUESTION: 1327

The doctor gave the patient a cursory examination.

- A. in-depth
- B. painful
- C. unnecessary
- D. superficial

Answer(s): D

QUESTION: 1328

Arcanum most nearly means _____.

- A. rare
- B. secret
- C. tangible
- D. false

Answer(s): B

QUESTION: 1329

Her answer was terse.

- A. defensive
- B. angry
- C. lengthy
- D. brief

Answer(s): D

QUESTION: 1330

The dulcet songs of the band got the attention of the audience.

- A. harmonious
- B. love
- C. jazzy
- D. loud

Answer(s): A

QUESTION: 1331

He was arrested on a misdemeanor charge.

- A. theft
- B. serious
- C. petty crime
- D. bogus

Answer(s): C

QUESTION: 1332

Embonpoint most nearly means _____.

- A. plump
- B. tall
- C. fast
- D. cold

Answer(s): A

QUESTION: 1333

He concocted a story about me.

- A. told
- B. rehearsed
- C. invented
- D. remembered

Answer(s): C

QUESTION: 1334

He spent his days searching fruitlessly for that chimera, his true self.

- A. personality
- B. enigma

- C. talent
- D. monster

Answer(s): D

QUESTION: 1335

Her former home was in Colorado.

- A. previous
- B. current
- C. second
- D. abandoned

Answer(s): A

QUESTION: 1336

Mulct most nearly means _____.

- A. complain
- B. play
- C. work
- D. defraud

Answer(s): D

QUESTION: 1337

My voice is strident.

- A. soft
- B. melodious
- C. harsh
- D. baritone

Answer(s): C

QUESTION: 1338

Raffish most nearly means _____.

- A. clean
- B. serene
- C. tawdry
- D. expensive

Answer(s): D

QUESTION: 1339

The abhorrent smell from the lake overpowered the picnickers gathered on the shore.

- A. strong
- B. pleasant
- C. offensive
- D. tantalizing

Answer(s): C

QUESTION: 1340

Regale most nearly means _____.

- A. pleasure
- B. rule
- C. pretend
- D. lecture

Answer(s): A

QUESTION: 1341

The water was calm that day with detritus slowly moving in the small eddies.

- A. fish
- B. lily pads
- C. plants
- D. debris

Answer(s): D

QUESTION: 1342

The Prime Minister was always cautious about leaving his redoubt in Belgrade.

- A. city
- B. stronghold
- C. house
- D. country

Answer(s): B

QUESTION: 1343

Mike was afraid he might be ostracized for stepping out of line.

- A. banished
- B. scolded
- C. assaulted
- D. arrested

Answer(s): A

QUESTION: 1344

The hotel was specifically designed for the wayworn traveler.

- A. lost
- B. weary
- C. demanding
- D. happy

Answer(s): B

QUESTION: 1345

The park has no showers and no potable water.

- A. usable
- B. clear
- C. drinkable
- D. tasty

Answer(s): C

QUESTION: 1346

Umbrage most nearly means _____.

- A. headfirst
- B. injury
- C. doubtful
- D. recklessly

Answer(s): B

QUESTION: 1347

Vie most nearly means _____.

- A. quest
- B. particular
- C. delayed
- D. contend

Answer(s): D

QUESTION: 1348

Sustenance most nearly means _____.

- A. food

- B. shelter
- C. fish
- D. reliant

Answer(s): A

QUESTION: 1349

Revelation most nearly means _____.

- A. sermon
- B. participate
- C. inside
- D. disclose

Answer(s): D

QUESTION: 1350

Puerile most nearly means _____.

- A. dangerous
- B. illegal
- C. childish
- D. automatic

Answer(s): C

QUESTION: 1351

Nomic most nearly means _____.

- A. old
- B. logical
- C. customary
- D. uninformed

Answer(s): C

QUESTION: 1352

Magnitude most nearly means _____.

- A. importance
- B. peculiar
- C. alone
- D. tantamount

Answer(s): A

QUESTION: 1353

His vapid presentation earned him a C in the class.

- A. mediocre
- B. plagiarized
- C. dull
- D. polished

Answer(s): C

QUESTION: 1354

Percil was popular at the meeting because of the extraneous information he provided.

- A. interesting
- B. exciting
- C. outside
- D. informative

Answer(s): C

QUESTION: 1355

She was often solicitous of her father's feelings.

- A. careful
- B. ignorant
- C. forgetful
- D. abusive

Answer(s): A

QUESTION: 1356

I could never get over her liquid blue, limpid eyes.

- A. bright
- B. clear
- C. attentive
- D. dull

Answer(s): B

QUESTION: 1357

The goal of the treaty is to develop international amity and reciprocal trade.

- A. agreement
- B. friendship
- C. standards
- D. understanding

Answer(s): B

QUESTION: 1358

He often bragged about the bravery of his favorite cohort.

- A. person
- B. teacher
- C. companion
- D. employee

Answer(s): C

QUESTION: 1359

Lapse most nearly means _____.

- A. drama
- B. deviation
- C. evil
- D. unhappy

Answer(s): B

QUESTION: 1360

Indigenous most nearly means _____.

- A. poor
- B. rich
- C. immigrant
- D. native

Answer(s): D

QUESTION: 1361

Illusive most nearly means _____.

- A. insignificant
- B. deceptive
- C. useful
- D. hidden

Answer(s): B

QUESTION: 1362

Hesitant most nearly means _____.

- A. slam
- B. pause
- C. foreclose
- D. end

Answer(s): B

QUESTION: 1363

Gravity most nearly means _____.

- A. planet
- B. relationship
- C. earn
- D. seriousness

Answer(s): D

QUESTION: 1364

Fondle most nearly means _____.

- A. passionate
- B. handle
- C. uncensored
- D. concise

Answer(s): B

QUESTION: 1365

Fete most nearly means _____.

- A. festival
- B. criticize
- C. approve
- D. eat

Answer(s): A

QUESTION: 1366

Encore most nearly means _____.

- A. play
- B. applause
- C. repetition
- D. excite

Answer(s): C

QUESTION: 1367

Diverse most nearly means _____.

- A. various
- B. hide
- C. nestle
- D. pastime

Answer(s): A

QUESTION: 1368

Detest most nearly means _____.

- A. anger
- B. hate
- C. surprise
- D. excite

Answer(s): B

QUESTION: 1369

Tim was known as a smart aleck, able to deliver acerbic one-liners with no effort.

- A. funny
- B. cheap
- C. sharp
- D. poetic

Answer(s): C

QUESTION: 1370

It took a great degree of inexorable overwhelming force to break into the cavern.

- A. strong
- B. unyielding
- C. acute
- D. powerful

Answer(s): B

QUESTION: 1371

Attendants were stationed at intervals, with the obvious intent to hector those who moved too slowly.

- A. hurry
- B. harass

- C. encourage
- D. note

Answer(s): B

QUESTION: 1372

Reggy was as gauche in this group of polite company as he always had been.

- A. funny
- B. entertaining
- C. tactless
- D. embarrassed

Answer(s): C

QUESTION: 1373

Confident most nearly means _____.

- A. assured
- B. positive
- C. intelligent
- D. educated

Answer(s): A

QUESTION: 1374

The manufacturing error was an aberration.

- A. deviation
- B. first of a trend
- C. intentional occurrence
- D. interrogatory

Answer(s): A

Explanation:

An aberration is a deviation or unusual occurrence, usually negative, and therefore hopefully not the first of a trend nor an intentional occurrence, and certainly not interrogatory (questioning) in nature.

QUESTION: 1375

I vanquished my opponent.

- A. advocate
- B. incumbent
- C. adversary
- D. neophyte

Answer(s): C

Explanation:

Opponent most nearly means adversary.

An advocate is one who speaks in favor of or on behalf of another; an incumbent is a current officeholder; and a neophyte is one who is new to a field of endeavor.

QUESTION: 1376

Napoleon often showed his audacity in battle.

- A. alacrity
- B. clemency
- C. boldness
- D. temerity

Answer(s): C

Explanation:

Here audacity means boldness; alacrity means speed or quickness of movement, clemency means mercy or forgiveness, and timidity means being shy or unsure.

QUESTION: 1377

Propinquity most nearly means _____.

- A. evenhandedness
- B. tendency
- C. skill at doing one's job
- D. nearness

Answer(s): D

Explanation:

Propinquity most nearly means nearness.

Tendency means a propensity for a certain behavior or characteristic, and skill at doing one's job could be expressed as proficiency.

QUESTION: 1378

Jonathan was unusually taciturn this morning.

- A. untalkative
- B. diplomatic
- C. overly detailed
- D. able to be touched or felt

Answer(s): A

Explanation:

Jonathan was unusually taciturn this morning, meaning he was untalkative. This is often confused with "tactful" a synonym for choice "diplomatic."

QUESTION: 1379

Amanda showed herself to be the epitome of grace and skill.

- A. top-level example of a quality, usually positive
- B. teacher or mentor
- C. one who carries a gift to give to others
- D. moving force or starting idea

Answer(s): A

Explanation:

Amanda showed herself to be the epitome of grace and skill, meaning she was a top-level example of a positive quality.

QUESTION: 1380

Heather's rapport with her superiors and coworkers was really something to see.

- A. retaliation for real or perceived injuries
- B. respect or status achieved through achievement or rank
- C. a harmonious or mutually trusting relationship
- D. love of mankind as exemplified by donations to charitable causes

Answer(s): C

Explanation:

A rapport is a harmonious or mutually trusting relationship. It is not a retaliation for real or perceived injuries (reprisal), a measure of respect or status achieved through achievement or rank (prestige), or a love of mankind as exemplified by donations to charitable causes (philanthropy).

QUESTION: 1381

Chester displayed a surprising amount of guile during his interviews about his alleged crime.

- A. feelings of remorse or shame
- B. cunning, deceit, duplicity
- C. the conscious use of words to convey the opposite of their literal meaning
- D. insulting or abusive speech

Answer(s): B

Explanation:

Chester displayed a surprising amount of guile during his interviews about his alleged crime, meaning that he used cunning, deceit, and duplicity. He may have felt guilt (feelings of remorse or shame); he may have used irony (the conscious use of words to convey the opposite of their literal meaning); and he may even have used

invective (insulting or abusive speech) - but that's not what the sentence says.

QUESTION: 1382

"Nearly all men can stand adversity, but if you want to test a man's character, give him power." - Abraham Lincoln.

- A. something added after the original
- B. soothing or pacifying words or actions
- C. hardship
- D. an unrelentingly strong desire

Answer(s): C

Explanation:

"Nearly all men can stand adversity, but if you want to test a man's character, give him power." In this quote from President Abraham Lincoln, adversity means hardship, not something added after the original (addendum), soothing or pacifying words or actions (appeasement), or an unrelentingly strong desire (addiction).

QUESTION: 1383

"Without an acquaintance with the rules of propriety, it is impossible for the character to be established." - Confucius.

- A. negative publicity
- B. correct or acceptable conduct
- C. strength, skill, superior ability
- D. stinginess or overdone desire to be thrifty

Answer(s): B

Explanation:

"Without an acquaintance with the rules of propriety, it is impossible for the character to be established." Confucius used the word propriety to mean correct or acceptable conduct, not negative publicity (notoriety), nor strength, skill, and superior ability (prowess), nor an overdone desire to be thrifty (parsimonious).

QUESTION: 1384

The winds are variable today.

- A. steady
- B. mild
- C. shifting
- D. strong

Answer(s): C

Explanation:

Variable means shifting.

QUESTION: 1385

Latent most nearly means _____.

- A. obvious
- B. hidden
- C. long-lasting
- D. ineffective

Answer(s): B

Explanation:

Latent most nearly means hidden out of the available choices.

QUESTION: 1386

"Glory is fleeting, but obscurity is forever." - Napoleon Bonaparte.

- A. complete ignorance, forgetfulness, or unawareness
- B. unknown, hidden, unacknowledged
- C. isolated place of comfort
- D. moral corruption

Answer(s): B

Explanation:

"Glory is fleeting, but obscurity is forever."

Napoleon Bonaparte, in this maxim, used obscurity to mean the quality of being unknown, hidden, unacknowledged - not an acceptable situation for the self-proclaimed Emperor of France.

He was not referring to complete ignorance, forgetfulness, or unawareness (the definition of oblivion), an isolated place of comfort (oasis), nor the quality of moral corruption (depravity).

QUESTION: 1387

"I have come to believe that the whole world is an enigma." - Umberto Eco.

- A. mystery
- B. miracle
- C. unobstructed view of a wide area
- D. something that represents the whole

Answer(s): A

Explanation:

"I have come to believe that the whole world is an enigma." This sentence tells us that Umberto Eco believes the world is a mystery.

QUESTION: 1388

This subject almost exceeds my comprehension.

- A. strong disgust
- B. planned actions
- C. ability to see
- D. understanding

Answer(s): D

Explanation:

Of the choices given, the nearest synonym for comprehension is understanding. A synonym for "strong disgust" could be revulsion; "planned actions" could be cast as intentions; and a synonym for the "ability to see" could be observation.

QUESTION: 1389

Flaunt most nearly means _____.

- A. lose
- B. show off
- C. compromise
- D. beat harshly

Answer(s): B

Explanation:

Flaunt most nearly means to show off.

QUESTION: 1390

"Human history becomes more and more a race between education and catastrophe." - H. G. Wells.

- A. conclusion
- B. scientific discovery
- C. danger
- D. disaster

Answer(s): D

Explanation:

In this quote from noted writer H.G. Wells, the choice that means the nearest to catastrophe is disaster.

QUESTION: 1391

"Cynical politicians may even foment conflicts among groups to advance their own power." - Martha Minow.

- A. instigate

- B. resolve
- C. lessen
- D. mitigate

Answer(s): A

Explanation:

"Cynical politicians may even foment conflicts among groups to advance their own power." In this quote from Harvard law professor Martha Minow, foment means the same as instigate.

QUESTION: 1392

Inspid most nearly means _____.

- A. unavoidable
- B. irascible
- C. dull, meaningless
- D. harmless, insignificant

Answer(s): C

Explanation:

Inspid most nearly means dull or meaningless.

QUESTION: 1393

Recalcitrant most nearly means _____.

- A. stubbornly defiant of authority or control
- B. repetitive, unnecessary
- C. firm, determined
- D. uncommunicative, restrained

Answer(s): A

Explanation:

Recalcitrant refers to someone who is stubbornly defiant of authority or control.

QUESTION: 1394

Mike's legacy was worthy of debate.

- A. an example handed down from the past
- B. magical skill
- C. unproven story
- D. efficiency or skill

Answer(s): A

Explanation:

Mike's legacy was worthy of debate. In this sentence, although we can't necessarily tell what

kind of legacy is meant, we do know that legacy means an example handed down from the past. He might have handed down his magical skill (legerdemain), but we can't tell that from just this short sentence.

Other perhaps easily confused terms might be legend for an unproven story, and efficacy (efficiency or skill).

QUESTION: 1395

"The graveyards are full of indispensable men." - Charles de Gaulle.

- A. undeniably true
- B. no longer living
- C. untiring
- D. absolutely essential

Answer(s): D

Explanation:

"The graveyards are full of indispensable men."

When Charles de Gaulle uttered this bit of irony, he was referring to men who believed themselves to be absolutely essential - but whom we have had to do without anyway.

The graveyards are in fact full of men who are "no longer living", but that isn't the definition for this word.

QUESTION: 1396

Ms. Arnold displayed a surprising amount of lethargy for one in her position.

- A. sluggishness
- B. frivolity
- C. disease
- D. intellectual ability

Answer(s): A

Explanation:

Lethargy means sluggishness or a seeming unwillingness or inability to move.

QUESTION: 1397

He attempted to assuage her concerns _____.

- A. add to
- B. to pacify by giving in to demands
- C. to soothe or comfort
- D. to waste away

Answer(s): C

Explanation:

He attempted to assuage her concerns, meaning to soothe or comfort.

"Add to" is the same as augment; when you pacify someone by giving in to their demands you appease them; and to waste away means to atrophy.

QUESTION: 1398

Vacillate most nearly means _____.

- A. to waste
- B. to waver between alternatives
- C. to treat something as holy
- D. to portray as evil

Answer(s): B

Explanation:

Of the choices given, vacillate most nearly means to waver between alternatives.

QUESTION: 1399

Retract most nearly means _____.

- A. to withdraw
- B. to examine closely
- C. to turn away from
- D. not theoretical

Answer(s): A

Explanation:

Retract most nearly means to withdraw.

QUESTION: 1400

The class was extremely tedious.

- A. shallow, not thorough
- B. advancing a particular point of view
- C. insultingly patronizing
- D. boring, overly detailed

Answer(s): D

Explanation:

The class was extremely tedious, meaning boring or overly detailed.

Superficial is another word for shallow or not thorough; tendentious refers to advancing a particular point of view; and someone who is supercilious is insultingly patronizing.

QUESTION: 1401

Marie has a temperate nature.

- A. hot-blooded, fiery
- B. moderate, restrained
- C. determined in pursuit of an objective
- D. malicious, full of hate

Answer(s): B

Explanation:

Marie has a temperate nature. In this case, temperate means moderate or restrained - the opposite of choice "hot-blooded, fiery."

When someone is determined in the pursuit of an objective they can be said to be tenacious, and someone who is malicious or full of hate could be described as virulent.

QUESTION: 1402

Censure most nearly means _____.

- A. to limit communication
- B. to insult or degrade
- C. to condemn severely for inappropriate behavior
- D. to waste away from lack of use

Answer(s): C

Explanation:

Censure most nearly means to condemn severely for inappropriate behavior. It is often confused with the word censor, which means "to limit communication".

QUESTION: 1403

Desecrate most nearly means _____.

- A. to ridicule or laugh at contemptuously
- B. to profane a holy place
- C. to speak out against
- D. to lessen or relieve discomfort or pain

Answer(s): B

Explanation:

Desecrate most nearly means to profane a holy place. "To ridicule or laugh at contemptuously", means to deride.

When one speaks out against something or someone, they can be said to denounce that something or someone; and when you "lessen or relieve discomfort or pain" you alleviate their discomfort.

QUESTION: 1404

A panacea is _____.

- A. an unreasonable fear of something
- B. a very small amount
- C. a statement that contradicts itself
- D. a cure for all problems or diseases

Answer(s): D

Explanation:

A panacea is a cure for all problems or diseases. An unreasonable fear of something is a phobia; a very small amount is a pittance; and a statement that contradicts itself is a paradox.

QUESTION: 1405

A rejoinder is _____.

- A. a person who is rescued from danger
- B. an answer or response
- C. a bacteria that benefits as well as harms the host
- D. a person who joins a group again

Answer(s): B

Explanation:

A rejoinder is an answer or response.

QUESTION: 1406

Clandestine most nearly means _____.

- A. esoteric
- B. relating to a cult-like organisation
- C. hidden
- D. provocative

Answer(s): C

Explanation:

Clandestine most nearly means hidden or secret.

QUESTION: 1407

A benevolent person is one who _____.

- A. wants the best for everyone
- B. is overly proud
- C. is lazy
- D. conceives evil schemes

Answer(s): A

Explanation:

A benevolent person is one who wants the best for everyone. Someone who is overly proud can be said to be arrogant, while someone who is lazy can be said to be indolent, and someone which receives evil schemes is malevolent.

QUESTION: 1408

A poseur is _____.

- A. a difficult question that requires deep thought
- B. someone who pretends to be something they are not
- C. a representative of others' viewpoints
- D. a person with intolerant attitudes

Answer(s): B

Explanation:

A poseur is someone who pretends to be something they are not. This is sometimes confused with the slang term poser, which means a difficult question.

QUESTION: 1409

He reached the pinnacle of his career.

- A. lowest point
- B. final days
- C. smallest salary
- D. summit

Answer(s): D

Explanation:

Pinnacle, like summit, means the highest point.

QUESTION: 1410

Brawl most nearly means _____.

- A. trap
- B. fight
- C. speak
- D. provide

Answer(s): B

Explanation:

Brawl, like fight, means a noisy quarrel.

QUESTION: 1411

She acted like a pauper.

- A. poor person
- B. slanderer
- C. patriot
- D. peculiar person

Answer(s): A

Explanation:

A pauper is an extremely poor person.

QUESTION: 1412

She has always been punctual.

- A. ajokester
- B. prompt
- C. correct
- D. prepared

Answer(s): B

Explanation:

Punctual, like prompt, means arriving or doing something exactly at the appointed time.

QUESTION: 1413

Segment most nearly means _____.

- A. secret
- B. hermit
- C. part
- D. seep

Answer(s): C

Explanation:

Segment, like part, means a portion.

QUESTION: 1414

Revere most nearly means _____.

- A. esteem
- B. regulate
- C. reverse
- D. hold

Answer(s): A

Explanation:

Revere, like esteem, means to regard someone or something with great respect.

QUESTION: 1415

The speaker's audience appreciated brevity.

- A. brilliance
- B. illustrations
- C. activity
- D. conciseness

Answer(s): D

Explanation:

Brevity, like conciseness, means of short duration.

QUESTION: 1416

Illogical most nearly means _____.

- A. coherent
- B. illusive
- C. fallacious
- D. real

Answer(s): C

Explanation:

Illogical, like fallacious, means senseless.

QUESTION: 1417

Excerpt most nearly means _____.

- A. exam
- B. extract
- C. eventual
- D. superfluous

Answer(s): B

Explanation:

Excerpt, like extract, means something picked out from the total.

QUESTION: 1418

The surface had a sheen.

- A. color
- B. luster
- C. scratch
- D. cover

Answer(s): B

Explanation:

Sheen, like luster, means shininess or brightness.

QUESTION: 1419

Divulge most nearly means _____.

- A. divide
- B. fatten
- C. dissolve
- D. tell

Answer(s): D

Explanation:

Divulge, like tell, means to disclose or make known.

QUESTION: 1420

The attorney could cite examples.

- A. cultivate
- B. change
- C. specify
- D. Grasp

Answer(s): C

Explanation:

Cite, like specify, means to bring information forward as proof.

QUESTION: 1421

I cannot allay your concerns.

- A. ease
- B. confirm
- C. reverse
- D. tempt

Answer(s): A

Explanation:

Allay, like ease, means to relieve.

QUESTION: 1422

The animal had vitality.

- A. liveliness
- B. viciousness
- C. cunning
- D. understanding

Answer(s): A

Explanation:

Vitality, like liveliness, means vigor or energy.

QUESTION: 1423

Defer most nearly means _____.

- A. ascertain
- B. define
- C. abase
- D. delay

Answer(s): D

Explanation:

Defer, like delay, means to postpone doing something.

QUESTION: 1424

I deplore your actions.

- A. approve
- B. applaud
- C. regret
- D. deprive

Answer(s): C

Explanation:

Deplore, like regret, means to feel or express disapproval.

QUESTION: 1425

Concur most nearly means _____.

- A. defeat
- B. agree
- C. confine
- D. truth

Answer(s): B

Explanation:

Concur, like agree, means to have the same opinion.

QUESTION: 1426

Refute most nearly means _____.

- A. refer
- B. disapprove
- C. disprove
- D. regard

Answer(s): C

Explanation:

Refute, like disprove, means to prove an argument or statement wrong.

QUESTION: 1427

She served copious amounts of food.

- A. unequal
- B. small
- C. uninviting
- D. large

Answer(s): D

Explanation:

Copious, like large, means an ample or plentiful supply.

QUESTION: 1428

Credulous most nearly means _____.

- A. fantastic
- B. credit
- C. gullible
- D. awe-inspiring

Answer(s): C

Explanation:

Credulous, like gullible, means able to believe too readily.

QUESTION: 1429

Random most nearly means _____.

- A. haphazard
- B. purpose
- C. anger

D. ramble

Answer(s): A

Explanation:

Random, like haphazard, means having no clear pattern.

QUESTION: 1430

State the salient points.

- A. legal
- B. conspicuous
- C. less obvious
- D. missed

Answer(s): B

Explanation:

Salient, like conspicuous, means prominent or pronounced.

QUESTION: 1431

Tangible most nearly means _____.

- A. illusory
- B. tangent
- C. concrete
- D. tangle

Answer(s): C

Explanation:

Tangible, like concrete, means capable of being touched.

QUESTION: 1432

Residue most nearly means _____.

- A. resident
- B. part
- C. rescue
- D. remainder

Answer(s): D

Explanation:

Residue, like remainder, means what is left after the removal of something.

QUESTION: 1433

Catastrophe most nearly means _____.

- A. violence
- B. disorder
- C. decision
- D. disaster

Answer(s): D

Explanation:

Catastrophe, like disaster, means a great calamity or tragedy.

QUESTION: 1434

He promised us a diagnosis.

- A. analysis
- B. decision
- C. dialect
- D. diagram

Answer(s): A

Explanation:

Diagnosis, like analysis, means the opinion derived from examining the nature of something.

QUESTION: 1435

Your reaction leaves me in a quandary.

- A. condition
- B. predicament
- C. certainty
- D. compunction

Answer(s): B

Explanation:

Quandary, like predicament, means a dilemma or state of uncertainty.

QUESTION: 1436

Stilted most nearly means _____.

- A. bored
- B. formal
- C. stuffed
- D. stifled

Answer(s): B

Explanation:

Stilted, like formal, means stiffly dignified.

QUESTION: 1437

The law says you must not encroach.

- A. enclose
- B. overreach
- C. trespass
- D. encounter

Answer(s): C

Explanation:

Encroach, like trespass, means to intrude.

QUESTION: 1438

Deter most nearly means _____.

- A. dissipate
- B. detain
- C. dispute
- D. prevent

Answer(s): D

Explanation:

Deter, like prevent, means to restrain someone from doing something.

QUESTION: 1439

Falter most nearly means _____.

- A. die
- B. fake
- C. hesitate
- D. calm

Answer(s): C

Explanation:

Falter, like hesitate, means to waver.

QUESTION: 1440

Our meal was a meager one.

- A. sparse
- B. rich

- C. fig
- D. massive

Answer(s): A

Explanation:

Meager, like sparse, means skimpy or scant.

QUESTION: 1441

Stifle most nearly means _____.

- A. sign
- B. suppress
- C. shoot
- D. stiff

Answer(s): B

Explanation:

Stifle, like suppress, means to restrict.

QUESTION: 1442

Dearth most nearly means _____.

- A. lack
- B. passing
- C. ground
- D. overabundance

Answer(s): A

Explanation:

Dearth, like lack, means scarcity.

QUESTION: 1443

Although Johnny's full-time job was regarded by many as menial, he liked the peace and solitude it offered.

- A. lowly
- B. boring
- C. unpleasant
- D. unrewarding

Answer(s): A

Explanation:

Something that is menial is subservient or lowly.

QUESTION: 1444

Emulate most nearly means _____.

- A. discipline
- B. imitate
- C. approve
- D. amaze

Answer(s): B

Explanation:

To emulate something is to strive to equal or imitate that thing.

QUESTION: 1445

Scrupulous most nearly means _____.

- A. Unorganized
- B. Tolerable
- C. Easy going
- D. Thorough

Answer(s): D

Explanation:

When one is scrupulous is meticulous or thorough.

QUESTION: 1446

Our computer system at work is archaic.

- A. worthy
- B. obsolete
- C. current
- D. trendy

Answer(s): B

Explanation:

To be archaic is to be out-of-date or obsolete.

QUESTION: 1447

Obstinate most nearly means _____.

- A. depressing
- B. destructive
- C. confident
- D. stubborn

Answer(s): D

Explanation:

To be obstinate is to be unyielding or stubborn.

QUESTION: 1448

Max's new cell phone worked great, but every now and then he would hear intermittent static.

- A. loud
- B. soft
- C. periodic
- D. vehement

Answer(s): C

Explanation:

When something is intermittent, it is periodic or starts and stops at intervals.

QUESTION: 1449

However most nearly means _____.

- A. Nevertheless
- B. Consequently
- C. Thus
- D. Fortunately

Answer(s): A

Explanation:

Nevertheless means nonetheless or in spite of, however.

QUESTION: 1450

Negligible most nearly means _____.

- A. Weak
- B. Significant
- C. Trivial
- D. Noteworthy

Answer(s): C

Explanation:

To be negligible is to be unimportant or trivial.

QUESTION: 1451

Vigilant most nearly means _____.

- A. Nonchalant
- B. Observant
- C. Virtuous
- D. Strenuous

Answer(s): B

Explanation:

To be vigilant is to be alert or observant.

QUESTION: 1452

Sam had been timid all of his life. When he decided to take up mountain climbing, his friends were incredulous.

- A. afraid
- B. heartbroken
- C. angry
- D. disbelieving

Answer(s): D

Explanation:

When one is incredulous, one is skeptical or disbelieving.

QUESTION: 1453

Dr. Martin has a unique technique he uses when performing a surgery.

- A. aggressive
- B. unparalleled
- C. respected
- D. antiquated

Answer(s): B

Explanation:

To be unique is to be one of a kind or unparalleled.

QUESTION: 1454

Ameliorate most nearly means _____.

- A. Persist
- B. Worsen
- C. Endorse
- D. Mend

Answer(s): D

Explanation:

To ameliorate is to recover, heal, or mend.

QUESTION: 1455

Perseverance most nearly means _____.

- A. popular
- B. happy
- C. persistent
- D. modest

Answer(s): C

Explanation:

To have perseverance is to be steadfast in your course or to be persistent.

QUESTION: 1456

Our company policy is that every person is accountable for his or her actions.

- A. responsible
- B. awarded
- C. thanked
- D. complimented

Answer(s): A

Explanation:

To be accountable is to be held answerable or responsible.

QUESTION: 1457

The board voted unanimously to approve the merger.

- A. independently
- B. undivided
- C. happily
- D. individual

Answer(s): B

Explanation:

When a group's does something unanimously, they do it undivided or undisputed.

QUESTION: 1458

Contemptuous most nearly means _____.

- A. exclusive
- B. secured
- C. common

D. scornful

Answer(s): D

Explanation:

When one is contemptuous, one is disdainful or scornful.

QUESTION: 1459

Deceive most nearly means _____.

- A. cheat
- B. defend
- C. pay
- D. disobey

Answer(s): A

Explanation:

When one deceives they swindle, trick, or cheat

QUESTION: 1460

Resigned most nearly means _____.

- A. Saddened
- B. Passive
- C. Excited
- D. Cold

Answer(s): B

Explanation:

To be resigned is to be compliant and accepting, or passive.

QUESTION: 1461

Philanthropy most nearly means _____.

- A. Greedy
- B. Generosity
- C. Selfish
- D. Respectful

Answer(s): B

Explanation:

An act of philanthropy is an act of charity or generosity.

QUESTION: 1462

After being arrested, the suspect was nondescript in his interrogation.

- A. ordinary
- B. special
- C. unique
- D. lethargic

Answer(s): A

Explanation:

Something that is nondescript is without distinction or ordinary.

QUESTION: 1463

The word synthesize most nearly means _____.

- A. divide
- B. combine
- C. proceed
- D. separate

Answer(s): B

QUESTION: 1464

The word fragility most nearly means _____.

- A. elasticity
- B. firmness
- C. sturdiness
- D. delicacy

Answer(s): D

QUESTION: 1465

The word dossier most nearly means _____.

- A. clues
- B. theory
- C. portfolio
- D. testimony

Answer(s): C

QUESTION: 1466

The word meticulous most nearly means _____.

- A. precise
- B. punctual

- C. negligent
- D. careless

Answer(s): A

QUESTION: 1467

The word most opposite in meaning to the word vitalize is _____.

- A. inspire
- B. prompt
- C. dishearten
- D. invigorate

Answer(s): C

QUESTION: 1468

The word sanctioned most nearly means _____.

- A. irregular
- B. banned
- C. patronized
- D. authorized

Answer(s): D

QUESTION: 1469

The word retract most nearly means _____.

- A. advance
- B. repeat
- C. cancel
- D. enforce

Answer(s): C

QUESTION: 1470

The word allusive most nearly means _____.

- A. graphic
- B. colorful
- C. indirect
- D. energetic

Answer(s): C

QUESTION: 1471

The word most opposite in meaning to the word bigotry is _____.

- A. moderateness
- B. discrimination
- C. bias
- D. fairness

Answer(s): D

QUESTION: 1472

The word advocacy most nearly means _____.

- A. promoting
- B. blocking
- C. prosecuting
- D. impeding

Answer(s): A

QUESTION: 1473

The word adobe most nearly means _____.

- A. headquarters
- B. consent
- C. brick
- D. underground

Answer(s): C

QUESTION: 1474

The word nullify most nearly means _____.

- A. confine
- B. void
- C. ratify
- D. free

Answer(s): B

QUESTION: 1475

The word restitution most nearly means _____.

- A. fine
- B. take
- C. damages
- D. stipend

Answer(s): C

QUESTION: 1476

The word perforate most nearly means _____.

- A. puncture
- B. please
- C. smooth
- D. excite

Answer(s): A

QUESTION: 1477

The word demurral most nearly means _____.

- A. disagreement
- B. liberating
- C. hastening
- D. agreement

Answer(s): A

QUESTION: 1478

The word crevasse most nearly means _____.

- A. ravine
- B. peak
- C. river
- D. juncture

Answer(s): A

QUESTION: 1479

The word intrinsic most nearly means _____.

- A. separated
- B. accidental
- C. layered
- D. built-in

Answer(s): D

QUESTION: 1480

The word predisposed most nearly means _____.

- A. unwilling

- B. prone to
- C. ready
- D. removed from

Answer(s): B

QUESTION: 1481

The word insignia most nearly means _____.

- A. dignity
- B. indecency
- C. trademark
- D. formality

Answer(s): C

QUESTION: 1482

The word quiescence most nearly means _____.

- A. suspend
- B. revive
- C. continue
- D. promote

Answer(s): A

QUESTION: 1483

The word stalwart most nearly means _____.

- A. shy
- B. loyal
- C. coward
- D. modest

Answer(s): B

QUESTION: 1484

The word circumvent most nearly means _____.

- A. avoid
- B. allow
- C. concise
- D. release

Answer(s): A

QUESTION: 1485

The word eminent most nearly means _____.

- A. distinguished
- B. dominant
- C. forthcoming
- D. immediate

Answer(s): A

QUESTION: 1486

The word pliable most nearly means _____.

- A. gentle
- B. inflexible
- C. mild
- D. manageable

Answer(s): D

QUESTION: 1487

She was conversant in Spanish, German, and English.

The word conversant most nearly means _____.

- A. versed
- B. ignorant
- C. canonical
- D. unskilled

Answer(s): A

QUESTION: 1488

The defendant asked for clemency in his sentencing.

The word clemency most nearly means _____.

- A. brevity
- B. no mercy
- C. leniency
- D. time

Answer(s): C

QUESTION: 1489

The harvest of grapes was extremely abundant this year.

The word abundant most nearly means _____.

- A. sparse

- B. plentiful
- C. bitter
- D. scarce

Answer(s): B

QUESTION: 1490

The driver swerved purblindly all over the road.

The word purblindly most nearly means _____.

- A. carefully
- B. willfully
- C. cautiously
- D. recklessly

Answer(s): D

QUESTION: 1491

Kevin's dad said he would contemplate the decision before giving an answer.

To contemplate most nearly means _____.

- A. observe
- B. disregard
- C. ponder
- D. discount

Answer(s): C

QUESTION: 1492

The painted wall did not quite have the luster she was looking for.

The word luster most nearly means _____.

- A. darkness
- B. matte
- C. opacity
- D. sheen

Answer(s): D

QUESTION: 1493

The amalgamation of the two companies was seen as almost inevitable.

The word amalgamation most nearly means _____.

- A. closure
- B. success
- C. profitability
- D. union

Answer(s): D

QUESTION: 1494

Susan was going to concoct a scheme to get him to pay for the bill.
To concoct most nearly means _____.

- A. vamp
- B. manipulate
- C. devise
- D. frame

Answer(s): C

QUESTION: 1495

The chef hoped the meal would be as savory as he had described it.
The word savory most nearly means _____.

- A. insipid
- B. pleasant
- C. enticing
- D. searing

Answer(s): B

QUESTION: 1496

The sergeant ordered him to bivouac duty.
The word bivouac most nearly means _____.

- A. building
- B. encampment
- C. pontoon
- D. concession

Answer(s): B

QUESTION: 1497

She was surprised by the terse reply to her question.
The word terse most nearly means _____.

- A. lengthy
- B. compact
- C. gentle
- D. abrupt

Answer(s): D

QUESTION: 1498

Terry had no objections to the dissolution of the contract agreement.
The word dissolution most nearly means _____.

- A. construction
- B. disintegration
- C. detachment
- D. degeneration

Answer(s): B

QUESTION: 1499

We commiserated with the families of the fallen soldiers.
The word commiserated most nearly means _____.

- A. sympathized
- B. cooperated
- C. participated
- D. capitalized

Answer(s): A

QUESTION: 1500

The house was built with a staunch foundation.
The word staunch most nearly means _____.

- A. rugged
- B. small
- C. stiff
- D. strong

Answer(s): D

QUESTION: 1501

His speech impediment was a factor in not getting the promotion.
The word impediment most nearly means _____.

- A. style
- B. eminence
- C. hindrance
- D. expression

Answer(s): C

QUESTION: 1502

The commentator made several remarks about the breaking news that surprised the listeners.

The word commentator most nearly means _____.

- A. orator
- B. passerby
- C. correspondent
- D. onlooker

Answer(s): C

QUESTION: 1503

She lounged on the couch, which upset her mother because she had chores to do.

The word lounged most nearly means _____.

- A. tilted
- B. reclined
- C. bounded
- D. jumped

Answer(s): B

QUESTION: 1504

The reporter concealed his sources.

The word concealed most nearly means _____.

- A. specified
- B. hid
- C. denied
- D. cited

Answer(s): B

QUESTION: 1505

Nobody questioned the veracity of his speech.

The word veracity most nearly means _____.

- A. intensity
- B. truthfulness
- C. intention
- D. deceitfulness

Answer(s): B

QUESTION: 1506

Jennifer thought the new dress was horrendous.

The word horrendous most nearly means _____.

- A. beautiful

- B. flattering
- C. daunting
- D. awful

Answer(s): D

QUESTION: 1507

Karen was hoping she could be one of the new members of the sorority.
The word sorority most nearly means _____.

- A. club
- B. initiation
- C. pageant
- D. charity

Answer(s): A

QUESTION: 1508

His short answer showed how naive he was on the topic.
The word naive most nearly means _____.

- A. experienced
- B. aware
- C. unknowing
- D. unaffected

Answer(s): C

QUESTION: 1509

The captain was going to sequester the troops because of their behavior.
The word sequester most nearly means _____.

- A. release
- B. commend
- C. seclude
- D. combine

Answer(s): C

QUESTION: 1510

My uncle always has a hankering for potato chips!
The word hankering most nearly means _____.

- A. distaste
- B. dislike
- C. yearning
- D. ambivalence toward

Answer(s): C

QUESTION: 1511

They commenced the contract on the first day of the new year.

The word commenced most nearly means _____.

- A. established
- B. closed
- C. started
- D. dissolved

Answer(s): C

QUESTION: 1512

The bottle was stogged in the sand.

The word stogged most nearly means _____.

- A. lying
- B. broken
- C. stuck
- D. on top of

Answer(s): C

QUESTION: 1513

The word bounteous most nearly means _____.

- A. beholden
- B. unaware
- C. withdrawn
- D. abundant

Answer(s): D

QUESTION: 1514

Dawn's schoolwork was turning into drudgery with each passing hour.

The word drudgery most nearly means _____.

- A. fascinating
- B. wastage
- C. tedium
- D. educational

Answer(s): C

QUESTION: 1515

After improving the soil with mulch and compost, Dwayne found his garden to be quite fecund. The word fecund most nearly means _____.

- A. slothful
- B. dormant
- C. dubious
- D. productive

Answer(s): D

QUESTION: 1516

The word verdant most nearly means _____.

- A. mirthful
- B. green
- C. truculent
- D. lovely

Answer(s): B

QUESTION: 1517

The word timorous most nearly means _____.

- A. gloating
- B. strident
- C. nervous
- D. bellicose

Answer(s): C

QUESTION: 1518

Tom had many good thoughts, but he was known to bloviate in his writing. The word bloviate most nearly means _____.

- A. blemish
- B. rant
- C. exhume
- D. transcribe

Answer(s): B

QUESTION: 1519

The word verisimilitude most nearly means _____.

- A. originality
- B. falsity

- C. authenticity
- D. causality

Answer(s): C

QUESTION: 1520

The word vacuous most nearly means _____.

- A. swollen
- B. blatant
- C. mindless
- D. tedious

Answer(s): C

QUESTION: 1521

The word tumescent most nearly means _____.

- A. waifish
- B. forgotten
- C. didactic
- D. swollen

Answer(s): D

QUESTION: 1522

Which word is the most opposite in meaning to the word venerable?

- A. vivacious
- B. unrespectable
- C. beloved
- D. wrathful

Answer(s): B

QUESTION: 1523

The word acquiescence most nearly means _____.

- A. foreclosure
- B. striation
- C. denial
- D. consent

Answer(s): D

QUESTION: 1524

The word effulgence most nearly means _____.

- A. splendor
- B. ugliness
- C. regionality
- D. pellucidity

Answer(s): A

QUESTION: 1525

The senator won the election because of the fervor of his speeches and debates.

The word most opposite in meaning to fervor is _____.

- A. bawdry
- B. vigor
- C. apathy
- D. flamboyance

Answer(s): C

QUESTION: 1526

The word neophyte most nearly means _____.

- A. balladeer
- B. beginner
- C. master
- D. craftsman

Answer(s): B

QUESTION: 1527

Ryan perjured himself in court, but his testimony was accepted nonetheless.

The word perjure most nearly means _____.

- A. sully
- B. eviscerate
- C. lie
- D. soil

Answer(s): C

QUESTION: 1528

The word ponderous most nearly means _____.

- A. blithe
- B. weighty
- C. lamentable

D. wizened

Answer(s): B

QUESTION: 1529

The word sapience most nearly means _____.

- A. uniqueness
- B. troublesomeness
- C. wisdom
- D. wispieness

Answer(s): C

QUESTION: 1530

The word tableau most nearly means _____.

- A. bundle
- B. troupe
- C. tabletop
- D. arrangement

Answer(s): D

QUESTION: 1531

Marge was valorous in her work repairing the house after the hurricane.

The word valorous most nearly means _____.

- A. perspicacious
- B. withered
- C. fearful
- D. stouthearted

Answer(s): D

QUESTION: 1532

The word winsome most nearly means _____.

- A. thoughtful
- B. wishful
- C. unremarkable
- D. attractive

Answer(s): D

QUESTION: 1533

The word prosaic most nearly means _____.

- A. truculent
- B. unimaginative
- C. latent
- D. vexatious

Answer(s): B

QUESTION: 1534

The word obtrude most nearly means _____.

- A. impose
- B. portend
- C. refund
- D. plot

Answer(s): A

QUESTION: 1535

The word mollify most nearly means _____.

- A. to enrage
- B. to present
- C. to bear
- D. to soothe

Answer(s): D

QUESTION: 1536

The word viscous most nearly means _____.

- A. cruel or violent
- B. a runny liquid
- C. thick or sticky
- D. relating to veins

Answer(s): C

QUESTION: 1537

The word ensconce most nearly means _____.

- A. nestle
- B. restrict
- C. destroy
- D. enslave

Answer(s): A

QUESTION: 1538

Abate most nearly means _____.

- A. Lessen
- B. Detest
- C. Bitter
- D. Proficient

Answer(s): A

Explanation:

Abate means to reduce in amount or degree - to lessen.

QUESTION: 1539

Brevity most nearly means _____.

- A. Long
- B. Briefness
- C. Funny
- D. Light

Answer(s): B

Explanation:

Brevity means a shortness of time or briefness.

QUESTION: 1540

Culpable most nearly means _____.

- A. Having integrity
- B. Innocent
- C. Touchable
- D. Blameworthy

Answer(s): D

Explanation:

Culpable means deserving blame or blameworthy.

QUESTION: 1541

Derelict most nearly means _____.

- A. Responsible
- B. Destroyed
- C. Abandoned

D. Inhabited

Answer(s): C

Explanation:

Derelict means to be left or deserted; abandoned.

QUESTION: 1542

Gregarious most nearly means _____.

- A. Sociable
- B. Shy
- C. Introverted
- D. Angry

Answer(s): A

Explanation:

Gregarious means sociable, outgoing and friendly.

QUESTION: 1543

The suite at the hotel was opulent.

- A. Clean
- B. Luxurious
- C. Spacious
- D. Smelly

Answer(s): B

Explanation:

Opulent means luxurious or wealthy.

QUESTION: 1544

The movie put her in a pensive mood.

- A. Excited
- B. Happy
- C. Thoughtful
- D. Angry

Answer(s): C

Explanation:

Pensive means thoughtful or dreamily.

QUESTION: 1545

The politician tried to placate his constituents.

- A. Satisfy
- B. Locate
- C. Influence
- D. Conceal

Answer(s): A

Explanation:

Placate means to appease or satisfy.

QUESTION: 1546

The principal wanted to rebuke the students.

- A. Reward
- B. Question
- C. Ignore
- D. Reprimand

Answer(s): D

Explanation:

Rebuke means to reprimand or express sharp disapproval.

QUESTION: 1547

The email was succinct.

- A. Lengthy
- B. Concise
- C. Confusing
- D. Late

Answer(s): B

Explanation:

Succinct means concise, terse or expressed in few words.

QUESTION: 1548

Amicable most nearly means _____.

- A. Disgruntled
- B. Friendly
- C. Stern
- D. Shy

Answer(s): B

Explanation:

Amicable means friendly, agreeable, peaceable.

QUESTION: 1549

Mundane most nearly means _____.

- A. Extraordinary
- B. Unique
- C. Tired
- D. Ordinary

Answer(s): D

Explanation:

Mundane means ordinary, common.

QUESTION: 1550

Parity most nearly means _____.

- A. Equality
- B. Difference
- C. Separate
- D. Twice

Answer(s): A

Explanation:

Parity means equal in amount or status.

QUESTION: 1551

Placid most nearly means _____.

- A. Agitated
- B. Cold
- C. Calm
- D. Ruffled

Answer(s): C

Explanation:

Placid means calm or peaceful

QUESTION: 1552

Rescind most nearly means _____.

- A. Send again
- B. Revoke

- C. Confirm
- D. Deny

Answer(s): B

Explanation:

Rescind means to revoke or repeal.

QUESTION: 1553

They are hoping for tangible improvements.

- A. Long lasting
- B. Speedy
- C. Real
- D. Subtle

Answer(s): C

Explanation:

Tangible means real or actual.

QUESTION: 1554

Venerate most nearly means _____.

- A. Discriminate
- B. Revere
- C. Trust
- D. Condemn

Answer(s): B

Explanation:

Venerate means to hold in deep respect or to revere.

QUESTION: 1555

The programs suffer from chronic underfunding.

- A. Temporary
- B. Severe
- C. Habitual
- D. Massive

Answer(s): C

Explanation:

Chronic means constant or habitual.

QUESTION: 1556

She possessed an indefatigable spirit.

- A. Tireless
- B. Enormous
- C. Quiet
- D. Humble

Answer(s): A

Explanation:

Indefatigable means tireless or not yielding to fatigue.

QUESTION: 1557

The six-year-old was precocious.

- A. Shy
- B. Rude
- C. Outgoing
- D. Advanced

Answer(s): D

Explanation:

Precocious means unusually advanced mental development.

Section 8: Mechanical Comprehension

QUESTION: 1558

An induction clutch works by _____.

- A. magnetism
- B. pneumatics
- C. hydraulics
- D. friction

Answer(s): A

Explanation:

An induction clutch is a magnetic clutch.

QUESTION: 1559

If a first class lever with a resistance arm measuring 2 feet and an effort arm measuring 8 feet are being used, what's the mechanical advantage?

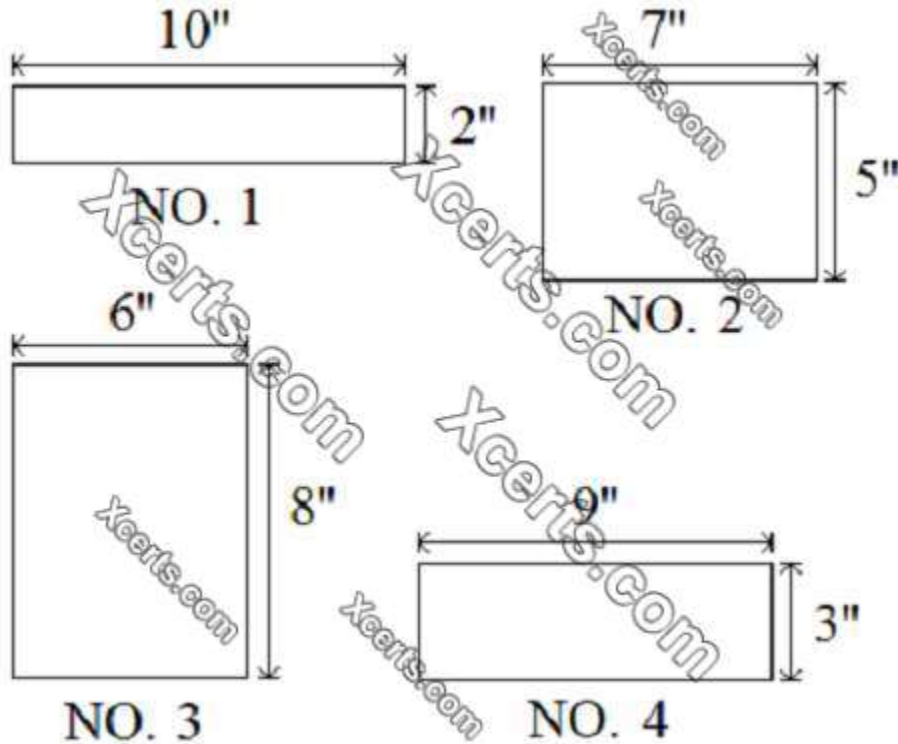
- A. 2
- B. 4
- C. 6
- D. 1

Answer(s): B

Explanation:

Mechanical advantage can be calculated as Length of Effort Arm ÷ Length of Resistance Arm.
 $MA = 8 \div 2 = 4$.

QUESTION: 1560



The bottoms of four boxes are shown above. The boxes all have the same volume. If postal regulations state that the sides of a box must meet a minimum height, which box is most likely to be too short to go through the mail?

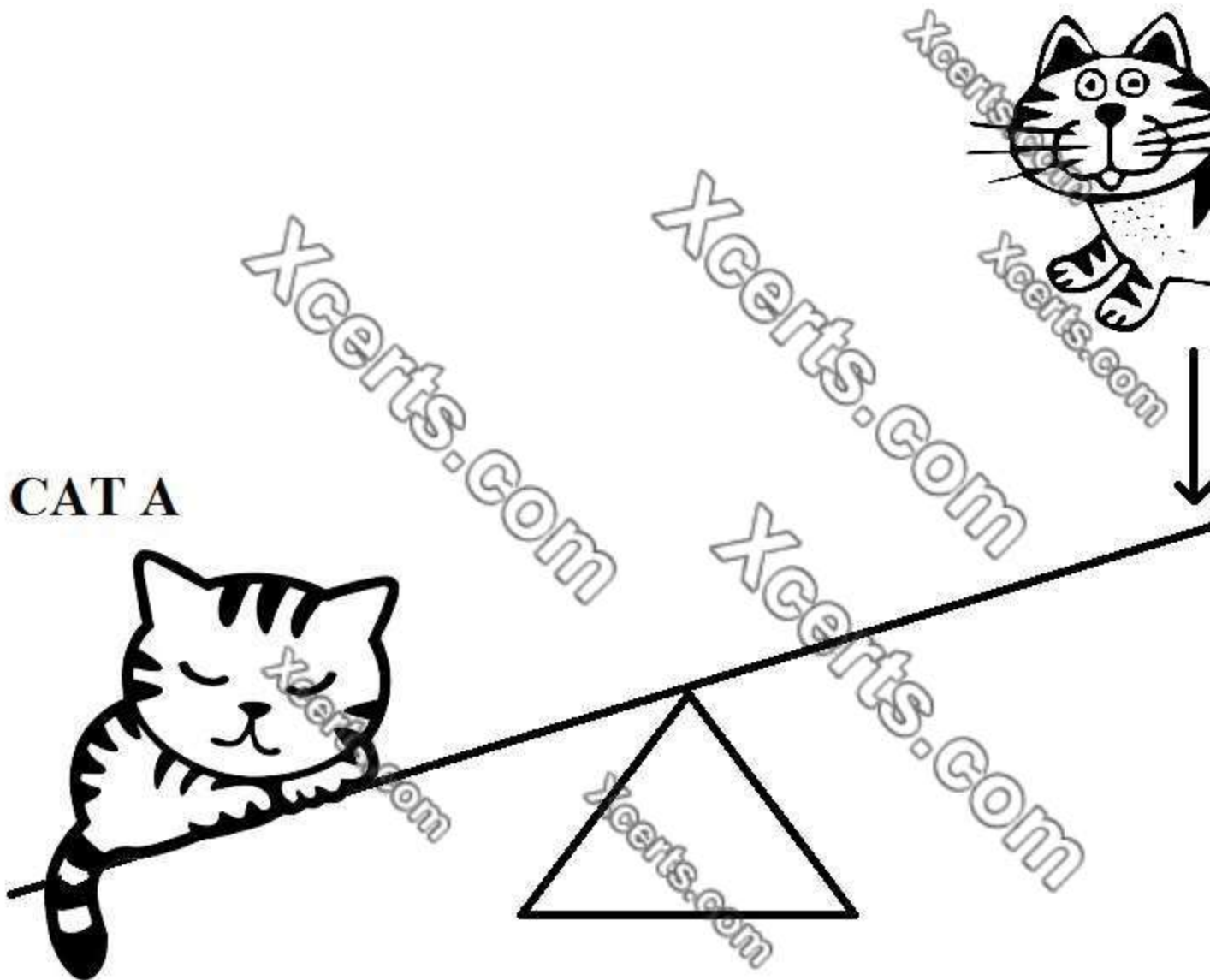
- A. NO. 4
- B. NO. 2
- C. NO. 1
- D. NO. 3

Answer(s): D

Explanation:

The box with the largest area on the bottom will have the shortest sides. If $\text{length} \times \text{width} \times \text{height} = \text{volume}$, and all the boxes have equal volume, then the sides must be shortest on the box with the largest area on the bottom. Calculate the area of each box bottom:
 NO. 1 = 20 square inches;
 NO. 2 = 35 square inches;
 NO. 3 = 48 square inches; and
 NO. 4 = 27 square inches.
 NO. 3, which has the largest area, will have the shortest sides.

QUESTION: 1561



Looking at the figure above, when Cat B lands on the seesaw, Cat A will _____.

- A. remain stationary
- B. hit the ground hard
- C. rise in the air quickly
- D. enter the stratosphere

Answer(s): C

Explanation:

Cat B landing on the seesaw will propel Cat A into the air.

QUESTION: 1562

Air weighs about 15 psi.

What's the amount of pressure (force) exerted on the top of your head, given a surface area of

24 inches?

- A. 360 pounds
- B. 625 pounds
- C. 5/8 pound
- D. 180 pounds

Answer(s): A

Explanation:

Power equals force divided by area in square inches ($P = F/A$).

This formula can also be stated as $F = A \times P$.

Substitute the known quantities.

$F = 15 \times 24 = 360$ pounds.

QUESTION: 1563

The force produced when a boxer's hand hits a heavy bag and "bounces" off it is called _____.

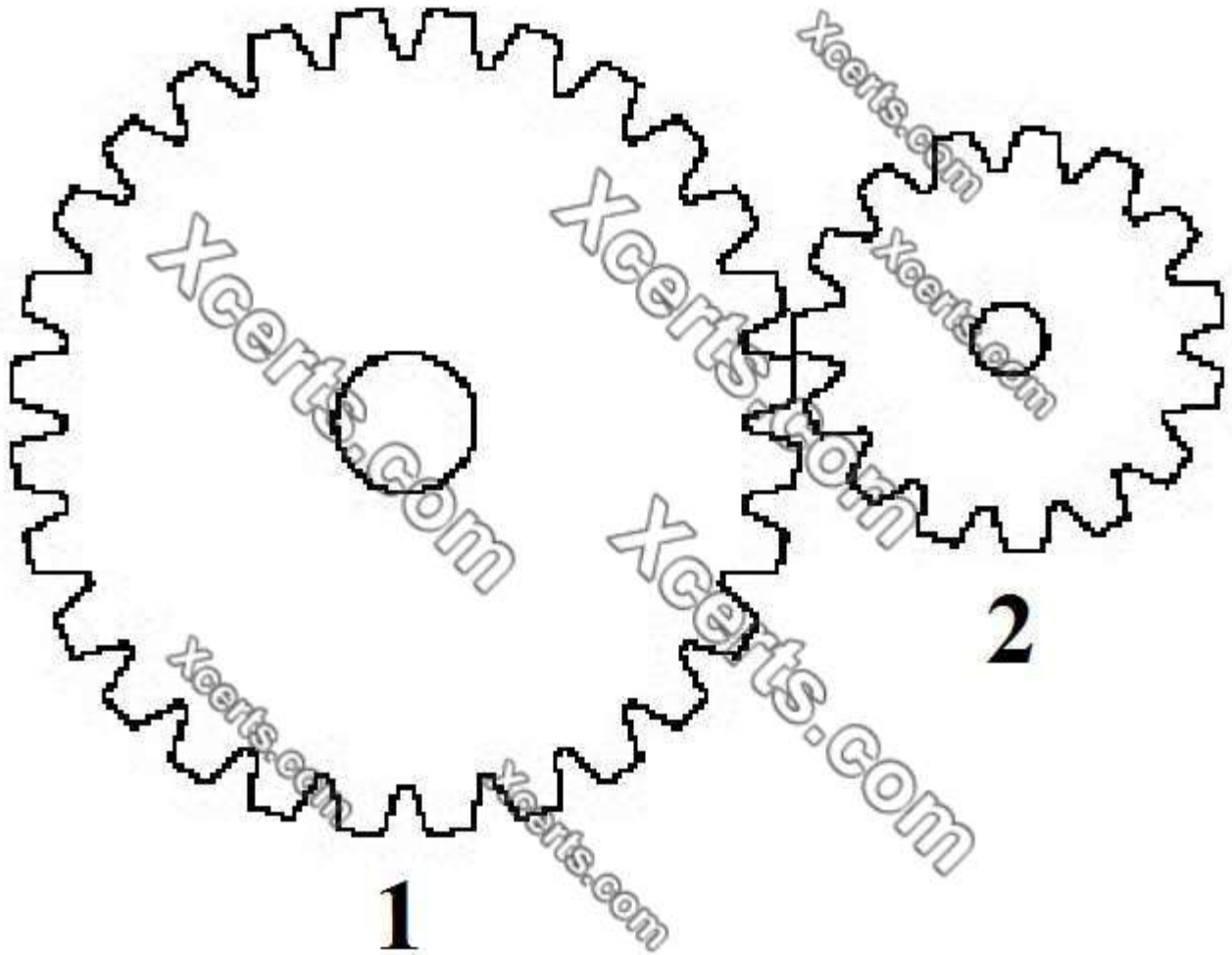
- A. static electricity
- B. magnetism
- C. recoil
- D. gravity

Answer(s): C

Explanation:

Recoil occurs when an object producing a force is kicked back.

QUESTION: 1564



In the figure above, if Gear 1 has 25 teeth and Gear 2 has 15 teeth, how many revolutions does Gear 2 make for every 10 revolutions Gear 1 makes?

- A. about $16\frac{2}{3}$
- B. 12
- C. about $\frac{1}{3}$ more
- D. about 20

Answer(s): A

Explanation:

To determine the answer, multiply the number of teeth Gear 1 has D and the number of revolutions it makes (R).

Divide that number by the number of teeth Gear 2 has d to determine the number of revolutions Gear 2 makes (r).

Because the gears are proportional, this formula will show you the ratio of teeth to revolutions.

$$r = DR/d$$

$$r = (25 \times 10)/15$$

$$r = 250/15, \text{ or } 16\frac{2}{3}, \text{ or } 16\frac{2}{3}$$

QUESTION: 1565

A cubic foot of water weighs about 62.5 pounds.

If an aquarium is 18 feet long, 10 feet deep, and 12 feet wide, what's the approximate pounds-per-square-inch pressure (psi) on the bottom of the tank?

- A. 2 psi
- B. 4 psi
- C. 5 psi
- D. 7 psi

Answer(s): B

Explanation:

You can determine the pressure of all that water by multiplying the volume of the aquarium by the weight of the water. $\text{Volume} = lwh$.

The bottom of the tank is 18 feet long by 12 feet wide by 10 feet high for a total volume of 2,160 cubic feet ($18 \times 12 \times 10$).

A cubic foot of water weighs approximately 62.5 pounds.

$2,160 \times 62.5$ gives an approximate pressure on the bottom of the tank of about 135,000 pounds over the entire surface area.

The surface area of the bottom of the tank is $\text{length} \times \text{width}$.

$216 \text{ inches } (18 \text{ feet} \times 12) \times 144 \text{ inches } (12 \text{ feet} \times 12) = 31,104$.

Dividing the pressure of 135,000 by the number of square inches of surface area gives an approximate PSI of 4.

QUESTION: 1566

Springs used in machines are usually made of _____.

- A. plastic
- B. bronze
- C. nylon fiber
- D. steel

Answer(s): D

Explanation:

Machine springs are usually made of steel although sometimes they're made of brass or other metal alloys.

QUESTION: 1567

A clutch is a type of _____.

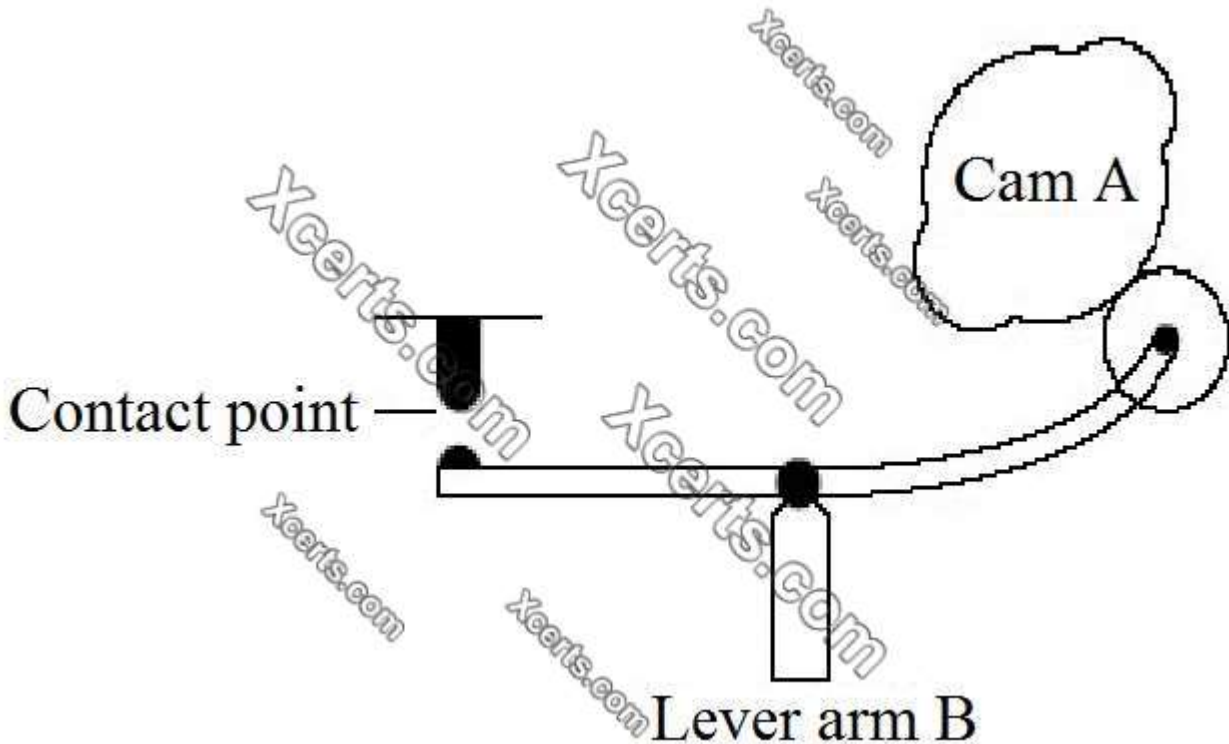
- A. universal joint
- B. coupling
- C. gear differential
- D. cam follower

Answer(s): B

Explanation:

Clutches connect and disconnect parts, so they're a type of coupling.

QUESTION: 1568



When Cam A completes one revolution, the lever will touch the contact point _____.

- A. once
- B. never
- C. four times
- D. twice

Answer(s): D

Explanation:

When the high point of the cam connects with the lever arm, the lever arm will touch the contact point.

Two high points on the cam mean the lever arm will touch the contact point twice with each revolution of the cam.

QUESTION: 1569

A single block-and-fall is called a _____.

- A. fixed pulley
- B. gun tackle
- C. runner
- D. sheave

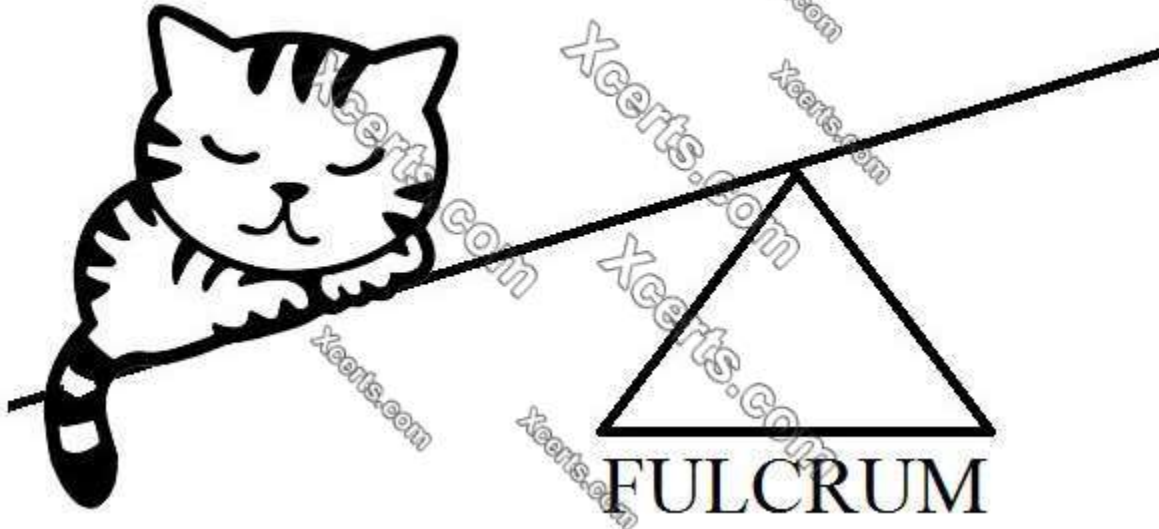
Answer(s): C

Explanation:

A single block-and-fall is called a runner.

QUESTION: 1570

CAT



In the figure above, if the fulcrum supporting the lever is moved closer to the cat, the cat will be _____.

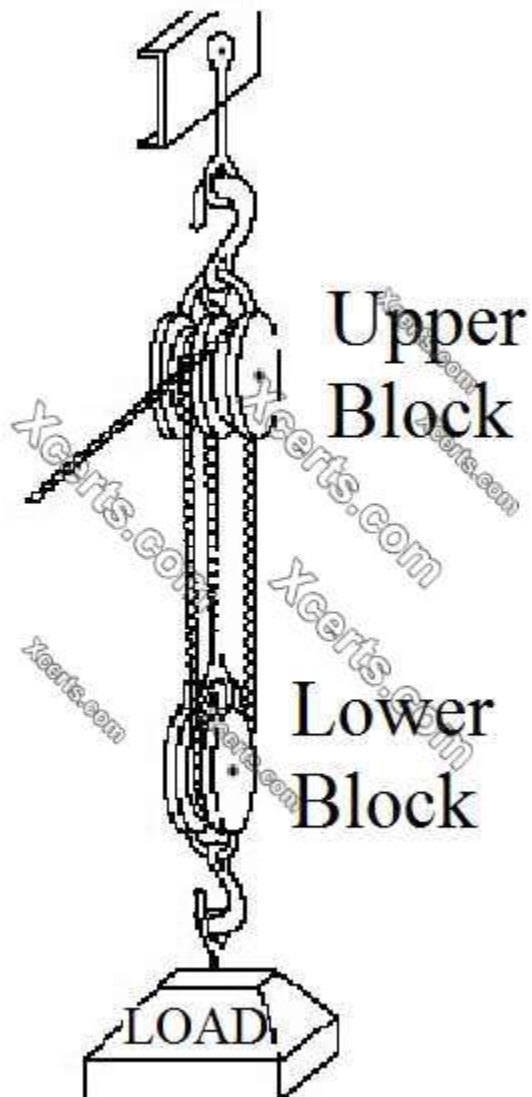
- A. easier to lift and will move higher
- B. harder to lift but will move higher
- C. easier to lift but will not move as high
- D. harder to lift and will not move as high

Answer(s): C

Explanation:

If the fulcrum is moved closer to the cat, the length of the effort arm of the lever will be increased, making the cat easier to raise, but the height to which the cat can be raised will be reduced.

QUESTION: 1571



The mechanical advantage of the block-and-tackle arrangement shown above is _____.

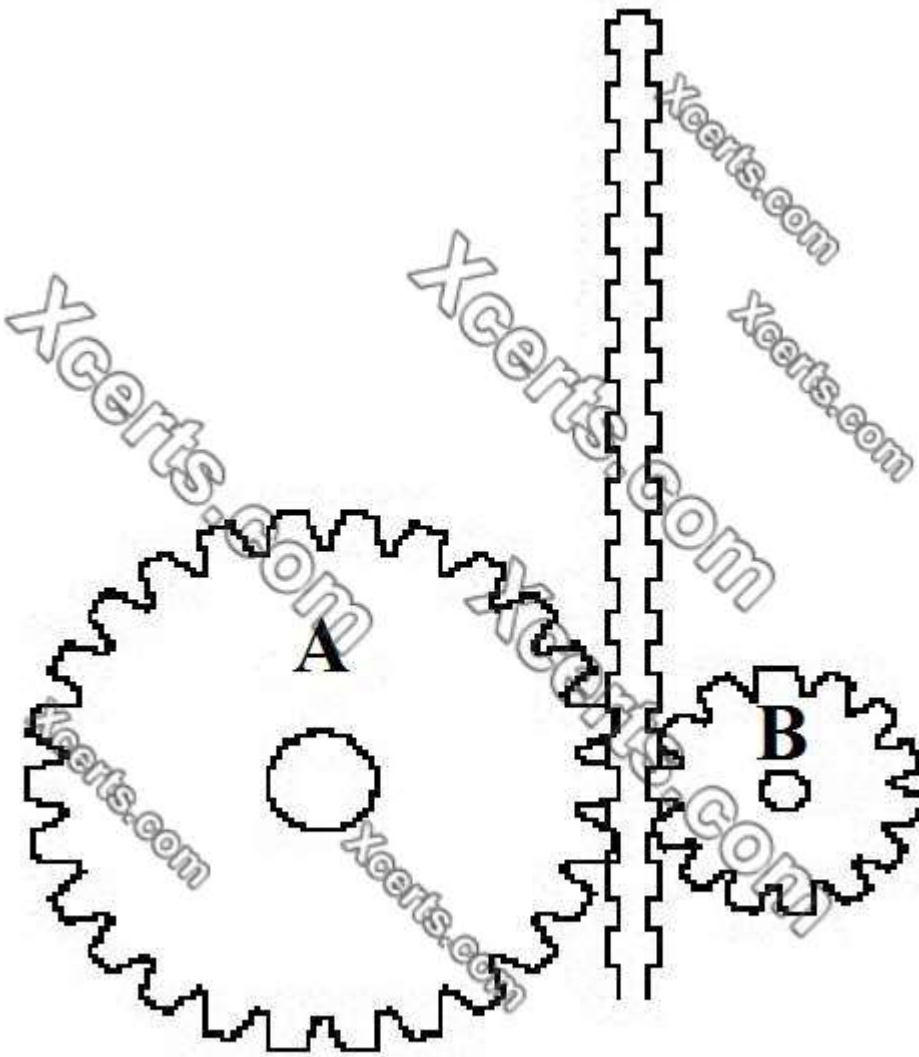
- A. 2
- B. 4
- C. 6
- D. 1

Answer(s): A

Explanation:

Because this block-and-tackle arrangement merely changes the direction of the pull, it has a mechanical advantage of only 2.

QUESTION: 1572



In the figure above, if the cogs move up the track at the same rate of speed, Cog A will _____.

- A. reach the top at the same time as Cog B
- B. reach the top after Cog B
- C. reach the top before Cog B
- D. have greater difficulty staying on track

Answer(s): C

Explanation:

The larger cog (Cog A) covers a greater linear distance in a given period of time.

QUESTION: 1573

If a house key, a wooden spoon, a plastic hanger, and a wool jacket are all the same temperature, which one feels the coldest?

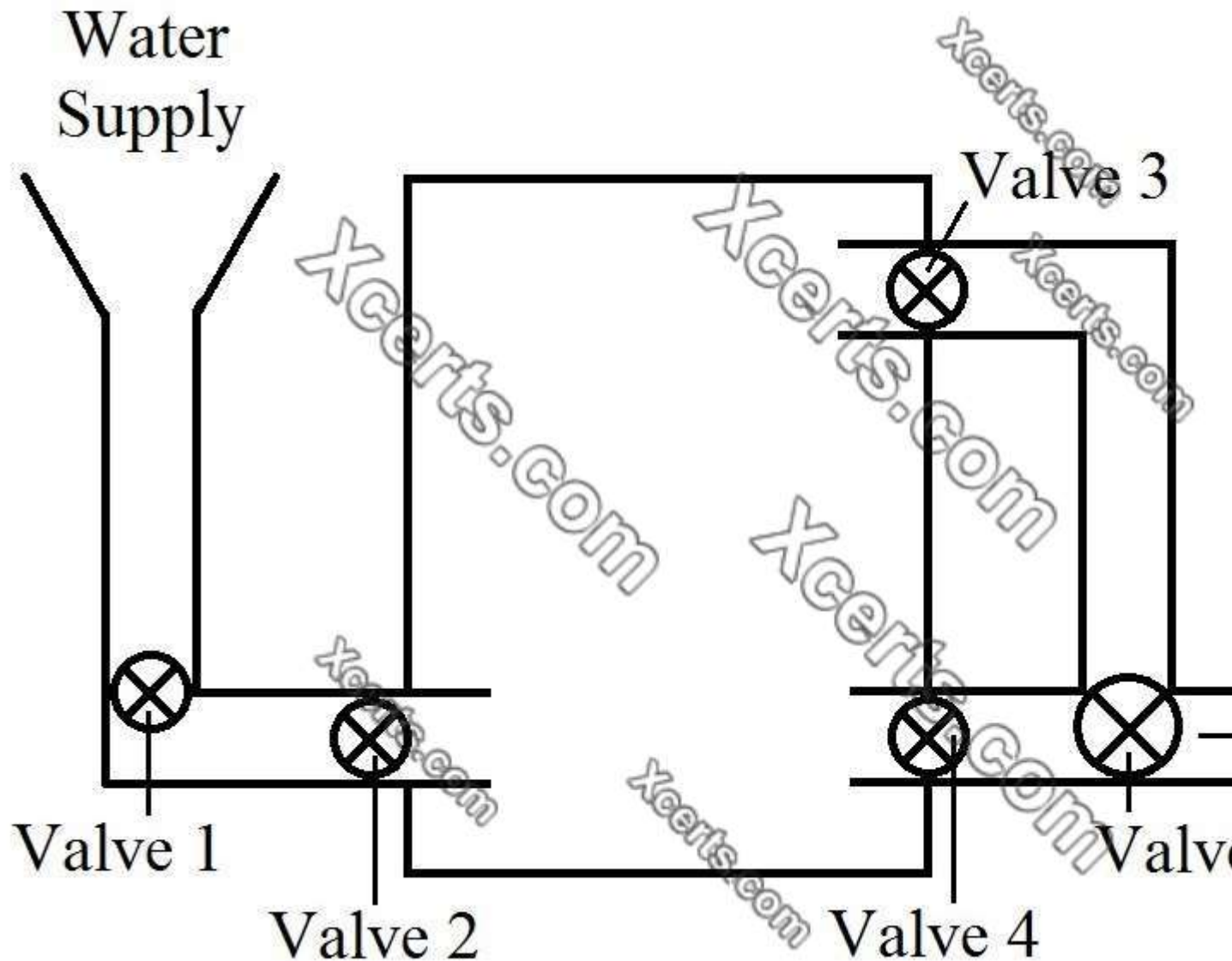
- A. key
- B. spoon
- C. hanger
- D. jacket

Answer(s): B

Explanation:

The key will feel coldest because metal is a better conductor than the other materials.

QUESTION: 1574



In the figure above, assume the valves are all closed.

To fill the tank, but to prevent it from filling entirely, which valves should be open?

- A. 1 and 2 only
- B. 1, 2, and 3 only

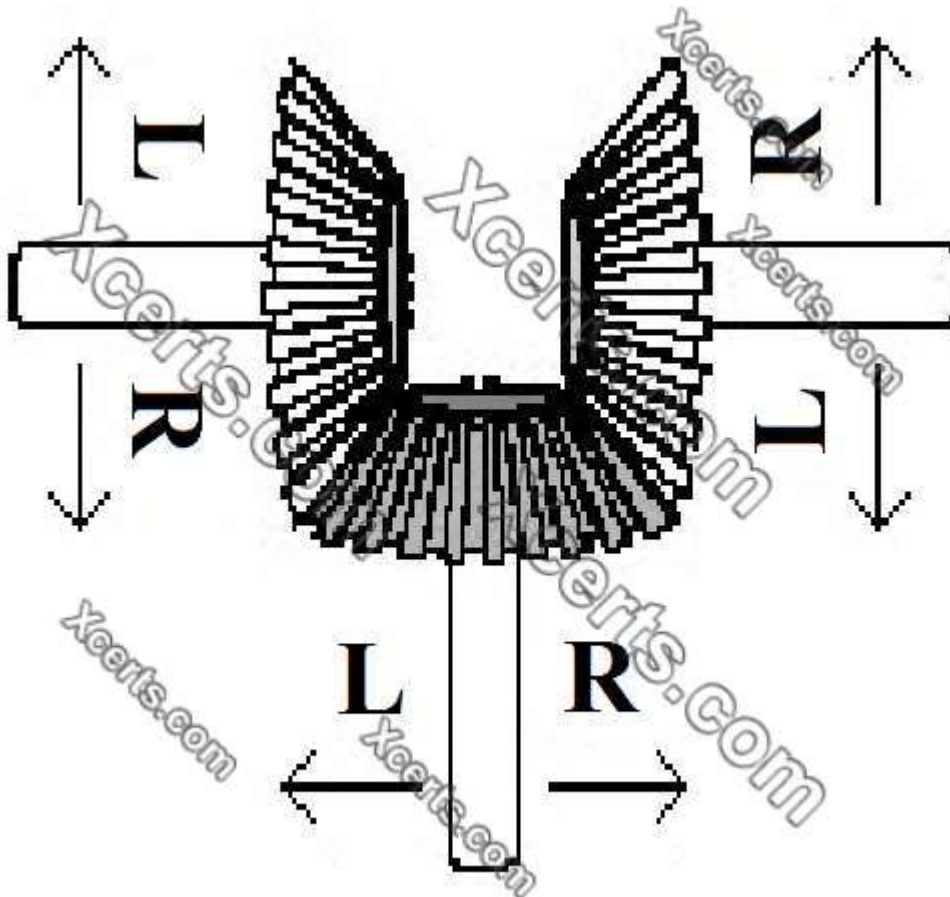
- C. 1, 2, and 4 only
- D. 1, 2, 3, and 5 only

Answer(s): D

Explanation:

Opening Valves 1 and 2 allows water to enter the tank.
 Opening Valves 3 and 5 prevents water from filling the tank entirely.
 Opening Valve 4 allows water to leave the tank.

QUESTION: 1575



If Gear A is turned to the left _____.

- A. Gear B turns to the right and Gear C turns to the left.
- B. Gear B turns to the left and Gear C turns to the left.
- C. Gear B turns to the right and Gear C turns to the right.
- D. Gear B turns to the left and Gear C turns to the right.

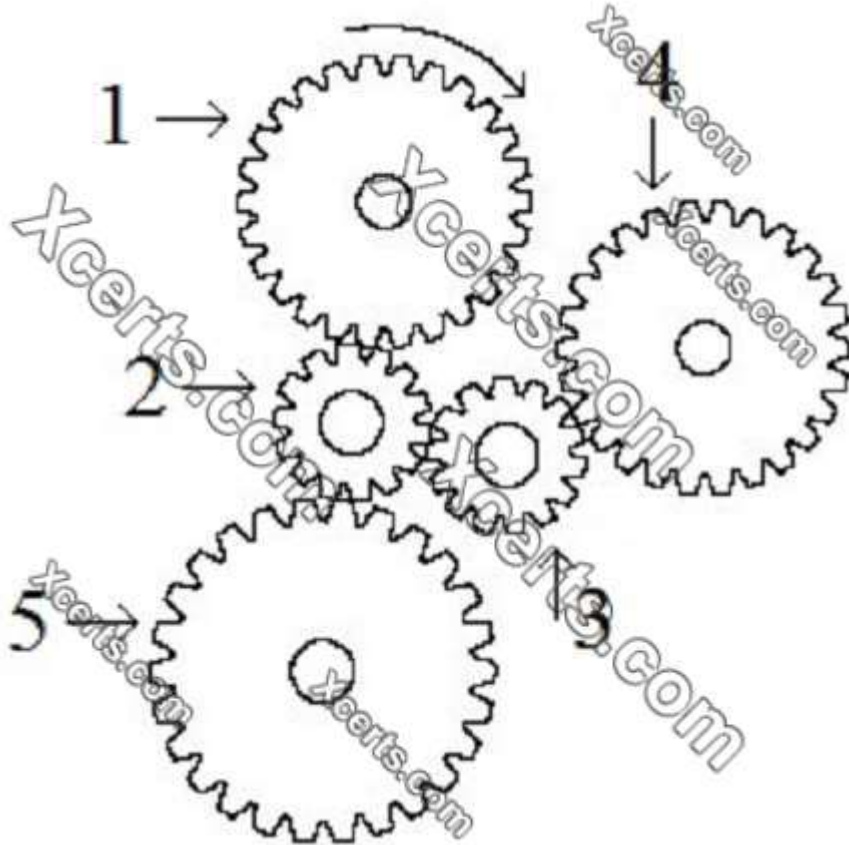
Answer(s): A

Explanation:

Gears with their teeth together in mesh turn in opposite directions.

Gear A turns Gear B in the opposite direction (right), and Gear B turns Gear C in the opposite direction (left).

QUESTION: 1576



If Gear 1 moves in a clockwise direction, which other gears also turn clockwise?

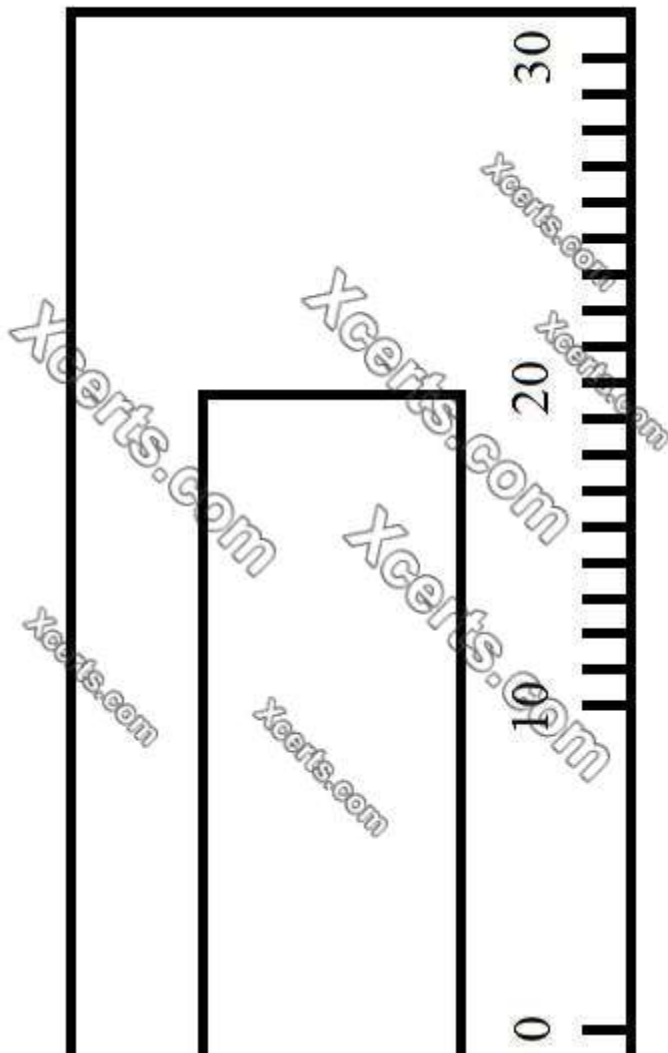
- A. 3 and 5
- B. 3, 4, and 5
- C. 2 and 5
- D. 3 and 4

Answer(s): A

Explanation:

Gears with their teeth together in mesh turn in opposite directions.
 Gear 1 turns clockwise. Gear 2, in mesh with Gear 1, turns counterclockwise.
 Gear 3, in mesh with Gear 2, turns clockwise.
 Gear 4, in mesh with Gear 3, turns counterclockwise.
 Gear 5, in mesh with Gear 2, turns clockwise.

QUESTION: 1577



The pressure gauge in the figure above shows a reading of _____.

- A. 15.0
- B. 19.5
- C. 21.0
- D. 23.0

Answer(s): C

Explanation:

The gauge shows a reading of 21.

QUESTION: 1578

A way to determine the amount of power being used is to _____.

- A. Multiply the amount of work done by the time it takes.
- B. Multiply the distance covered by the time it takes to move a load.

- C. Divide the amount of work done by 550 pounds per second.
- D. Divide the amount of work done by the amount of time it takes.

Answer(s): D

Explanation:

The formula for determining power is $\text{Power} = \text{Work} \div \text{Time}$.

QUESTION: 1579

A wood tool, a silver tool, and a steel tool are placed in boiling water for cleaning. Which tool gets the hottest?

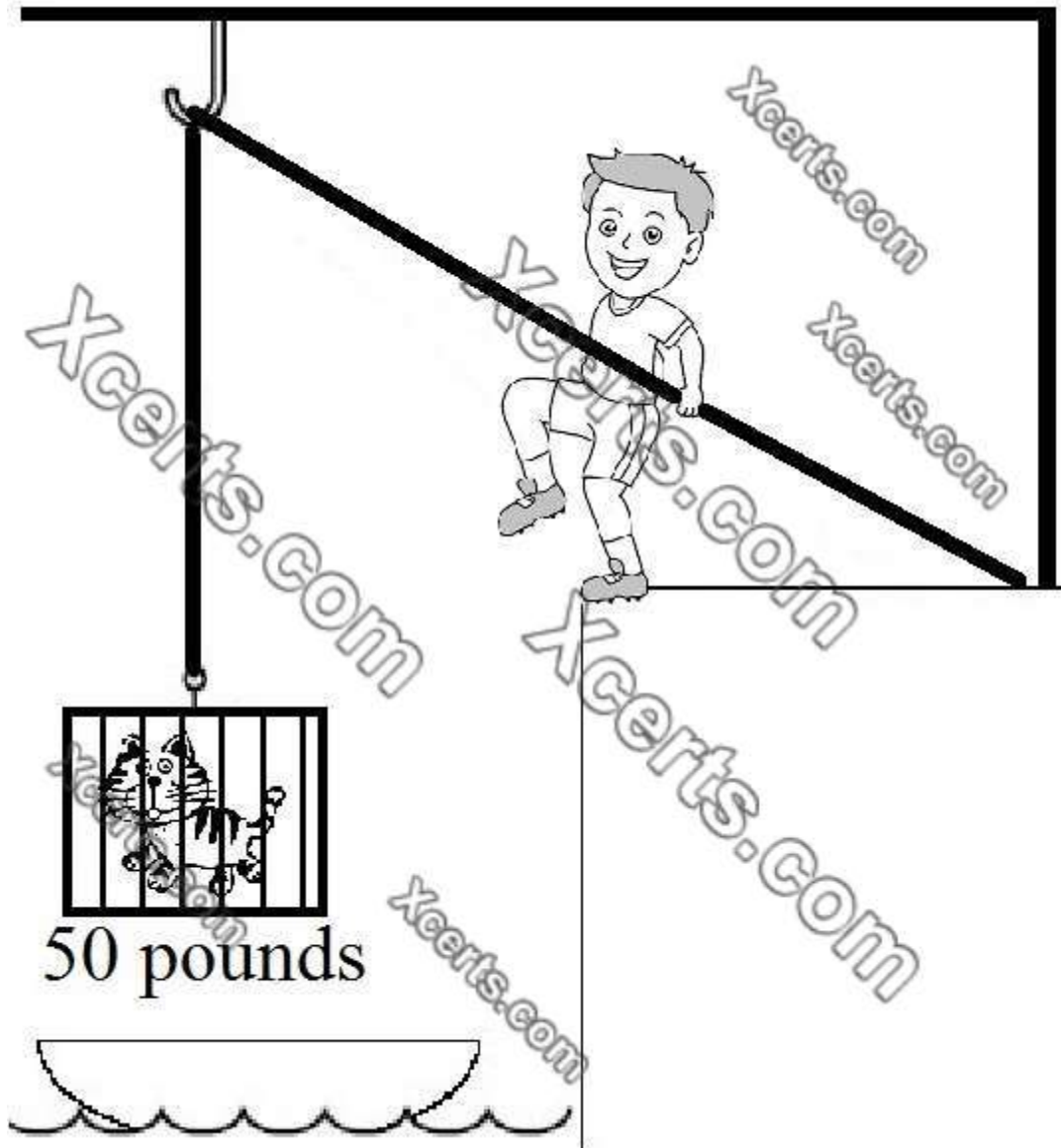
- A. steel
- B. wood
- C. silver
- D. All three are equally hot.

Answer(s): C

Explanation:

Silver is the best conductor, so it becomes hottest.

QUESTION: 1580



A runner is being used in the figure shown.

How much effort (disregard friction, wind resistance, and the weight of the pulley and the rope) is using the cat lover lifting the 50-pound crate (with cat)?

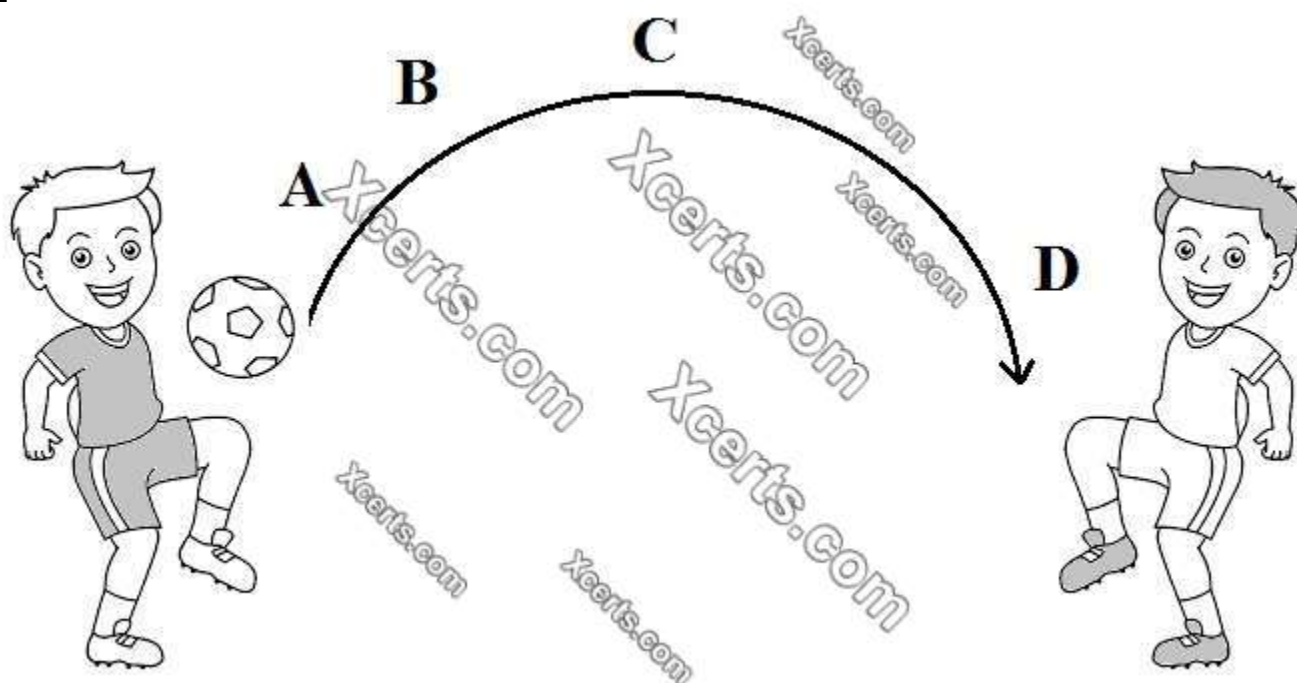
- A. 50-pound effort
- B. 100-pound effort
- C. 25-pound effort
- D. 10-pound effort

Answer(s): A

Explanation:

Stationary pulleys give no mechanical advantage, so effort equals the weight of the crate or 50 pounds.

QUESTION: 1581



In the figure above, at what point was the ball traveling most slowly?

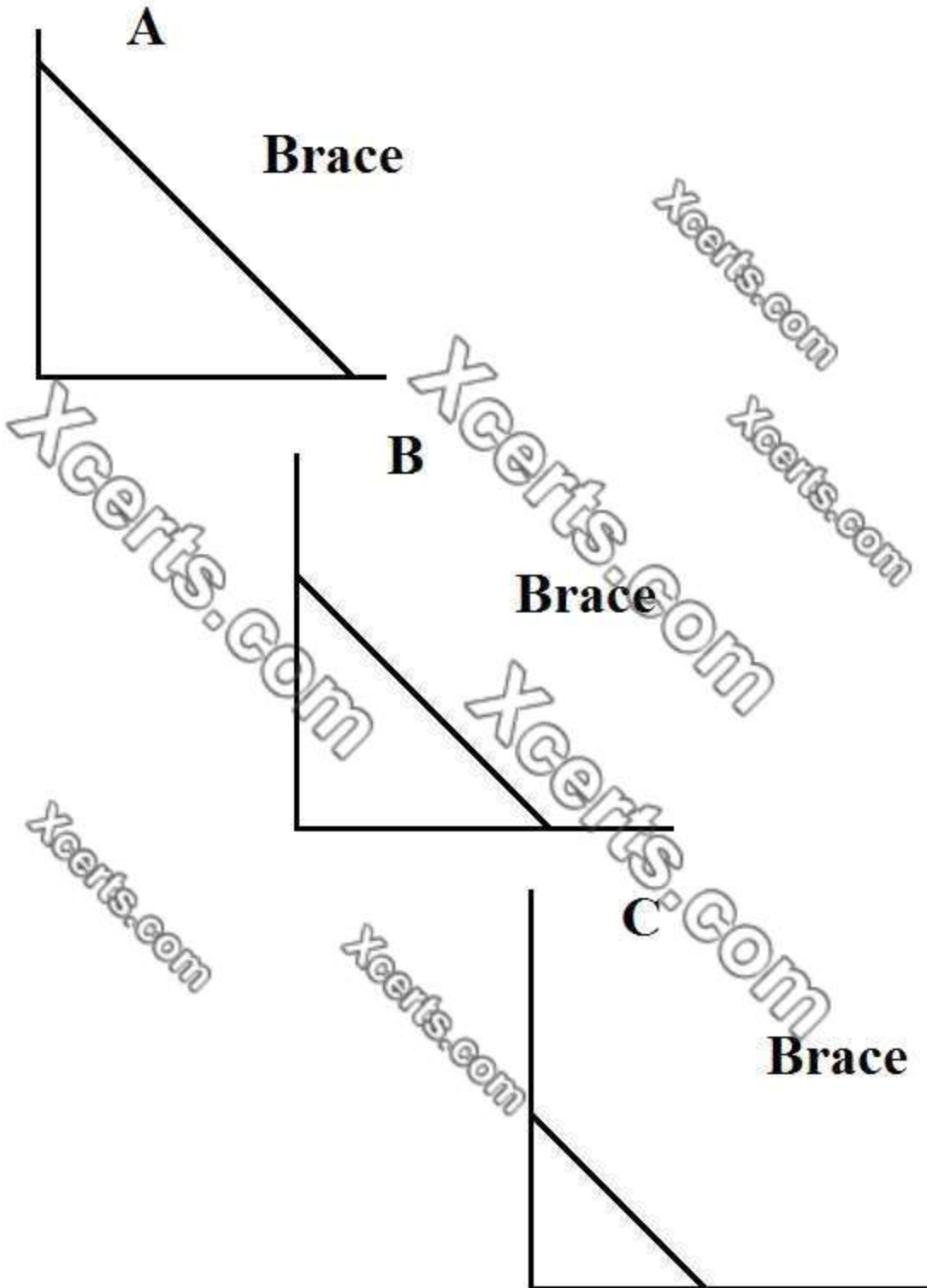
- A. A
- B. B
- C. C
- D. D

Answer(s): C

Explanation:

At the height of the arc, the ball has no upward momentum, so it goes the slowest at that point.

QUESTION: 1582



In the figure above, which angle is braced most solidly?

- A. A
- B. B
- C. C

D. All are braced equally solidly.

Answer(s): A

Explanation:

The brace on Angle A covers more area of the angle, so it's more solidly braced.

QUESTION: 1583

A simple pulley gives a mechanical advantage of _____.

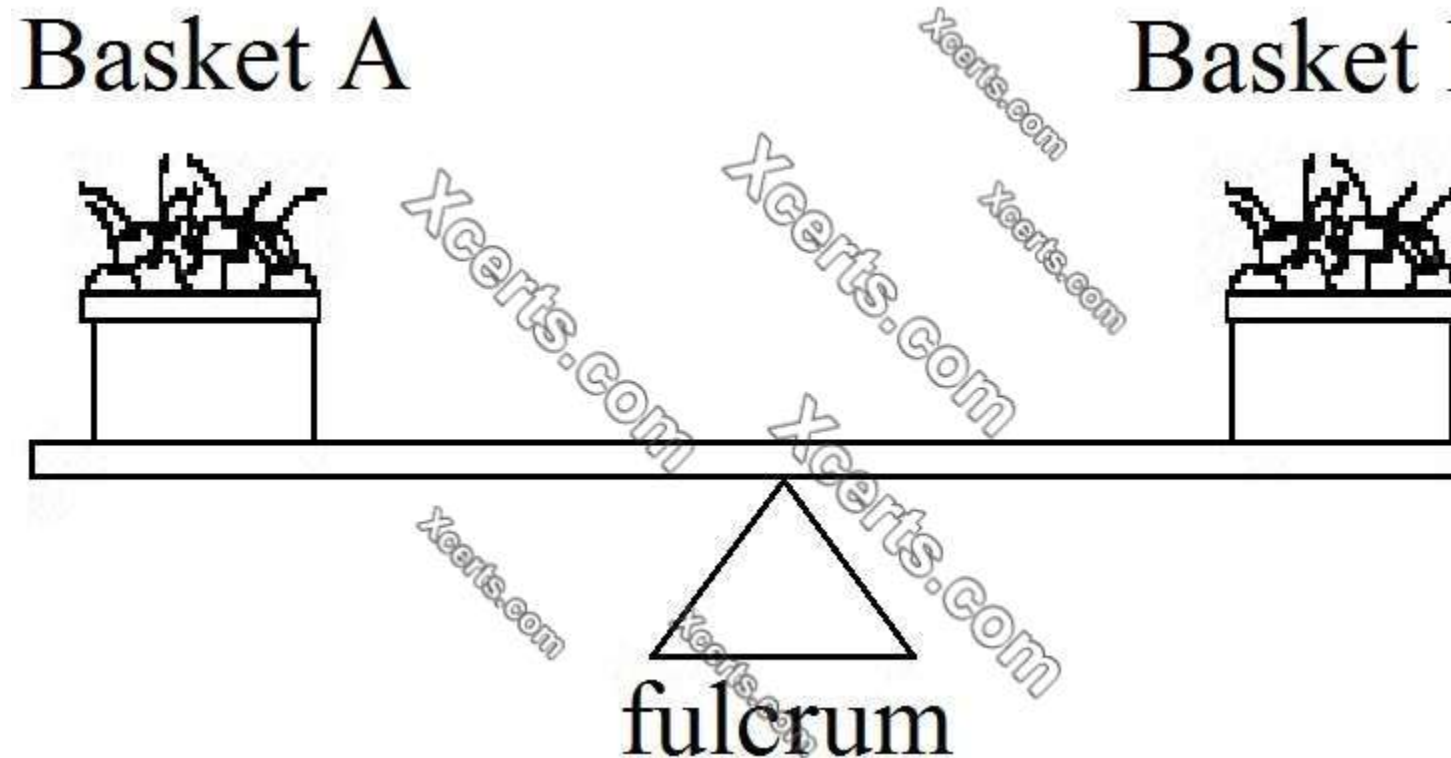
- A. 2
- B. 3
- C. 1
- D. unknown

Answer(s): C

Explanation:

A simple pulley gives no mechanical advantage, although it does make work easier by spreading out the work needed over several tries. So the mechanical advantage is 1.

QUESTION: 1584



The baskets are balanced on the arm in the figure above. If cherries are removed from Basket B, to rebalance the arm _____.

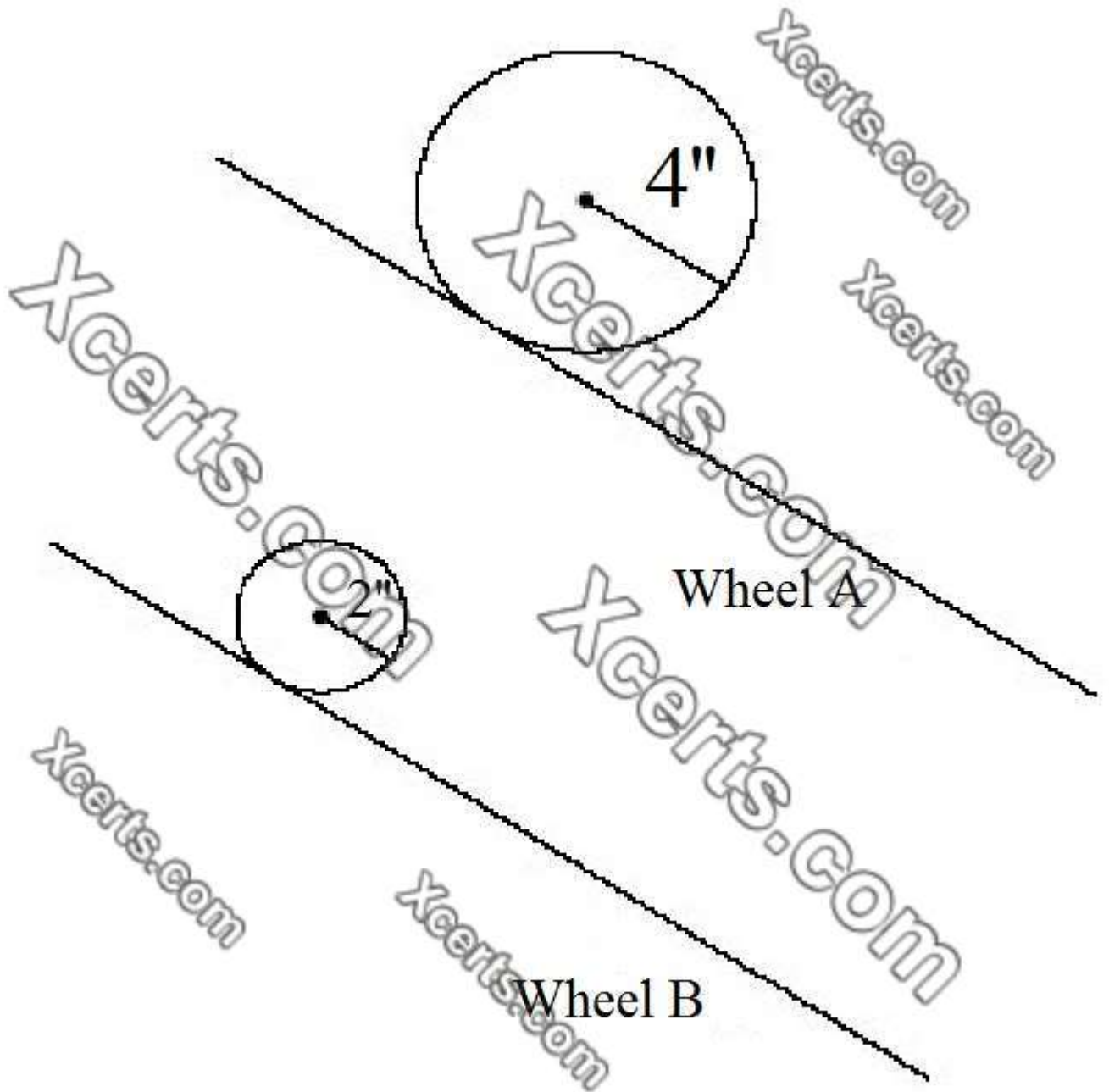
- A. The fulcrum will have to be moved to the right.
- B. Basket B will have to be moved to the right.
- C. Basket A will have to be moved to the left.
- D. Basket A will have to be moved to the right.

Answer(s): D

Explanation:

Moving Basket A to the right counterbalances the loss of cherries from Basket B.

QUESTION: 1585



If both Wheel A and Wheel B revolve at the same rate in the figure above, Wheel A will cover a linear distance of 12 feet _____.

- A. faster than Wheel B
- B. slower than Wheel B
- C. in about the same time as Wheel B
- D. half as quickly as Wheel B

Answer(s): A

Explanation:

Wheel B has to make more revolutions to cover the same ground as Wheel A, so it will cover the distance more slowly.

QUESTION: 1586

If a force of 200 pounds is exerted over an area of 10-square inches, what's the psi?

- A. 10
- B. 15
- C. 20
- D. 200

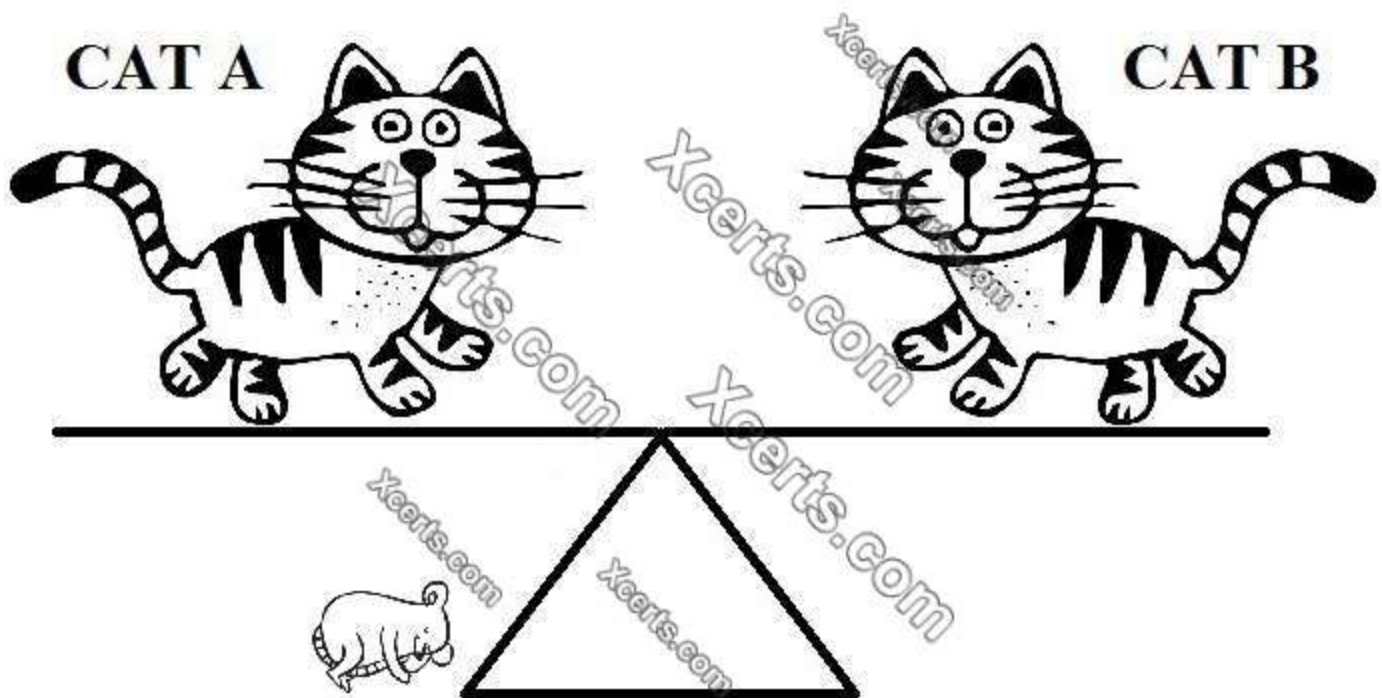
Answer(s): C

Explanation:

You can calculate psi as $\text{Pressure} = \text{Force}/\text{Area}$

A. So, in this problem, $P = 200/10 = 20$.

QUESTION: 1587



In the figure above, if Cat A moves toward the middle of the seesaw to get a better look at the mouse, Cat B will _____.

- A. remain stationary
- B. move toward the ground
- C. rise in the air
- D. instigate a cat fight

Answer(s): B

Explanation:

The Cat B will move towards the ground

QUESTION: 1588

If a ramp measures 6 feet in length and 3 feet in height, an object weighing 200 pounds requires how much effort to move using the ramp?

- A. 200 pounds
- B. 100 pounds
- C. 50 pounds
- D. 300 pounds

Answer(s): B

Explanation:

The formula to determine mechanical advantage of an incline plane is $\text{Length of Ramp} \div \text{Height of Ramp} = \text{Weight of Object} \div \text{Effort}$.

Plugging in the numbers gives you:

$$6/3 = 200/E$$

$$6E = 600$$

$$E = 100$$

QUESTION: 1589

A micrometer is used to measure _____.

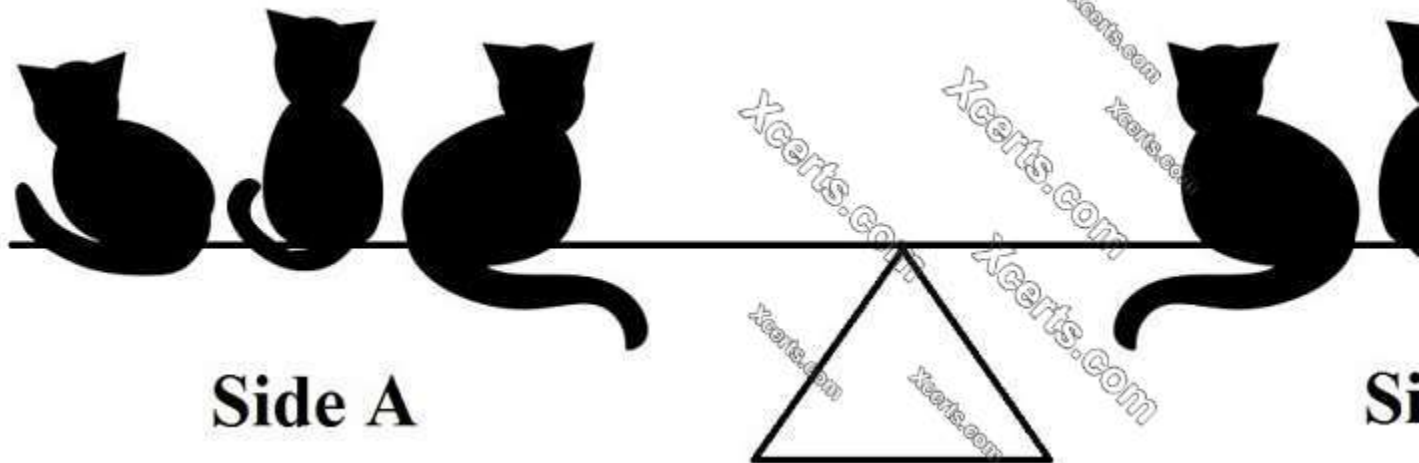
- A. small changes in temperature
- B. changes in psi
- C. thicknesses to a few thousandths of an inch
- D. objects invisible to the unaided eye

Answer(s): C

Explanation:

Micrometers measure very small but not microscopic objects.

QUESTION: 1590



If a cat is removed from Side B of the seesaw, what happens to the cats on Side A?

- A. The cat will never move from Side B until it's good and ready.
- B. The cats on Side A will move up in the air.
- C. The cats on Side A will move toward the ground.
- D. Nothing will happen.

Answer(s): C

Explanation:

Reducing the weight on Side B (by removing the cat) will cause Side A to move toward the ground.

QUESTION: 1591

The force produced when two objects rub against each other is called _____.

- A. gravity
- B. recoil
- C. magnetism
- D. friction

Answer(s): D

Explanation:

Objects rubbing together produce friction.

QUESTION: 1592

Normally, atmospheric pressure is approximately _____.

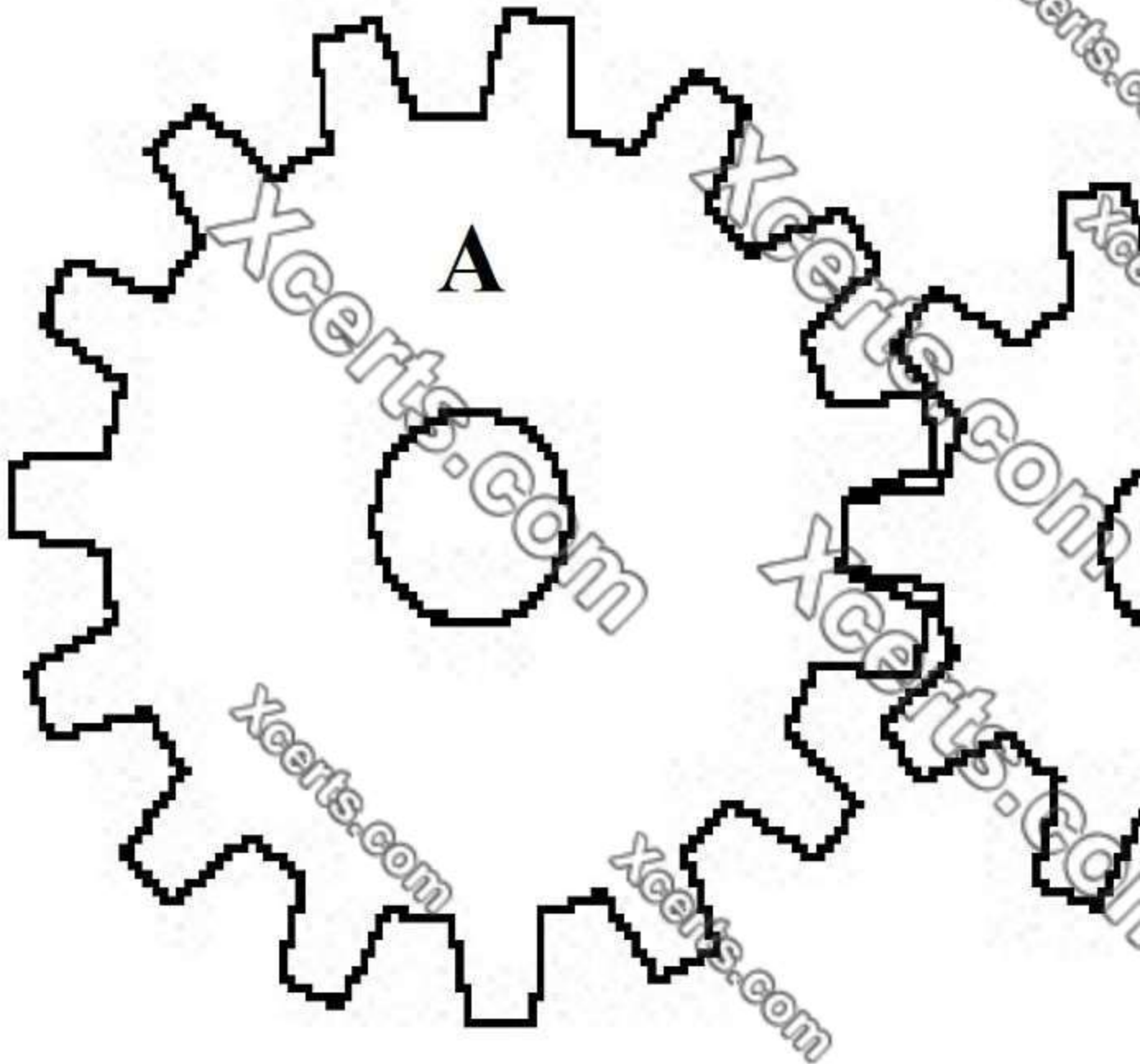
- A. 14.7 psi
- B. 23.2 psi
- C. 7.0 psi
- D. 10.1 psi

Answer(s): A

Explanation:

"Normal" atmospheric pressure is 14.7 psi.

QUESTION: 1593



For Gear A and Gear B to mesh properly in the figure above _____.

- A. They must be the same size.
- B. They must turn at different rates.
- C. They must both turn in the same direction.
- D. Their teeth must be of equal size.

Answer(s): D

Explanation:

Gears of unequal size can mesh properly as long as their teeth are of equal size.

QUESTION: 1594

Torsion springs _____.

- A. produce a direct pull
- B. exert no pull
- C. produce a twisting action
- D. coil but do not uncoil

Answer(s): C

Explanation:

Torsion springs coil or uncoil and produce a twisting action, not a direct pull.

QUESTION: 1595

To move a 400-pound crate from the floor of a warehouse to the bed of a truck 4-feet off the ground, the most efficient device to use is a _____.

- A. lever
- B. incline plane
- C. fixed pulley
- D. jackscrew

Answer(s): B

Explanation:

To move a heavy object a few feet in height, the incline plane is the most efficient device (of those listed) to use.

QUESTION: 1596

Water in an engine can cause damage in winter weather because _____.

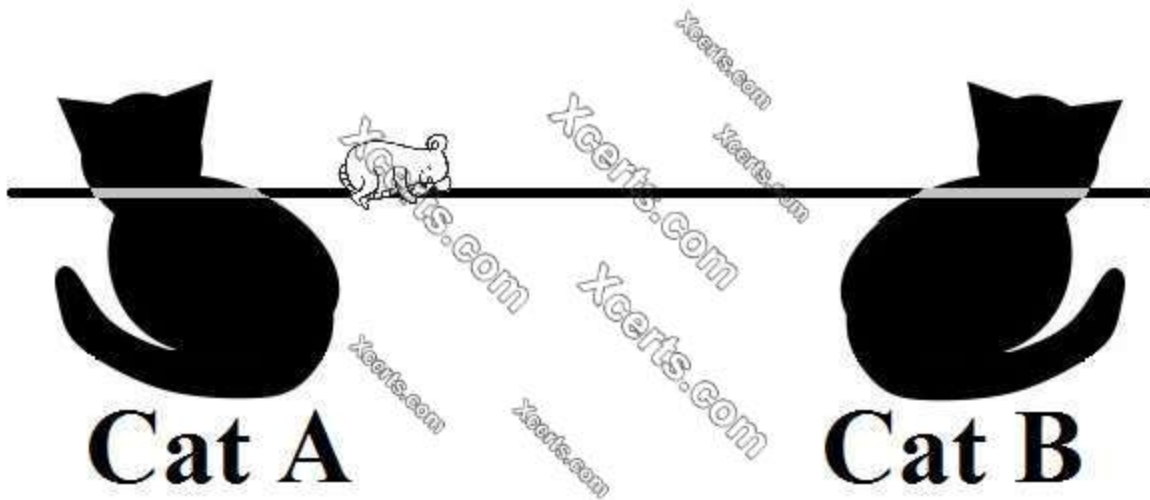
- A. it can vaporize
- B. water expands when it freezes
- C. ice is heavier than water
- D. cold water creates more steam than warm water

Answer(s): B

Explanation:

Water expands when it freezes, possibly damaging engine components.

QUESTION: 1597



The weight of the load is being carried on the backs of the two cats shown in the figure. Which cat is carrying the most weight?

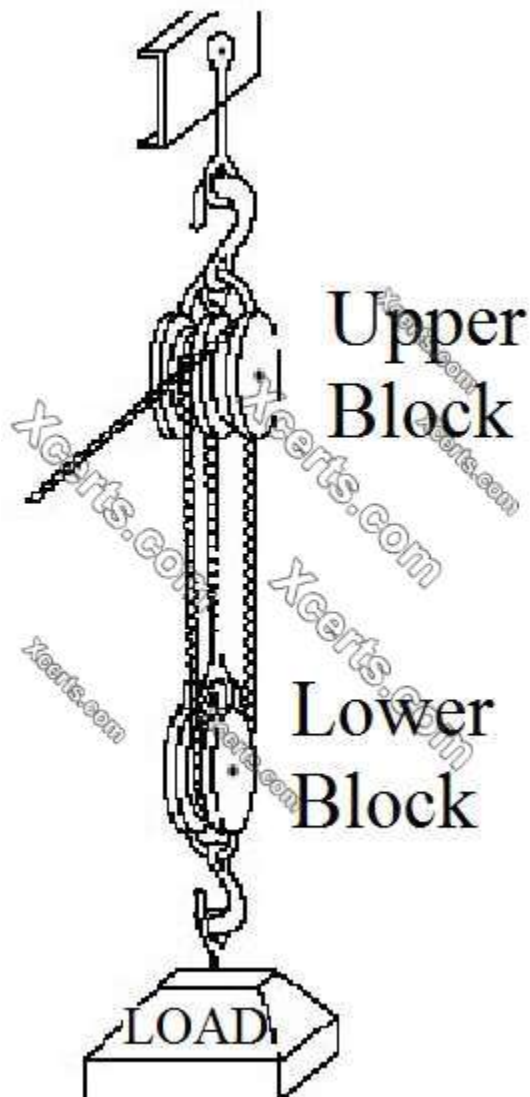
- A. Cat A
- B. Cat B
- C. Both are carrying an equal amount of weight.
- D. It can't be determined without more information.

Answer(s): A

Explanation:

The load is closer to Cat A, so he's carrying the greater portion of the weight.

QUESTION: 1598



When the block-and-tackle arrangement shown in the figure above is used to lift a load, all the following parts remain stationary EXCEPT _____.

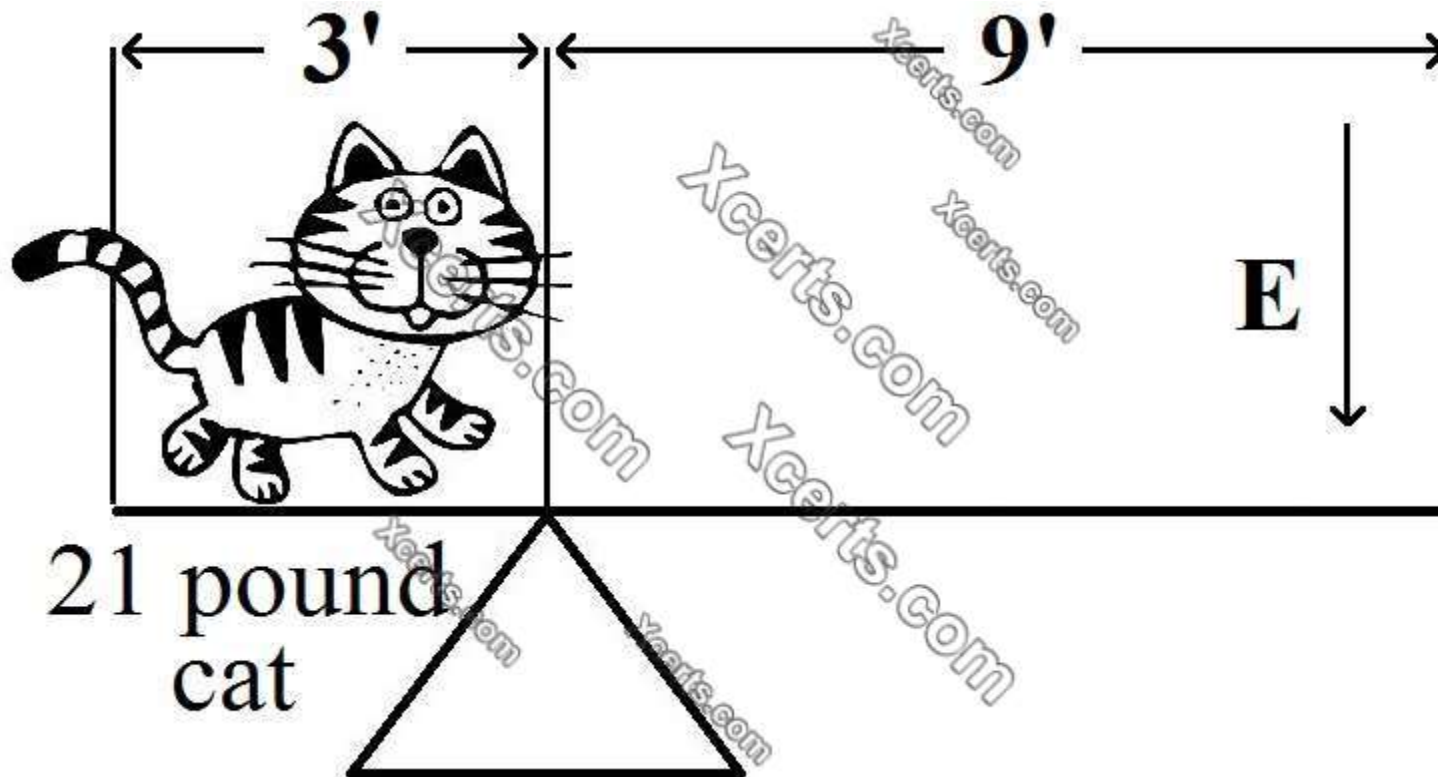
- A. the upper hook
- B. the upper block
- C. the lower block
- D. none; all the parts move

Answer(s): C

Explanation:

All the listed parts remain stationary except the lower block.

QUESTION: 1599



In the figure shown above, what effort E must be applied to lift the cat?

- A. 7.0 pounds
- B. 9.0 pounds
- C. 21.0 pounds
- D. 10.5 pounds

Answer(s): A

Explanation:

Apply the leverage formula: Length of Effort Arm divided by Length of Resistance Arm = Resistance Force divided by Effort Force. Or:

$$9 \div 3 = 21 \div E \text{ (effort force)}$$

$$3 = 21 \div E$$

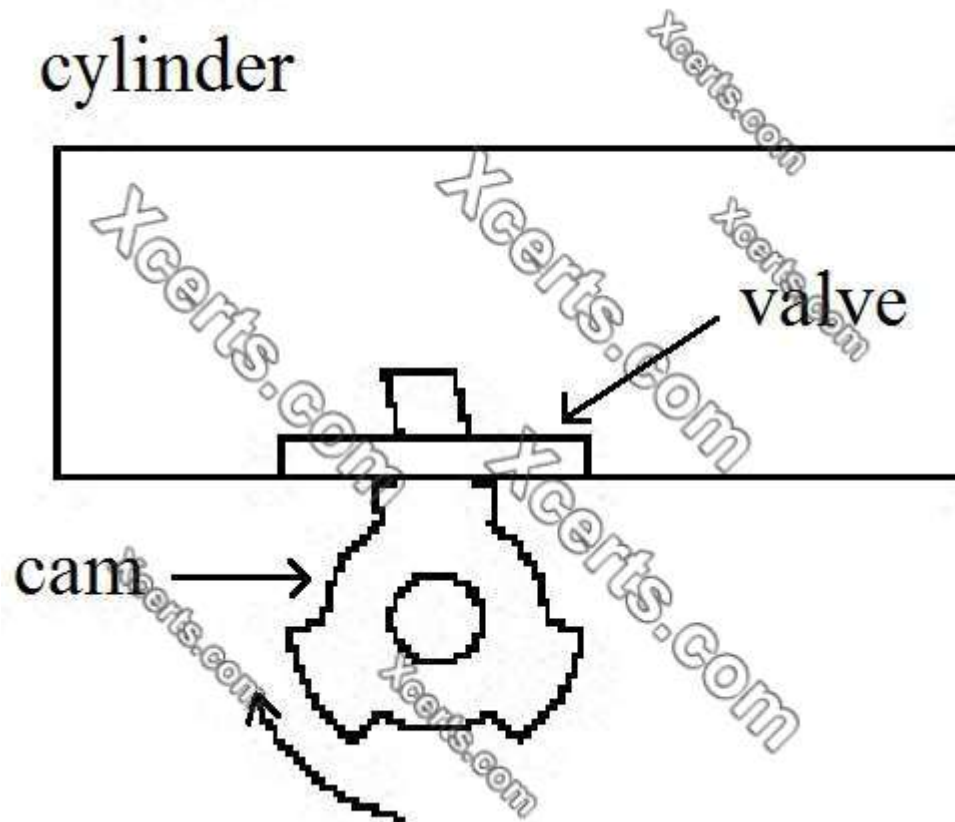
$$3E = 21 \div E \times E$$

$$3E = 21$$

$$3E \div 3 = 21 \div 3$$

$$E = 7$$

QUESTION: 1600



In the figure above, for each complete revolution the cam makes, how many times will the valve open?

- A. 1
- B. 6
- C. 3
- D. 2

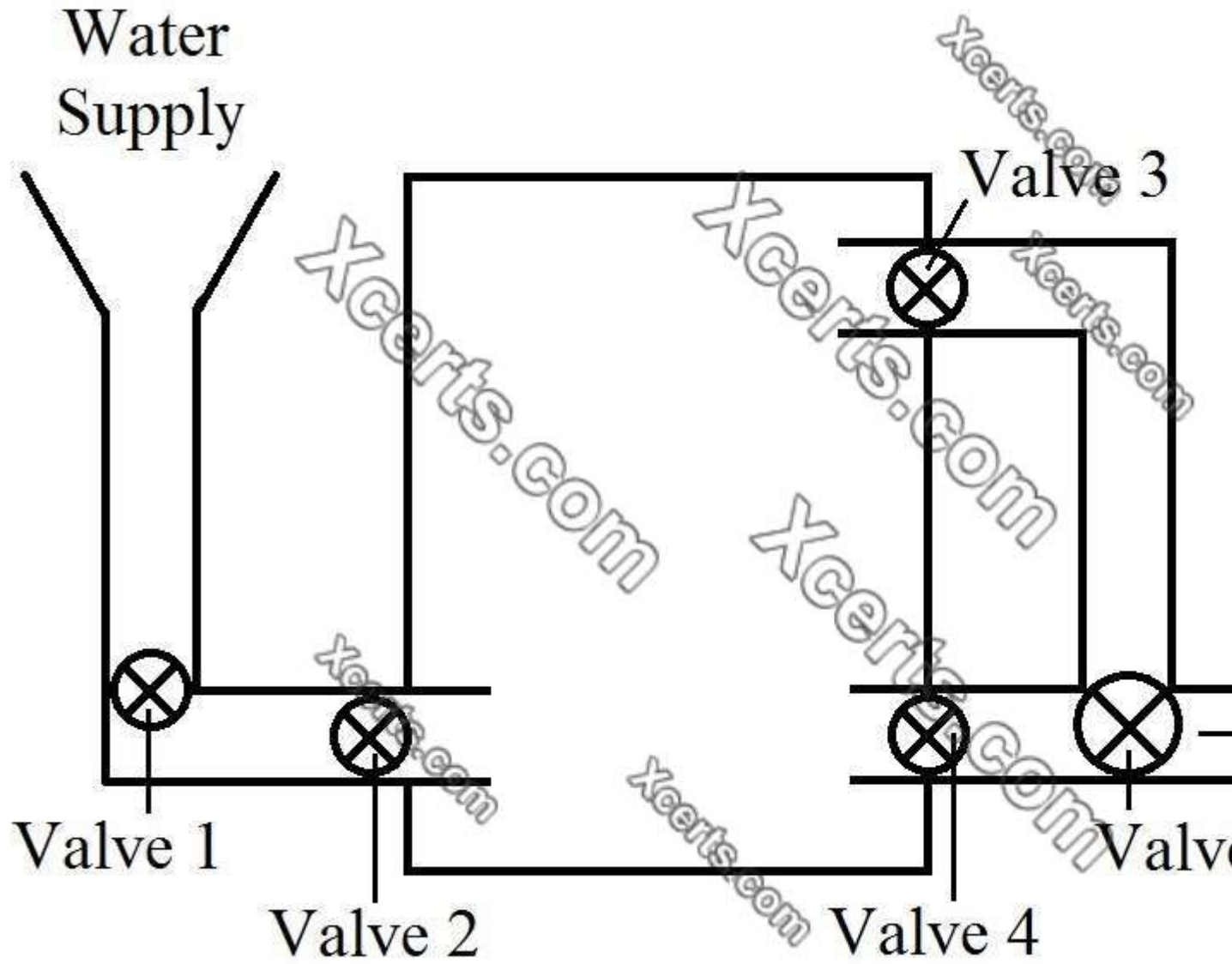
Answer(s): C

Explanation:

The valve will open each time a high point of the cam hits it.

The cam has three high points, so the valve will open three times per revolution.

QUESTION: 1601



In the figure above, assume the valves are all open.
Which valves need to be closed for the tank to fill up completely?

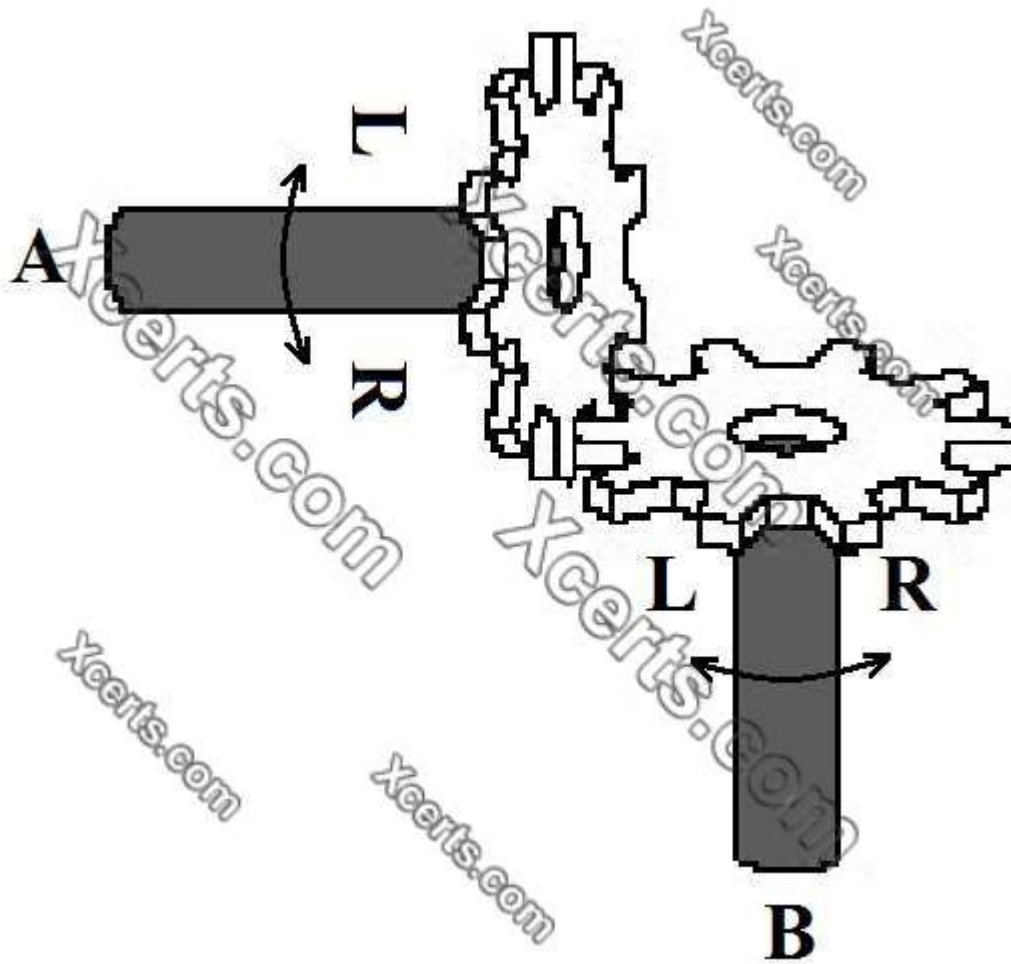
- A. 3 and 4 only
- B. 3, 4, and 5
- C. 2, 3, and 4
- D. 4 only

Answer(s): A

Explanation:

Closing only Valves 3 and 4 keeps the water from leaving the tank.

QUESTION: 1602



If Gear A turns left in the figure above, Gear B _____.

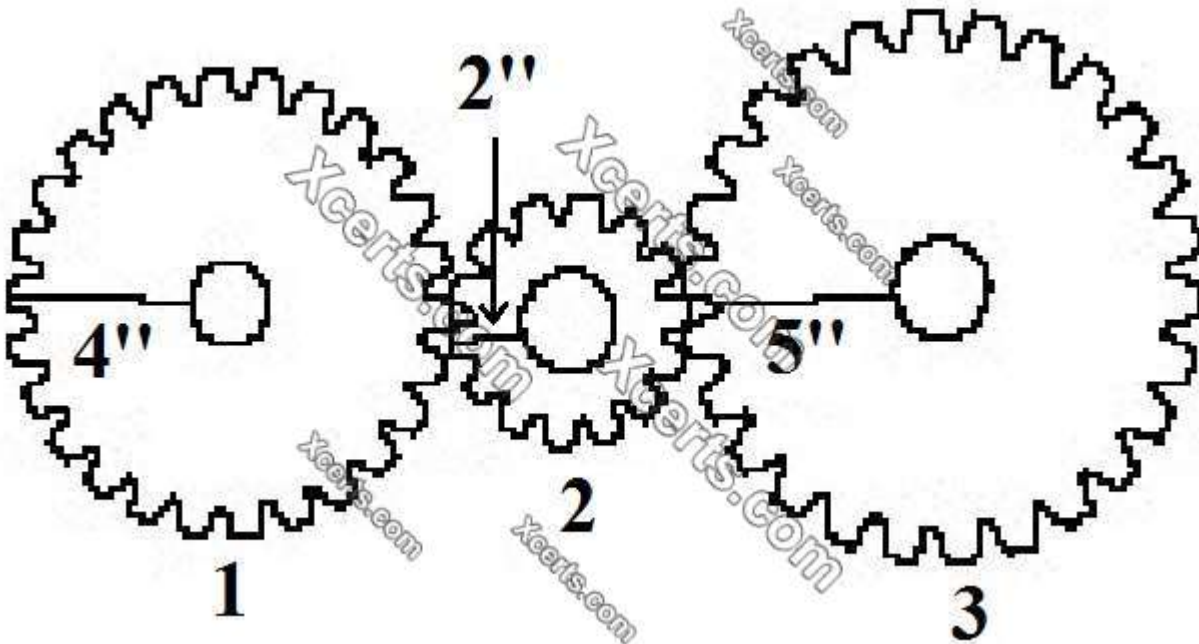
- A. won't turn
- B. turns left
- C. turns right
- D. It can't be determined.

Answer(s): C

Explanation:

Gears in mesh always turn in opposite directions.

QUESTION: 1603



If Gear 1 makes 10 complete clockwise revolutions per minute in the figure above, then _____.

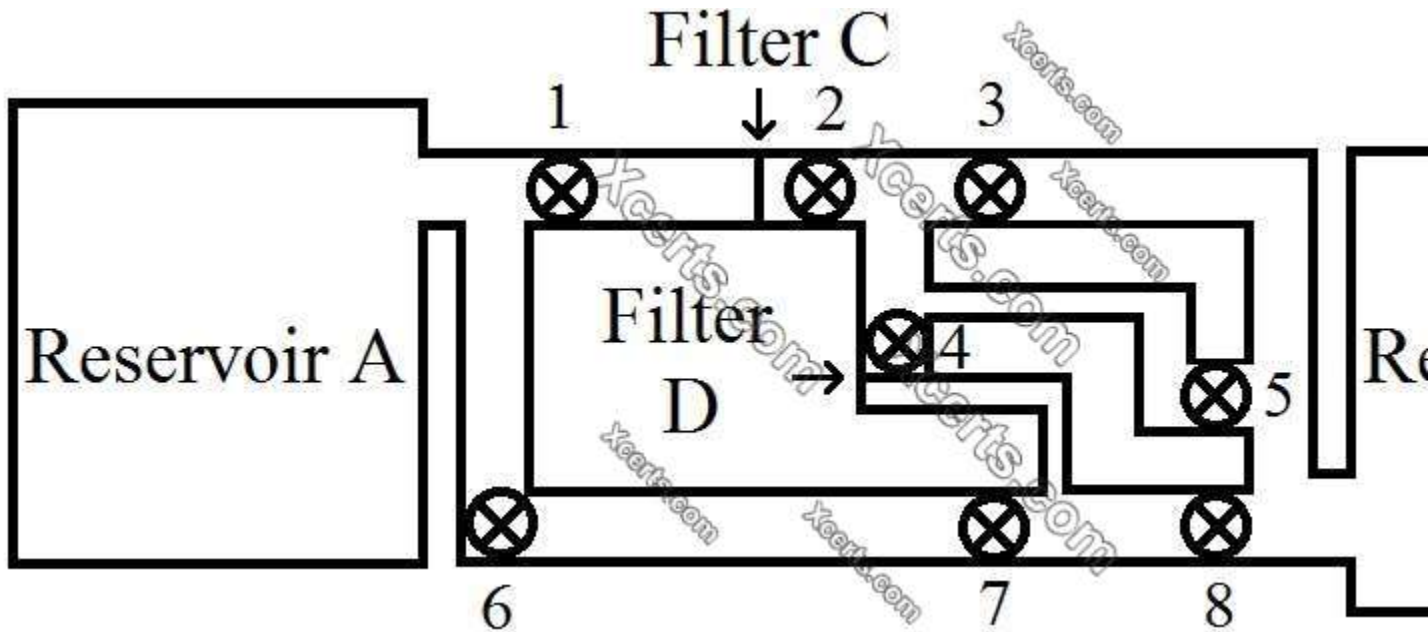
- A. Gear 2 makes 10 complete clockwise revolutions per minute.
- B. Gear 2 makes 20 complete counterclockwise revolutions per minute.
- C. Gear 2 makes 5 complete counterclockwise revolutions per minute.
- D. Gear 3 keeps Gear 2 from making any revolutions.

Answer(s): B

Explanation:

If Gear 1 turns at 10 rpm, then Gear 2, which is half the size, turns twice as fast at a rate of 20 rpm.

QUESTION: 1604



For the fuel to travel from Reservoir A to Reservoir B, passing through Filters C and D on the way, which valves must be open?

- A. 1, 2, 4, and 8
- B. 1, 2, and 3
- C. 6, 7, and 8
- D. 4, 6, and 7

Answer(s): A

Explanation:

Opening Valves 1, 2, 4, and 8 allows the fuel to travel through the filters.
 Opening Valves 1, 2, and 3 doesn't allow the fuel to travel through the filters.
 Opening Valves 6, 7, and 9 doesn't allow the fuel to travel through the filters.
 Opening Valves 4, 6, and 7 will allow fuel to travel through the filters but not to Reservoir B.

QUESTION: 1605

A yellow flame on a gas furnace indicates _____.

- A. Everything is fine.
- B. The fuel-air mixture is too rich.
- C. The fuel-air mixture is too lean.
- D. The gas pressure is too low.

Answer(s): B

Explanation:

A yellow flame indicates too much fuel or not enough air.
 More air should be allowed to enter and mix with the gas.
 Thus, the fuel-air mixture is too rich.

QUESTION: 1606

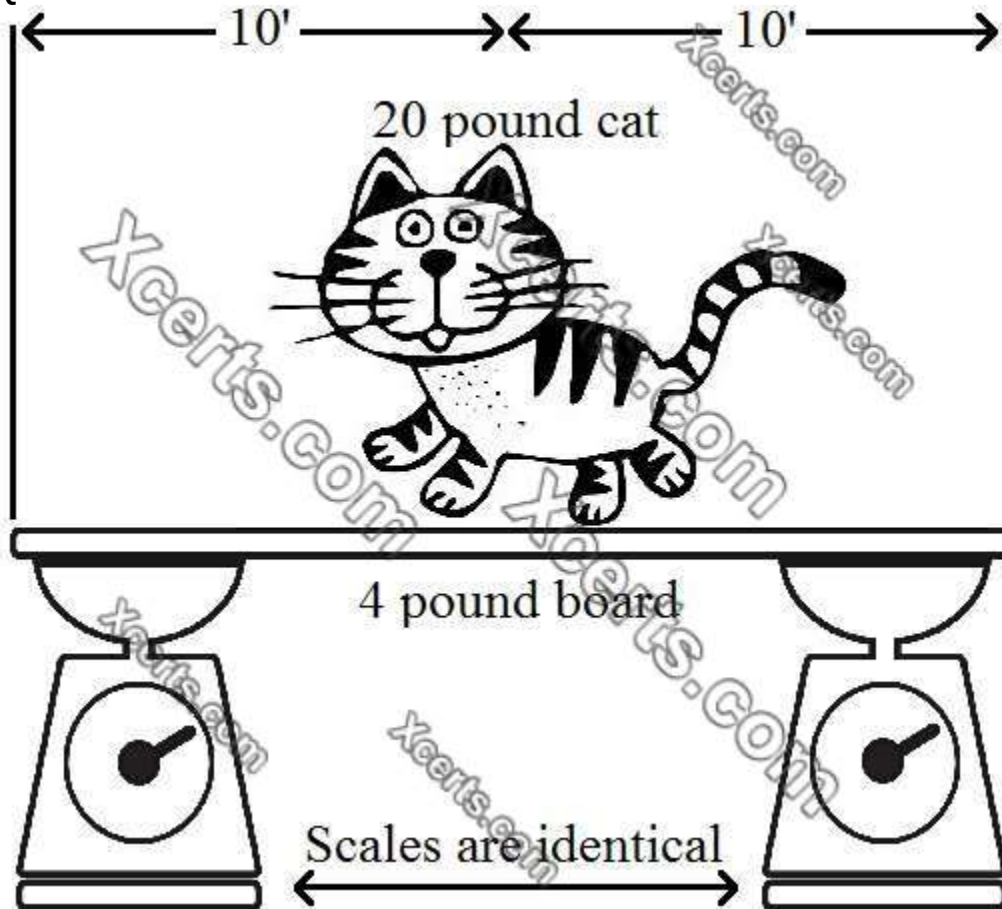
If a water tank on a toilet keeps overflowing, the problem is probably a _____.

- A. defective float
- B. clogged pipe
- C. crimped chain
- D. improper seal

Answer(s): A

Explanation:

The float measures the water level in the tank. If the tank overflows, the float is probably defective.

QUESTION: 1607

In the figure shown above, the board holds the weight of the cat. The board is placed on two identical scales.

Each scale reads _____.

- A. 24
- B. 10
- C. 12

D. 40

Answer(s): C

Explanation:

The 20-pound cat and the 4-pound board weigh 24 pounds total or, divided by 2, 12 pounds per scale.

QUESTION: 1608

Helical gears have _____.

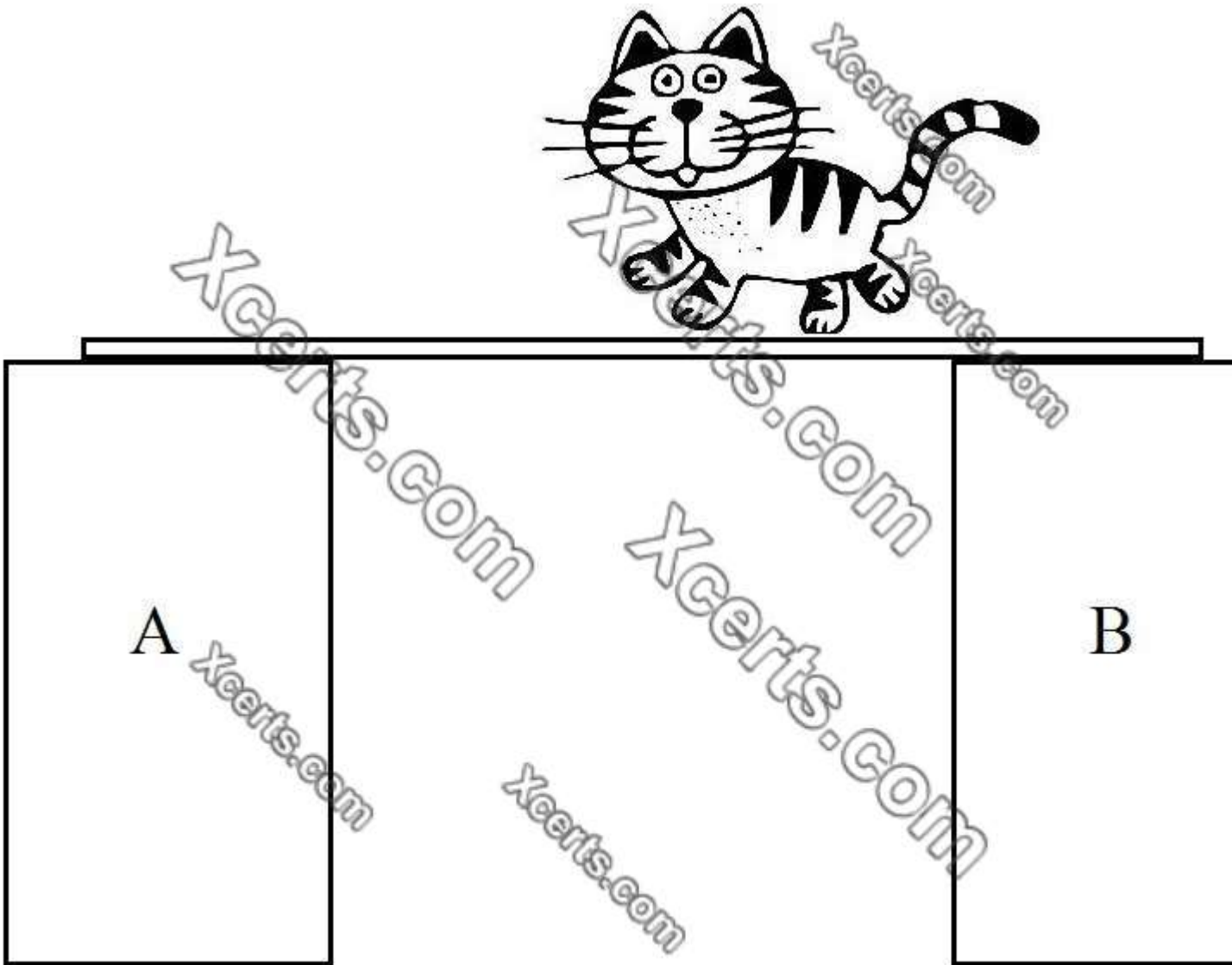
- A. straight teeth
- B. slanted teeth
- C. teeth of unequal size
- D. no advantage over spur gears

Answer(s): B

Explanation:

The teeth of helical gears are slanted.

QUESTION: 1609



In the figure above, which pillar supports the greater load of the cat?

- A. Pillar A
- B. Pillar B
- C. Both pillars support the cat equally.
- D. It's impossible to determine from the information given.

Answer(s): B

Explanation:

The cat is closer to Pillar B, so Pillar B bears more weight.

QUESTION: 1610

Wheel A has a diameter of 10 feet. Wheel B has a diameter of 8 feet.

If both wheels revolve at the same rate, Wheel B will cover a linear distance of 16 feet

_____.

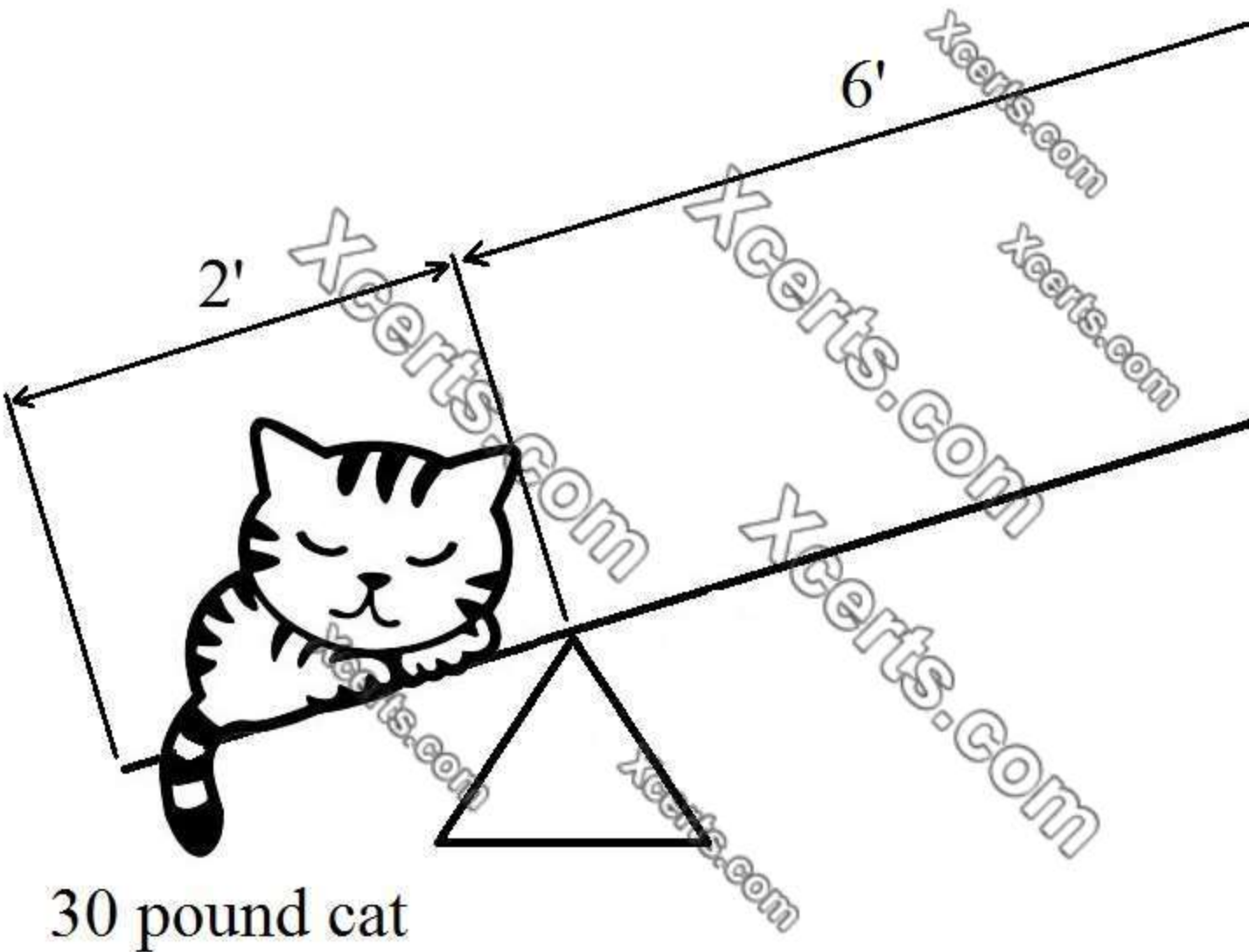
- A. at the same time as Wheel A
- B. more slowly than Wheel A
- C. in twice the time as Wheel A
- D. faster than Wheel A

Answer(s): B

Explanation:

Wheel B has to make more revolutions than Wheel A to cover the same amount of distance, so it will go slower.

QUESTION: 1611



What effort must be used to lift a 30-pound cat (see the figure above) using a first-class lever? (Don't include the weight of the lever in your calculations.)

- A. 10 pounds

- B. 15 pounds
- C. 50 pounds
- D. 5 pounds

Answer(s): A

Explanation:

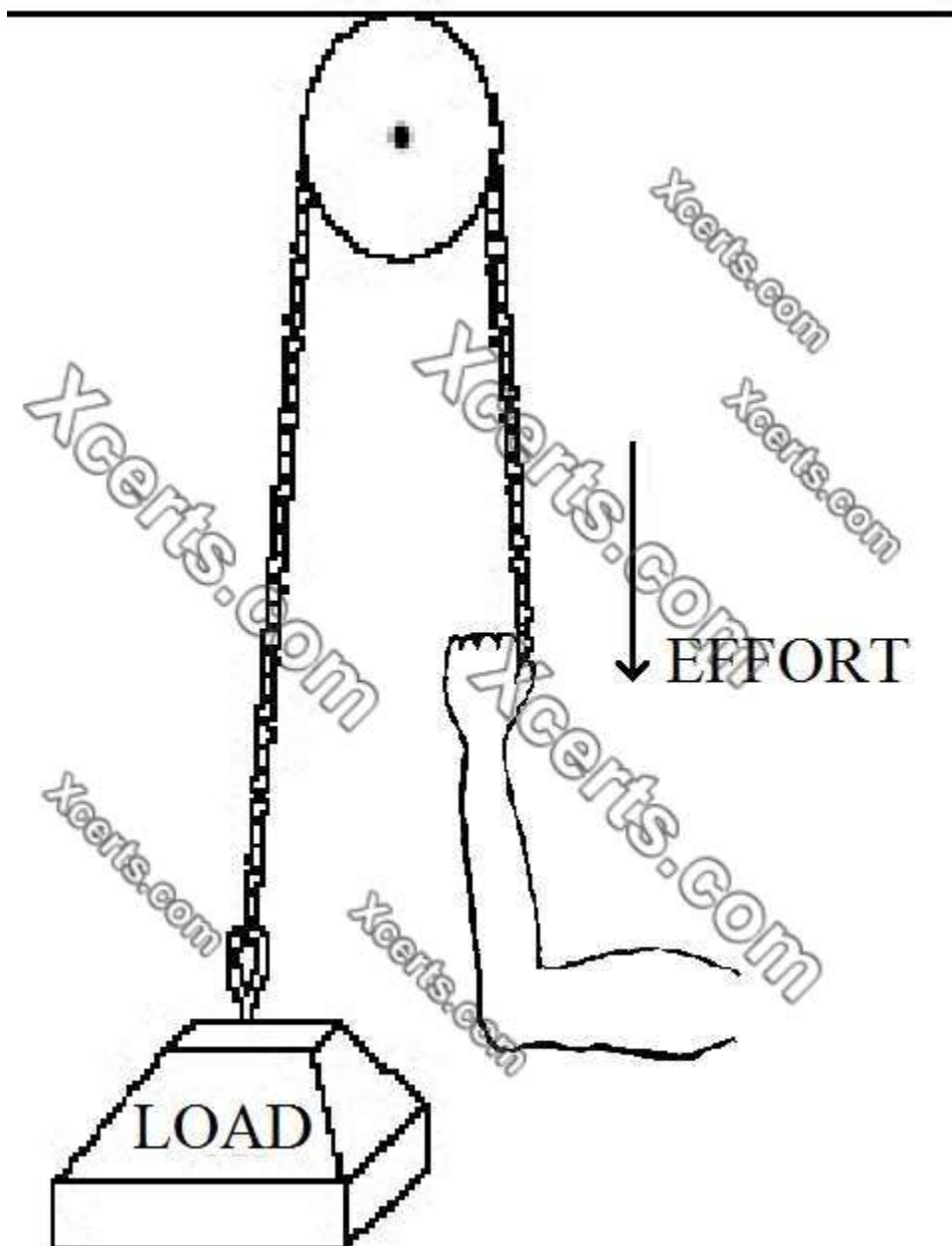
E stands for effort needed. 30 (weight of the cat) $\times 2$ (length of resistance arm) $= x \times 6$ (length of effort arm).

Do a little multiplication, and you get $60 = 6x$.

To isolate x , divide each side by 6 :

$60 \div 6 = 6x \div 6$, or $10 = x$.

QUESTION: 1612



What mechanical advantage does the block and tackle arrangement in the above figure give?

- A. 1
- B. 3
- C. 2
- D. 4

Answer(s): A

Explanation:

A fixed, simple pulley gives no mechanical advantage, so its mechanical-advantage number is 1.

QUESTION: 1613

If a ramp is 8-feet long and 4-feet high, how much effort is required to move a 400-pound object up the ramp?

- A. 35 pounds
- B. 150 pounds
- C. 800 pounds
- D. 200 pounds

Answer(s): D

Explanation:

The formula to determine the mechanical advantage of an incline plane is Length of Ramp ÷ Height of Ramp = Weight of Object ÷ Effort.

$$8 \div 4 = 400 \div E$$

$$8/4 = 400/E$$

$$8E = 1,600$$

$$8E/8 = 1,600/8$$

$$E = 200$$

QUESTION: 1614

33,000 foot-pounds of work done in one minute is called _____.

- A. a job for an enlisted soldier
- B. one horsepower
- C. 330 psi
- D. meaningful force

Answer(s): B

Explanation:

Scientists agree that 33,000 foot-pounds per minute is one horsepower.

QUESTION: 1615

A 130-pound woman is wearing shoes with high heels that measure 1-inch square.

If the woman is standing on one heel, what psi does the heel exert as it rests on the ground? (Disregard atmospheric pressure from your calculations.)

- A. 130
- B. 65
- C. 260
- D. 11

Answer(s): A

Explanation:

Power = Force/Area.

$$P = 130/1 = 130.$$

QUESTION: 1616

The force that causes clothes from the dryer to stick together is called _____.

- A. gravity
- B. magnetism
- C. friction
- D. static electricity

Answer(s): D

Explanation:

Static electricity causes materials to "stick" together this way.

QUESTION: 1617

An aneroid barometer measures _____.

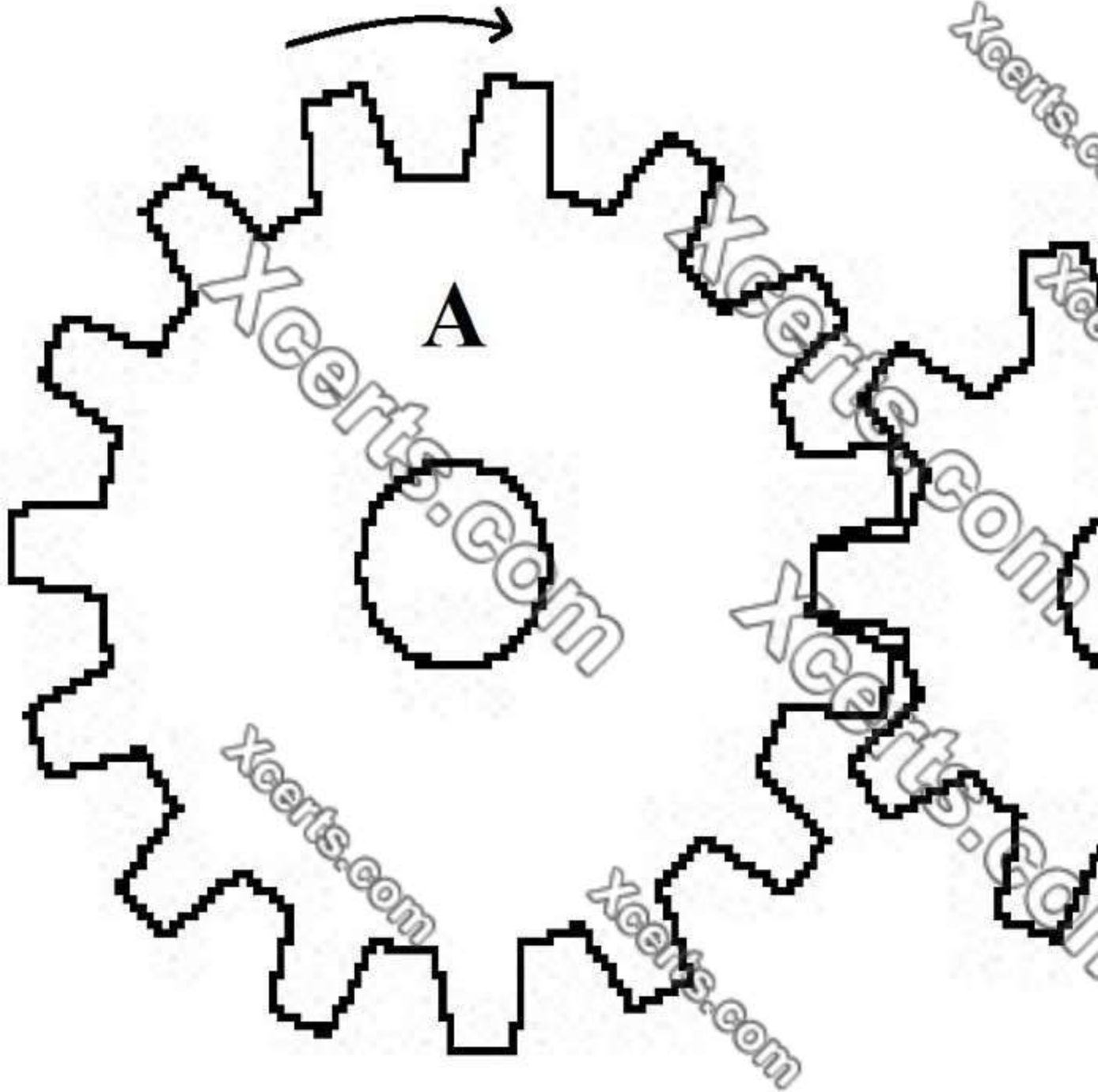
- A. atmospheric pressure
- B. water pressure
- C. hydraulic-fluid pressure
- D. the ambient temperature

Answer(s): A

Explanation:

An aneroid barometer measures atmospheric pressure.

QUESTION: 1618



If Gear A is revolving in a clockwise manner, as seen in the figure above, Gear B _____.

- A. remains stationary
- B. revolves in a clockwise manner
- C. revolves in a counterclockwise manner
- D. turns more slowly than Gear A

Answer(s): C

Explanation:

Meshed gears always turn in opposite directions.

QUESTION: 1619

Springs are used for the following purposes EXCEPT _____.

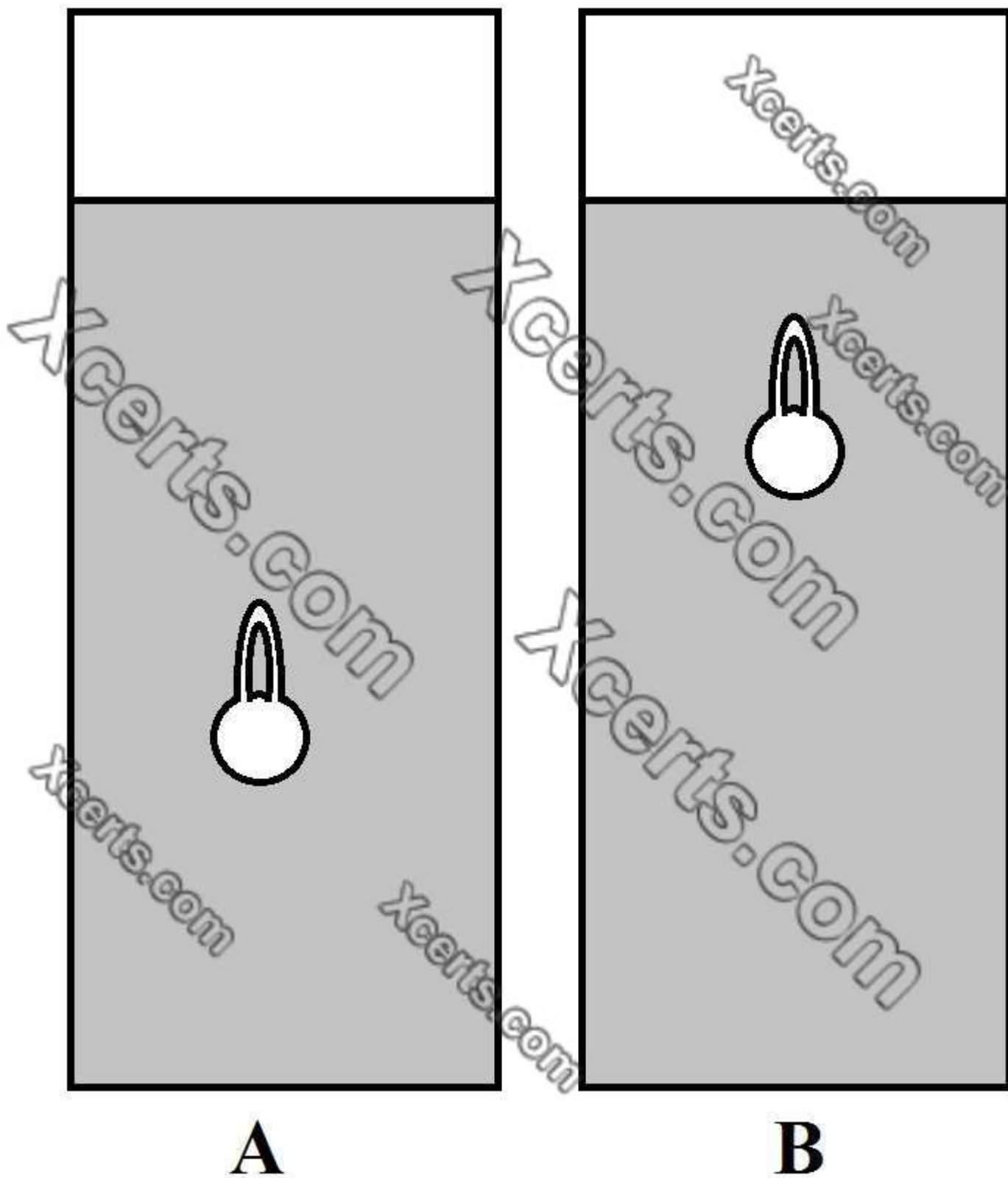
- A. to store energy for part of a mechanical cycle
- B. to force a mechanical component to maintain contact with another component
- C. to reduce shock or impact
- D. to increase the weight of a mechanism

Answer(s): D

Explanation:

Springs are used for all the listed purposes except to add weight.

QUESTION: 1620



The floats in Tubes A and B measure specific gravity.
Which tube contains the liquid with the higher specific gravity?

- A. Tube A
- B. Tube B
- C. It can't be determined.
- D. Both Tube A and Tube B have the same specific gravity.

Answer(s): B

Explanation:

Specific gravity is a comparison between the weight of a liquid and the weight of water.
The liquid with the higher specific gravity will have a float that rises higher.

QUESTION: 1621

Universal joints are used to _____.

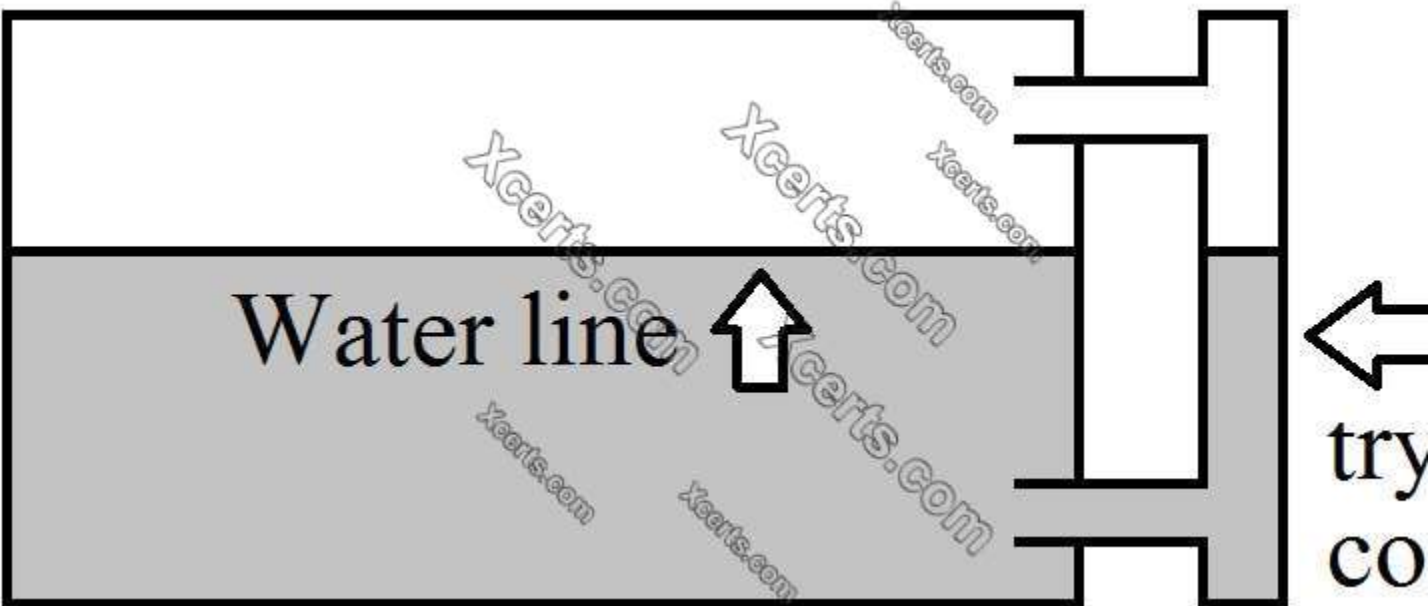
- A. connect ball bearings
- B. fix two shafts so they don't pivot or rotate
- C. connect shafts in a U-shape
- D. couple two shafts set at different angles

Answer(s): D

Explanation:

Universal joints are used to connect shafts that aren't in the same plane.

QUESTION: 1622



The try-cock in the schematic above measures _____.

- A. temperature of water
- B. pressure of water
- C. pressure of steam buildup
- D. level of water

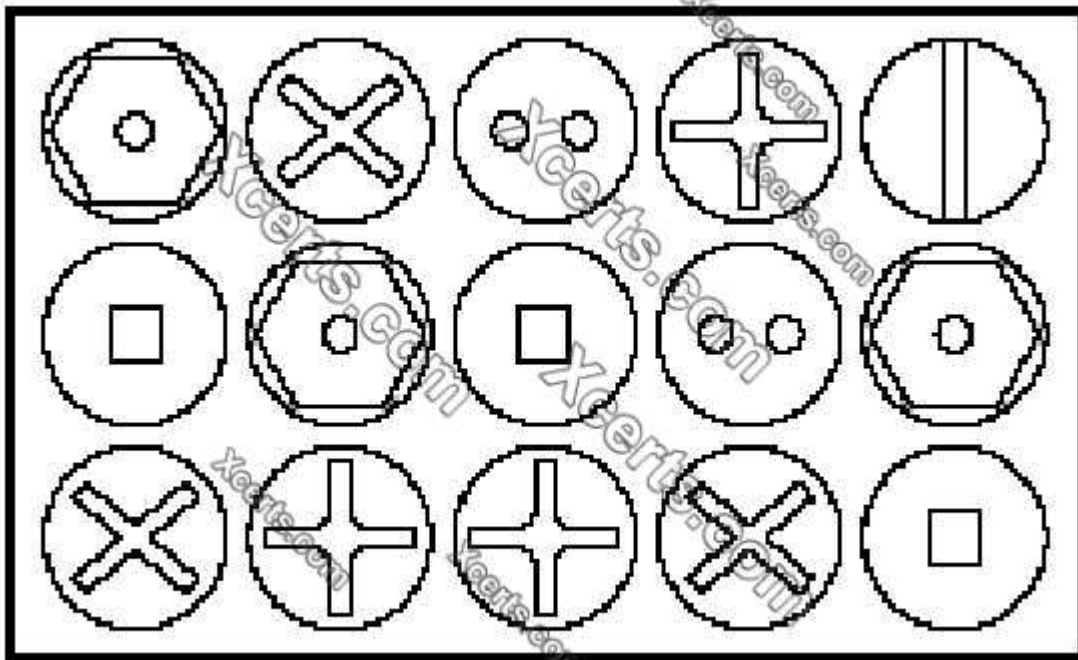
Answer(s): D

Explanation:

Try-cocks measure water level.

Water seeks a level throughout a system, so in the schematic, the try-cock correctly indicates the water level.

QUESTION: 1623



The steel plate above is held in place by different machine screws, each indicated by different symbols.

How many different types of machine screws have been used?

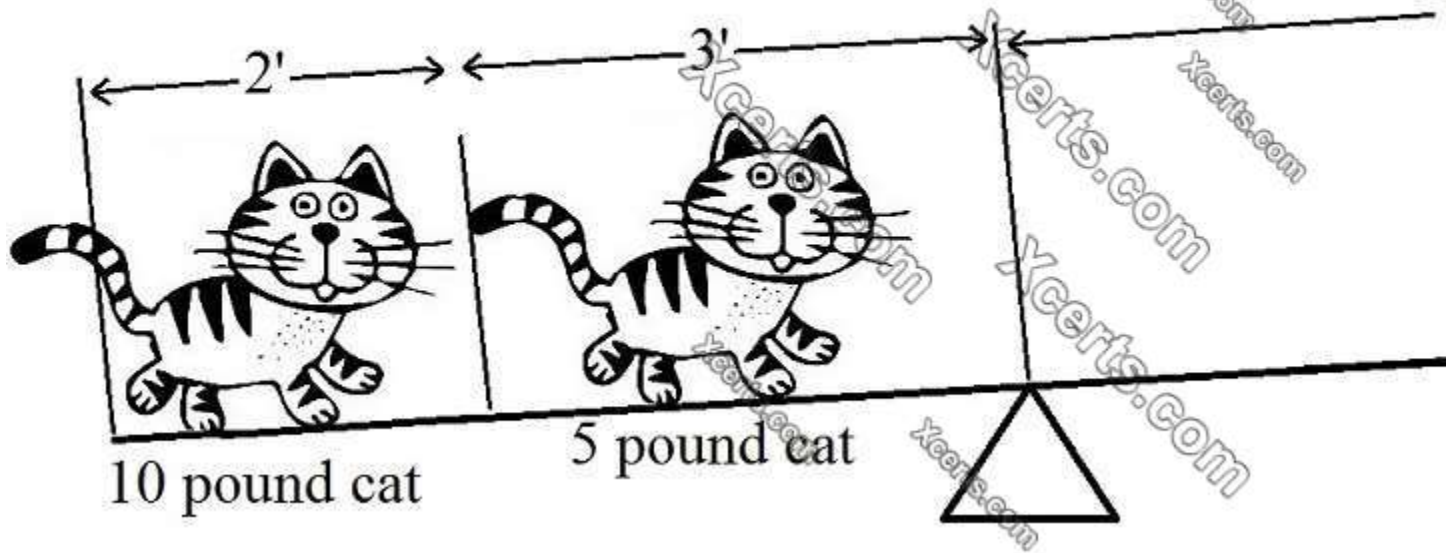
- A. 6
- B. 15
- C. 5
- D. 9

Answer(s): A

Explanation:

There are 6 different symbols, so 6 different types of machine screws were used.

QUESTION: 1624



The amount of force (F) needed to balance the lever in the figure above is most nearly

- A. 15 pounds
- B. 13 pounds
- C. 7.5 pounds
- D. 20 pounds

Answer(s): B

Explanation:

To determine the amount of force exerted by the cats, first multiply the length of the resistance arm (as it applies to the cat) by the weight of each cat and add the products together.

The 10-pound cat is supported by the entire weight of the resistance arm, so $5 \times 10 = 50$.

The 5-pound cat is being supported by 3 feet of the resistance arm, so $3 \times 5 = 15$.

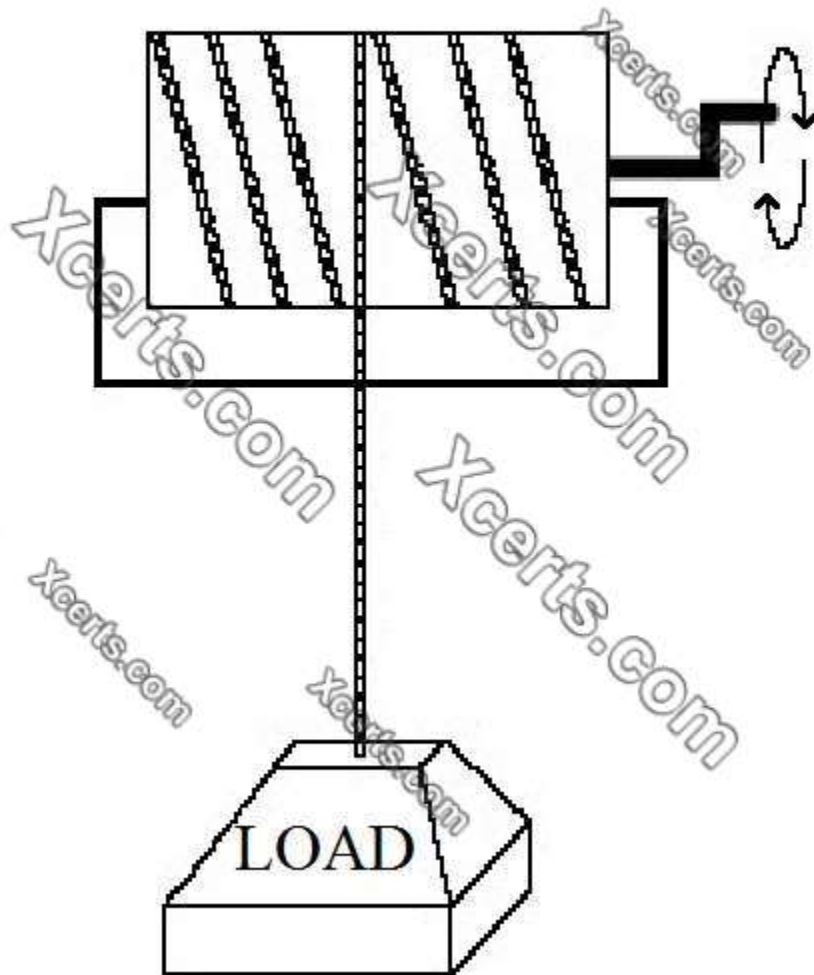
Add them up: $50 + 15 = 65$.

This number is equal to the length of the resistance arm times effort (force) or $65 = 5F$.

To isolate F, divide both sides by 5: $65 \div 5 = 5F \div 5$ or $13 = F$.

QUESTION: 1625

Drum circumference 24 inches



With one complete revolution of the cable winch shown above, the load will move _____.

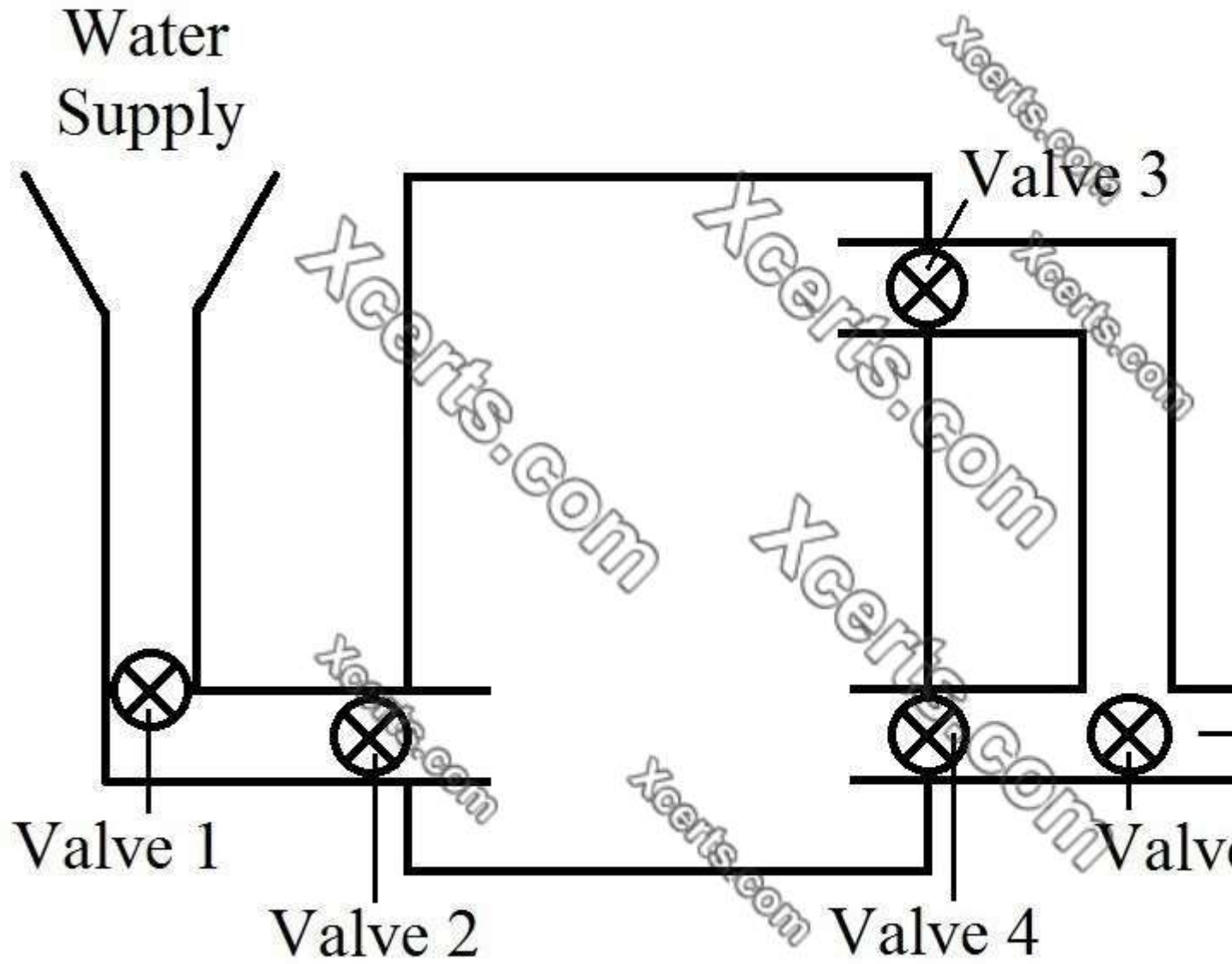
- A. 12 inches
- B. 6 inches
- C. 24 inches
- D. 36 inches

Answer(s): C

Explanation:

One revolution of the winch will move the weight 24 inches, the circumference of the winch drum.

QUESTION: 1626



In the figure above, assume the valves are all closed. Which valves need to be open to fill the tank entirely?

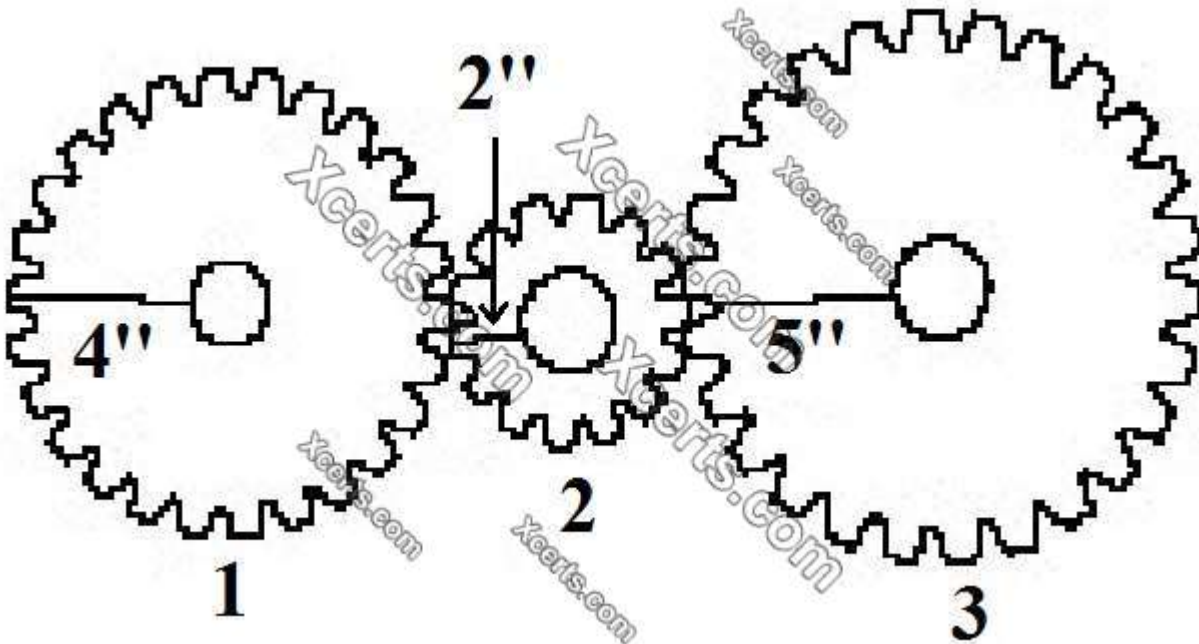
- A. 1 and 2 only
- B. 1 only
- C. 1, 2, and 3
- D. 2 only

Answer(s): A

Explanation:

Valves 1 and 2 need to be open to fill the tank.

QUESTION: 1627



If Gear 1 in the figure above makes 10 complete clockwise revolutions per minute, then _____.

- A. Gear 2 makes 2 clockwise revolutions per minute.
- B. Gear 3 makes 8 clockwise revolutions per minute.
- C. Gear 3 makes 30 clockwise revolutions per minute.
- D. Gear 3 makes 9 counterclockwise revolutions per minute.

Answer(s): B

Explanation:

Gear 1 makes 10 clockwise revolutions per minute.

Gear 2, which is half the size, makes 20 counterclockwise revolutions per minute. (The number of revolutions it makes is inversely proportional to its difference in size.)

Gear 2 is half the size of Gear 1, so to determine the number of revolutions it makes, multiply the number of revolutions Gear 1 makes by the inverse of $1/2$: $10 \times 2/1$ (or just 2) = 20.

Gear 3 is 2.5 times the size of Gear 2. In other words, it is $5/2$ the size of Gear 2.

To determine the number of revolutions Gear 3 makes, multiply the inverse of $5/2$ by the number of revolutions Gear 2 makes: $2/5 \times 20$.

This can be stated as $20/1$ (the number of revolutions Gear 2 makes per minute) $\times 2/5$ (the fraction of revolutions Gear 3 makes) = $40/5$ or 8 revolutions per minute.

QUESTION: 1628

A gear and pinion have a ratio of 4 to 1.

If the gear makes 200 revolutions per minute, the speed of the pinion is _____.

- A. 50 rpm
- B. 800 rpm
- C. 400 rpm
- D. 200 rpm

Answer(s): B

Explanation:

The pinion turns 4 times as often as the gear: $4 \times 200 = 800$.

QUESTION: 1629

The gas gauge in an auto relies on what mechanical device to measure the amount of gas in the tank?

- A. ball and cock
- B. automatic valve
- C. float
- D. mechanical switch

Answer(s): C

Explanation:

A float indicates the level of liquid in a container.

QUESTION: 1630

Using a runner gives you a mechanical advantage of _____.

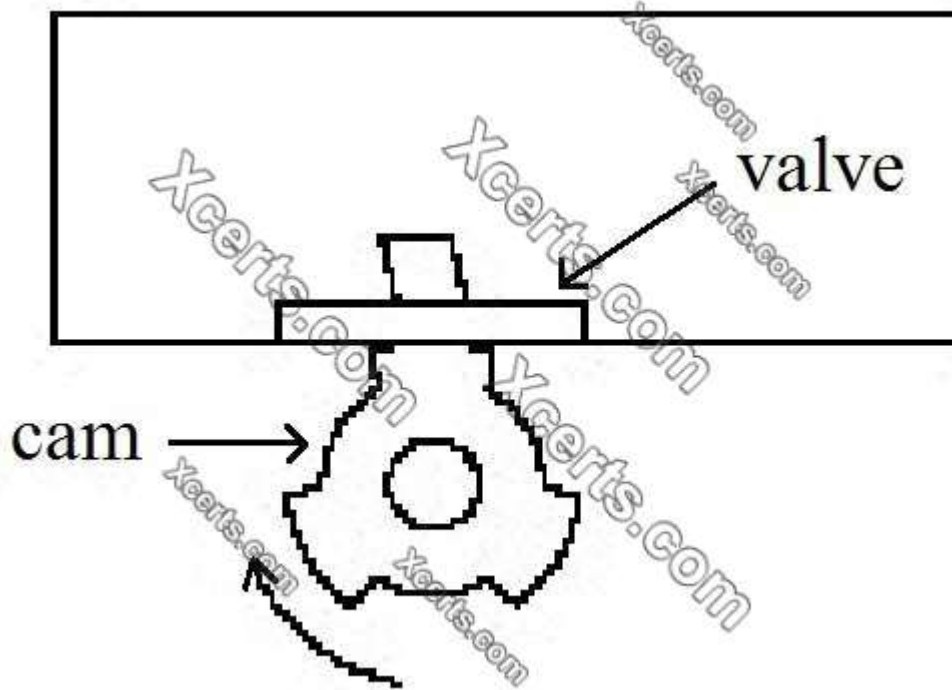
- A. 4
- B. 2
- C. 3
- D. 1

Answer(s): B

Explanation:

Using a runner (a single, moveable pulley) gives a mechanical advantage of 2.

QUESTION: 1631



For the valve shown in the figure above to open once each second, the cam must revolve at a rate of _____.

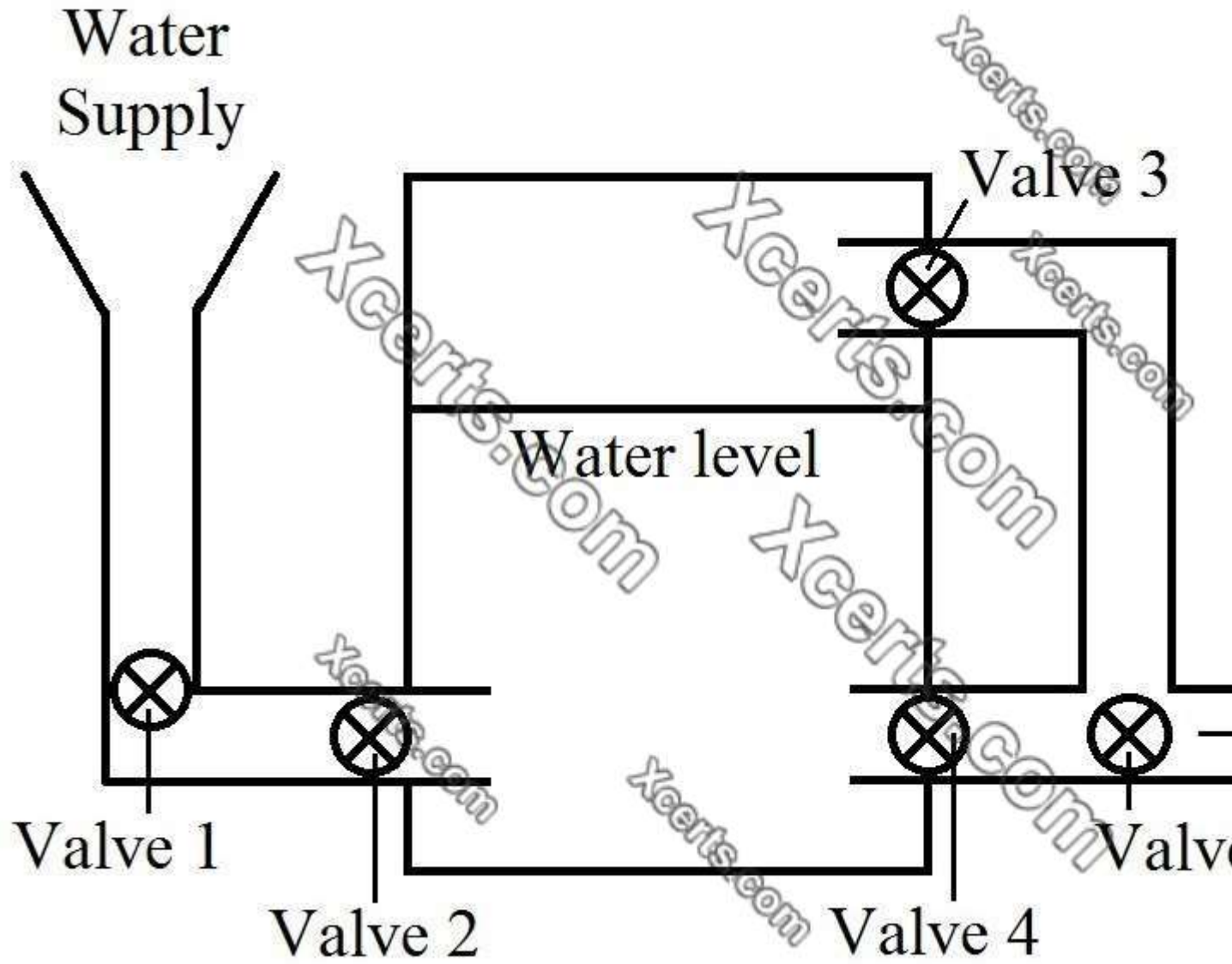
- A. 6 rpm
- B. 10 rpm
- C. 15 rpm
- D. 3 rpm

Answer(s): A

Explanation:

Because 60 seconds comprise a minute, the valve must open 60 times per minute. The cam will open the valve 10 times per revolution, so $60 \div 10 = 6$. The cam must make 6 revolutions per minute to raise the valve 60 times per minute.

QUESTION: 1632



The figure above represents a water tank.
Which of the following statements is not true?

- A. If Valves 1 and 2 are open and Valves 3, 4, and 5 are closed, the tank will eventually overflow.
- B. If all valves are open, the water remains at a constant level as long as the rate of intake is equal to the rate of discharge.
- C. Water in the tank will rise if Valves 1 and 2 are open and Valves 3 and 4 are closed.
- D. The tank will empty entirely if Valves 1 and 2 are closed and Valves 4 and 5 are open.

Answer(s): D

Explanation:

Because Valve 4 is above the bottom of the tank, some water will remain in the tank below the level of the valve, so the tank will never be completely empty.

QUESTION: 1633

Heat is a form of _____.

- A. calories
- B. energy
- C. beneficence
- D. motion

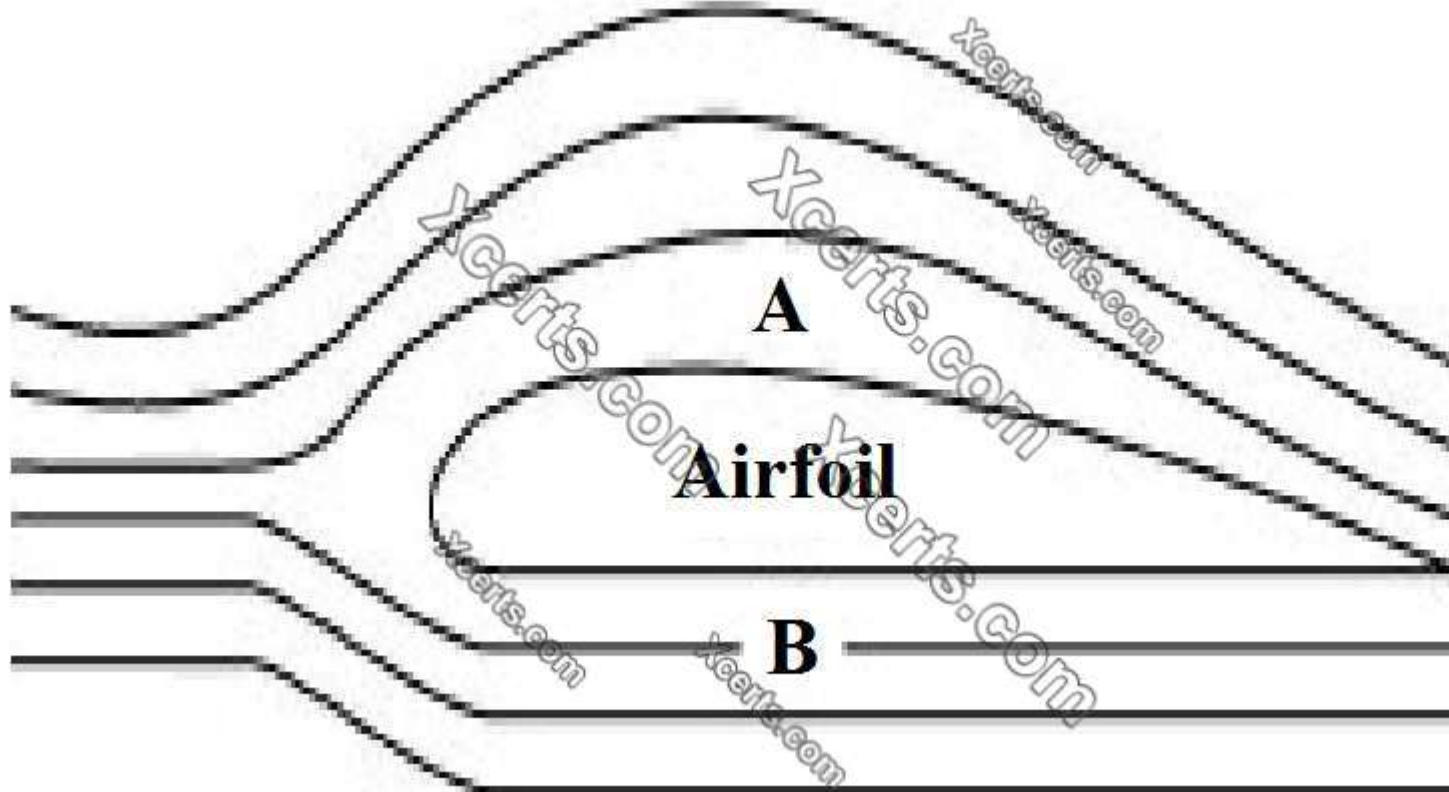
Answer(s): B

Explanation:

Heat is a form of energy - Calories measure heat, beneficence is an attitude or emotion, and motion is something that can cause heat.

QUESTION: 1634

The lift caused by an aircraft's wings is primarily due to _____.



- A. a decrease in pressure on the upper side of the wing (A)
- B. a decrease in pressure on the bottom side of the wing B
- C. a vacuum created under the wing at point B
- D. an increase in pressure on the upper side of the wing (A)

Answer(s): A

Explanation:

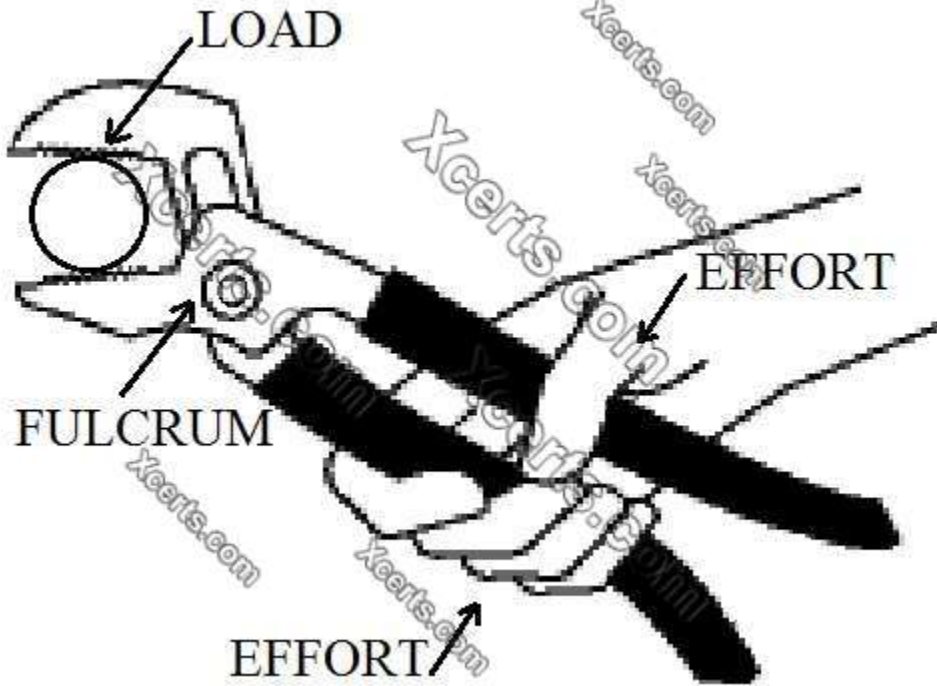
When oncoming air meets the leading edge of the airfoil (wing), some of it goes over the top of

the airfoil and some of it flows underneath.

The air flowing over the top of the wing has to go further than the air underneath, because it must join together again with the air that flowed under the wing at the far (back) side of the wing - physical laws act together to prevent or minimize vacuums in most cases.

QUESTION: 1635

Pliers are an example of a _____.



- A. first-class lever
- B. second-class lever
- C. third-class lever
- D. fourth-class lever

Answer(s): A

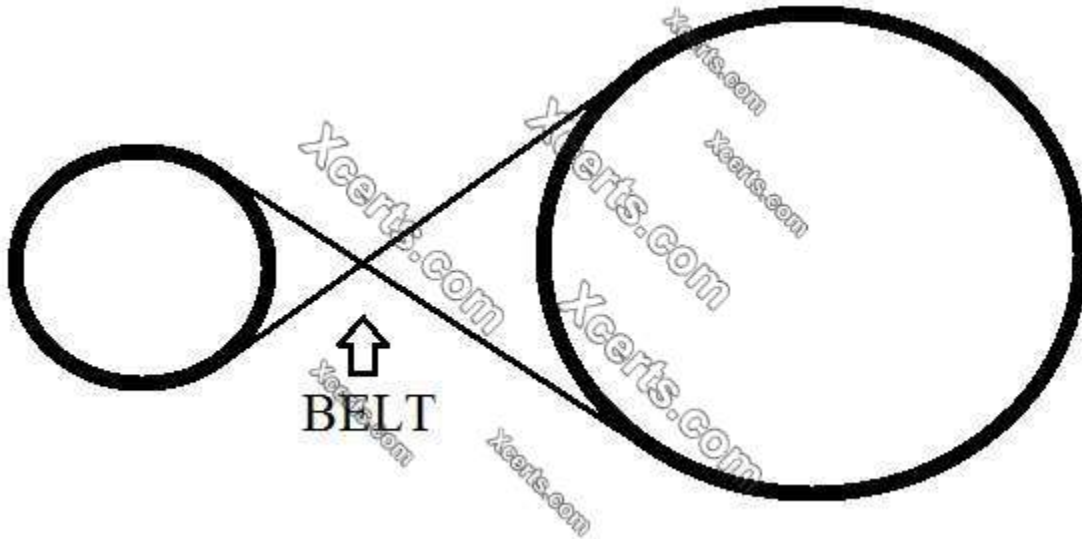
Explanation:

Pliers are an example of a first-class lever because the fulcrum is positioned between the effort (where your hand grips the pliers) and the load (the object that is being held or squeezed by the pliers).

QUESTION: 1636

The wheels below are connected by a belt as shown.

If the larger wheel makes two revolutions, how many revolutions will the smaller wheel make?



- A. Less than one
- B. One
- C. Two
- D. More than two

Answer(s): D

Explanation:

We are not told the sizes of the two wheels, but we can see that one is larger than the other. If the two wheels are connected by a belt, the small wheel will be forced to turn faster and complete more turns than the larger wheel.

QUESTION: 1638

If the wheel on the left turns clockwise, which way will the wheel on the right turn?

- A. Clockwise
- B. Counterclockwise
- C. Counter-counterclockwise
- D. Unable to tell from the information given

Answer(s): B

Explanation:

If the wheel on the left turns clockwise, the wheel on the right will turn counterclockwise.

QUESTION: 1639

How can a gear system derive the greatest amount of mechanical advantage when a seven-tooth gear drives a _____?

- A. 29-tooth gear
- B. 5-toothgear

- C. 24-tooth gear
- D. 17-tooth gear

Answer(s): A

Explanation:

The greater the difference between the number of teeth of the two meshed gears, the greater the torque or mechanical advantage.

QUESTION: 1640

An above-ground suburban water tank holds 540 gallons of water when full.

If the tank starts out full and is drained at a rate of 60 gallons per minute, how many seconds will it take to empty out halfway?

- A. 54
- B. 240
- C. 270
- D. 480

Answer(s): C

Explanation:

540 gallons divided by 60 gallons a minute means that it takes nine minutes to drain the tank completely.

Half of that time is 4.5 minutes; 4.5 multiplied by 60 seconds equals 270 seconds.

QUESTION: 1641

Which of the following is not a simple machine?

- A. Lever
- B. Inclined plane
- C. Axe
- D. Pulley

Answer(s): C

Explanation:

An axe is a compound machine: the handle is a lever, and the head is two inclined planes (each side of the blade edge).

QUESTION: 1642

If you want to hammer in a heavy nail, it might be a good idea to use a hammer with a _____.

- A. lighter head
- B. heavy head
- C. longer handle
- D. shorter handle

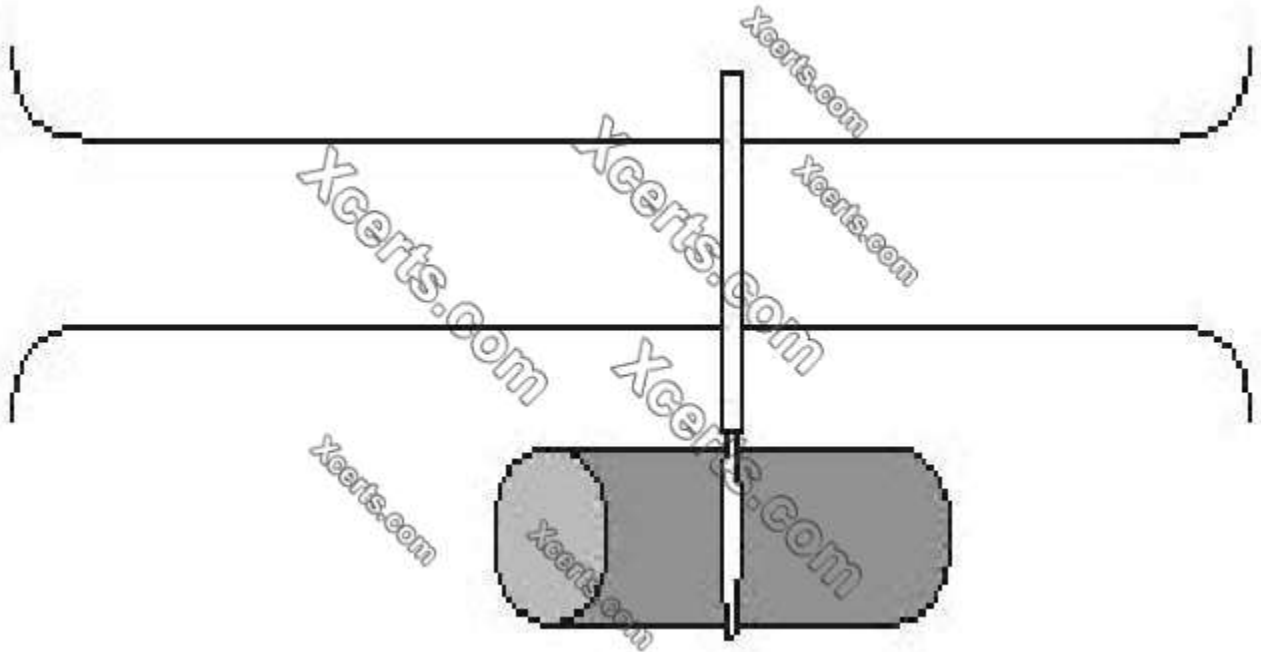
Answer(s): C

Explanation:

A longer handle - a longer lever - allows the hammerhead to reach a higher speed, thereby producing more striking force.

QUESTION: 1643

If you drag a log across the road, one way to reduce the amount of force needed to pull it would be to _____.



- A. use a thicker rope
- B. pull harder
- C. cross at an angle
- D. lubricate the road

Answer(s): D

Explanation:

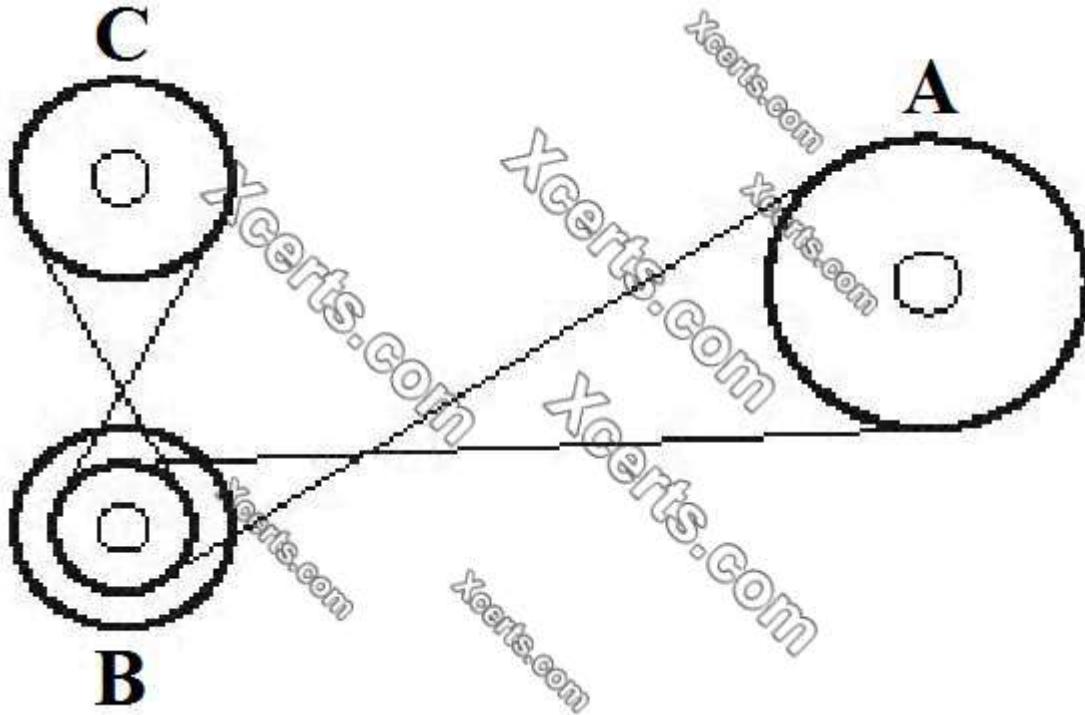
If you drag a log across the road, one way to reduce the amount of force needed to pull it would be to lubricate the road.

Two things contribute to the force needed to pull the log across the road: the weight (mass) of the log itself, and the friction between the log and the road.

Lubricating the road will reduce the friction and therefore reduce the force needed to pull the log across the road.

QUESTION: 1644

Presume that Pulley C is rotating clockwise.
What direction will Pulley A rotate?



- A. Clockwise
- B. Counterclockwise
- C. Insufficient information to answer
- D. Pulley A will not rotate

Answer(s): A

Explanation:

By rotating clockwise. Pulley C will cause Pulley B to rotate counterclockwise, which will in turn cause Pulley A to rotate clockwise.

QUESTION: 1645

The rate at which work is done is called _____.

- A. Power
- B. Effort
- C. Watts
- D. Horsepower

Answer(s): A

Explanation:

The English unit of power is foot pound per second. It takes 550 foot pounds to make 1 horsepower. The SI unit is the joule per second or watt. A horsepower is 746 watts. The rate at which work is done is called power. Power is work done per unit of time.

QUESTION: 1646

Forces existing in pairs are sometimes called _____.

- A. action and interaction
- B. reaction and interaction
- C. action and reaction
- D. friction and gravity

Answer(s): C

Explanation:

A force cannot be exerted unless there is something there to push back. Forces exist in pairs. For every action there is a reaction.

QUESTION: 1647

When forces act in pairs they are _____.

- A. equal in magnitude and opposite in direction
- B. equal in magnitude and in the same direction
- C. equal in magnitude
- D. unequal in magnitude

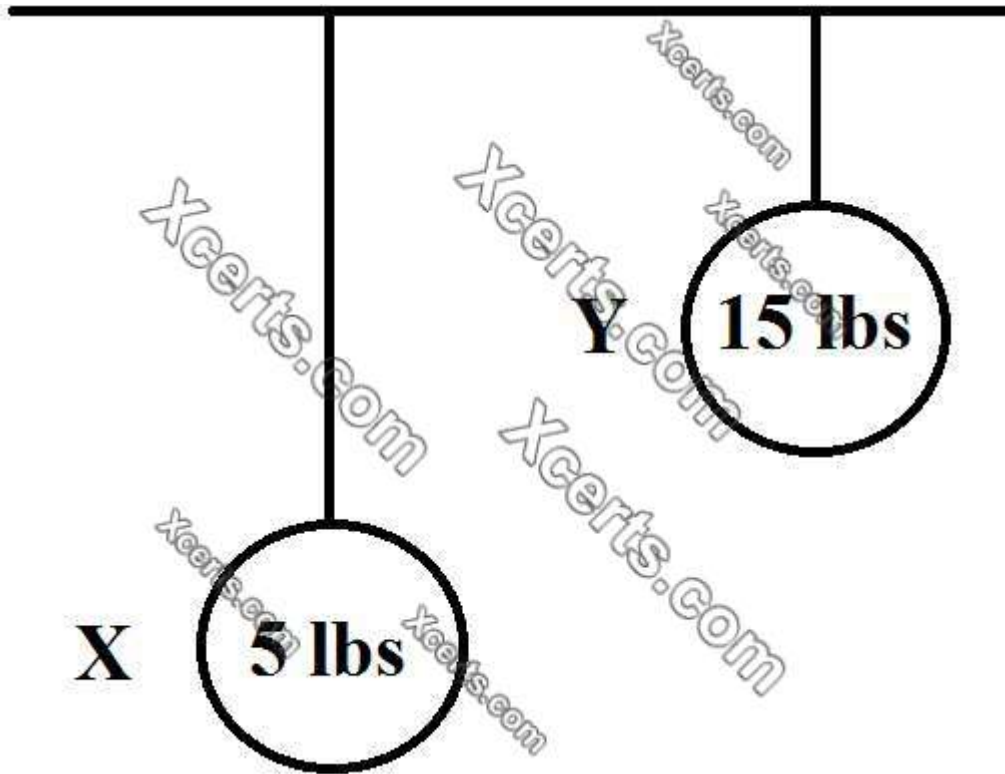
Answer(s): A

Explanation:

Since it is impossible to have one force unless it acts as one of a pair of forces, it means that these forces must be equal in magnitude and opposite in direction.

QUESTION: 1648

Which pendulum takes less time to make one complete back-and-forth swing?



- A. X
- B. Y
- C. Both take the same amount of time.
- D. There is not enough information to calculate the answer.

Answer(s): B

Explanation:

The length of time it takes for one all back and forth swing depends on the length of the string, not the weight at the end of it.

QUESTION: 1649

When you heat water that is confined in a closed container so that the steam cannot escape, the pressure inside the container _____ and the temperature of the boiling water _____.

- A. increases, decreases
- B. increases, increases
- C. decreases, decreases
- D. decreases, increases

Answer(s): B

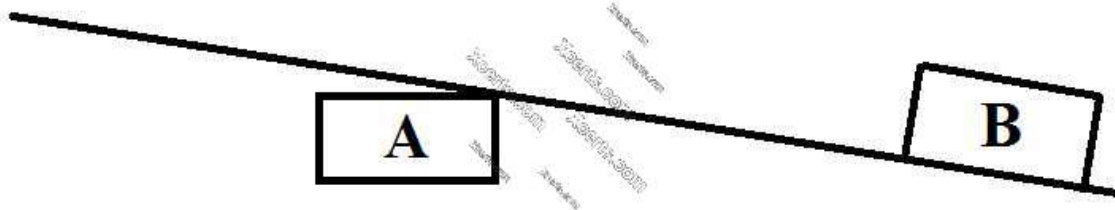
Explanation:

Heating an enclosed container of boiling water increases the pressure of the water vapor (steam) inside the

container and increases the temperature of the water.

QUESTION: 1650

If Block A, on which the lever is resting, is moved farther away from Block B _____.



- A. it will be harder to lift Block B but it will go higher
- B. it will be easier to lift Block B and it will go higher
- C. it will be harder to lift Block B and it will not be lifted to the same height
- D. it will be easier to lift Block B but it will not be lifted as high

Answer(s): A

Explanation:

It will be harder to lift Block B but it will go higher.

If Block A under the beam serving as the lever is moved away from Block B on top of the lever beam, the moment for the force exerted will increase, since the fulcrum is now closer to the force; this will make Block B harder to lift.

However, the height to which Block B on the end of the lever beam can be raised increases the further away Block A is moved from Block B.

QUESTION: 1651

For two gears to mesh properly, they must _____.

- A. rotate at the same speed
- B. rotate in the same direction
- C. have the same size teeth
- D. have the same number of teeth

Answer(s): C

Explanation:

For two gears to mesh properly, they must have the same size teeth.

Different size gears will, by definition, rotate at different speeds even when their driving axles are moving at the same speed.

Also, interlocking or meshing gears rotate in opposite directions.

QUESTION: 1652

Kinetic energy is _____.

- A. energy stored in chemicals, such as in a battery.

- B. energy in moving electrons, as in an electric current.
- C. energy in a moving object.
- D. energy that can be released under certain conditions.

Answer(s): C

Explanation:

Kinetic energy is energy in a moving object.

Energy stored in chemicals, such as in a battery, is chemical energy; energy in moving electrons, as in an electric current, is electric energy; and energy that can be released under certain conditions is potential energy.

QUESTION: 1653

The amount that a machine (simple or complex) multiplies your force or effort is called _____.

- A. load distance
- B. mechanical advantage
- C. resistance effect
- D. acceleration

Answer(s): B

Explanation:

The amount that a machine (simple or complex) multiplies your force or effort is called mechanical advantage.

Load distance is how far the load moves when force is applied, and acceleration is the change in velocity or speed in a certain direction.

QUESTION: 1654

The mechanical advantage of two meshing gears is calculated by dividing _____.

- A. the number of teeth on the driving gear by the number of teeth on the driven gear.
- B. the number of teeth on the driven gear by the number of teeth on the driving gear.
- C. the number of teeth on the driving gear by the rotational speed.
- D. the diameter of the driven gear by the number of teeth on the driving gear.

Answer(s): B

Explanation:

The mechanical advantage of two meshing gears is calculated by dividing the number of teeth on the driven gear by the number of teeth on the driving gear.

QUESTION: 1655

A compound machine is one in which _____.

- A. a battery provides the force
- B. one simple machine rotates around the second one

- C. two or more simple machines work together
- D. the mechanical advantages can be made to balance each other out

Answer(s): C

Explanation:

A compound machine is one in which two or more simple machines work together.

QUESTION: 1656

When meshing or interlocking gears have different numbers of teeth, _____.

- A. relative speeds depend on the speed of the drive aide for both
- B. both gears will rotate at the same speed
- C. the gear with more teeth will rotate fester than the gear with fewer teeth
- D. the gear with fewer teeth will rotate fester than the gear with more teeth

Answer(s): D

Explanation:

When meshing or interlocking gears have different numbers of teeth, the gear with fewer teeth will rotate faster than the gear with more teeth.

If two gears are meshed, both cannot be driven or something will break - one must be driving and one must be driven.

QUESTION: 1657

Cranks are used to _____.

- A. convert motion in a circle into motion in a straight line
- B. convert motion in a straight line into motion in a circle
- C. attach to rotating shafts to push separate pieces in a given direction
- D. transfer power between rotating shafts of different diameters

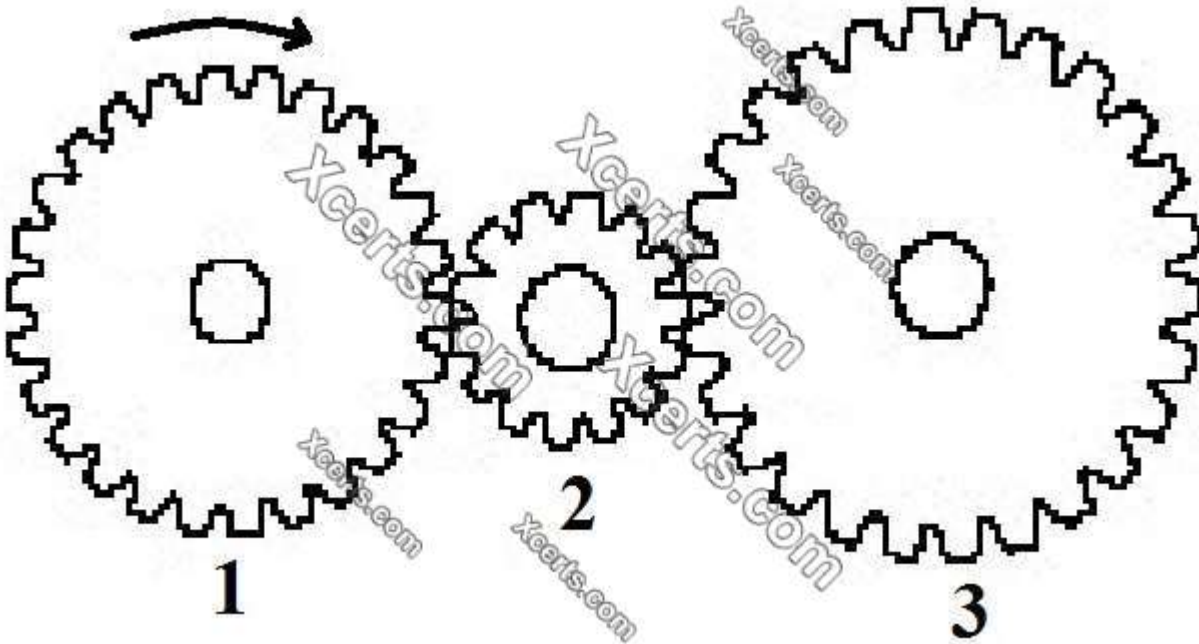
Answer(s): B

Explanation:

Cranks are used to convert motion in a straight line into motion in a circle, such as in a bicycle's pedals and pistons in an automotive engine.

QUESTION: 1658

Which of the other gears is moving in the opposite direction from Gear 1?



- A. Gear 2
- B. Gear 3
- C. Both Gears 2 and 3
- D. Neither Gear 2 or 3

Answer(s): A

Explanation:

Meshed gears turn in opposite directions.

Gear 1 is turning clockwise; therefore.

Gear 2 is turning counterclockwise (the opposite direction from Gear 1) and Gear 3 is turning clockwise (the same direction as Gear 1).

QUESTION: 1659

Force is something that can change the velocity of an object by making it _____.

- A. start or stop
- B. speedup
- C. slow down
- D. all of the above

Answer(s): D

Explanation:

Force is something that can change the velocity of an object by making it start or stop, speed up, or slow down.

This means that all the answers given are correct.

QUESTION: 1660

The brakes on your car use the same force that stops your car if you just let it coast. This force is called _____.

- A. velocity
- B. gravity
- C. friction
- D. newton

Answer(s): C

Explanation:

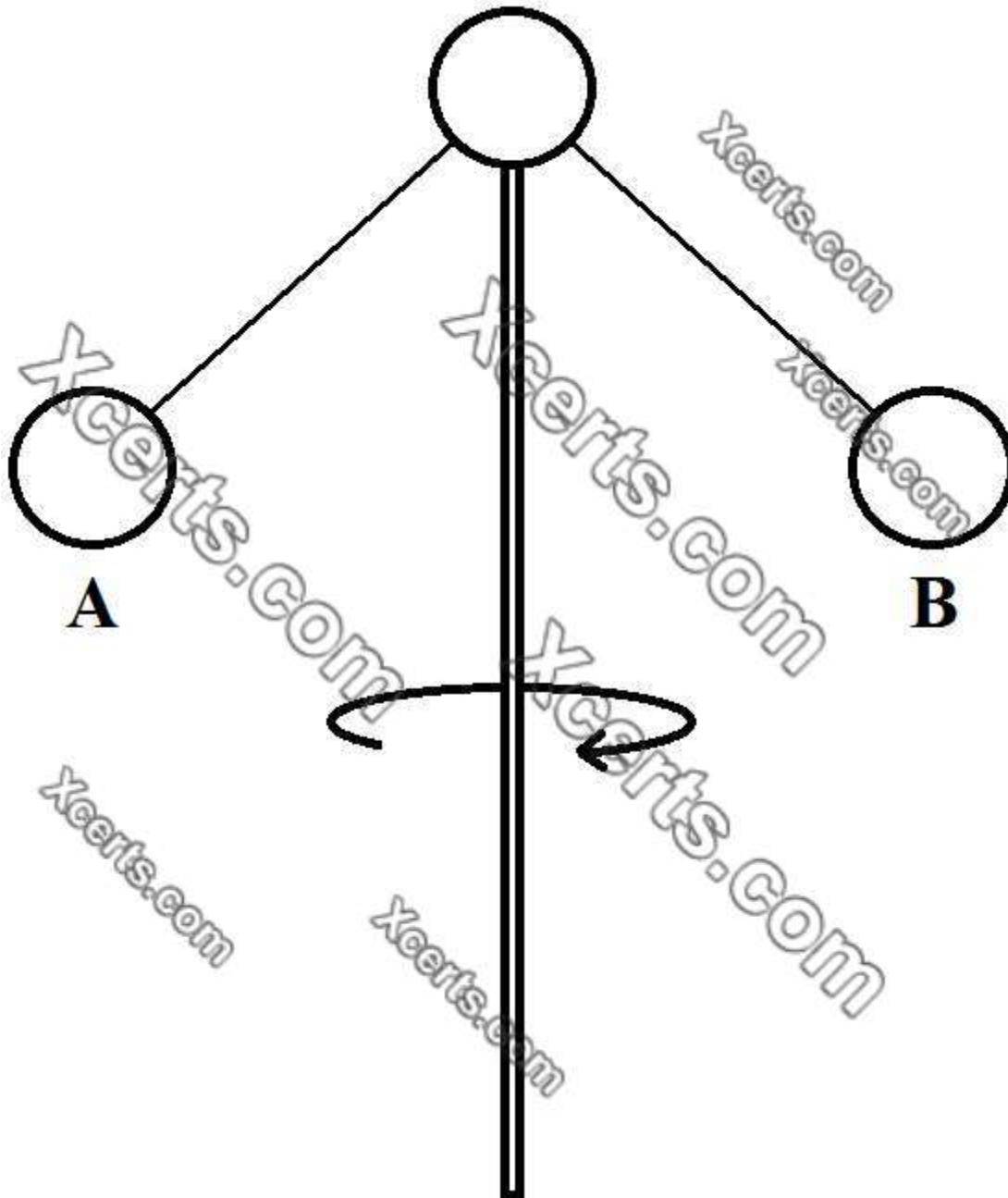
Friction is used to stop a car.

The brakes rub against the wheel drum or the disc to cause the car to stop.

This rubbing is creating friction.

QUESTION: 1661

As the central shaft in the illustration below spins faster in a clockwise direction, the balls labeled A and B will _____.



- A. move outward and downward
- B. move outward and upward
- C. move up
- D. move down

Answer(s): B

Explanation:

Centrifugal force from the spinning shaft, regardless of direction, will cause the balls to move outward, away from the shaft; the tension on the strings holding them will result in the balls moving upward.

QUESTION: 1662

If you drop something, the force acting on it is _____.

- A. gravity
- B. friction
- C. elastic recoil
- D. equilibrium

Answer(s): A

Explanation:

Gravity is the force that pulls things down whenever they are dropped.

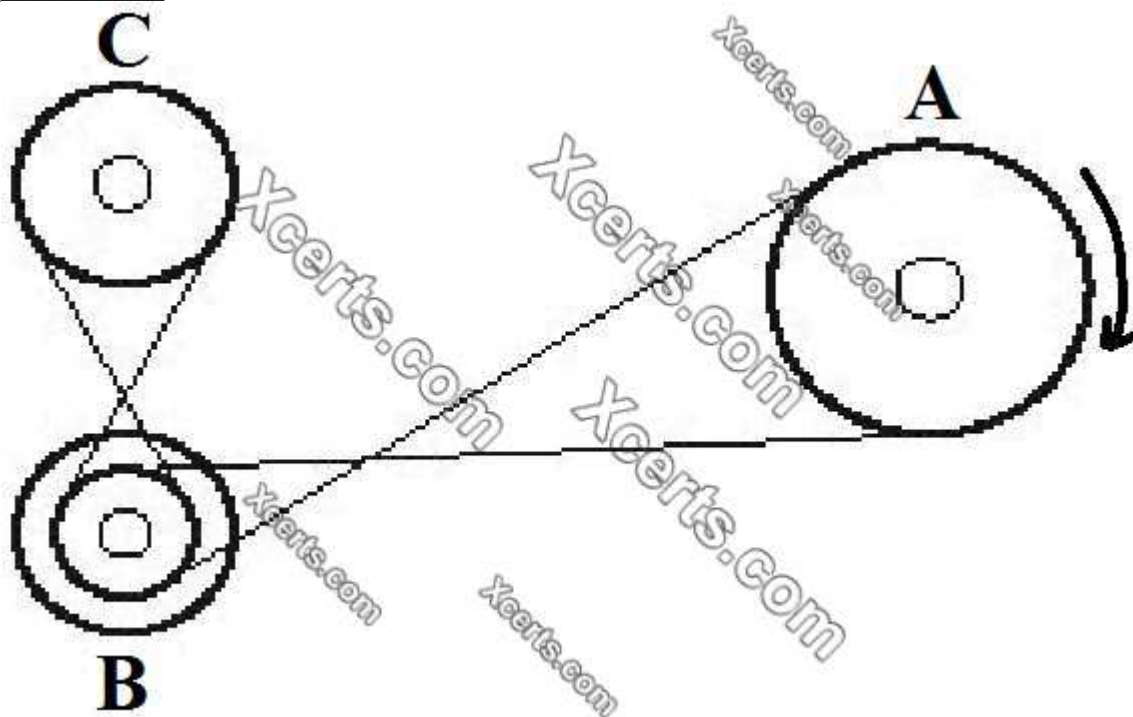
It is said that Isaac Newton "discovered" the concept of gravity as an identifiable force while sitting under an apple tree and the obvious happened.

Gravity is created by the mass of the earth.

The pull of gravity is much less (about one-sixth) on the moon than on earth.

QUESTION: 1663

In the illustration below, if Pulley A is rotating in the direction indicated, then Pulley C will _____.



- A. rotate in the same direction as Pulley B
- B. rotate in the same direction as Pulleys A and B
- C. rotate in the opposite direction of Pulleys A and B
- D. rotate in the same direction as Pulley A

Answer(s): D

Explanation:

Pulley C will rotate in the same direction as Pulley A.

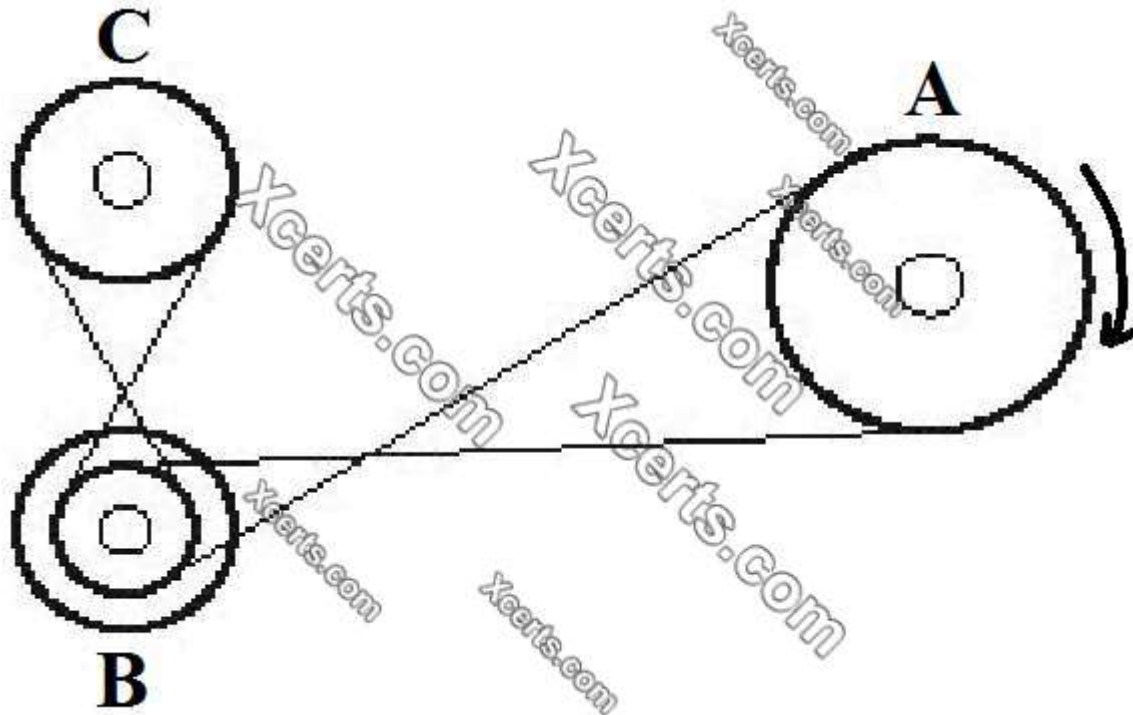
Pulley A causes Pulley B to rotate in the opposite direction from Pulley A, and Pulley B causes Pulley C to rotate in the opposite direction from Pulley B.

Since there are only two directions of rotation available.

Pulley C's direction of rotation is the same as Pulley A.

QUESTION: 1664

In the pulley system shown, which pulley will rotate the fastest?



- A. Pulley A
- B. Pulley B
- C. Pulley C
- D. They will all rotate at the same speed.

Answer(s): B

Explanation:

Pulley B will rotate the fastest. Just like meshing gears, the smaller the pulley in a system, the faster it rotates.

QUESTION: 1665

Weight is directly proportional to the acceleration due to _____.

- A. friction

- B. elastic recoil
- C. viscous drag
- D. gravity

Answer(s): D

Explanation:

Weight is not a fixed property of an object; it varies with location.
Your weight is slightly less at the North Pole than it is at the equator.
Your weight is much less on the moon than on earth.
Everything weighs less where the acceleration due to gravity is smaller.
Weight is directly proportional to local acceleration due to gravity.

QUESTION: 1666

The force that acts in an upward direction on anything submerged in a liquid or a gas is called _____.

- A. friction
- B. viscous drag
- C. gravity
- D. buoyancy

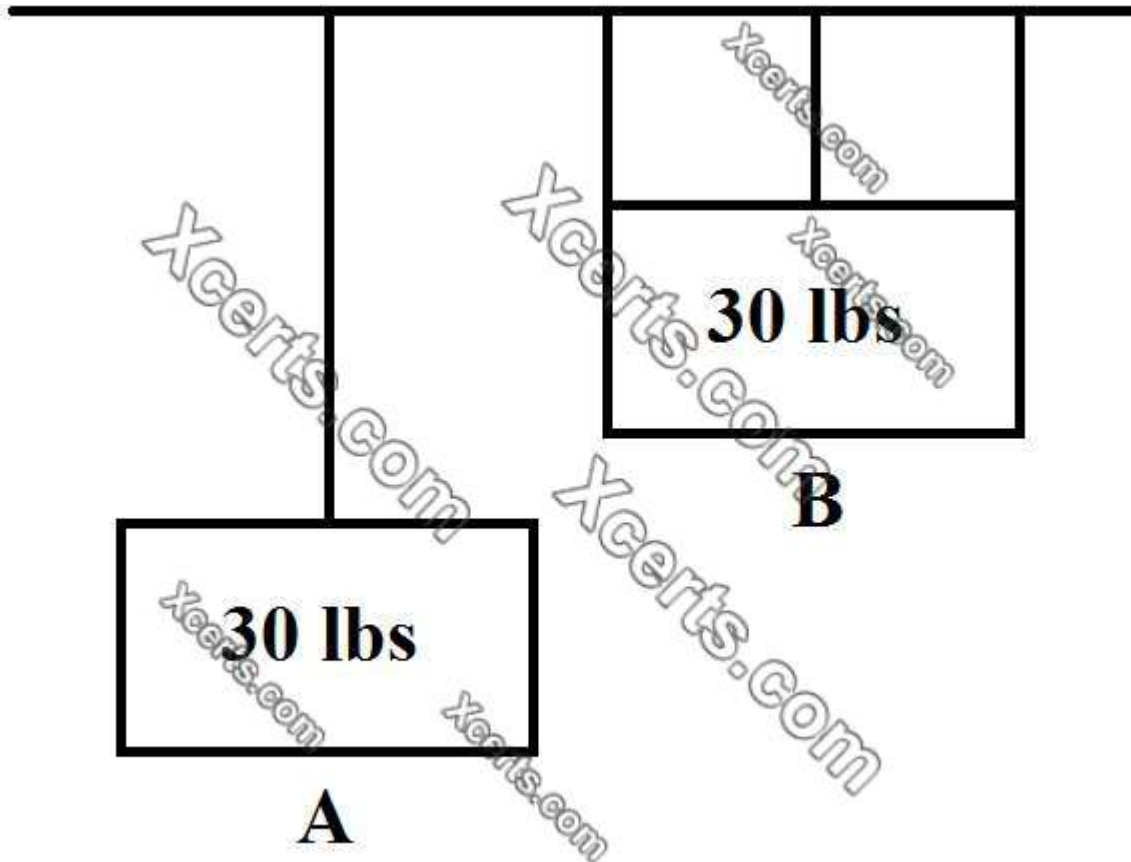
Answer(s): D

Explanation:

Buoyancy is the force that makes ships float and helium-filled balloons rise.
This means that the force that acts in an upward direction on anything submerged in a liquid or a gas is called buoyancy.

QUESTION: 1667

In the illustration below, two 30-lbs blocks are attached to the ceiling using ropes, as shown.
Which of the following statements is true?



- A. All the ropes are under equal amounts of tension.
- B. The rope supporting Block A is under 2x the tension of the ropes holding Block B.
- C. The ropes supporting Block B are under $\frac{1}{3}$ of the tension of the rope holding Block A.
- D. The rope holding Block A is under $\frac{1}{3}$ of the tension of the ropes holding Block B.

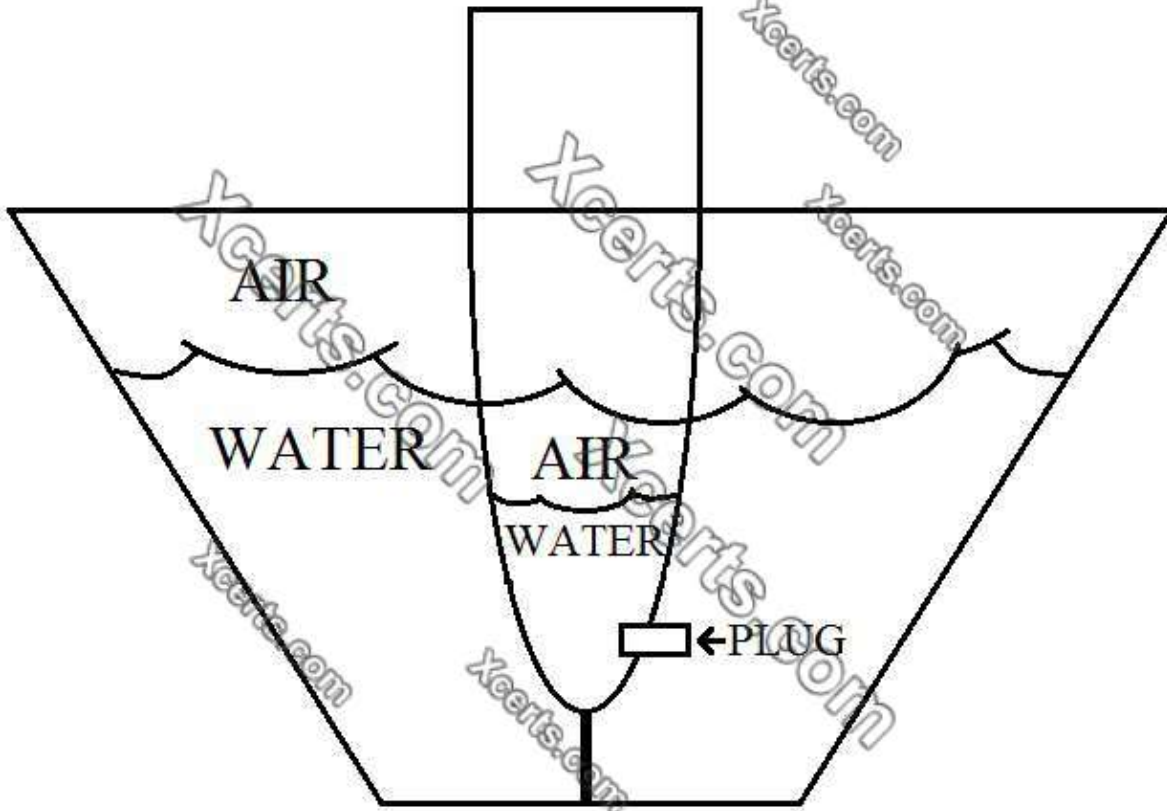
Answer(s): C

Explanation:

There are three ropes supporting Block B; therefore, they are under $\frac{1}{3}$ the tension as the rope supporting Block A.

QUESTION: 1668

When the plug in the tube is removed, water will flow _____.



- A. into the tube
- B. out of the tube
- C. neither direction
- D. impossible to tell

Answer(s): A

Explanation:

When the plug is removed, water will flow into the tube to equalize the water level both inside and outside the tube.

QUESTION: 1669

If an object is in equilibrium, it is said to be _____.

- A. at rest
- B. moving
- C. at rest or moving at a constant speed and in a straight line
- D. moving upward only

Answer(s): C

Explanation:

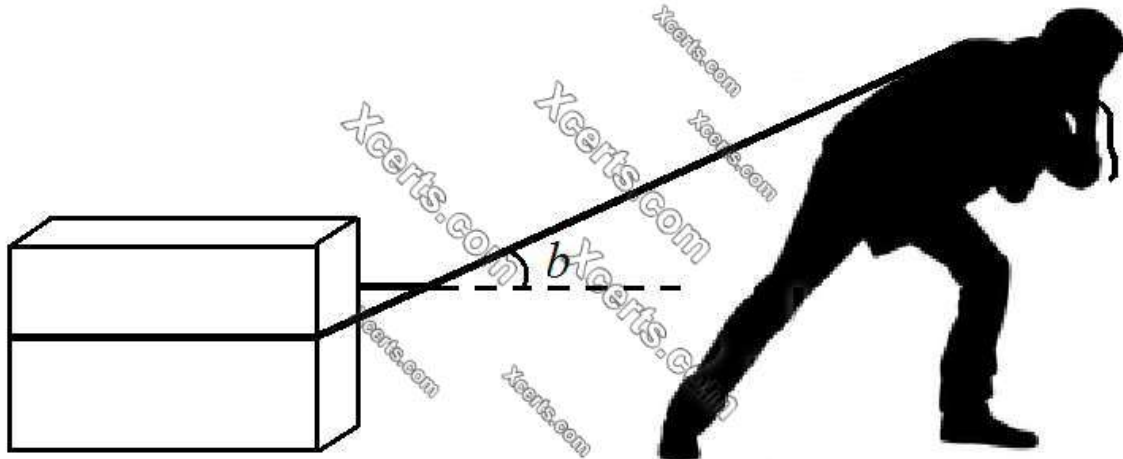
An object in equilibrium may or may not be at rest.

If two or more forces act, their effects may eliminate each other.

When this condition of equilibrium is reached, there is no net force and the velocity does not change.

Equilibrium is reached when the object is at rest or moving at a constant speed and in a straight line.

QUESTION: 1670



In the figure, the angle b is important, since when it is _____.

- A. 0 degrees the entire force is dragging the box
- B. 90 degrees the entire force is lifting the box
- C. both lifting and dragging between 10 degrees and 90 degrees
- D. all of the above

Answer(s): D

Explanation:

The angle of the rope determines if the box is being pulled along the floor or being lifted from the floor.

That means it can be both lifted and pulled along at any angle that is more than 0 degrees and less than 90 degrees.

QUESTION: 1671

A good example of a simple machine is a _____.

- A. manually operated pencil sharpener
- B. doorknob
- C. light switch
- D. all of the above

Answer(s): D

Explanation:

Simple machines are devices that make work easier.

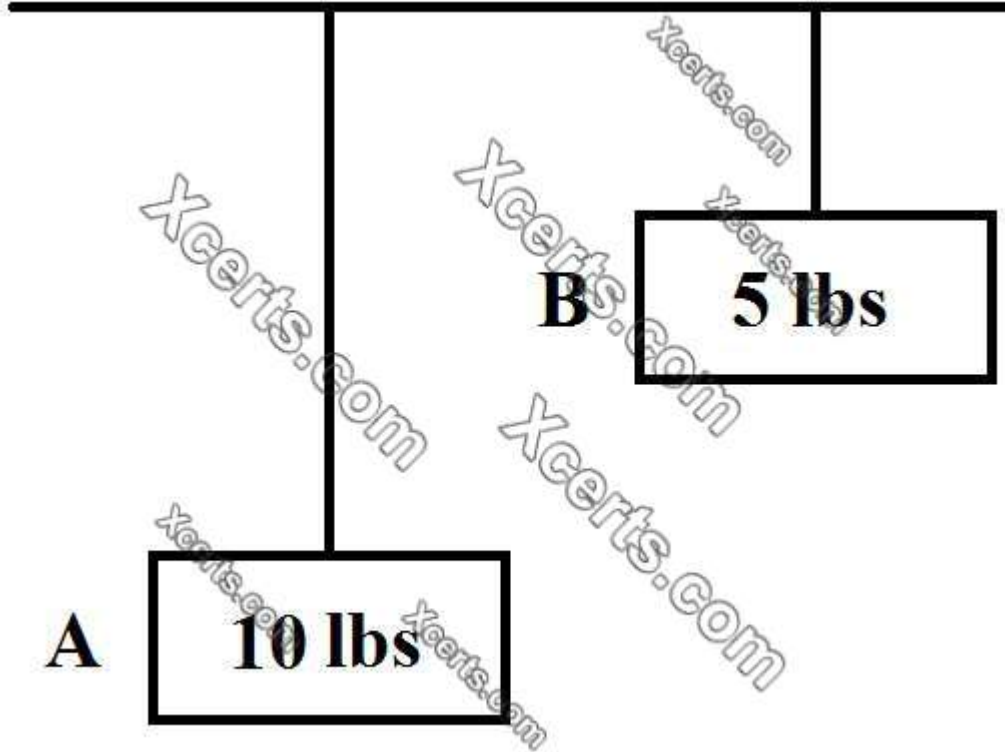
Some examples of simple machines are levers, hydraulic jacks, loading ramps, vises, and

machines that spin - winches and gears.

Door knobs, light switches, and manually operated pencil sharpeners are also simple machines.

QUESTION: 1672

Which weight exerts more pull on the horizontal bar from which both weights hang by strings as shown?



- A. Block A
- B. Block B
- C. Both exert the same pull
- D. Insufficient information to solve this problem

Answer(s): A

Explanation:

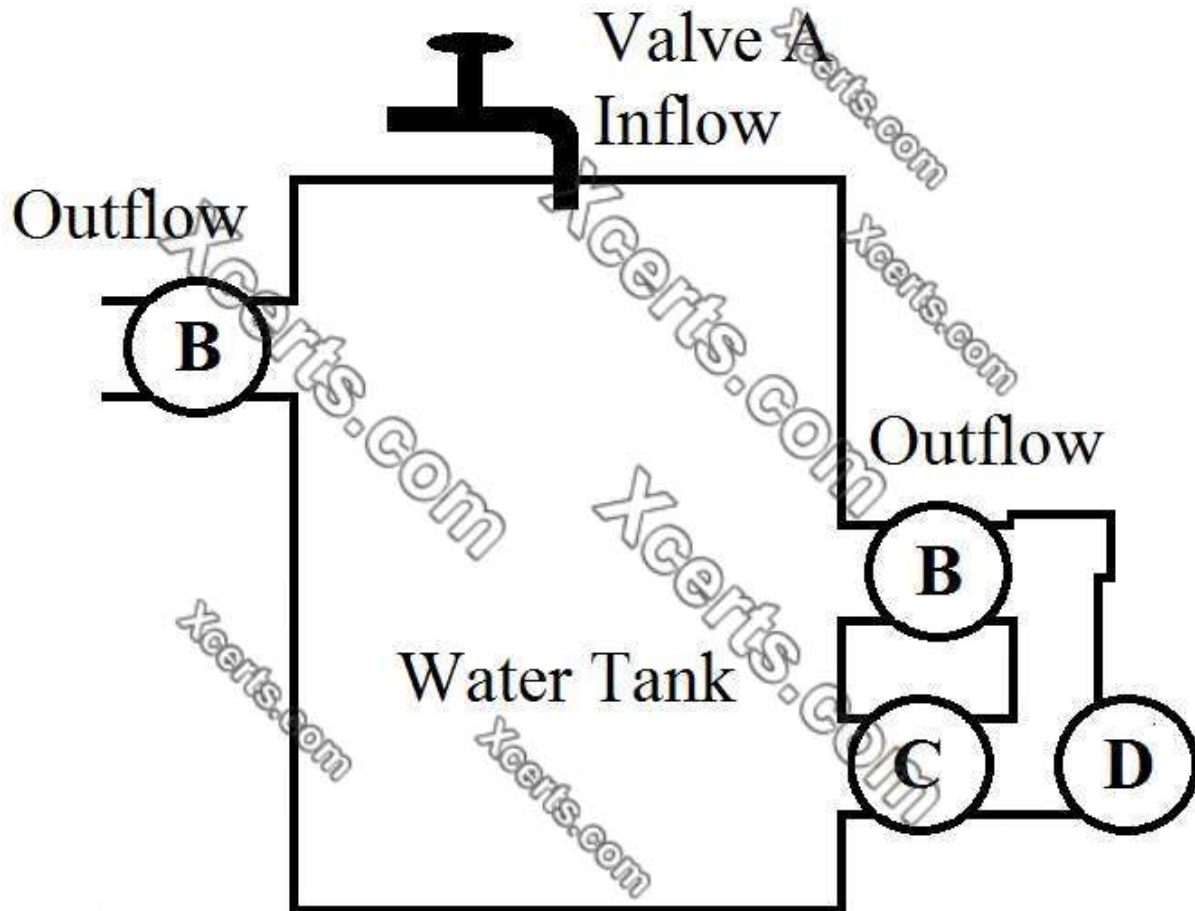
The string holding the 10-lb. weight exerts more pull on the bar.

The fact that that string is longer does not change how much tension is on it due to the weight at its end.

QUESTION: 1673

In the water system shown below, assume that the main tank starts empty and that all the valves are closed.

In order for the tank to fill about halfway and keep that level, which valves have to be open?



- A. Valves A, B, and C
- B. Valves A, B, and D
- C. Valves A, E, and B
- D. Valves A, E, and C

Answer(s): B

Explanation:

Valves A, B, and D must be open for the tank to fill halfway and maintain that level. Water flows in through Valve A (meaning that any choice not including Valve A is automatically incorrect), and flows out when the tank is half full through Valve B. However, for that water to leave the system. Valve D must also be open.

QUESTION: 1674

Work input is always more than work output because of _____.

- A. gravity
- B. buoyancy
- C. friction
- D. tension

Answer(s): C

Explanation:

Efficiency is affected by friction. It is a ratio of how much effort went into a job compared to what was produced or outputted.

Therefore, work input is more than the work output. Friction accounts for the loss in efficiency.

QUESTION: 1675

The ratio between work output and work input is called _____.

- A. efficiency
- B. work
- C. leverage
- D. mass

Answer(s): A

Explanation:

In order to find out how well work was performed and how much effort was expended in getting the job done, the term efficiency is used to describe the ratio of work input to work output.

QUESTION: 1676

The amount of magnification that a machine produces is referred to as the _____.

- A. effort arm
- B. work
- C. mechanical advantage
- D. efficiency

Answer(s): C

Explanation:

By using a machine, it is possible to multiply the amount of effort expended to get a job done. Mechanical advantage is the amount of magnification, or the ratio of load to effort.

QUESTION: 1677

A ramp is a device commonly used to aid in _____.

- A. lifting
- B. lowering weights
- C. high-efficiency situations
- D. loading

Answer(s): A

Explanation:

A ramp is a device commonly used to aid in lifting.
The ramp is an inclined plane.

The work output of an inclined plane is the work that would have to be done to lift the load directly.

The work input is the actual force exerted in pushing the load up the ramp times the length of the ramp.

An ideal mechanical advantage of an inclined plane is equal to its length divided by its height.

QUESTION: 1678

A screw consists of a single continuous spiral wrapped around a _____.

- A. piston
- B. cylinder
- C. helix
- D. square

Answer(s): B

Explanation:

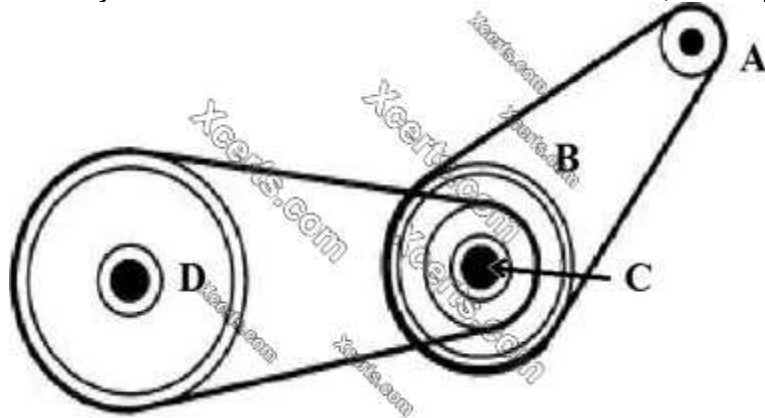
A screw is a simple device. It has been used for centuries as a means of lifting water out of wells and over the banks of irrigation canals.

A screw consists of a single continuous spiral wrapped around a cylinder.

The threads are cut into a rod in a spiral and make it possible to use the holding power of the screw in fastening objects together.

QUESTION: 1679

If Pulley A is the driver and turns counterclockwise, which pulley will turn the slowest?



- A. Pulley A
- B. Pulley B
- C. Pulley C
- D. Pulley D

Answer(s): D

Explanation:

The pulley that is the largest will turn most slowly.

QUESTION: 1680

The mechanical advantage of a winch is the ratio of the radius of the crank to the radius of the _____.

- A. arm
- B. lever
- C. fulcrum
- D. shaft

Answer(s): D

Explanation:

The principle of a winch is not much different from that of a lever.

Since the crank and the shaft turn together, the torque exerted by the effort (the force on the handle) must be equal to the torque exerted by the load (the tension in the rope).

The mechanical advantage then is the ratio of the radius of the crank to the radius of the shaft.

QUESTION: 1681

The ideal mechanical advantage of a gear is the ratio of the number of teeth in the load gear to the number of teeth in the _____.

- A. effort gear
- B. loaded gear
- C. lifting gear
- D. spiral gear

Answer(s): A

Explanation:

In mechanical devices, gears are used to change torque.

The ideal mechanical advantage of a gear is the ratio of the number of teeth in the load gear to the number of teeth in the effort gear.

QUESTION: 1682

Which mechanical components are typically used between a wheel and an axle to reduce friction?

- A. springs
- B. hinges
- C. bearings
- D. levers

Answer(s): C

Explanation:

A set of bearings is typically a set of small metal balls packed in a groove and lubricated with grease or oil.

The wheel rubs against one side of each ball, and the axle rubs against the other side.

The net effect is that the wheel rolls much more easily.

QUESTION: 1683

A block of steel has a density of 0.29 pounds per cubic inch.
If the block has dimensions of 1 inch by 1 inch by 3 inches, what is its weight?

- A. 0.29 pounds
- B. 0.58 pounds
- C. 0.87 pounds
- D. 1.0 pounds

Answer(s): C

Explanation:

The volume of the block can be calculated by multiplying its length by its width by its height, or 1 times 1 times 3, which equals 3 cubic inches.

The weight is the density multiplied by its volume, which is 3 cubic inches multiplied by 0.29 pounds per cubic inch, which equals 0.87 pounds.

QUESTION: 1684

Which principle of mechanical motion is used in the design of a roller coaster?

- A. momentum
- B. friction
- C. acceleration
- D. all of the above

Answer(s): D

Explanation:

Acceleration must be considered in designing the maximum rise of the first hill.

Momentum must be considered to ensure the train gets back to the starting point, since it has no motor.

Friction must be considered in the design of the braking system.

QUESTION: 1685

Which term below best describes the OPPOSITE of an increase in speed?

- A. rotation
- B. deceleration
- C. friction
- D. velocity

Answer(s): B

Explanation:

Deceleration means slowing down.

QUESTION: 1686

A valve is used to perform which of the following functions?

- A. increase the temperature of a liquid
- B. control the flow of a liquid
- C. aid evaporation of a liquid
- D. decrease the density of a liquid

Answer(s): B

Explanation:

Valves are placed in piping systems and can be opened or closed in order to control the flow of liquids or gasses.

QUESTION: 1687

The sprocket on a bicycle is most similar to which of the simple mechanical devices listed below?

- A. a gear
- B. a spring
- C. a lever
- D. all of the above

Answer(s): A

Explanation:

A bicycle sprocket is the classic example of a gear, which is defined as a toothed wheel or cylinder that meshes with a chain or with another toothed element to transfer energy or motion.

QUESTION: 1688

A pump is typically used to accomplish which of the following tasks?

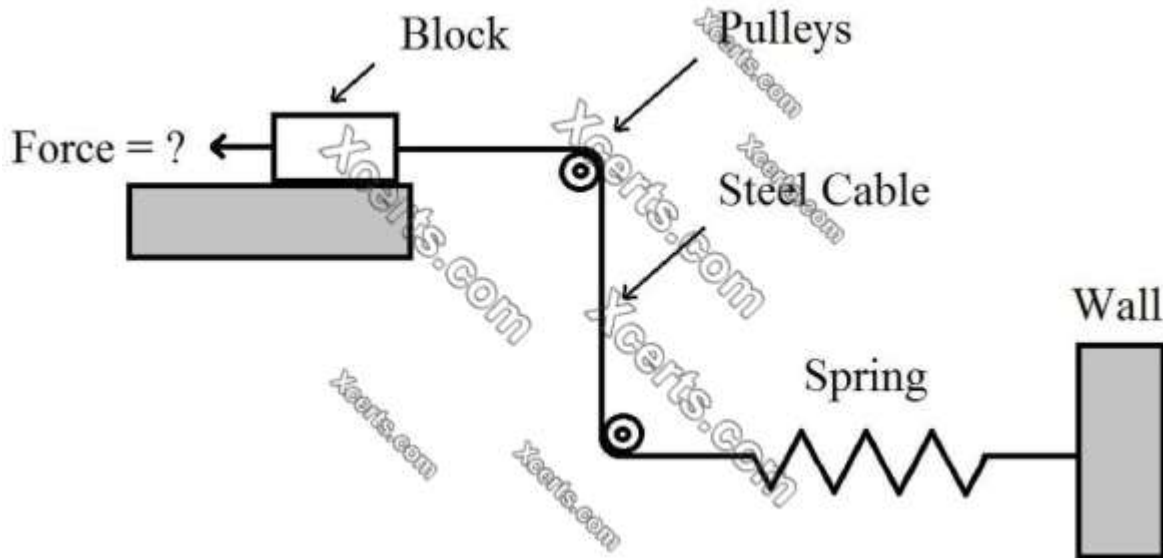
- A. to clarify liquids
- B. to separate liquids
- C. to move liquids downhill
- D. to move liquids uphill

Answer(s): D

Explanation:

Pumps are not used to move liquids downhill because liquids will flow downhill by gravity. Also, pumps do not separate or clarify liquids; instead, they agitate and mix liquids.

QUESTION: 1689



In the diagram shown above, the spring is very stiff and can be stretched 1 inch by a pulling force of 100 pounds.
How much force must be applied to the block in order to move the box 3.5 inches to the left?

- A. 150 pounds
- B. 250 pounds
- C. 350 pounds
- D. 450 pounds

Answer(s): C

Explanation:

3.5 inches multiplied by 100 pounds per inch equals 350 pounds.

QUESTION: 1690

A concrete beam has a maximum strength of 3,000 psi (pounds per square inch). In an experiment, a 500- pound weight is placed in the center of the beam, and the stress in the beam is measured to be 1,000 psi.

If the stresses in the beam continue to increase at the same rate with added weight, how much additional weight can be added to the same location on the beam before the beam will break?

- A. 500 pounds
- B. 1,000 pounds
- C. 1,500 pounds
- D. 3,000 pounds

Answer(s): B

Explanation:

The first 500 pounds generated 1,000 psi of stress in the beam; therefore, 500 more pounds will increase the total stress to 2,000 psi.

Another 500 pounds will increase the stress to 3,000 psi, which we are told is the maximum strength of the beam.

Therefore, the maximum additional load that can be applied to this beam before it breaks is 1,000 pounds.

QUESTION: 1691

An elevator is most similar to which of the following mechanical devices?

- A. a spring
- B. a hydraulic jet
- C. a lever
- D. a crane

Answer(s): D

Explanation:

An elevator is simply a crane that raises and lowers people.

QUESTION: 1692

There are three beams that are each 10 feet long and all of the same size. One is made of wood, another of steel, and the third of concrete.

If identical loads are applied to these three beams, which of the following will occur?

- A. The steel beam will deflect less than the wood beam.
- B. The wood beam will deflect less than the concrete beam.
- C. The wood beam will deflect less than the steel beam.
- D. The concrete beam will deflect more than the other two.

Answer(s): A

Explanation:

The steel beam will deflect less than the wood beam.

Choice "The steel beam will deflect less than the wood beam" is not correct since the concrete beam will deflect less than the wood beam - concrete is stiffer than wood.

QUESTION: 1693

What mechanical motion principle do the brakes in a car or on a bicycle use?

- A. friction
- B. centrifugal force
- C. momentum
- D. acceleration

Answer(s): A

Explanation:

When the brakes are applied, the brake pads rub on the wheel and the frictional force of this rubbing slows down the vehicle.

QUESTION: 1694

The center of gravity of a baseball bat would be best described as _____.

- A. near the skinny end
- B. at the top
- C. near the fat end
- D. near the grip

Answer(s): C

Explanation:

The center of gravity is the place on an object where there is equal weight on either side.

QUESTION: 1695

Which material is best suited for use as a boat anchor?

- A. foam
- B. metal
- C. wood
- D. glass

Answer(s): B

Explanation:

Foam and wood float and would therefore make poor boat anchors.
Glass is not strong enough to be used as a boat anchor.

QUESTION: 1696

What would happen to a balloon full of air if you moved it from above a water surface to ten feet below the water surface?

- A. The balloon would explode.
- B. The volume of the balloon would decrease.
- C. The volume of the balloon would increase.
- D. The volume of the balloon would stay the same.

Answer(s): B

Explanation:

The volume of the balloon would decrease.
The pressure of the water would press inward on the balloon and cause it to shrink in volume.

QUESTION: 1697

What is Newton's Third Law of Motion?

- A. What goes up must come down.

- B. For every action, there is an equal and opposite reaction.
- C. Reactions are a result of large actions.
- D. Mass cannot be destroyed.

Answer(s): B

QUESTION: 1698

The term equilibrium is defined as _____.

- A. the state in which similar forces attract, causing a swell of movement in a single direction
- B. the principle describing the notion that every action must have a reaction
- C. the state in which opposing forces are balanced
- D. the presence of motion in the present day which serves as a means of understanding movement as a whole

Answer(s): C

QUESTION: 1699

What is the definition and the danger of recoil?

- A. To recoil is to move quickly and efficiently, and it is not dangerous but desired.
- B. To recoil is to feel revolted or disgusted, and it is dangerous because it creates job dissatisfaction.
- C. To recoil is to respond in fear, and it is dangerous because engineers and workers need to remain in control. Losing control could result in serious injuries on the job.
- D. Recoil is the concept of "kicking back" or leaping back after being released. It may be dangerous due to the possibility of injury upon having equipment "leap back" rapidly.

Answer(s): D

QUESTION: 1700

Another word or phrase for false gravity is _____.

- A. tension
- B. friction
- C. centrifugal force
- D. Newton's Law

Answer(s): C

QUESTION: 1701

The phrase "to adopt power" means _____ when working.

- A. to increase speed and efficiency to become your absolute best
- B. to use a forceful tone when discussing work with a coworker
- C. to focus energy on the lesser objects before moving on to larger instincts
- D. to work authoritatively and with a slim, fit physique

Answer(s): A

QUESTION: 1702

A _____ should be applied if gear shafts are not parallel to one another.

- A. shifting gear
- B. third gear
- C. bevel gear
- D. stopping gear

Answer(s): C

QUESTION: 1703

What is the advantage of using wheels and axles?

- A. Wheels and axles provide greater power, enabling workers to achieve greater results with less effort.
- B. Wheels and axles are responsible for bearing the brunt of difficult lifting.
- C. Wheels and axles are not beneficial.
- D. Wheels and axles create a more effective conveyor belt effect.

Answer(s): A

QUESTION: 1704

What is the function of a vise in mechanical applications?

- A. In mechanical applications, vises are used to arrest motion.
- B. Vises are used to suspend quick movements, slowing the effects of conveyor belts.
- C. In mechanical applications, vises are designed to assist movement.
- D. Vises are designed to function as a source of stability, much like how legs function beneath a table.

Answer(s): A

QUESTION: 1705

A hydraulic jack performs the task of moving objects by _____.

- A. utilizing oil; because oil cannot be compressed, pressure is placed upon the oil, resulting in a lifting motion
- B. pressing together two interfacing parts to increase height
- C. using a pulley system to lift items
- D. using a corkscrew-like effect to function

Answer(s): A

QUESTION: 1706

Which of these statements about force is correct?

- A. The amount of force required to move an object depends on the frictional forces between the object and its surroundings and must be calculated entirely based on individual situations.
- B. The amount of force required to move an object does not depend on frictional conditions and cannot and will not exceed the weight of the object in question.
- C. The amount of force required to move an object depends only on the weight of the object in question.
- D. Force is best used in levered applications.

Answer(s): A

QUESTION: 1707

The purpose of mechanical comprehension is _____.

- A. to inform workers on the proper use and function of machines
- B. to determine how to adequately use mechanic's tools
- C. to learn the use of various mechanisms
- D. to describe the training undergone by car mechanics

Answer(s): A

QUESTION: 1708

What is the primary difference between distillation and evaporation?

- A. Distillation separates or purifies liquids, while evaporation changes the form of a substance.
- B. One is intentional in scope, while the other occurs during other practices.
- C. Distillation involves the use of implements, while evaporation only requires heat.
- D. Distillation and evaporation have no primary difference; they are the same process.

Answer(s): A

QUESTION: 1709

What is the state known as equilibrium?

- A. Equilibrium describes the action of two similar forces coming together.
- B. Equilibrium describes the state in which two different items possess equal force, resulting in balance.
- C. Equilibrium describes the action of two opposing forces coming together in a crash.
- D. Equilibrium describes the state created by the coming together of parallel forces.

Answer(s): B

QUESTION: 1710

What purpose does an inclined plane serve?

- A. An inclined plane is a tilted surface designed to aid in supporting or transporting a load.
- B. Inclined planes are not useful implements and have been replaced by more effective methods of transportation.
- C. Inclined planes are tilted surfaces designed to assist in suspension.
- D. An inclined plane is intended to aid only in unloading merchandise.

Answer(s): A

QUESTION: 1711

If two workers are carrying a single load with the assistance of a board or plane, how should the load be placed to ensure an entirely equitable division of effort?

- A. The load should be placed the same length from one side as the load is long; i.e. if the load is 3 feet wide, it should be placed 3 feet from the edge of the plane.
- B. The load should be placed directly in the middle of the board or plane to ensure equal labor division between workers.
- C. The cargo should be held directly, rather than placing strain on a board or plane.
- D. The load should be placed closer to one end to prevent bowing of the plane.

Answer(s): B

QUESTION: 1712

A string of six gears exists, with each gear turning in conjunction with one another. Which gears operate in the same rotation?

- A. Gears 1, 2, and 3
- B. Gears 1, 3, and 5
- C. Gears 3, 4, and 5
- D. Gears 2, 4, and 5

Answer(s): B

QUESTION: 1713

Which is the best description of centrifugal force?

- A. the force responsible for the changes made to water during distillation.
- B. the response of bodies with inertia to the straight-moving or curved behavior of their container.
- C. the force by which bodies are propelled forward.
- D. the relationship between two opposing forces.

Answer(s): B

QUESTION: 1714

What is the definition of a "concurrent force"?

- A. A force system requiring concurrent movement.

- B. A force system described as having multiple sources of force.
- C. A system requiring all systems to pass through the same point.
- D. A series of forces requiring simultaneous movement.

Answer(s): C

QUESTION: 1715

How does oil function in a machine with a series of moving parts?

- A. Oil functions as a cleaning agent, allowing all moving parts to remain free of dirt and grime.
- B. Oil functions as a source of lubrication, allowing moving parts to function as a single unit.
- C. Oil serves as a fuel for moving parts in a machine.
- D. Oil provides a source of power when paired with fire.

Answer(s): B

QUESTION: 1716

Which of the following materials serves as a conductor?

- A. Metal
- B. Wood
- C. Acrylic
- D. Paper

Answer(s): A

QUESTION: 1717

To move an object across a surface, which system requires little input from the user, yet creates extreme force?

- A. A conveyor belt, as those found in factories.
- B. Any form of mechanical device.
- C. wheel and axle system, as found in vehicles.
- D. A pulley system, as might be found in warehouses and shipyards.

Answer(s): C

QUESTION: 1718

What determines the speed of a pulley system?

- A. The material used as the belt (such as leather and rubber).
- B. The size of the pulleys as a whole.
- C. The size and ratio of all pulleys involved.
- D. The length of the rope or belt used in the pulley system.

Answer(s): C

QUESTION: 1719

Pulley systems require the synchronized movement of all pulleys.
How can pulleys escape this rule?

- A. To violate this rule, pulley systems may be used in conjunction with gears.
- B. This rule cannot be effectively violated.
- C. To violate this rule, pulley system belts must be reversed.
- D. Pulleys using twisted belts may move in opposing directions.

Answer(s): D

QUESTION: 1720

What are the three components of a lever?

- A. Resistance arm and effort arm
- B. A point of origin and solid plane
- C. A wide plane, fulcrum, and effort arm
- D. Fulcrum, resistance arm, and effort arm

Answer(s): D

QUESTION: 1721

What determines the speed of an output gear?

- A. None of these (Gear speed is not capable of alteration.)
- B. The size of gears.
- C. The speed of the input gear and the ratio of the number of teeth on each gear.
- D. The ratio of companion gears.

Answer(s): C

QUESTION: 1722

What are the potential functions of gear systems?

- A. Gears may be used to change direction in machinery and can be used to increase speed.
- B. Gears possess a single function: transmitting motion.
- C. The potential functions of gears are: changes in direction, speed changes (both increases and decreases), and motion transmission.
- D. Gears are primarily used for making changes in speed, but may also be used to instigate motion in machinery.

Answer(s): C

QUESTION: 1723

If a mechanical device is reported to have an advantage of 2, what does this mean?

- A. The number "2" describes the effectiveness of the machine.
- B. This means the device allows users to complete twice the amount of work with the same amount of effort used to complete a task without assistance.
- C. The number refers to the speed of the device in question.
- D. This means that the device requires twice the amount of effort to complete the same amount of work.

Answer(s): B

QUESTION: 1724

An "inclined plane" is a commonly used implement and has many applications, including handicap access.

What is the most common name for an inclined plane?

- A. A pulley system
- B. A lever
- C. A ramp
- D. A series of gears

Answer(s): C

QUESTION: 1725

How can workers increase the mechanical advantage of pulley systems?

- A. Using longer stretches of rope.
- B. Using a higher platform for the load in need of lifting.
- C. Using additional pulleys to further distribute weight.
- D. Decreasing load size and weight.

Answer(s): C

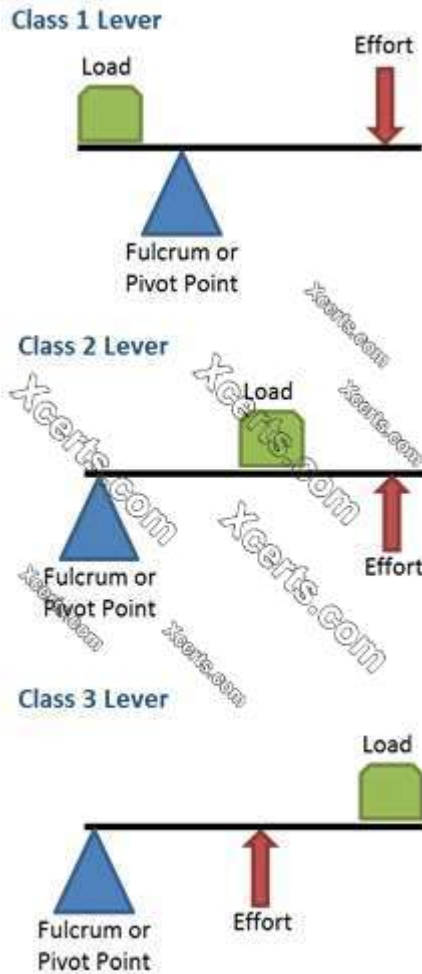
QUESTION: 1726

The best definition for drag is _____.

- A. a responsive force that affects the movement of an object through a fluid
- B. a responsive force that only affects movement on a solid
- C. a responsive force that only affects movement in air
- D. the internal stretch force of a material

Answer(s): A

QUESTION: 1727



Which of the following is an example of a Class 1 Lever?

- A. Pliers
- B. Wheelbarrow
- C. Tongs
- D. Stapler

Answer(s): A

Explanation:

Pliers are an example of a Class 1 lever (actually 2 class 1 levers). Wheelbarrow and stapler are class 2. Tongs are class 3.

QUESTION: 1728

Which of the following will happen when a mass of air contracts?

- A. The air remains at the same temperature
- B. The air gets colder
- C. The air gets warmer
- D. The air moves faster

Answer(s): B

Explanation:

Air is a gas. According to the gas laws, gasses will get colder when they contract.

QUESTION: 1729

Which of the following would feel hottest at one end if the other end were placed over a fire?

- A. A metal skewer
- B. A wooden stick
- C. A plastic rod
- D. A rubber stick

Answer(s): A

Explanation:

Of the four materials, metal is the best conductor of heat, so a metal skewer would feel the hottest.

QUESTION: 1730

If you use a lever to lift a 200 pound weight with an effort of 40 pounds, with is the mechanical advantage?

- A. 4
- B. 0.2
- C. 8,000
- D. 5

Answer(s): D

Explanation:

Mechanical advantage = load/effort = $200/40 = 5$

QUESTION: 1731

Water flows into a container at a rate of 140 gallons per minute. The container has a small opening at the bottom that drains water at a rate of 1 gallon per second. How long will it take to fill the container to 240 gallons?

- A. 2 min
- B. 3 min
- C. 4 min
- D. Not enough information

Answer(s): B

Explanation:

The opening drains water at 1 gallon/second which is equivalent to 60 gallons/min.

Therefore, the net gain of water is 140 gallons/min - 60/gallons/min = 80 gallons/min.
To fill 240 gallons at a rate of 80 gallons/min will take 3 minutes.

QUESTION: 1732

Which of the following statements is true for water flowing from a 4 inch pipe to a smaller 2 inch pipe?

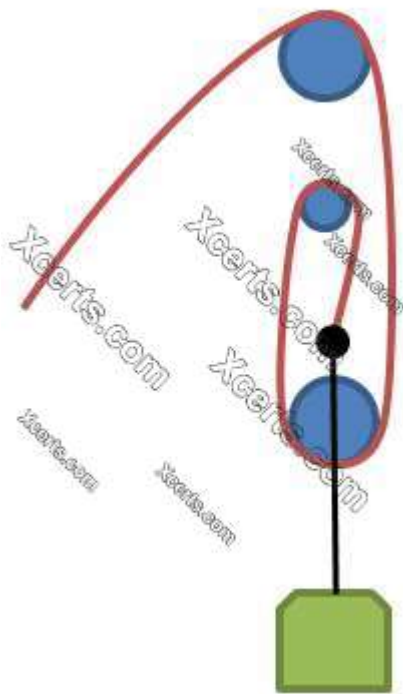
- A. The same amount of water passes through both the 4 inch pipe and the 2 inch pipe.
- B. The pressure is greater in the 2 inch pipe.
- C. More water passes through the 4 inch pipe than through the 2 inch pipe.
- D. The pressure is lower in the 4 inch pipe.

Answer(s): A

Explanation:

Water cannot be compressed. Therefore, the same amount of water must be flowing through both the 4 inch pipe and the 2 inch pipe.

QUESTION: 1733



What is the mechanical advantage of this pulley system?

- A. 6
- B. 4
- C. 3
- D. 2

Answer(s): C

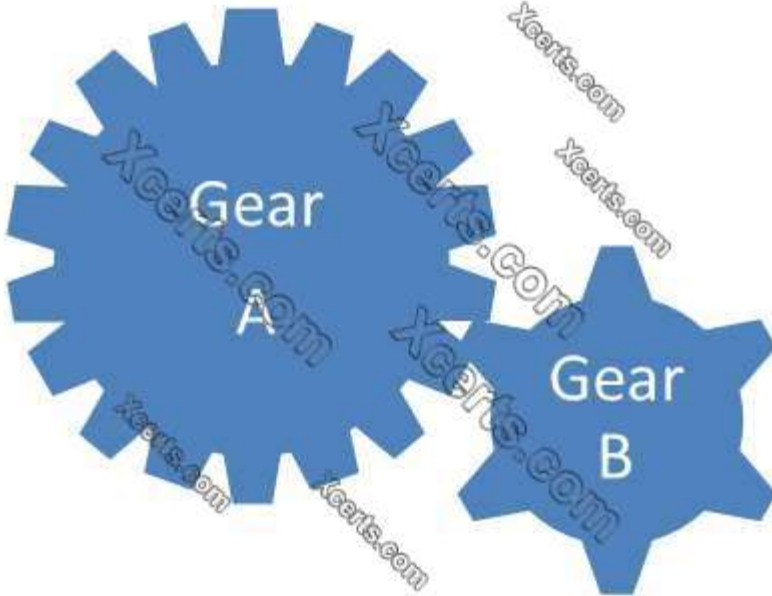
Explanation:

In a pulley system, you calculate the mechanical advantage by counting up the number of supporting ropes/ strands that support the weight.

In this case, there are 3 supporting strands that support the load, so the mechanical advantage is 3.

QUESTION: 1734

In the figure below, if gear A turns 15 times, how many revolutions will gear B make?



- A. 15 revolutions
- B. 5 revolutions
- C. 30 revolutions
- D. 45 revolutions

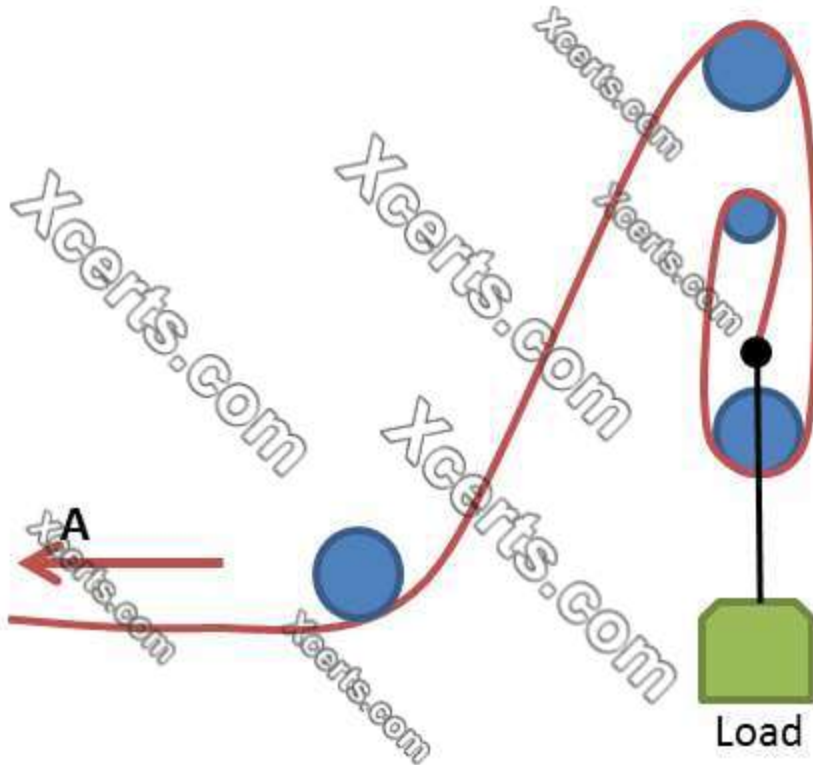
Answer(s): D

Explanation:

Gear A has 18 teeth and gear B has 6 teeth.

Therefore, for every 1 revolution of gear A, gear B will have 3 revolutions.

QUESTION: 1735



Pulling on the rope at point A will lift the load.
How far will you have to pull the rope at point A to lift the load 15 feet?

- A. 15 feet
- B. 45 feet
- C. 30 feet
- D. 60 feet

Answer(s): B

Explanation:

The mechanical advantage of the pulley is 3 since there are three ropes supporting the load. Therefore, the distance at point A needs to be $3x$ or $3 \times 15 = 45$ ft.

QUESTION: 1736

In the figure below, if gear 1 turns counterclockwise (as shown by the arrow), which of the gears will also turn in the same direction?



- A. 2 and 4
- B. 2 and 3
- C. 3 and 4
- D. 4

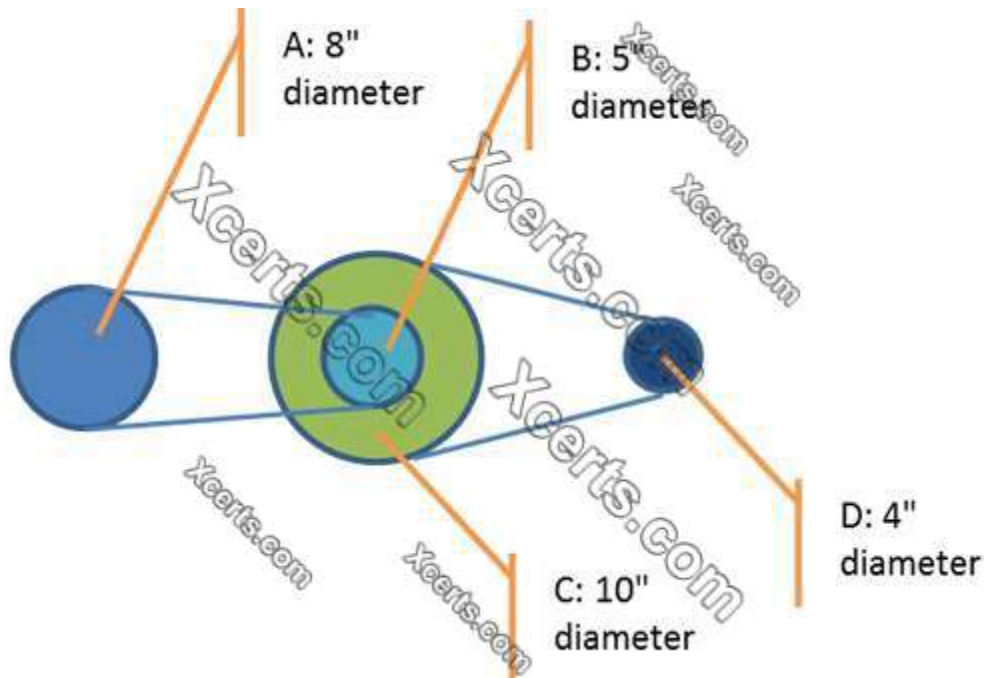
Answer(s): C

Explanation:

Gear 1 turns counterclockwise causing gear 2 to turn clockwise which results in gear 3 and 4 to turn counterclockwise.

QUESTION: 1737

In the pulley system below, how fast does pulley D rotate, if pulley A rotates at 50 rpm?



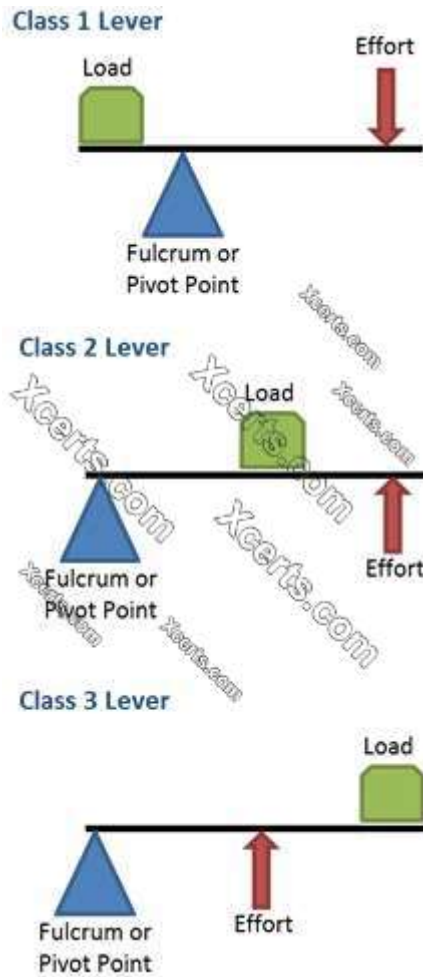
- A. 100 rpm
- B. 50 rpm
- C. 200 rpm
- D. 64 rpm

Answer(s): C

Explanation:

A larger pulley causes a smaller pulley to rotate faster by the ratio of their diameters. If pulley A rotates at 50 rpm, then pulley B rotates at $50 \times 8/5 = 80$ rpm. Since C is directly connected to B it rotates at the same rate (80 rpm). Finally, pulley D will rotate at $80 \text{ rpm} \times 10/4 = 200$ rpm

QUESTION: 1738



Which of the following is an example of a Class 2 Lever?

- A. Fishing rod
- B. Bottle opener
- C. See-Saw
- D. Scissors

Answer(s): B

Explanation:

A bottle opener is a single class 2 lever.
 A see-saw and scissors are class 1 levers.
 A Fishing rod is a class 3 lever.

QUESTION: 1739

Which of the following is not a type of simple machine?

- A. Pulley
- B. Spring
- C. Lever
- D. Screw

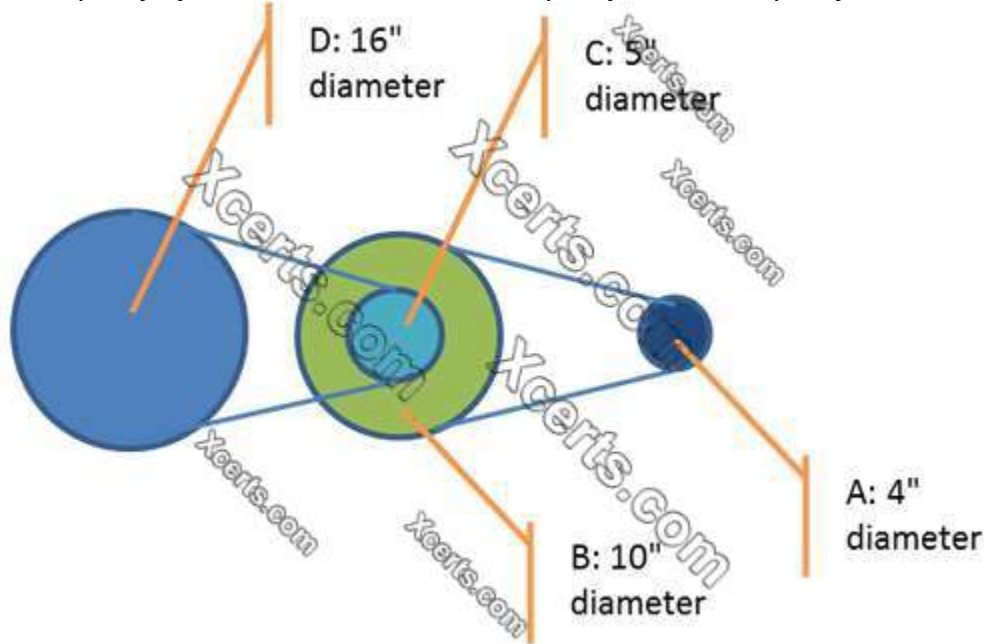
Answer(s): B

Explanation:

The six types of simple machines are: pulley, lever, screw, wheel and axle, wedge, inclined plane.

QUESTION: 1740

In the pulley system below, how fast does pulley D rotate, if pulley A rotates at 100 rpm?



- A. 25 rpm
- B. 50 rpm
- C. 100 rpm
- D. 12.5 rpm

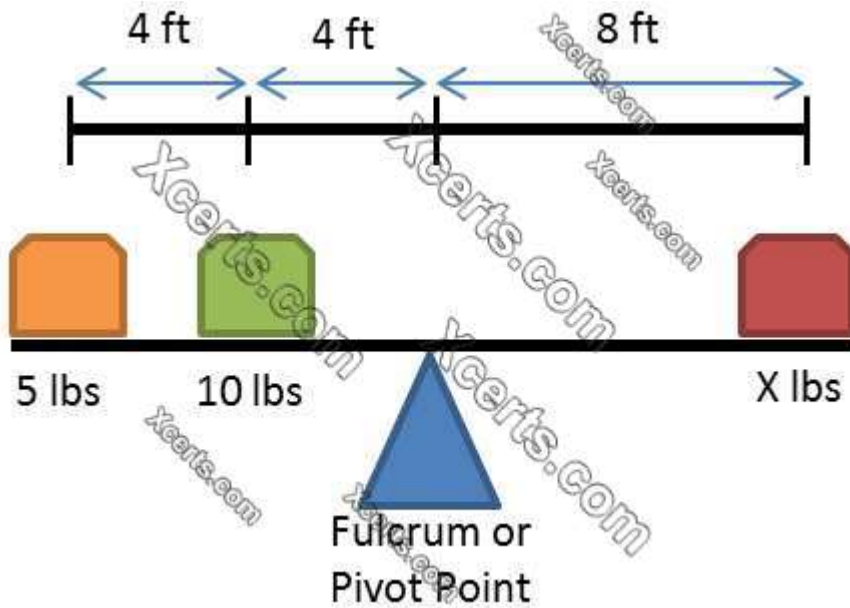
Answer(s): D

Explanation:

A larger pulley causes a smaller pulley to rotate faster by the ratio of their diameters. If pulley A rotates at 100 rpm, then pulley B rotates at $100 \times 4/10 = 40$ rpm. Since C is directly connected to B it rotates at the same rate (40 rpm). Finally, pulley D will rotate at $40 \text{ rpm} \times 5/16 = 12.5$ rpm

QUESTION: 1741

What weight (X) is needed to balance the two loads shown in the figure below?



- A. 15 lbs
- B. 10 lbs
- C. 5 lbs
- D. 7.5 lbs

Answer(s): B

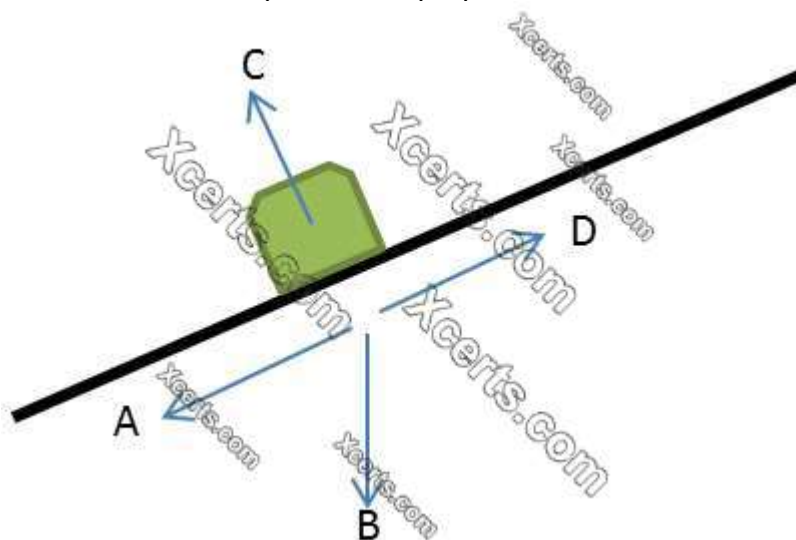
Explanation:

For the weights to balance, the sum of the forces must balance.

Therefore, $(5 \text{ lbs} \times 8 \text{ ft}) + (10 \text{ lbs} \times 4 \text{ ft}) = X \text{ lbs} \times 8 \text{ ft}$ or $40 \text{ lb-ft} + 40 \text{ lb-ft} = 8X \text{ lb-ft}$ or $X = 10 \text{ lbs}$

QUESTION: 1742

The diagram below represents a load sliding down an incline. Which of the forces (A,B,C, or D) represent friction?



- A. A
- B. B
- C. C
- D. D

Answer(s): D

Explanation:

Friction is represented by the force the resists the load moving down the hill.

QUESTION: 1743

On a bicycle there are 30 teeth on the gear sprocket connected to the pedals, and 10 teeth on the sprocket connected to the rear wheels.

If you pedal the bike 12 times, how many times will the rear wheel turn?

- A. 4 times
- B. 12 times
- C. 24 times
- D. 36 times

Answer(s): D

Explanation:

Since there are 30 teeth on the pedal sprocket, but only 10 teeth on the rear wheel, each revolution of the pedal will turn the rear wheel 3 times.

So 12 revolutions will turn the rear wheel 36 times.

QUESTION: 1744

An axe is an example of what type of simple machine.

- A. Lever
- B. Inclined plane
- C. Wedge
- D. None of the above

Answer(s): C

Explanation:

An axe is a classic example of a wedge.

QUESTION: 1745

When a football is thrown upwards in the air, what causes it to arc back towards the ground?

- A. Centrifugal force
- B. Gravity
- C. Friction
- D. Wind resistance

Answer(s): B

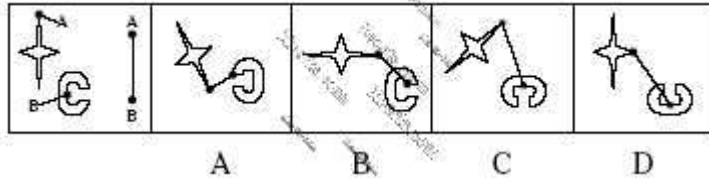
Explanation:

Gravity is the force that causes objects to be pulled back towards the earth.

Section 9: Assembling Objects

QUESTION: 1746

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

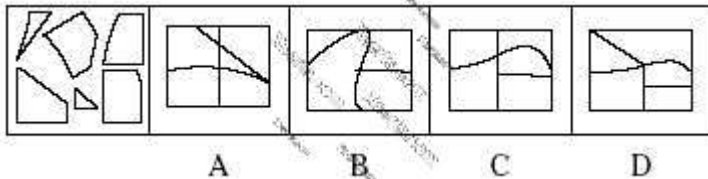


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1747

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

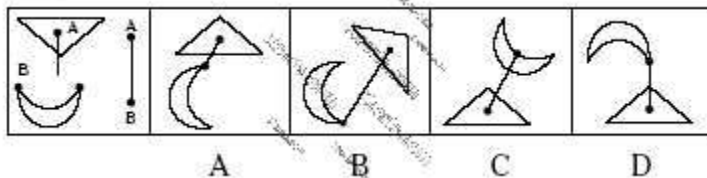


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1748

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



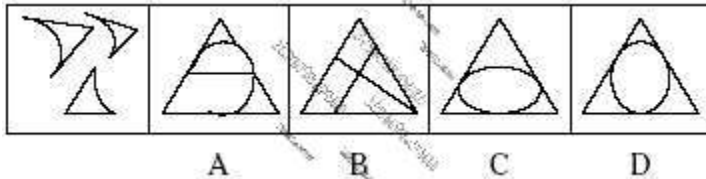
- A. Option A
- B. Option B

- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1749

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

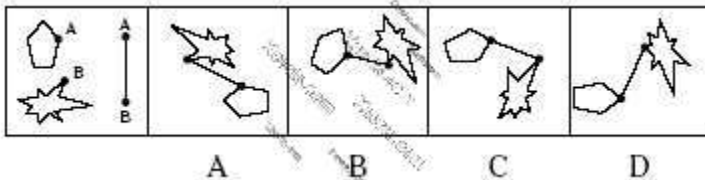


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1750

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

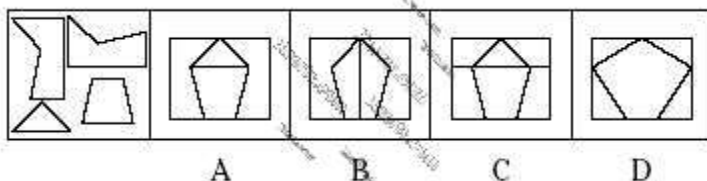


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1751

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

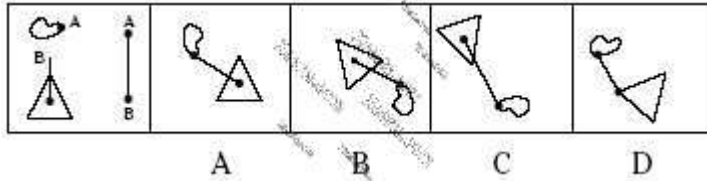


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1752

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

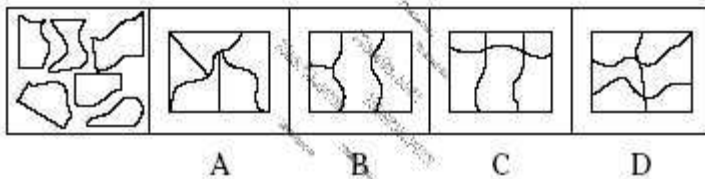


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1753

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

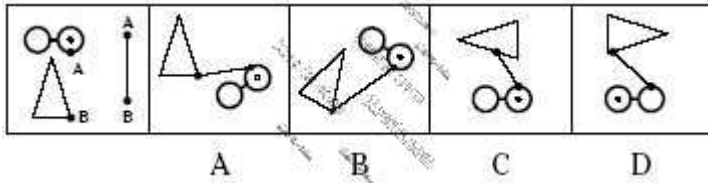


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1754

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

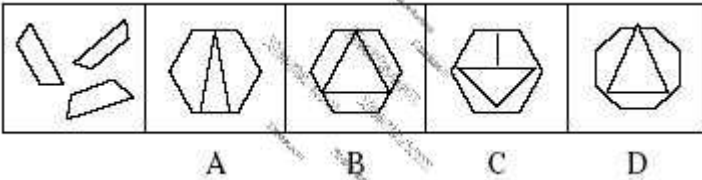


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1755

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

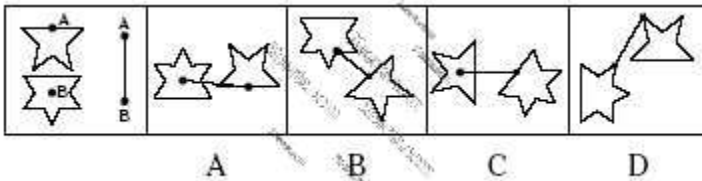


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1756

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

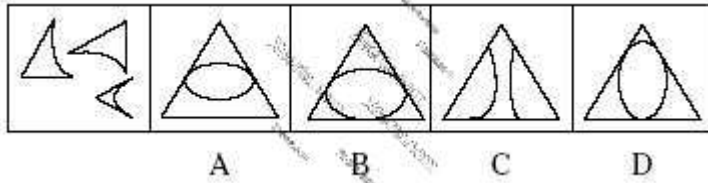


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1757

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

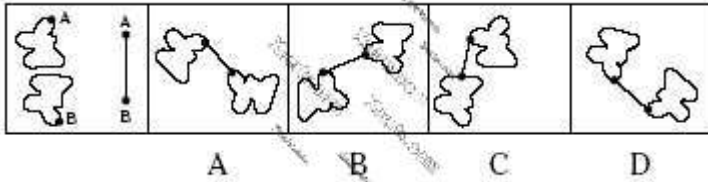


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1758

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

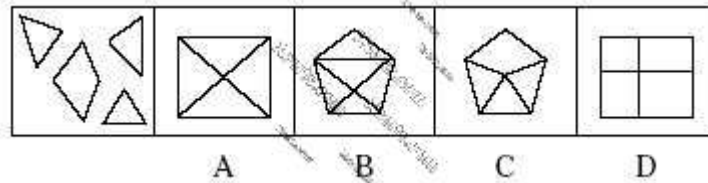


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1759

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

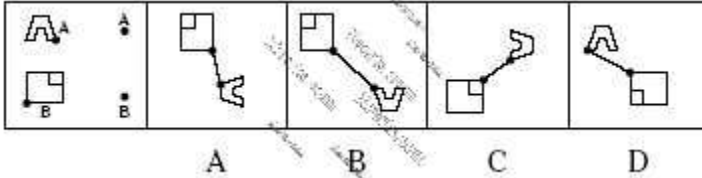


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1760

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

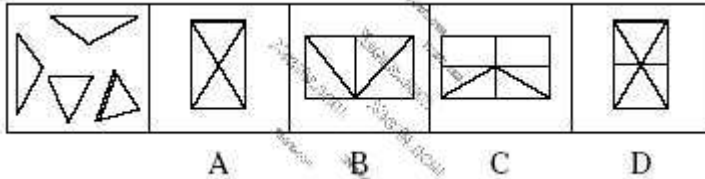


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1761

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

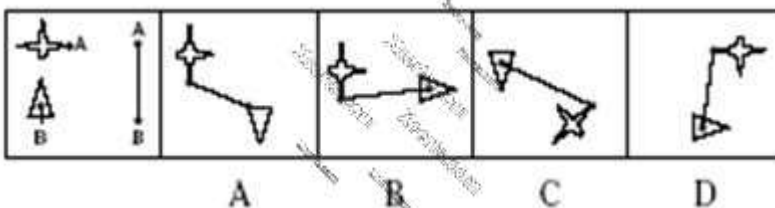


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1762

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



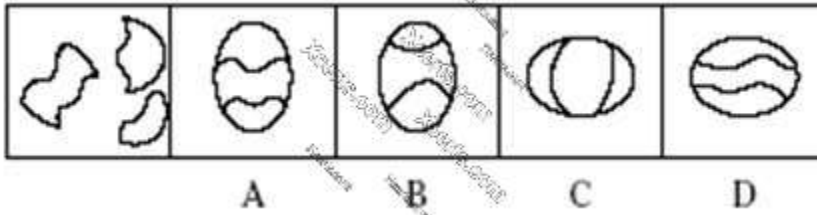
- A. Option A

- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1763

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

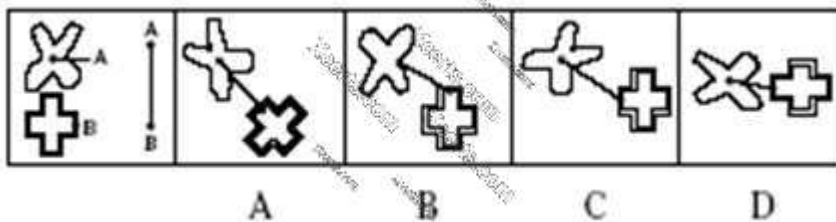


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1764

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

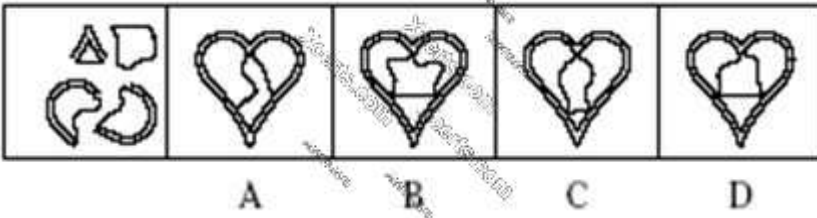


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1765

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

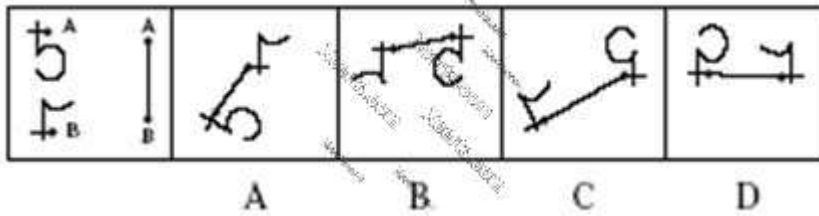


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1766

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

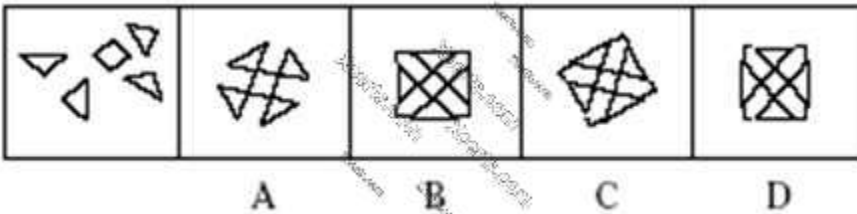


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1767

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

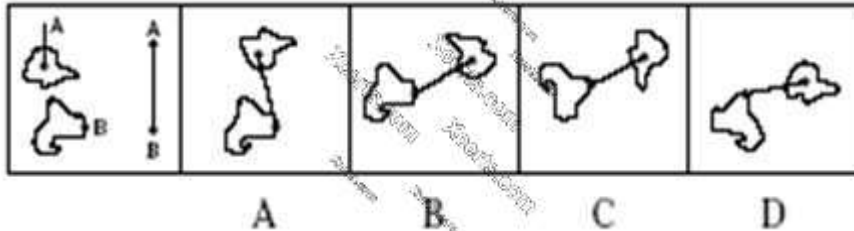


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1768

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

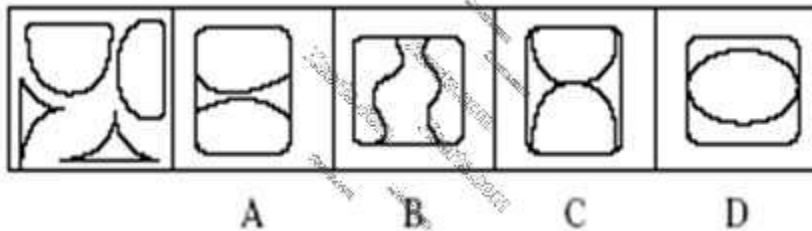


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1769

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

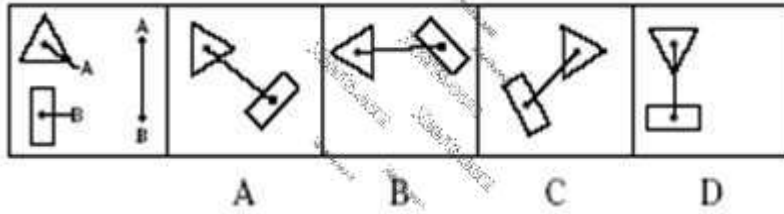


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1770

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

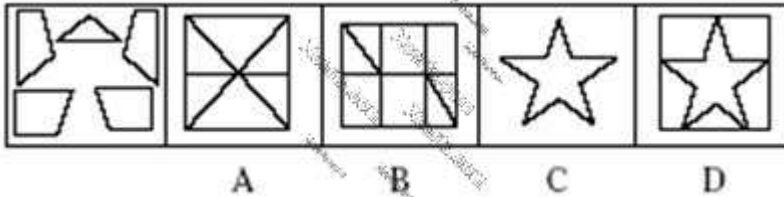


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1771

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

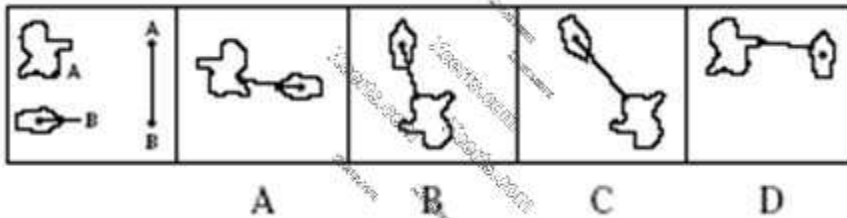


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1772

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

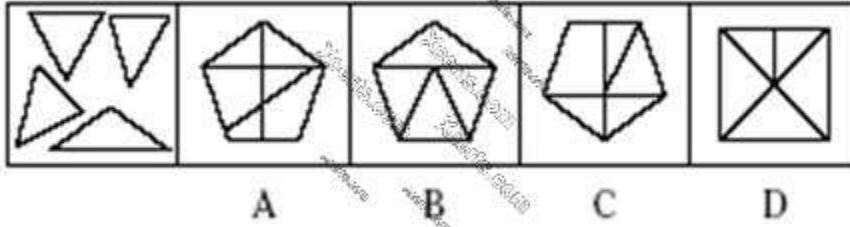


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1773

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

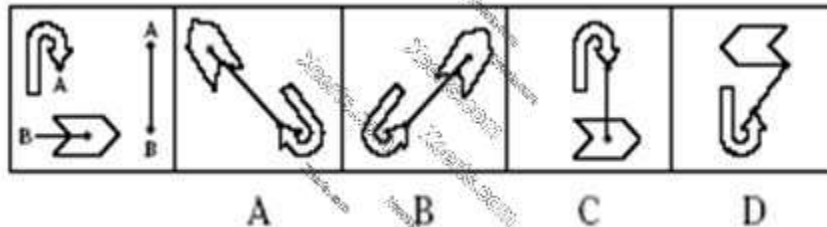


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1774

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

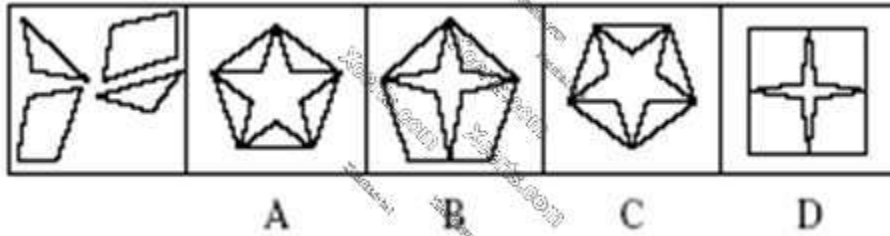


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1775

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

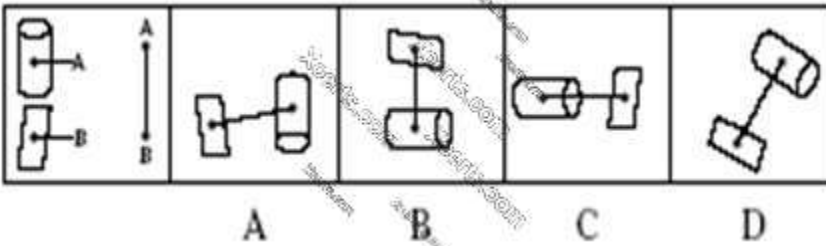


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1776

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

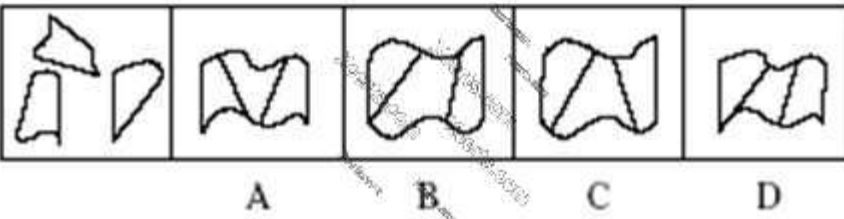


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1777

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



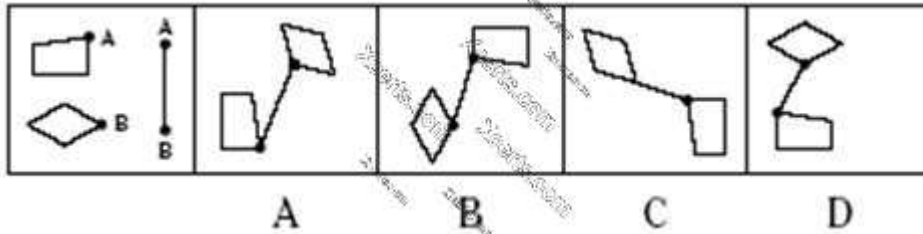
- A. Option A
- B. Option B
- C. Option C

D. Option D

Answer(s): D

QUESTION: 1778

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



A. Option A

B. Option B

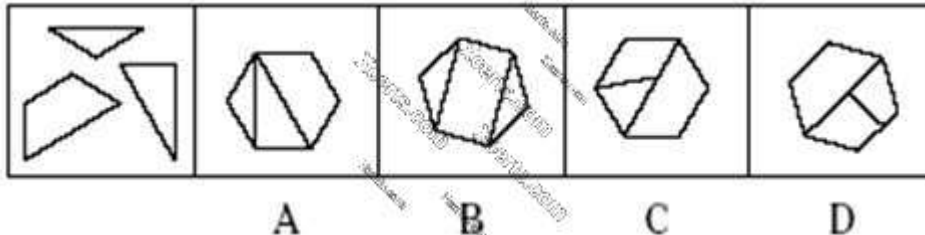
C. Option C

D. Option D

Answer(s): C

QUESTION: 1779

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



A. Option A

B. Option B

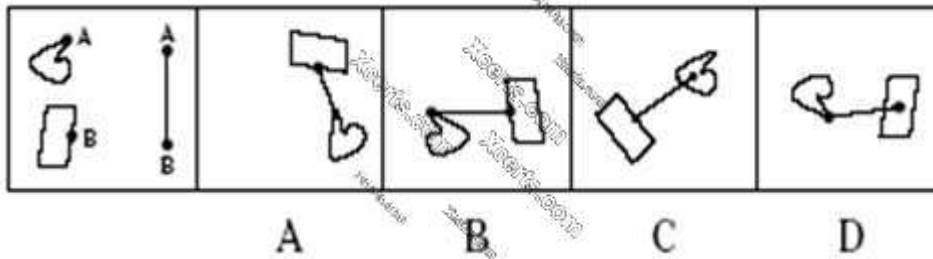
C. Option C

D. Option D

Answer(s): A

QUESTION: 1780

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

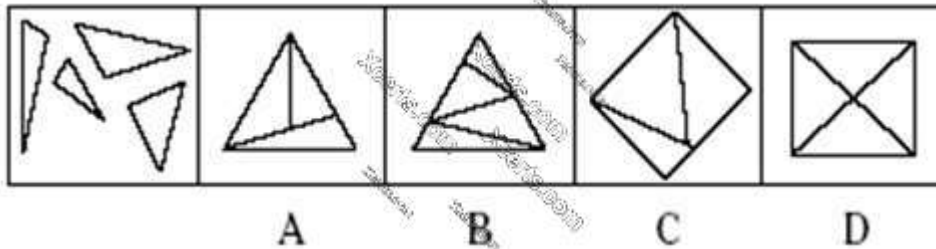


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1781

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

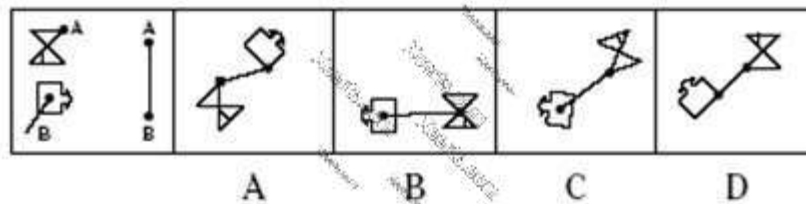


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1782

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



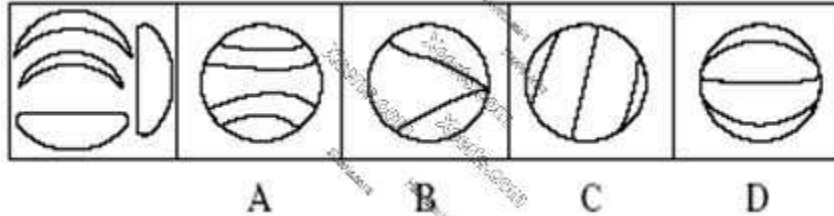
- A. Option A
- B. Option B
- C. Option C

D. Option D

Answer(s): C

QUESTION: 1783

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

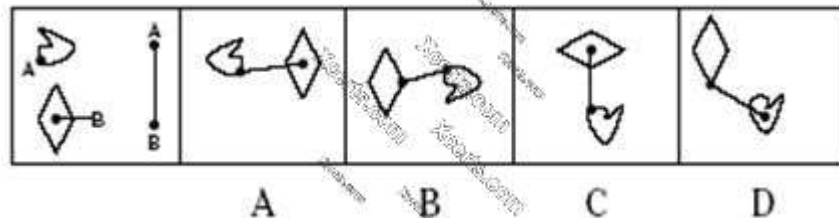


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1784

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

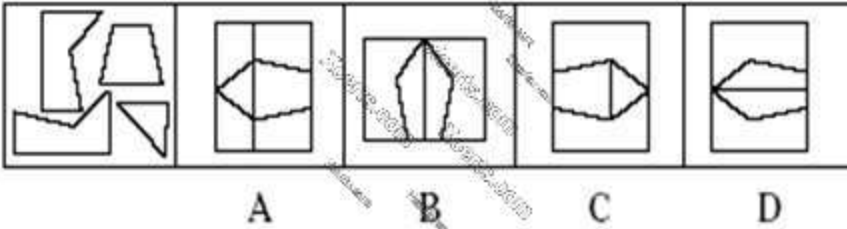


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1785

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

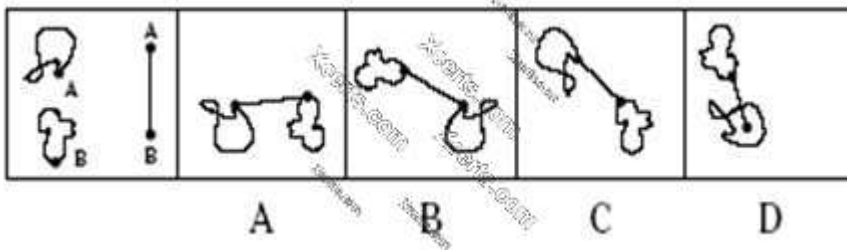


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1786

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

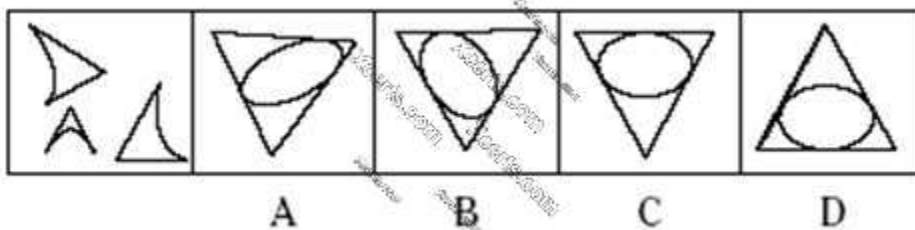


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1787

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



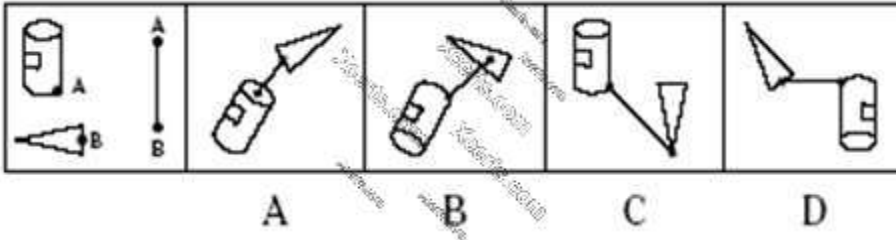
- A. Option A
- B. Option B
- C. Option C

D. Option D

Answer(s): A

QUESTION: 1788

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

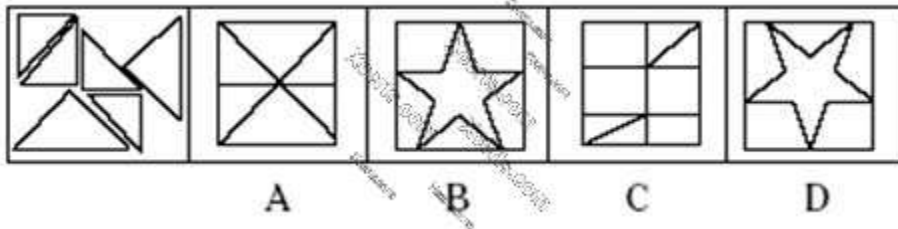


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1789

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

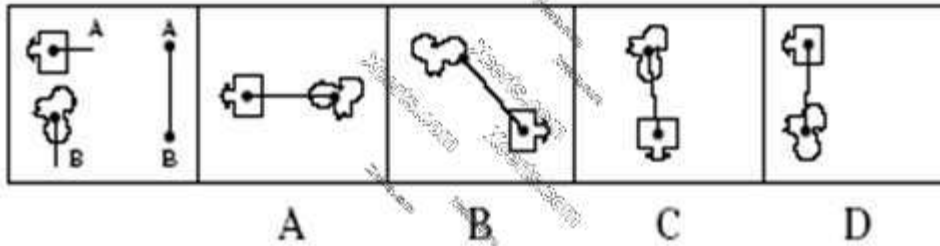


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1790

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

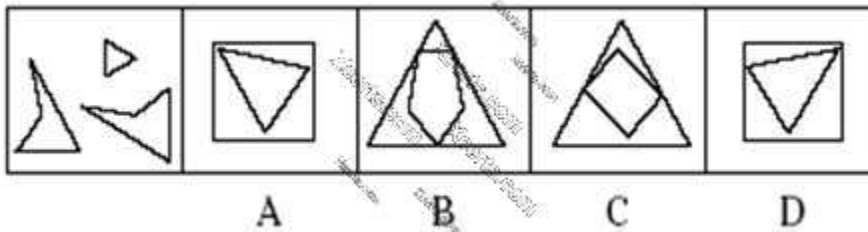


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1791

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

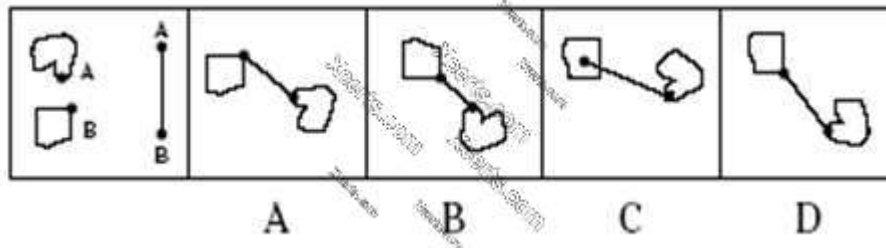


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1792

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



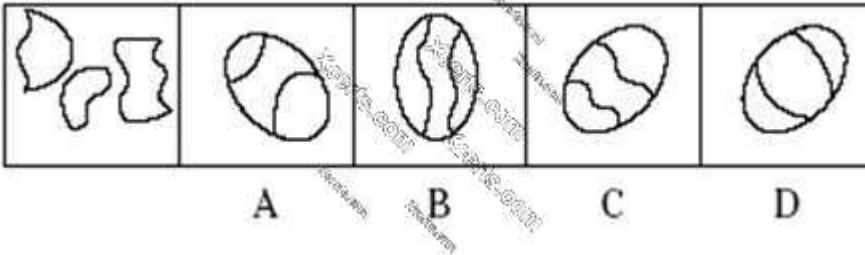
- A. Option A
- B. Option B
- C. Option C

D. Option D

Answer(s): D

QUESTION: 1793

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



A. Option A

B. Option B

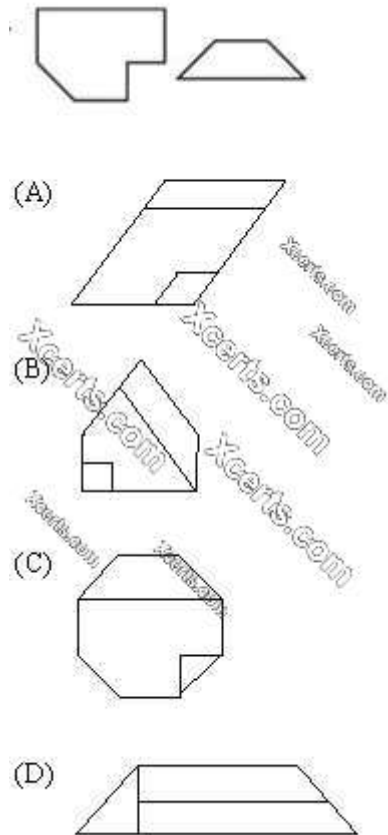
C. Option C

D. Option D

Answer(s): C

QUESTION: 1794

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

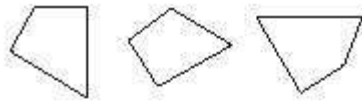


- A. Option A
- B. Option B
- C. Option C
- D. Option D

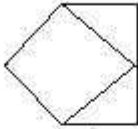
Answer(s): C

QUESTION: 1795

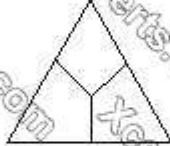
Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



☐ (A)



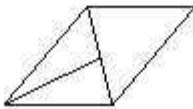
☒ (B)



☐ (C)



☐ (D)

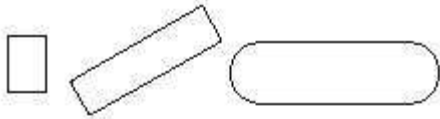


- A. Option A
- B. Option B
- C. Option C
- D. Option D

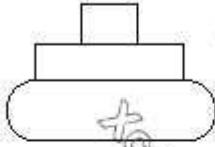
Answer(s): B

QUESTION: 1796

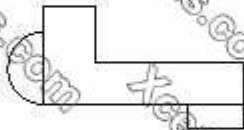
Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



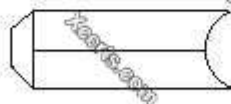
☐ (A)



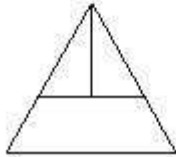
☐ (B)



☐ (C)



☐ (D)



A. Option A

B. Option B

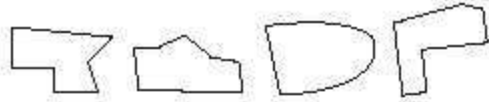
C. Option C

D. Option D

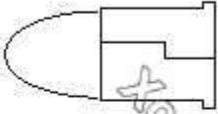
Answer(s): A

QUESTION: 1797

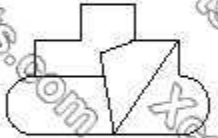
Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



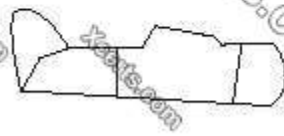
☐ (A)



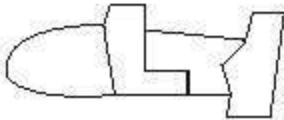
☐ (B)



☐ (C)



☐ (D)

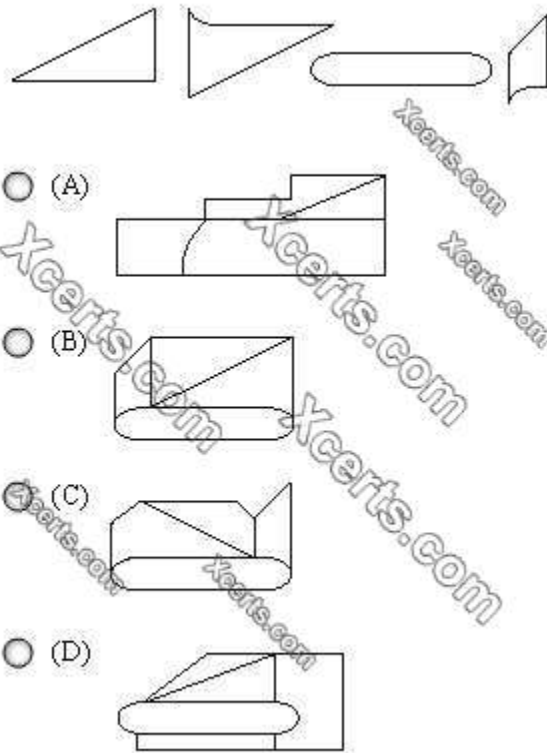


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1798

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

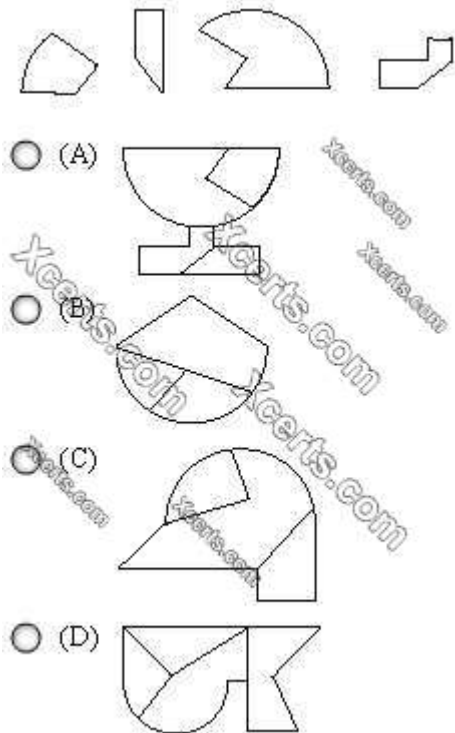


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1799

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

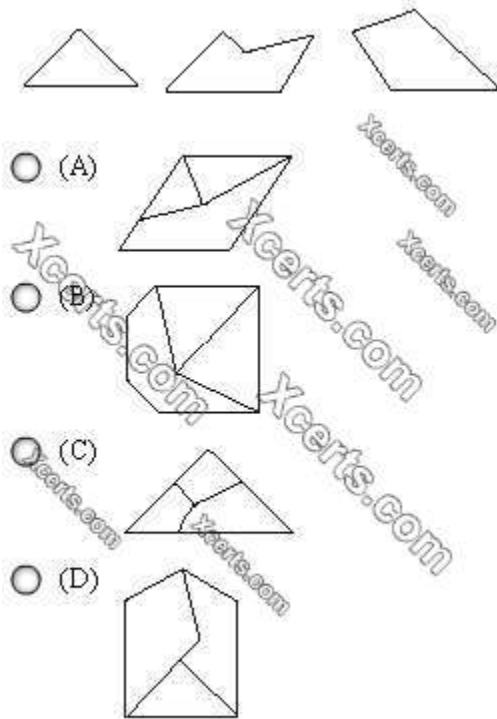


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1800

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

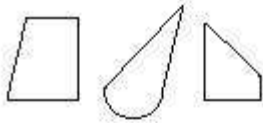


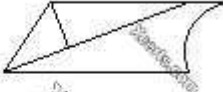
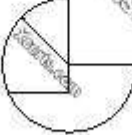
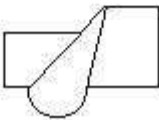
- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1801

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



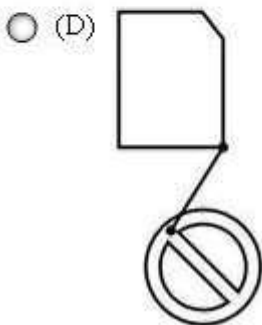
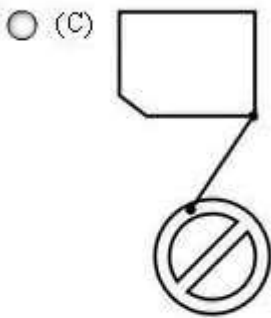
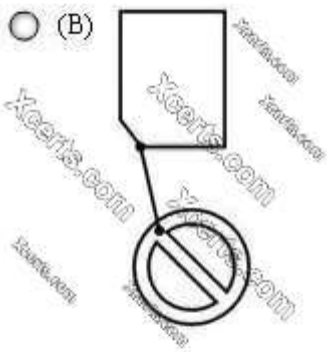
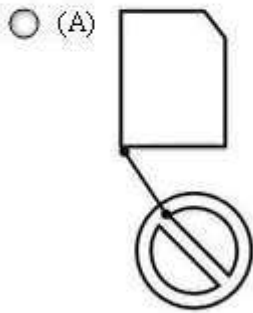
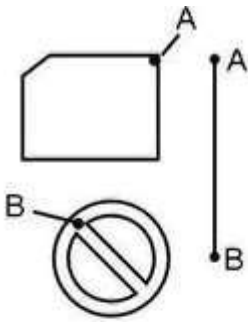
- ☐ (A) 
- ☐ (B) 
- ☐ (C) 
- ☐ (D) 

- A. Option A
B. Option B
C. Option C
D. Option D

Answer(s): D

QUESTION: 1802

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

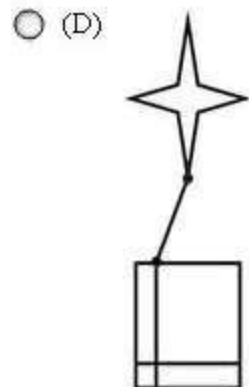
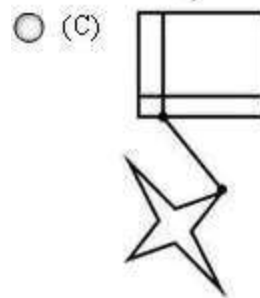
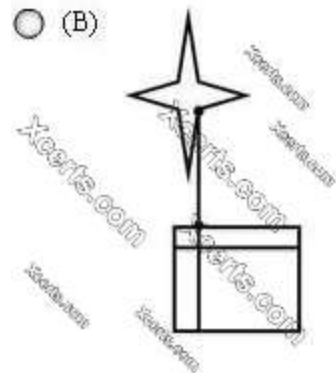
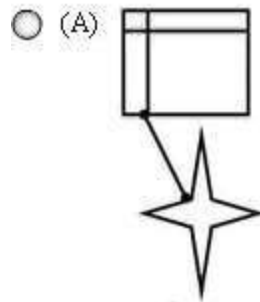
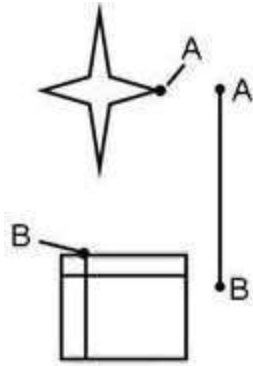


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1803

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

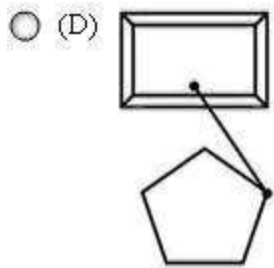
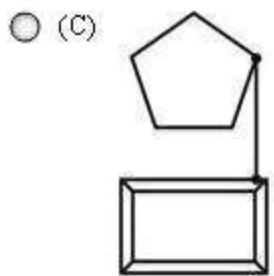
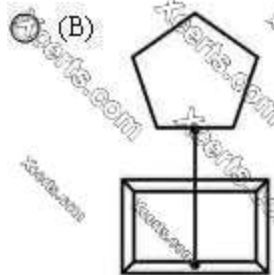
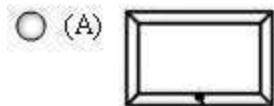
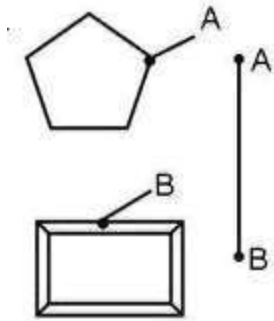


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1804

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



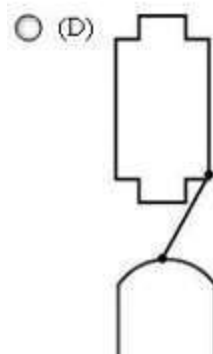
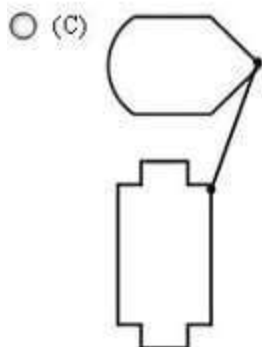
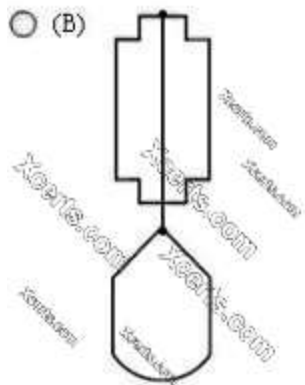
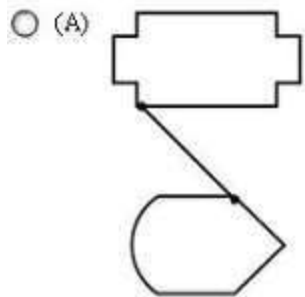
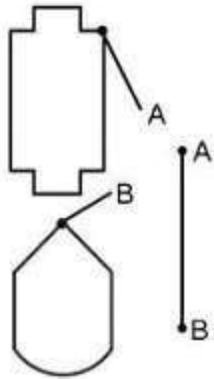
A. Option A

- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1805

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

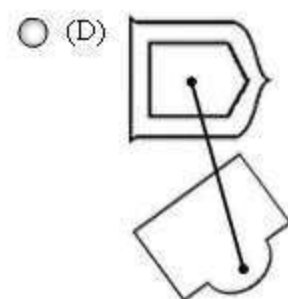
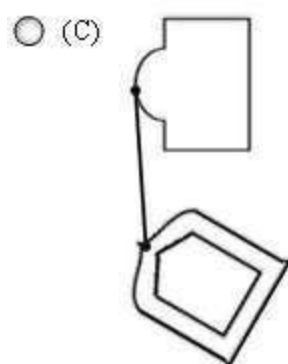
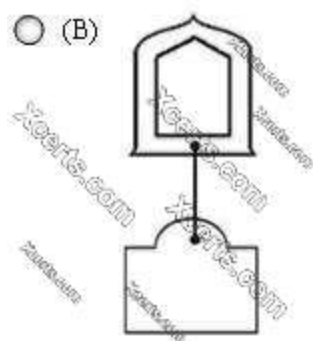
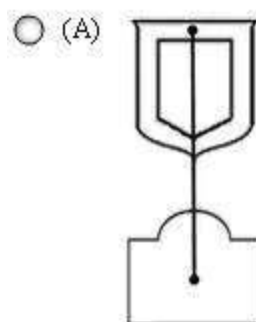
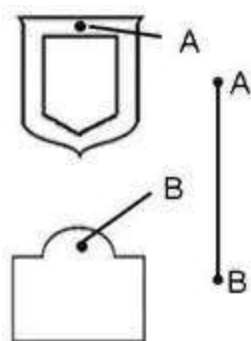


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1806

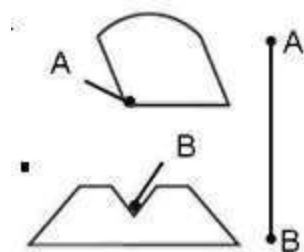
Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

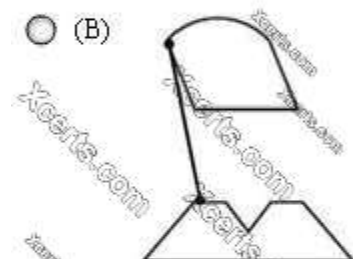
QUESTION: 1807



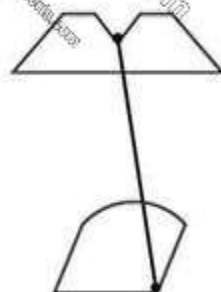
☐ (A)



☐ (B)



☐ (C)



☐ (D)

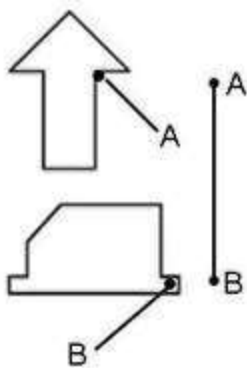


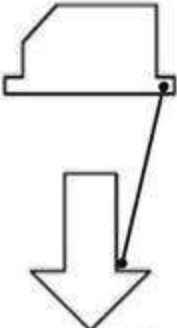
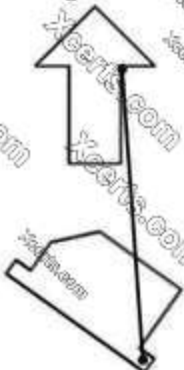
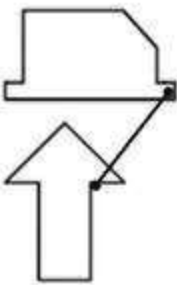
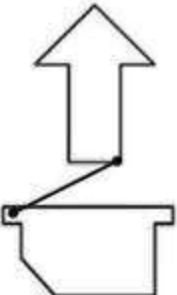
- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1808

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



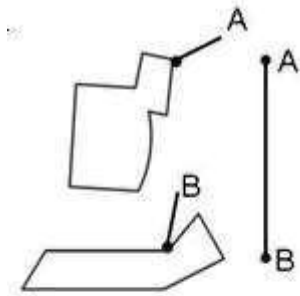
- ☐ (A) 
- ☐ (B) 
- ☐ (C) 
- ☐ (D) 

- A. Option A
- B. Option B
- C. Option C
- D. Option D

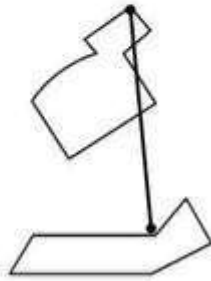
Answer(s): B

QUESTION: 1809

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



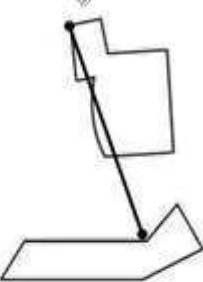
☐ (A)



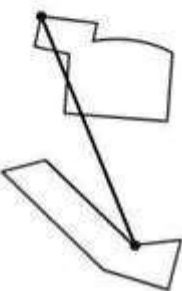
☐ (B)



☐ (C)



☐ (D)



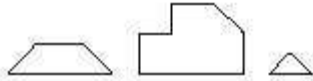
A. Option A

- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1810

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



- ☐ (A)
- ☐ (B)
- ☐ (C)
- ☐ (D)

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1811

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



- ☐ (A)
- ☐ (B)
- ☐ (C)
- ☐ (D)

- A. Option A
B. Option B
C. Option C
D. Option D

Answer(s): A

QUESTION: 1812

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



- ☐ (A)
- ☐ (B)
- ☐ (C)
- ☐ (D)

- A. Option A
B. Option B
C. Option C
D. Option D

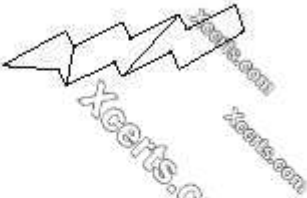
Answer(s): A

QUESTION: 1813

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



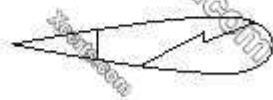
☐ (A)



☐ (B)



☐ (C)



☐ (D)

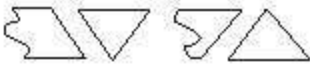


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1814

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



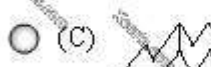
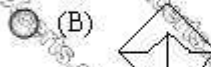
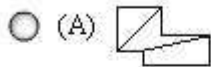
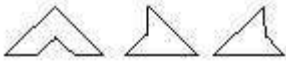
- ☐ (A)
- ☐ (B)
- ☐ (C)
- ☐ (D)

- A. Option A
B. Option B
C. Option C
D. Option D

Answer(s): B

QUESTION: 1815

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

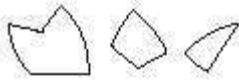


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1816

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



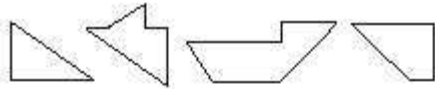
- ☐ (A)
- ☐ (B)
- ☐ (C)
- ☐ (D)

- A. Option A
B. Option B
C. Option C
D. Option D

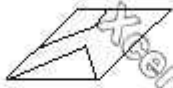
Answer(s): B

QUESTION: 1817

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



☐ (A)



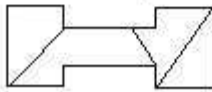
☐ (B)



☐ (C)



☐ (D)

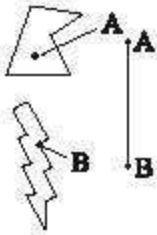


- A. Option A
- B. Option B
- C. Option C
- D. Option D

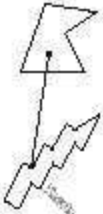
Answer(s): D

QUESTION: 1818

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



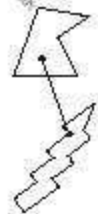
☐ (A)



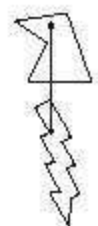
☐ (B)



☐ (C)



☐ (D)

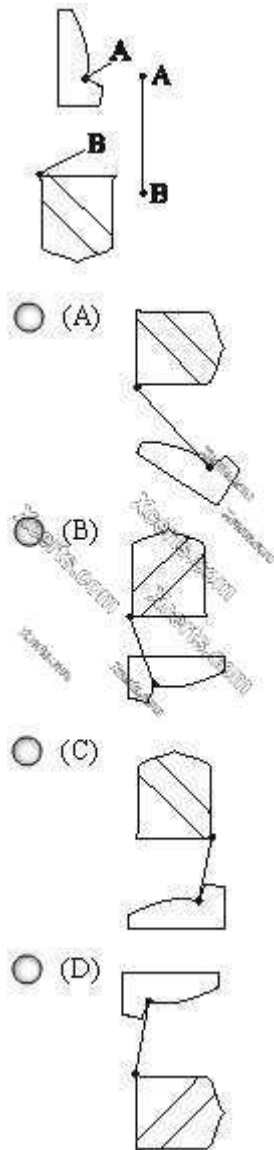


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1819

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

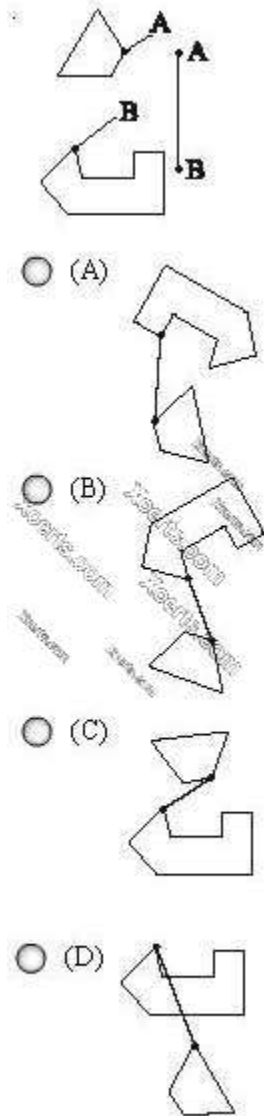


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1820

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

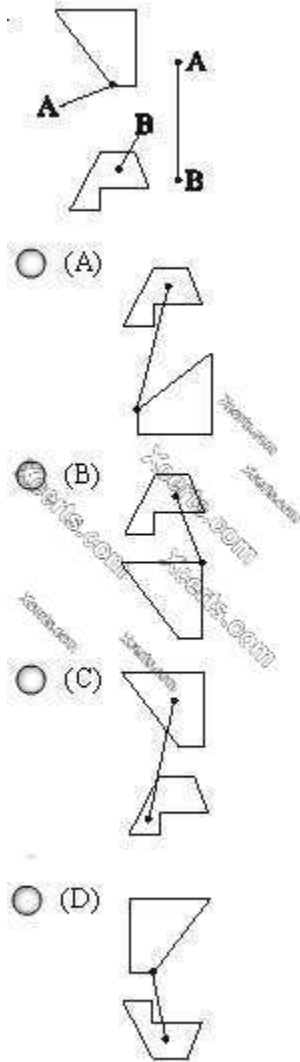


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1821

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

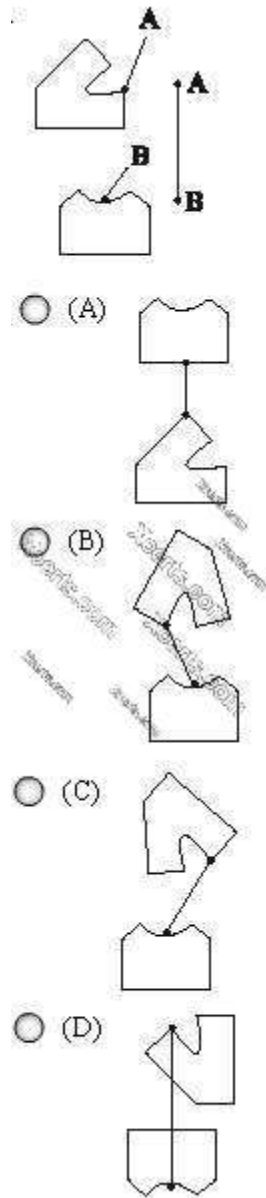


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1822

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

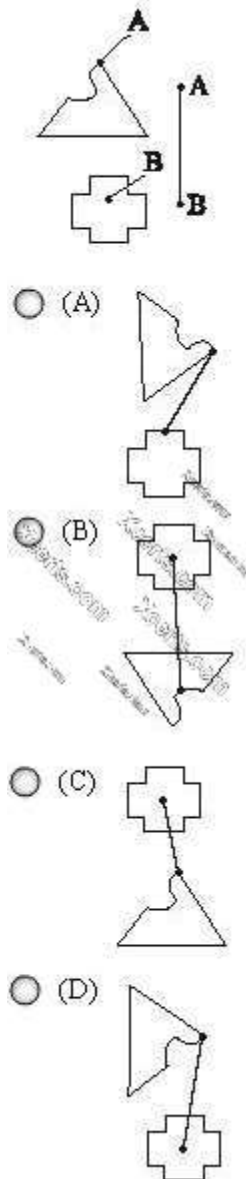


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1823

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.

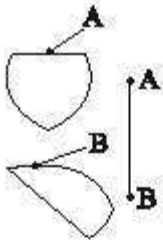


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1824

Determine which of the choices best solves the problem shown in the first picture. The problem is presented in the first drawing and the remaining four drawings are possible solutions.



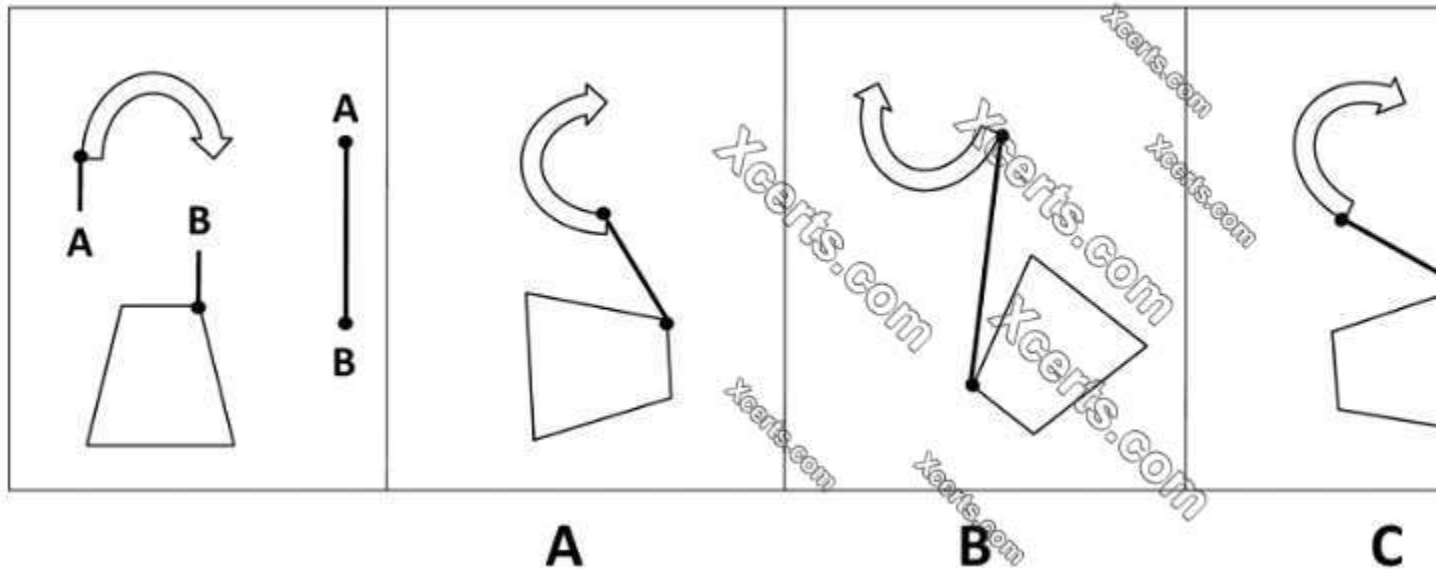
- ☐ (A)
- ☒ (B)
- ☐ (C)
- ☐ (D)

- A. Option A
B. Option B
C. Option C
D. Option D

Answer(s): A

QUESTION: 1825

Which of these images best solves the problem in the first picture?

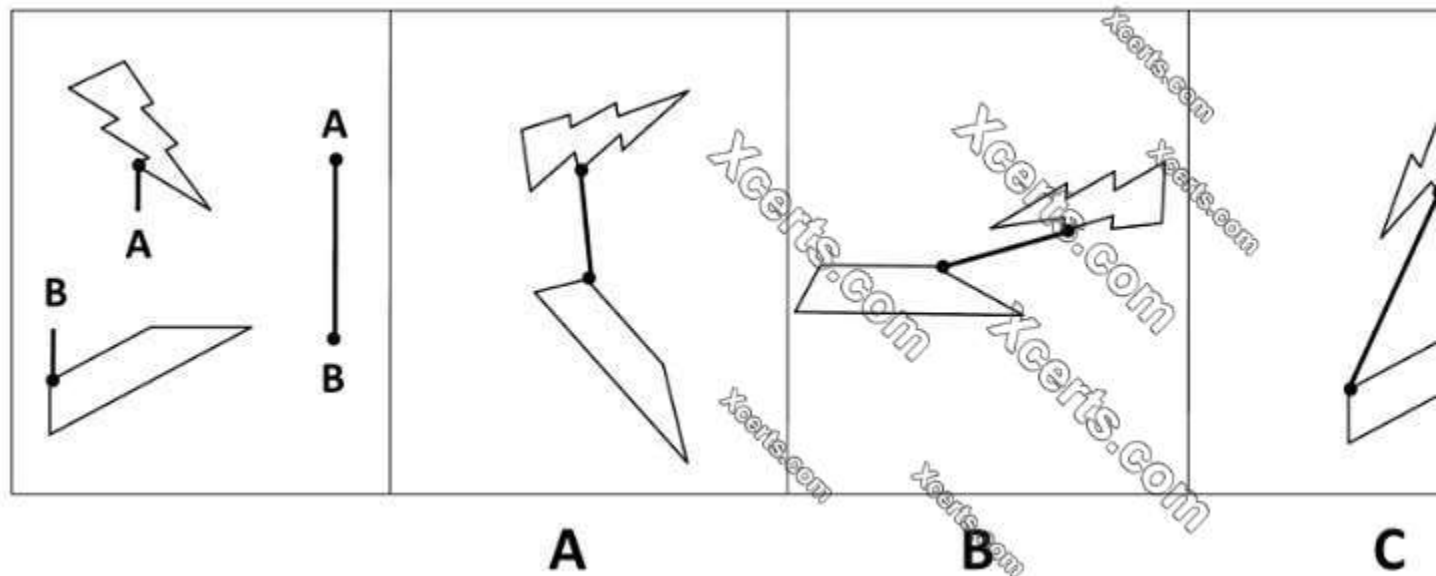


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1826

Which of these images best solves the problem in the first picture?



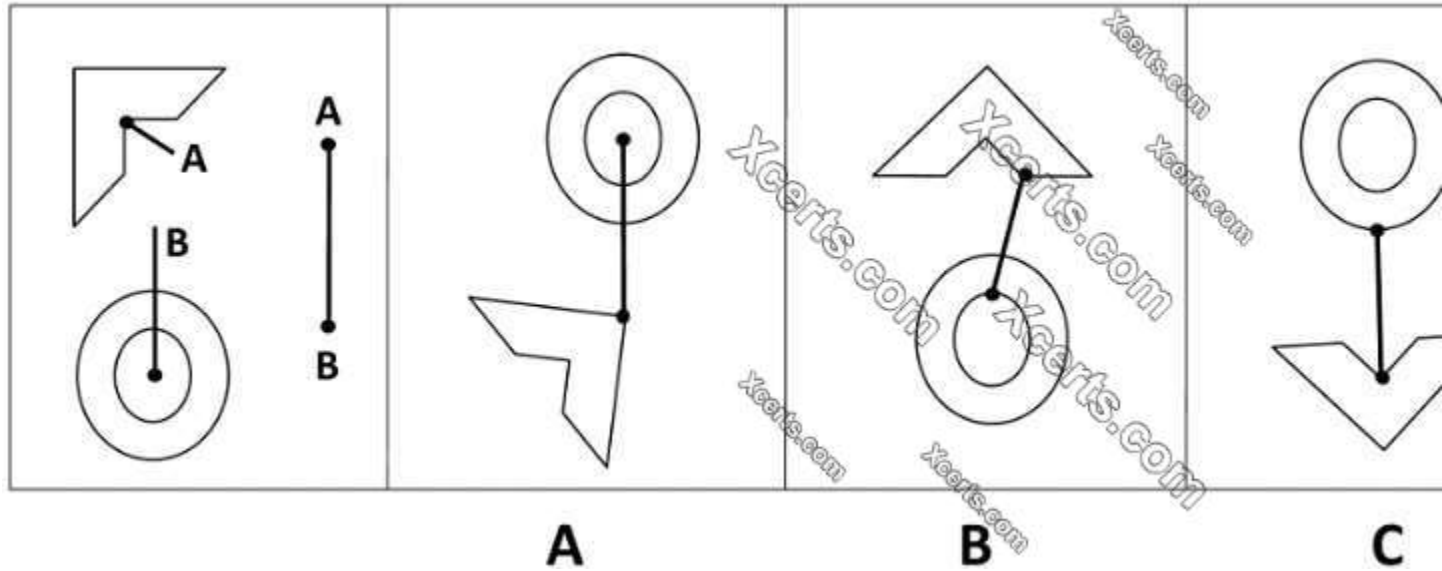
- A. Option A
- B. Option B
- C. Option C

D. Option D

Answer(s): D

QUESTION: 1827

Which of these images best solves the problem in the first picture?

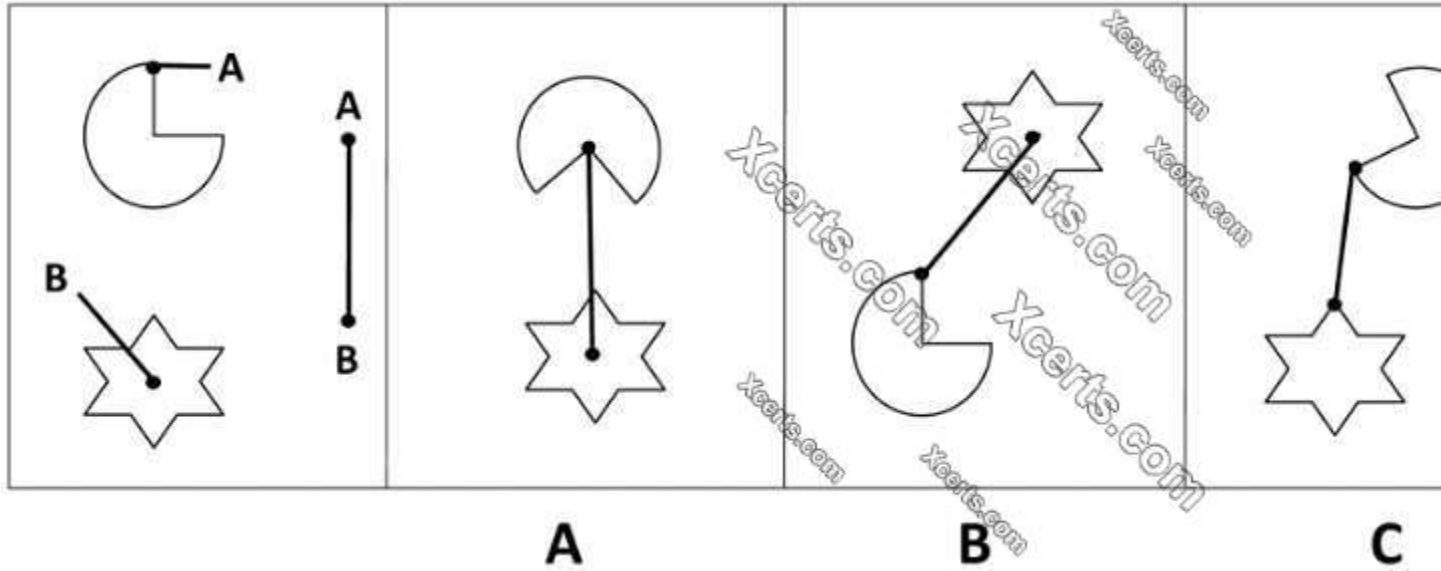


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1828

Which of these images best solves the problem in the first picture?

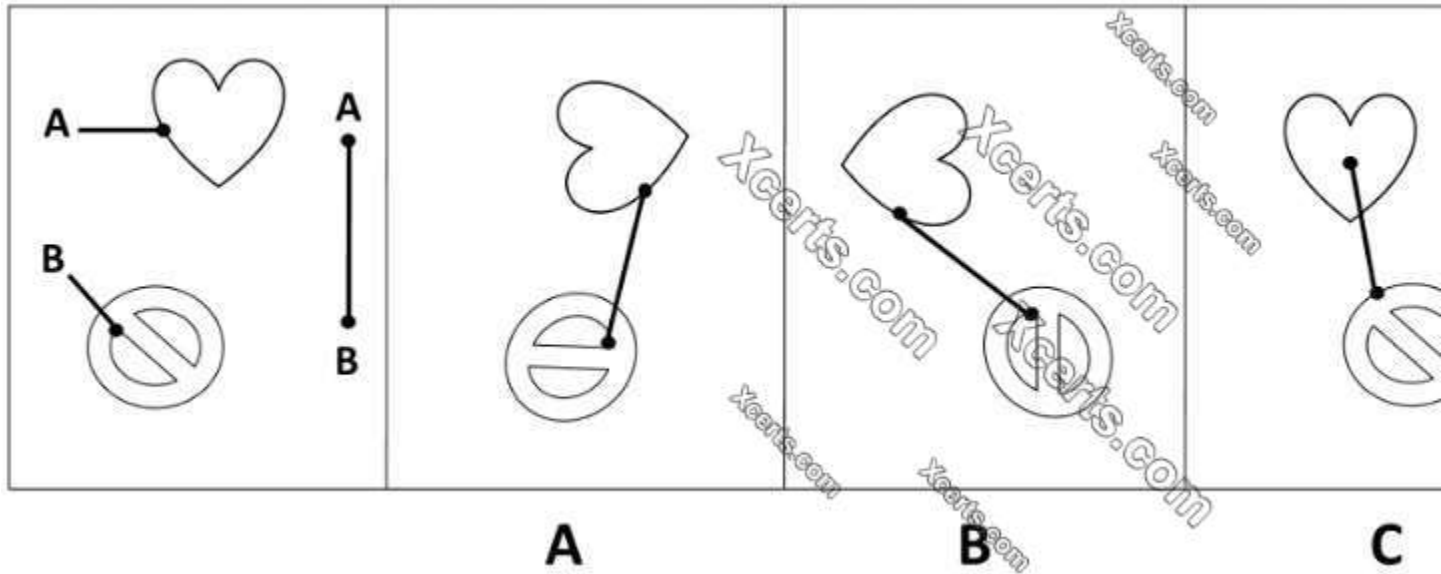


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1829

Which of these images best solves the problem in the first picture?

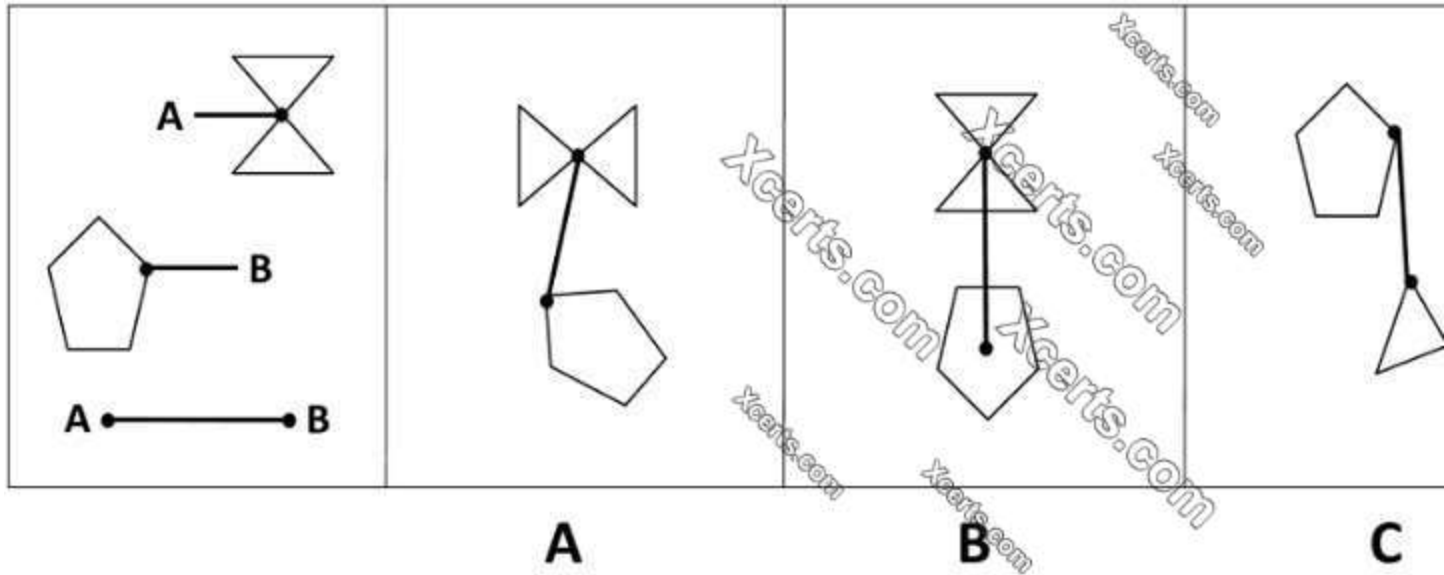


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1830

Which of these images best solves the problem in the first picture?

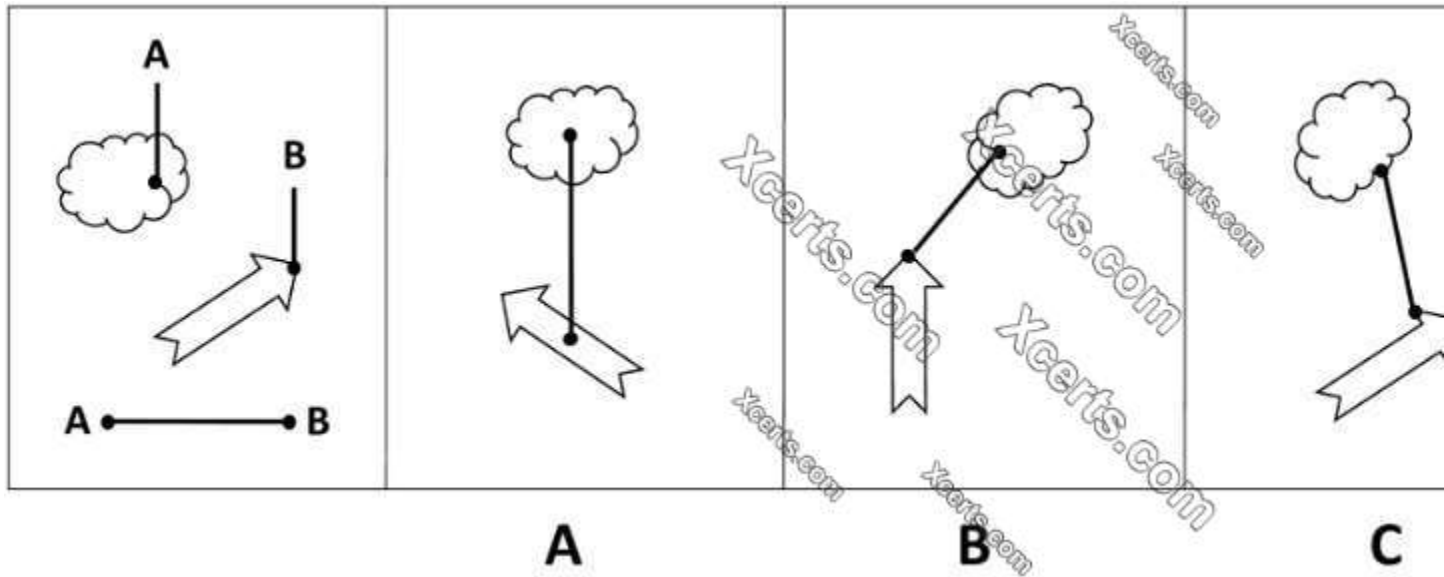


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1831

Which of these images best solves the problem in the first picture?

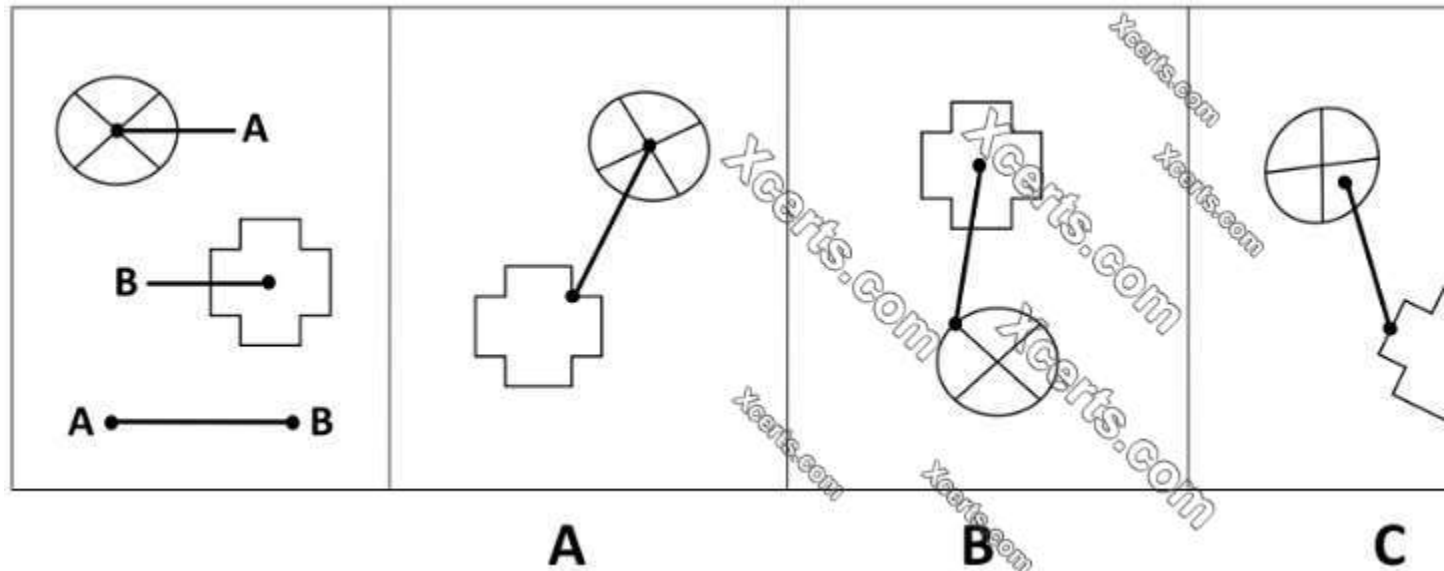


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1832

Which of these images best solves the problem in the first picture?

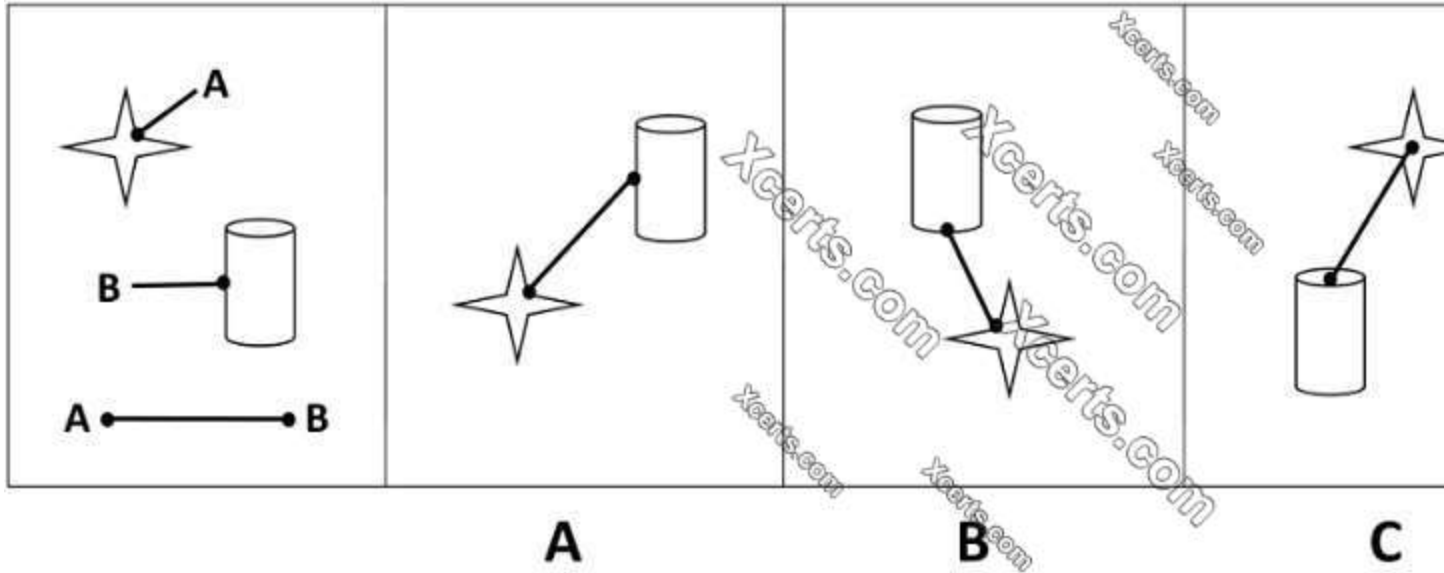


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1833

Which of these images best solves the problem in the first picture?

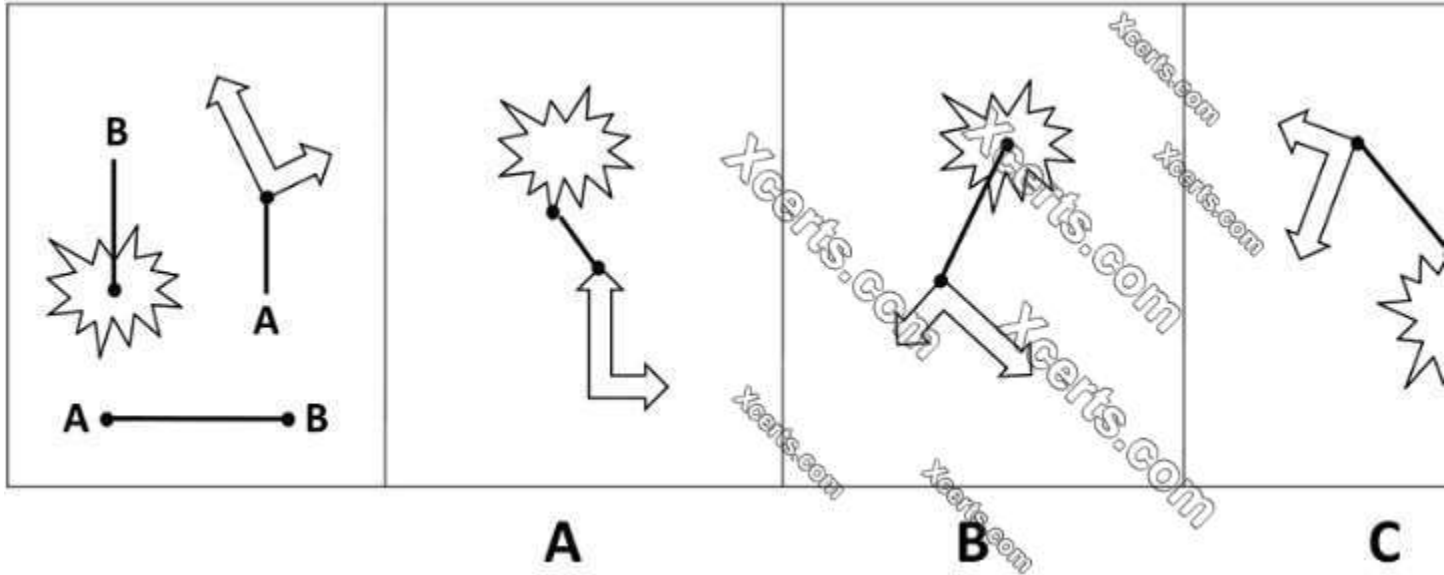


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1834

Which of these images best solves the problem in the first picture?

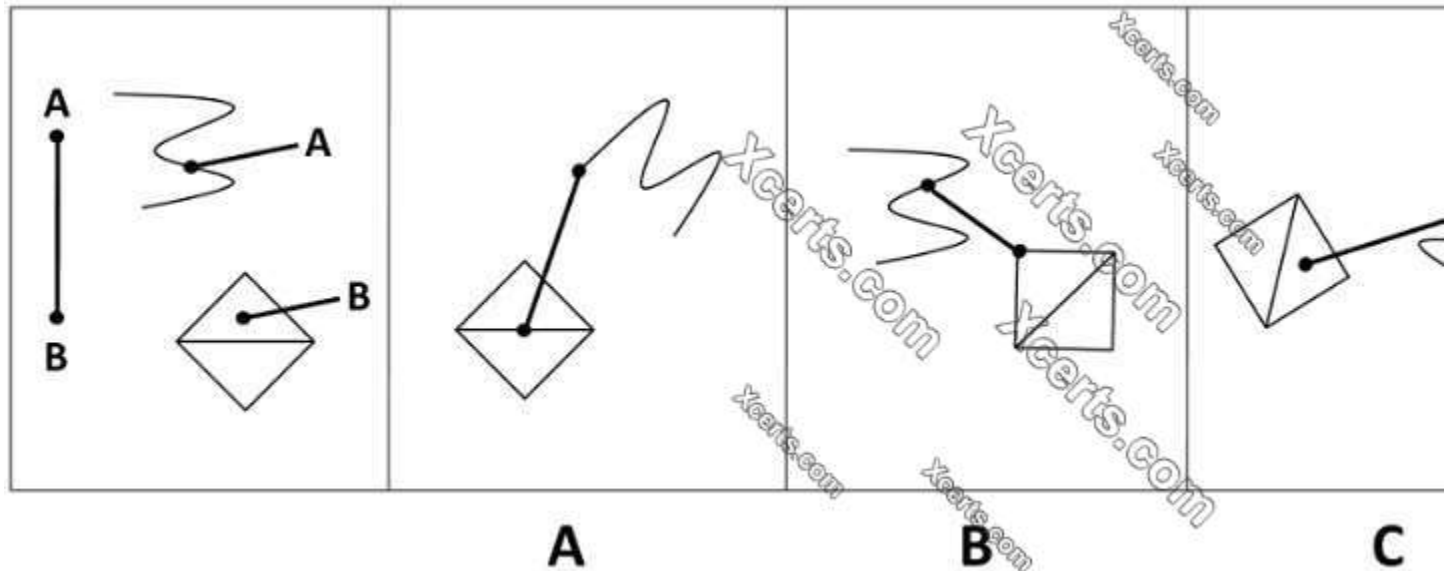


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1835

Which of these images best solves the problem in the first picture?

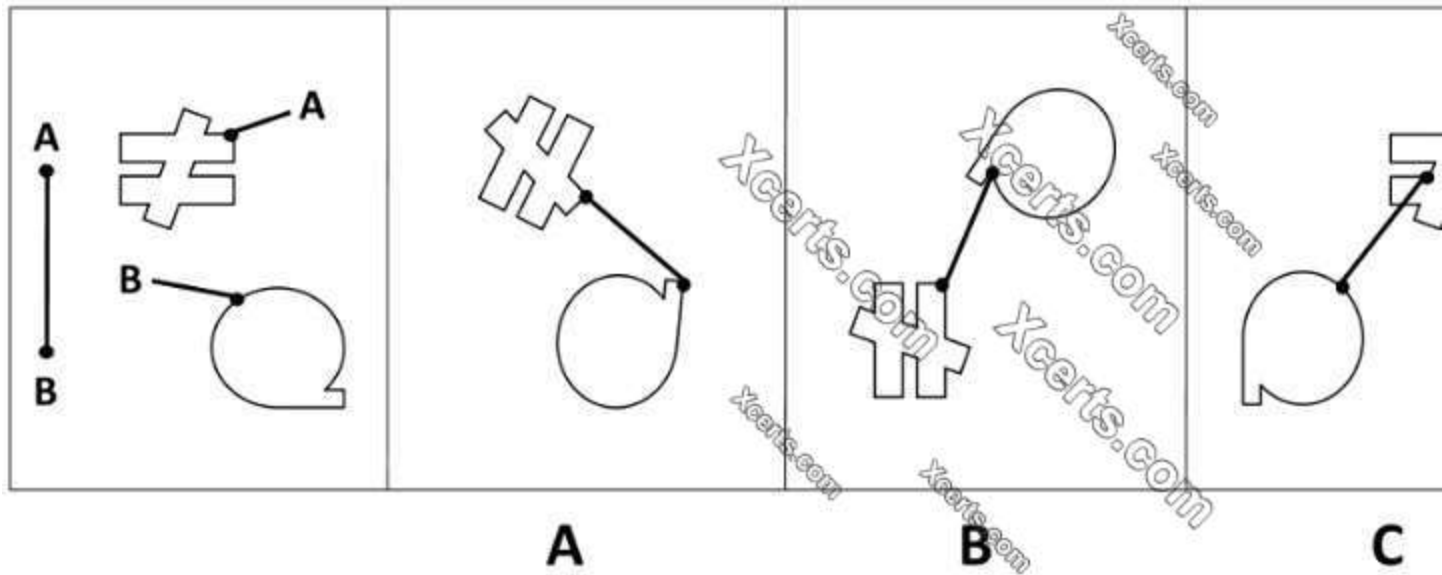


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1836

Which of these images best solves the problem in the first picture?

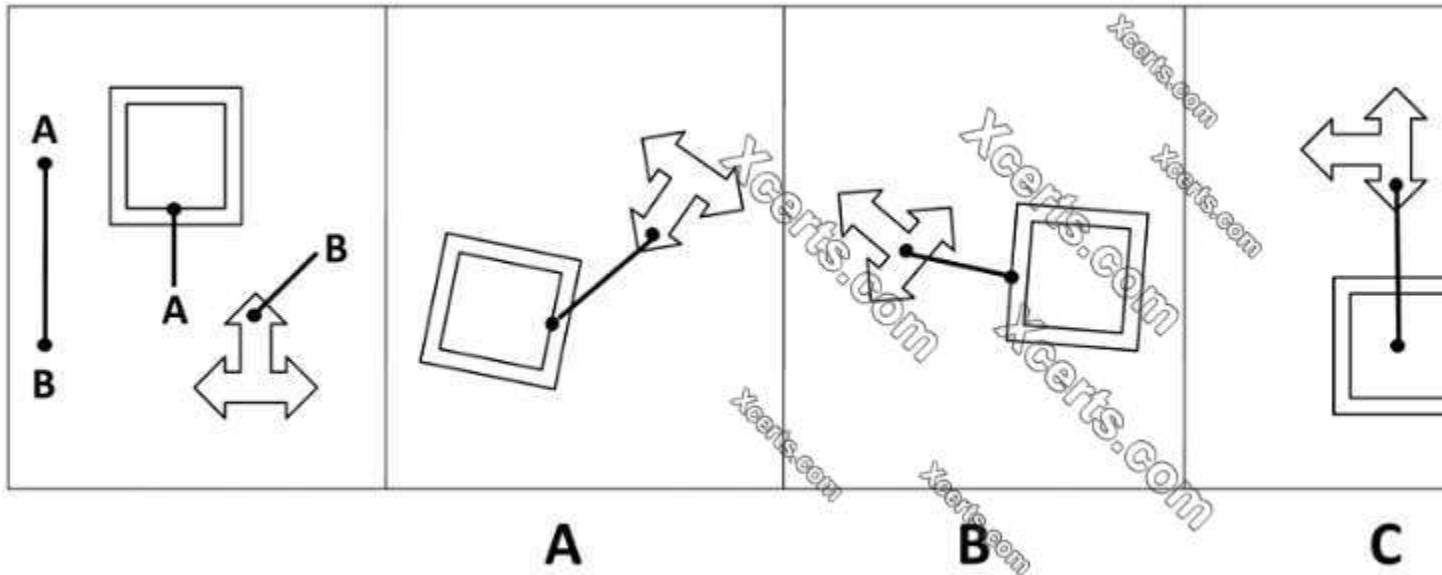


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1837

Which of these images best solves the problem in the first picture?

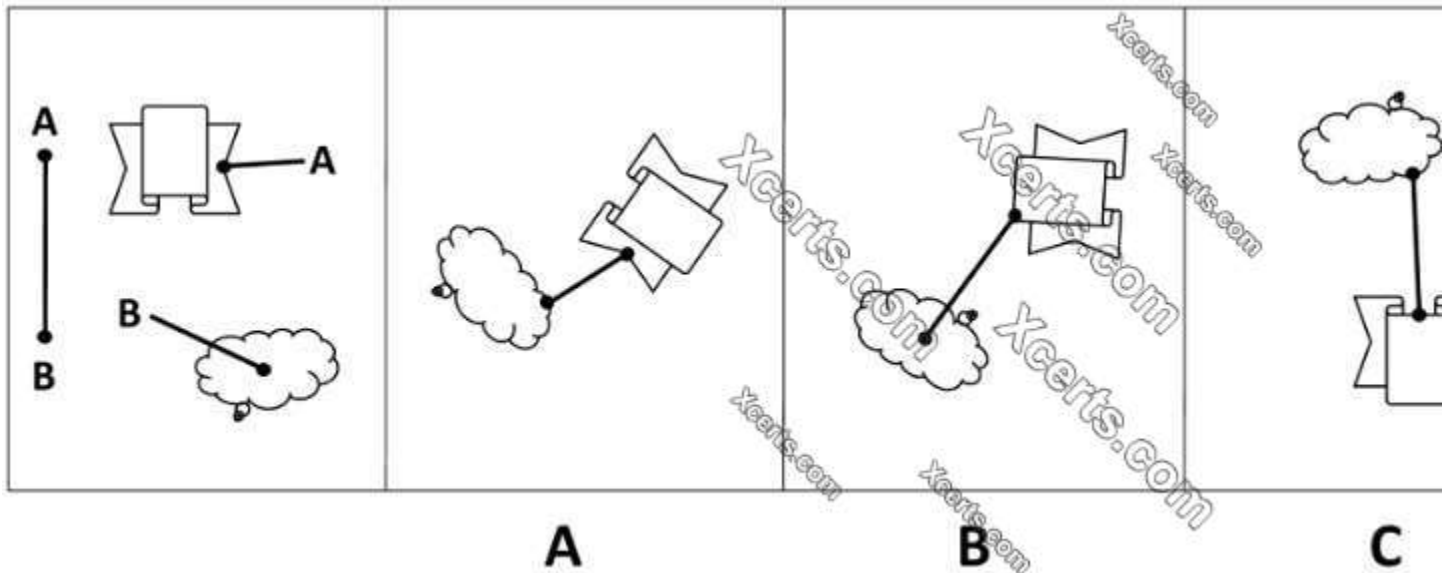


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1838

Which of these images best solves the problem in the first picture?

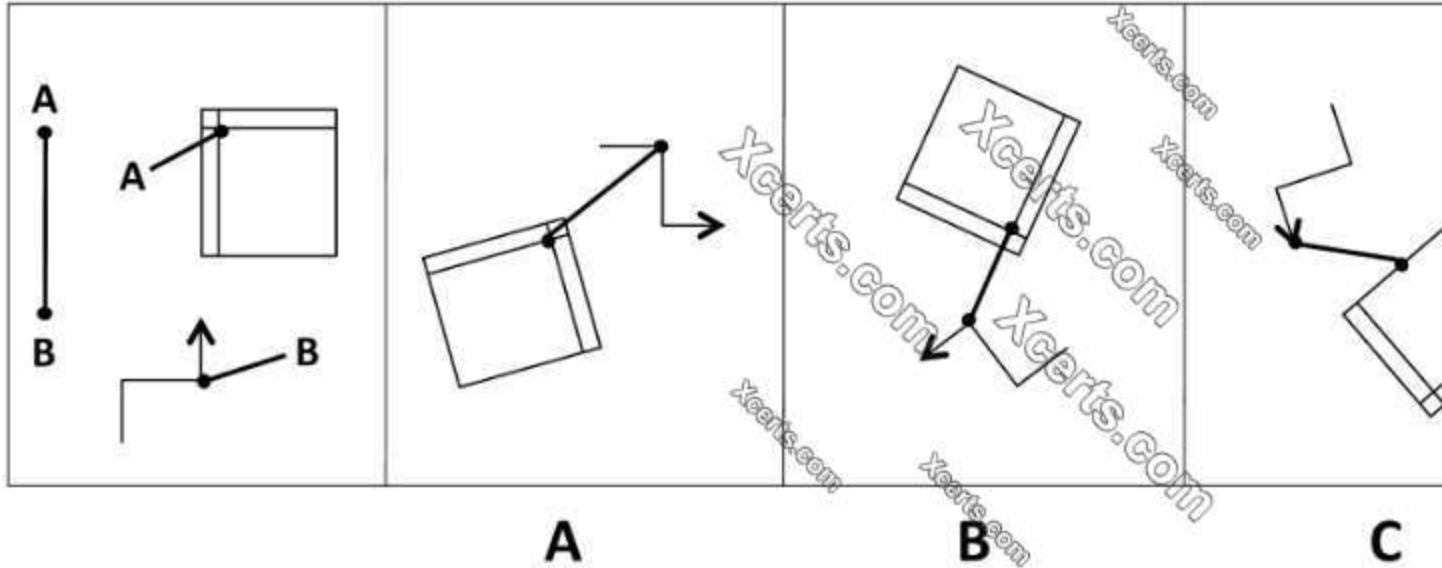


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1839

Which of these images best solves the problem in the first picture?

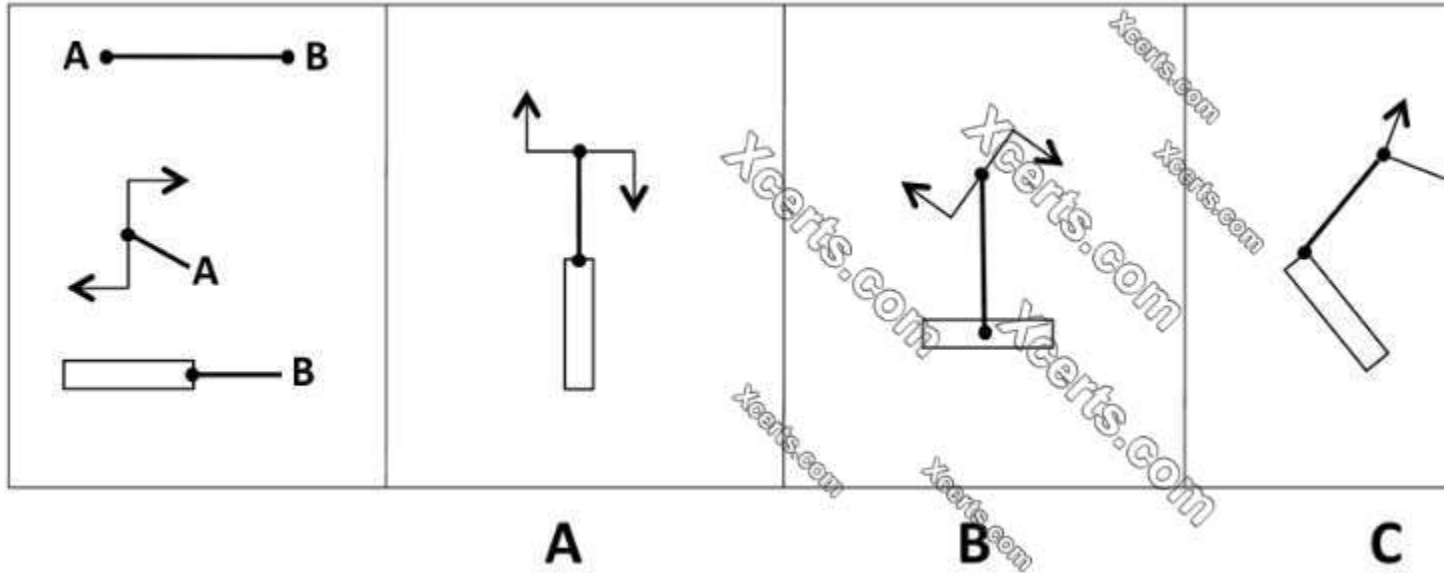


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1840

Which of these images best solves the problem in the first picture?

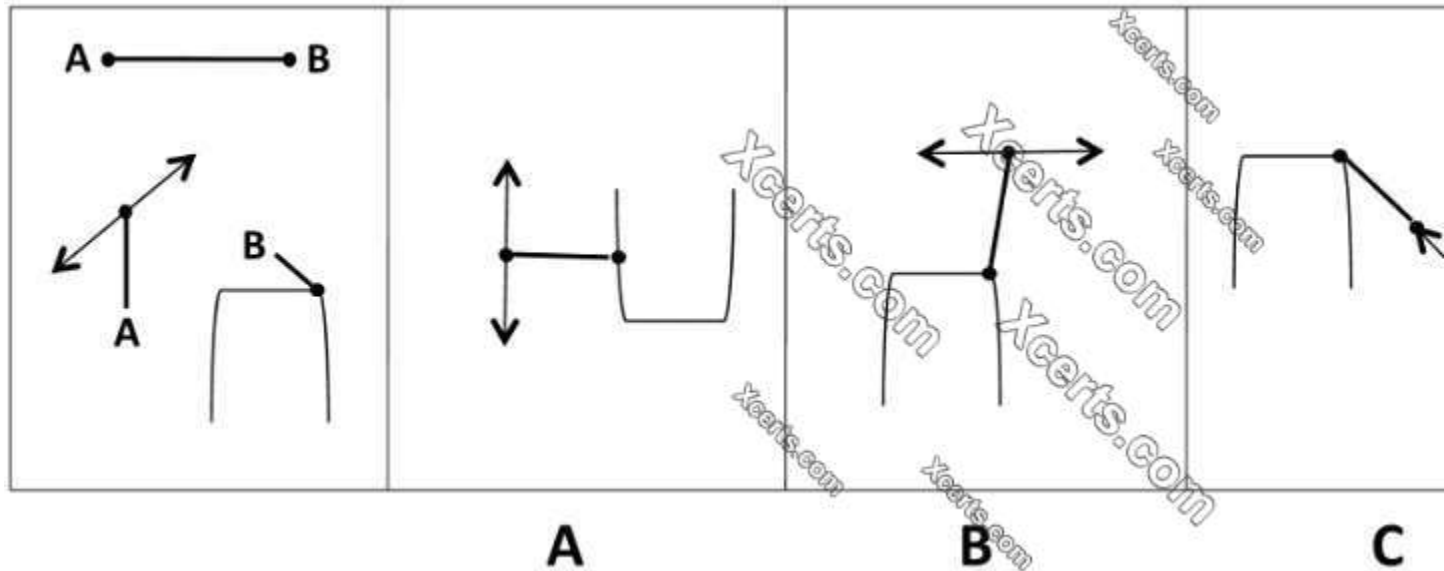


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1841

Which of these images best solves the problem in the first picture?

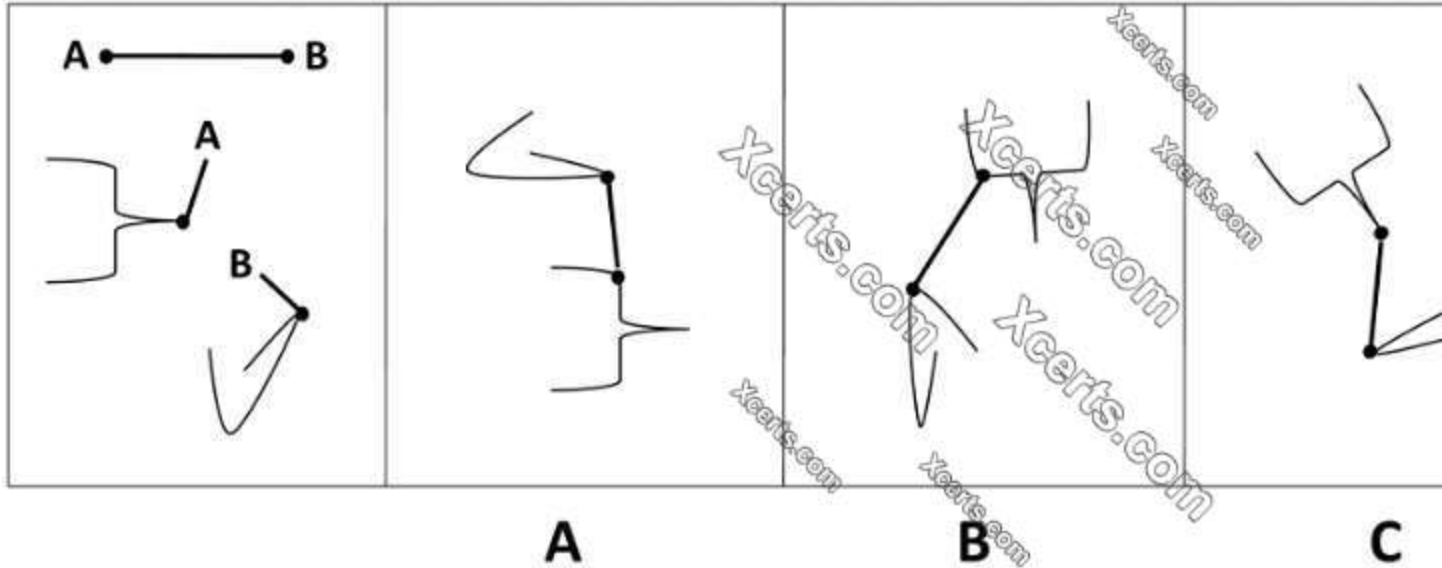


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1842

Which of these images best solves the problem in the first picture?

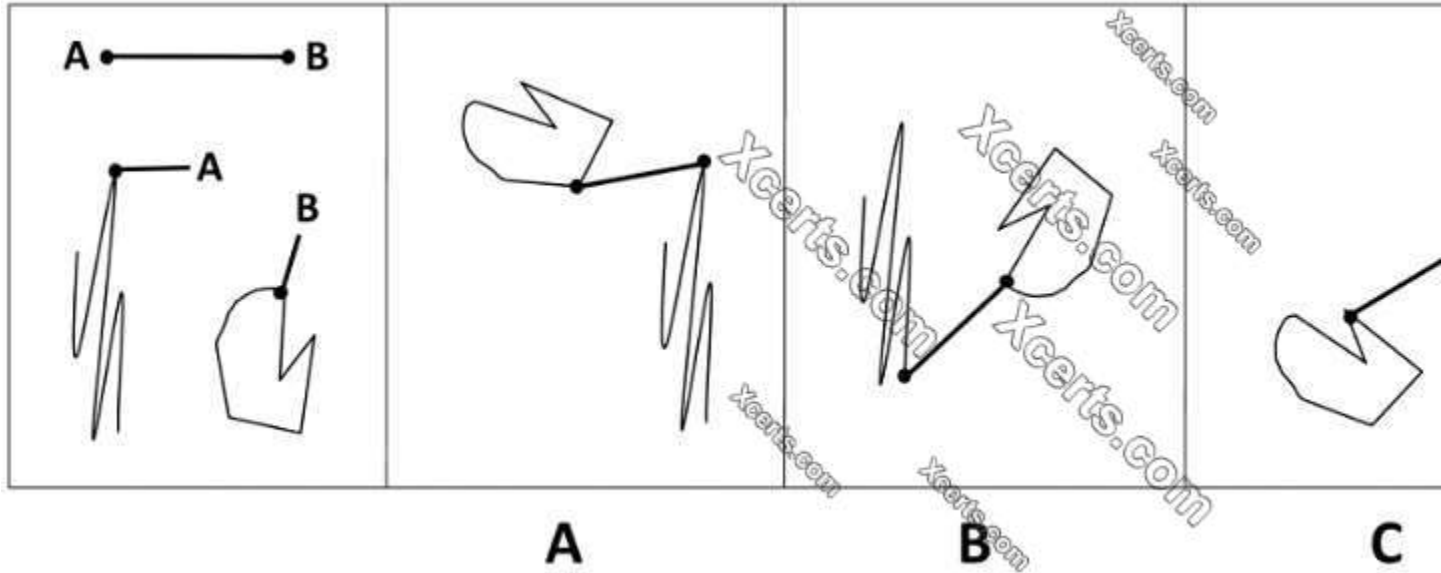


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1843

Which of these images best solves the problem in the first picture?

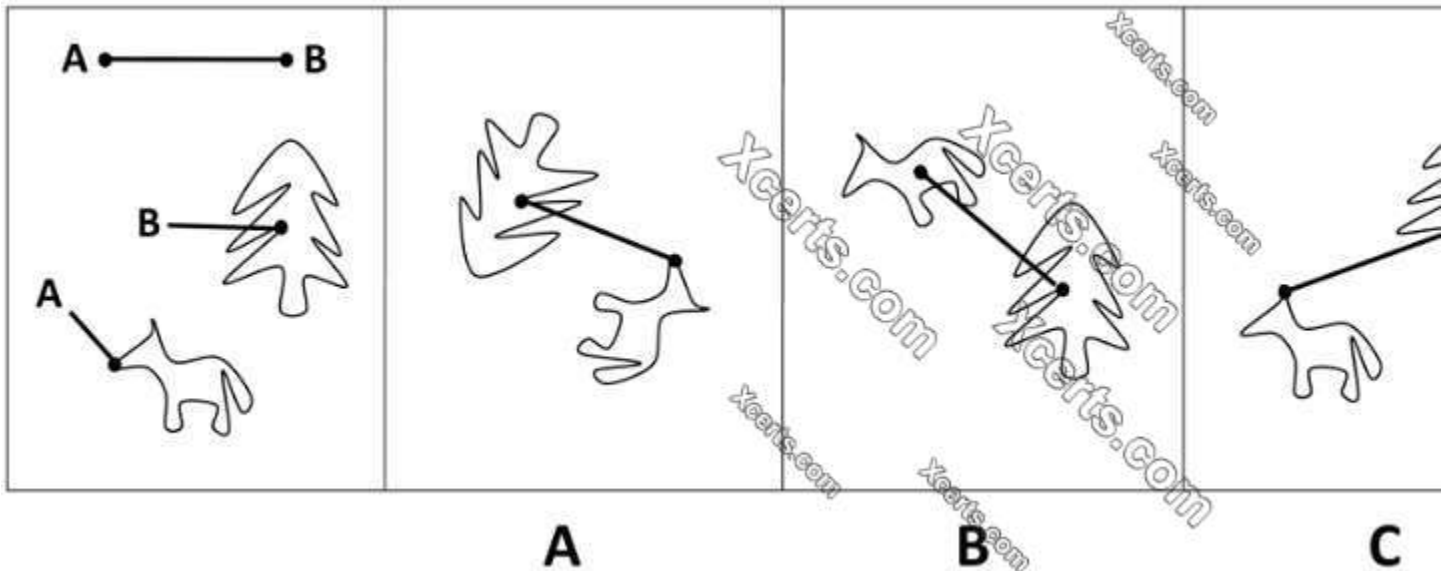


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1844

Which of these images best solves the problem in the first picture?

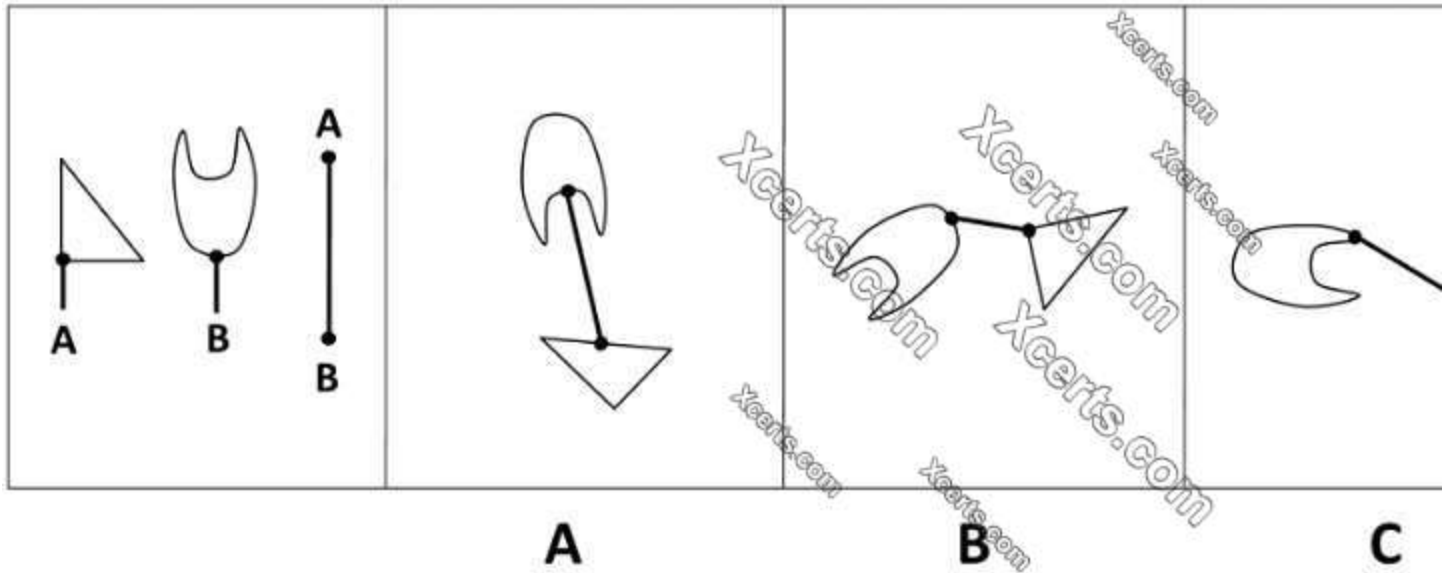


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1845

Which of these images best solves the problem in the first picture?

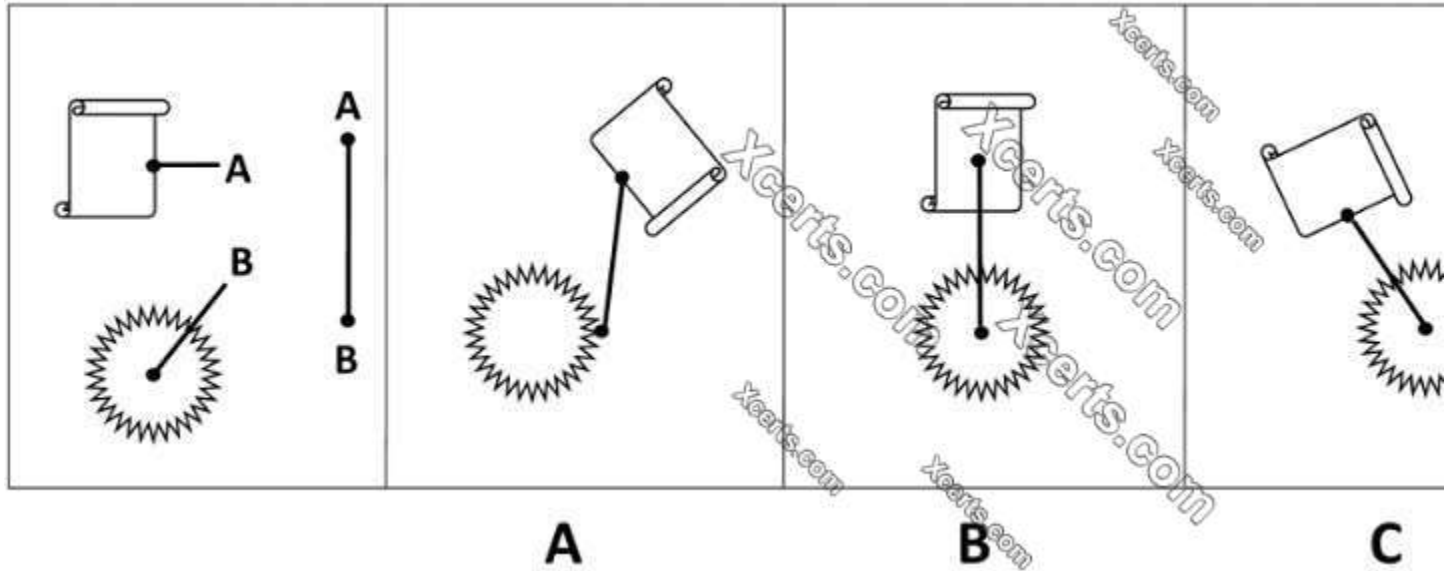


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1846

Which of these images best solves the problem in the first picture?

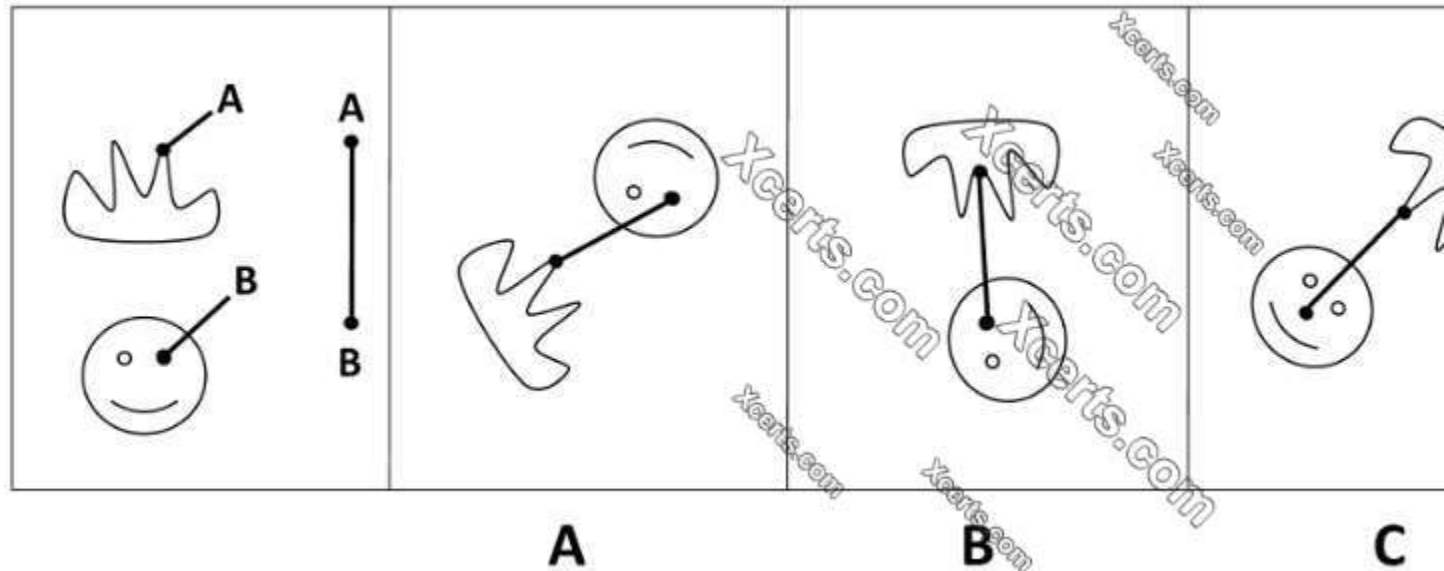


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1847

Which of these images best solves the problem in the first picture?

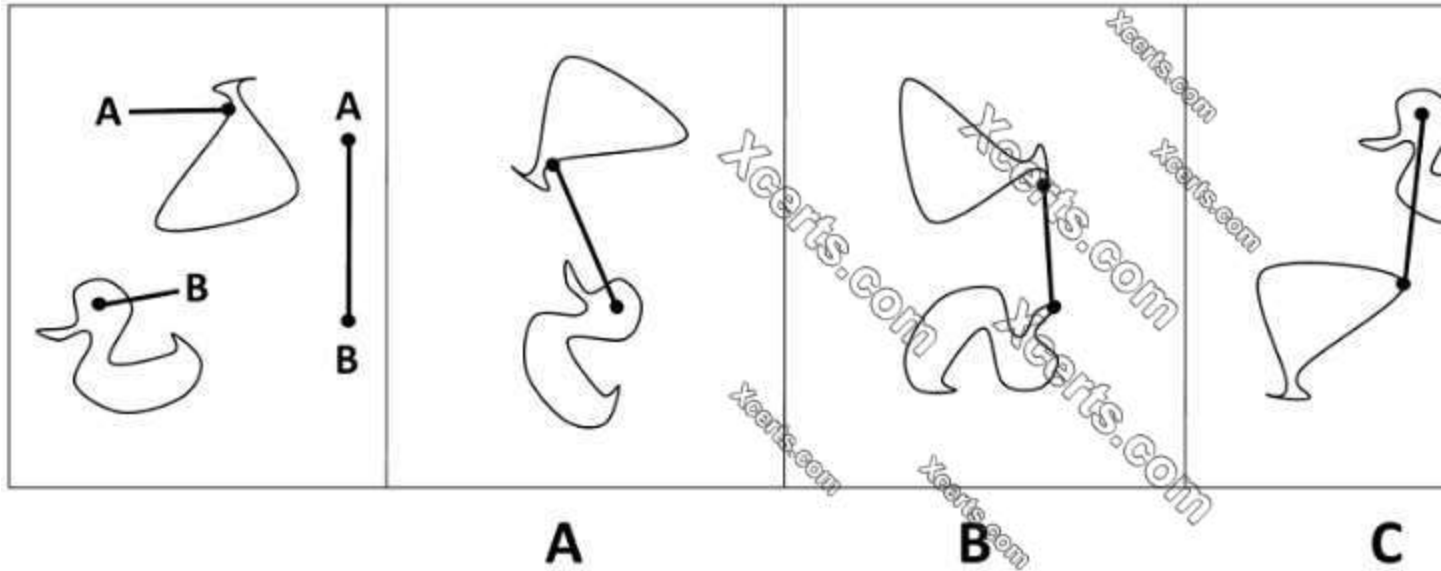


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1848

Which of these images best solves the problem in the first picture?

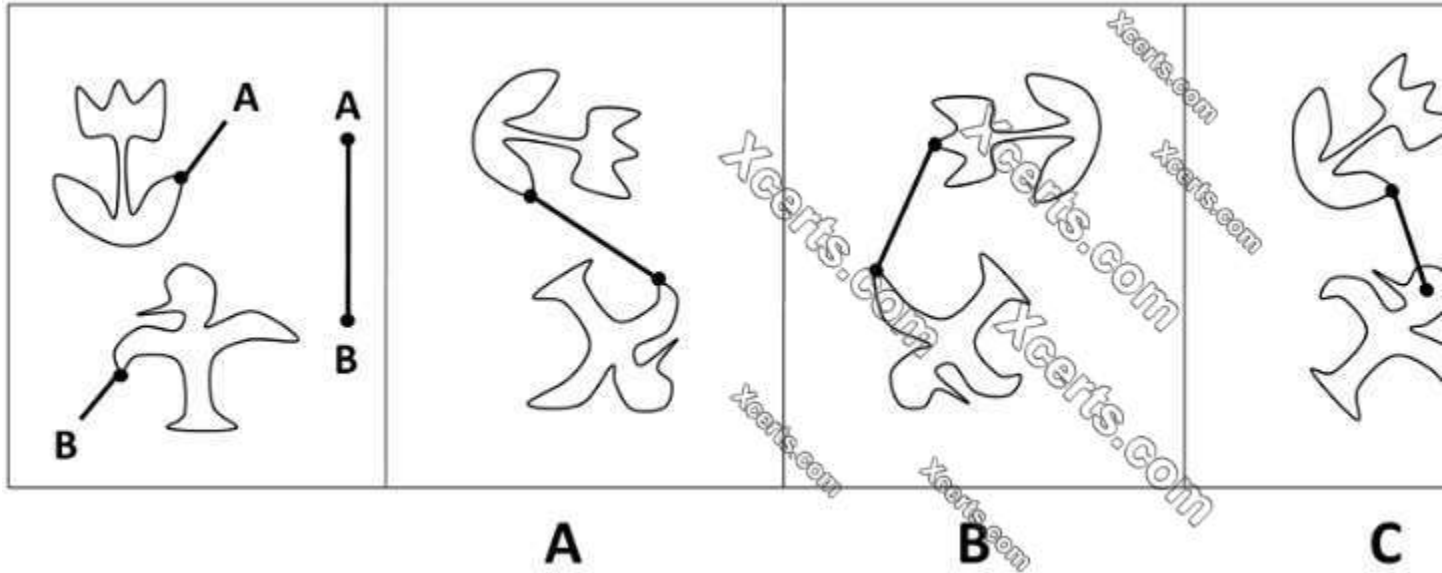


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1849

Which of these images best solves the problem in the first picture?

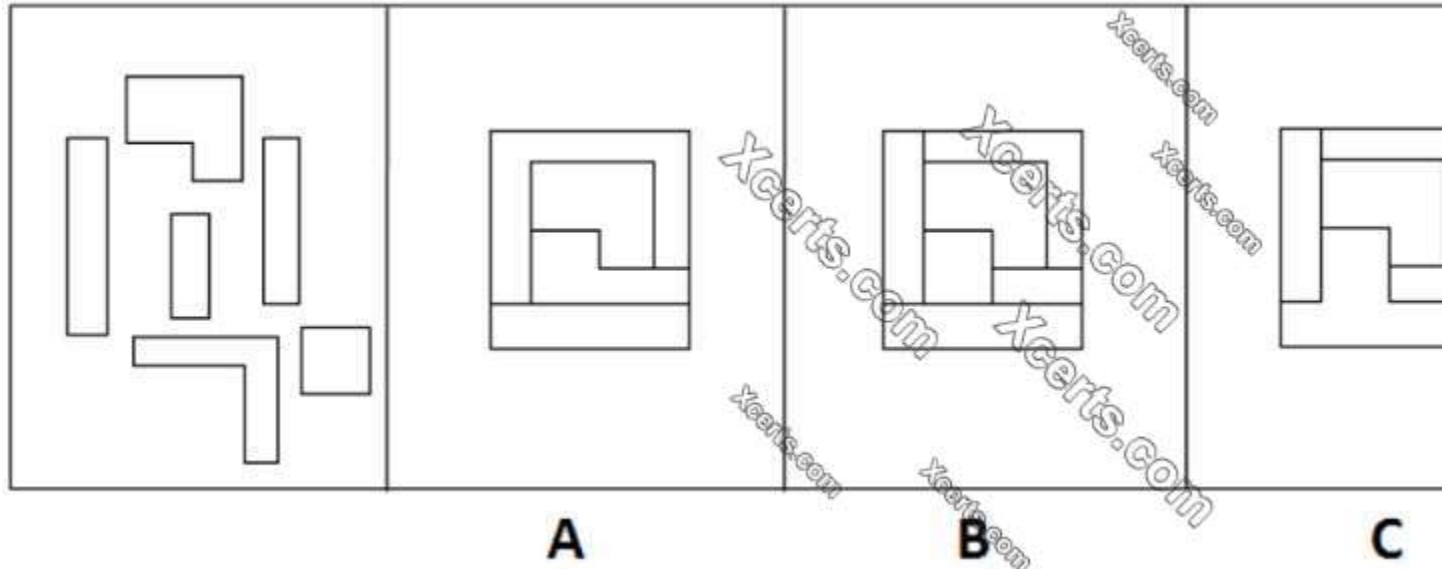


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1850

Which of these images best solves the problem in the first picture?

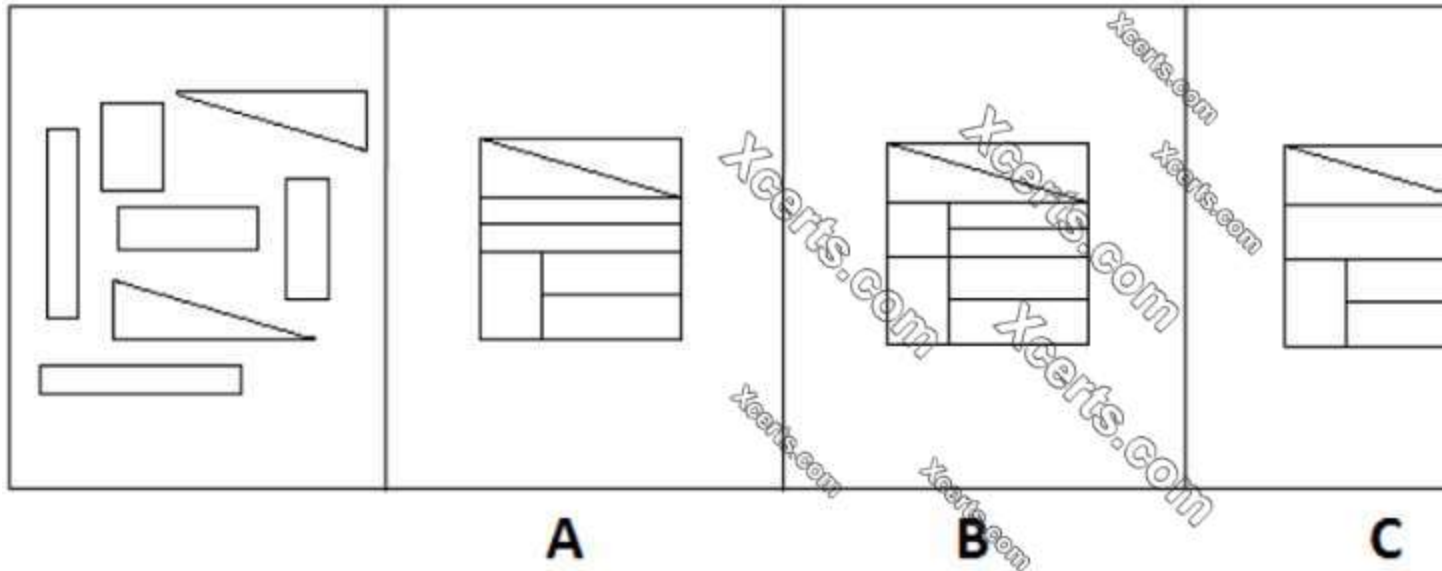


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1851

Which of these images best solves the problem in the first picture?

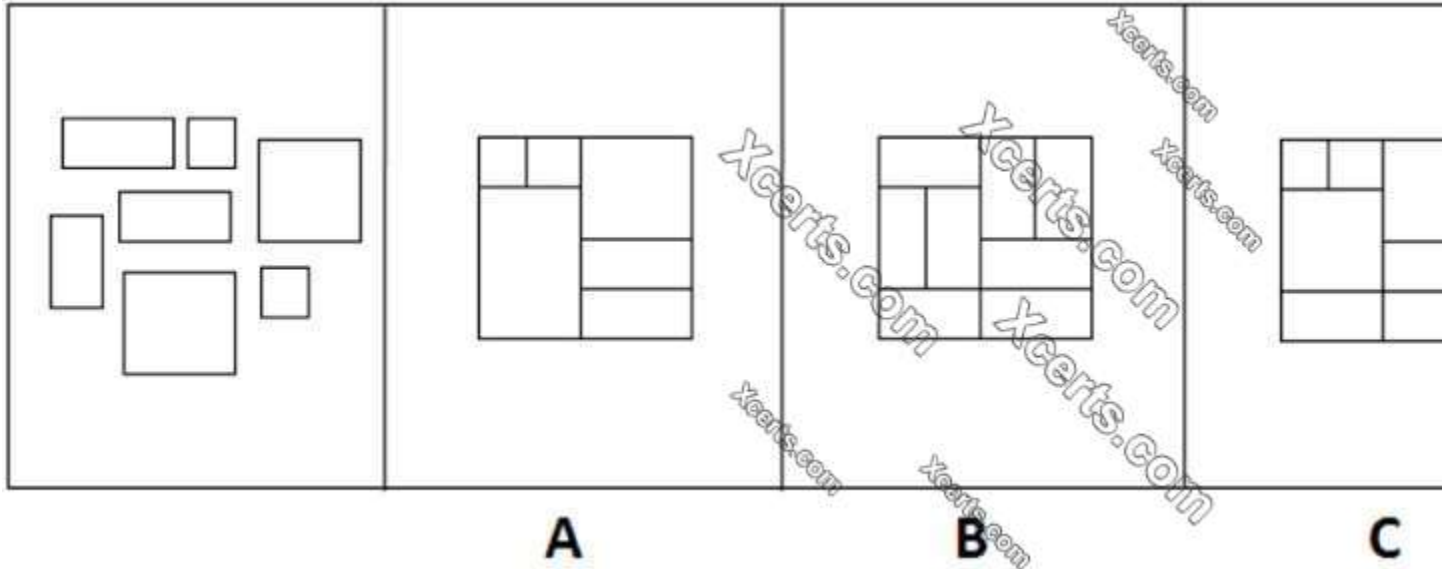


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1852

Which of these images best solves the problem in the first picture?

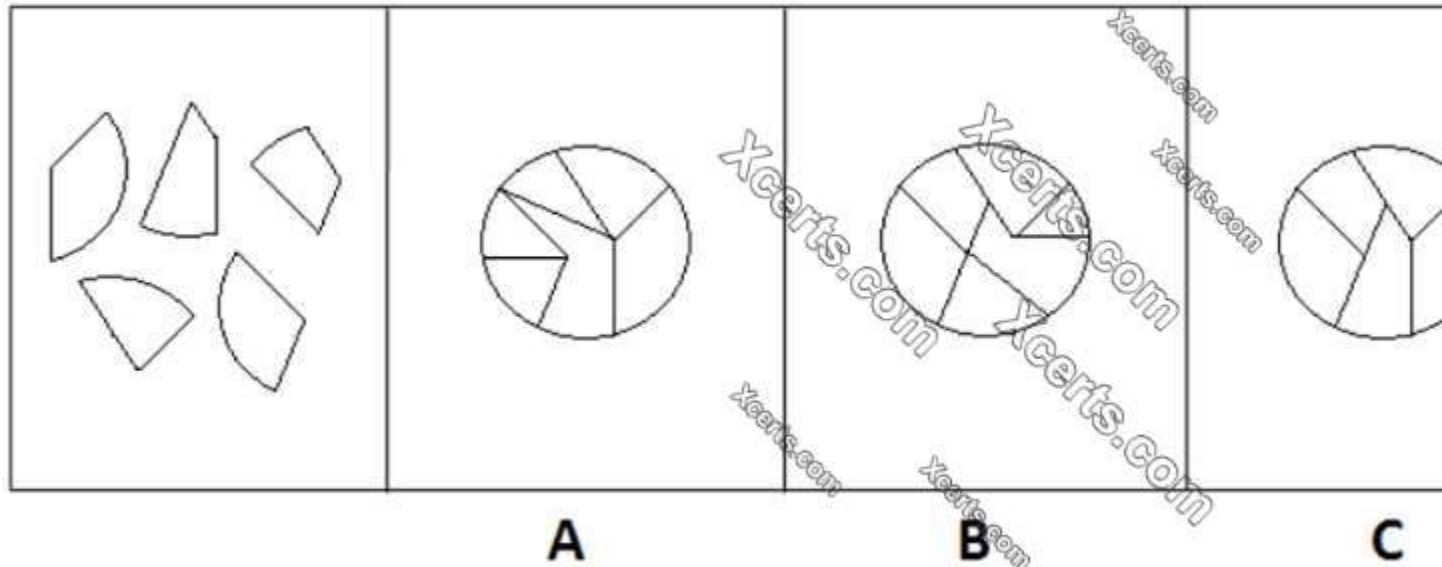


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1853

Which of these images best solves the problem in the first picture?

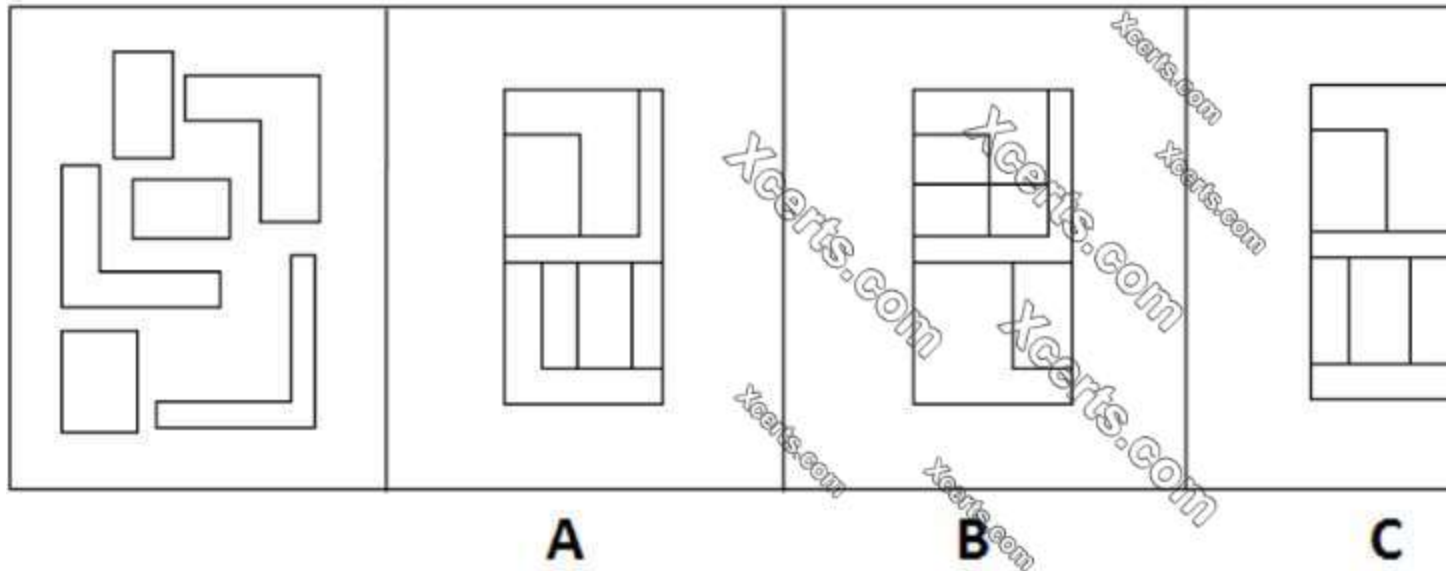


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1854

Which of these images best solves the problem in the first picture?

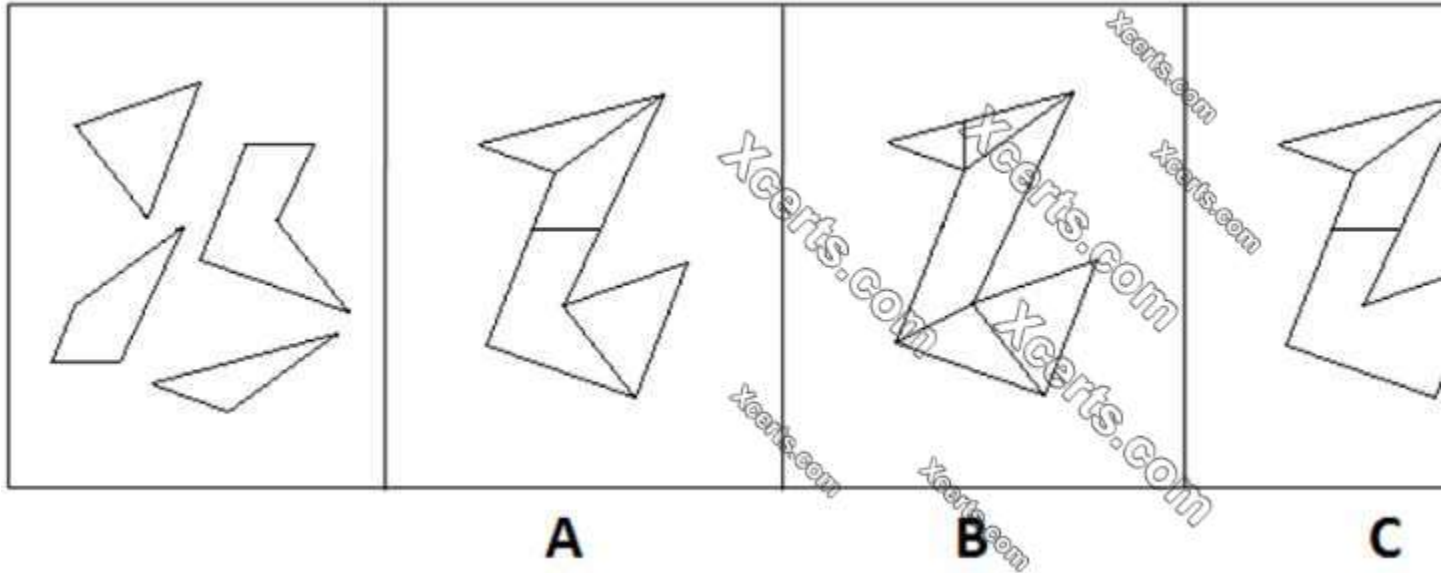


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1855

Which of these images best solves the problem in the first picture?

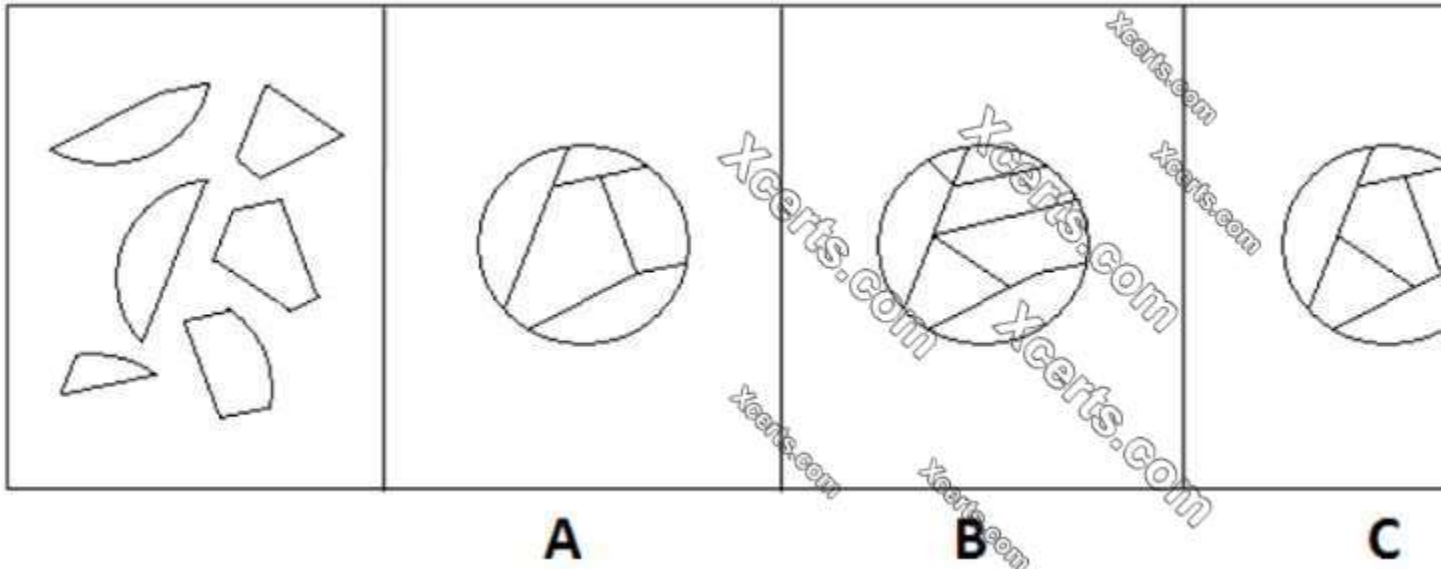


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1856

Which of these images best solves the problem in the first picture?

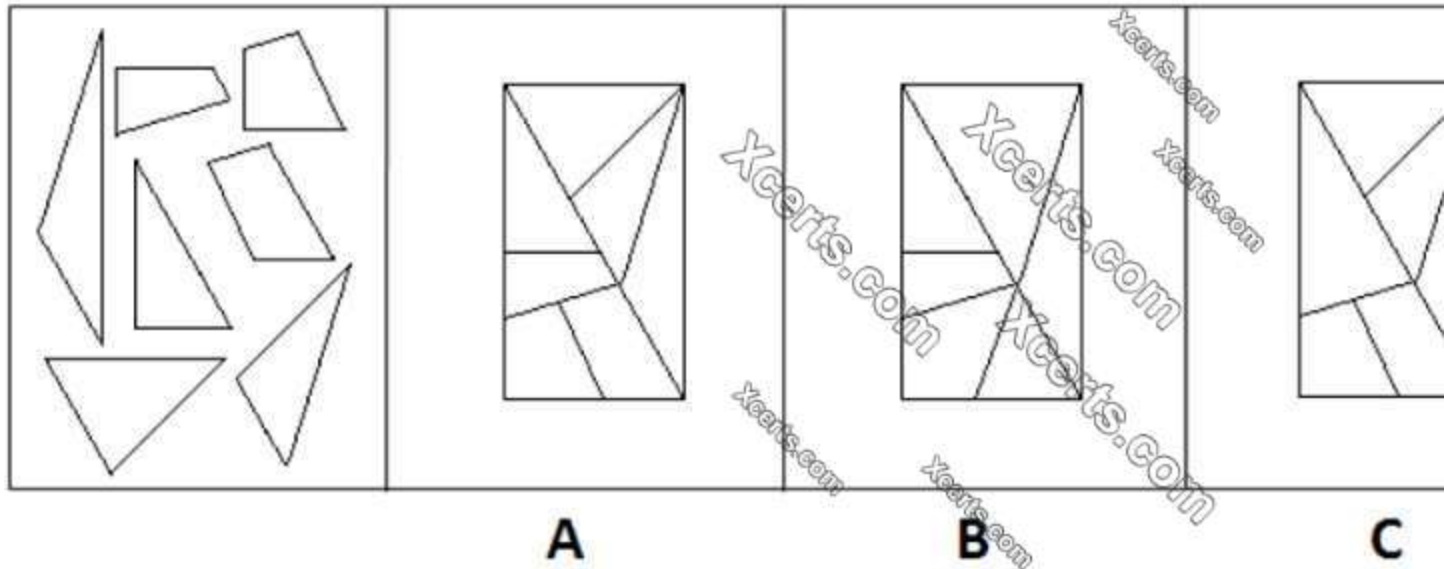


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1857

Which of these images best solves the problem in the first picture?

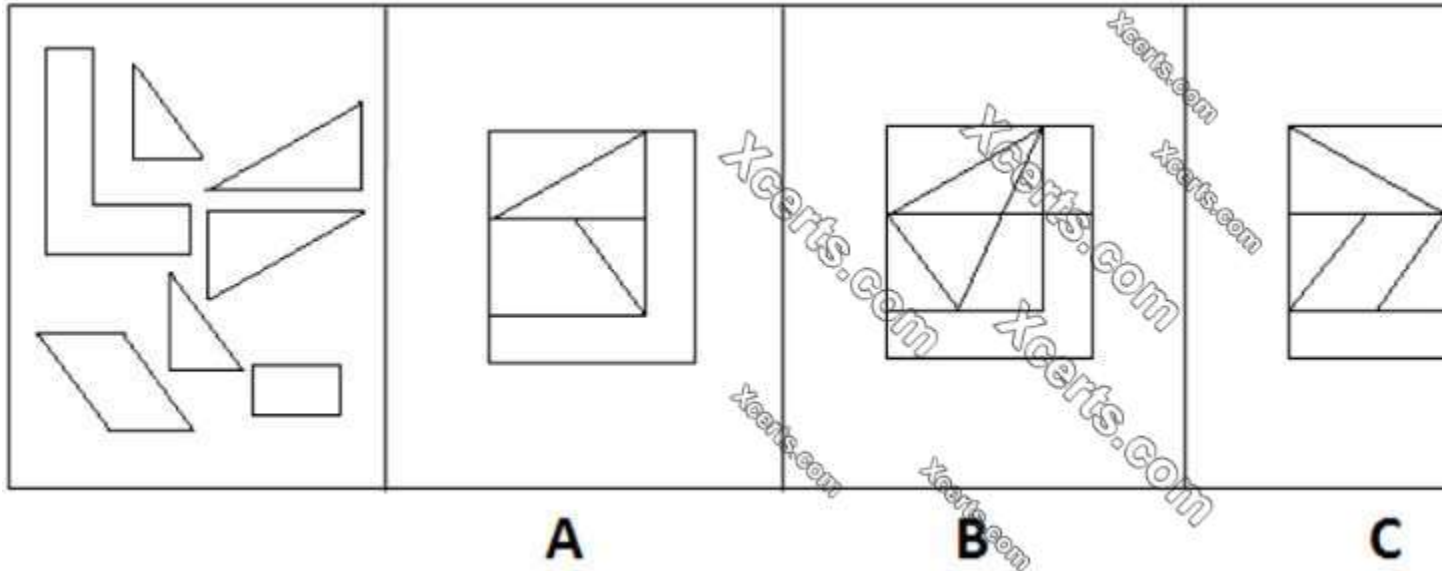


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1858

Which of these images best solves the problem in the first picture?

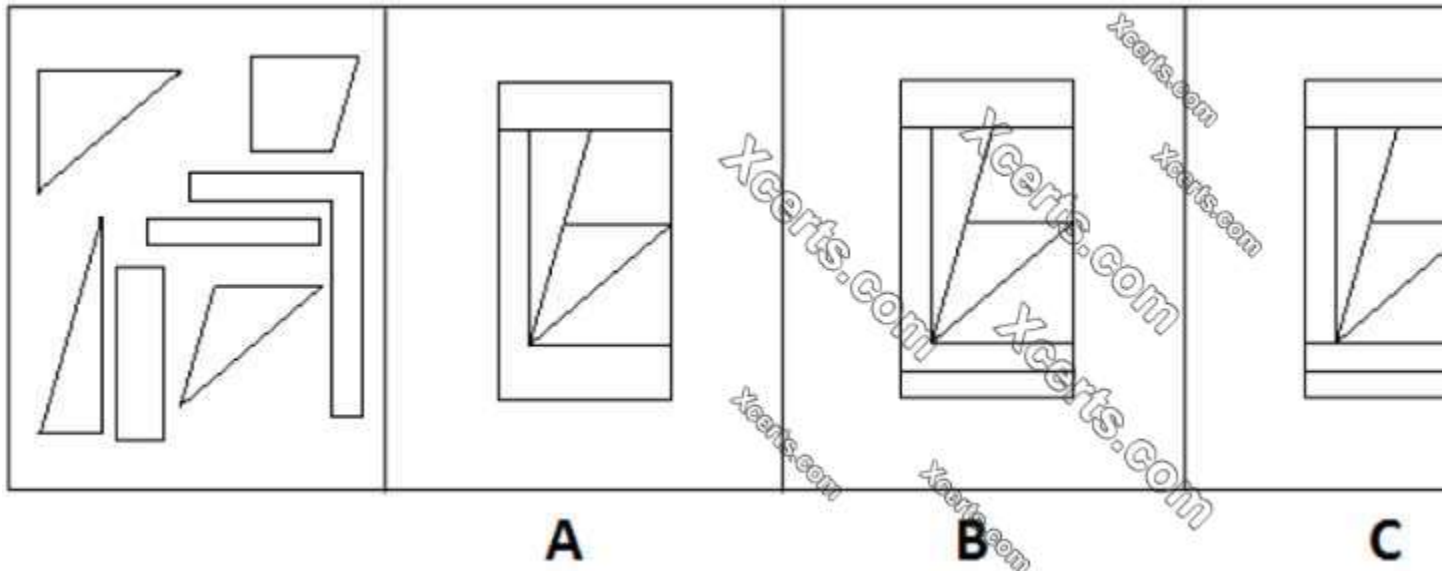


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1859

Which of these images best solves the problem in the first picture?

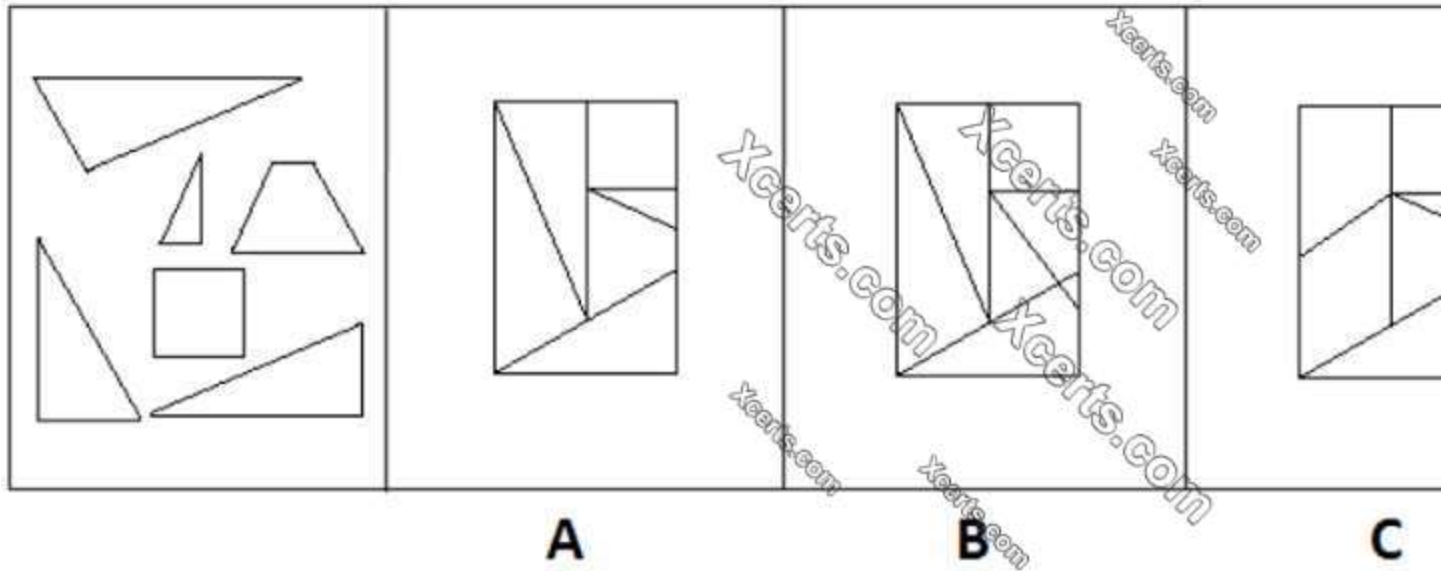


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1860

Which of these images best solves the problem in the first picture?

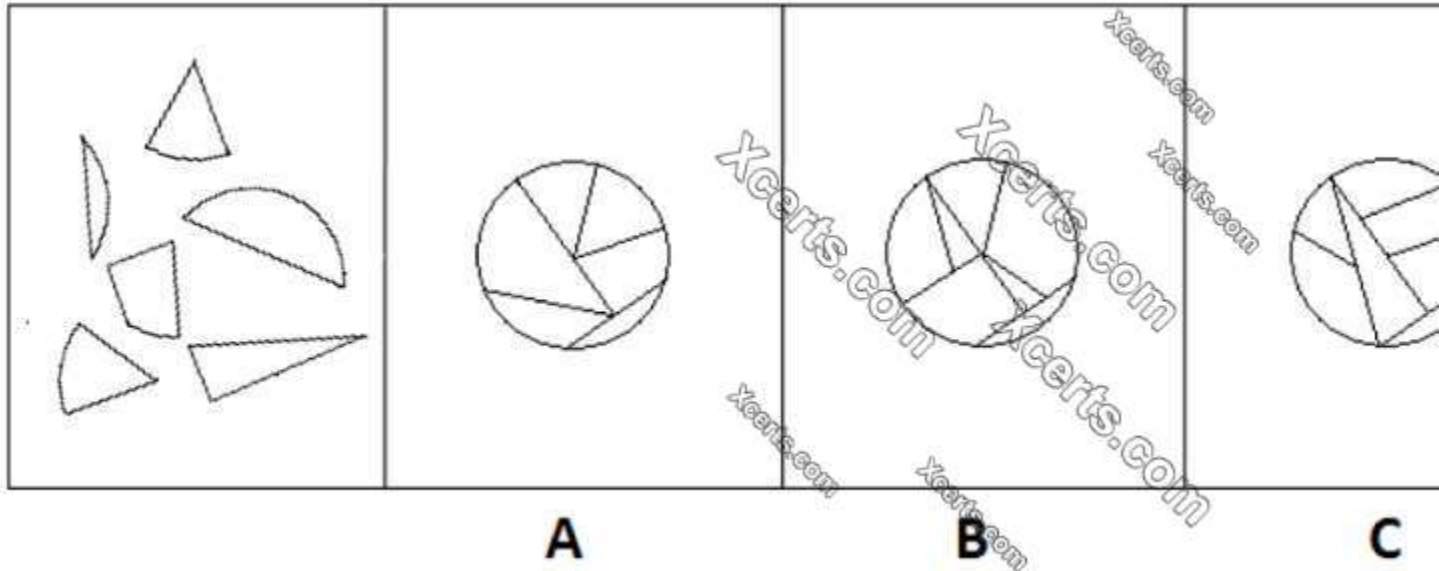


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1861

Which of these images best solves the problem in the first picture?

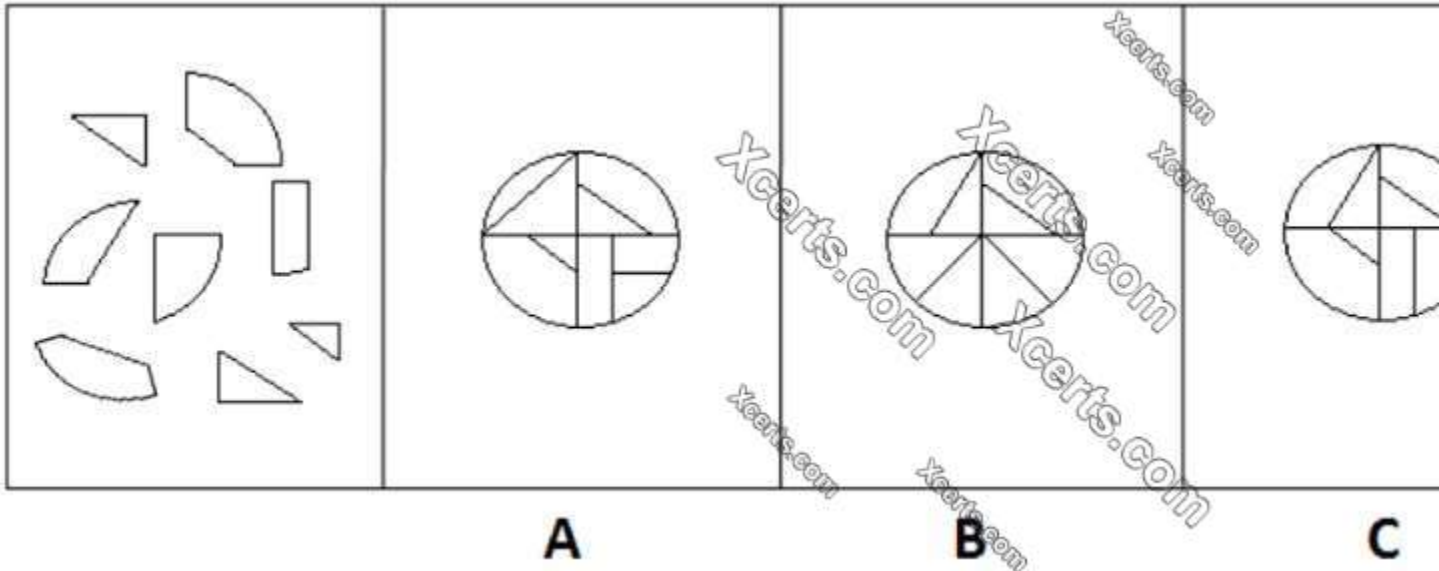


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1862

Which of these images best solves the problem in the first picture?

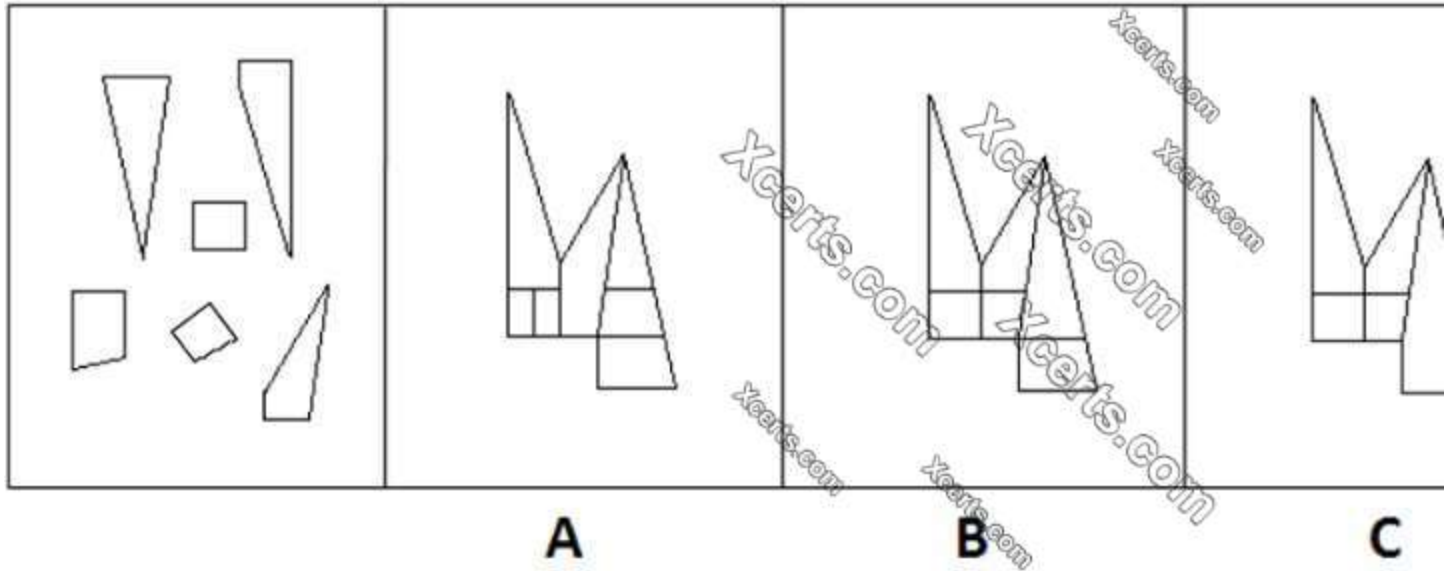


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1863

Which of these images best solves the problem in the first picture?

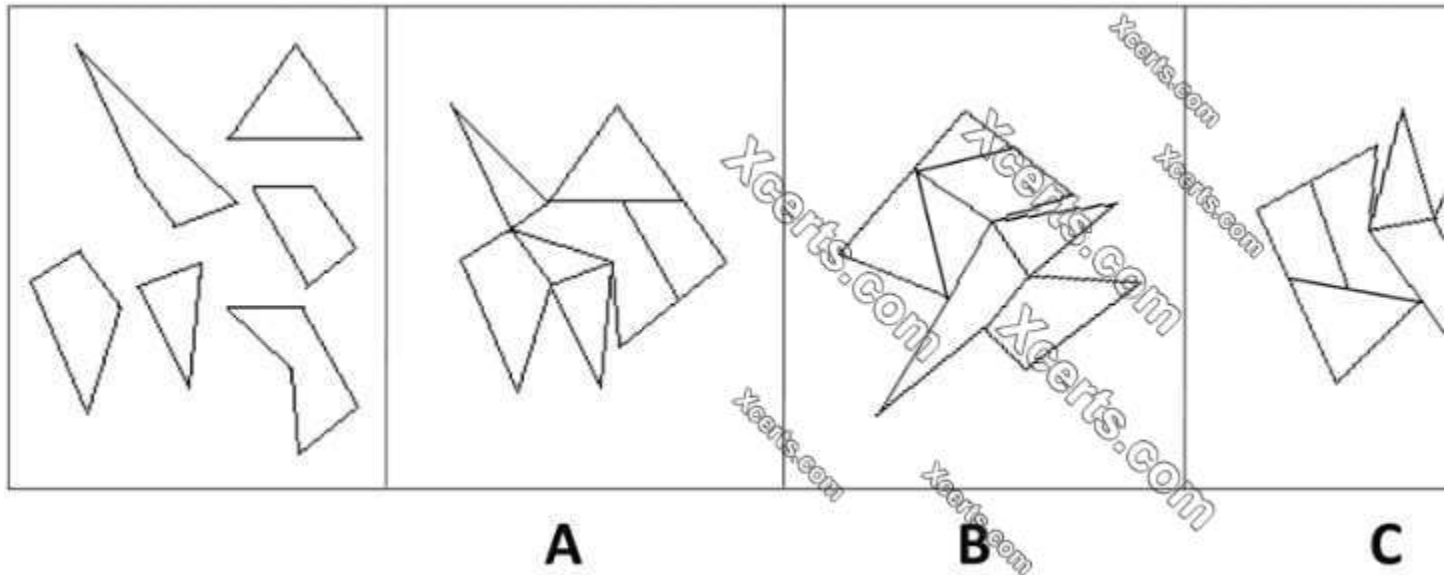


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1864

Which of these images best solves the problem in the first picture?

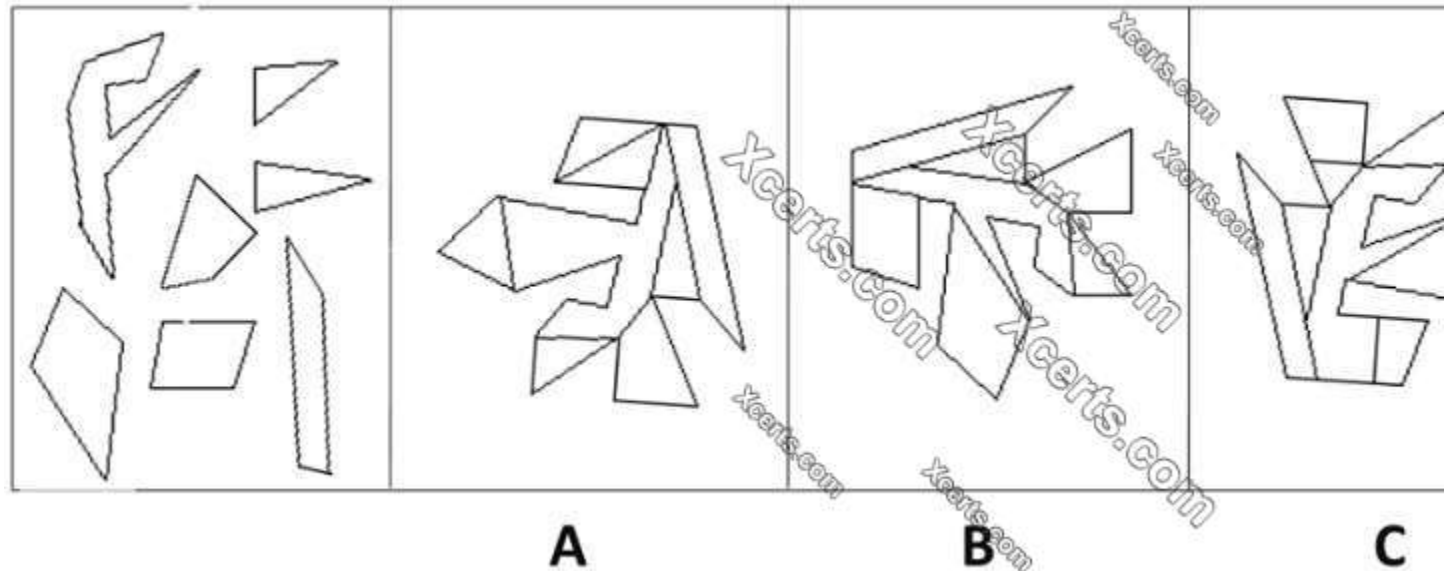


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1865

Which of these images best solves the problem in the first picture?

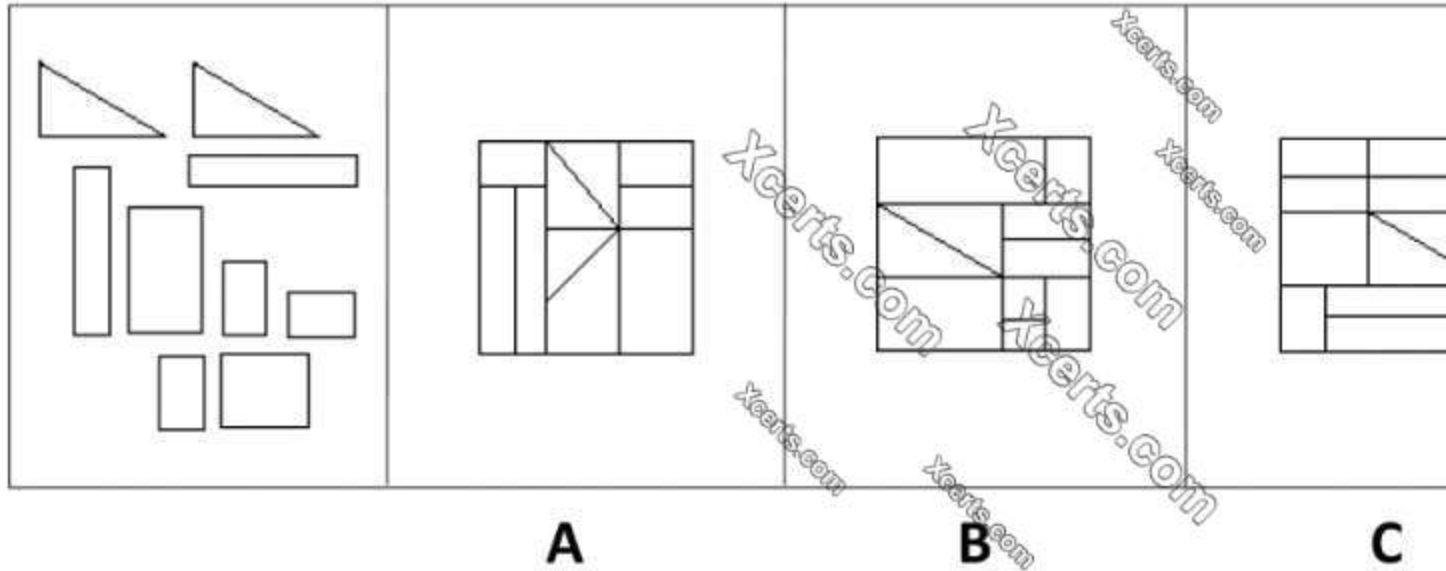


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1866

Which of these images best solves the problem in the first picture?

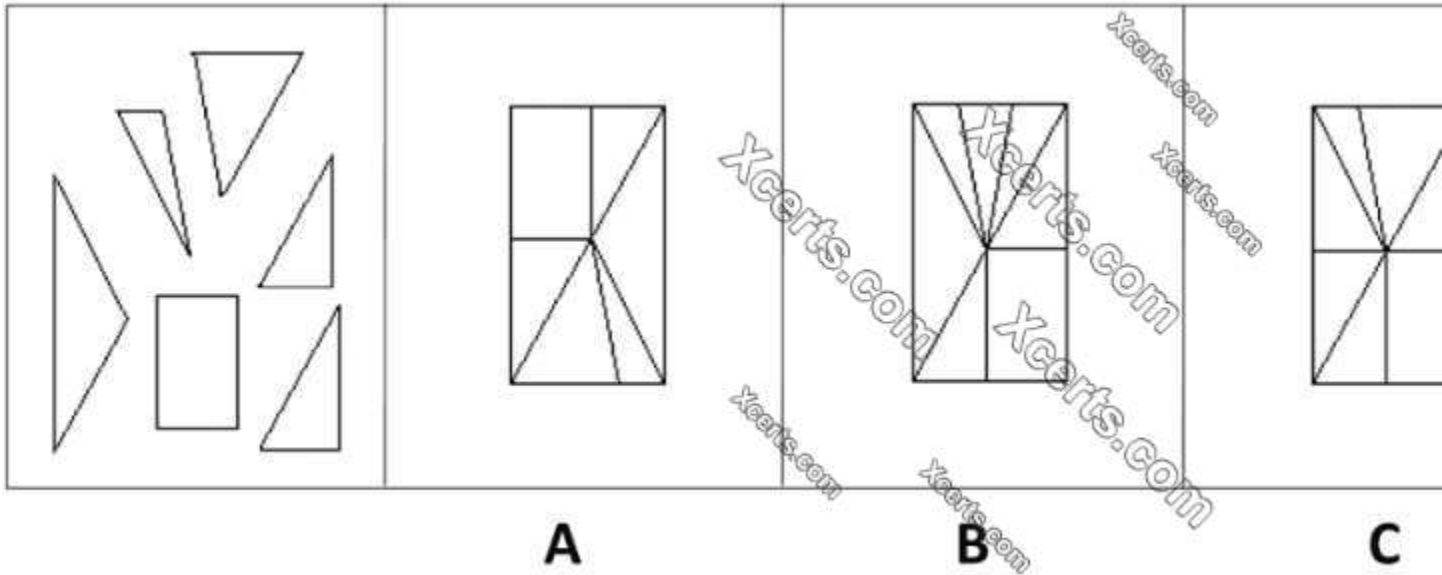


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1867

Which of these images best solves the problem in the first picture?

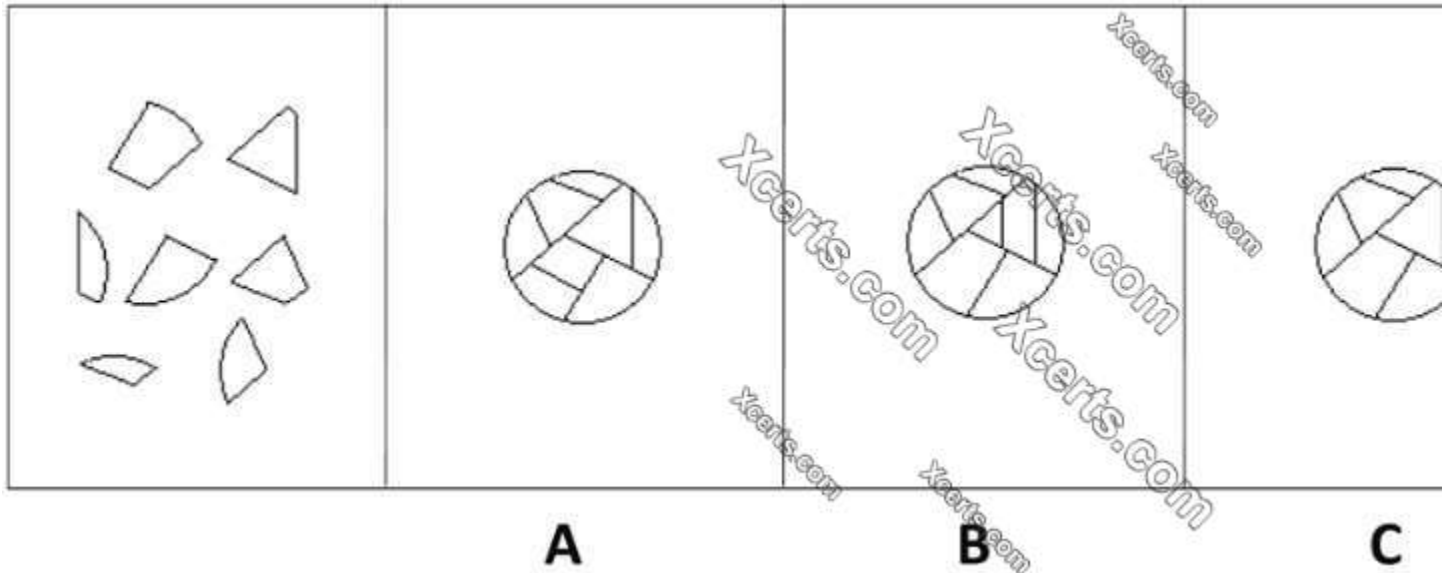


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1868

Which of these images best solves the problem in the first picture?

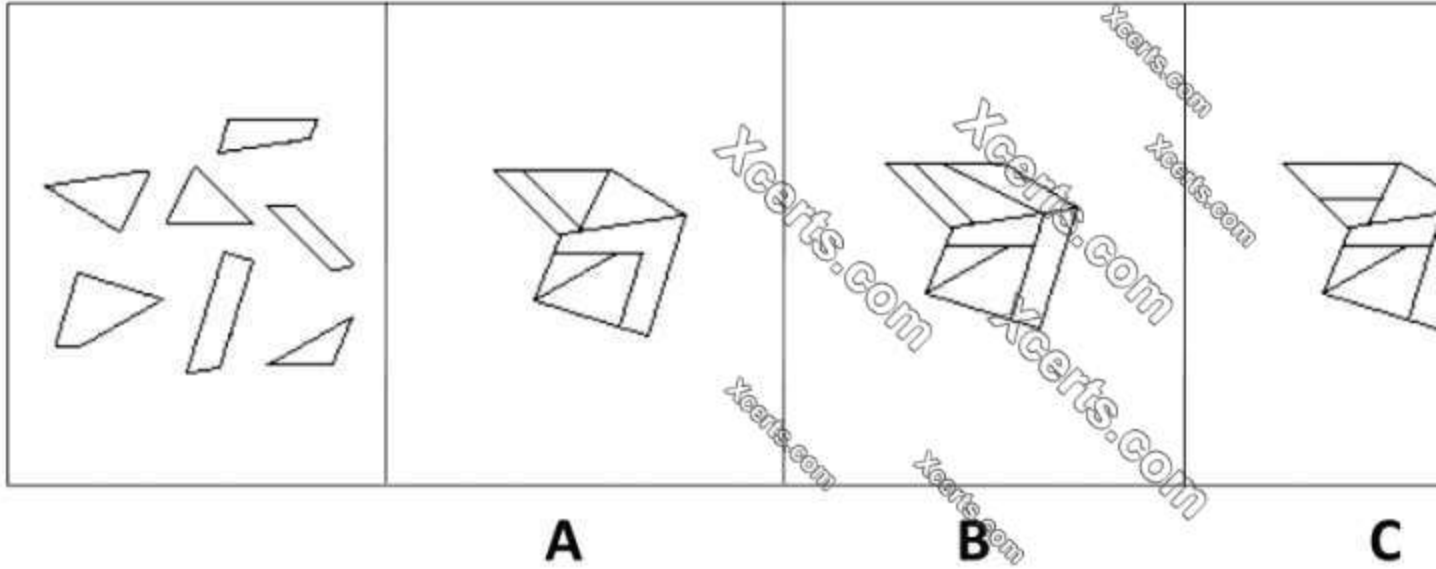


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1869

Which of these images best solves the problem in the first picture?

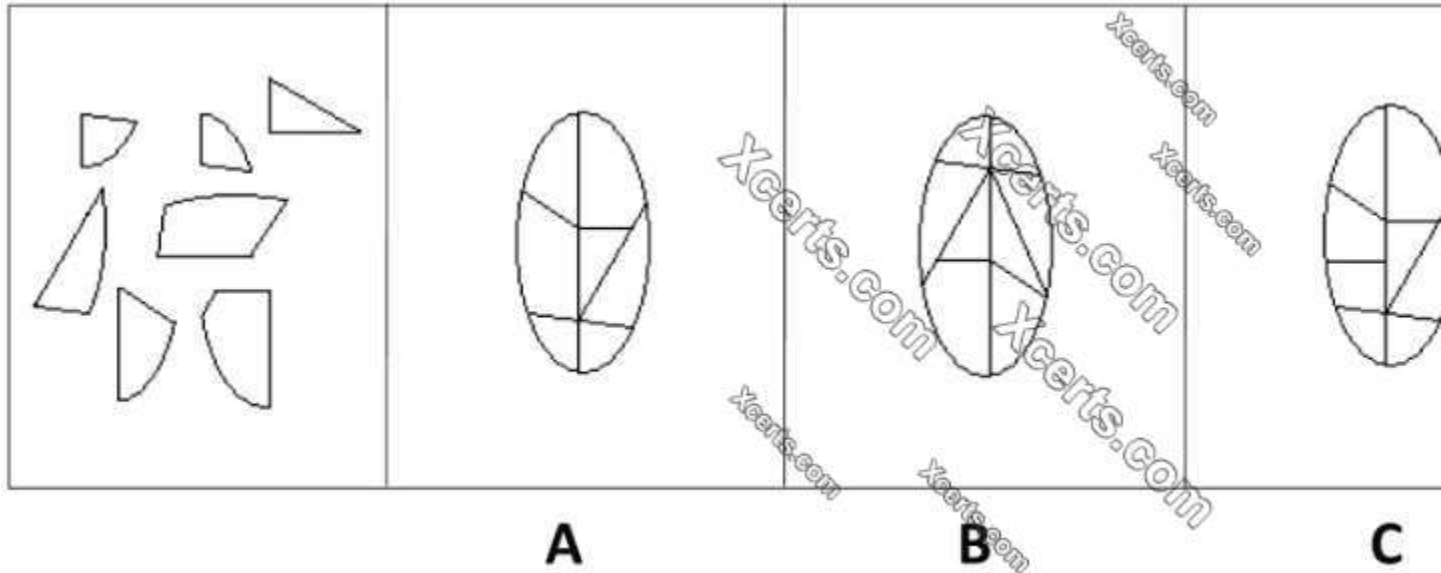


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1870

Which of these images best solves the problem in the first picture?

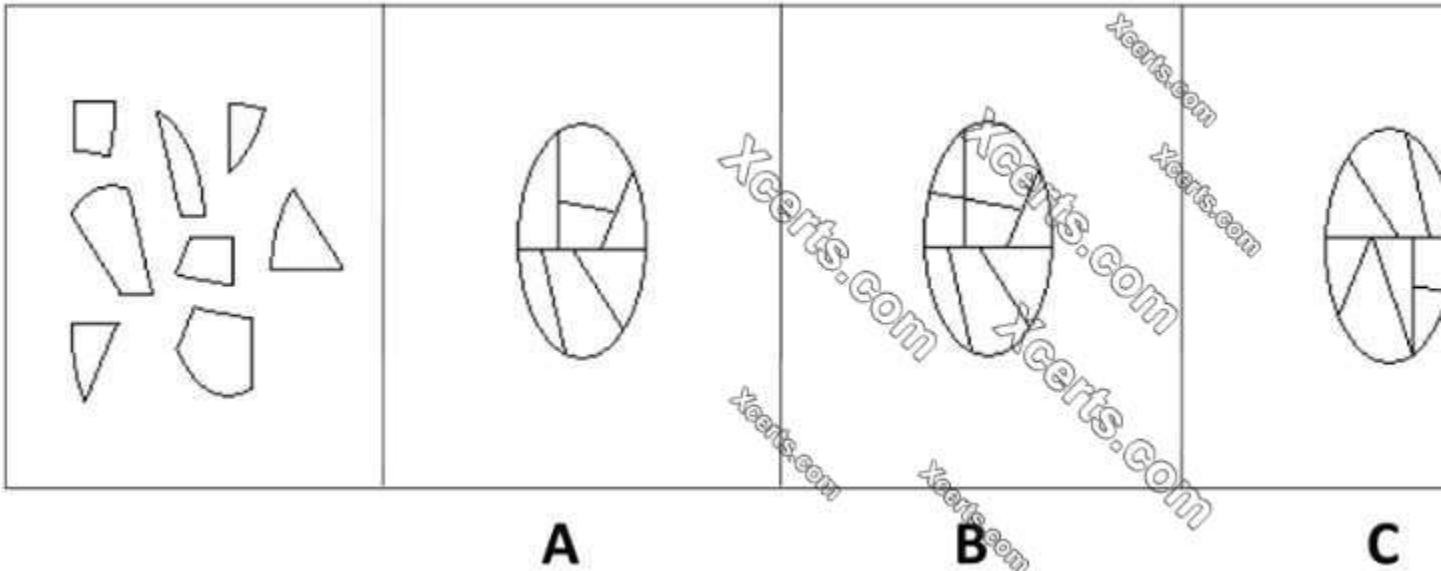


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1871

Which of these images best solves the problem in the first picture?

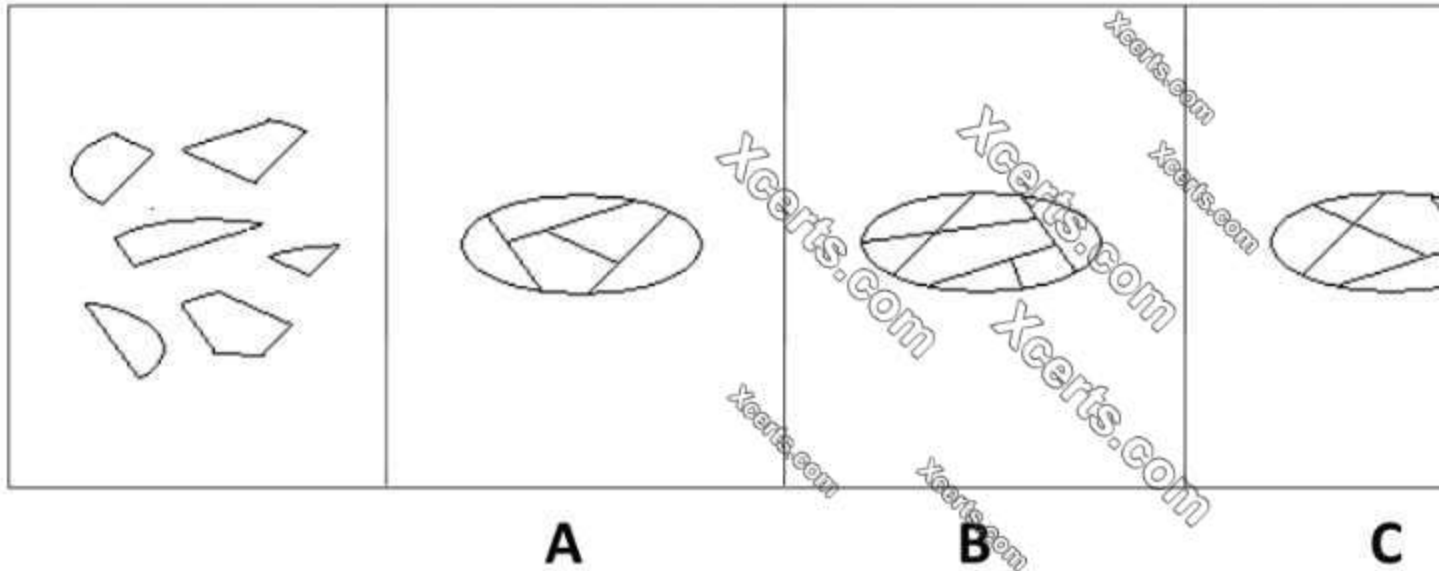


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1872

Which of these images best solves the problem in the first picture?

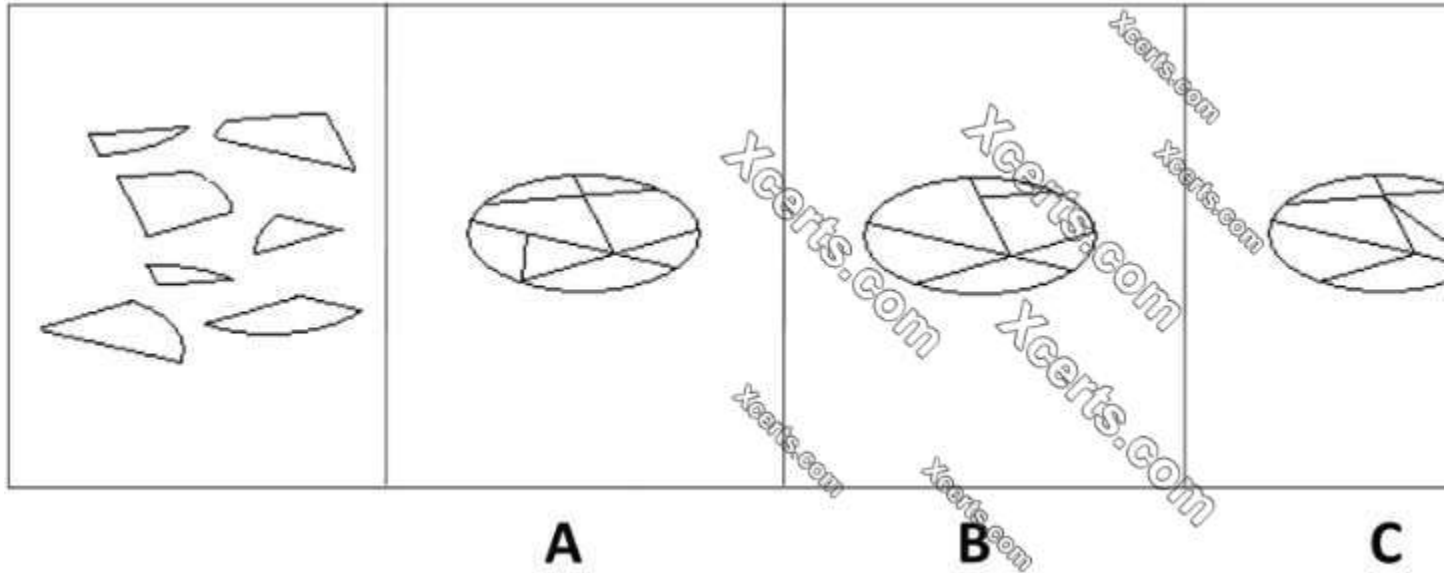


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1873

Which of these images best solves the problem in the first picture?

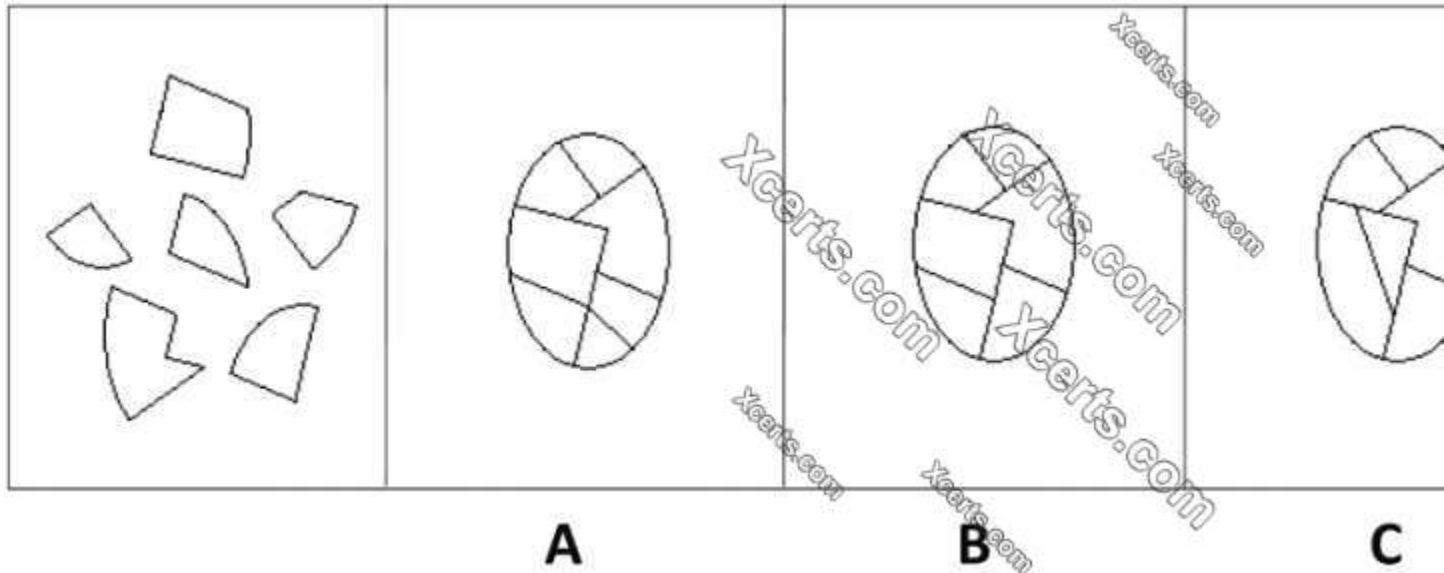


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1874

Which of these images best solves the problem in the first picture?

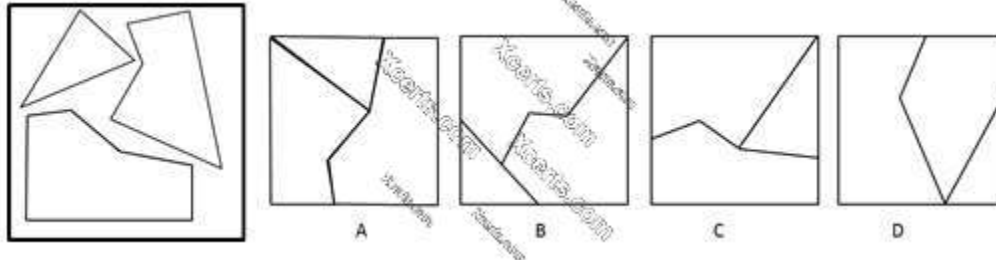


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1875

Which figure best shows how the objects in the left box will appear if they are fit together?

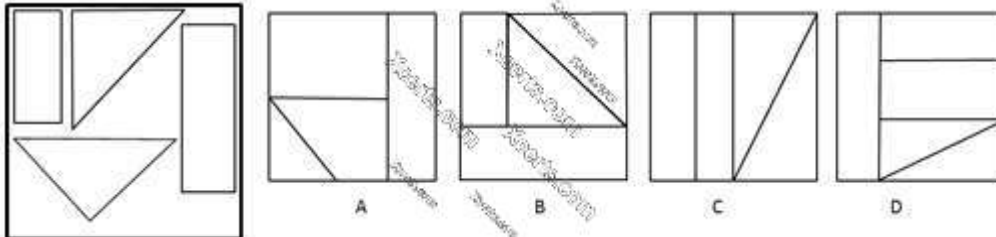


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1876

Which figure best shows how the objects in the left box will appear if they are fit together?

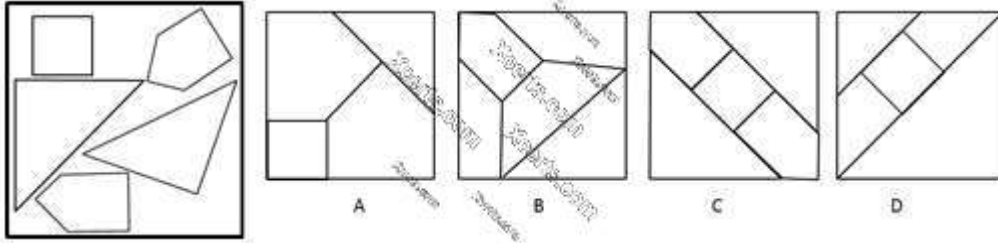


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1877

Which figure best shows how the objects in the left box will appear if they are fit together?

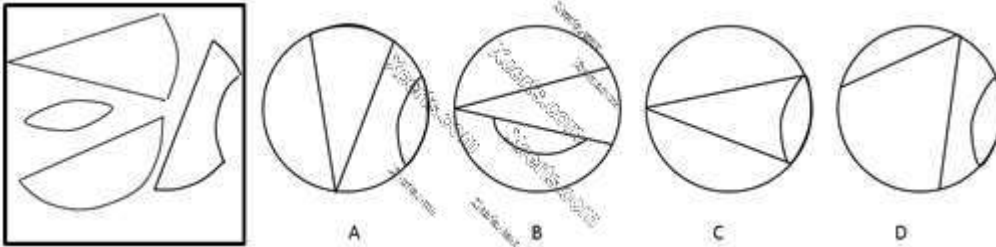


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1878

Which figure best shows how the objects in the left box will appear if they are fit together?

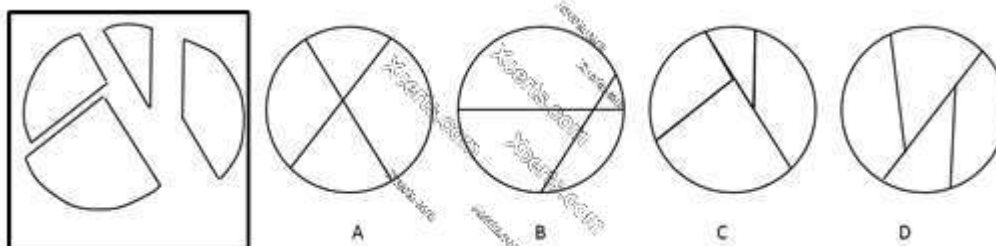


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1879

Which figure best shows how the objects in the left box will appear if they are fit together?

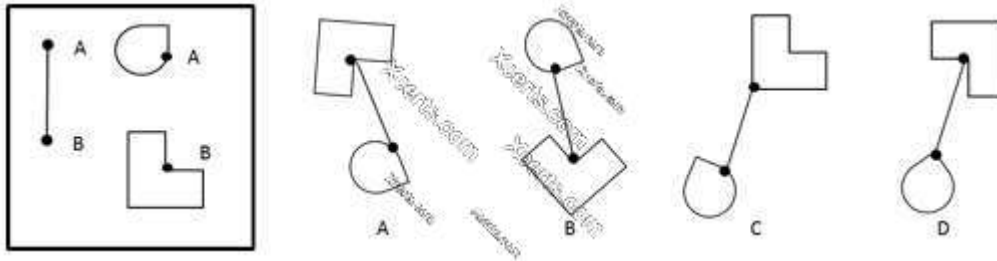


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1880

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

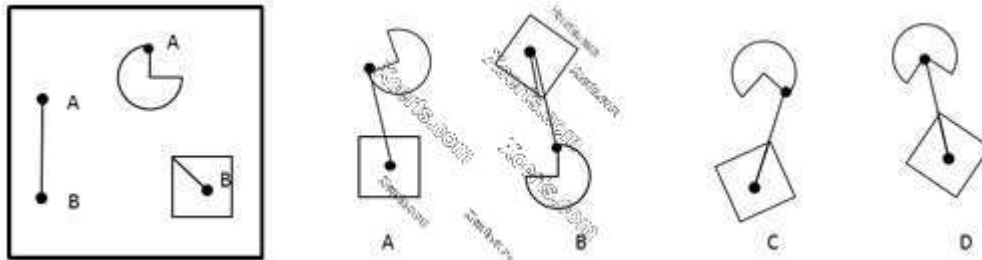


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1881

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

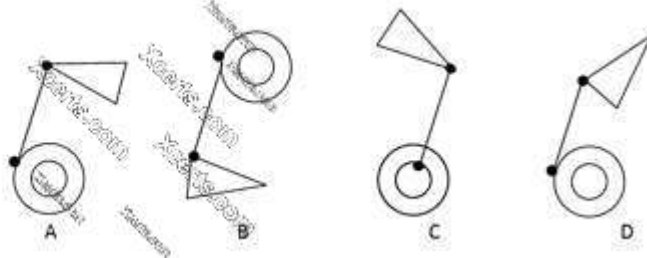
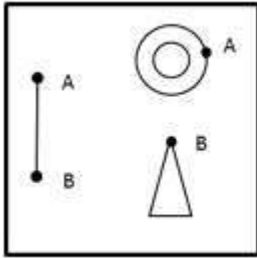


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1882

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

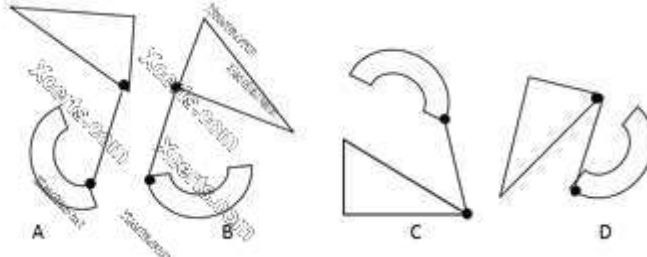
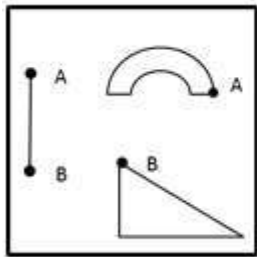


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1883

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

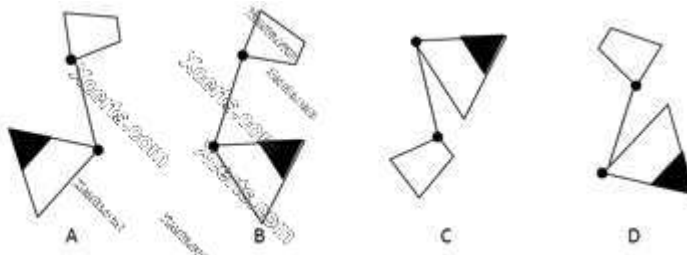
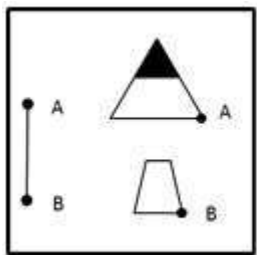


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1884

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?



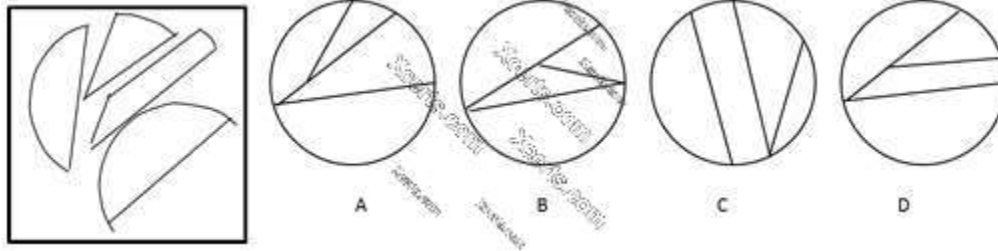
- A. Option A
- B. Option B

- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1885

Which figure best shows how the objects in the left box will appear if they are fit together?

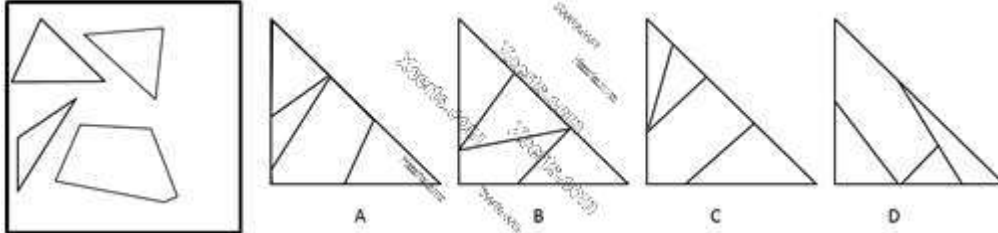


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1886

Which figure best shows how the objects in the left box will appear if they are fit together?

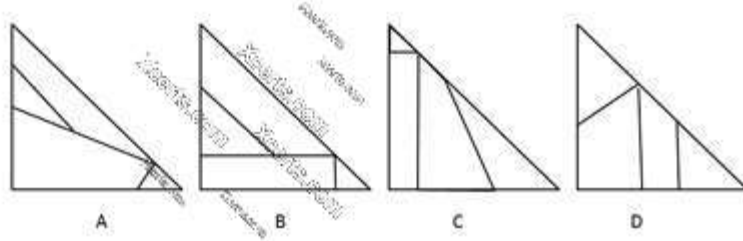
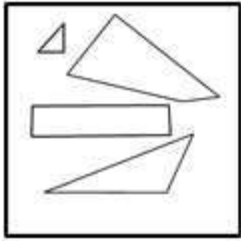


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1887

Which figure best shows how the objects in the left box will appear if they are fit together?

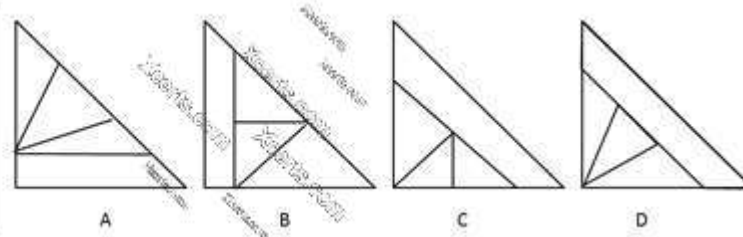
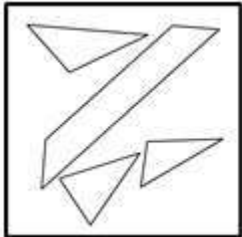


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1888

Which figure best shows how the objects in the left box will appear if they are fit together?

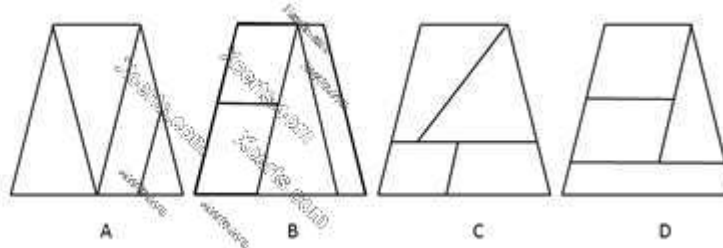
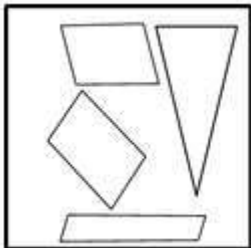


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1889

Which figure best shows how the objects in the left box will appear if they are fit together?

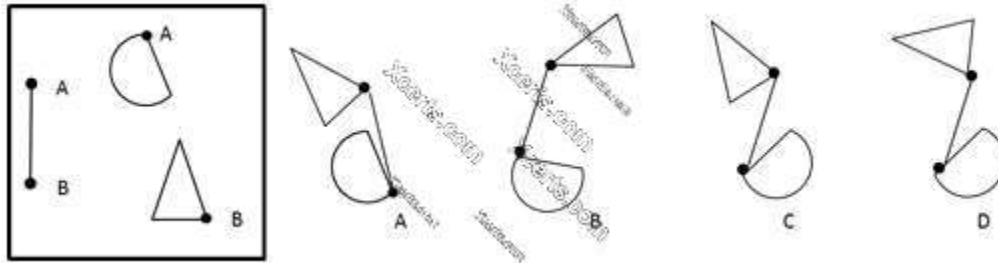


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1890

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

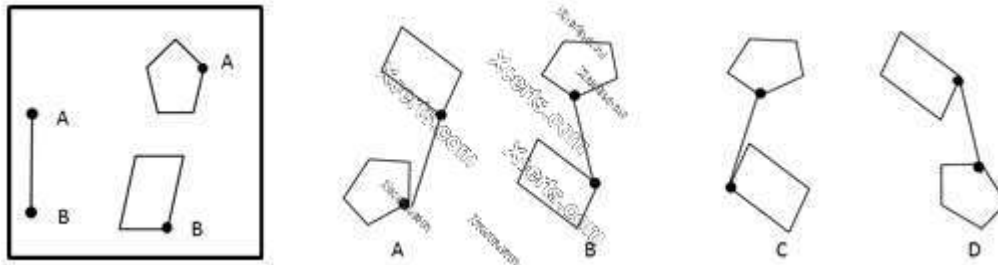


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): C

QUESTION: 1891

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

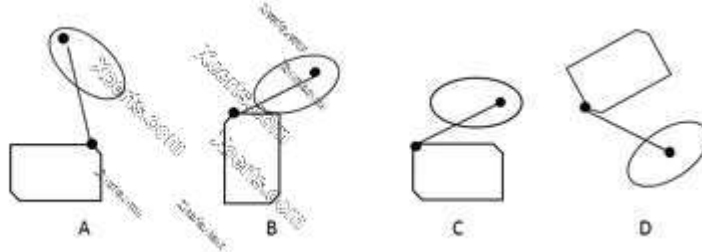
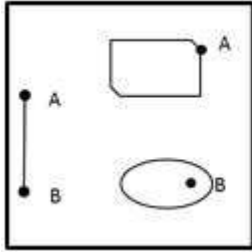


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): D

QUESTION: 1892

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

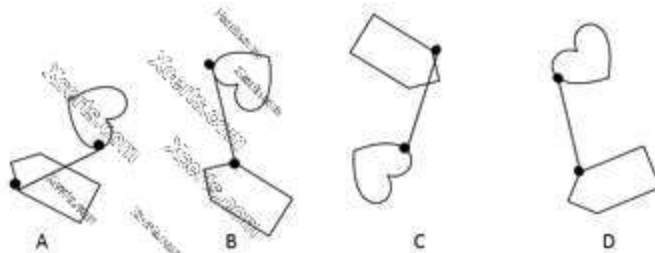
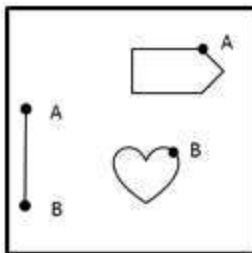


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): B

QUESTION: 1893

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?

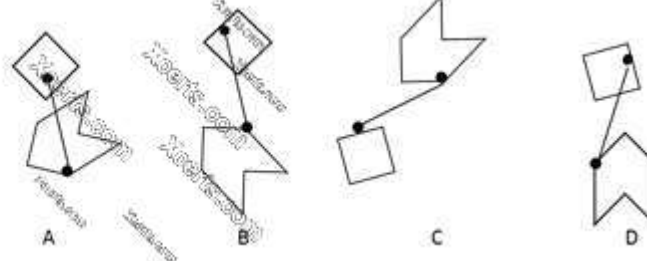
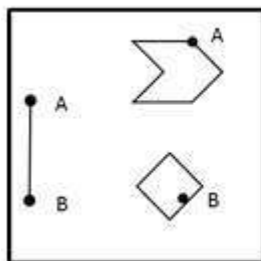


- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer(s): A

QUESTION: 1894

Which figure best shows how the objects in the left box will touch if the letters for each object are matched?



- A. Option A
- B. Option B

- C. Option C
- D. Option D

Answer(s): D